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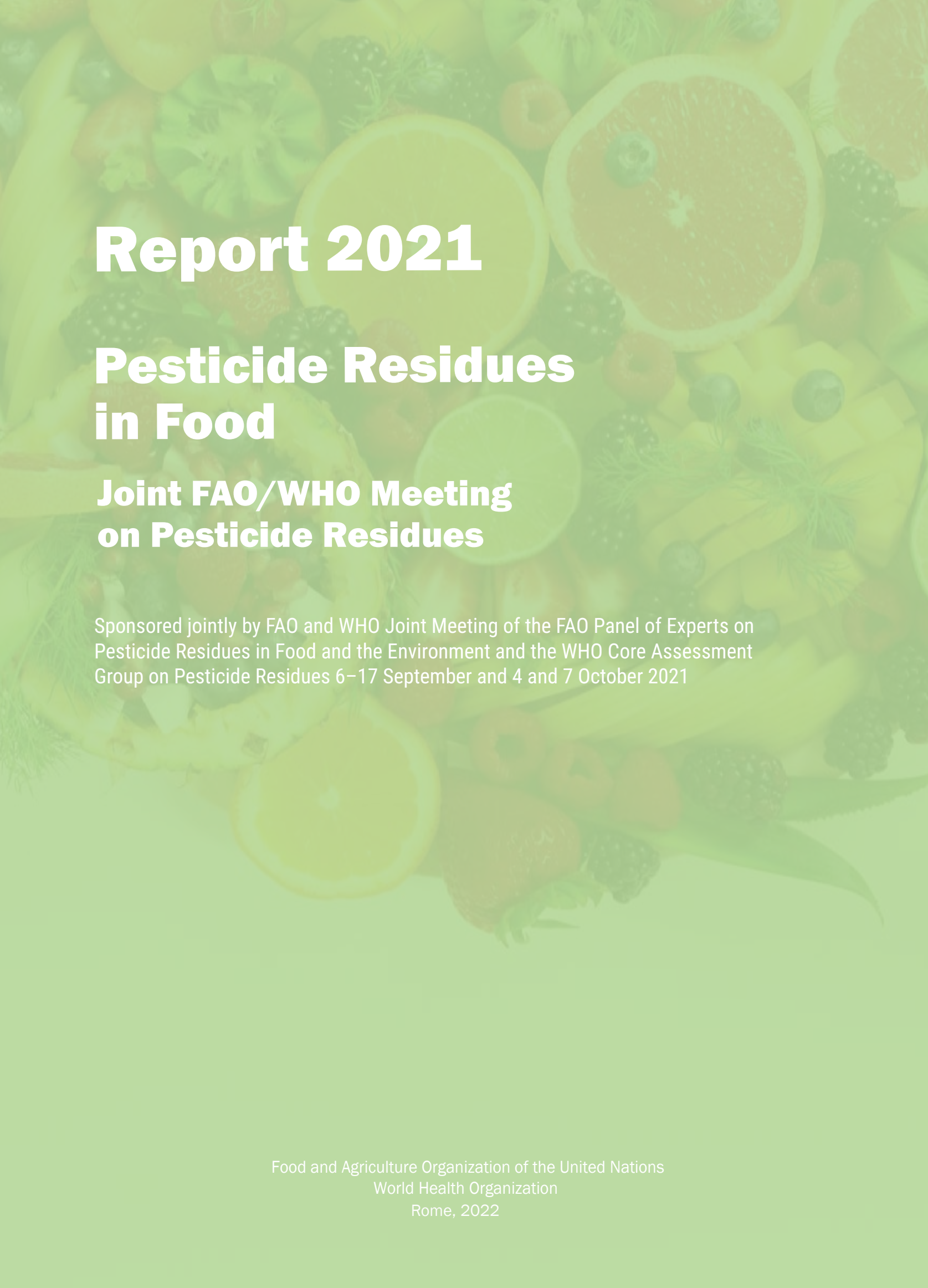


REPORT 2021

Pesticide residues in food

**Joint FAO/WHO Meeting
on Pesticide Residues**





Report 2021

Pesticide Residues in Food

Joint FAO/WHO Meeting on Pesticide Residues

Sponsored jointly by FAO and WHO Joint Meeting of the FAO Panel of Experts on Pesticide Residues in Food and the Environment and the WHO Core Assessment Group on Pesticide Residues 6–17 September and 4 and 7 October 2021

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R, residue and analytical aspects; T, toxicological evaluation

* New compound

** Evaluated within the periodic review programme of the Codex Committee on Pesticide Residues

List of participants

2021 Joint FAO/WHO Meeting on Pesticide Residues

FAO Experts

- Dr Christos Anagnostopoulos**, Benaki Phytopathological Institute, Athens, Greece
- Professor Eloisa Dutra Caldas**, Faculty of Health Sciences, University of Brasilia, Brazil
- Dr Julian Cudmore**, Health and Safety Executive, York, United Kingdom
- Dr Michael Doherty**, United States Environmental Protection Agency, Arlington, VA, USA. **(Chair)**
- Mr Jonathon Giordano**, United States, Environmental Protection Agency, Arlington, VA, USA
- Dr Jochen Heidler**, Federal Institute for Risk Assessment, Berlin Germany
- Dr Hidetaka Kobayashi**, Ministry of Agriculture, Forestry and Fisheries, Tokyo Japan.
- Mr Makoto Irie**, Food and Agricultural Materials Inspection Center, Tokyo Japan
- Ms Monica Le**, Pest Management Regulatory Agency, Health Canada, Ottawa, Ontario, Canada
- Dr Arion Leahigh**, United States Environmental Protection Agency, Washington, DC, USA
- Professor Mi-Gyung Lee**, Andong National University, Gyeongsangbuk-do, Republic of Korea
- Mr David Lunn**, Ministry for Primary Industries, Wellington, New Zealand
- Dr Dugald MacLachlan**, Australian Government Department of Agriculture, Water and the Environment, Canberra, ACT 2601, Australia
- Ms Karin Mahieu**, National Institute of Public Health and Environment, Bilthoven, The Netherlands
- Mr Daniel Poflowski**, Australian Pesticides and Veterinary Medicines Authority, Kingston ACT, Australia
- Mr Christian Sieke**, Federal Institute for Risk Assessment, Berlin, Germany
- Ms Monique Thomas**, Pest Management Regulatory Agency, Health Canada, Ottawa, Ontario, Canada
- Ms Trijntje van der Velde-Koerts**, Centre for Nutrition, Prevention and Health Services Bilthoven, The Netherlands
- Dr Yukiko Yamada**, International Food Safety Consultant, Adjunct Professor, Azabu University; Guest Scholar, National Institute of Health Sciences; Tokyo 1070062, Japan
- Dr Guibiao Ye**, Institute for the Control of Agrochemicals, Ministry of Agriculture and Rural Affairs, Beijing People's Republic of China

WHO Experts

- Mr Davide Arcella**, European Food Safety Authority (EFSA) Parma, Italy
- Ms Janis Baines**, Stirling, ACT 2611, Australia
- Professor Alan R. Boobis**, National Heart & Lung Institute Imperial College London, United Kingdom
- Dr Jessica Broeders**, Board for the Authorisation of Plant Protection Products and Biocides (Ctgb) Ede, The Netherlands
- Dr Marloes Busschers**, Courcenay Mardore (Thizy-les-Bourgs), France
- Dr Tamara Coja**, Austrian Agency for Health and Food Safety, Vienna, Austria

Dr Rhian Cope, Australian Pesticides and Veterinary Medicines Authority, Kingston ACT, Australia

Dr Amélie Crépet, French Agency for Health and Safety, Maisons-Alfort, France

Dr Ian Dewhurst, Leavening, North Yorkshire YO17 9SA, United Kingdom (Co- Chair)

Dr Thorhallur I. Halldorsson, University of Iceland, Reykjavik, Iceland

Dr Debabrata Kanungo, Food Safety Authority of India, Faridabad, Haryana, India

Dr Sheila Logan, Australian Pesticides and Veterinary Medicines Authority, Kingston ACT, Australia

Dr Kimberley Low, Pest Management Regulatory Agency, Ottawa, Ontario, Canada

Dr Elizabeth Mendez, United States Environmental Protection Agency, Washington DC, United States of America

Prof Angelo Moretto, University of Padova, Occupational Health Unit, Padova University Hospital
Via Giustiniani 2, 35128 Padova, Italy

Dr Pasquale Mosesso, Università degli Studi della Tuscia, Viterbo, Italy

Dr Lars Niemann, German Federal Institute for Risk Assessment, Berlin, Germany

Dr Prakashchandra V. Shah, Brookeville MD 20833, United States of America

Dr Luca Tosti, Department of Biomedical and Clinical Sciences - University of Milan, Milano, Italy

Dr Gerrit Wolterink, National Institute for Public Health and the Environment, Bilthoven, The Netherlands

Dr Midori Yoshida, Food Safety Commission of Japan, Tokyo, Japan

Dr Juerg Zarn, Federal Food Safety and Veterinary Office, Bern, Switzerland

JMPR Secretariat

Mr Kevin Bodnaruk, West Pymble, NSW 2073, Australia (*FAO Editor*)

Dr Jeevan Khurana, Lyneham, ACT 2602, Australia (*FAO Editor*)

Mr Soren Madsen, Department of Food Safety and Zoonoses (FOS), World Health Organization, 1211 Geneva 27, Switzerland (*WHO JMPR Secretariat*)

Ms Ngai Yin Ho, World Health Organization, Geneva, Switzerland (*WHO JMPR Secretariat*)

Dr Russell Parry, Shrewsbury SY2 6HZ, United Kingdom (*WHO Editor*)

Ms Yong Zhen Yang, Plant Production and Protection Division, Food and Agriculture Organization of the United Nations, Rome, Italy (*FAO JMPR Secretariat*)

Abbreviations

AChE	acetylcholinesterase
AD	administered dose
ADI	acceptable daily intake
ADME	absorption, distribution, metabolism, excretion
AGF	aspirated grain fractions
AR	applied radioactivity
as	as received
ARfD	acute reference dose
AUC	area under the concentration–time curve
BBCH	Biologische Bundesanstalt, Bundessortenamt und Chemische Industrie
BUN	blood urea nitrogen
bw	body weight
CAC	Codex Alimentarius Commission
CAR	constitutive androstane receptor
CAS	Chemical Abstracts Service
CCPR	Codex Committee on Pesticide Residues
cGAP	critical GAP
C _{max}	maximum concentration
DAA	days after application
DALA	days after last application
DAT	days after treatment
DM	dry matter
DMSO	dimethyl sulfoxide
DT ₅₀	time required for 50% dissipation of the initial concentration
dw	dry weight
EFSA	European Food Safety Authority
eq	equivalent(s)
ESI	electrospray ionization
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
GAP	good agricultural practice
GC-ECD	gas chromatography – electron capture detector
GC-MS	gas chromatography – mass spectrometry
GC-NPD	nitrogen-phosphorus detectors

GD	gestation day
GECDE	global estimate of chronic dietary exposure
GEMS	Global Environment Monitoring System – Food Contamination Monitoring and Assessment Programme
GI	gastrointestinal tract
GLP	good laboratory practice
HPLC	high performance liquid chromatography
HR	highest residue level in the edible portion of a commodity
HR-P	highest residue level in a processed commodity
IC ₅₀	half-maximal inhibitory concentration
IEDI	International Estimated Daily Intake
IENTI	International Estimate of Short-Term Dietary Intake
ILV	independent laboratory validation
ISO	International Organization for Standardization
IUPAC	International Union of Pure and Applied Chemistry
JECFA	Joint FAO/WHO Expert Committee on Food Additives
JMPR	Joint FAO/WHO Meeting on Pesticide Residues
LC ₅₀	median lethal concentration
LC-MS/MS	liquid chromatography-tandem mass spectrometry
LD ₅₀	median lethal dose
LLNA	local lymph node assay
LOAEL	lowest-observed-adverse-effect level
LOD	limit of detection
LOQ	limit of quantification
LP	large portion
MOA	mode of action
MRL	maximum residue limit
NOAEC	no-observed-adverse-effect concentration
NOAEL	no-observed-adverse-effect level
OECD	Organisation for Economic Co-operation and Development
PBI	plant-back interval
PC	post coitum
PF	processing factor
PHI	pre-harvest interval
Po	post-harvest
ppm	parts per million

PXR	pregnane X receptor
QuEChERS	quick, easy, cheap, effective, rugged and safe
QSAR	quantitative structure–activity relationship
RAC	raw agricultural commodity
RBC	red blood cell
RSD	relative standard deviation
RTI	re-treatment interval
SC	suspension concentrate
SDHI	succinate dehydrogenase inhibitor
SPE	solid phase extraction
STMR	supervised trials median residue
STMR-P	supervised trials median residue in a processed commodity
$T_{1/2}$	half-life
T3	triiodothyronine
T4	thyroxine
TLC	thin layer chromatography
T_{max}	time to reach maximum concentration
TPO	thyroid peroxidase
TRR	total radioactive residues
TSH	thyroid stimulating hormone
TTC	threshold of toxicological concern
UK	United Kingdom
USA	United States of America
WHO	World Health Organization

Use of JMPR reports and evaluations by registration authorities

Most of the summaries and evaluations contained in this report are based on unpublished proprietary data provided for use by JMPR in making its assessments. A registration authority should not grant a registration on the basis of an evaluation unless it has first received authorisation for such use from the owner of the data provided for the JMPR review or has received the data on which the summaries are based, either from the owner of the data or from a second party that has obtained permission from the owner of the data for this purpose.

Pesticide residues in food

Report of the 2021 Joint FAO/WHO Meeting on Pesticide Residues

1. Introduction

A Joint Meeting of the Food and Agriculture Organization of the United Nations (FAO) Panel of experts on Pesticide Residues in Food and the Environment and the World Health Organization (WHO) Core assessment Group on Pesticide Residues (JMPR) was held virtually, from 6 to 17 September and the 4th and 7th of October, 2021.

The FAO Panel Members held its pre-meetings in biweekly virtual preparatory sessions from July to September. The WHO Core Assessment Group had organized several preparatory sessions during 2020 and 2021. The Meeting involved 45 participants from five continents spanning most time zones.

The FAO JMPR Secretariat, Ms YongZhen Yang, welcomed all the experts, noting the number of new participants. She expressed her appreciation for the amount of preparatory work already undertaken by the experts in preparing documents and their participation in the virtual pre-meetings with their spare time in the early morning or during the night.

Mr Soren Madsen, WHO JMPR Secretariat, also welcomed the participants noting that the virtual format would make detailed and complex technical discussions challenging and that short concise interventions would be helpful in optimizing the outcome of the meeting. Particularly given the reduced meeting time of a total of 30 hours available for panel discussions during the Meeting, necessitated by the multiple time zones in which experts reside.

During the meeting, the FAO Panel of Experts on Pesticide Residues in Food was responsible for reviewing residue and analytical aspects of the pesticides under consideration, including data on their metabolism, fate in the environment and use patterns, and for estimating the maximum levels of residues that might occur as a result of use of the pesticides according to good agricultural practice. The methodologies are described in detail in the FAO Manual on the submission and evaluation of pesticide residue data for the estimation of maximum residue levels in food and feed (2016) hereafter referred to as the FAO manual. The WHO Core Assessment Group on Pesticide Residues was responsible for reviewing toxicological and related data in order to establish acceptable daily intakes (ADIs) and acute reference doses (ARfDs), where necessary and possible.

The Meeting evaluated 15 pesticides, including five new compounds and two compounds that were re-evaluated for toxicity or residues, or both, within the periodic review programme of the Codex Committee on Pesticide Residues (CCPR). The Meeting established ADIs and ARfDs, estimated maximum residue levels and recommended them for use by CCPR, and estimated supervised trials median residue (STMR) and highest residue (HR) levels as a basis for estimating dietary exposures.

The Meeting also estimated the dietary exposures (both acute and long-term) to the pesticides reviewed and, on this basis, performed a dietary risk assessment in relation to the relevant ADI and where necessary the ARfD. Cases in which ADIs or ARfDs may be exceeded, if they occur, are clearly indicated in order to facilitate the decision-making process by CCPR.

In addition, the Meeting considered a number of general issues addressing procedures for the evaluation and risk assessment of pesticide residues in food.

2. General considerations

2.1 Benefits and challenges to virtual JMPR meetings

The Meeting acknowledged the significant additional efforts made by the FAO and WHO secretariats to overcome challenges and logistical complexity for the 2021 JMPR meeting.

With restrictions still in place due to the ongoing COVID-19 pandemic, the 2021 JMPR was conducted in a virtual environment. As with the Extra 2021 Meeting, which was also held virtually, organizers needed to accommodate a 16-hour range in time zones for the individual experts. The meeting was originally scheduled for a two-week period, with meeting times set for three hours per day in an attempt to minimise unsocial hours across time zones, although that was frequently extended on an as-needed basis. At the end of the scheduled two-week period, additional discussion was needed on a number of topics, and two additional days were added. The follow-up meeting days were scheduled after a two-week break to allow participants time to advance the work under consideration.

Whereas the Extra Meeting focused on new uses, which did not involve establishing health-based guidance values or determining residue definitions, this Meeting focused on new compounds and compounds under the periodic review programme, for which more extensive reviews and discussions were required. The consideration of new compounds and those under periodic review are the most difficult and time-intensive topics that the JMPR handles as part of its regular business.

The WHO group has had positive experiences over the last 3 years in the use of virtual video/teleconference pre-meeting discussions and recognises their value as an additional tool to help streamline and enhance the efficiency of the physical meeting. The Meeting agreed that the virtual meeting format may be more beneficial for facilitating meeting planning and initiating pre-meeting discussions to advance decision making for certain topics than for conducting in-depth reviews of new compounds or periodic reviews. The Meeting noted the time commitments devoted to pre-meeting preparations and peer review even with the current reduced agenda were significant.

The Meeting agreed that there are some advantages to the virtual meeting format, including the absence of time lost in transit. While not aware of the details, the Meeting assumed that a virtual meeting can be accomplished at a lower financial cost to meeting organisers with a particular reduction in travel costs compared to an in-person meeting. The Meeting considered that this was at the expense of output, with the current Meeting considering 15 compounds on the agenda where 12 detailed evaluations were conducted for residues, toxicology or both, while for comparison a typical JMPR agenda would have in the region of 35 agenda items on average, so a significant reduction on a typical JMPR agenda¹. It was recognised that there was increased pressure on meeting participants resulting in additional costs for individuals and (where relevant) their organisations to cover the additional time commitments.

The virtual meeting environment presented many challenges, including the need for adequate and mutually compatible IT resources (for individual experts and FAO/WHO), competing demands for expert's time and attention, limited time to discuss issues and reach consensus, lack of side discussions (e.g. over lunch), and in many cases significant personal sacrifice for experts needing to work outside of normal business hours. These aspects were noted by the 2021 Extra Meeting, which elaborated on these challenges in more detail².

The Meeting agreed that conducting business by a virtual meeting platform provides a realistic way to accomplish some aspects of the work that needs to be addressed and is better than not meeting

¹ Agenda items: JMPR 2019 (29), 2018 (38) and 2017 (38)

² Section 2.1 of the JOINT FAO/WHO Meeting on Pesticide Residues Summary Report Acceptable Daily Intakes, Acute Reference Doses, Residue Definitions, Recommended Maximum Residue Limits, Supervised Trials Median Residue values and other values recorded by the 2021 Extra Meeting 17–21 May and 7–11 June 2021; Issued July 2021; <http://www.fao.org/3/cb5358en/cb5358en.pdf>

when there are extraordinary circumstances that prevent an in-person meeting. Overall, however, the virtual format is not favourable to the efficient completion of much of the work of the JMPR, especially aspects requiring in depth scientific discussions involving a number of contributors. A comparison of the agenda for this Meeting versus other recent annual meetings clearly demonstrates the restricted amount of work that JMPR was able to complete via the virtual format. The Meeting reiterated the conclusion from the 2021 Extra Meeting that “... continuation of the online-only meeting format is expected to give only limited benefits which overall are outweighed by counterproductive aspects which do not aid future JMPR decision making.”

2.2 International estimate of short-term intakes (IESTI) equations

The Joint FAO/WHO Meeting on Pesticide Residues (JMPR) derives acute dietary exposure estimates to carry out dietary risk assessments for compounds where an acute reference dose (ARfD) for a pesticide has been considered necessary and, since 1999, has used international estimate of short-term intake (IESTI) equations to make those estimates. Following its initial development, the methodology has been modified several times by the JMPR, but not all modifications have been adopted at a national level.

Following the EFSA/FAO/WHO workshop held in 2015 where modifications to the IESTI equations were discussed (EFSA 2015), the workshop outcomes were considered by the 2015 JMPR and a recommendation was made to CCPR that an FAO/WHO working group be established to compare results from the current and the proposed IESTI equations. In 2016, CCPR initiated an assessment of the IESTI equations in terms of their advantages, challenges, impact on risk management, consumer protection goals and trade. Four working papers on the IESTI equations have been considered by subsequent CCPR meetings. CCPR52 forwarded the most recent discussion paper (CCPR52 CX/PR 21/52/15) to the JMPR for consideration of the following topics:

- *Benefits/advantages and challenges of the current IESTI methodology* – consider a possible way forward to address the challenges identified in Table 2 of the discussion paper on issues that fall under the remit of JMPR.
- *Benchmarking of IESTI calculations against probabilistic exposure estimates* – consider comments submitted in response to CL 2021/42-PR and the final version of the acute probabilistic exposure assessment published in the paper by Crépet *et al.* (2021).
- *Review of the parameters of the IESTI equations: findings of FAO/WHO and published in peer reviewed literature* – discuss the need for developing further guidance on how to derive certain input values such as large portion, unit weight and the variability factor (LP, U, U_e, VF).
- *Information on bulking and blending relevant for IESTI Case 3* – The information should support discussions in JMPR to decide whether the list of commodities for which the exposure calculation is performed according to IESTI Case 3 needs to be revised.

Benefits/advantages and challenges of the current IESTI methodology

The CCPR working paper recognized the benefits of the current IESTI equation:

“they are transparent, easy to undertake, reproducible and use conservative assumptions that take national food consumption patterns into account. From a risk management perspective use of the IESTI equations provides a clear answer as to whether there is an expected risk of acute exposures to a pesticide residue exceeding the relevant ARfD due to consumption of specific commodities for which an MRL is recommended.”

The challenges identified were:

‘the level of uncertainty in data for variables used in the IESTI equations, difficulties in risk communication and the fact that national MRLs may not be fully harmonised with Codex MRLs

(CXLs)'.

The 2019 JMPR considered the IESTI equations as appropriate and fit for the purpose of estimating acute dietary exposure as part of its evaluation of pesticide residues. The Meeting acknowledged that the quality of the input information could be improved. Changes to the IESTI equations proposed by different countries, such as the removal of the unit weight (U_e) parameter and use of the MRL as the residue level instead of the highest residue from the field trial data (HR), could be further considered by the JMPR, pending output from the proposed FAO/WHO working group described below.

Conservative assumptions are used in IESTI equations to ensure that all populations considered in the risk assessment are sufficiently protected. For pesticides with an ARfD, the IESTI is calculated separately for each pesticide residue/commodity combination where a maximum residue level is estimated by the JMPR. Selection of the appropriate IESTI equation (Cases 1, 2a, 2b, 3) depends on the unit weight of raw fruit or vegetable commodities, pesticide treatment (pre- or post-harvest) and the level of bulking and blending of the commodity premarket. These are simple deterministic calculations in that they utilize single data points for food consumption and pesticide residue concentration rather than distributional data, and information cannot be provided on the distribution of dietary exposures nor can the uncertainty of the exposure estimate be quantified (FAO/WHO 2020).

Benchmarking of IESTI calculations against probabilistic exposure estimates

To investigate the extent to which the current IESTI equations provide sufficient consumer protection, results from a separate FAO/WHO research study were considered. This study by Crépet *et al.* (2021) was undertaken to benchmark the IESTI equations against probabilistic acute exposure estimates. The Meeting noted that to date the CCPR, as risk managers, have not nominated a specific level of protection to be met in pesticide residue risk assessments undertaken by the JMPR.

Crépet *et al.* (2021) assessed acute dietary exposure to pesticide residues from all commodities likely to contain the residue by undertaking a probabilistic dietary exposure assessment. The study was based on national food consumption and residue monitoring data, which served as a second tier, 'real world' estimate of the actual acute dietary exposure. Forty-seven pesticides with Codex MRLs (CXLs), an ARfD and the same residue definition for enforcement and dietary risk assessment were initially selected for inclusion in the study. Acute dietary exposure estimates for adults and children in eight countries were reported for 38 pesticides with adequate residue monitoring data.

The aim of the study was not to provide a comprehensive assessment for countries in all regions. Crépet *et al.* (2021) demonstrated that it is possible to conduct a more refined acute dietary exposure assessment that incorporated contributions from all commodities containing the residue of interest, despite some of the food consumption data sets for the countries included being incomplete.

Crépet *et al.* (2021) indicated that the probability of acute dietary exposures exceeding the relevant ARfD was null for all countries and populations considered (whole population, consumers only at the 97.5th percentile of exposure), even with a very conservative scenario based on assumed 100% usage of each pesticide in all foods in which it was permitted for use. Based on the probabilistic acute risk assessment results, the study authors concluded there was no appreciable risk to the population of adults or children in the eight countries studied. The results of the benchmarking study are supported by previous research studies, for example a study using US EPA pesticide residue monitoring data (Cleveland *et al.* 2019) where the study authors concluded that:

"alternate methodology choices are not expected to impact the large margins observed between the probabilistic estimates and the IESTI equations or to change the overall conclusion that existing IESTI equations are conservative and health-protective."

A level of protection (LoP) analysis was also performed in Crépet *et al.* (2021) using the same consumption data as for the acute dietary exposure estimates, but assuming that all food consumed contained pesticide residues at the Codex MRL for each of the 47 pesticides selected. The LoP was defined by the study authors as the percentage of person $\times$ days with acute exposure estimates from all

food sources at or below the ARfD when the residue occurs at the level of the MRL, which is highly unlikely to occur. In this analysis the LoP was almost always 100%, with only a few exceptions for a small number of age groups in some countries (see Table 5 in Crépet *et al.* 2021). The Meeting concluded that based on the very conservative assumptions used in the model, this would not constitute a public health concern.

The Meeting confirmed the 2019 JMPR conclusion that, based on the benchmarking study report, the current IESTI equations were considered protective for acute risk.

The Meeting noted that probabilistic estimates of acute dietary exposure would generally be used post-regulation to verify that pesticide standards such as the Codex MRLs are providing acceptable levels of consumer protection. The Meeting considered that it would be inappropriate to undertake probabilistic assessments pre-regulation because there are only limited analytical results available from supervised field trials.

Review of the parameters of the IESTI equations

In terms of providing guidance on input parameters for IESTI equations, the Meeting proposed an FAO/WHO expert group be established to ensure the most appropriate and scientifically robust input parameters (LP, U, VF) are used in IESTI assessments. This FAO/WHO expert group could, for example, discuss the following issues:

- Use of the LP_{bw} (g/kg body weight) calculated from single consumer day dietary records, expressed on an individual body weight basis rather than the LP (g/day) and generic body weight data.
- Determination of the minimum number of food consumption data points required for the derivation of a 97.5th percentile food consumption value with a 95th confidence interval for consumers of the commodity can be derived, noting research studies indicate that at least 120 data points are needed (Hamilton *et al.* 2004, EFSA 2014, [Ambrus and Szenczi-Cseh 2017](#)).
- Development of options for deriving the LP or LP_{bw} for infrequently consumed foods where the food consumption data do not support the derivation of a valid high-percentile consumption amount for a single food owing to a small data set (FAO/WHO, 2020).
- Confirmation of standard data formats for the submission of consumption and body weight data at an individual record level (rather than summary data) by countries to FAO/WHO, including details on how food consumption data for raw agricultural commodities have been derived from records of foods actually consumed.
- Development of agreed procedures for linking the food codes used in the WHO food consumption database (FoodEx2 food classification system) to the Codex food classification system, including use of recipes to disaggregate composite foods to ingredients, so that residue levels can be assigned in a consistent manner.
- Development of options for deriving and using unit weights of raw commodities, including exploration of the option of removing this parameter from the IESTI equations. In practice, as many countries do not submit unit weight data, the unit weight is often derived by JMPR from limited data for a few countries.
- Review of the available literature on residue variability in crops. JMPR currently supports the use of a single variability factor of three as it is based on a substantial amount of research data from a number of countries and pesticide applications in supervised trials that measured pesticide residues in single units in a number of different crops (Hamilton *et al.*, 2004; EFSA 2015; [Ambrus and Szenczi-Cseh 2017](#)).

One of the modifications to the current IESTI equations outlined in the 2015 EFSA/WHO/FAO report was to replace the HR level with the proposed MRL, which would result in a more conservative estimate than using the HR. The Meeting acknowledged that the limited number of residue data points

submitted in supervised trial studies contribute to a level of uncertainty in the HR variable. In contrast, the MRL takes into account the distribution of all selected field trial residue values as calculated by the OECD MRL calculator (OECD 2020). The Meeting noted that the overall impact of modifying several parameters in the IESTI equations has been investigated in several studies (Breyse *et al.* 2018, Richter *et al.* 2018, van der Velde-Koerts *et al.* 2018). These indicate that using MRLs as the high residue level, in combination with other modifications, such as removing the unit weight parameter from the Case 2a equation, compensated each other numerically to some extent. Research into the impact of substituting the MRL for the STMR/STMR-P value in the IESTI Case 3 equation indicates that the current IESTI Case 3 equation makes a conservative assumption about the residue level in bulked and/or blended commodities, and is adequate for consumer protection purposes. Substituting the MRL for the STMR/STMR-P value would result in a potentially unnecessary and appreciable over-estimation of acute dietary exposure for these commodities (van der Velde-Koerts *et al.*, 2018). The Meeting noted that the impact of changing the IESTI equation as outlined in the 2015 EFSA/WHO/FAO report would be larger for those compound/commodity combinations where the acute dietary exposure estimate approaches the level of the ARfD.

Information on bulking and blending relevant for IESTI Case 3

For assessing acute dietary exposure to pesticide residues in most commodities the JMPR uses the highest residue (HR) derived from residue studies. Exceptions to this are commodities that are: [1] treated pre-harvest and are sufficiently bulked and blended from multiple producers prior to consumption, and [2] processed commodities which are sufficiently bulked/blended prior to or during processing.

For these commodities, the use of a central-tendency residue level is more appropriate for assessing exposure, and the JMPR uses a median residue level (STMR or, for processed commodities, STMR-P). In 2018, CCPR50 agreed:

“to gather relevant information on bulking and blending, in order to feed into the risk assessors’ work through the JMPR Secretariat”

as part of ongoing work to evaluate the need for changes to the equations used to derive International estimated short-term intake (IESTI). Subsequently, CCPR52 (2021) agreed to provide:

“information on bulking and blending relevant for IESTI Case 3 [...] to JMPR for further evaluation/consideration. The information should support discussions in JMPR to decide whether the list of commodities for which the exposure calculation is performed according to IESTI Case 3 needs to be revised”.

The Meeting welcomed the opportunity to review the data on bulking and blending and to incorporate any findings into its dietary risk assessment practices. However, only a listing of the available data was made available to the Meeting. This issue will be considered when the data have been provided to the JMPR.

Conclusion

The Meeting confirmed the 2019 JMPR conclusion that, on the basis of the probabilistic benchmark study of acute dietary exposures for high consumers (97.5th percentile of consumer-only exposure), the current IESTI equations used as part of JMPR risk assessments are fit for the purpose of ensuring consumer protection and provide confidence that adoption of recommended MRLs is not expected to result in a public health concern.

The Meeting noted that the modifications to the IESTI equations discussed at the 2015 EFSA/FAO/WHO meeting are not expected to change the conclusions of the risk assessment in terms of consumer protection, but introduce an additional degree of conservatism based on the benchmarking analysis. The Meeting further noted that the absence of quantitative consumer protection goals clearly

formulated by CCPR does pose a challenge for determining the appropriate level of conservatism of the IESTI equation.

The Meeting proposed that FAO/WHO establish an expert working group to develop guidance that ensures the most appropriate and scientifically robust data for the input parameters is available for use in IESTI equations, and to further consider the impact of possible modifications to the IESTI equations in relation to the unit weight and residue level parameters.

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2.3 First considerations on a possible need for amendments to EHC 240 guidance on appropriate use of toxicological historical control data (HCD)

The meeting discussed a fundamental set of principles that, it was proposed, needed to be established before guidance for harmonized use of HCD could be drafted. Among other things, the discussions pointed to the need for a detailed delineation of the possible causes of study-to-study variability, which may be due to different study conditions and thus different influencing factors. Based on such a consolidated cause–effect analysis, criteria should be developed that enable decisions regarding in which circumstances and for which endpoints the use of HCD can serve as a basis for assessment, and the circumstances/endpoints where it cannot.

The Meeting was informed about an ongoing project launched by the European Food Safety Authority (EFSA; scheduled to last until Autumn 2022), the aim of which is to collate all relevant information on stakeholder experience, knowledge and understanding of the use and interpretation of HCD when evaluating toxicity studies. The FAO/WHO Joint Meeting on Pesticide Residues (JMPR), as one of the relevant stakeholders, will be approached to participate in the planned survey and in the workshop, to be held in Spring 2022.

It was agreed that the JMPR will actively participate in the above mentioned survey and the workshop, and that the possibility of developing JMPR Guidance on HCD would be discussed at the next Meeting, taking into account progress of the EFSA project.

2.4 Guidance on the assessment and interpretation of non-linear dispositional kinetics

The 2019 Joint FAO/WHO Meeting on Pesticide Residues (JMPR) recognised that further work was needed to accommodate the increasing use of kinetically-derived maximum dose (KMD) as an alternative to maximum tolerated dose (MTD). Accordingly, the Joint Secretariat convened a group of experts to undertake preparatory work for the development of guidance on the assessment and interpretation of non-linear dispositional kinetics and the KMD-based evaluation of pesticide residues.

In guideline toxicology studies, chemicals (including pesticides) are evaluated using a dose-selection protocol that includes either MTD or a limit dose, designed to maximize the detection of any toxicity due to the treatment, observed in experimental. The introduction of concurrent in-life toxicokinetics into repeat-dose studies has revealed that in toxicity studies the dispositional kinetics at high dose levels may exhibit dose-related non-linearity. Information derived at such dose levels may be less useful, more difficult to interpret in relation to human exposures to chemicals, and may not be compatible with modern animal welfare considerations.

Non-linear kinetics can result from the saturation of absorption, distribution, metabolism or excretion, or any combination of such saturations. Adding to this complexity, non-linear kinetics can apply to the parent, its metabolites or both.

The KMD approach, which is based on evidence of dose-associated dispositional non-linear kinetics (dose non-proportionality), is now being used by sponsors as an alternative to the MTD approach for setting the dose range and top dose in animal toxicity studies. The KMD approach is likely to contribute particularly to the evaluation of the carcinogenicity observed at high doses and the results of teratogenicity studies conducted using oral gavage. As a result, the Meeting agreed that guidance on the integration of the KMD approach into JMPR evaluations is needed, and that such guidance should not focus specifically on KMD but more generally on the interpretation of non-linearity in the dispositional kinetics of pesticides.

Such guidance on the interpretation of toxicology studies is needed to increase the scientific quality, consistency and transparency of JMPR assessments. Currently it is proposed that any guidance on these issues should focus on JMPR's terms of reference, which relate to the risks from residues of pesticides in food, rather than hazard classification. A partial list of questions that the proposed guidance could consider includes:

- How should non-linearity be interpreted and evaluated?
- What is the minimum data base need to sufficiently evaluate issues pertaining to non-linearity?
- Does the presence of non-linearity sufficiently justify not investigating the effect of higher doses, and should the proposed electronic working group recommend a cut-off based on the occurrence of non-linear kinetics?
- Should non-linear absorption, non-linear target tissue exposure and other non-linear events be considered separately?
- Can guidance be provided on the design and interpretation of in vitro evidence of non-linearity in kinetic processes?

The Meeting agreed the composition of an electronic working group and charged the group to prepare draft guidance for discussion at a future JMPR meeting.

2.5 Recommendations for use of leafy vegetables to extrapolate residues to the Subgroup 027A Herbs (herbaceous plants).

Some delegations at the Fifty-second Session of the CCPR expressed concerns that the 2019 JMPR had utilised residues in mustard greens to extrapolate to herbs rather than using leaf lettuce and spinach, the representative commodities recommended by CCPR in “Principles and Guidance on the Selection of Representative Commodities for the Extrapolation of Maximum Residue Limits for Pesticides to Commodity Groups (CXG 84-2012)”.

The Meeting recalled that CXG 84-2012 allows for the use of alternative representative commodities provided these can be justified. The following provides the justification used by the JMPR in selecting mustard greens to extrapolate residues to the Sub-group 027A Herbs.

The principles used for the selection of representative commodities listed in CXG 84-2012 are:

- A representative commodity is most likely to contain the highest residues.
- A representative commodity is likely to be major in terms of production and/or consumption.
- A representative commodity is most likely similar in morphology, growth habit, pest problems and edible portion to the related commodities within a group or subgroup.

To provide an evidence-based justification for extrapolation, a review was conducted of the residue potential of crops in the Subgroup 027A, Herbs (herbaceous plants).

Residues of foliar applied pesticides are to a large extent governed by the initial spray deposits which in turn depend on a number of plant parameters including the relative surface area of the leaves and stems, the wettability of the leaf surfaces (waxy surface versus hairy surface etc.) as well as crop morphology.

Residues on the day of application of foliar sprays provide a good indication of relative residue potential for different commodities, with the ranking of residue potential largely preserved with increasing time after application even with relative differences in growth dilution within a group or subgroup and the potential impact on residues at longer post-application intervals.

A measure of the initial spray deposits can be gained by collating residue levels in the commodities on the day of application following a single spray. To expand the database, the Meeting considered that data from trials where more than one spray had been applied could be used provided there was sufficient evidence to conclude that the earlier spray did not contribute significantly to the observed residue. The Meeting utilised JMPR evaluations in the period 1993 to 2019 and supplemented these with other publicly available information such as published scientific papers and EU draft Assessment Reports to assemble a database of initial residue levels normalised to an application rate of 1 kg ai/ha.

As only a small number of trials were located, where decline information was available this was used to correct for the contribution to the terminal residue (day 0) from earlier sprays and thus further expand the database.

A summary of the initial residue deposits for the different commodities is shown in Figure 1 in the form of box-plots. The boxes cover 50% of values (25th to 75th percentiles) while the whiskers cover 95% of values with the median represented by the dark horizontal lines.

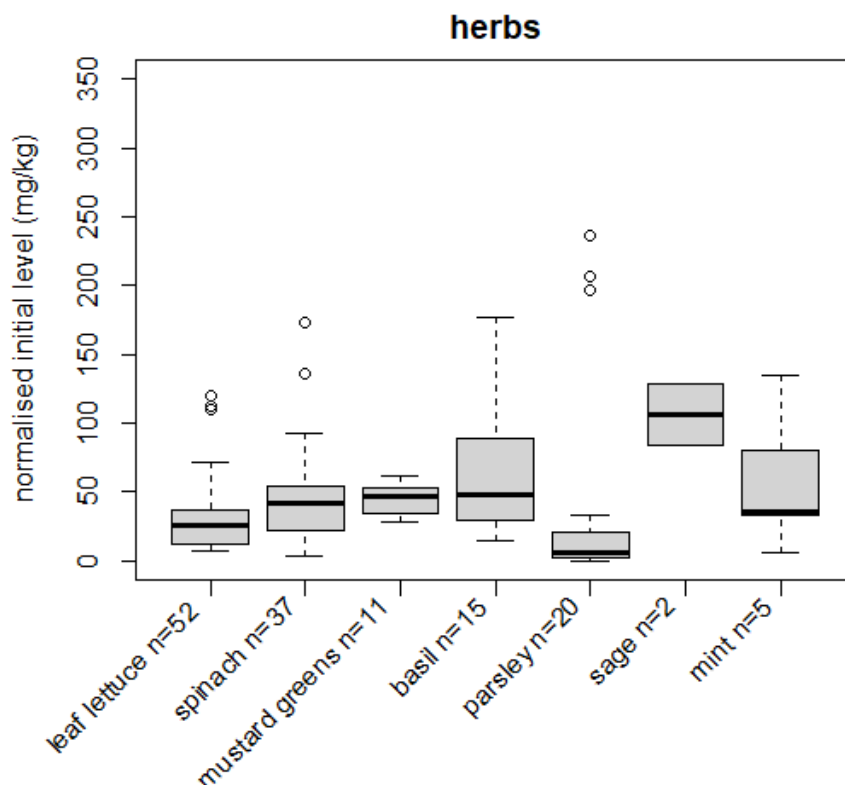


Figure 1 Initial residue (normalised to an application rate of 1 kg ai/ha) for herbs and for leaf lettuce, spinach and mustard greens

Table 1 provides a summary of the mean and median normalised residues observed for each crop.

Table 1 Day-0 residues normalised for application rate

Commodity	n	Median (mg/kg)	Mean (mg/kg)
Basil	15	47.9	64.5
Mint	5	36.3	58.0
Parsley	20	5.7	38.5
Sage	2	-	106
Leaf lettuce	52	26.4	31.3
Spinach	37	42.0	46.5
Mustard greens	11	46.9	44.9

Basil: clethodim²⁰, cyazofamid¹⁴, lambda-cyhalothrin¹⁴, oxathiapiprolin¹⁵, spirotetramat²⁰

Mint: abamectin¹⁵, clethodim²⁰, spirotetramat²⁰

Parsley: acetamiprid², azoxystrobin¹⁹, chlorpyrifos-methyl¹⁸, cypermethrin¹⁹, deltamethrin¹⁹, difenoconazole¹⁹, dimethoate¹⁸, imidacloprid², iprodione¹⁸, lambda-cyhalothrin¹⁹, metalaxyl-M^{18,19}, permethrin¹⁸, pirimicarb¹⁹, propargite¹⁸, spinosad¹⁹, spirotetramat²¹, tebuconazole¹⁹, thiacloprid¹⁹

Sage: clethodim²⁰, spirotetramat²⁰

Leaf lettuce: beta-cypermethrin²³, butocarboxin³, chlorantraniliprole¹¹, chlorothalonil²³, chlorpyrifos^{22,23}, cyantraniliprole¹³, alpha-cypermethrin¹¹, zeta-cypermethrin¹¹, deltamethrin²³, dimethoate^{6,11,23}, fluopyram¹², iprodione⁵, pirimicarb¹⁰, thiamethoxam¹², trichlorfon²³

Mustard greens: Spinosad⁸, thiamethoxam¹²

Spinach: chlorantraniliprole¹¹, chlorpyrifos¹⁷, alpha-cypermethrin¹¹, deltamethrin⁹, diazinon⁴, dimethoate⁶, fluazifop-P-butyl¹⁷, lambda-cyhalothrin¹⁷, metalaxyl¹⁷, methomyl⁸, mevinphos⁷, thiamethoxam¹², thiophanate-methyl¹⁶

Normalised day-0 residues for single residue trials on coriander¹, dill¹, marjoram²⁴ and thyme²⁴ were 13.4, 21.1, 43.6 and 51.2 mg/kg, respectively.

The available trials on basil, parsley, sage and mint demonstrate that residue levels in herbs from Subgroup 027A are closer to those of mustard greens than those of leaf lettuce. The Meeting confirmed that mustard greens and spinach are suitable representative crops for extrapolation from a leafy vegetable to Subgroup 027A Herbs.

The JMPR encourages the CCPR and its members to consider the above evidence-based approach in the selection of representative crops for establishing MRLs.

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3. Responses to specific concerns raised by the Codex Committee on Pesticide Residues (CCPR)

3.1 Afidopyropen (312)

Concern #1 – On the inclusion of M007 in the residue definition of risk assessment of plant commodities

Background

Afidopyropen was evaluated as a new compound by the 2019 JMPR for toxicity and residues.

The following residue definitions were established for afidopyropen in plant commodities:

For compliance with the MRL: *afidopyropen*

For dietary risk assessment: *sum of afidopyropen + M007, expressed as afidopyropen.*

The toxicity of the dimer M440I007 (M007) was evaluated by the 2019 JMPR. It was concluded that *“The dimer was not considered to be of greater toxicity than the parent. The ADI and ARfD cover both parent and the dimer.”*

The current Meeting received a concern form from the Delegation of the United States of America (USA), relating to the residue definition for dietary risk assessment for plant commodities. The delegation noted that addition of M440I007 in the risk assessment definition does not reflect the known significantly lower toxicity of the M440I007 metabolite; it is very conservative to add the parent and M440I007 metabolite and then assess against the parent health-based guidance values (especially for the ARfD, where there is no evidence of acute toxicity). A clearer explanation addressing this point was requested from the JMPR.

Comments by the JMPR

Despite the claim from the manufacturer that this metabolite is less toxic than the parent, the WHO concluded that M007 is “of similar toxicity” to the parent. The WHO based this conclusion on the common finding of changes in the myocardium. These changes were also seen at the highest dose tested in the short-term study with M007, but the authors didn’t do a histopathological assessment of the lower doses. Consequently, the WHO could not establish whether this effect did not occur at lower levels and subsequently could only conclude that M007 is “of similar toxicity”. No new information on the toxicity of the dimer was provided to the current Meeting with the concern form.

The dimer is not observed in the rat metabolism and as such, the toxicity of the dimer is not “covered by parent”. NOAELs/LOAELs for parent (in mg/kg bw) and for M007 (in mg/kg bw) for the critical effects are very close (same effect at the same dose). Since in this case both parent and the dimer are of similar toxicity (in terms of dose per kg bw), the dimer does not need to be normalized to parent equivalents as is usually done for metabolites that are covered by parent.

As such, the residue definition proposed by the 2019 JMPR as *“Sum of afidopyropen + M007, expressed as afidopyropen”* was imprecise and should have been *“Sum of afidopyropen + M007”*. In this case, stoichiometric correction to parent compound equivalents ($\times 2$) and for molecular weight ($\times 0.5$) would result in an overall factor of 1 and lead to a simple summation of residue concentrations.

The toxicity data provided on the dimer M007 are insufficient to conclude that it is of lower toxicity than the parent. To justify the use of a lowering potency factor or even exclusion of the dimer from the residue definition, supportive data are needed. A histopathological examination of the relevant (myocardial) tissues for the mid and low-dose groups could prove that M007 is “of less toxicity”. The manufacturer is kindly asked to submit this data.

The Meeting decided to rephrase the residue definition for dietary risk assessment for plant commodities to *the sum of afidopyropen + dimer of [(3R,6R,6aR,12S,12bR)-3-[(cyclopropanecarbonyl)oxy]-6,12-dihydroxy-4,6a,12b-trimethyl-11-oxo-9-(pyridin-3-yl)-1,3,4,4a,5,6,6a,12,12a,12b-decahydro-2H,11H-naphtho[2,1-b]pyrano[3,4-e]pyran-4-yl]methyl rac-cyclopropanecarboxylate (M007).*

The change in the residue definition for dietary risk assessment does not impact the previous estimates of dietary exposure calculated by the 2019 JMPR because in the previous evaluation the residue concentrations were added without further adjustment in agreement with the corrected expression of the definition.

Concern #2 – On the low MRL for milk

Background

A second concern was raised by the Delegation of the USA regarding the very low maximum residue level recommendation drafted for milk at 0.001 mg/kg. It was indicated that, though supported by current methods, this value is very low for practical monitoring in trade. A typical default of 0.01(*) mg/kg would be better harmonized with enforcement practices and more useful to the international trading community and it was recommended that this 0.01(*) mg/kg CXL value, be used instead.

Response of the JMPR

The JMPR acknowledges that the maximum residue level for milk is low but is indeed supported by current analytical methods. Furthermore, the use of (very) low maximum residue levels in milk is observed for many other active substances. The JMPR always recommends maximum residue levels based on expected residues in food or feed commodities and the technical capabilities of validated analytical methodologies.

3.2 Benomyl (069), carbendazim (072), thiophanate-methyl (077)

The Delegation of the European Union (EU) submitted a concern form at the Fifty-second CCPR relating to benomyl, carbendazim and thiophanate-methyl is in reference to these substances' status in terms of priority listing and scheduling for periodic review by the JMPR. The JMPR acknowledges receipt of the concern form, however, priority listing and scheduling is the responsibility of the CCPR. The JMPR refers this matter to the CCPR for resolution.

3.3 Chlorothalonil (081)

Concern #1 Concern was raised by the European Union (EU)

A potential public health concern was raised by the European Union (EU) about a number of metabolites of chlorothalonil which meant the consumer risk assessment could not be finalized and new toxicological studies had been submitted to the EU which had not been evaluated by JMPR. According to an evaluation by the European Food Safety Authority (EFSA) published in 2018, the specific concerns related to:

- EFSA was unable to conclude on toxicological reference values for metabolite R182281 (SDS-3701), there being concerns over potential genotoxicity;
- Metabolite R613636 was formed under processing conditions and genotoxic potential could not be excluded by EFSA;
- Metabolite R417888, a soil metabolite, had not been investigated in rotational crop residue trials and genotoxic potential could not be excluded by EFSA.

The EU has set an ARfD for chlorothalonil of 0.05 mg/kg body weight (bw) per day based on body weight loss at the start of dosing at 10 mg/kg bw per day in a study of developmental toxicity in rabbits. The previous EU ARfD was 0.6 mg/kg bw per day.

JMPR last reviewed chlorothalonil for toxicology in 2009, establishing an ADI of 0–0.02 mg/kg bw based on the NOAEL for kidney toxicity in a two-year study in rats, and an ARfD of 0.6 mg/kg bw, based on the NOAEL for kidney toxicity in acute toxicity studies in rats.

In addition, the 2010 JMPR evaluated data on soil metabolite R611965 and concluded it was of lower toxicity than chlorothalonil and covered by the ADI and ARfD for chlorothalonil.

In the 2019 Extra JMPR the dietary risk from metabolites R613636 and R417888 was considered.

The EU concern form does not identify the types of new toxicological studies submitted, nor whether they had identified effects critical to the consumer risk assessment that were not covered by the existing database. As the new EU ADI and ARfD are based on studies evaluated previously by JMPR, it appears unlikely that the new toxicology studies would be critical to the consumer risk assessment.

Regarding the specific concerns:

The 2009 JMPR evaluated an extensive database on SDS-3701 concluding that it was more toxic than chlorothalonil and established an ADI of 0–0.008 mg/kg bw and an ARfD of 0.03 mg/kg bw. A positive result in an in vitro chromosomal aberration study was not confirmed in vivo and the Meeting concluded that SDS-3701 was not genotoxic.

The 2019 Extra JMPR considered genotoxicity data and data on the formation of R613636. Results were negative in two gene mutation tests. Although a positive response was found in the chromosomal aberration test R613636 was negative for genotoxicity in an in vivo mouse micronucleus test. The Meeting concluded that R613636 was unlikely to be genotoxic in vivo and could be assessed using the TTC approach in Cramer class III.

Metabolite R613636 was primarily found in processed foods and is formed from parent chlorothalonil during sterilization at 120 °C (but not during pasteurization or cooking). The metabolite was assessed under the TTC approach as a Cramer class III compound. Since the current international estimated dietary intake (IEDI) model does not sufficiently address the percentage of processed foods in the daily diet, a conservative approach was selected by multiplying the maximum IEDI of 9.3 µg/kg bw for parent chlorothalonil by a factor of 0.23 (representing the percentage of total radioactive residues [TRR] recovered as R613636 in simulated hydrolysis studies). In the absence of more detailed data on the consumption of sterilized foods, the Meeting decided that the gap between the maximum IEDI for R613636 of 2.37 µg/kg bw and the Cramer class III threshold of 1.5 µg/kg bw is small and it would be very unlikely that the majority of foods (> 50%) would be subject to sterilization treatment.

The 2021 Meeting concluded that R613636 is unlikely to present a public health concern.

The 2019 Extra JMPR Meeting considered genotoxicity, toxicity and formation data on R417888. It gave a positive response in the chromosomal aberration assay without metabolic activation and in one mouse lymphoma assay with metabolic activation, but it was negative in a repeat mouse lymphoma assay at higher concentrations. R417888 was negative for genotoxicity in vivo in a mouse micronucleus test and for unscheduled DNA synthesis. The Meeting concluded R417888 was unlikely to be genotoxic in vivo and would be covered by the ARfD and ADI of chlorothalonil because of its lower acute and repeat-dose toxicity in comparison with the parent compound.

Metabolite R417888 was not identified in the residue data package received by the Meeting for the last periodic review. It was noticed that in the EU re-assessment report an additional confined rotational crop study by Rizzo (2005) was described, showing residues for “metabolites R611965 and R417888, accounting for 59%, 66% and 51% TRR for the 30, 120 and 365 plant-back intervals” in carrot roots. However, both compound co-eluted in the thin-layer chromatography (TLC) system used, and specific analysis revealed that “the majority of the residue was due to metabolite R611965 (51% TRR,

0.14 mg/kg)" in the 30-days sample. The toxicity of R611965 itself was also considered by the JMPR 2019 Extra Meeting, which concluded that "it would be covered by the ADI and ARfD of chlorothalonil, but noted that it is at least 10 times less potent than chlorothalonil." Consequently, the relevance of R417888 to consumer exposure was considered low. Although the data on R417888 were missing, its potentially small contribution to the TRR in rotational crops also suggests low significance for consumer exposure.

The JMPR 2009 monograph cites body weight loss at 20 mg/kg bw per day as the basis for the NOAEL in the rabbit study, used by the EU to derive its ARfD, but JMPR did not consider this finding applicable to the derivation of the ARfD.

The Meeting concluded that, based on the information presented in the EU documentation, the potential public health concerns raised by the EU over dietary exposures to chlorothalonil and its metabolites had not been substantiated and that they did not merit any review in advance of the normal periodic review.

Concern #2 Concern raised by the United Kingdom acute intake assessment for the metabolite R613636 in cranberry

Background

Chlorothalonil was reviewed by the JMPR in 2009 (T) and 2010 (T, R) within the periodic review program and evaluated for an additional use in cranberry at the 2019 Extra JMPR. The 2010 Meeting identified a hydrolysis product, R613636, that formed primarily during conditions simulating sterilisation (pH 6, 120 °C, 20 minutes; approximately 23% of total residue), with lesser amounts formed during conditions similar to baking, brewing, and boiling (pH 5, 100 °C, 60 minutes; approximately 3.4% of total residue). Furthermore, the 2010 Meeting noted that temperature, and not pH, appears to be the key variable in the formation of R613636.

The United Kingdom submitted a concern form at the Fifty-second CCPR stating that the exposure estimated for R613636 from cranberry exceeded the threshold of toxicological concern for Cramer class III, that the overall chronic exposure to R613636 from all commodities had not been addressed, and that the acute exposure to R613636 from cranberry had not been addressed.

Comment by the JMPR

The 2019 Extra Meeting agreed that R613636 could be assessed using the TTC approach as a Cramer Class III compound (1.5 µg/kg per day). As the consumption data within the IEDI model used to assess long-term dietary exposure does not allow specifically for assessment of sterilised foods, the Meeting decided to apply the factor of 0.23 to the maximum IEDI of 9.33 µg/kg bw for chlorothalonil to assess exposure to R613636. The estimated exposure was 2.37 µg/kg bw. While this estimate is greater than the threshold of 1.5 µg/kg per day, the Meeting noted that the estimate assumes that all foods are sterilised, and that it is very unlikely that all foods would be subjected to such high-temperature (120°C) treatment. The 2019 Extra Meeting concluded that long-term exposure to R613636 was unlikely to present a public health concern.

The current Meeting received processing studies on barley, wheat grain, cabbage, beans with pod (*Phaseolus vulgaris*) and tomato that include analysis for R613636. A cursory review of the results indicates that in all cases, residues of R613636 in processed commodities subject to heating were < 0.01 mg/kg and that overall it was only observed in one sample of wheat germ at 0.02 mg/kg. As the IEDI inputs for parent chlorothalonil were generally much greater than 0.02 mg/kg, it is likely that the new data will result in a much lower exposure estimate than was made by the 2019 Extra Meeting; however, time and resource constraints did not allow for a full evaluation of the data by the current Meeting. The Meeting agreed to evaluate the new data on processed commodities and to re-assess exposure to R613636 at its next meeting.

With regard to acute exposure to R613636 in cranberry juice, the 2019 Extra Meeting, in keeping with then-current practice, did not make an acute assessment by TTC due to lack of an agreed-upon threshold for assessing acute exposure. The current Meeting examined the acute exposure to R613636 in cranberry juice. The STMR and HR for cranberry are 3.0 and 7.7 mg/kg, respectively. A processing study on cranberry was not available. The Meeting agreed to apply the processing factors of 0.14 for unpasteurised grape juice and 0.26 for raisins (2010 JMPR) to derive estimates for cranberry juice and dried cranberry, respectively. As no processing factors were available to refine residue estimates for other processed cranberry commodities, the Meeting used the STMR/HR from the RAC. Residue estimates of chlorothalonil for all processed commodities were then adjusted to R613636 using the factor of 0.23 discussed above. Residues of R613636 used in the acute assessment for cranberry were: raw with skin, 0 mg/kg; canned/preserved, 1.77 mg/kg ($7.7 \text{ mg/kg} \times 0.23$); dried, 0.46 mg/kg ($7.7 \text{ mg/kg} \times 0.26 \times 0.23$); juice, 0.0966 mg/kg ($3 \text{ mg/kg} \times 0.14 \times 0.23$); jam, sauce/puree, and unspecified processed, 0.69 mg/kg ($3 \text{ mg/kg} \times 0.23$). The Meeting noted that (1) none of the processed cranberry commodities experience conditions similar to sterilisation; therefore, these are very conservative estimates and (2) the new data on residues of R613636 will likely support lower estimates than those calculated above.

The resulting maximum IESTI for cranberry (all commodities) was 3.51 µg/kg bw. A single-exposure TTC for Cramer class III compounds of 5 µg/kg bw was proposed by the European Food Safety Authority (EFSA 2012). The Meeting considered that this is precautionary and appropriate for use in assessing acute intakes of R613636. As a result, the acute exposure to R613636 in cranberry commodities is not expected to be a public health concern.

References

EFSA (2012). Scientific opinion on the evaluation of the toxicological relevance of pesticide metabolites for dietary risk assessment. EFSA Panel on Plant Protection Products and their Residues (PPR). *EFSA J.*, 10(7):2799 [187 pp.]. doi:10.2903/j.efsa.2012.2799.

3.4 Chlorpyrifos (017) and Chlorpyrifos-methyl (090)

Chlorpyrifos

A potential public health concern was raised by the European Union (EU) about the potential for chlorpyrifos to be genotoxic, damage DNA and affect neurodevelopment in children. Based on a statement published in 2019 by the European Food Safety Authority (EFSA), the concern form was triggered by studies in the published literature and new evaluations of an unpublished developmental neurotoxicity study. The concern form noted that the last JMPR review of chlorpyrifos for toxicology was in 1999.

The 2019 JMPR Meeting had been aware of the availability of new information on chlorpyrifos and the outcomes of the EFSA review (see Section 2.6 of the JMPR 2019 Report). The 2019 Meeting strongly recommended that given the 20-year gap since chlorpyrifos was last reviewed by the JMPR and the magnitude of potential concerns identified by the EU, chlorpyrifos be prioritized for periodic re-evaluation. It was noted that aspects of epidemiology should be included.

Given the concerns identified by the 2019 JMPR the current Meeting urged that chlorpyrifos should be re-evaluated as soon as possible. It was noted that findings from recent epidemiology studies would need to be assessed. The Meeting noted that CCPR has scheduled chlorpyrifos and chlorpyrifos-methyl for periodic evaluation by the 2024 JMPR.

The JMPR Joint Secretariats are currently investigating the most efficient ways to re-evaluate chlorpyrifos and chlorpyrifos-methyl for toxicology and residues, taking into account the size and complexity of their dossiers, and the aspects they have in common.

Chlorpyrifos-methyl

A potential public health concern was raised by the European Union (EU) about the potential for chlorpyrifos-methyl to be genotoxic, damage DNA and affect neurodevelopment in children. Based on a statement published in 2019 by the European Food Safety Authority (EFSA), the concern form was triggered by studies in the published literature and findings in the developmental neurotoxicity studies with chlorpyrifos-methyl and the closely related compound chlorpyrifos. The concern form noted that the last JMPR review of chlorpyrifos-methyl for toxicology was in 2001.

The Meeting noted that CCPR has scheduled chlorpyrifos and chlorpyrifos-methyl for periodic evaluation by the 2024 JMPR. The JMPR Joint Secretariats are currently investigating the most efficient ways to re-evaluate chlorpyrifos and chlorpyrifos-methyl for toxicology and residues, taking into account the size and complexity of their dossiers, and the aspects they have in common.

3.5 Fluensulfone (265)

Background

Fluensulfone was evaluated by the 2019 JMPR for additional uses in a range of commodities, including pome fruit and citrus.

The current Meeting received a concern form from the Delegation of the USA, relating to the proposed maximum residue level for pome fruit and also on the decision not to calculate a processing factor for citrus juice.

For pome fruit, the Delegation of the USA advised that in one of the pear trials used for estimating the maximum residue level, the reported residue values were incorrect, and that based on the corrected values, a higher maximum residue limit should be estimated.

For citrus juice, the Delegation of the USA proposed that since detectable residues of the BSA metabolite of fluensulfone were present in orange juice, processing factors for total residues (parent plus BSA) could be calculated from the two processing studies, and since the higher of these factors was very similar to that calculated for apple juice, the apple juice processing factor should be considered the appropriate processing factor for calculating the MRL for citrus/orange juice.

Comments by the JMPR

Pome fruit

For pome fruit, the 2019 JMPR estimated a maximum residue level of 0.2 mg/kg for fluensulfone (sum of fluensulfone + BSA metabolite, expressed as fluensulfone) in pome fruit (except persimmon, Japanese) based on the residues reported in 16 trials on apples and 8 trials on pears matching the critical GAP (USA) for a pre-flowering soil application of 3.92 kg ai/treated ha (broadcast, banded or by chemigation).

The current Meeting reviewed the study reports of the pome fruit residue trials and confirmed that in one of the pear trials (Ref: R-35572: PR-WA), there was a transcription error in the 2019 Fluensulfone Evaluation Table 13. The corrected values (after rounding) for this trial are reported below and the residue selected for maximum residue level estimation is underlined:

Table 1 Residues in pear trial Ref: R-35572: PR-WA

POME FRUIT Country, Year Location (variety) Reference	Application				DALA	Matrix	Residues, mg/kg [mean]		
	No.	Type	kg ai/ treated ha	Water (L/ha)			Fluensulfone	BSA	Total ^a
GAP: USA	2	Chemigation	3.92		Up to flower bud swell and/or after harvest (max 3.92 kg ai/ha/year)				
PEAR									
USA, 2015 Ephrata, WA (D'Anjou) R-35572	1	Band spray	4.00	18947	114	Pear	ND, ND [ND]	0.13, 0.15 [0.14]	0.2, 0.24 [0.22]
					121 (NCH)		ND, ND [ND]	0.13, 0.14 [0.13]	0.21, 0.22 [0.21]
					128		ND, ND [ND]	0.14, 0.13 [0.13]	0.22, 0.21 [0.22]
					135		ND, ND [ND]	0.11, 0.12 [0.12]	0.18, 0.2 [0.19]

ND = <0.003 mg/kg, with a value of 0.01 mg/kg used for calculating Total residues and a value of 0 mg/kg used for dietary exposure estimation

^a Total residues, expressed as parent = Sum of fluensulfone + (BSA×1.53) mg/kg.

The corrected data sets for total residues in apples and pears from trials matching the critical GAP for fluensulfone on pome fruit are:

Apples: < 0.025 (10), 0.028 (3), 0.031, 0.037 and 0.16 mg/kg (n = 16).

Pears: < 0.025 (4), 0.026, 0.11, 0.17 and 0.22 mg/kg (n = 8).

Noting that the residues arising from early season soil applications to apple and pear trees were not statistically different (Kruskal-Wallis), the Meeting agreed to estimate a group maximum residue level based on a combined total residue data set of: < 0.025 (14), 0.026, 0.028 (3), 0.031, 0.037, 0.11, 0.16, 0.17, and 0.22 mg/kg (n = 24). Corresponding residues of fluensulfone (parent only) were all below the detection limit.

The Meeting estimated a maximum residue level of 0.3 mg/kg for fluensulfone (fluensulfone+BSA metabolite), an STMR of 0 mg/kg and an HR of 0 mg/kg for fluensulfone (parent only) in pome fruit (except persimmon, Japanese) to replace the previous recommendation.

Because the STMR and the HR remain unchanged, no refinement of the dietary exposure estimation was needed. Based on the 2019 JMPR conclusion that any uptake of the metabolite MeS from permanent crops would be insignificant, the current Meeting considered it unnecessary to revisit the Cramer class III TTC assessment for MeS (2-Methylsulfonylthiazole).

Citrus juice

For citrus juice, the 2019 JMPR reviewed the two citrus processing studies evaluated by the 2017 JMPR, where fluensulfone was applied as a soil-irrigated treatment matching the critical GAP but at an exaggerated (2×) rate of 8.1 kg ai/ha and agreed that since fluensulfone residues were not detected in whole fruit (RAC), processing factors could not be calculated.

The current Meeting noted that for apple, plum and grape commodities, where fluensulfone residues were also not detectable in the RAC or the processed commodities, fluensulfone residues in fruit from field trials matching GAP were also <LOQ, and the processing studies reflecting this 'no parent residue' situation were able to be used to estimate processing factors based on the transfer of the BSA metabolite residues.

However, for citrus, since measurable residues of fluensulfone (up to about 0.05 mg/kg) were observed in supervised field trials matching GAP, the citrus processing studies were not suitable for estimating processing factors as they did not address the behaviour of fluensulfone during processing.

The Meeting confirmed the 2019 JMPR conclusion that a processing factor for citrus juice could not be calculated.

3.6 Metconazole (313)

Background

Metconazole was evaluated as a new compound by the 2019 JMPR and maximum residue levels were estimated for a range of commodities. In evaluating metconazole residues in wheat, the 2019 JMPR concluded that no maximum residue level could be estimated due to an insufficient number of trials matching the GAP with regards to the PHI.

The current Meeting received a concern raised by the Delegation of the USA that only four residue trials with pre-harvest intervals (PHIs) longer than 21 days were considered to approximate the GAP PHI of 30 days $\pm$ 25%, despite PHIs in most of the trials being 20–22 days. It was noted that the 2019 JMPR applied a tolerance of $\pm$ 25% to the PHI parameter itself, rather than the expected residue concentrations.

Comments by the JMPR

The current Meeting re-evaluated the decline trials for wheat, as well as for rye, barley and oat. For wheat, two decline trials were provided. The first trial (R05047) did not show a typical decline pattern and was considered inconclusive. However, the second trial (R05050) showed declining residues. The decline rate was estimated by using 14 DALT data as the starting point following normalisation. Wheat samples taken at 19 DALT or later are not expected to result in a greater than 25% change in the residue concentration compared to the label PHI of 30 days. In addition, with consideration of decline trials for rye (R05147), barley (R05156) and oat (R05132), the mean relative decline showed that a $\pm$ 25% change in the residue concentration is not expected for samples collected between 12 and 48 DALT (Figure 1).

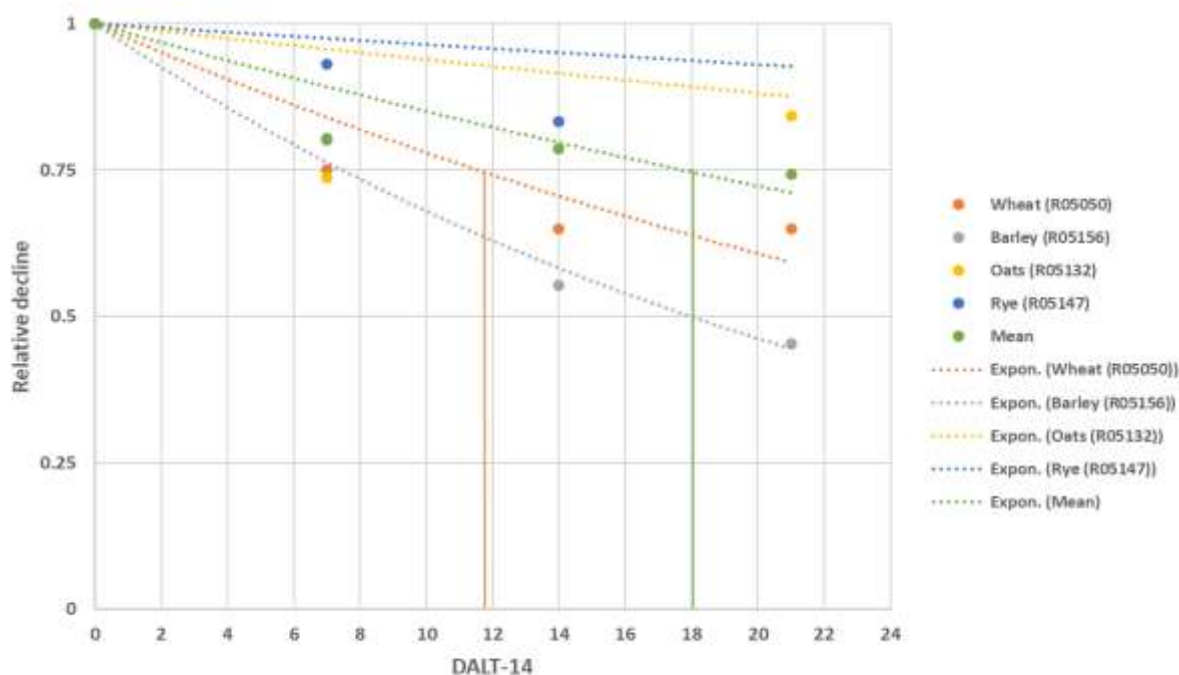


Figure 1 Relative decline of metconazole in wheat, barley, oat and rye

The Meeting noted that the expected change in residue concentration is less than 25% for samples taken at PHIs of 20–22 days compared to the 30-day label PHI and decided to use these trials to estimate a maximum residue level for wheat.

The ranked order of residues in wheat following applications of metconazole approximating the GAP, for estimating maximum residue levels and dietary risk assessment, were (n = 15): 0.011, 0.013, 0.015, 0.019, 0.023, 0.027, 0.028, 0.035, 0.036, 0.040, 0.048, 0.051, 0.054, 0.094, and 0.096 mg/kg.

The Meeting estimated a maximum residue level of 0.15 mg/kg and an STMR of 0.035 mg/kg for metconazole in wheat and decided to extrapolate its recommendation to triticale.

The fate of metconazole residues has been examined under conditions simulating household and commercial processing of wheat. These trials were evaluated by the 2019 JMPR. Estimated processing factors for the commodities considered at this Meeting are summarised below.

Table 1 Estimated processing factors for maximum residue level and dietary exposure estimations for processed wheat commodities according to the residue definition metconazole (sum of cis and trans isomer)

Crop	Residue (mg/kg) in RAC			Processed commodity	Individual PF	Median or best estimate PF	Residue (mg/kg) in processed commodity		
	MRL	STMR	HR				MRL-P	STMR-P	HR-P
Wheat	0.15	0.035	-	Coarse bran	1.6, 1.8, 2.0, 2.1	1.9	0.3	0.067	
			-	Whole meal flour	0.39, 0.60, 0.88, 0.91	0.74	-	0.026	
			-	Flour	0.18, 0.21, 0.24, 0.42	0.23	-	0.008	
			-	Germ	0.64, 0.92, 1.1, 1.2	1.0	-	0.035	
			-	Bread	0.44, 0.58, 0.63, 0.70	0.61	-	0.021	-

The addition of wheat, triticale and their processed commodities that may be used as animal feed items did not significantly change the livestock dietary burden. The Meeting agreed that a revision of the previously estimated maximum residue level recommendations for animal commodities was unnecessary.

RECOMMENDATIONS

On the basis of the data from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI and IESTI assessments.

Dietary risk assessment

Long-term dietary exposure

The ADI for metconazole is 0–0.04 mg/kg bw. The International Estimated Daily Intakes (IEDIs) for metconazole were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs ranged from 0–2% of the maximum ADI. The Meeting concluded that long-term dietary exposure to residues of metconazole from uses considered by the JMPR is unlikely to present a public health concern.

Acute dietary exposure

The ARfD for metconazole is 0.04 mg/kg bw. The International Estimate of Short Term Intakes (IESTIs) for metconazole were calculated for the food commodities and their processed commodities for which HRs/HR-Ps or STMRs/STMR-Ps were estimated by the current Meeting and for which consumption data were available. The results are shown in Annex 4 of the 2021 JMPR Report.

The IESTIs varied from 0–3% of the ARfD for children and 0–1% of the ARfD for the general population. The Meeting concluded that acute dietary exposure to residues of metconazole from uses considered by the current Meeting is unlikely to present a public health concern.

References

FAO and WHO. 2019. Pesticide Residues in Food - Report 2021 – Joint FAO/WHO Meeting on Pesticide Residues. Rome.

3.7 Propiconazole (160)

A potential public health concern was raised by the European Union (EU) about a number of aspects of propiconazole, which had resulted in differences between JMPR and EU in respect of the ADI and ARfD, the residue definition and consideration of metabolites. According to communication from the European Food Safety Authority (EFSA), the concern form related to the following:

- the ADI and ARfD values established by EFSA are lower than those of the JMPR;
- EFSA was unable to conclude on the toxicity of some metabolites;
- EFSA was unable to conclude on endocrine disrupting potential;
- EFSA was unable to conclude on the residue definition and consumer dietary intake assessment;
- acute intake concerns were cited with respect to some CXLs.

JMPR reviewed propiconazole in 2004, establishing an ADI of 0–0.07 mg/kg (bw) (not 0.7 mg/kg bw as stated in the concern form), based on the NOAEL from a two-generation study of reproductive toxicity, and an ARfD of 0.3 mg/kg bw, based on the NOAEL from a rat developmental toxicity study, applying a safety factor of 100.

Regarding the specific concerns

- 1a. The EU ADI of 0.04 mg/kg bw per day is derived from the NOAEL from a two-year study of chronic toxicity and carcinogenicity. The JMPR Meeting noted that from the values in Tables B-6.5-3 and 4 of the EU RAR this NOAEL is based on slight (< 5%) reductions in adrenal weights in males and slight (< 10%) reductions in body weight gain in females at some time points, but not over the entire duration of the study. The JMPR considers such slight reductions in adrenal weight and body weight gains, in the absence of any related findings, as not adverse. If the EU have scientific evidence to support the consideration of such minor changes as adverse, it would be helpful if this could be made available for consideration by JMPR.
- 1b. The EU ARfD of 0.1 mg/kg bw is derived from the same study and NOAEL (30 mg/kg bw per day) as that of the JMPR but applying a safety factor of 300 to maintain a margin of 900 to the LOAEL for developmental effects (90 mg/kg bw per day). The JMPR 2004 Meeting of considered that the margin between the ARfD of 0.3 mg/kg bw and the LOAEL for the severe effect of cleft palate and maternal toxicity at 300 mg/kg bw per day was adequate.
2. The EU concern form does not identify for which metabolites conclusions on toxicity cannot be made, exactly why the residue definition cannot be finalized nor which CXLs present acute intake concerns. Without this information JMPR cannot comment on these generic EU concerns. The JMPR has noted that the list of end-points for the EFSA conclusion cites that a number of metabolites of propiconazole yielded negative Ames tests, indicating they are unlikely to be DNA reactive, and therefore conclusions on them could be made based on a threshold of toxicological concern (TTC) assessment.
3. The Meeting noted that, within the EU legislative framework, endocrine disruption is a hazard identification process with direct risk management consequences, while JMPR includes these aspects as part of the risk assessments as a means to understand mode of action for certain apical effects, if relevant. The JMPR 2004 Meeting concluded that the available database on propiconazole was adequate to characterize the potential hazards to fetuses, infants and children.
4. The Meeting concluded that based on the information presented in the EU documentation, the concerns identified about dietary exposures to propiconazole and metabolites were insufficiently precise, and no conclusions could be made as to whether they represent a public health concern. It should also be noted that in the

absence of clear marker components, the JMPR decided to define the residue both for MRLs and for the estimation of the dietary exposure as “propiconazole plus all metabolites convertible to 2,4-dichlorobenzoic acid, expressed as propiconazole”. This common moiety approach covers the majority of residues for parent propiconazole and its metabolites in plant and animal commodities and represents a conservative estimate on the exposure in terms of propiconazole equivalents.

The Meeting concluded that based on the information presented in the EU documentation, the potential concerns identified about dietary exposures to propiconazole and its metabolites were not substantiated and did not merit any review in advance of the normal periodic review.

4. Dietary exposure assessment for pesticide residues in food

Dimethoate, ethoxyquin, guazatine, isoprothiolane, mefentrifluconazole and tetranilprole were considered by the current Meeting for toxicology only and no dietary risk assessment was conducted.

The current Meeting agreed to withdraw all previous recommendations made in terms of metalaxyl-M and to make new recommendations in terms of metalaxyl. The risk assessment for metalaxyl, therefore, covers the considered uses of metalaxyl-M.

4.1 Chronic dietary exposure

At the current Meeting, an international estimated daily intake (IEDI) was calculated for each compound for which an acceptable daily intake (ADI) was established. The IEDI was calculated by multiplying the median concentrations of residues by the average daily per capita consumption of treated commodities. The concentrations were supervised trials median residues [STMRs] and/or supervised trials median residues in a processed commodity [STMR-Ps]. The per capita food consumption amounts were estimated using the 17 Global Environment Monitoring System – Food Contamination Monitoring and Assessment Programme (GEMS/Food) cluster diets. A detailed description of the method is included in the Environmental Health Criteria 240 (EHC 240) monograph.³

These IEDIs were expressed as a percentage of the maximum ADI for a 55 kg or 60 kg person, depending on the region covered by each cluster diet (Table 1). The spreadsheet application is available at ied-calculation-vs04-17clusters.xlsx (live.com).

The detailed calculations of the chronic (long-term) dietary exposure assessments are given in Annex 3.

Table 1 Summary of chronic (long-term) dietary exposure assessments (IEDI)

CCPR code	Compound name	ADI (mg/kg bw)	Range of IEDI, as % of the maximum ADI
193	Fenpyroximate	0–0.005	4–20
202	Fipronil	0–0.0002	30–170 (G15 diet - 170)
319	Flutianil	0–0.8	0
138	Metalaxyl ^a	0–0.08	0–1
313	Metconazole	0–0.04	0–2
292	Pendimethalin	0–0.1	0–1
321	Pyrasulfotole	0–0.01	1–2
322	Pyrziflumid	0–0.02	0–8
323	Spiropidion	0–0.02	0–10

ADI: acceptable daily intake; bw: body weight; CCPR: Codex Committee on Pesticide Residues; IEDI: international estimated daily intake.

^a the assessment covers use of metalaxyl and metalaxyl-M

Metconazole and fluensulfone were evaluated by the 2019 JMPR. The current Meeting reconsidered a previous decision and decided to estimate additional maximum residue levels for metconazole and thus also conducted a dietary risk assessment for this compound. The current Meeting

³ FAO/WHO (2009). Principles and methods for the risk assessment of chemicals in food. A joint publication of the Food and Agriculture Organization of the United Nations and the World Health Organization. Geneva: World Health Organization (Environmental Health Criteria 240; <http://www.who.int/foodsafety/publications/chemical-food/en/>).

revised a maximum residue level estimation for fluensulfone, however there was no change to the STMR or HR and no refinement of the dietary exposure estimation was needed.

Possible refinement when the IEDI exceeds the ADI

The Meeting concluded that the long-term dietary exposure to residues of fipronil from uses considered by the Meeting may present a public health concern. Mammalian meat, mammalian fat and milks including dairy products were the main contributors to estimated dietary exposure for the G15 diet (IEDI was 170% of the maximum ADI), with residues in these food products due to residues in animal feed, noting rice straw and fodder makes the highest contribution to the calculated livestock dietary burden. The chronic dietary exposure for fipronil was also estimated using national dietary survey food consumption data rather than the WHO cluster diets which are used in the IEDI, as described below.

Chronic dietary exposure estimates using national dietary survey data

Food consumption data from national dietary surveys were used as the basis for estimating mean and high consumer chronic dietary exposure to pesticide residues for the general population and different age/sex groups. The global estimate of chronic dietary exposure (GECDE) model, developed by JECFA (veterinary drugs) in 2011⁴, was used for estimating potential high consumer dietary exposure to pesticide residues for population subgroups of toxicological concern, such as women of childbearing age, infants, toddlers and young children. The GECDE model is based on summary statistics derived from individual food consumption data from representative national surveys and takes account of consumption of one commodity at a high level (for consumers of that commodity only) plus consumption of the remaining commodities at a population mean level. This model is intended to be used when raw individual dietary records are not available for use in risk assessments, which is often the case for evaluations undertaken by JMPR i.e. the GECDE serves as a proxy for deriving a high percentile dietary exposure value from a distribution of dietary exposures obtained by using the raw data.

Food consumption data suitable for use in the GECDE model are available in the WHO Chronic Individual Food Consumption – summary statistics (CIFOcOss) database, which contains individual and summary food consumption data derived from national surveys that have two or more records per survey participant. For use in chronic dietary exposure assessments, results for each individual are averaged over the number of days of the survey prior to deriving population statistics. For many countries, different population subgroups are surveyed separately over different time periods and may use different approaches. In these cases it is not possible to estimate dietary exposure for the population as a whole.

In the mean and high consumer chronic dietary exposure estimates, the STMR residue levels were assigned to each food with a maximum residue level recommendation for the pesticide. In some cases, dilution factors were required to correct the STMR for the raw commodity (e.g. tea leaves) to a level relevant to the food reported as consumed in the dietary survey (e.g. tea beverage) and processing factors were applied where relevant. Overall, conservative assumptions were made in the calculations where maximum residue levels were recommended for grains, mammalian meats, poultry meats, milks and dairy products to ensure the use of these commodities in mixed foods was captured appropriately (e.g. if an maximum residue level for wheat was recommended, the STMR/STMR-P was applied to derivative products including wheat flour and bread but for mixed foods, such as fine bakery products that have many other ingredients, the proportion of wheat flour was assumed to be 50%). The results are summarized in Table 2.

Generally, the results using national survey data to estimate mean chronic dietary exposure to a specified pesticide residue for the general population were similar to the IEDI results, except for fipronil where estimated mean dietary exposures were 38–236% of the maximum ADI for the general population

⁴ FAO/WHO (2012). Joint FAO/WHO Expert Meeting on Dietary Exposure Assessment Methodologies for Residues of Veterinary Drugs. Final Report including Report of Stakeholder Meeting. Geneva: World Health Organization (http://www.fao.org/fileadmin/user_upload/agns/pdf/jecfa/Dietary_Exposure_Assessment_Methodologies_for_Residues_of_Veterinary_Drugs.pdf).

(IEDI 30–170% of the maximum ADI). As expected, chronic dietary exposure estimates for the high consumer (GECDE model) were all higher than the corresponding IEDI for the pesticides considered, by up to a factor of 6. For six of the pesticides considered the estimated high consumer dietary exposures were less than the maximum ADI. For fipronil and fenpyroximate, high consumer estimates exceeded the relevant maximum ADI for one or more population groups.

The long elimination half-lives of fipronil and fipronil sulfone (the major chemical form in the body) means that intermittent and repeated exposures to fipronil dietary residues over any duration should not exceed the maximum ADI. The mean dietary exposure for each population subgroup all exceeded the maximum ADI. Shorter-than-lifetime estimated high consumer dietary exposures for each age group considered exceeded the maximum ADI for fipronil, with the infants and toddlers having the highest range (159–1014% of the maximum ADI). For this age group milks including infant formula and other dairy products were the main food contributor, but for other age groups a range of foods were major contributors to the high consumer dietary exposure estimates (> 50%) in one or more of the estimates for the countries and age groups considered, for example milks and dairy products, pig fat, beef, beef offal, yeast leavened pastries, basil and semolina. The Meeting concluded that results using national food consumption data confirmed the finding from the IEDI calculation that long-term dietary exposure to residues of fipronil from uses considered by the Meeting may present a public health concern.

As the ADI for fenpyroximate was based on a potentially acute effect the risk from intermittent and shorter-than-lifetime exposure was assessed. The mean dietary exposure for each population subgroup were all less than the maximum ADI. Shorter-than-lifetime estimated high consumer dietary exposures exceeded the maximum ADI at the top end of the range for children and adolescents (1–210% of the maximum ADI) and infants and toddlers (8–118% of the maximum ADI). There were a number of commodities that were major contributors to high consumer dietary exposure estimates (> 50%) in one or more of the estimates for the countries and age groups considered, including kumquats, stems/stalks eaten as vegetables, cherries (sweet), blueberries, celery, bilberries, chayote fruits, watermelons, aubergines, raspberries, apple, soft drink with fruit, fruit juice, fennel, teas and offal. The Meeting concluded that long-term dietary exposure to residues of fenpyroximate from uses considered by the Meeting may present a public health concern for high consumers.

For pyrasulfotole, data for one country was excluded from the dietary exposure assessment. For this country sorghum and products were the major contributor to dietary exposure. However, the amount of sorghum reported as being consumed in this national survey (> 100 g flour/kg bw per day) is highly unlikely from a physiological point of view (equivalent to consumption of > 6 kg flour/day for a 60 kg adult or > 1.5 kg/day flour for a young child of 15 kg bodyweight). The Meeting noted this requires further investigation of the data set and method used to code foods by the WHO and the relevant country.

Table 2 Chronic dietary exposure estimates using national dietary survey data ^b

Pesticide	ADI (mg/kg bw)	IEDI (% maximum ADI)	Mean dietary exposure General population (% maximum ADI)	High consumer dietary exposure (GECDE ^a) (% maximum ADI)
Fenpyroximate	0-0.005	4-20 All	4-22 All 2-22 All adults 2-23 Adults, female 1-50 Children & adolescents 4-44 Infants & toddlers	NA 2-75 All adults 2-73 Adults, female 1-210 Children & adolescents 8-118 Infants & toddlers
Fipronil	0-0.0002	30-170 All	38-236 All 15-220 All adults 15-223 Adults, female 54-693 Children & adolescents	NA 15-610 All adults 15-534 Adults, female 74-863 Children & adolescents

Pesticide	ADI (mg/kg bw)	IEDI (% maximum ADI)	Mean dietary exposure General population (% maximum ADI)	High consumer dietary exposure (GECDE ^a) (% maximum ADI)
			108-561 Infants & toddlers	159-1014 Infants & toddlers
Flutianil	0-0.8	0 All	0 All 0 All adults 0 Adults, female 0 Children & adolescents 0 Infants & toddlers	NA 0-0.1 All adults 0-0.1 Adults, female 0-0.3 Children & adolescents 0-0.2 Infants & toddlers
Metalaxyl ^c	0-0.08	0-1 All	0-2 All 0-1 All adults 0-1 Adults, female 0-3 Children & adolescents 0 Infants & toddlers	NA 0-13 All adults 0-13 Adults, female 0-30 Children & adolescents 0-17 Infants & toddlers
Metconazole	0-0.04	0-2 All	NA	NA
Pendimethalin	0-0.1	0-1 All	0-1 All 0-1 All adults 0-1 Adults, female 0-2 Children & adolescents 0-2 Infants & toddlers	NA 0-2 All adults 0-2 Adults, female 0-4 Children & adolescents 1-5 Infants & toddlers
Pyrasulfotole	0-0.01	1-2 All	1-5 All 0-3 All adults 0-9 Adults, female 1-7 Children & adolescents 1-13 Infants & toddlers	NA 0-8 All adults 0-23 Adults, female 2-19 Children & adolescents 3-56 Infants & toddlers
Pyraziflumid	0-0.02	0-8 All	0-8 All 0-11 All adults 0-9 Adults, female 0-10 Children & adolescents 0-17 Infants & toddlers	NA 0-42 All adults 0-43 Adults, female 0-87 Children & adolescents 0-67 Infants & toddlers
Spiropidion	0-0.02	0-10 All	2-10 All 1-10 All adults 1-12 Adults, female 0-15 Children & adolescents 1-15 Infants & toddlers	NA 1-29 All adults 1-31 Adults, female 0-39 Children & adolescents 3-42 Infants & toddlers

ADI: acceptable daily intake; bw: body weight; CIFOCSs: Chronic Individual Food Consumption summary statistics; GECDE: global estimate of chronic dietary exposure; IEDI: international estimated daily intake; NA Not assessed

^a For each national survey in the CIFOCSs database, for the GECDE model for a specified age group, a high percentile dietary exposure was first calculated for each commodity with an assigned supervised trials median residue (STMR): if there were more than 180 consumers of a commodity, a 97.5th percentile dietary exposure for consumers only was derived; if there were more than 60 but fewer than 181 consumers, a 95th percentile dietary exposure was derived; if there were more than 30 but fewer than 61 consumers, a 90th percentile dietary exposure was derived; and if there were more than 10 but fewer than 31 consumers, a median dietary exposure was derived. If there were fewer than 11 consumers, only the mean dietary exposure for the whole population was derived for that Codex commodity code.

^b The most recent national surveys for each country were sourced from the WHO CIFOCSs database for 37 countries (45 surveys in total): whole population, 11 surveys; adults, 29 surveys; adult women, 35 surveys; children and adolescents (aged 3–17 years), 25 surveys; infants (aged 0–11 months) and toddlers (aged 1–3 years), 19 surveys.

^c the assessment covers use of metalaxyl and metalaxyl-M

4.2 Acute dietary exposure

At the current Meeting, an international estimate of short-term intake (IESTI) was calculated for compounds for which an acute reference dose (ARfD) was established. For each relevant food commodity, the highest expected residue (highest residue in the edible portion of a commodity [HR] or highest residue in a processed commodity [HR-P]) and the highest large portion (LP) data for the general population (all ages) and children (6 years and under) were used for the calculation of the IESTI. The LP data are derived from national dietary survey data by WHO. For bulked and blended commodities, the STMR or STMR-P is used as the residue level in the IESTI calculation. A description of the method is included in EHC 240.

These IESTI results are expressed as a percentage of the ARfD in Table 3. The spreadsheet application is available at [iesti_calculation21_model.xlsx \(live.com\)](https://live.com/iesti_calculation21_model.xlsx). The detailed calculations of acute dietary exposure are given in Annex 4.

Metconazole and fluensulfone were evaluated by the 2019 JMPR. The current Meeting reconsidered a previous decision and decided to estimate additional maximum residue levels for metconazole and thus also conducted a dietary risk assessment for this compound. The current Meeting revised a maximum residue level estimation for fluensulfone, however there was no change to the STMR or HR and no refinement of the dietary exposure estimation was needed.

The current or previous Meetings agreed that ARfDs for flutianil and pyrasulfotole were unnecessary. Therefore, for these compounds, an acute dietary exposure assessment was not conducted.

Table 3 Summary of acute dietary exposure assessments (IESTI)

CCPR code	Compound name	ARfD (mg/kg bw)	Commodity (maximum % of ARfD)	Exceeding: population, (country)
193	Fenpyroximate	0.005	0–1410 Peeled mandarin 140 Orange 130 Apple 170 Pear 190 Cherries 240 Apricot 110 Nectarine 220 Peach 260 Blackberries 300 Raspberries 260 Raspberry juice 200 Blueberries 270 Currants 290 Currant juice 460 Gooseberries 140 Bitter melon 110 Chayote 200 Cucumber 190 Angled loofah 110	Child (China) Child (Australia) Child (China) Child (Netherlands) Child (Denmark) Child (Germany) Toddler (Netherlands) Child (Japan) Toddler (UK) Child (Ireland) Child (Germany) Child (Canada) General population (Australia) Child (Netherlands) Child (Germany) General population (China) Child (China) Child (Czech) Child (Thailand)

CCPR code	Compound name	ARfD (mg/kg bw)	Commodity (maximum % of ARfD)	Exceeding: population, (country)
			Smooth loofah 170 Snake gourd 110 Summer squash 150 Melon 160 Watermelon 370 Tomato 130 Eggplant 160 Phaseolus beans with pods 110 Cardoon 190 Cardoons cooked 260 Celery 1410 Celery cooked 850 Celery canned 120 Celery juice 220 Bulb fennel 530 Bulb fennel cooked 1140 Rhubarb 680 Rhubarb sauce 140	Child (China) Child (Thailand) Child (US) Child (Belgium) Child (Canada) Child (China) Child (China) Child (Canada) Adult (Italy) General population (Netherlands) Child (China) General population (Netherlands) Child (Netherlands) General population (Australia) Child (Australia) Child (Netherlands) Toddler (UK) Child (Netherlands)
202	Fipronil	0.03	0–20	
138	Metalaxyl ^a	0.5	0–100	
313	Metconazole	0.04	0–3	
292	Pendimethalin	1	0–1	
322	Pyraziflumid	2	0–3	
323	Spiropidion	0.3	0–60	

ARfD: acute reference dose; bw: body weight; CCPR: Codex Committee on Pesticide Residues; IESTI: international estimate of short-term intake. ^a the assessment covers use of metalaxyl and metalaxyl-M

Possible refinement when the IESTI exceeds the ARfD

The current Meeting revised the ARfD for fenpyroximate from 0.01 mg/kg bw to 0.005 mg/kg bw and conducted a new acute dietary risk assessment for all new recommendations and those made by the 2017 and 2018 JMPRs.

For recommendations of the current Meeting, it was concluded that the estimated acute dietary exposure to residues of fenpyroximate for the consumption of commodities from the subgroups of Mandarins (FC 0003), Oranges, sweet, sour (FC 0004), Cane berries (FB 2005), Bush berries (FB 2006), Cucumbers and Summer squash (VC 2039), and Stems and Petioles (VS 2080) may present a public health concern.

Based on the revised ARfD, the current Meeting confirmed the 2017 JMPR conclusion that the estimated acute dietary exposure to residues of fenpyroximate for the consumption of commodities from FS 0013 Subgroup of cherries, FS 0247 Peach and VC 0432 Watermelon may present a public health

concern. Alternative GAP data were available for plums, so the 2017 JMPR exceedances noted for FS 0014 Plums and dried plum no longer exist.

The current Meeting also concluded, based on the revised ARfD, that the estimated acute dietary exposure to residues of fenpyroximate for the consumption of commodities FP 0226 Apple, FP 0230 Pear, FS 0240 Apricot, VC 0046 Melons (except watermelon), VO 2045 Subgroup of Tomatoes, VO 2046 Subgroup of Eggplants and VP 2060 Subgroup of Beans with pods may present a public health concern.

As no alternative GAP data were available to the Meeting to estimate lower HR values for these commodities (except for plums) no refinement of the acute dietary exposure estimates was possible.

The ARfD for fenpyroximate was based on repeat dose studies, further information to determine if the effect was also relevant to single doses might permit refinement of the ARfD.

The Meeting recognized that the IESTI for fenpyroximate may be refined if new GAP data become available in the future.

5. Evaluation of data for acceptable daily intake and acute reference dose for humans, maximum residue levels and supervised trials median residue values

5.1 Dimethoate (027)

TOXICOLOGY

Dimethoate is the International Organization for Standardization (ISO)-approved common name for *O,O*-dimethyl *S*-[2-(methylamino)-2-oxoethyl] phosphorodithioate (IUPAC), with the Chemical Abstracts Service number 60-51-5, and CIPAC number 59.

Dimethoate is an anticholinesterase organophosphate insecticide, having contact and systemic action. It is used against a broad range of insects in agriculture and also for the control of the housefly.

Dimethoxon (also known as omethoate), the oxygen analogue metabolite of dimethoate, appears to play a dominant role in the toxicity of dimethoate towards insects and mammals.

The Joint FAO/WHO Meeting on Pesticide Residues (JMPR) evaluated dimethoate for its toxicological effects in 1963, 1965, 1967, 1984, 1987, 1996, 2003 and 2019. The 2019 Meeting established an acceptable daily intake (ADI) for dimethoate of 0–0.001 mg/kg body weight (bw) on the basis of the no-observed-adverse-effect level (NOAEL) of 0.1 mg/kg bw per day for the inhibition of erythrocyte acetylcholinesterase (AChE) activity in female pups on postnatal day 21 at 0.5 mg/kg bw in a developmental special study. The study was designed to assess the effects of dimethoate on AChE activity in pregnant rats, pre-weaning rats and young adult rats. A safety factor of 100 was applied. The 2019 Meeting also established the ARfD for dimethoate of 0.02 mg/kg bw on the basis of an overall NOAEL of 2 mg/kg bw based on inhibition of erythrocyte AChE activity in an acute neurotoxicity study and a special study in pre-weaning females and young adult females, with an overall LOAEL of 3 mg/kg bw. A safety factor of 100 was applied. The 2019 Meeting noted that the proposed ADI and ARfD were consistent with the data from the human volunteer studies in which there was an apparent NOAEL of about 0.2 mg/kg bw per day in studies with volunteers receiving single or repeated doses.

In considering the metabolites of dimethoate, the 2019 Meeting noted the conclusion of the 1996 JMPR:

"Omethoate has been extensively investigated for mutagenicity in vitro and in vivo. The Meeting concluded that it has clear mutagenic potential but that the weight of the evidence observed in vivo was negative; however, the positive result obtained in the mouse spot test could not be completely disregarded."

At the 1996 Meeting the ADI for omethoate was withdrawn. Although a number of new studies had been conducted focussing on the inhibition of AChE by omethoate, no new genotoxicity studies were provided to the 2019 Meeting. Consequently the 2019 Meeting was unable to complete the assessment of omethoate with respect to its mutagenic potential. Apart from omethoate, a number of plant metabolites have been identified, including:

- dimethoate carboxylic acid,
- *O*-desmethyl dimethoate carboxylic acid,
- *O*-desmethyl dimethoate,
- *N*-desmethyl dimethoate,
- *O*-desmethyl omethoate,
- *O*-desmethyl-*N*-desmethyl omethoate,
- *O*-desmethyl omethoate carboxylic acid,

- hydroxymethoate,
- isodimethoate,
- O-desmethyl isodimethoate and
- hydroxydimethoate (and its glucose conjugate).

Dimethoate carboxylic acid is a major rat metabolite, therefore the Meeting concluded that the toxicity of this metabolite would be covered by that of dimethoate. As for the remaining metabolites, although the desmethylated metabolites are less potent inhibitors of AChE, the Meeting noted that they are also likely to retain the moiety conferring genotoxic activity. Therefore, in the absence of genotoxicity studies, no conclusion was drawn as to the genotoxic potential of these metabolites.

The sponsor had indicated that new genotoxicity studies on metabolites were being performed, and would be submitted for consideration at the 2020 JMPR. Hence the metabolites of dimethoate, including omethoate, were on the agenda for the 2020 JMPR. Some data were submitted for metabolites, however the sponsor indicated that although further new studies had been performed, the results were still not ready for submission.

No further information from the above studies on metabolites of dimethoate were received in time for the September 2021 meeting.

In view of the above, and in the absence of complete study results, the Meeting re-affirmed the decision of the 2019 JMPR regarding the metabolites of dimethoate including omethoate.

5.2 Ethoxyquin (035)

Ethoxyquin was scheduled for review by the current Meeting as part of the periodic review program of the Codex Committee on Pesticide Residues (CCPR). Upon receipt of the toxicological dossier from the sponsor the Meeting noted that the available toxicological database was very limited and concluded that it was insufficient as a basis to establish an ADI and ARfD. A toxicological evaluation of ethoxyquin was not performed.

5.3 Fenpyroximate (193)

TOXICOLOGY

Fenpyroximate is the International Organization for Standardization (ISO)-approved common name for *tert*-butyl (*E*)- α -(1,3-dimethyl-5-phenoxy-1H-pyrazol-4-yl)methyleneaminoxy)-*p*-toluate (IUPAC), with Chemical Abstracts Service number 134098-61-6 (*E* isomer) and 111812-58-9 (unspecified stereochemistry).

Fenpyroximate is a phenoxypyrazole acaricide for application to leaves infested with phytophagous mites. Fenpyroximate acts by inhibiting mitochondrial NADH-coenzyme Q reductase of the electron transport chain in mites.

Fenpyroximate was evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) in 1995, when an acceptable daily intake (ADI) of 0–0.01 mg/kg body weight (bw) was established, based upon a no-observed-adverse-effect level (NOAEL) of 1 mg/kg bw per day in a 104-week study in rats and a safety factor of 100. The critical effect in that study was a reduction in body weight gain and plasma protein concentration. Fenpyroximate was re-evaluated by JMPR 2004 in order to determine an acute reference dose (ARfD). An ARfD of 0.01 mg/kg bw was established based on the lowest-observed-adverse-effect level (LOAEL) of 2 mg/kg bw per day for the marginal induction of diarrhoea at the beginning of the 13-week toxicity study in dogs, and using a safety factor of 200 since no NOAEL was identified in this study. The ARfD was re-evaluated by JMPR 2007. The 2007 Meeting established an ARfD of 0.02 mg/kg bw based on the NOAEL of 2 mg/kg bw identified from a study of fenpyroximate where diarrhoea was observed after a single dose in dogs, and using a safety factor of 100. The 2017 Meeting withdrew the existing ARfD and established a new ARfD of 0.01 mg/kg bw on the basis of the LOAEL of 2 mg/kg bw for the induction of diarrhoea seen in a newly submitted single bolus gavage study, and a 13-week study of toxicity in dogs. A safety factor of 200 was used since no NOAEL was identified. The 2017 Meeting also stated that it was unclear whether the diarrhoea was the result of direct irritant or the pharmacological effect of fenpyroximate. In the database that was available, however, histopathological examination of the gastrointestinal tract had not revealed any evidence of irritation. The JMPR 2017 Meeting re-affirmed the ADI of 0–0.01 mg/kg bw established at JMPR 1995. The 2017 Meeting concluded that the ADI and ARfD can be applied to metabolites M-1, M-3, M-5, M-21, M-22 and Fen-OH.

Fenpyroximate was reviewed by the present Meeting under the periodic review programme of Codex Committee on Pesticide Residues (CCPR) due to a number of proposed additional uses: citrus, banana, celery, cranberry, summer squash, watermelon, bean [succulent shelled], blueberry, plum, apricot, and peach. New studies included a pathological evaluation of tissues from a single-bolus dose study in dogs, an incidence report of diarrhoea in beagle dogs at Life Science Research lab and its analysis, genotoxicity studies with metabolites, sensitivity analysis of selected studies previously reviewed, newly reported pharmacology studies with the parent compound, two published metabolism studies in rats, together with acute and mutagenicity studies on impurities found in fenpyroximate.

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with relevant national or international test guidelines unless otherwise specified. A search of the open literature did not find any additional relevant publications.

Biochemical aspects

In a published study, a metabolite which had not been previously identified was found when liver microsomes were incubated with fenpyroximate; this was referred to as R-UL-1 (1-hydroxymethyl-1-methylethyl- (*E*)- α -(1,3-dimethyl-5-phenoxy-1H-pyrazol-4-yl)methyleneaminoxy)-*p*-toluate). The metabolite R-UL-1 (detected during an early phase in vitro study) is an unstable intermediate, especially in relation to *t*-butyl ester oxidation. The metabolite R-UL-1 converts to 2-hydroxy-2-methylpropyl (*E*)- α -(1,3-dimethyl-5-phenoxy-1H-pyrazol-4-yl)methyleneaminoxy)-*p*-toluate under neutral and basic conditions.

In a published study, it was suggested that selectivity of fenpyroximate for spider mites as against non-target organisms, especially mammals, could be attributable to a species-specific detoxification, described as ester hydrolysis via microsomal hydroxylation, followed by intramolecular transesterification.

Toxicological data

In repeat-dose toxicity studies, bradycardia, diarrhoea and decreased body weight in dogs, and decreased body weight and liver effects in rats, were the key findings.

In a 13-week toxicity study in rats using dietary fenpyroximate concentrations of 0, 20, 100 or 500 ppm (equal to 0, 1.3, 6.57 and 35.2 mg/kg bw per day for males, 0, 1.65, 8.29 and 38.6 mg/kg bw per day for females) the NOAEL was 20 ppm (equal to 1.3 mg/kg bw per day) based on decreased body weight and body weight gains, and reduced feed consumption in both sexes at 100 ppm (equal to 6.57 mg/kg bw per day).

In a single-dose oral toxicity study in dogs, fenpyroximate was administered as a bolus gavage dose of 0, 2 or 5 mg/kg bw. As diarrhoea was seen at all dose levels, the NOAEL could not be identified.

In a dose escalation oral toxicity study in the same dogs, fenpyroximate was administered via bolus gavage dose of 2, 5, 20 and 25 mg/kg bw. The NOAEL was 2 mg/kg bw per day based on diarrhoea seen at 5 mg/kg bw per day.

In a 90-day oral toxicity study in dogs using oral (capsule) fenpyroximate doses of 0, 2, 10 or 50 mg/kg bw per day, the LOAEL was 2 mg/kg bw per day, the lowest dose tested, based on slight bradycardia and an increased incidence of diarrhoea in both sexes (weekly observations), and in females only, reduced feed consumption, body weight and body weight gain and increased emaciation and torpor.

In a one-year oral toxicity study in dogs using oral (capsule) fenpyroximate doses of 0, 0.5, 1.5, 5.0 or 15.0 mg/kg bw per day, the NOAEL was 0.5 mg/kg bw per day based on bradycardia observed in male dogs at the LOAEL of 1.5 mg/kg bw per day.

Re-examination of gastrointestinal tract tissues from a single-bolus study by Harvey in 2006 showed no microscopic findings that suggested local or systemic fenpyroximate-related effects had been noted in the oesophagus or gastrointestinal tract following the administration of a 2 or 5 mg/kg bw single-bolus gavage dose to beagle dogs.

Fenpyroximate was tested for genotoxicity in an adequate range of assays, both in vitro and in vivo. No evidence of genotoxicity was found.

The 2017 Meeting concluded that fenpyroximate was unlikely to be genotoxic, and this was confirmed by the 2021 Meeting.

A special study conducted in mice, rats, rabbits and guinea pigs aimed at evaluating pharmacological parameters with fenpyroximate indicated that the test substance inhibited the electron transport system of mitochondria, lowered body weight, temperature, blood pressure and heart rate, caused changes in electrocardiograms and respiratory excitation, and caused non-specific effects on biological functions at the lethal dose and dose levels close to the lethal dose.

Toxicological data on metabolites and/or degradates

Metabolite M-1

Fenpyroximate metabolite M-1 (the Z-isomer) was negative for mutagenicity according to in vivo mouse micronucleus assay and comet assay.

Metabolite NH-4

The fenpyroximate impurity, 1,3-dimethyl-5-phenoxy-4-pyrazolecarbaldehyde oxime (NH-4), was shown to have an acute oral median lethal dose (LD₅₀) in rats of 881 mg/kg, and was negative in the reverse mutation assay.

Metabolite NN-9

The fenpyroximate impurity, *tert*-butyl 4-chloromethylbenzoate (NN-9), was shown to have an acute oral LD₅₀ value in rats of >5000 mg/kg bw. It showed weak positive mutagenicity in reverse mutation assays, was negative in the chromosomal aberration test, and negative in the *in vivo* mouse micronucleus assay.

Metabolite NN-10

The fenpyroximate impurity, 1,3-dimethyl-5-phenoxy-4-pyrazolecarbonitrile (NN-10), was shown to have an acute oral LD₅₀ in rats of 3741 mg/kg. It was negative in bacterial reverse mutation assays.

Metabolite NN-14

The fenpyroximate impurity, *tert*-butyl α -4-[(5-chloro-1,3-dimethylpyrazol-4-yl)methyleneamino oxymethyl]benzoate (NN-14), was shown to have an acute oral LD₅₀ value in rats of >5000 mg/kg bw and to be negative in the bacterial reverse mutation assay.

Metabolite R-UL-1

Metabolite R-UL-1 was not identified in rat metabolism studies. It is plausible that Fen-OH and/or R-UL-1 occur as unstable intermediate metabolites in the rat when the parent is oxidized at its *t*-butyl moiety to give M-22, or hydrolysed at its *t*-butyl moiety to give M-3. No toxicological data were available on Fen-OH or R-UL-1, but an additional hydroxy group usually renders a molecule more likely to undergo further phase II metabolism and be more rapidly excreted.

The Meeting concluded that the toxicity of plant and livestock metabolites M-1, M-3, M-5, M-21, M-22, Fen-OH and R-UL-1 would be covered by the acceptable daily intake (ADI) established for fenpyroximate.

Microbiological data

No data was available.

Human data

No new data was available.

Toxicological evaluation

The Meeting withdrew the previous ADI and established a new ADI of 0–0.005 mg/kg bw on the basis of a NOAEL of 0.5 mg/kg bw per day with a LOAEL of 1.5 mg/kg bw per day for bradycardia seen in males in a one-year study of toxicity in dogs. A safety factor of 100 was used.

The Meeting withdrew the previous ARfD and established a new ARfD of 0.005 mg/kg bw on the basis of a NOAEL of 0.5 mg/kg bw per day, with a LOAEL of 1.5 mg/kg bw per day for bradycardia seen in males in a one-year study of toxicity in dogs. A safety factor of 100 was used.

Although it was not clear from the study design whether or not the bradycardia already occurs after a single dose, the effect was used as the basis for an ARfD in a cautious and conservative approach.

The new ADI and ARfD can be applied to M-1, M-3, M-5, M-21, M-22, Fen-OH and R-UL-1.

A toxicological monograph addendum was prepared.

Addendum to levels relevant to risk assessment of fenpyroximate

Species	Study	Effect	NOAEL	LOAEL
Dog	One-year studies of toxicity ^a	Toxicity	0.5 mg/kg bw per day	1.5 mg/kg bw per day

^a Capsule administration

Acceptable daily intake (ADI) applies to fenpyroximate and to M-1, M-3, M-5, M-21, M-22, Fen-OH and R-UL-1, expressed as fenpyroximate.

0–0.005 mg/kg bw

Acute reference dose (ARfD) applies to fenpyroximate and to M-1, M-3, M-5, M-21, M-22, Fen-OH and R-UL-1, expressed as fenpyroximate.

0.005 mg/kg bw

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Addendum to critical end-points for setting guidance values for exposure to fenpyroximate**Short-term studies of toxicity**

Target/critical effect	bradycardia (dog)
Lowest relevant oral NOAEL	0.5 mg/kg bw per day (dog)

Genotoxicity

No evidence of genotoxicity

Studies on toxicologically relevant metabolites

M-1	Ames test, mammalian cytogenetic assay in Chinese hamster lung cells, mouse lymphoma assay, micronucleus test and comet assay: all negative
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Summary

	Value	Study	Safety factor
ADI	0–0.005 mg/kg bw ^a	One-year study of toxicity in dogs	100
ARfD	0.005 mg/kg bw ^a	One-year study of toxicity in dogs	100

^a Applies to fenpyroximate and to M-1, M-3, M-5, M-21, M-22, Fen-OH and R-UL-1, expressed as fenpyroximate

RESIDUE AND ANALYTICAL ASPECTS

Fenpyroximate is a miticide/insecticide. It controls all motile stages of mites by inhibiting cellular respiration in the mitochondrion of cells which results in rapid cessation of all biological activities including feeding and reproduction.

Fenpyroximate was first evaluated by the JMPR in 1995 and then in 1999 and 2010 for additional uses, and in 2004 and 2007 for toxicology. Fenpyroximate was scheduled for periodic re-evaluation of residues and toxicology by the 2017 JMPR. The 2018 JMPR re-evaluated the lactating goat metabolism studies and the residue definitions for animal commodities. The 2017 JMPR confirmed the ADI of 0–

0.01 mg/kg bw derived by the 1995 JMPR and established a new ARfD of 0.01 mg/kg bw. The 2021 Meeting received new toxicological data and established an ADI of 0.005 mg/kg and an ARfD of 0–0.005 mg/kg bw ARfD. The 2021 JMPR confirmed that the ADI and ARfD can be applied to metabolites M-1, M-3, M-5, M-21, M-22 and Fen-OH, and in addition R-UL-1. The 2017 and 2018 JMPR recommended the following residue definitions:

Definition of the residue for compliance with the MRL for plant commodities: *Fenpyroximate*.

Definition of the residue for dietary risk assessment for plant commodities: *Sum of parent **fenpyroximate** and tert-butyl (Z)- α -(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneamino-oxy)-p-toluate (its **Z-isomer M-1**), expressed as fenpyroximate.*

Definition of the residue for compliance with the MRL and dietary risk assessment for animal commodities: *Sum of **fenpyroximate**, 2-hydroxymethyl-2-propyl (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)-methylenaminooxymethyl]benzoate (**Fen-OH**), and (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneaminooxymethyl]benzoic acid (**M-3**), expressed as fenpyroximate.*

The residue is fat-soluble.

Fenpyroximate was scheduled by the Codex Alimentarius Commission for evaluation of additional uses by the 2021 JMPR. The Meeting received additional information from the manufacturer on the method of residue analysis, stability in stored analytical samples, use patterns, supervised residue trials and fate of residues during processing. Furthermore, the manufacturer provided additional information on the metabolite Fen-OH and the Z-isomer M-1, and requested reconsideration of the residue definition for animal commodities.

Animal metabolism

The 2017 JMPR farm animal dietary burden calculations were based on total residues (sum of fenpyroximate and its Z-isomer M-1), although M-1 was not part of the residue definition for animal commodities such as meat or milk. In the metabolism studies with lactating goats (evaluated by the 2017 and 2018 JMPR), fenpyroximate and M-1 were detected in tissues at low levels, if at all, and thus M-1 was not included in the residue definition for animal commodities. Due to the nature of these metabolism studies with dosing of the E-isomer (fenpyroximate), the Meeting was unable to conclude on the conversion from M-1 to fenpyroximate.

The current Meeting received a simulated gastric fluid hydrolysis study, which showed that [pyrazole-3-¹⁴C] fenpyroximate rapidly declined at pH 1.2 at 37 °C with a half-life (DT₅₀) of 11.7 days. The major dissipation pathway of fenpyroximate was hydrolysis towards M-6 and M-3, which accounted for 86.9% TAR and 10.0% TAR at its maximum concentration, respectively. The Z-isomer M-1, produced through isomerisation, accounted for 6.4% TAR at its maximum concentration.

The study further showed that the Z-isomer M-1 can be converted to the E-isomer fenpyroximate at pH 1.2, 37 °C. The isomerisation rate from M-1 to fenpyroximate was greater than the rate from fenpyroximate to M-1, resulting in an approximate fenpyroximate to M-1 ratio of about 10 at equilibrium state within 6 hours of incubation, regardless of the starting isomers.

The current Meeting confirmed that if livestock are exposed to residues of M-1 in the diet it can contribute to the marker residues (fenpyroximate + M-3) and to the total residues (fenpyroximate + Fen-OH + R-UL-1 + M-3) in animal commodities such as meat and milk. The farm animal dietary burden calculations for both MRL setting and dietary risk assessment should therefore be based on the sum of fenpyroximate and M-1.

Methods of analysis

The Meeting received additional information on analytical methods for fenpyroximate, M-1 and M-3 in plant matrices.

LC-MS/MS methods RES/RAM/004, RES/RAM/005 were provided for lemons, oranges and orange processed commodities where residues were extracted with acetonitrile. The extract was cleaned-up as for the multi-residue QuEChERS method. The LOQ was 0.01 mg/kg.

Another LC-MS/MS method (WO 9.163 and modifications thereof) was provided for cane berries, blueberries, banana, summer squash, succulent beans without pods and celery, where residues were extracted with acetone/water (4:1, v/v) and diluted. The LOQ was 0.01 mg/kg.

The Meeting concluded that these LC-MS/MS methods were sufficiently validated and are suitable to determine fenpyroximate and M-1 in plant commodities. LC-MS/MS method RES/RAM/005 is also suitable to determine M-3 in plant commodities.

LC-MS/MS method AJW/03/01 was provided for plums and plum processed commodities, where residues were extracted with ethyl acetate. The LOQ was 0.01 mg/kg. Limited validation suggests this method is suitable to determine fenpyroximate and M-1 in plum processed commodities.

GC-NPD methods (R-4107, 1057W and 1872W) were provided for oranges and orange processed commodities, where residues were extracted with acetone/water and cleaned up by liquid-liquid partition and chromatographic methods. The LOQ was 0.05 mg/kg for method R-4107 and 0.01 mg/kg for methods 1057W and 1872W. The Meeting concluded that these methods are valid for the determination of fenpyroximate and M-1 in orange whole fruit, orange juice and orange molasses.

Orange oil and orange dried pulp analysed by these GC-NPD methods suffered from matrix interferences and matrix effects shown by average recoveries outside the 70–120% limits, relative standard deviations above 20% and levels in control samples above the reported LOQ depending on the clean-up steps performed. The Meeting concluded that method 1872W was suitable for the determination of fenpyroximate and M-1 in orange oil and orange dried pulp; method 1057W in orange dried pulp; and method R-4107 for none of the two matrices.

Stability of pesticide residues in stored analytical samples

The Meeting received additional information on storage stability of fenpyroximate and its Z-isomer M-1 in plant commodities.

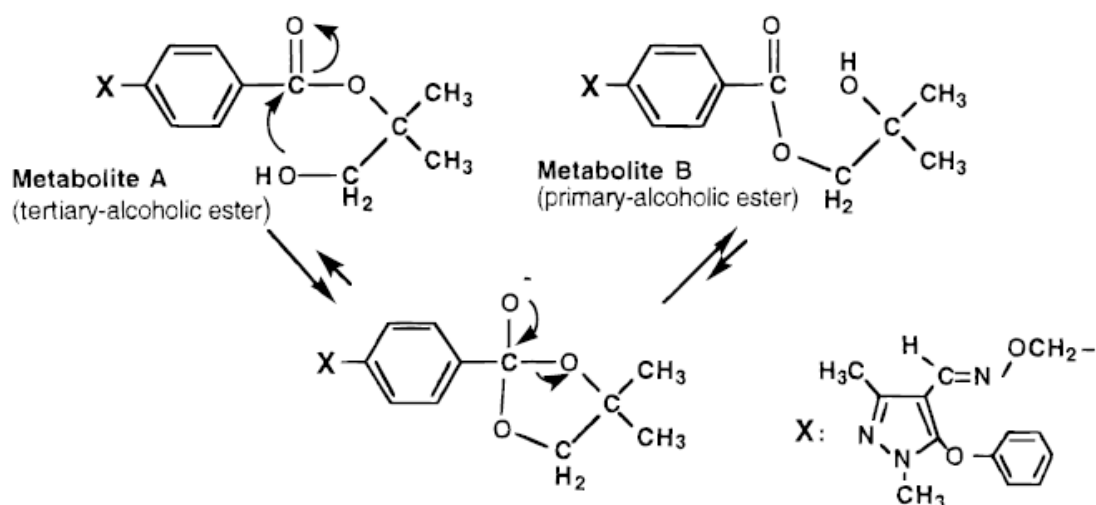
Storage stability studies provided to the current and previous Meetings showed that fenpyroximate and its Z-isomer M-1 were stable at -20 °C for at least 20 months in high water commodity groups (lettuce, celery, apple, summer squash, plum), at least 18 months in high acid commodity groups (oranges, grapes, blueberries), at least 9 months in high starch commodity groups (potato), at least 9 months in high oil commodity groups (sunflower seed), at least 19 months in banana, at least 24 months in dry hops and at least 18 months in dried plums and at least 7 months in orange juice.

The Meeting agreed that the demonstrated storage stability for high acid or high water commodity groups can be exchanged, since the 2017 JMPR evaluation showed that fenpyroximate and M-1 do not dissociate under acid or base conditions (no dissociation constant). As there were no storage issues with any of the commodity groups, the Meeting agreed that the storage period of at least 20 months can be extended to orange juice, plum juice, orange marmalade, plum jam, plum puree, orange molasses, orange dried pulp and orange oil.

The demonstrated storage stability of at least 20 months for high water commodity groups covered the residue sample storage intervals used in the supervised field trials on citrus, plums, blueberries, summer squashes, succulent bean seeds and celery and in the processing studies on orange juice, plum juice and plum puree. The demonstrated storage stability of 19 months in banana, 20 months for high water commodity groups and 18 months in dried plums did not fully cover the 20, 21 and 20 month storage periods in the banana trials, caneberry trials, and dried plums processing study, respectively. The Meeting considered this one to two month extension of the storage period to have no impact on the residue levels and decided that the selected trials could be used for to derive a maximum residue level or processing factor.

Definition of the residue

The Meeting received a request from the manufacturer to reconsider the residue definitions for animal commodities because of the instability of the metabolite Fen-OH.



Metabolite Fen-OH (metabolite A) was shown to be unstable in buffer solutions of neutral to alkaline pH as it underwent an intramolecular transesterification to its isomer R-UL-1 (metabolite B). Calculated half-lives were 171, 41 and 2.1 min at pH 7, 8 and 9, respectively.

The 2018 JMPR indicated that Fen-OH was the major residue in milk, fat and muscle in the two goat metabolism studies, albeit at low residue levels, but represented only a minor proportion of the residue in liver and kidney (3–5% TRR). Removing Fen-OH from the residue definition for MRL compliance for animal commodities would reduce the marker residue from 6–58% to 3–33% TRR for milk (TRR = 0.013–0.033 mg eq/kg), 91–114% to 33–40% TRR for muscle (TRR = 0.024–0.027 mg eq/kg) and 67–85% to 50% TRR for fat (TRR = 0.082–0.138 mg eq/kg). As Fen-OH is not stable and fenpyroximate (parent) and M-3 are found in milk, muscle and fat at sufficient quantities, the Meeting agreed that Fen-OH should not be included in the residue definition for enforcement purposes. The Meeting decided to change the residue definition for compliance with the MRL for animal commodities to:

Sum of fenpyroximate and M-3, expressed as fenpyroximate

The residue is fat-soluble.

Metabolite Fen-OH was shown to undergo an intramolecular transesterification to its isomer R-UL-1 *in-vitro* in plasma and in rat liver S9 systems in the presence of di-isopropyl fluorophosphate (DFP) which inhibited carboxy-esterase activity and thus prevents the forming of metabolite M-3. The formation of R-UL-1 *in-vivo* is suggestive as the analytical conditions used in all livestock metabolism studies submitted were such that the spots/peaks ascribed to Fen-OH or R-UL-1 could be Fen-OH alone, R-UL-1 alone or the sum of Fen-OH plus R-UL-1 as the two compounds could not be separated from each other. However, if a metabolite is seen *in-vitro* it is likely that it would be produced *in-vivo*, especially if the metabolite is formed very early in the pathway. The *in-vivo* forming of Fen-OH is proven indirectly by the presence of M-22 in rat and livestock metabolism studies, as M-22 can only be formed if Fen-OH is formed. Indirect evidence for the forming of R-UL-1 *in-vivo* is not available, as M-3 is the metabolic product of both Fen-OH and R-UL-1 and both compounds convert rapidly to M-3. The spots/peaks ascribed to Fen-OH and/or R-UL-1 were sufficiently stable to be able to quantify them in the goat metabolism study (27% TRR milk, 74% TRR muscle, 35% TRR fat), albeit at low levels (0.006–0.029 mg eq/kg).

The Meeting concluded that the spots/peaks ascribed to Fen-OH or R-UL-1 in the goat metabolism study were not found in rat. It is plausible that Fen-OH and/or R-UL-1 occur as unstable intermediate metabolites in the rat when the parent is oxidized at its t-butyl moiety to give M-22 or hydrolysed at its t-butyl moiety to give M-3. No toxicological data were available on Fen-OH or R-UL-1, but an additional

hydroxy group usually renders a molecule more likely to undergo further phase II metabolism and be more rapidly excreted. The 2017 JMPR therefore concluded that the ADI and ARfD of fenpyroximate also apply to Fen-OH, and similarly the current Meeting concluded that the ADI and ARfD of fenpyroximate also apply to R-UL-1.

In view of the instability of Fen-OH and its conversion into R-UL-1, their significant occurrence in matrices of animal origin and applicability of the ADI and ARfD of fenpyroximate to both compounds, the Meeting decided to change the residue definition for dietary risk assessment for animal commodities to: Sum of fenpyroximate, Fen-OH, R-UL-1 and M-3, expressed as fenpyroximate.

Results of supervised residue trials on crops

The Meeting received additional supervised residue trial data for the use of fenpyroximate on lemons, oranges, plums, blackberries, raspberries, blueberries, banana, summer squash, succulent seeds of common beans, succulent seeds of cowpeas, succulent seeds of Lima beans and celery.

Total residues were calculated by addition of fenpyroximate and M-1 concentrations. Residues below the LOQ were taken as at the LOQ. When both fenpyroximate and M-1 were below the LOQ, the total residue was taken as the sum of LOQs of each analyte and each commodity.

Group of Citrus fruits

The Meeting received new authorised labels from the USA on the citrus fruit group, where the PHI was modified from 14 days to 3 days. These labels supersede the US labels on the citrus fruit group used by the 2017 JMPR. The new critical GAP for the citrus fruit group from the USA consists of two foliar applications at 0.23 kg ai/ha with a retreatment interval of 14 days and a PHI of 3 days.

Trials conducted in the USA with an application rate of $2 \times 0.22\text{--}0.25$ kg ai/ha, a retreatment interval of 12–15 days and a PHI of 2–4 days approximated the new critical US GAP.

Lemon whole fruit: residues of fenpyroximate (n = 9): 0.24, 0.25, 0.28, 0.30, 0.30, 0.30, 0.32, 0.36, 0.41 mg/kg with an individual highest residue of 0.47 mg/kg, total residues of fenpyroximate and M-1 (n = 9): 0.27, 0.27, 0.34, 0.35, 0.37, 0.40, 0.41, 0.48, 0.51 mg eq/kg with an individual highest residue of 0.59 mg eq/kg.

Orange whole fruit: residues of fenpyroximate (n = 12) residues were: 0.076, 0.097, 0.14, 0.16, 0.16, 0.19, 0.20, 0.26, 0.31, 0.32, 0.39, 0.42 mg/kg with an individual highest residue of 0.42 mg/kg. Total residues of fenpyroximate and M-1 (n = 12) 0.10, 0.11, 0.17, 0.19, 0.20, 0.22, 0.23, 0.30, 0.39, 0.39, 0.47, 0.47 mg eq/kg with an individual highest residue of 0.48 mg eq/kg.

Grapefruit whole fruit: residues of fenpyroximate (n = 6) were: 0.10, 0.13, 0.15, 0.15, 0.20, 0.24 mg/kg with an individual highest residue of 0.28 mg/kg; Total residues of fenpyroximate and M-1 (n = 6) were: 0.11, 0.15, 0.18, 0.20, 0.25, 0.28 mg eq/kg with an individual highest residue of 0.32 mg eq/kg.

As supervised residue trials on mandarins did not comply with the new critical GAP and extrapolation to the group of citrus fruits requires data on the major crop mandarins, the Meeting decided to estimate separate recommendations for the subgroups.

Separate trials were available, where the residue distribution between the peel and flesh of lemons (n = 15), mandarins (n = 14), satsuma mandarins (n = 10), tangors (n = 16), oranges (n = 64) and natsudaidai (n = 8) were analysed. As there was no effect of citrus species or pre-harvest interval, all data were put together (n = 127), resulting in a residue ratio for flesh to whole fruit of 0.23 for total residues to calculate the STMR and HR.

The Meeting decided to withdraw its previous recommendation of 0.6 mg/kg for the Group of Citrus fruits and estimated:

- a maximum residue level of 1 mg/kg for fenpyroximate (based on whole fruit), an STMR of 0.085 mg eq/kg and an HR of 0.14 mg eq/kg for total residues of fenpyroximate and M-1 in the flesh of the subgroup of lemons and limes, and decided to extrapolate these to the subgroup of mandarins;
- a maximum residue level of 0.7 mg/kg for fenpyroximate (based on whole fruit), an STMR of 0.052 mg eq/kg and an HR of 0.11 mg eq/kg for total residues of fenpyroximate and M-1 in the flesh of the subgroup of oranges, sweet, sour;
- a maximum residue level of 0.5 mg/kg for fenpyroximate in whole fruit, an STMR 0.044 mg eq/kg and an HR of 0.074 mg eq/kg for total residues of fenpyroximate and M-1 in the flesh of the subgroup of pummelos.

For the purpose of deriving STMR-Ps and HR-Ps for processed commodities, an STMR-RAC and HR-RAC (respectively) for total residues of fenpyroximate and M-1 of 0.37 and 0.59 mg eq/kg for the subgroup of lemons and limes (whole fruit) and the subgroup of mandarins, 0.225 and 0.48 mg/kg for the subgroup of oranges, sweet, sour (whole fruit) and 0.19 and 0.32 mg eq/kg for the subgroup of pummelos (whole fruit), were estimated.

The Meeting noted that acute dietary exposure assessment showed that residues in the subgroup of mandarins exceeded the ARfD of 0.005 mg/kg bw, at 140% for peeled mandarins for Chinese children. No alternative GAP was available.

The Meeting further noted that acute dietary exposure assessment showed that residues in the subgroup of oranges exceeded the ARfD of 0.005 mg/kg bw, at 130% for oranges for Australian children. No alternative GAP was available.

Subgroup of plums

The 2017 JMPR estimated a maximum residue level, STMR and HR of 0.8, 0.155 and 0.33 mg/kg, respectively, for plums based on the critical GAP for the USA (stone fruit group) with a PHI of 7 days. The Meeting noted that the acute dietary exposure in dried plums exceeded the acute reference dose (ARfD) of 0.01 mg/kg bw at 270% of ARfD for children (Australia).

A label from Spain for plums was provided to the current Meeting as an alternative GAP consisting of a single foliar application at 0.077 kg ai/ha with a PHI of 14 days.

Trials conducted in Europe (France, Germany, Italy, Spain) with an application rate of 0.070–0.078 kg ai/ha and a PHI of 14 days were within 25% of the alternative Spanish GAP.

The resulting fenpyroximate residues (n = 14) were: < 0.01, < 0.01, 0.010, 0.010, 0.010, 0.010, 0.010, 0.020, 0.020, 0.020, 0.020, 0.025, 0.030, 0.030 mg/kg.

The resulting total residues of fenpyroximate and M-1 (n = 14) were: < 0.02, < 0.02, 0.020, 0.020, 0.020, 0.020, 0.020, 0.030, 0.030, 0.030, 0.030, 0.035, 0.040, 0.040 mg eq/kg.

The Meeting noted that the GAP for plums in Spain covers all the commodities in the subgroup of plums and that plums are the representative crop for the subgroup of plums. The Meeting decided to withdraw its previous recommendation for plums of 0.8 mg/kg and to replace it with a maximum residue level of 0.05 mg/kg for fenpyroximate and an STMR of 0.025 mg eq/kg and HR of 0.040 mg eq/kg for total residues of fenpyroximate and M-1 for the subgroup of plums.

For the purpose of deriving maximum residue levels for processed commodities, the Meeting recommended a highest residue in the RAC for fenpyroximate of 0.030 mg/kg for the subgroup of plums.

Subgroup of cane berries

The 2017 JMPR estimated a maximum residue level for raspberries based on a label from Austria. The current Meeting received a more critical GAP for cane berries from the USA (cane berry subgroup). The

registered use of cane berries in the USA allows two foliar applications at 0.12 kg ai/ha with a retreatment interval of 14 days and a PHI of 1 day.

Trials on blackberries and raspberries conducted in the USA with an application rate of 0.11–0.12 kg ai/ha, a retreatment interval of 13–15 days and a PHI of 1 day approximated this US GAP. The resulting residues were:

Blackberries: residues of fenpyroximate (n = 3): 0.43, 0.79, 0.92 mg/kg, total residues of fenpyroximate and M-1 (n = 3): 0.52, 0.84, 1.1 mg eq/kg

Raspberries, red, black: residues of fenpyroximate (n = 4): 0.19, 0.48, 1.1, 1.1 mg/kg, total residues of fenpyroximate and M-1 (n = 4): 0.24, 0.57, 1.2, 1.3 mg eq/kg, with a highest individual sample residue of 1.4 mg eq/kg.

The Meeting noted that median residues following two foliar applications to blackberries and raspberries are within a 5-fold range. The Meeting further noted that blackberries and/or raspberries are representative crops for the subgroup of cane berries and the US label is for cane berries. The Meeting decided to combine the data to estimate a maximum residue level for the subgroup of cane berries.

The combined residue data of fenpyroximate (n = 7) were: 0.19, 0.43, 0.48, 0.79, 0.92, 1.1, 1.1 mg/kg. The combined residue data of fenpyroximate and M-1 (n = 7) were: 0.24, 0.52, 0.57, 0.84, 1.1, 1.2, 1.3 mg eq/kg with a highest sample residue of 1.4 mg eq/kg.

The Meeting decided to withdraw its previous recommendation for raspberries of 0.2 mg/kg and estimated a maximum residue level of 3 mg/kg for fenpyroximate and an STMR of 0.84 mg eq/kg and an HR of 1.4 mg eq/kg for total residues of fenpyroximate and M-1 in the subgroup of cane berries.

The Meeting noted that acute dietary exposure assessment showed that residues in caneberries exceeded the ARfD of 0.005 mg/kg bw, at 300% for blackberries for UK toddlers and 260% for raspberries for Irish children, 200% for raspberry juice for German children. No alternative GAP was available.

Subgroup of bush berries

The critical GAP for bush berries is from the USA and consists of two foliar applications at 0.12 kg ai/ha with a retreatment interval of 14 days and a PHI of 1 day.

Trials on blueberries conducted in the USA with an application rate of 0.10–0.12 kg ai/ha, a retreatment interval of 13–15 days and a PHI of 1 day approximated this critical US GAP. The residues of fenpyroximate (n = 8) in blueberries were: 0.23, 0.38, 0.57, 0.57, 0.65, 0.67, 0.82, 0.78 mg/kg and total residues of fenpyroximate and M-1 in blueberries (n = 8) were: 0.32, 0.48, 0.76, 0.79, 0.81, 0.87, 0.99, 0.99 mg eq/kg with a highest sample residue of 1.2 mg eq/kg.

The Meeting noted that blueberries are a representative crop for the subgroup of bush berries. The Meeting estimated a maximum residue level of 2 mg/kg for fenpyroximate, and an STMR of 0.80 mg eq/kg and an HR of 1.2 mg eq/kg for total residues of fenpyroximate and M-1 in the subgroup of bush berries.

The Meeting noted that acute dietary exposure assessment showed that residues in bushberries exceeded the ARfD of 0.005 mg/kg bw, at 270% for blueberries for Canadian children, 290% for currants for Australian children, 460% for currant juice for Dutch children, and 140% for gooseberries for German children. No alternative GAP was available.

Banana

The critical GAP for banana is from the USA and consists of two foliar applications at 0.12 kg ai/ha with a retreatment interval of 7 days and a PHI of 7 days.

Trials on bananas conducted in the USA and Puerto Rico with an application rate of 0.11–0.12 kg ai/ha, a re-treatment interval of 6–7 days and a PHI of 7–8 days approximated this critical US GAP. The residues were:

Unbagged bananas: fenpyroximate (n = 3): 0.053, 0.14, 0.39 mg/kg and total residues of fenpyroximate and M-1 in whole fruit (n = 3) were: 0.10, 0.19, 0.52 mg eq/kg with a highest sample residue of 0.59 mg eq/kg.

Bagged bananas: fenpyroximate (n = 2): < 0.05, < 0.05 mg/kg and total residues of fenpyroximate and M-1 in whole fruit (n = 2) were: < 0.1, < 0.1 mg eq/kg.

The Meeting noted that unbagged bananas resulted in higher residues. Banana is a major crop and based on current practice, the Meeting considered three trials for unbagged bananas insufficient to estimate a maximum residue level for banana.

Subgroup of cucumbers and summer squashes

The 2017 JMPR estimated a maximum residue level for cucumber based on a label for cucumber only from the USA and a maximum residue level for summer squash based on a label for summer squash only from Germany. The current Meeting received an extended GAP from the USA for the squash/cucumber subgroup allowing two foliar applications at 0.12 kg ai/ha with a retreatment interval of 14 days and a PHI of 1 day. This US GAP was previously used by the 2017 JMPR to select cucumber trials, while it is a more critical GAP than that previously considered for summer squash.

Trials on cucumber and summer squash conducted in the USA with an application rate of 0.10–0.12 kg ai/ha, a retreatment interval of 13–17 days and a PHI of 1 day approximated this critical US GAP. The residues were:

Cucumber (taken from 2017 JMPR) residues of fenpyroximate (n = 7): < 0.05, 0.063, 0.065, 0.08 (2), 0.11 and 0.17 mg/kg and residues of fenpyroximate and M-1 in whole fruit (n = 7): < 0.1, 0.113, 0.115, 0.13(2), 0.16 and 0.22 mg eq/kg with a highest sample residue of 0.24 mg eq/kg.

Squash, summer, residues of fenpyroximate (n = 4): < 0.05, < 0.05, 0.065, 0.066 mg/kg and total residues of fenpyroximate and M-1 in whole fruit (n = 4): < 0.1, < 0.1, 0.12, 0.12, mg eq/kg with a highest sample residue of 0.17 mg eq/kg. One trial was not considered as the sample size was too low (6 fruits, weighing 0.91 kg).

The Meeting noted that the median residues following two foliar applications to cucumber and squash, summer, were within a 5-fold range. The Meeting further noted that cucumber and squash, summer, are representative commodities for the subgroup of cucumbers and summer squashes and the US label covers this subgroup. The Meeting decided to combine the data to estimate a maximum residue level for the subgroup of cucumbers and summer squashes.

The combined residue data for fenpyroximate (n = 11) were: < 0.05, < 0.05, < 0.05, 0.063, 0.065, 0.065, 0.066, 0.080 (2), 0.11 and 0.17 mg/kg. The combined total residue data for fenpyroximate and M-1 in whole fruit (n = 11) were: < 0.1, < 0.1, < 0.1, 0.113, 0.115, 0.12, 0.12, 0.13(2), 0.16 and 0.22 mg eq/kg with a highest sample residue of 0.24 mg eq/kg.

The Meeting decided to withdraw its previous recommendation for cucumber of 0.3 mg/kg and squash, summer, of 0.06 mg/kg and estimated a maximum residue level of 0.3 mg/kg for fenpyroximate, and an STMR of 0.12 mg eq/kg and an HR of 0.24 mg eq/kg for total residues of fenpyroximate and M-1 for the subgroup of cucumbers and summer squashes.

The Meeting noted that acute dietary exposure assessment showed that residues in the subgroup of cucumbers and summer squashes exceeded the ARfD of 0.005 mg/kg bw, at 110% for bitter melon for the Chinese general population, 200% for chayote for Chinese children, 190% for cucumbers for Czechian children, 110% for angled loofah for Thai children, 170% for smooth loofah for Chinese children, 110% for snake gourd for Thai children, 150% for summer squashes for US children. No alternative GAP was available.

Subgroup of succulent beans without pods

The critical GAP for succulent beans without pods is from the USA (beans, succulent) allowing two foliar applications at 0.12 kg ai/ha with a re-treatment interval of 14 days and a PHI of 1 day.

Trials on common bean, cowpea and Lima bean conducted in the USA with an application rate of $2 \times 0.11\text{--}0.12$ kg ai/ha, a retreatment interval of 13–15 days and a PHI of 1 day approximated this critical US GAP. The residues were:

Common bean – succulent seeds residues of fenpyroximate (n = 2): < 0.05 (2) mg/kg and total residues of fenpyroximate and M-1 (n = 2): < 0.1 (2) mg eq/kg.

Cowpea – succulent seeds residues of fenpyroximate (n = 1): < 0.05, mg/kg and total residues of fenpyroximate and M-1 (n = 1): < 0.1 mg eq/kg.

Lima bean – succulent seeds residues of fenpyroximate (n = 2): < 0.05 (2) mg/kg and total residues of fenpyroximate and M-1 (n = 2): < 0.1 (2) mg eq/kg. A third trial on Lima beans conducted at 3×0.12 kg ai/ha, a retreatment interval of 13–15 days and a PHI of 1 day confirmed residues at < 0.05 mg/kg for fenpyroximate and < 0.1 mg eq/kg for total residues of fenpyroximate and M-1.

The Meeting noted that succulent seeds from common bean, cowpea and Lima beans are representative for the subgroup of succulent beans without pods and the US label covers this subgroup. The Meeting decided to combine the data to estimate a maximum residue level for the subgroup of succulent beans without pods.

The combined residue data of fenpyroximate (n = 5) were: < 0.05 (5) mg/kg. The combined total residue data of fenpyroximate and M-1 (n = 5) were: < 0.1 (5) mg eq/kg.

The Meeting estimated a maximum residue level of 0.05* mg/kg for fenpyroximate, and an STMR of 0.1 mg eq/kg and an HR of 0.1 mg eq/kg for total residues of fenpyroximate and M-1 for the subgroup of succulent beans without pods.

Subgroup of stems and petioles

The critical GAP for stems and petioles is from the USA (leaf petiole vegetable subgroup) allowing two foliar applications at 0.12 kg ai/ha with a retreatment interval of 14 days and a PHI of 1 day.

Trials on celery conducted in the USA with a dose rate of $2 \times 0.10\text{--}0.12$ kg ai/ha, a retreatment interval of 14–15 days and a PHI of 1 day approximated this critical US GAP. The residues of fenpyroximate in celery (n = 6) were: 0.39, 0.49, 0.60, 0.62, 0.73, 1.8 mg/kg. Total residues of fenpyroximate and M-1 were (n = 6): 0.61, 0.79, 0.83, 0.86, 0.94, 2.1 mg eq/kg.

The Meeting noted that celery is a representative crop for the subgroup of stems and petioles and the US label covers this subgroup.

The Meeting estimated a maximum residue level of 3 mg/kg for fenpyroximate, and an STMR of 0.845 mg eq/kg and HR of 2.1 mg eq/kg for total residues of fenpyroximate and M-1 for the subgroup of stems and petioles.

The Meeting noted that acute dietary exposure assessment showed that residues in stems and petioles exceeded the ARfD of 0.005 mg/kg bw, at 190% for cardoons for Italian adults; 260% for cooked cardoons for the Dutch general population, 1410% for celery for Chinese children, 850% for cooked celery for the Dutch general population, 120% for canned celery for Dutch children; 220% for celery juice for the Australian general population, 530% for bulb fennel for Australian children, 1140% for cooked bulb fennel for Dutch children, 680% for rhubarb for UK toddlers and 140% for rhubarb sauce for Dutch children. No alternative GAP was available.

Fate of residues during processing

Processing studies on oranges and plums provided to the 2017 JMPR were reassessed as some of the processing data were not taken into account by the 2017 JMPR.

The Meeting decided to withdraw the STMR-Ps of 0.0048, 0.011, 6.5 and 0.78 mg/kg for orange juice, orange molasses, orange oil and orange dried pulp, respectively, and the HR-P of 1.5 mg/kg for orange dried pulp and estimated processing factors, STMR-Ps and HR-Ps as listed below.

The processing factors for succulent beans with pods, derived by the 2017 JMPR, were extrapolated to succulent beans without pods (i.e. succulent bean seeds). The processing factors for oranges were extrapolated to the subgroup of lemons and limes, the subgroup of oranges, the subgroup of mandarins, and the subgroup of pummelos and grapefruits.

Table 1 Processing factors, STMR-Ps and HR-Ps for the sum of fenpyroximate and M-1, expressed as fenpyroximate, and used for dietary risk assessment and livestock dietary burden calculation

Raw commodity	Processed commodity	Individual processing factors ^c	Median or best estimate processing factor	STMR-P = $STMR_{RAC} \times PF$ (mg/kg)	HR-P = $HR_{RAC} \times PF$ (mg/kg)
lemons mandarins oranges; pummelos	fresh, pasteurized or canned citrus juice	<0.037, <0.043, <u>0.062</u> , 0.099, <0.12, <0.22, <0.24	0.099	$0.37 \times 0.099 = 0.037$ $0.37 \times 0.099 = 0.037$ $0.225 \times 0.099 = 0.022$ $0.19 \times 0.099 = 0.019$	-
	canned citrus marmalade	0.037, 0.062	0.0495	$0.37 \times 0.0495 = 0.018$ $0.225 \times 0.0495 = 0.011$ $0.19 \times 0.0495 = 0.0094$	-
	citrus dried peel oil	138, 176	157	$0.37 \times 157 = 58$ $0.225 \times 157 = 35$ $0.19 \times 157 = 30$	-
	citrus dried pulp	3.6, <u>5.0</u> , 8.2	5.0	$0.37 \times 5.0 = 1.8^b$ $0.225 \times 5.0 = 1.1$ $0.19 \times 5.0 = 0.95$	
plums	dried plums (prunes)	1.7, 2.0, 2.8	2.0	$0.025 \times 2.0 = 0.050$	$0.040 \times 2.0 = 0.080$
	pasteurized plum juice	<0.50, <0.25	0.50	$0.025 \times 0.50 = 0.012$	-
	plum jam	<0.50, 0.50	0.50	$0.025 \times 0.50 = 0.012$	-
	plum puree	<0.50, <0.50	0.50	$0.025 \times 0.50 = 0.012$	-
succulent beans without pods	cooked succulent beans without pods	0.47, 0.73 ^a	0.60 ^a	$0.1 \times 0.60 = 0.060$	$0.1 \times 0.60 = 0.060$
	canned succulent beans without pods	0.42, 0.46 ^a	0.44 ^a	$0.1 \times 0.44 = 0.044$	$0.1 \times 0.44 = 0.044$

^a taken from 2017 JMPR succulent beans with pods

^b used in livestock dietary burden calculation

^c processing factors based on sum of fenpyroximate and M-1

Table 2 Processing factors, for fenpyroximate (parent only) used for maximum residue level estimation

Raw commodity	Processed commodity	Individual processing factors ^a	Mean or best estimate processing factor
lemons mandarins oranges pummelos	citrus dried peel oil	137, 175	156
	citrus dried pulp	3.5, 4.8, 7.4	4.8
plums	dried plums (prunes)	1.7, 2.3, 3.3	2.3

^a processing factors based on parent only

As residues in citrus dried peel oil, citrus dried pulp and dried plums were higher than residues in the RAC, the Meeting decided to estimate maximum residue levels for these commodities and to withdraw the MRL of 25 mg/kg for orange oil. Because feed commodities need to be expressed as dry matter, the maximum residue level for dried pulp on as received basis was multiplied by 100/90 assuming a dry matter content of 90%.

Using the estimated maximum residue level of 1 mg/kg for the subgroup of lemons and limes and applying the processing factor of 156, and 4.8 (and multiplied by 100/90), the Meeting estimated a maximum residue level of 150, and 6 mg/kg for fenpyroximate in the subgroup of lemons and limes dried peel oil and dried pulp (as dry matter), respectively. The Meeting decided to extrapolate the maximum residue level of 150 mg/kg to the subgroup of mandarins, oil.

Using the estimated maximum residue level of 0.7 mg/kg for the subgroup of oranges and applying the processing factor of 156 and 4.8 (and multiplied by 100/90), the Meeting estimated a maximum residue level of 100 and 4 mg/kg for fenpyroximate in the subgroup of oranges, dried peel oil and dried pulp (as dry matter), respectively.

Using the estimated maximum residue level of 0.5 mg/kg for the subgroup of pummelos and applying the processing factor of 156 and 4.8 (and multiplied by 100/90), the Meeting estimated a maximum residue level of 80 and 3 mg/kg for fenpyroximate in the subgroup of pummelos dried peel oil and dried pulp (as dry matter), respectively.

Using the estimated maximum residue level of 0.05 mg/kg for the subgroup of plums and applying the processing factor of 2.3, the Meeting estimated a maximum residue level of 0.15 mg/kg for fenpyroximate in the subgroup of plums, dried (prunes).

Residues in animal commodities

Farm animal feeding studies

As the current Meeting decided to change the residue definitions for animal commodities, the residues from the dairy cow feeding study were reassessed. Feeding studies or metabolism studies on laying hens were not available.

The 2017 JMPR received information on dairy cows dosed with fenpyroximate at the equivalent of 1.0, 3.0 and 10 ppm in the feed for 28 consecutive days.

Milk, muscle and fat

The GC-NPD method used in the dairy cow feeding study on milk, muscle and fat converts fenpyroximate, Fen-OH and R-UL-1 to M-3 (combined residue). M-3 was quantified separately in muscle and fat, but was not analysed for in milk. The residues for dietary exposure are based on the sum of the combined residues (covering fenpyroximate, Fen-OH and R-UL-1) and M-3 and are thus properly addressed. As fenpyroximate per se is not analysed in these matrices, the residues for enforcement are equal to those for dietary risk assessment and the residues for enforcement will be overestimated in milk, muscle and fat as they include Fen-OH and R-UL-1.

Total residues of fenpyroximate+Fen-OH+R-UL-1 in milk from the high dose group reached a plateau of 0.017 mg eq/kg at 3 days after the first dose. Total residues of fenpyroximate+Fen-OH+R-UL-1 in milk at 3–28 days from the high and medium dose group were 0.006–0.022 (mean 0.013) mg eq/kg and < 0.005–0.011 (mean 0.005) mg eq/kg. The low dose group was not analysed. Metabolite M-3 was not analysed for but is not expected in milk based on the metabolism studies in lactating goats.

Total residues of fenpyroximate+Fen-OH+R-UL-1+M-3 in muscle from the high, medium and low dose groups were 0.024–0.059 (median 0.048) mg eq/kg, 0.012–0.017 (median 0.015) mg eq/kg and < 0.01 mg/kg for all samples from the low dose group.

Total residues of fenpyroximate+Fen-OH+R-UL-1+M-3 in fat from the high, medium and low dose groups were 0.046–0.169 (median 0.115) mg eq/kg, 0.025–0.083 (median 0.066) mg eq/kg and 0.010–0.018 (median 0.015) mg eq/kg. The median value for the medium dose rate was corrected and differs from that reported by the 2018 JMPR.

Liver and kidney

The LC-MS/MS method used in the feeding study on liver and kidney, quantified fenpyroximate and M-3 separately, but did not analyse Fen-OH (and R-UL-1). The 2018 JMPR concluded that a scaling factor of 1.1 should be applied to the estimates of dairy cow liver and kidney residues for the purpose of dietary exposure. The current Meeting concluded that the same scaling factor could be applied to accommodate both Fen-OH and R-UL-1 as analytical methods used in the metabolism studies cannot discriminate between the two compounds.

Total residues of fenpyroximate+M-3 in liver from the high, medium and low dose groups were 0.71–0.911 (median 0.81) mg eq/kg, 0.28–0.42 (median 0.37) mg eq/kg and 0.16–0.22 (median 0.19) mg eq/kg. Residues in the feeding study need to be multiplied by 1.1 to take Fen-OH and R-UL-1 into account.

Total residues of fenpyroximate+M-3 in kidney from the high, medium and low dose groups were 0.359–0.459 (median 0.414) mg eq/kg, 0.24–0.36 (median 0.30) mg eq/kg, and 0.18–0.23 (median 0.20) mg eq/kg. Residues in the feeding study need to be multiplied by 1.1 to take Fen-OH and R-UL-1 into account. The median value for the medium dose rate was corrected and differs from that reported by the 2018 JMPR.

Farm animal dietary burden

For the additional uses considered by the current Meeting, only dried citrus pulp is relevant for farm animal dietary burden calculation. The median total residue for dried citrus pulp is 1.8 mg eq/kg based on the lemon data.

The farm animal dietary burden was recalculated based on exposure to total residues (fenpyroximate plus M-1) in raw and processed feed commodities and excluding bean forage in the Australian animal feed rations. Exposure was based on bean forage (for USA, EU, Japan only), maize, maize aspirated grain fractions, maize forage, maize fodder, apple wet pomace, grape wet pomace and dried citrus pulp. Exposure from tomato wet pomace (median residue 0.05 mg eq/kg based on tomato), potato culls, potato process waste and potato dried pulp (median and highest residue 0 mg eq/kg based on potato), field corn milled by-products (median residue 0.0032 mg eq/kg based on maize flour), field corn hominy meal, corn gluten feed and corn gluten meal (median residue 0.0013 mg eq/kg based on maize meal) was also taken into account and it was noted that these commodities were not included in the 2017 JMPR dietary burden calculations.

The dietary burdens, estimated using the OECD diets listed in Appendix IX of the 2016 edition of the FAO manual (OECD 2018 feed calculator, version October 2019), are presented in Annex 6 and summarised below. The refined calculation of farm animal dietary burden for beef cattle and dairy cattle for Australia is shown in parentheses. The high dietary burden calculated by the 2017 JMPR for layer poultry were based on outdated EU diets, where maize fodder was still included. The dietary burden for poultry was therefore recalculated.

Table 3 Farm animal dietary burden (ppm fenpyroximate equivalents of dry matter diet) based on total residues (fenpyroximate and M-1, expressed as fenpyroximate)

	US-Canada		EU		Australia		Japan	
	Max	Mean	Max	mean	max	Mean	max	Mean
Beef cattle	0.997	0.627	3.67	1.38	14.8 (3.93)	5.79 (1.87)	0.00949	0.00949
Dairy cattle	1.92	0.858	6.97 ^{①②}	2.87 ^{③④}	16.5 (3.93)	6.34 (1.87)	1.63	0.481
Broilers	0.00961	0.00961	0.00926	0.00926	0.000341	0.000341	0.00795	0.00795
Layers	0.00961 ^⑤	0.00961 ^⑥	0.00926	0.00926	0.000341	0.000341	0.00947	0.00947

① Highest maximum beef or dairy cattle dietary burden suitable for MRL and HR estimates for mammalian meat

② Highest maximum dairy cattle dietary burden suitable for MRL estimates for mammalian milk

③ Highest mean beef or dairy cattle dietary burden suitable for STMR estimates for mammalian meat.

④ Highest mean dairy cattle dietary burden suitable for STMR estimates for milk.

⑤ Highest maximum poultry dietary burden suitable for MRL and HR estimates for poultry meat and eggs

⑥ Highest mean poultry dietary burden suitable for STMR estimates for poultry meat and eggs

Animal commodity maximum residue levels

Cattle

As the residues in the feeding study and the dietary burden calculations were re-assessed by the current Meeting, the animal commodity residue levels were re-assessed as well. The calculation used to estimate maximum residue levels, STMR and HR values for cattle matrices is shown below.

Table 4 Estimated maximum residue levels, STMR and HR values for cattle matrices

	Feed level ppm) for milk residues	Residues (mg/kg) in milk	Feed level (ppm) for tissue residues	Residues (mg/kg) in:			
				Muscle	Liver	Kidney	Fat
MRL (mg eq/kg), beef or dairy cattle, based on fenpyroximate+ M-3							
Feeding study	3	0.005 ^a	3	0.017 ^a	0.42	0.36	0.083 ^a
	10	0.013 ^a	10	0.059 ^a	0.91	0.46	0.17 ^a
Maximum dietary burden (total residues) and MRL estimate	6.97	0.0095 ^a	6.97	0.041 ^a	0.70	0.42	0.13 ^a
HR (mg eq/kg), beef or dairy cattle, based on fenpyroximate+FenOH+R-UL-1+M-3							
Feeding study	3	0.005	3	0.017	0.42×1.1 =0.46	0.36×1.1 =0.40	0.083
	10	0.013	10	0.059	0.911×1.1 =1.00	0.459×1.1 =0.50	0.17
Maximum dietary burden (total residues) and HR estimate			6.97	0.041	0.77	0.46	0.13
STMR (mg eq/kg), beef or dairy cattle based on fenpyroximate+FenOH+R-UL-1+M3							
Feeding study	1	<0.005	1	< 0.01	0.19×1.1 =0.21	0.20×1.1 =0.22	0.015
	3	0.005	3	0.015	0.37×1.1 =0.41	0.30×1.1 =0.33	0.066
Mean dietary burden (total residues) and STMR estimate	2.87	0.005	2.87	0.015	0.40	0.32	0.063

^a Residues overestimated as they contain Fen-OH and R-UL-1

The Meeting estimated a maximum residue level of 0.01 mg/kg for fenpyroximate+M3 for milk, of 0.2 mg/kg for mammalian meat (fat), 0.2 mg/kg for mammalian fats and 0.8 mg/kg for edible offal (mammalian) to replace its previous recommendations. Based on the sum of fenpyroximate, Fen-OH, R-UL-1 and M3, the Meeting estimated an STMR of 0.005 mg eq/kg for milk, 0.015 mg eq/kg for mammalian muscle, 0.063 mg eq/kg for mammalian fats and 0.40 mg eq/kg for edible offal (mammalian). The Meeting estimated an HR of 0.041 mg eq/kg for mammalian muscle, 0.13 mg eq/kg for mammalian fats and 0.77 mg eq/kg for edible offal (mammalian).

Since neither a feeding study nor metabolism study on laying hens was available, the Meeting was unable to estimate maximum residue levels, STMRs and HRs for poultry commodities. As the maximum dietary burden for poultry was approximately 0.01 ppm, which is equivalent to a feed intake of < 0.004 mg/kg bw per day (based on 130 g feed/day and 1.9 kg bw for layers), the trigger value to submit hen metabolism and feeding studies is not exceeded. Therefore, dietary exposure will not be affected.

RECOMMENDATIONS

On the basis of the data obtained from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI, GECDE and IESTI assessments.

Definition of the residue for compliance with the MRL for plant commodities: *Fenpyroximate*.

Definition of the residue for dietary risk assessment for plant commodities and for dietary burden calculations: Sum of parent **fenpyroximate** and tert-butyl (Z)-α-(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneamino-oxy)-p-toluate (**its Z-isomer M-1**), expressed as fenpyroximate.

Definition of the residue for compliance with the MRL for animal commodities: Sum of **fenpyroximate** and (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneamino-oxy)methyl]benzoic acid (**M-3**), expressed as fenpyroximate.

The residue is fat-soluble.

Definition of the residue for dietary risk assessment for animal commodities: Sum of **fenpyroximate**, 2-hydroxymethyl-2-propyl (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methylenamino-oxy)methyl]benzoate (**Fen-OH**), 2-hydroxy-2-methylpropyl (E)-α-(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneamino-oxy)-p-toluate (**R-UL-1**) and (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneamino-oxy)methyl]benzoic acid (**M-3**), expressed as fenpyroximate.

DIETARY RISK ASSESSMENT

Long-term dietary exposure

The current Meeting revised the ADI for fenpyroximate to 0–0.005 mg/kg bw. The International Estimated Daily Intakes (IEDIs) for fenpyroximate were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the JMPR (including exposure to residues from cherries, peaches, watermelons, eggplants and tomatoes for which the 2017 JMPR indicated that the ARfD was exceeded). The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs ranged from 4–20% of the maximum ADI.

Shorter than lifetime and high consumer long-term dietary exposure

The Global Estimates of Chronic Dietary Exposure (GECDEs) for fenpyroximate were estimated for male and female, adults and children.

As the ADI for fenpyroximate was based on a potentially acute effect the risk from intermittent and shorter-than-lifetime exposure was assessed. The mean dietary exposure for each population subgroup were all less than the maximum ADI. Shorter-than-lifetime estimated high consumer dietary exposures exceeded the maximum ADI at the top end of the range for children and adolescents (1–210% of the maximum ADI) and infants and toddlers (8–118% of the maximum ADI). There were a number of commodities that were major contributors to high consumer dietary exposure estimates (> 50%) in one or more of the estimates for the countries and age groups considered, including kumquats, stems/stalks eaten as vegetables, cherries (sweet), blueberries, celery, bilberries, chayote fruits, watermelons, aubergines, raspberries, apple, soft drink with fruit, fruit juice, fennel, teas and offal.

The Meeting concluded that long-term dietary exposure to residues of fenpyroximate from uses considered by the Meeting may present a public health concern for high consumers.

Acute dietary exposure

The current Meeting revised the ARfD for fenpyroximate to 0–0.005 mg/kg bw.

As the current Meeting revised the ARfD for fenpyroximate, a new acute dietary risk assessment for all recommendations made by the 2017 and 2018 JMPRs was conducted in addition to those commodities considered by the current Meeting.. The International Estimate of Short Term Intakes (IESTIs) for fenpyroximate were calculated for the food commodities and their processed commodities for which HRs/HR-Ps or STMRs/STMR-Ps were estimated by the 2017, 2018 and 2021 Meetings and for which consumption data were available. The results are shown in Annex 4 of the 2021 JMPR Report.

The IESTIs were less than 100% of the ARfD, except for:

- Peeled mandarins (140% for Chinese children)
- Oranges (130% for Australian children)
- Apple (170% for Chinese children)
- Pear (190% for Dutch children)
- Cherries (240% for Danish children)
- Apricot (110% for German children)
- Nectarine (220% for Dutch toddlers)
- Peach (260% for Japanese children)
- Blackberries (300% for UK toddlers),
- Raspberries (260% for Irish children), raspberry juice (200% for German children)
- Blueberries (270% for Canadian children),
- Currants (290% for Australian general population), currant juice (460% for Dutch children),
- Gooseberries (140% for German children)
- Bitter melon (110% for Chinese general population)
- Chayote (200% for Chinese children)
- Cucumbers (190% for Czechian children)
- Angled loofah (110% for Thai children)
- Smooth loofah (170% for Chinese children)
- Snake gourd (110% for Thai children)
- Summer squashes (150% for US children)

Melons (160% for Belgian children)

Watermelons (370% for Canadian children)

Tomato (130% for Chinese children)

Eggplants (160% for Chinese children)

Phaseolus beans with pods (110% for Canadian children)

Cardoons (190% for Italian adults); cardoons cooked (260% for Dutch general population),

Celery (1410% for Chinese children), cooked celery (850% for Dutch general population), canned celery (120% for Dutch children), celery juice (220% for Australian general population),

Bulb fennel (530% for Australian children); cooked bulb fennel (1140% for Dutch children);

Rhubarb (680% for UK toddlers), rhubarb sauce (140% for Dutch children).

The Meeting concluded that acute dietary exposure to residues of fenpyroximate in commodities where the ARfD is exceeded may present a public health concern.

5.4 Fipronil (202)

TOXICOLOGY

Fipronil (M&B 46030) is the International Organization for Standardization (ISO)–approved name for 5-amino-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-4-[(*RS*)-(trifluoromethyl)sulfinyl]-1H-pyrazole-3-carbonitrile (IUPAC), for which the Chemical Abstracts Service (CAS) number is 120068-37-3.

Fipronil was previously evaluated by the FAO/WHO Joint Meeting on Pesticide Residues (JMPR) in 1997 and 2000. It was reviewed by the present Meeting as part of the periodic review programme of the Codex Committee on Pesticide Residues (CCPR).

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with current test guidelines, unless otherwise noted. Additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Biochemical aspects

Fipronil exhibits dose- and dose regimen-dependent, but sex-independent, oral absorption in rats; oral systemic bioavailability is estimated at 39–53%. Following oral absorption, fipronil undergoes rapid and extensive first-pass metabolism. As a result fipronil sulfone (M&B 46136) is the predominant systemically circulating and tissue chemical form found representing >90% of systemic exposure following oral dosing. Fipronil and its first-pass metabolites were extensively distributed and persistent in lipophilic tissues (up to 55% of the administered radioactivity at 168 hours post dosing). The terminal elimination half-life ($t_{1/2}$) for fipronil-derived radioactivity was up to 175 hours. The major route of elimination was via faeco-biliary excretion. Enterohepatic cycling is extensive.

Mammalian metabolism of fipronil has four main features:

- the dominant metabolic pattern is rapid first-pass hepatic metabolism to fipronil sulfone (M&B 46136), the predominant in vivo circulating and tissue metabolite;
- rats convert fipronil to fipronil sulfone (M&B 46136) more quickly than humans;
- the metabolism of fipronil by rats is more extensive compared with metabolism in other species, including humans;
- a large number of minor metabolites are also generated in vivo.

Several of these minor metabolites, and the metabolic pathways that generate them, are not fully characterized. Enzymes of the CYP3A family (CYP3A4 in humans, CYP3A2 and CYP3A1 in rats) predominantly catalyse the conversion of fipronil to fipronil sulfone. Fipronil sulfone is an inhibitor of the CYP2C subfamily. Enantioselective metabolism of fipronil was not toxicologically significant in laboratory mammals.

Toxicological data

The acute oral median lethal dose (LD_{50}) of fipronil was ≥ 49 mg/kg body weight (bw) and the dermal LD_{50} was ≥ 354 mg/kg bw. The inhaled median lethal concentration (LC_{50}) of fipronil was ≥ 0.36 mg/L. Fipronil was not irritating to skin under aqueous conditions but was slightly irritating to the skin under lipophilic conditions. Fipronil was slightly irritating to the eyes in rabbits and was not skin sensitizing in the guinea pig.

In a battery of secondary and safety pharmacology studies in mice, oral gavage dosing of fipronil induced convulsions and clinical signs of neurotoxicity at ≥ 30 mg/kg bw (NOAEL 10 mg/kg bw), reduced pentylenetetrazol seizure threshold at ≥ 10 mg/kg bw (NOAEL 3 mg/kg bw), and pupillary dilation and inhibition of gastrointestinal transport at 30 mg/kg bw (NOAEL 10 mg/kg bw). No effects on the

electroconvulsion threshold, body temperature, or hind limb traction were noted at doses up to 30 mg/kg bw. Fipronil at doses up to 30 mg/kg bw was not haemolytic.

Fipronil at concentrations up to 215.4 µg/mL was not phototoxic in an appropriately validated in vitro 3T3 NRU phototoxicity test.

In repeat-dose toxicity studies on mice, rats and dogs, the key effects were on the central nervous system, the liver and body weight parameters. Oral short-term toxicity studies with fipronil were conducted in the rat and the dog and consisted of four-week studies, ninety-day studies and, in the dog, two 1-year studies. Based on a human-relevance mode of action (MOA) evaluation, the effects of fipronil on the hypothalamic–pituitary–thyroid axis appeared secondary to its properties as a CAR and PXR receptor agonist. These effects are not, however, relevant to humans exposed to fipronil-derived residues in the diet.

No short-term studies of toxicity in the mouse were supplied. In a four-week oral toxicity study rats were fed diets incorporating fipronil at 0, 25, 50, 100, 200, or 400 ppm (equal to 0, 3.4, 6.9, 13, 24, and 45 mg/kg bw per day for males, 0, 3.5, 6.7, 13, 25, and 55 mg/kg bw per day for females). The NOAEL was 50 ppm (equal to 6.7 mg/kg bw per day) due to reduced body weight gain during the first week of treatment at ≥100 ppm (equal to 13 mg/kg bw per day).

In a 13-week oral toxicity study rats were fed diets incorporating fipronil at 0, 1, 5, 30 or 300 ppm (equal to 0, 0.07, 0.33, 1.9 and 20 mg/kg bw per day for males, 0, 0.07, 0.37, 2.3 and 24 mg/kg bw per day for females). The NOAEL was 30 ppm (equal to 1.9 mg/kg bw per day) due to increased body fat deposition occurring in males dosed at 300 ppm (equal to 20 mg/kg bw per day).

In a 4–6 week oral repeat-exposure toxicity study, dogs were capsule dosed with fipronil at 1 or 20 mg/kg bw per day for four weeks; other dogs were dosed at 0 or 10 mg/kg bw per day for six weeks. The NOAEL was 1 mg/kg bw per day due to clinical signs consistent with neurotoxicity at 10 mg/kg bw per day.

In a 13-week oral repeat-exposure toxicity study, dogs were capsule dosed with fipronil at 0, 0.5, 2 or 10 mg/kg bw per day. The NOAEL was 0.5 mg/kg bw per day due to reduced body weight gain at 2 mg/kg bw per day.

In a one-year oral repeat-exposure toxicity study, dogs were capsule dosed with fipronil at 0, 0.2, 2.0 or 5.0 mg/kg bw per day. The NOAEL was 0.2 mg/kg bw per day due to clinical signs consistent with neurotoxicity at 2.0 mg/kg bw per day.

In a one-year dietary exposure toxicity study, dogs were fed fipronil in food at concentrations that achieved doses equal to 0, 0.075, 0.3, 1 or 3 mg/kg bw per day; the highest dose was reduced to 2 mg/kg bw per day after 38 days. The NOAEL was 0.3 mg/kg bw per day due to clinical signs consistent with neurotoxicity at 1 mg/kg bw per day.

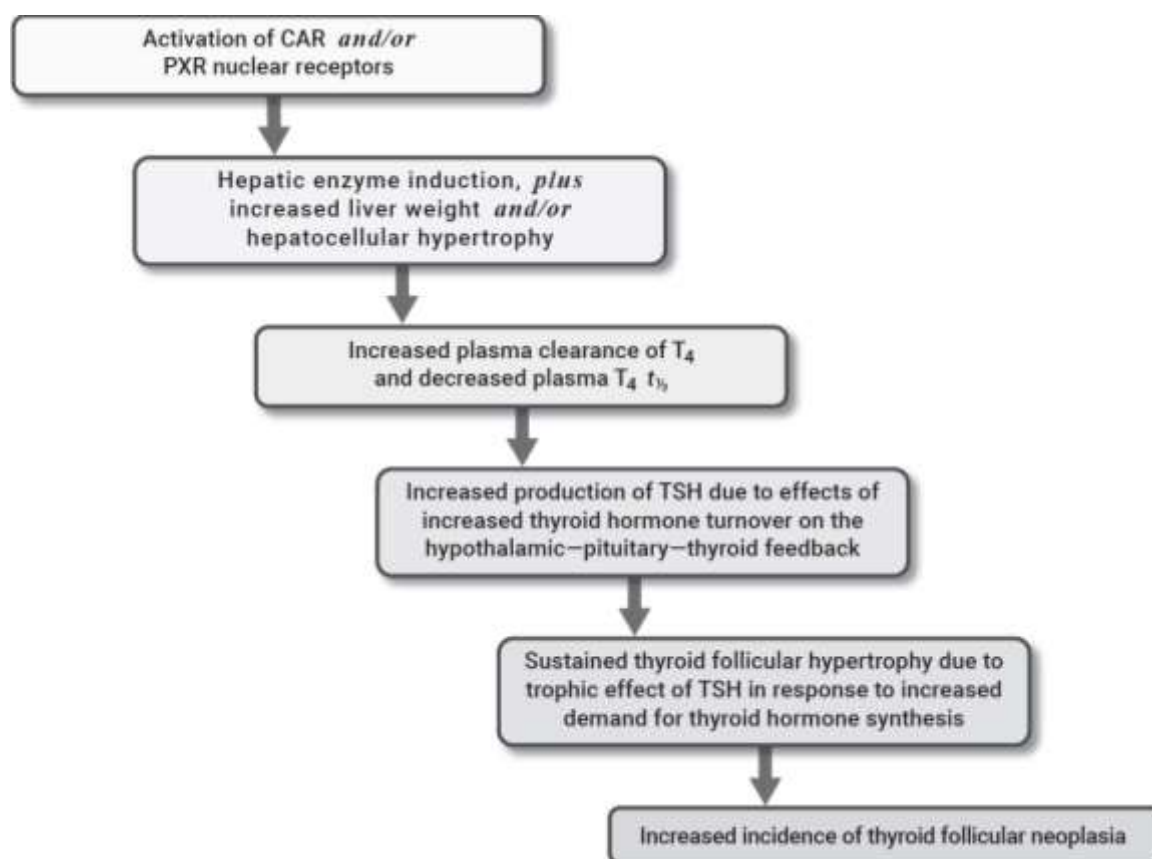
Based on the combined findings of the 13-week and one-year studies the overall NOAEL for repeat-dose toxicity in the dog was 0.3 mg/kg bw per day based on the occurrence of clinical signs of neurotoxicity occurring at doses of ≥1 mg/kg bw per day.

In a 78-week combined chronic toxicity and carcinogenicity study, mice were fed diets containing fipronil at 0, 0.1, 0.5, 10 or 30 ppm (equal to 0, 0.011, 0.055, 1.2 and 3.4 mg/kg bw per day in males, 0, 0.012, 0.063, 1.2 and 3.6 mg/kg bw per day in females). An additional cohort was fed a diet containing 60 ppm. The dose in mg/kg bw per day was not determined because the animals in this last cohort either died during the first nine weeks of exposure or were euthanized after 10 weeks of treatment. The NOAEL for the study was 0.5 ppm (equal to 0.055 mg/kg bw per day) due to an increased incidence of hepatic microvesicular periportal vacuolation in males and decreased body weight gain over the first 13 weeks of the study, both at 10 ppm (equal to 1.2 mg/kg bw per day). Fipronil was not carcinogenic in mice.

In an 89/91 week combined chronic toxicity and carcinogenicity study, rats were given dietary concentrations of 0, 0.5, 1.5, 30 or 300 ppm of fipronil (equal to 0, 0.019, 0.059, 1.3 and 13 mg/kg bw

per week in males, 0, 0.025, 0.078, 1.6 and 17 mg/kg bw per day in females). The NOAEL for toxicity was 0.5 ppm (equal to 0.019 mg/kg bw per day) due to clinical signs consistent with neurotoxicity at 1.5 ppm (equal to 0.059 mg/kg bw per day). The NOAEL for carcinogenicity was 30 ppm (equal to 1.3 mg/kg bw per day), based on an increase in the incidence of thyroid follicular cell neoplasia at 300 ppm (equal to 13 mg/kg bw per day).

Fipronil has phenobarbital-like CAR and PXR receptor agonist properties with oral exposure in rats inducing hepatic CYP1A1, CYP2B1, CYP2B2, CYP2E1, CYP3A1, CYP3A4, the carboxylesterases Ces2 and Ces6, the phase II enzymes UDP-glucuronosyltransferase 1A1 (UGT1A1), SULT1B1 and glutathione S-transferase alpha 2 (GSTA2), and the membrane transporters ATP binding cassette subfamily C Member 2 (Abcc2), Abcc3, Abcg5, Abcg8, and the solute carrier organic anion transporters 1a1 (Slco1a1) and Slco1a4. ToxCast high throughput screening evaluation confirmed that subcytotoxic concentrations of fipronil showed CAR receptor agonist behaviour. Based on a human relevance MOA evaluation, the effects of fipronil on the rat hypothalamic–pituitary–thyroid axis and induction of thyroid follicular epithelial neoplasia are consistent with the following mode of action:



Evaluating this from a human relevance standpoint, compared to rats humans are insensitive to this MOA. Thus, the effects of fipronil on the rat hypothalamic–pituitary–thyroid axis (and associated thyroid follicular epithelial cell neoplasia) have no relevance to humans exposed to fipronil at the levels that occur in the diet.

The Meeting concluded that fipronil is carcinogenic in rats due to a MOA that is not relevant to humans. Fipronil is not carcinogenic in mice.

Fipronil was tested in an adequate range of validated in vitro and in vivo genotoxicity assays. No evidence of genotoxicity was found.

The Meeting concluded that fipronil is unlikely to be genotoxic.

In view of the lack of genotoxicity, the absence of carcinogenicity in mice, and the fact that the thyroid tumors observed in rats occurred via a MOA considered not relevant to humans, the Meeting concluded that fipronil is unlikely to pose a carcinogenic risk to humans following exposure in the diet.

In a rat multigenerational reproductive toxicity study, rats were fed fipronil at dietary concentrations of 0, 3, 30 or 300 ppm (equal to 0, 0.25, 2.5 and 26 mg/kg bw per day in males, 0, 0.27, 2.7 and 28 mg/kg bw per day in females) continuously for two generations. The NOAEL for parental toxicity, offspring toxicity and reproductive toxicity was 30 ppm (equal to 2.5 mg/kg bw per day) due to the presence of parental effects (deaths, convulsions, hepatotoxicity, reduced body weight gain, reduced food consumption), reproductive effects (reduced mating performance, reduced fertility index) and developmental effects (reduced pup body weight gain, delayed tooth eruption, delayed pinna unfolding and reduced pre- and post-natal viability) at 300 ppm (equal to 26 mg/kg bw per day).

In a developmental toxicity study in rats, dose levels of 0, 1, 4 or 20 mg/kg bw per day of fipronil were administered by oral gavage from days 6 to 15 post coitum (PC). The NOAEL for maternal toxicity was 4 mg/kg bw per day due to reduced body weight gain at 20 mg/kg bw per day. The NOAEL for embryo/fetal toxicity was 20 mg/kg bw per day, the highest dose tested.

In a prenatal toxicity study in rabbits, dose levels of 0, 0.1, 0.2, 0.5 or 1.0 mg/kg bw per day were administered by oral gavage from days 6–19 PC. NOAEL for maternal toxicity was 0.2 mg/kg bw per day due to reduced body weight gain at 0.5 mg/kg bw per day. The NOAEL for embryo/fetal toxicity was 1.0 mg/kg bw per day, the highest dose tested.

In an acute neurotoxicity study rats were given a single oral gavage dose of 0, 0.5, 5.0 or 50.0 mg/kg bw. The NOAEL for the study was 0.5 mg/kg bw due to the presence of clinical signs consistent with neurotoxicity at 5.0 mg/kg bw per day.

In a second acute neurotoxicity study rats were administered a single oral gavage dose of 0, 2.5, 7.5 or 25 mg/kg bw of fipronil. The NOAEL for the study was 2.5 mg/kg bw due to decreased hind leg splay in males at 7.5 mg/kg bw per day. Taking into account the different dose ranges evaluated in the two studies of acute neurotoxicity the overall NOAEL for the two studies was 2.5 mg/kg bw.

In a 90-day neurotoxicity study rats were administered fipronil in the diet at concentrations of 0, 0.5, 5.0 or 150 ppm (equal to 0, 0.03, 0.3 and 8.9 mg/kg bw per day in males, 0, 0.04, 0.35 and 11 mg/kg bw per day in females). The NOAEL was 150 ppm (equal to 8.9 mg/kg bw per day), the highest dose tested.

In a postnatal developmental neurotoxicity study in rats, fipronil was administered continuously via the diet at dietary concentrations of 0, 0.5, 10 or 200 ppm (equal to 0, 0.05, 0.91 and 15 mg/kg bw per day) from day 6 PC to day 10 of lactation. The NOAEL for maternal toxicity, postnatal developmental neurotoxicity and developmental toxicity was 10 ppm (equal to 0.91 mg/kg bw per day).

In an oral, daily repeat-dose study of neurotoxicity, four female dogs were given 20 mg/kg bw per day of fipronil until signs of neurotoxicity were observed. Clinical signs consistent with neurotoxicity were observed after ≥ 6 days of dosing and signs of neurotoxicity persisted for up to 10 days following the cessation of dosing. There were no neuroanatomical abnormalities.

Toxicological data on metabolites and degradates

Fipronil sulfide (M&B 45950, fipronil thioether)

Fipronil sulfide (M&B 45950) is present in plants, livestock, milk and the environment.

Following oral dosing of rats with fipronil sulphide, the major route of excretion of radioactivity was the faeces ($\geq 66\%$ of the administered radioactivity). Absorbed radioactivity was widely distributed to the tissues but was highest in the fat.

Fipronil sulfide retains the $-\text{CN}$ group present on fipronil, but a $-\text{SCF}_3$ side chain replaces the parent's $-\text{SOCF}_3$ side chain. These features were associated with an approximately three- to five-fold

reduction in the IC_{50} for rat brain $GABA_A$ -gated chloride channel binding compared to that of fipronil. Notably, and in contrast to fipronil, clinical signs consistent with neurotoxicity were not observed in short-term repeated oral exposure studies in rats and dogs treated with fipronil sulfide.

The acute oral median lethal dose (LD_{50}) for fipronil sulfide was ≥ 69 mg/kg bw (similar to fipronil) and its dermal LD_{50} was $>500 < 4000$ mg/kg bw. Fipronil sulfide was not irritant to the skin of rabbits but was a slight eye irritant in this species.

In a 13-week toxicity study rats were administered fipronil sulfide in the diet at concentrations of 0, 10, 25, 50 or 300 ppm (equal to 0, 0.69, 1.8, 3.5 and 21 mg/kg bw per day in males, 0, 0.81, 2.2, 4.1 and 25 mg/kg bw per day in females). The NOAEL was 3.5 mg/kg bw per day due to effects on body weight, damaged vibrissae and nasal discharge observed at 21 mg/kg bw per day.

In a 28-day oral toxicity study dogs were capsule dosed with 0, 1, 5 or 15 mg/kg bw per day of fipronil sulfide. The NOAEL was 5 mg/kg bw per day due to reduced body weight gain at 15 mg/kg bw per day. Fipronil sulfide did not induce bacterial reverse mutations or chromosomal aberrations in human lymphocytes in vitro.

The Meeting concluded that fipronil sulfide is unlikely to be genotoxic.

Overall, the available data indicates that fipronil sulfide has similar acute toxicity to fipronil, has similar toxicity following repeated oral exposure (in rats 13-week oral LOAELs for fipronil sulfide and fipronil were 21 and 20 mg/kg bw per day, respectively) and a lower propensity to induce fipronil-like neurotoxicity. Fipronil sulfide has toxicological relevance in relation to dietary exposure to fipronil residues in the diet, is regarded as being equitoxic with fipronil (although likely having reduced neurotoxic potency) and is covered by the health-based guidance values for fipronil.

Fipronil sulfone (M&B 46136)

Fipronil sulfone (M&B 46136), the dominant systemic and tissue metabolite of fipronil in all evaluated mammalian species, is found in plants, livestock and the environment.

Oral bioavailability in rats was estimated to be $\geq 84\%$. Fipronil sulfone is widely distributed and persists in the tissues ($>35\%$ of fipronil sulfone and its metabolites remaining at 168 hours post dose). Up to 63% of fipronil sulfone and its metabolites are excreted via the faeco-biliary route. The dominant chemical species present in faeces was fipronil sulfone (up to 57% of the administered dose). Up to 10% of fipronil sulfone and its metabolites were excreted in urine. The dominant metabolite in urine was the 5-amino-glucuronide conjugate UMET/3 (up to 7% of the dose).

In fipronil sulfone the $-SOCF_3$ side chain of fipronil is replaced with $-SO_2CF_3$. This is associated with a rat brain $GABA_A$ -gated chloride binding IC_{50} about 2.5-fold to 5.8-fold lower than that of fipronil. Being the dominant chemical form in all tissues, fipronil sulfone is likely the main systemic driver of the neurotoxic effects seen with fipronil.

The acute oral LD_{50} of fipronil sulfone was ≥ 184 mg/kg bw (similar to fipronil) and the dermal LD_{50} was $>500 < 4000$ mg/kg bw. Fipronil sulfone was not irritant to the skin of rabbits but was a mild eye irritant in this species.

Fipronil sulfone did not induce bacterial reverse mutations or chromosomal aberrations in mammalian cells in vitro.

The Meeting concluded that fipronil sulfone (M&B 46136) is unlikely to be genotoxic.

Overall, the available data indicates that fipronil sulfone is likely responsible for the neurotoxic effects associated with the exposure of mammals to fipronil. Accordingly, based on this and the fact that it accounts for most of the dose ($>90\%$) in rats, it is toxicologically relevant and is covered by the health-based guidance values for fipronil.

***Fipronil desulfinyl (M&B 46513),
5-amino-1-(2,6-dichloro-4-(trifluoromethyl)phenyl)-4-trifluoromethylpyrazole-3-carbonitrile, a
photometabolite***

Fipronil desulfinyl (M&B 46513) is found in plants and livestock. Fipronil desulfinyl is also an environmental metabolite produced by soil metabolism as well as soil and aqueous photolysis.

Following oral dosing of rats fipronil desulfinyl was relatively well absorbed and extensively metabolized by oxidations followed by sulfate and amino acid conjugations. Fipronil desulfinyl was widely distributed, particularly to lipophilic tissues, and was persistent in the carcass (up to 17% of the administered radioactivity remaining at 168 hours post dose). Elimination was primarily in the faeces. The chemical identification of its various polar urinary and fecal metabolites (sulfate, glucuronide, cysteine and cysteine + glycine conjugates) is incomplete.

Fipronil desulfinyl retains fipronil's -CN group, but the -SOCF₃ sidechain present in fipronil is replaced with a -CF₃ group. This change results in a three- to five-fold decrease in the rat brain GABA_A-gated chloride binding IC₅₀ when compared with that of fipronil.

The acute oral LD₅₀ of fipronil desulfinyl was ≥15 mg/kg bw (about six-fold lower than fipronil) and the dermal LD₅₀ was >2000 mg/kg bw. The key clinical signs observed in the rat acute oral toxicity studies for fipronil desulfinyl were consistent with neurotoxicity: increased reactivity to noise at the LOAEL of ≥3 mg/kg bw, reduced motor activity at ≥10 mg/kg bw, convulsions at 30 mg/kg bw.

In a 90-day toxicity study mice were fed diets containing fipronil desulfinyl at 0, 0.5, 2 or 10 ppm (equal to 0, 0.08, 0.32, and 1.7 mg/kg bw per day in males, 0, 0.11, 0.43 and 2.2 mg/kg bw per day in females). The NOAEL was 2 ppm (equal to 0.32 mg/kg bw per day) due to mortality and neurotoxicity observed at 1.7 mg/kg bw per day.

In a 28-day toxicity study rats were administered fipronil desulfinyl in their diet at concentrations of 0, 0.5, 3, 30 or 100 ppm (equal to 0, 0.04, 0.23 and 2.2 mg/kg bw per day in males, 0, 0.04, 0.24 and 2.3 mg/kg bw per day in females). All animals fed at 100 ppm died before the completion of the study. The NOAEL was 3 ppm (equal to 0.23 mg/kg bw per day) due to the effects on body weight and neurotoxicity observed at 30 ppm (equal to 2.2 mg/kg bw per day).

In a 90-day toxicity study rats were fed diets containing fipronil desulfinyl at 0, 0.5, 3, 10 or 30 ppm (equal to 0, 0.029, 0.18, 0.59 and 1.8 mg/kg bw per day in males, 0, 0.035, 0.21, 0.71 and 2.1 mg/kg bw per day in females). The NOAEL was 0.5 ppm (equal to 0.029 mg/kg bw per day) due to the presence of clinical signs consistent with neurotoxicity at 3 ppm (equal to 0.18 mg/kg bw per day).

In a 28-day toxicity study dogs were fed diets containing fipronil desulfinyl at 0, 27, 80 or 270 ppm. All animals dosed at ≥80 ppm died before the completion of the study. The LOAEL was 27 ppm (equal to 1 mg/kg bw per day; the lowest dose tested) due to the evidence of neurotoxicity at this dose level.

In a 90-day toxicity study dogs were fed diets containing 0, 3.5, 9.5 or 35 ppm (equal to 0, 0.1, 0.27 and 0.95 mg/kg bw per day in males, 0, 0.1, 0.29 and 1.05 mg/kg bw per day in females) of fipronil desulfinyl for 90 days. The NOAEL was 9.5 ppm (equal to 0.27 mg/kg bw per day) due to clinical signs consistent with neurotoxicity at 35 ppm (equal to 0.95 mg/kg bw per day).

In a 24-month combined chronic toxicity and carcinogenicity study, rats were fed diets containing fipronil desulfinyl at 0, 0.5, 2 or 10 ppm, equal to 0, 0.025, 0.098 and 0.797 mg/kg bw per day in males. In females this was equal to 0, 0.032, 0.127, and 0.733 mg/kg bw per day. However, due to excessive mortality amongst high-dose females over the first 26 weeks their dose was reduced to 6 ppm (equal to 0.546 mg/kg bw per day) for the remainder of the study. The NOAEL for toxicity was 0.5 ppm (equal to 0.025 mg/kg bw per day) due to the presence of clinical signs consistent with neurotoxicity at 2 ppm (equal to 0.098 mg/kg bw per day). No evidence of treatment-related neoplastic changes was seen.

Fipronil desulfinyl did not induce bacterial reverse mutations or forward mutations in mammalian cells or chromosomal aberrations in mammalian cells in vitro.

The Meeting concluded that fipronil desulfinyl (M&B 46513) is unlikely to be genotoxic.

In a developmental toxicity study rats were orally gavage dosed with 0, 0.2, 1.0 or 2.5 mg/kg bw per day of fipronil desulfinyl over days 6–15 PC. The maternal NOAEL was 0.2 mg/kg bw per day due to effects on maternal body weight at 1.0 mg/kg bw per day. The NOAEL for developmental toxicity was 1.0 mg/kg bw per day due to the presence of reduced fetal weights at 2.5 mg/kg bw per day.

In an acute oral neurotoxicity study rats were administered single oral gavage doses of 0, 0.5, 2.0 or 12 mg/kg bw of fipronil desulfinyl in corn oil. The NOAEL for general toxicity and neurotoxicity was 2 mg/kg bw due to effects on body weight gain, hind leg splay, body temperature, motor activity and righting reflex at 12 mg/kg bw.

Overall the available database on fipronil desulfinyl is very extensive and indicated that it is approximately equitoxic with fipronil, has a similar toxicological profile to fipronil and is toxicologically relevant to exposure of fipronil-derived residues in the diet. It is covered by the health-based guidance values for fipronil.

M&B 45897

5-amino-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-1H-pyrazole-3-carbonitrile

Metabolite M&B 45897 is not a food or feed residue. It is an aerobic metabolite in soil.

Following oral dosing of rats, radioactivity derived from M&B 45897 was rapidly absorbed, and widely distributed to the tissues, with a terminal elimination half-life of about 115 hours. It was eliminated in the urine (about 52% over 168 h) and in the faeces (37% over 168 h). The major chemical form present in urine was unchanged M&B 45897 (about 27% of the urine radioactivity). A multi-component chromatographic peak designated UMET/4 was responsible for about 21% of the radioactivity present in urine. Other metabolites were present at <1% of the administered dose. Up to ten M&B 45897-derived metabolites were present in faeces, the most significant being FMET/1, which accounted for about 12% of the administered dose.

M&B 45897 lacks the –SOCF₃ side chain present on fipronil, but retains the –CN group. This is consistent with its high IC₅₀ for rat brain GABA_A-gated chloride binding, its low acute oral toxicity in rats, and a low propensity for fipronil-like neurotoxicity.

The acute oral (LD₅₀) of M&B 45897 was >2000 mg/kg bw and its dermal LD₅₀ was >2000 mg/kg bw. M&B 45897 was not irritant to the skin of rabbits but was a slight eye irritant in this species.

M&B 45897 did not induce bacterial reverse mutations, did not induce chromosomal aberrations in human lymphocytes in vitro and did not induce erythrocyte bone marrow micronuclei in mice in vivo. The Meeting concluded that M&B 45897 is unlikely to be genotoxic.

As M&B 45897 lacks the neurotoxic potential of fipronil, exposure in the diet should be compared with the Cramer class III threshold of toxicological concern, 1.5 µg/kg bw per day.

M&B 46400

5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-1H-pyrazole-4-carboxylic acid

M&B 46400 is found after the heat treatment of food at 120°C, as well as in livestock. It is also a soil and water hydrolysis product of the photometabolite fipronil desulfinyl (M&B 46513).

No data was submitted for this metabolite. M&B 46400 retains the –CN group of fipronil, but replaces fipronil's –SOCF₃ side chain with a –CO₂H group. Although definitive data is lacking, the known neurotoxicity structure activity relationships of this family of compounds predict that M&B 46400 is likely to have a low propensity for neurotoxicity.

M&B 46400 is a major metabolite in urine (>5% of the administered dose) after exposure of rats to M&B 46513. Therefore it can be assumed that M&B 46400 is also formed in significant amounts in the S9 mix reactions performed as part of the in vitro genotoxicity studies with M&B 46513. Consequently, the genotoxicity data for M&B 46513 is also applicable for M&B 46400. On this basis the Meeting concluded that M&B 46400 is unlikely to be genotoxic.

As M&B 46400 lacks the neurotoxic potential of fipronil, exposure in the diet should be compared with the Cramer class III threshold of toxicological concern, 1.5 µg/kg bw per day.

RPA 104615

potassium 5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-1H-pyrazole-4-sulfonate; a photometabolite

RPA 104615 is found in plants and is an environmental metabolite produced by soil and aqueous photolysis.

RPA 104615 retains the –CN group of fipronil, but replaces fipronil's –CF₃ moiety on the side chain with a –SO₃[–]. This is associated with high IC₅₀ value for rat brain GABA_A-gated chloride binding, and a high acute oral LD₅₀.

The acute oral LD₅₀ of RPA 104615 was >2000 mg/kg bw.

In a 28-day toxicity study rats were fed diets containing 0, 50, 500, 5000 or 10 000 ppm of RPA 104615 (equal to 0, 4.5, 45.7, 459 and 916 mg/kg bw per day in males, 0, 4.7, 50.4, 487 and 950 mg/kg bw per day in females). The NOAEL was 500 ppm (equal to 45.7 mg/kg bw per day) due to increased prothrombin time and increased serum triglycerides at 5000 ppm (equal to 459 mg/kg bw per day).

RPA 104615 did not induce bacterial reverse mutations and did not induce chromosome aberrations in human lymphocytes in vitro.

The Meeting concluded that RPA 104615 is unlikely to be genotoxic.

Due to the minimal toxicity of RPA 104615 at doses up to 459 mg/kg bw per day, exposure to RPA 104615 in the diet is not of toxicological concern.

RPA 105048

5-amino-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-(trifluoromethyl)-1H-pyrazole-3-carboxamide

RPA 105048 is a major (present at ≥10%) residue in food following heating at 120°C, as well as an environmental hydrolysis product of fipronil desulfinyl (M&B 46513).

RPA 105048 substitutes a –CF₃ side-chain for the –SOCF₃ side chain present in fipronil, and replaces the –CN moiety with a –CONH₂ group. Although definitive data is lacking, the known neurotoxicity structure activity relationships of this family of compounds predict that RPA 105048 is likely to have a low propensity for neurotoxicity.

RPA 105048 did not induce bacterial reverse mutations in vitro.

The Meeting concluded that RPA 105048 is unlikely to be genotoxic.

As RPA 105048 lacks the neurotoxic potential of fipronil, exposure in the diet should be compared with the Cramer class III threshold of toxicological concern, 1.5 µg/kg bw per day.

RPA 105320

5-amino-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-(trifluoromethylsulfonyl)-1H-pyrazole-3-carboxamide

RPA 105320 is found in plants and is also an environmental residue produced by soil hydrolysis.

RPA 105320, lacks fipronil's –CN group and has a –SOCF₃ side chain. This is associated with high IC₅₀ values for rat brain GABA_A-gated chloride binding (>10 000 nM) and a high acute oral LD₅₀ in rats. RPA 105320 is unlikely to cause fipronil-like neurotoxicity.

The acute oral LD₅₀ of RPA 105320 was >2000 mg/kg bw.

RPA 105320 did not induce bacterial reverse mutations in vitro.

The Meeting concluded that RPA 105320 is unlikely to be genotoxic.

As RPA 105320 lacks the neurotoxic potential of fipronil, exposure in the diet should be compared with the Cramer class III threshold of toxicological concern, 1.5 µg/kg bw per day.

RPA 108058

RPA 108058 is found in livestock that have been dosed with fipronil desulfinyl (M&B 46513).

No data was submitted for this metabolite. RPA 108058 has substantial structural differences from fipronil and is unlikely to produce fipronil-like neurotoxicity.

Based on QSAR evaluation, RPA 108058 is not predicted to be genotoxic. RPA 108058 is also a metabolite of fipronil amide (RPA 200766), which did not induce bacterial reverse mutations or chromosomal aberrations in mammalian cells in vitro.

The Meeting concluded that RPA 105320 is unlikely to be genotoxic at levels likely to occur in the diet.

As RPA 108058 is considered to lack the neurotoxic potential of fipronil, dietary exposure should be compared with the Cramer class III threshold of toxicological concern per day, **1.5 µg/kg bw per day**.

RPA 200761

5-amino-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-4-[(trifluoromethyl)sulfinyl]-1H-pyrazole-3-carboxylic acid

RPA 200761 is found in plants.

RPA 200761 was detected in the bile of rats orally dosed with fipronil; the fraction of administered radioactivity was not reported. Compared with fipronil it lacks the –CN group but possesses a –SOCF₃ side chain. This is associated with high IC₅₀ values for rat brain GABA-gated chloride channel binding (>10 000 nM) and a high acute oral LD₅₀ in rats).

The acute oral LD₅₀ of RPA 200761 was >2000 mg/kg bw.

RPA 200761 is a metabolite of fipronil amide (RPA 200766). Fipronil amide did not induce bacterial reverse mutations, or chromosomal aberrations in mammalian cells in vitro.

The Meeting concluded that RPA 200761 is unlikely to be genotoxic at levels likely to occur in the diet.

As RPA 200761 lacks the neurotoxic potential of fipronil, exposure in the diet should be compared with the Cramer class III threshold of toxicological concern, 1.5 µg/kg bw per day.

Fipronil amide, fipronil carboxamide (RPA 200766),

5-amino-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-4-[(trifluoromethyl)sulfinyl]-1H-pyrazole-3-carboxamide

Fipronil amide (RPA 200766) is found in plants and livestock. It is an environmental metabolite formed by soil metabolism and hydrolysis.

Fipronil amide was detected in the faeces and urine (approximately 2%) of rats orally dosed with fipronil. Compared with fipronil it replaces the –CN group with a –CONH₂ group, and also has a –SOCF₃

side chain. This is associated with high IC_{50} values for rat brain GABA_A-gated chloride binding (>10 000 nM) and a high acute oral LD_{50} in rats.

The acute oral LD_{50} of fipronil amide (RPA 200766) was >2000 mg/kg bw.

In a 28-day toxicity study, rats were fed diets containing 0, 50, 500, 5000 or 15 000 ppm (equal to 0, 3.8, 38, 385 and 1087 mg/kg bw per day in males, 0, 4.4, 44, 387 and 1063 mg/kg bw per day in females) of fipronil amide. The NOAEL was 50 ppm (equal to 3.8 mg/kg bw per day) due to the presence of adverse adrenal zona fasciculata vacuolation at 500 ppm (equal to 38 mg/kg bw per day).

Fipronil amide did not induce bacterial reverse mutations, or chromosomal aberrations in mammalian cells in vitro.

The Meeting concluded that fipronil amide (RPA 200766) is unlikely to be genotoxic.

Overall, the available data indicates that fipronil amide (RPA 200766) is likely to be less toxic than fipronil, but has toxicological relevance to exposure of fipronil-derived residues in the diet. It is covered by the health-based guidance values for fipronil.

Overall conclusions on metabolites

The Meeting concluded that fipronil sulfide (M&B 45950), fipronil sulfone (M&B 46136), fipronil desulfinyl (M&B 46513) and fipronil amide (RPA 200766) are toxicologically relevant to exposure to fipronil-derived residues in the diet and are of equal or less potency than the parent. Risk assessment should be performed using the health-based guidance values for fipronil.

RPA 104615 was of considerably lower toxicity than fipronil and other metabolites and was considered not to be toxicologically relevant.

Exposure in the diet to the remaining fipronil-derived metabolites should be compared with the Cramer class III threshold for toxicological concern, 1.5 µg/kg bw per day.

Microbiological data

Fipronil, at the level of exposure encountered in food, is unlikely to result in adverse effects on the human gut microbiome.

Human data

No cases of human occupational disease associated with the production, transportation, formulation and packaging of fipronil-containing materials have been reported.

In terms of occupational exposure, one study examined thyroid function among workers involved in production of fipronil-containing veterinary drugs. Fipronil was detected in serum of around 20% of workers while fipronil sulfone was detected in all participants. Eighteen workers had abnormal serum thyroid levels of which seven had elevated thyroid stimulating hormone (TSH). Serum fipronil and fipronil sulfone were not associated with thyroid hormone levels, with the exception of a modest inverse relationship between fipronil sulfone and TSH (Spearman's $\rho \sim -0.2$). Taken together this study found no suggestion for a link between occupational exposure to fipronil and thyroid function.

In terms of background exposed population, one small study ($n = 59$) from South Korea quantified fipronil sulfone in maternal serum and cord blood sampled at delivery. The concentrations in these samples were highly correlated (Pearson's $r \sim 0.75$), suggesting efficient placental transfer. An inverse correlation of fipronil sulfone with T3 and freeT3 concentrations in cord blood was observed, but no association was found for other thyroid hormones. Fipronil sulfone in cord blood was also inversely associated with Apgar score five minutes after birth. Given the cross-sectional nature of these analyses and the small sample size, no conclusion on possible associations with newborn thyroid hormones or Apgar score can be drawn.

The human clinical toxidrome associated with suicidality-associated ingestion of fipronil is consistent with its neurotoxicological MOA (clinical signs of central neurotoxicity including tonic-clonic convulsions). In general, such poisonings are responsive to GABA_A agonist drugs (benzodiazepines, barbiturates).

The Meeting concluded that the existing database on fipronil was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting established an ADI of 0–0.000 2 mg/kg bw per day based on the NOAEL of 0.019 mg/kg bw per day in the combined chronic toxicity and carcinogenicity study in rats and using a safety factor of 100.

The Meeting established an ARfD (for the general population) of 0.03 mg/kg bw, on the basis of the group NOAEL of 3 mg/kg bw per day in the acute neurotoxicity study in rats and the safety pharmacology studies (pentylenetetrazol seizure threshold) in mice. A safety factor of 100 was used. Because of the long elimination half-lives of fipronil and fipronil sulfone (the major chemical form in the body) intermittent and repeated exposures to fipronil dietary residues over any duration should not exceed the ADI.

A toxicological monograph was prepared.

Levels relevant to risk assessment of fipronil

Species	Study	Effect	NOAEL	LOAEL
Mouse	78-week study of toxicity and carcinogenicity ^a	Toxicity	0.5 ppm, equal to 0.6 mg/kg bw per day	10 ppm, equal to 1.2 mg/kg bw per day
		Carcinogenicity	30 ppm, equal to 3.6 mg/kg bw per day ^c	-
	Acute pentylenetetrazol seizure threshold ^b	Neurotoxicity	3 mg/kg bw	10 mg/kg bw
Rat	Acute neurotoxicity study ^b	Neurotoxicity	2.5 mg/kg bw	5.0 mg/kg bw
	Two-year studies of toxicity and carcinogenicity ^{a, d}	Toxicity	0.5 ppm, equal to 0.019 mg/kg bw per day	1.5 ppm, equal to 0.06 mg/kg bw per day
		Carcinogenicity	300 ppm, equal to 16.8 mg/kg bw per day ^c	-
	Two-generation study of reproductive toxicity ^a	Reproductive toxicity	30 ppm, equal to 2.5 mg/kg bw per day	300 ppm, equal to 26 mg/kg bw per day
		Parental toxicity	30 ppm, equal to 2.5 mg/kg bw per day	300 ppm, equal to 26 mg/kg bw per day
		Offspring toxicity	30 ppm, equal to 2.5 mg/kg bw per day	300 ppm, equal to 26 mg/kg bw per day
	Developmental toxicity study ^b	Maternal toxicity	4 mg/kg bw per day	20 mg/kg bw per day
		Embryo and fetal toxicity	20 mg/kg bw per day ^c	-
Rabbit	Developmental toxicity study ^b	Maternal toxicity	0.2 mg/kg bw per day	0.5 mg/kg bw per day
		Embryo and fetal toxicity	1.0 mg/kg bw per day ^c	-
Dog	13-week and one-year studies of toxicity ^{d, f}	Toxicity	0.3 mg/kg bw per day	1 mg/kg bw per day

Species	Study	Effect	NOAEL	LOAEL
<i>Fipronil sulfide (M&B 45950)</i>				
Rat	13-week study of toxicity ^a	Toxicity	3.5 mg/kg bw per day	21.5 mg/kg bw per day
<i>Fipronil sulfone (M&B 46136)</i>				
Major systemic active metabolite that is responsible for >90% of systemic exposure following oral dosing				
<i>Fipronil desulfinyl (M&B 46513, photometabolite)</i>				
Rat	Two-year studies of toxicity and carcinogenicity ^{a, d}	Toxicity	0.5 ppm, equal to 0.03 mg/kg bw per day	2 ppm, equal to 0.1 mg/kg bw per day
		Carcinogenicity	10 ppm, equal to 0.8 mg/kg bw per day ^c	-
<i>Fipronil amide (RPA 200766)</i>				
Rat	28-day study of toxicity ^a	Toxicity	3.8 mg/kg bw per day	38.2 mg/kg bw per day

^a Dietary administration

^b Gavage administration

^c Highest dose tested

^d Two or more studies combined

^e Lowest dose tested

^f Capsule administration

Acceptable daily intake (ADI) applies to fipronil, fipronil sulfide (M&B 45950), fipronil sulfone (M&B 46136), fipronil amide (RPA 200766) and fipronil desulfinyl (M&B 46513) expressed as fipronil

0–0.000 2 mg/kg bw

Acute reference dose (ARfD) applies to fipronil, fipronil sulfide (M&B 45950), fipronil sulfone (M&B 46136), fipronil amide (RPA 200766) and fipronil desulfinyl (M&B 46513) expressed as fipronil

0.03 mg/kg bw

Because of the long elimination half-lives of fipronil and fipronil sulfone (the major chemical form in the body) intermittent and repeated exposures to fipronil dietary residues over any duration should not exceed the ADI.

Information that would be useful for the continued evaluation of the compound

No additional data is required.

Critical end-points for setting guidance values for exposure to [compound]

Absorption, distribution, excretion and metabolism in mammals

Rate and extent of oral absorption	Rats: low to high (estimated 34 to 90%) and was dose- and dose regimen-dependent (but sex independent) Rabbits: high (estimated 80%)
Dermal absorption	Rat: low (<0.4% to <3%); dose-dependent
Distribution	Rat: rapid and extensive with accumulation in lipophilic tissues;

Potential for accumulation	Rabbit: rapid and extensive with accumulation in lipophilic tissues; Significant potential for accumulation. Rat: present in the tissues for at least 99 days post dose; mean adipose tissue elimination $t_{1/2}$ of up to 545 ± 56 h following repeated oral dosing. Retention of administered radioactivity in the tissues was dose-dependent and was higher with lower doses Rabbit: present in the tissues for at least 75 days post dose; blood elimination $t_{1/2}$ up to about 49.5 days
Rate and extent of excretion	Rat: following a single oral dose the radioactivity mean terminal elimination $t_{1/2}$ was about 175 h, likely due to the slow release of fipronil sulfone from lipophilic tissues; considerable enterohepatic recirculation occurs Rabbit: excretion via urine and faeces was protracted, likely reflecting the slow release of radioactivity from lipophilic tissues
Metabolism in animals	Rat: fipronil underwent rapid and extensive first-pass metabolism to fipronil sulfone (major systemic metabolite); minor metabolites included fipronil conjugates (mostly glucuronides); the conjugates were rapidly excreted in urine whereas fipronil sulfone was the major component (90% of radiolabel) in tissues (especially in lipophilic tissues) Humans: based on in vitro data, conversion from fipronil to fipronil sulfone is slower in humans than in rats
Toxicologically significant compounds in animals and plants	Fipronil, fipronil sulfone, fipronil sulfide, fipronil amide and fipronil desulfinyl

Acute toxicity

Rat, LD ₅₀ , oral	92 mg/kg bw
Rat, LD ₅₀ , dermal	>2000 mg/kg bw
Rabbit, LD ₅₀ , dermal	354 mg/kg bw
Rat, LC ₅₀ , inhalation	0.36 mg/L
Rabbit, dermal irritation	Aqueous conditions: not irritating Lipophilic conditions: slightly irritating
Rabbit, ocular irritation	Slightly irritating
Guinea pig, dermal sensitization	Not sensitizing (Buehler three-fold induction)

Short-term studies of toxicity

Target/critical effect	CNS neurotoxicity
Lowest relevant oral NOAEL	0.3 mg/kg bw per day (dog)
Lowest relevant dermal NOAEL	5 mg/kg bw per day (rabbit; occlusive exposure)
Lowest relevant inhalation NOAEC	No data.

Long-term studies of toxicity and carcinogenicity

Target/critical effect	CNS neurotoxicity
Lowest relevant NOAEL	0.2 mg/kg bw per day (rat)
Carcinogenicity	Carcinogenic in rats ^a Thyroid carcinogen in rats secondary to CAR xenosensor activation-associated increased plasma clearance of T4 and T4 $t_{1/2}$: this mode of action is not relevant to humans

Genotoxicity		Unlikely to be genotoxic
Reproductive toxicity		
Target/critical effect		Increased mortality, convulsions, reduced body weight gain and food consumption, reduced mating index, reduced fertility index, reduced viability index, delayed tooth eruption and pinna unfolding, and hepatotoxicity
Lowest relevant parental NOAEL		2.5 mg/kg bw per day (rat)
Lowest relevant offspring NOAEL		2.5 mg/kg bw per day (rat)
Lowest relevant reproductive NOAEL		2.5 mg/kg bw per day (rat)
Developmental toxicity		
Target/critical effect		Maternal: reduced body weight gain Embryo/fetal: no adverse effects
Lowest relevant maternal NOAEL		0.1 mg/kg bw per day (rabbit)
Lowest relevant embryo/fetal NOAEL		1 mg/kg bw per day; highest dose tested (rabbit)
Neurotoxicity		
Acute neurotoxicity NOAEL		3 mg/kg bw (rat)
Subchronic neurotoxicity NOAEL		9 mg/kg bw per day; highest dose tested (rabbit)
Developmental neurotoxicity NOAEL		0.9 mg/kg bw per day (rat)

Immunotoxicity	No direct data, but no evidence of immunotoxicity in the general toxicity studies
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Studies on toxicologically relevant metabolites

Fipronil desulfinyl	Acute oral LD₅₀: 18 mg/kg bw (rat) 28-day NOAEL: 0.2 mg/kg bw per day (rat) 90-day NOAEL: 0.03 mg/kg bw per day (rat) 53/104-week combined chronic toxicity/carcinogenicity NOAEL: 0.025 mg/kg bw per day; not carcinogenic (rat) Not genotoxic (bacterial reverse mutation, chromosomal aberration and mammalian cell gene mutation assays, in vitro) Developmental toxicity (rat): Maternal NOAEL: 0.2 mg/kg bw per day Developmental NOAEL: 1 mg/kg bw per day; highest dose tested Acute neurotoxicity NOAEL: 2 mg/kg bw
Microbiological data	Not data relevant to humans

Human data

No deaths attributable to fipronil exposure alone have been reported. No cases of human occupational disease associated with the production, transportation, formulation and packaging of fipronil containing materials have been reported. While fipronil exposure was inversely correlated with blood thyroid hormones in people with occupational exposure to fipronil during veterinary drug manufacture, there were no effects on blood TSH concentrations detected. While most cases of fipronil acute neurotoxicity associated with suicidal ideation are minor and have generally good clinical outcomes, serious (non-fatal) cases of convulsive/status epilepticus disorders have been reported which in some cases have responded poorly to treatment with benzodiazepines. While deaths due to suicidal ideation-associated ingestion of fipronil alone have not been reported, a death due to fipronil plus endosulfan ingestion has been reported.

^a Does not pose a carcinogenic risk to humans via exposure from the diet

Summary

	Value	Study	Safety factor
ADI	0–0.000 2 mg/kg bw per day ^a	Two-year studies of toxicity (rat)	100
ARfD	0.03 mg/kg bw	Studies of the acute neurotoxicity of fipronil (rat) Acute pentylenetetrazol seizure threshold (rat)	100

^a Applies to fipronil, fipronil sulfide (M&B 45950), fipronil sulfone (M&B 46136), fipronil amide (RPA 200766) and fipronil desulfinyl (M&B 46513) expressed as fipronil

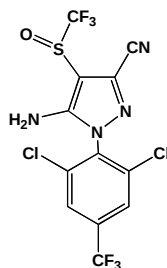
RESIDUE AND ANALYTICAL ASPECTS

Fipronil is a broad-spectrum insecticide that belongs to the phenylpyrazole chemical family and is used for the control of a wide range of crop, public hygiene, amenity, and veterinary pests. It was evaluated by

JMPR in 1997 (T), 2000 (T), 2001 (R) and 2016 (R). At the Fifty-first Session of the CCPR (2019), it was scheduled for periodic evaluation by the 2020 JMPR, which was postponed to the 2021 JMPR.

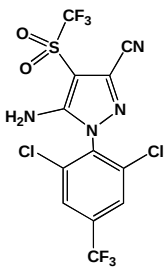
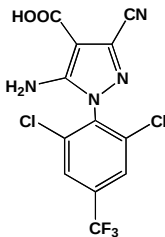
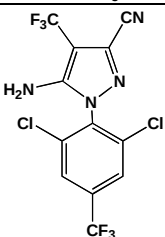
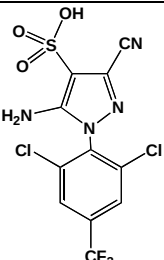
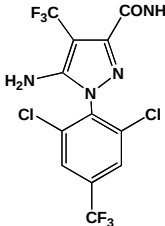
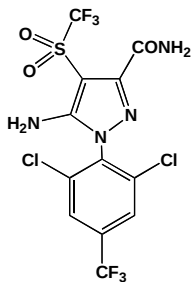
The Meeting received information on the metabolism of fipronil and its photometabolite fipronil-desulfinyl in lactating goats and laying hens, the metabolism of fipronil in cabbage, potato, sugar beet, sunflower, cotton, maize, rice and wheat and follow crops, methods of residue analysis, freezer storage stability, GAP information, supervised residue trials on a range of crops as well as livestock feeding studies (lactating cow, laying hen).

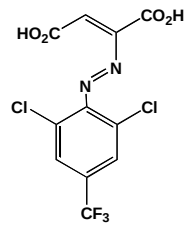
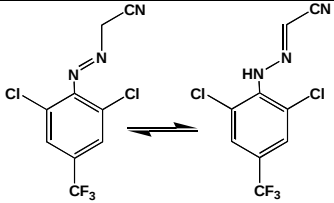
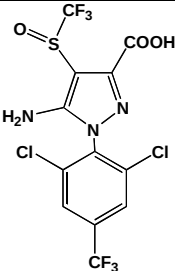
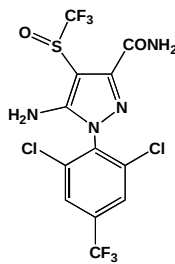
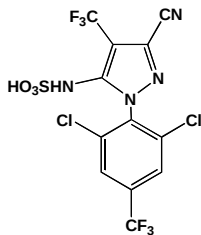
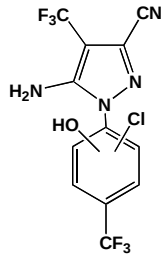
Fipronil is 5-amino-1-(2,6-dichloro- α,α,α -trifluoro-p-tolyl)-4-[(trifluoromethyl)sulfinyl]-1H-pyrazole-3-carbonitrile.



The following abbreviations are used for the major metabolites discussed below:

Code Names, MW	Chemical Name	Chemical Structure
MB45897	5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)pyrazole	
MB45950 Fipronil-sulfide Fipronil-thioether	5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylthiopyrazole	
MB46126 Fipronil-sulfide-amide		

Code Names, MW	Chemical Name	Chemical Structure
MB46136 Fipronil-sulfone	5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfonylpyrazole	
MB46400	5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)pyrazole-4-carboxylic acid	
MB46513 Fipronil-desulfinyl	5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylpyrazole	
RPA104615 Fipronil-sulfonate	5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)pyrazole-4-sulfonic acid	
RPA105048 Fipronil-carboxamide	5-amino-3-carbamoyl-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylpyrazole	
RPA105320 Fipronil-sulfone-amide	5-amino-3-carbamoyl-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfonylpyrazole	

Code Names, MW	Chemical Name	Chemical Structure
RPA106889	5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)pyrazole-3,4-dicarboxylic acid	
RPA108058		
RPA200761 Fipronil-acid Fipronil-carboxy	5-amino-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfinylpyrazole-3-carboxylic acid	
RPA200766 Fipronil-amide	5-amino-3-carbamoyl-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfinylpyrazole	
Fipronil-N-sulfate		
Fipronil-mono-dechloro, monohydroxy		
2,6-dichloro-4-(trifluoromethyl)-aniline	2,6-dichloro-4-(trifluoromethyl)-aniline	

Based on the information on physical and chemical properties, fipronil is not volatile and is lipid soluble with a log P_{ow} of around 4. Fipronil is hydrolytically stable at environmental pH. Aqueous photolysis is likely to be a major degradation pathway of fipronil in the environment.

Metabolism

The Meeting received studies on the metabolism of fipronil in plants, laboratory animals as well as lactating goats and laying hens. Metabolism studies for the photo-metabolite fipronil-desulfinyl (MB46513) in lactating goat and laying hen were also made available to the Meeting. The studies on laboratory animals were evaluated by the WHO Core Assessment Group.

The metabolism of fipronil in plants, animals and soils was investigated using [phenyl- ^{14}C]-fipronil and of fipronil-desulfinyl (MB46513) in animals using [phenyl- ^{14}C]-fipronil desulfinyl.

Some reports were also located in the scientific literature that are relevant to plant metabolism. Fipronil consists of a racemic mixture of enantiomers. The stereoisomers of chiral pesticides may show selective metabolism in plants which can result in residues with a stereoisomer composition different from that in the applied product. Different stereoisomers can also exhibit different toxicological properties.

Plant metabolism

In a study of the stereoisomer composition of fipronil residues following application to Chinese cabbage crops, the degradation of the two enantiomers of fipronil in Chinese cabbage was enantioselective. R-(-)-fipronil degraded faster than S-(+)-fipronil, resulting in the relative enrichment of S-(+)-form in residue with the fraction of the S-(+)-enantiomer reaching 0.74 by 10 days after application. Contrary to what was observed with Chinese cabbage, in tea the fraction of the S-(+)-enantiomer remained close to 0.5 (range 0.465–0.485) in leaves, shoots and processed green tea suggesting that racemic-fipronil did not undergo significant enantioselective degradation in tea crops. In another report, the stereoselective metabolism in water hyacinth (*Eichhornia crassipes*), the fraction of the S-(+)-enantiomer ranged from 0.52 to 0.72.

There is no consistent picture for the stereo-selective metabolism of fipronil in plants, though where relative enrichment occurs it results in an increased proportion of the S-(+)-enantiomer.

Fipronil is typically used for three different situations:

- As a seed treatment
- Application to soil, prior to, or shortly after planting/sowing
- As a broadcast spray to the growing crop, i.e., post-emergence

The Meeting received plant metabolism studies with ^{14}C -fipronil following its use as a seed treatment, as a soil application and as a foliar application. The crops studied are representative of leafy vegetables (cabbage), root and tuber vegetables (potato, sugar beet), cereals (maize, rice, wheat) and oilseeds (sunflower, cotton).

Soil/Seed treatments

Sugar beet

The metabolic fate of ^{14}C -fipronil in sugar beet plants maintained in a greenhouse was examined following an in-furrow application of a WG-formulation to soil with seeds sown directly onto the treated soil. Plants were harvested after 2, 4 and 6 months.

Following application at 0.2 kg ai/ha, the majority of the ^{14}C in the whole plant was located in beet leaves (90.6–93.6%) with lower amounts in roots (6.4–9.4%).

The extractability of ^{14}C with acetone followed by methanol/water was good for sugar beet leaf and roots (> 73% TRR)

Fipronil parent was either not present or only present at low levels in leaves. The major component of ^{14}C identified in leaves was fipronil-sulfone (MB46136; 0.18–0.23 mg eq/kg, 30.1–35.4% TRR), followed by fipronil-sulfone-amide (RPA105320; 0.11–0.12 mg eq/kg, 17.3–19.0% TRR) with small amounts of fipronil-sulfide (MB45950; 0.017–0.029 mg eq/kg, 2.5–4.5% TRR), and MB45897; (0.01–0.019 mg eq/kg, 1.3–2.9% TRR) also detected.

In roots, fipronil parent accounted for 7.5–16.3% TRR. Fipronil-sulfone (MB46136; 0.028–0.053 mg eq/kg, 58.7–76.1% TRR) was again the main component of the ^{14}C . Fipronil-amide (RPA200766; 0.002–0.003 mg eq/kg, 4.3–5.5% TRR) was also detected.

Maize

Two studies were available on the metabolism of fipronil in maize (maintained in greenhouse) following soil application. In the first, ^{14}C -fipronil was applied as a soil application of a WG-formulation at an application rate of 0.41 kg ai/ha. Seeds were sown directly onto the granule treated soil. Forage was collected 42 days after sowing, fodder at 106 days and grain at 98 days.

Only a small amount (4.3%) of the soil applied radioactivity was taken up by the maize plants. The majority of the radioactivity taken up by the plant remained in the forage or fodder, with less than 7.5% of the radioactivity translocating to the grain.

The extractability of ^{14}C with acetonitrile and acetonitrile/water was good for forage (> 75% TRR) and fodder (> 93% TRR) and with acetonitrile and water was good for grain (> 67% TRR). Methanol extracted an additional 29% TRR from grain.

In forage, fipronil (0.08 mg/kg, 38.1% TRR), fipronil-amide (RPA200766; 0.03 mg eq/kg, 12.4% TRR), and fipronil-sulfone (MB46136; 0.02 mg eq/kg, 8.6% TRR) were the main components of ^{14}C .

In fodder, major components of ^{14}C were fipronil (0.40 mg/kg, 10.9% TRR), fipronil-acid (RPA200761; 0.28 mg eq/kg, 7.6% TRR), fipronil-amide (RPA200766; 0.93 mg eq/kg, 25.1% TRR), fipronil-sulfide (MB45950; 0.06 mg eq/kg, 1.6% TRR), and fipronil-sulfone (MB46136; 0.99 mg eq/kg, 26.8% TRR).

In grain, initial extraction identified fipronil as the major residue (0.08 mg/kg, 38.5% TRR) however, analysis on the same sample after two years incorporating different extraction solvents, improved clean-up and chromatography identified a conjugate of fipronil-amide (RPA200766, 0.14 mg eq/kg, 87.5% TRR) as the major component. The conjugate was tentatively identified as a malonic acid conjugate.

In the second study, ^{14}C -fipronil was applied as a pre-emergent soil application of a WG-formulation at an application rate of 0.146 kg ai/ha with seeds sown directly onto the treated soil. Forage was collected 35 days after sowing, fodder at 106 days and grain at 90 days.

At mature harvest, 0.81% of the applied radioactivity had been taken up by and distributed into the above ground portions of the plants that were grown to maturity. In the whole plant, fodder accounted for 95.1% TRR with 1.8% TRR located in the cob and 3.1% TRR in the grain.

Forage contained 0.11 mg eq/kg, fodder 0.51 mg eq/kg, the cob 0.025 mg eq/kg, and grain 0.013 mg eq/kg.

The extractability of ^{14}C using Procedure 1 (acetonitrile and acetonitrile/water) was good for forage (> 131% TRR) and fodder (> 86% TRR) and was poor for grain (> 57% TRR). Overall, extractability was improved using Procedure 2 (acetonitrile/water) which extracted 107% TRR from forage, 74% TRR for fodder and 95% TRR from grain.

In forage the main components of ^{14}C identified were fipronil at 0.044 mg/kg (39% TRR), fipronil-amide (RPA200766; 0.033 mg eq/kg, 30% TRR), fipronil-sulfone (MB46136; 0.013 mg eq/kg, 11.6% TRR), and fipronil-acid (RPA200761; 0.012 mg eq/kg, 10% TRR).

In fodder, fipronil (0.061 mg/kg, 12% TRR), fipronil-amide (RPA200766; 0.19 mg eq/kg, 38% TRR), and fipronil-sulfone (MB46136; 0.082 mg eq/kg, 16% TRR) were the main components with low levels of

fipronil-acid (RPA200761; 0.008 mg eq/kg, 1.6% TRR) and fipronil-sulfone-amide (RPA105320; 0.007 mg eq/kg, 1.4% TRR) also detected.

In grain, fipronil-amide (RPA200766; 0.008 mg eq/kg, 60% TRR) was the only component detected.

Conclusions regarding the identity of the major component of ^{14}C in grain in the two maize studies differed. Depending on the extraction and quantification methods used, the main component of the ^{14}C was variously identified as fipronil, fipronil-amide or a conjugate of fipronil-amide.

Rice

The metabolism of ^{14}C -fipronil in paddy rice plants grown outdoors was studied. Twenty days after rice seedlings were transplanted into soil covered by 5–10 cm water, the plots were treated with a granular formulation at 0.05 kg ai/ha. Immature plants were sampled at 32 days after the granular treatment. At harvest samples were separated into roots, straw, ears, panicles, husk, brown rice, polished rice and bran.

Extractability of ^{14}C residues with methanol was good at 74–85% TRR for the various rice commodities.

Fipronil parent accounted for 6.3–15.3% TRR in root, plant, straw samples and 25.4% TRR in brown rice. Major metabolites in root, plant and straw were fipronil-sulfide (MB45950), fipronil-sulfone (MB46316) and fipronil-desulfinyl (MB46513) at 6.4–24.5, 5.0–17.3 and 11.7–23.4% TRR respectively. In brown rice, the major metabolites were fipronil-amide (RPA200766; 0.0013 mg eq/kg, 12.1% TRR) and fipronil-sulfide (MB45950; 0.00025 mg eq/kg, 4.1% TRR). In polished rice the major components of ^{14}C were fipronil-amide (RPA200766; 0.0009 mg eq/kg, 22.8% TRR), fipronil (0.0007 mg/kg, 17.6% TRR) and fipronil-sulfide (MB45950; 0.0002 mg eq/kg, 5.1% TRR).

Wheat

The metabolism of ^{14}C -fipronil was studied on wheat (variety Consort) maintained in a greenhouse. Fipronil (^{14}C) was applied as a seed treatment of a WG-formulation at an application rate of 0.054 kg fipronil per 100 kg seed. The seeds were planted in a sandy silt loam soil at a rate of 250 kg seed/ha. Immature plants were harvested at Zadoks growth stages Z22–33 and Z61 with final harvest at Z91. The mature plants were separated into chaff, grain, straw and stubble.

The extractability of ^{14}C with acetonitrile/acetone (8.8% TRR) and methanol/water (73.9% TRR) was good for grain (combined > 82% TRR).

At harvest, fipronil represented the major component of the ^{14}C in grain at 40% TRR (0.0066 mg/kg) with fipronil-acid (RPA200761) the only other component identified, representing 16.4% TRR (0.0027 mg eq/kg).

Cotton

The metabolism of ^{14}C -fipronil on cotton plants (variety Delta Pine-90) maintained in a greenhouse was studied. An application of fipronil at a rate of 0.224 kg ai/ha was made over the seed furrow. Mature plants were sampled 140 days after planting. The leaves, stems and unopened bolls were collected as one sample. The opened bolls were separated into boll parts (bracts and other samples that make up the boll excluding lint and seed) and lint with seed. Seed cotton was later ginned to separate the seed and lint.

Following soil application, 98% of the ^{14}C taken up was associated with foliage with 2% in bolls and almost no radioactivity in seed or lint. Foliage contained 2.2 mg eq/kg, bolls 0.16 mg eq/kg, lint 0.02 mg eq/kg and seed < 0.01 mg eq/kg.

More than 76% of the TRR was extracted from the foliage and bolls with a sequence of organic solvents followed by acid and base digestion. Lint was digested with cellulase; insufficient radioactivity was present for identification of metabolites. Seed was not extracted, as ^{14}C levels were < 0.01 mg eq/kg.

In the foliage, fipronil accounted for 0.01 mg/kg (0.3% TRR), with significant components of the ^{14}C fipronil-amide (RPA200766; 0.9 mg eq/kg, 41.5% TRR), fipronil-sulfone (MB46136; 0.15 mg eq/kg, 6.9% TRR), and fipronil-acid (RPA200761; 0.17 mg eq/kg, 8.1% TRR).

In bolls, fipronil (0.02 mg/kg, 12% TRR) and fipronil-amide (RPA200766; 0.03 mg eq/kg, 21% TRR) were the main components of ^{14}C identified.

Sunflower

The metabolism of ^{14}C -fipronil on sunflower plants (variety Gala) maintained in a greenhouse was investigated. Fipronil (^{14}C) was applied as an in-furrow soil application of a WG-formulation at an application rate of 0.2 kg ai/ha. Seeds were sown directly onto the treated soil. Immature plants were collected one month after sowing and mature plants at harvest when plants were separated into samples of leaves, stalk, head and seeds.

The distribution of radioactivity in the whole plant was leaves, 82.9%, stalk 12.6%, head 1.7%, and seeds 2.9%.

Solvent extraction with acetone followed by methanol/water released > 65% TRR from the various sunflower samples.

Unchanged fipronil (0.48 mg/kg; 30% TRR) was the main component of ^{14}C in leaves. Fipronil-sulfone (MB46136; 0.22 mg eq/kg, 14% TRR) and fipronil-amide (RPA200766; 0.11 mg eq/kg 7.1% TRR) were also significant components. Numerous minor metabolites were found but none exceeded 0.05 mg eq/kg.

The stalk also contained unchanged fipronil (4.3% TRR) with fipronil-sulfone (MB46136; 1.5% TRR) a significant component. The head contained no components above 0.01 mg eq/kg. The metabolite pattern in the extracts of seeds (0.029 mg eq/kg, 93.7% TRR) contained 14 components, all at levels below 0.01 mg eq/kg. Fipronil and closely related metabolites were not present in the seeds.

In a separate study on sunflowers maintained outdoors under a fiberglass roof, ^{14}C -fipronil was applied as a seed treatment at an application rate of 0.5 kg ai/100 kg seeds, equivalent of 0.03 kg ai/ha. Immature plants were sampled at BBCH 53 (48 days after sowing) and BBCH 65 (57 days after planting). Mature plants were harvested 98 days after planting and separated into seeds (including hulls) and remaining plant material (leaves, stalk and seed head).

Of the ^{14}C applied to planted seeds, only 4.4% was taken up into the entire plant. The majority of the ^{14}C taken up was distributed to the plant material (leaves, stalk and seed head) at 93.9% of plant ^{14}C with seed at harvest accounting for 6.1%.

Extractability of ^{14}C with the solvent systems used (acetonitrile followed by acetonitrile/water) was good at > 77% TRR.

Fipronil parent compound was the major component of ^{14}C residues in leaves and stalks (64-77% TRR) but was not detected in florets. The other significant components, those present at $\geq 10\%$ parent, in leaves and stalks, were fipronil-amide (RPA200766; 0.4-10.2% TRR) and fipronil-sulfone (MB46136; 0.7-6.1% TRR). The only component identified in florets was fipronil-amide (RPA200766; 33% TRR).

Components of ^{14}C identified in seed were fipronil (0.0005 mg/kg, 1.7% TRR), fipronil-sulfonate (RPA104615; 0.0005 mg eq/kg, 1.7% TRR), fipronil-amide (RPA200766; 0.016 mg eq/kg, 5.2% TRR) and fipronil-acid (RPA200761; 0.0005 mg eq/kg, 1.7% TRR).

Some literature studies also reported on the metabolism of ^{14}C -fipronil in sunflowers. In one study germinated seeds (first roots and cotyledons) were cultivated either in a liquid medium (hydroponically) or on soil. The relative humidity was maintained between 65 and 80%, allowing for a high transpiratory rate of the seedlings. Ungerminated seeds exposed to fipronil solutions did not absorb any significant fipronil, while significant absorption occurred on germination and growth of the seedlings. Fipronil uptake and transport were associated with an intense water movement inside the xylem toward the leaves. The

^{14}C content of sunflower leaves at the end of the experiment showed the presence of a large amount of fipronil and metabolites, mainly fipronil-sulfide (MB45950) and fipronil sulfone (MB46136).

In a separate literature study, the behaviour of fipronil in soil surrounding treated seed was reported. The quantity of fipronil in the seed coat (437 $\mu\text{g}/\text{seed}$) decreased continuously during the cultivation period. At the end of the 6-month cultivation period, 42% of all phenylpyrazole compounds remained in the seed coat. Fipronil was poorly mobile in soil, and at the end of the study it was mostly concentrated in the soil layer close to the seed (3.24 mg/kg soil). Starting from the seed-coating, a fipronil concentration gradient was measured in the soil up to 11 cm from the seed. Degradation in the soil occurred at a moderate rate, such that after 6 months of cultivation, only 51% of the fipronil originally in the seed-coating was found in the soil, about 7% having been absorbed by the sunflower plant, and 42% remaining in the seed coat. The predominant metabolites produced in the soil were fipronil-sulfone (MB46136), fipronil-sulfide (MB45950) and fipronil-amide (RPA200766). In contrast, the photoproduct, fipronil-desulfinyl (MB46513) was barely detected.

In another study, when sunflower seeds were coated with ^{14}C -fipronil and allowed to grow in a greenhouse, after 83 days only small amounts of ^{14}C were detected in the inflorescence (0.6 %AR) and were identified as fipronil, fipronil-desulfinyl (MB46513), fipronil-sulfide (MB45950), fipronil-sulfone (MB46136) and fipronil-amide (RPA200766).

Foliar treatment

Cabbage

The uptake and metabolism of ^{14}C -fipronil applied in a WG formulation to greenhouse grown cabbage plants was studied following two foliar applications at 14-day intervals, each at an application rate of 0.2 kg ai/ha. Cabbage heads and outer leaves were collected 0 and 14 days after the first application and 0, 3, 7, 10, 14 and 21 days after the second application.

The extractability of ^{14}C with acetonitrile:acetone followed by acetonitrile was good for cabbage (> 90% TRR).

Fipronil parent compound was the main component of ^{14}C at harvest, 21 DALA (0.68 mg/kg, 51.6% TRR). The photodegradation products, fipronil-desulfinyl (MB46513; 0.17 mg eq/kg, 13% TRR) and fipronil-sulfonate (RPA104615 0.21 mg eq/kg, 16% TRR) and the hydrolysis product fipronil-amide (RPA200766; 0.16 mg eq/kg, 12.5% TRR) are significant components of ^{14}C found in cabbage plants at harvest 21 DALA.

Potato

The metabolism of ^{14}C -fipronil in potato (var Kennebec) was reported following foliar application of a WG formulation (20% w/w) applied two times, with 21-day intervals and at 0.112 kg ai/ha (2×0.112 kg ai/ha) or 0.56 kg ai/ha (2×0.56 kg ai/ha). The plants were maintained outdoors. The first application was made 7 weeks after planting (plants 7 to 25 cm high) with the second 10 weeks after planting. Potato tubers were collected 28 days after the second application.

The extractability of potato tuber ^{14}C with acetonitrile:acetone was good with 79% TRR extracted for the 2×0.112 kg ai/ha experiment and 95% for the 2×0.56 kg ai/ha experiment.

The low ^{14}C residue levels detected in tubers at the $5\times$ rate (0.02 mg eq/kg) suggest that fipronil and its metabolites are poorly translocated from the foliage to the tubers.

In tubers the major component of the identified ^{14}C is fipronil (0.0036 mg/kg, 18.1% TRR). Other components were the oxidation product fipronil-sulfone (MB46136; 0.0013 mg eq/kg, 6.6% TRR) and the photolysis product fipronil-desulfinyl (MB46513; 0.0032 mg eq/kg, 15.8% TRR). The sulfone undergoes further transformation to fipronil-sulfonate (RPA104615; 0.002 mg eq/kg, 9.6% TRR) followed by loss of the sulfonic acid group to form MB45897 (0.0013 mg eq/kg, 6.4% TRR).

Rice

The metabolism of ^{14}C -fipronil was studied in paddy rice plants (cv Supanburi 60) grown outdoors. Twenty days after rice seedlings were transplanted into soil covered by 5–10 cm water, the plants were treated with two foliar sprays at 30-day intervals, at 0.017 and 0.043 kg ai/ha respectively. Immature plants were sampled at 1 day after the second foliar application. At harvest (42 days after the second spray) samples were separated into roots, straw, ears, panicles, husk, brown rice, polished rice and bran.

Extractability of ^{14}C residues with methanol was good at 74–94% TRR for the various rice samples.

Following foliar application, fipronil parent accounted for the greater proportion of the ^{14}C at 24.9–80.0% TRR in root, plant, straw samples and 51.6% TRR in brown rice (0.012 mg/kg). Major metabolites in root, plant and straw were fipronil-sulfide (MB45950; 9.3–16.8% TRR), fipronil-sulfone (MB46316; 1.7–14.3% TRR) and fipronil-desulfinyl (MB46513; 7.1–19.1% TRR). In brown rice the major metabolite was fipronil-amide (RPA200766; 0.0023 mg eq/kg, 9.5% TRR).

Cotton

The metabolism of ^{14}C -fipronil on cotton plants (variety Delta Pine-90) maintained in a greenhouse was reported. Two foliar applications of ^{14}C -fipronil were made at a rate of 0.1 kg ai/ha. The first foliar application was conducted before first bloom when the cotton was approximately 23 to 41 cm high and the second later in the season at approximately the square or soft boll stage, 49 days before harvest. The leaves, stems and unopened bolls were collected as one sample. The opened bolls were separated into boll parts (bracts and other samples that make up the boll excluding lint and seed) and lint with seed. Seed cotton was later ginned to separate the seed and lint.

Following foliar application, the distribution of radioactivity in the whole plant was 97% in foliage, 2.7% in boll parts, < 0.1% in seed and 0.3% in lint. The foliage contained 8.32 mg eq/kg, bolls 0.81 mg eq/kg, lint 0.1 mg eq/kg and seed < 0.01 mg eq/kg.

Extractability of ^{14}C with acetonitrile/acetone and acetonitrile/water was good at 88–91% TRR for foliage and boll parts.

The compounds identified in the foliage were fipronil at 4.4 mg/kg (53% TRR), fipronil-sulfone (MB46136; 0.55 mg eq/kg, 6.6% TRR), fipronil-acid (RPA200761; 1.3 mg eq/kg, 15.5% TRR) with only low levels of the photodegradate fipronil-desulfinyl (0.01 mg eq/kg, 0.1% TRR).

In bolls, the same main metabolites were identified, but in different proportions: fipronil (0.52 mg/kg, 64% TRR), fipronil-sulfone (MB 46136; 0.11 mg eq/kg, 14% of the TRR), fipronil-acid (RPA200761; < 0.01 mg eq/kg, 0.3% TRR), and fipronil-desulfinyl (0.01 mg eq/kg, 0.9% TRR).

Only two ^{14}C components were identified in lint above 0.01 mg eq/kg, fipronil and fipronil-sulfone (MB 46136). Low levels (< 0.01 mg eq/kg) of fipronil-desulfinyl (MB46513), fipronil-acid (RPA200761), and fipronil-sulfide (MB45950) were also confirmed in lint; five additional unknowns were observed, all at levels below 0.01 mg eq/kg.

Radioactivity in seeds was too low to permit identification.

In summary, for seed treatments and soil-incorporated uses of fipronil, metabolism proceeds mainly by oxidation to fipronil-sulfone (MB46136) and hydrolysis to fipronil-amide (RPA200766). Small amounts of fipronil-acid (RPA200761) and the reduction product, fipronil-sulfide (MB45950) were also detected.

In contrast, for foliar applied fipronil, in addition to oxidation to fipronil-sulfone (MB46136), reduction to fipronil-sulfide (MB45950), and hydrolysis to fipronil-amide (RPA200766) and fipronil-acid (RPA200761), the photodegradation products fipronil-desulfinyl (MB46513) and fipronil-sulfonate (RPA104615) were identified as significant residues. The main residues were consistently parent fipronil

and fipronil-desulfinyl (MB46513) together with fipronil-amide (RPA200766) and fipronil-acid (RPA200761).

While the photometabolite fipronil-desulfinyl (MB46513) is observed following soil (soil or seed treatments) and foliar applications of fipronil, it is typically found in higher amounts following foliar treatment.

A number of metabolites observed in the plant metabolism studies were not observed in laboratory animal (rat) studies and were: fipronil-desulfinyl (MB46513), fipronil-sulfonate (RPA104615) and fipronil-sulfone-amide (RPA105320).

Rotational crop metabolism

Confined rotational crop studies

Three confined rotational crop metabolism studies were made available to the Meeting.

In a confined rotational crop study conducted on an outdoor plot with lettuce, radish, wheat and sorghum, bare sandy loam soil was treated with ^{14}C -fipronil at the equivalent of 0.165 kg ai/ha (0.4× maximum seasonal rate). The sprayed soil was covered with 2.5–5 cm of untreated soil and crops sown 30 (lettuce, carrot, sorghum), 153 (wheat) and 365 (lettuce, radish, sorghum) days after the soil application.

Fipronil parent compound and its oxidation derivative fipronil-sulfone (MB46136) were observed in most samples. Fipronil-sulfide (MB45950) was generally not detected (except as 0.004 mg eq/kg in carrot roots sown 30 days after application (DAA)).

Most samples also contained degradation products resulting from the modification of the nitrile group of fipronil, the major one (i.e. present at > 10% parent) being fipronil-amide (RPA200766; 0.067 mg eq/kg, i.e. 40% TRR in straw from wheat sown 153 DAA but < 0.01 mg eq/kg in other samples).

Metabolites fipronil-acid (RPA200761; sorghum forage, wheat straw), fipronil-amide (RPA200766; wheat straw) and fipronil-sulfone (MB46136; wheat straw) were detected at a concentration of > 0.01 mg eq/kg in some matrices. Fipronil-sulfone-amide (RPA105320; carrot leaf, wheat straw), fipronil-sulfonate (RPA104615; carrot leaf and root, sorghum forage and stover) and fipronil-carboxamide (RPA105048; wheat straw) were also identified in some of the samples, but the concentration was < 0.01 mg eq/kg.

A second study with confined rotational crops was conducted on an outdoor plot of sandy loam soil which had been treated with ^{14}C -fipronil at 0.369 kg ai/ha (0.9× maximum seasonal rate). The sprayed soil was left exposed to simulate the situation following foliar use of fipronil. Rotational crops were sown after various intervals of ageing, lettuce, radish and sorghum at 30 DAA, wheat, mustard, lettuce and radish at 150 DAA and lettuce, radish, wheat and sorghum at 365 DAA.

Highest ^{14}C residues were detected in leafy samples with lower levels in roots and grain.

Fipronil was identified in 30 DAA lettuce leaf, radish leaf, radish root and sorghum forage; in 150 DAA wheat forage and wheat straw; and in 365 DAA lettuce leaf, but at levels < 0.01 mg/kg. The photolysis product fipronil-desulfinyl (MB46513) and fipronil-sulfone (MB46136) were found in 30 DAA lettuce leaf, radish leaf and root, and sorghum forage; in 150 DAA wheat forage and straw; and in 365 DAA lettuce leaf. Fipronil-sulfone (MB46136) was also identified in 30 DAA sorghum forage and 365 DAA sorghum fodder.

Fipronil-desulfinyl (MB46513) levels were < 0.01 mg eq/kg in all cases except one (wheat straw 150 DAA). Fipronil-sulfone (MB46136) levels were < 0.01 mg eq/kg in all but two samples (30 DAA sorghum fodder and 150 DAA wheat straw).

Metabolites fipronil-acid (RPA200761), fipronil-amide (RPA200766) and fipronil-sulfonate (RPA104615) were detected at a concentration of > 0.01 mg eq/kg in some matrices. Fipronil-sulfone-

amide (RPA105320) and fipronil-carboxamide (RPA105048) were also identified in some of the samples, but the concentration was < 0.01 mg eq/kg.

In a third study, rotational crops were sown on an outdoor plot of sandy loam soil which had been treated with ^{14}C -fipronil at 0.176 kg ai/ha (0.4× maximum seasonal rate). Due to disease and cold weather, several crops had to be completely replanted or bare spots reseeded at later time points in order to obtain enough plant matrix to analyse. The sprayed soil was left exposed to simulate the situation following foliar use of fipronil. Rotational crops were sown after various intervals of ageing, lettuce 31 and 83 DAA, carrot 31, 49 and 83 DAA and sorghum at 31, cut and regrown at 137 DAA, lettuce 138 and 245 DAA, carrot 138 DAA and sorghum at 138, 166 and 254 DAA and lettuce, carrot and sorghum at 378 DAA.

Overall, forage and straw contained higher levels of radioactivity compared to roots and grain.

The only ^{14}C components present at levels ≥ 0.01 mg eq/kg were in sorghum straw and were fipronil-acid (RPA200761, 0.01-0.024 mg eq/kg) at 31 DAA and 378 DAA samples and fipronil-sulfonate (RPA104615, 0.006-0.017 mg eq/kg) at 31 DAA.

Fipronil was detected in lettuce (31 DAA), carrot root (31 and 138 DAA) and sorghum forage (31 and 138 DAA), and fipronil-sulfone (MB46136) in lettuce (31 DAA), carrot root (31, 138 and 378 DAA), sorghum forage (31 and 138 DAA) and sorghum straw (31 and 138 DAA).

Other metabolites detected were fipronil-amide (RPA200766), fipronil-desulfinyl (MB46513) and fipronil-sulfide (MB45950).

In summary, the metabolism of fipronil in rotated crops was similar for all crop types and proceeded by oxidation to fipronil-sulfone (MB46136) and fipronil-sulfonate (RPA104615), hydrolysis to fipronil-amide (RPA200766) and fipronil-acid (RPA200761) and reduction to fipronil-sulfide (MB45950). The photodegradate fipronil-desulfinyl (MB46513) was a minor component of the residue in rotated crops. Comparison with primary crop metabolism studies shows that the pathway in rotational crops is consistent with that in primary crops following seed treatment or soil application.

Animal metabolism

The Meeting received animal metabolism studies on rats, lactating goats and laying hens dosed with fipronil and separate studies following dosing with fipronil-desulfinyl (MB46513).

Rats

Metabolism of fipronil in rats was evaluated by the WHO Core Assessment Group of the 2021 JMPR. Metabolites identified in rats included: MB45897, fipronil-sulfide (MB45950), fipronil-sulfone (MB46136), fipronil-acid (RPA200761) and fipronil-amide (RPA200766). Following dosing with the photometabolite fipronil-desulfinyl (MB46513), MB46400 and RPA105046 were identified.

Lactating goats

Lactating goats were orally dosed by gavage once daily for seven consecutive days with ^{14}C -fipronil at doses equivalent to 0.05, 2 or 10 ppm in the diet and sacrificed within 23.5 hours of the last dose.

By 23.5 hours after the last dose, the majority of the ^{14}C was recovered in faeces (17.8–64.2% of the administered dose (AD)) and urine (ND–6.6% AD). Milk accounted for 0.86–4.6% AD while tissues accounted for 7.4–25.4% AD. The material balance was 50.3–83.3% AD and the poor accountability may in part be explained by radioactivity remaining in the gastrointestinal tract not being reported.

Residues in milk did not reach a plateau by the end of the study suggesting fipronil residues are not rapidly eliminated following dosing.

TRR were highest for fat, followed by liver; kidney and muscle had low residues; for the 10 ppm dose levels TRR were 1.9 mg eq/kg in fat, 0.86 mg eq/kg in liver, 0.15 mg eq/kg in kidney and 0.08 mg eq/kg in muscle.

Extractability of milk and tissues with acetonitrile was good at > 86% TRR.

Parent fipronil was detected in all tissues and milk accounting for a lower proportion of radioactivity in muscle, fat and milk of the 2 ppm animal (22–37% TRR) compared to the 10 ppm animal (60–73% TRR). Fipronil was a small component of the ^{14}C in liver and kidney (1.5–6.7% TRR) where the major component was fipronil-sulfone (MB46136), accounting for 53–75% TRR. Other significant metabolites were fipronil-amide (RPA200766) at levels up to 18% TRR in liver and fipronil-sulfide (MB45950) accounting for up to 12% TRR in milk.

Laying hens

The metabolism of ^{14}C - fipronil was studied in laying hens. Hens were dosed orally *via* intubation, once a day for a total of 28 days, with ^{14}C -fipronil at doses equivalent to 0.05, 2 or 10 ppm in the diet. Hens were sacrificed 23.5 hours after the final dose.

Residues in eggs from the different dose rates continued to rise during experiment and did not reach a plateau.

By 23.5 hours after the last dose, the majority of the ^{14}C was recovered in excreta (28.4–41.7% AD). Eggs accounted for 14.7–18.1% AD while tissues accounted for 0.65–5.4% AD. Levels of ^{14}C were highest in fat then liver with only low levels in muscle. For the 10 ppm dose group TRR levels were, fat 56 mg eq/kg, liver 4.9 mg eq/kg and muscle 0.73 mg eq/kg.

Extractability of egg yolks and skin/fat (acetonitrile, methanol), muscle and omental fat (acetonitrile) was very good at > 81% TRR for the 10 ppm dose group. In liver samples digested with pepsin/0.1M HCl for 72 hours, extractability was > 74% TRR.

Parent fipronil was detected at low levels (< 2.6% TRR) in fat, liver and egg samples and not detected in muscle. Fipronil-sulfone (MB46136) accounted for the majority of radioactivity (> 73% TRR) in all tissues and eggs (whole egg basis). Unidentified components accounted for < 1% TRR.

Fipronil-desulfinyl (MB46513)

Metabolism studies were also carried out in lactating goats and laying hens with the photo-degradation product fipronil-desulfinyl (MB46513).

Lactating goat

The metabolism and distribution of ^{14}C -fipronil-desulfinyl (MB46513) was investigated in three lactating goats following repeated oral administration by gelatine capsule. Fipronil-desulfinyl was administered twice daily on seven consecutive days at dose levels equivalent to 0.056, 2.4 or 8.3 ppm in the feed. Animals were euthanized approximately 23 hours after the last dose

By 23 hours after the last dose, the majority of the ^{14}C was recovered in faeces (19.5–49.7% AD) and urine (3.22–7.05% AD). Milk accounted for 0.96–5.28% AD while tissues accounted for 13.7–39.3% AD. Residues in milk did not reach a plateau by the end of dosing.

For the 8.3 ppm dose group, TRR were highest for liver (2.81 mg eq/kg), followed by fat (2.21–2.68 mg eq/kg). In comparison kidney and muscle had low residues (0.468 and 0.180 mg eq/kg, respectively).

Extractability of milk (acetonitrile, methanol) and kidney, fat and muscle was very good at > 86% TRR. In liver samples digested with pepsin/0.1M HCl for 72 hours, extractability was 91% TRR.

Unchanged fipronil-desulfinyl was detected in all tissues and milk accounting for > 49% TRR (49–94% TRR). Low levels of the deaminated ring-opened derivatives of RPA106889 and RPA105048, and ring-opened metabolite RPA108058 were tentatively identified based on metabolism observed in the laying hen.

Laying hen

The metabolism of ^{14}C -fipronil-desulfinyl (MB46513) in laying hens was also investigated. Hens were dosed orally *via* intubation, once a day for a total of fourteen days, with ^{14}C -fipronil-desulfinyl at doses equivalent to 0.05, 2 or 10 ppm in the diet. Hens were sacrificed 23 hours after the final dose.

By 23 hours after the last dose, the majority of the ^{14}C was recovered in excreta (53.4–71.4% AD). Eggs accounted for 4.2–6.7% AD while tissues accounted for 4.0–6.3% AD. At the 10 ppm dose, ^{14}C residues were 8.8 mg eq/kg in fat, 4.1 mg eq/kg in liver and up to 0.60 mg eq/kg in muscle.

Residues in eggs from the different dose rates continued to rise during the experiment.

Extractability of egg yolks and skin/fat (acetonitrile, methanol), egg whites, muscle and omental fat (acetonitrile) was very good at > 81% TRR.

Parent fipronil-desulfinyl (MB46513) was the major component of radioactive residues accounting for > 56% TRR in eggs and tissues other than liver. For liver, fipronil-desulfinyl while the major component of radioactivity accounted for only 13.8% TRR.

The deaminated ring-opened derivatives of RPA106889 and RPA105048, and ring-opened metabolite RPA108058, mono-dechloro, mono-hydroxy fipronil-desulfinyl (MB46513), the sulfate conjugate of fipronil-desulfinyl, the glucuronide conjugate of fipronil-desulfinyl and MB46400 were tentatively identified but at levels < 6% TRR for each. Unidentified components individually accounted for < 7.1% TRR.

In summary, consistent with the lipophilic nature of fipronil and its metabolites, residues in livestock commodities were highest in lipid rich matrices (fat, yolk). Parent fipronil and fipronil-sulfone (MB46136, goats and hens), fipronil-sulfide (MB45950, goats) and fipronil-amide (RPA200766, goats) were the principal compounds detected.

In studies with fipronil-desulfinyl, the main component of the residues in livestock commodities is unmetabolized fipronil-desulfinyl, with highest levels also in lipid rich tissues.

Environmental fate

The Meeting received aqueous and soil photolysis, aqueous hydrolysis and aerobic soil studies for fipronil.

In laboratory aerobic soil degradation studies on fipronil the major soil degradates were fipronil-amide (RPA200766), fipronil-acid (RPA200761), fipronil-sulfide (MB45950), fipronil-sulfone (MB46136) and fipronil-desulfinyl (MB46513). The laboratory DT_{50} values for fipronil degradation in different soils were 14–700 days, for fipronil-amide (RPA200766) 107–149 days, fipronil-acid (RPA200761) 42–139 days, fipronil-sulfide (MB45950) 89–224 days, fipronil-sulfone (MB46136) 185–278 days and fipronil-desulfinyl (MB46513) 47–98 days.

In field dissipation studies, the DT_{50} values for fipronil degradation ranged from 0.13 to 223 days with a median of 20 days indicating fipronil is non-persistent to moderately persistent. Residues of fipronil-sulfone increased following annual applications of fipronil.

Fipronil and metabolites (MB45950, MB46136, MB46513) are stable to hydrolysis (aqueous and soil) at environmental pHs, however aqueous photolysis is fast and may be a significant route of degradation with DT_{50} values < 1 day for fipronil, MB45950, MB46136 and MB46513. The DT_{50} for soil photolysis ranged from 3.8 to 49 days indicating soil photolysis may contribute to the degradation of fipronil in the environment.

Degradation of fipronil in water/sediment is fast and fipronil is not expected to persist in water/sediment systems. DT_{50} values for water/sediment studies ranged from 12.3 to 28.5 days.

Field rotational crop studies

The persistence in soil and potential uptake of fipronil and degradates by plants was further evaluated in field studies.

In a field rotational crop study, the magnitude of fipronil and metabolites in succeeding crops following a single spray application of a WG formulation of fipronil to bare ground at 0.34 kg ai/ha (0.85× maximum seasonal rate) was investigated at two sites in the USA, California and North Carolina. Rotational crops were planted at 30 ± 2 (radish, soybean, collard greens, sorghum), 120 ± 2 and 240 ± 2 (radish, winter peas, collard greens, winter wheat) and 365 ± 2 (radish, soybean, collard greens, sorghum) DAA.

Residues of fipronil and metabolites were < LOQ (0.005 mg/kg) in all samples from the North Carolina site and < LOQ for fipronil and fipronil-sulfide (MB45950) at the California site.

Fipronil-sulfone (MB46136) was detected in samples of lettuce (0.008–0.012 mg/kg) and sorghum straw (0.021–0.024 mg/kg) for the 31 DAA samples, in wheat straw (0.013–0.016 mg/kg) for the 239 DAA samples and sorghum straw (0.013–0.020 mg/kg) for the 367 DAA samples.

Fipronil-desulfinyl (MB46513) was detected in radish tops, cowpea forage, lettuce, sorghum forage and sorghum straw (0.007–0.033 mg/kg) for the 31 DAA samples, in wheat straw (0.020–0.023 mg/kg) for the 119 DAA samples, cowpea forage and wheat straw (0.005–0.026 mg/kg) for 239 DAA samples and sorghum straw (0.10–0.018 mg/kg) for the 367 DAA samples.

In another study, fipronil and its metabolites were determined in maize planted into fipronil treated soil fields in Belgium and France. In the spring of each year for four years, fipronil was sprayed at a nominal rate of 0.2 kg ai/ha (0.5× maximum seasonal rate) onto the soil surface and incorporated to about 5 cm depth prior to planting maize seed. Maize silage and grain samples were analysed for fipronil and its metabolites, MB45950, MB46136 and MB46513.

Fipronil and its metabolites (including fipronil-desulfinyl MB46513) were <LOQ in maize grain and silage (0.002 mg/kg for maize silage and 0.001 mg/kg for maize grain).

In a study of similar design, residues of fipronil and fipronil-sulfide (MB45950), fipronil-sulfone (MB46136) and fipronil-desulfinyl (MB46513) in maize, sugar beet and wheat planted into fipronil treated soil fields in Belgium and France were investigated. The trials were carried out over four seasons. In the autumn of each year in France and in spring in Belgium, fipronil was sprayed at a nominal application rate of 0.2 kg ai/ha (0.5× maximum seasonal rate) onto the soil surface and incorporated prior to planting seed, wheat at the trial in France and maize/sugar beet alternating on a yearly basis in Belgium. Wheat straw and grain (BBCH 89), maize silage (BBCH 83–85) and grain (BBCH 89), and sugar beet leaves and roots (BBCH 49) were collected and analysed.

In maize, fipronil and its metabolites were < LOQ (grain: < 0.001 mg/kg; silage and straw: < 0.002 mg/kg for each analyte). Residues of fipronil and fipronil-sulfone (MB46136) were occasionally detected in wheat straw but not grain and fipronil-sulfone (MB46136) in sugar beet roots and tops.

In summary, at the maximum use pattern considered by the Meeting, soil application at up to 0.4 kg ai/ha (potatoes), residues of fipronil and fipronil-sulfone (MB46136) may occasionally be expected in rotated crops. Median residues of fipronil and fipronil-sulfone (MB46136) in rotated crops were all < LOQ.

Methods of analysis

The Meeting received information on analytical methods for fipronil in plant and animal matrices. The methods are suitable for analysis of fipronil and metabolites in plant and fipronil and metabolites in animal matrices.

In more recent methods, fipronil and its metabolites MB46136, MB46513 and MB45950 were extracted from homogenised plant matrices using a mixture of methanol and water. For clean-up a

liquid/liquid partition against dichloromethane was used. The final determination of fipronil and its metabolites fipronil-sulfone (MB46136), fipronil-desulfinyl (MB46513) and fipronil-sulfide (MB45950) was performed by LC-MS/MS. The LOQs for plant commodities are typically 0.005 mg/kg for each compound while for animal commodities they are 0.0005 mg/kg for tissues and milk for the individual compounds. The Meeting concluded that the presented methods were sufficiently validated and are suitable to measure fipronil and metabolites in plant and animal commodities.

Older methods, used in some of the residue trials, utilised acetonitrile or acetonitrile/water (75:25 v/v) for the extraction step with clean-up typically on a C₁₈ SPE column. Quantification was by GC-ECD or GC-MS with LOQs ranging from 0.01 to 0.05 mg/kg.

As reported by the 2001 JMPR, a multi-residue method, DF S19 modified (extraction using water/acetone), was validated for determination of fipronil, fipronil-sulfide (MB45950), fipronil-sulfone (MB46136), fipronil-desulfinyl (MB46513) and fipronil-amide (RPA200766) in a variety of crops with a typical LOQ of 0.002 mg/kg for each analyte.

Stability of pesticide residues in stored analytical samples

The Meeting received additional information on storage stability of fipronil and metabolites in raw/processed plant commodities.

Fipronil and metabolites are stable in a range of high-water commodities (broccoli at least 11.6 months, cabbage 11 months, cauliflower 12.4 months), high oil (cottonseed at least 13.2 months), and high starch (potato at least 25 months, maize grain 28.4 months).

Fipronil and metabolites are stable in animal commodities for at least 3 months frozen storage.

The demonstrated stability intervals on frozen storage encompass the duration of storage in the residue trials evaluated by the Meeting.

Definition of the residue

Plant commodities

The metabolism of fipronil was similar in the submitted crops (soil/seed treatment: sugar beet, maize, rice, wheat, cotton, sunflower; foliar: cabbage, potato, rice, cotton, sunflower).

Following soil/seed treatment, parent fipronil was detected in all matrices except sugar beet leaves and maize grain. Components detected at levels $\geq 10\%$ parent were fipronil-sulfone (MB46136) at levels 0.2–23 $\times$ fipronil (6.9–60% TRR) though not detected in cereal or oilseed grain, fipronil-amide (RPA200766) at levels 0.3–138 $\times$ fipronil (4.1–60.4% TRR) though not detected in wheat grain, fipronil-acid (RPA200761) at levels 0.1–27 $\times$ fipronil (1.6–16.4% TRR) but not detected in rice commodities, sugar beet, cottons bolls or maize grain. Fipronil-sulfide (MB45950) was present at levels 1.2 $\times$ fipronil (14.8% TRR) in rice straw but was generally not detected or present at low levels in other commodities (0.2 $\times$ fipronil, 4.1% TRR brown rice, 3.9% TRR sugar beet leaves, 0.1 $\times$ fipronil, 1.7% TRR maize fodder). Further metabolites detected but at low levels or only in a single commodity were: fipronil-sulfonate (RPA104615) brown rice (0.1 $\times$ fipronil, 2.1% TRR), sunflower seed (1.0 $\times$ fipronil, 1.7% TRR); MB45897 sugar beet leaves (1.8% TRR); fipronil-sulfone-amide (RPA105320) sugar beet leaves (19% TRR); fipronil amide conjugate maize grain (87.5% TRR) though identification of this metabolite is questionable.

In contrast, following foliar application, parent fipronil was detected in all matrices except sunflower seed and accounted for 18.1–64.4% TRR. Other major components of the ¹⁴C residue were fipronil-sulfone (MB46136; 0.1–0.4 $\times$ fipronil, 0.8–13.6% TRR) though not detected in rice grain or sunflower seed, the photometabolite fipronil-desulfinyl (MB46513; 0.3–0.9 $\times$ fipronil, 0.1–19.0% TRR) but not detected in sunflower seed, fipronil-amide (RPA200766; 0.2 $\times$ fipronil, 1.6–12.5% TRR) though not detected in potato tubers or cotton commodities, fipronil-acid (RPA200761; 0.3 $\times$ fipronil, 0.3–17.5% TRR) but not detected in potato, cabbage or rice commodities. Fipronil-sulfide (MB45950) was 0.2 $\times$ fipronil (9.3% TRR) and 0.1 $\times$ fipronil (4.3% TRR) in rice straw and grain, respectively. Fipronil-sulfonate

(RPA104615) accounted for 1.8% TRR in brown rice, 0.5× fipronil (9.6% TRR) in potato tubers and 0.3× fipronil (16.4% TRR) in cabbage heads.

MB45897 was detected at low levels and accounted for 0.4× fipronil (6.4% TRR) and 0.1× fipronil (3.6% TRR) in potato tubers and brown rice, respectively.

In studies on rotational crops reflecting uptake from soil, residues of fipronil, fipronil-acid (RPA200761), fipronil-amide (RPA200766), fipronil-sulfone (MB46136), fipronil-desulfinyl (MB46513), fipronil-sulfonate (RPA104615) and fipronil-sulfone-amide (RPA105320) were sometimes detected at levels above 0.005 mg/kg, a typical LOQ for analytical methods, and may occur in rotational (follow) crops.

Fipronil and metabolites fipronil-sulfone (MB46136), fipronil-sulfide (MB45950) and fipronil-desulfinyl (MB46513) are the most significant residues among the commodities investigated and validated analytical methods are available for their determination.

In supervised field trials, fipronil and fipronil-sulfone (MB46136) were the most frequently detected residues.

As the combination of fipronil and fipronil-sulfone (MB46136) occurs in most crops that have detectable residues, the Meeting decided the residue definition for compliance with MRLs in plants should be the sum of fipronil and fipronil-sulfone (MB46136), expressed as fipronil.

In deciding which compounds should be included in the residue definition for risk assessment for plant commodities the Meeting considered the likely occurrence of the compounds present at 10% TRR of fipronil or the sum of fipronil and fipronil-sulfone (MB46136) and the toxicological properties of the candidates. Compounds considered were fipronil-sulfide (MB45950), fipronil-desulfinyl (MB46513), fipronil-amide (RPA200766), fipronil-acid (RPA200761), fipronil-sulfone-amide (RPA105320) and fipronil-sulfonate (RPA104615).

Of those compounds that, based on toxicological properties, are assumed to be covered by the fipronil HBGVs:

Fipronil-sulfide (MB45950) levels following seed treatment/soil application were 1.2× fipronil in rice straw (0.015 mg eq/kg), 0.2× in brown rice (0.0002 mg eq/kg), 3.9% TRR in sugar beet leaves (0.024 mg eq/kg), 0.1× fipronil levels in maize fodder (0.06 mg eq/kg) and following foliar application 0.2× fipronil in rice straw (0.023 mg eq/kg), 0.1× fipronil in brown rice (0.001 mg eq/kg).

Fipronil-desulfinyl (MB46513) following seed treatment/soil application was present at 1.9× fipronil levels in rice straw (0.023 mg eq/kg) and following foliar application 0.9× fipronil in potato tubers (0.003 mg eq/kg), 0.3× fipronil in cabbage heads (0.17 mg eq/kg), 0.5× fipronil in rice straw (0.047 mg eq/kg), 1.9% TRR in brown rice (0.00046 mg eq/kg), 0.1% TRR in cotton plants (0.008 mg eq/kg) and 0.9% TRR in cotton hulls (0.007 mg eq/kg).

Fipronil-amide (RPA200766) following seed treatment/soil application was present at 0.4× fipronil levels in rice straw (0.005 mg eq/kg), 0.5× fipronil in brown rice (0.0007 mg eq/kg), 0.3× fipronil in sugar beet roots (0.003 mg eq/kg), 4.1% TRR in sugar beet leaves (0.025 mg eq/kg), 138× fipronil in cotton plants (0.90 mg eq/kg), 1.7× fipronil in cotton hulls (0.034 mg eq/kg), 3.1× fipronil in sunflower seed (0.0016 mg eq/kg), 0.3–0.8× fipronil in maize forage (0.027–0.033 mg eq/kg), 2.1–3.2× fipronil in maize fodder (0.19–0.94 mg eq/kg), tentatively identified at 60.4% TRR in maize grain (0.008 mg eq/kg) and following foliar application was present at 0.2× fipronil levels in cabbage heads (0.16 mg eq/kg), 1.6% TRR in rice straw (0.004 mg eq/kg) and 0.2× fipronil in brown rice (0.002 mg eq/kg).

The Meeting considered it was not necessary to include fipronil-amide (RPA200766) in the residue definition as it was only occasionally detected in the field trials and the sum of fipronil, fipronil-sulfone (MB46136), fipronil-sulfide (MB45950) and fipronil-desulfinyl (MB46513) accounts for a high proportion of the residues of compounds covered by the health-based guidance values for fipronil.

Fipronil-sulfonate (RPA104615) is not of toxicological concern and is not considered further.

The Meeting agreed that the residue definition for dietary risk assessment should be the sum of fipronil, fipronil-sulfide (MB45950), fipronil-sulfone (MB46136) and fipronil-desulfinyl (MB46513) (expressed as fipronil).

The Meeting also considered fipronil-sulfone-amide (RPA105320) and fipronil-acid (RPA200761) could be assessed using the threshold of toxicological concern approach.

Animal commodities

Regarding the residue definition for livestock commodities, the metabolism of fipronil in lactating goats and laying hens was qualitatively similar. Fipronil was a major component of the residue in the goat but a minor component in the hen metabolism studies, perhaps reflecting the long duration between last dosing and slaughter in the hen study (goat: muscle 22.2%, fat 31.6%, kidney 6.7%, liver 5.4%, milk 26.5%; hen: egg yolk 2.7%, egg white 3.8%, fat 1.7%, liver 1.0%). The predominant residue in the lactating goat and laying hen metabolism studies was fipronil-sulfone (MB46136) (goat: muscle 60.8%, fat 56%, kidney 59.7%, liver 64.6%, milk 62.3%; hen: egg yolk 95.3%, egg white 95.2%, muscle 99.8%, fat 96.7%, liver 97.9%). Small amounts of fipronil-sulfide (MB45950) were also detected in the goat metabolism study (goat: muscle 7.8%, fat 7.2%, kidney 3.1%, liver 3.4%, milk 9.2%).

Fipronil-sulfone (MB46136) is present in milk, eggs and all tissues in the lactating goat and laying hen metabolism studies and would be suitable for monitoring compliance.

Animals are also potentially exposed through the diet to residues of the photometabolite fipronil-desulfinyl (MB46513) as well as fipronil-amide (RPA200766) and fipronil-acid (RPA200761). Supervised residue trial data where fipronil-amide (RPA200766) was analysed indicate negligible residues of RPA200766 in livestock feed and as such the livestock dietary burden for RPA200766 is also estimated to be negligible. Based on the plant metabolism studies, conclusions regarding fipronil-amide (RPA200766) should also apply to fipronil-acid (RPA200761). The livestock dietary burden for fipronil-desulfinyl (MB46513) is calculated to be lower than the sum of fipronil and fipronil-sulfone (MB46136) and as such residues of fipronil-desulfinyl are expected to be lower than fipronil and fipronil-sulfone (MB46136) in livestock commodities.

Methods are available for the determination of fipronil and fipronil-sulfone (MB46136) in tissues, milk and eggs.

The Meeting agreed the residue for compliance monitoring for tissues, milk and eggs should be the sum of fipronil and fipronil-sulfone (MB46136) (expressed as fipronil).

In deciding which compounds should be included in the residue definition for risk assessment, the Meeting considered the likely occurrence of the compounds and the toxicological properties of the candidates: fipronil, fipronil-sulfone (MB46136), fipronil-sulfide (MB45950) and fipronil-desulfinyl (MB46513) as well as fipronil-amide (RPA200766) and fipronil-acid (RPA200761).

Using estimates of livestock dietary burden for each of the compounds and assuming the same transfer rate to animal commodities as for fipronil, the Meeting noted that the metabolites fipronil-sulfone (MB46136), fipronil-desulfinyl (MB46513) and fipronil-sulfide (MB45950) could make a significant contribution to overall consumer exposure when compared with estimates for the sum of fipronil and fipronil-sulfone (MB46136). The Meeting considered that it was not necessary to include fipronil-amide (RPA200766) in the residue definition for risk assessment for livestock commodities as it was only occasionally detected in the field trials on feed commodities and a majority of the residues of compounds covered by the health-based guidance values for fipronil are accounted for by the sum of fipronil, fipronil-sulfone (MB46136), fipronil-sulfide (MB45950) and fipronil-desulfinyl (MB46513).

The Meeting agreed that the residue definition for dietary risk assessment for tissues and milk should be the sum of fipronil, fipronil-sulfide (MB45950), fipronil-sulfone (MB46136) and fipronil-desulfinyl (MB46513) (expressed as fipronil).

Consideration of metabolites using TTC approach

The Meeting also considered MB45897, MB46400, RPA105048, fipronil-sulfone-amide (RPA105320), RPA108058 and fipronil-acid (RPA200761) could be assessed using the threshold of toxicological concern for Cramer Class III compounds of 1.5 µg/kg bw per day.

The estimated exposure based on metabolism studies, resulted in the following maximum long-term exposures:

MB45897	0.0176	µg/kg bw per day,
MB46400	0.0000067	µg/kg bw per day,
fipronil-carboxamide (RPA105048)	0.0199	µg/kg bw per day,
fipronil-sulfone-amide (RPA105320)	0.0252	µg/kg bw per day,
RPA108058	0.0196	µg/kg bw per day,
fipronil-acid (RPA200761)	0.0642	µg/kg bw per day.

The estimated exposures are below the threshold of toxicological concern for Cramer Class III compounds. The Meeting concluded that MB45897, MB46400, fipronil-carboxamide (RPA105048), fipronil-sulfone-amide (RPA105320), RPA108058 and fipronil-acid (RPA200761) were unlikely to present a dietary exposure concern from the uses evaluated by the current Meeting.

The Meeting recommended the following residue definitions for fipronil:

Definition of the residue for compliance with the MRL for plant and animal commodities: sum of *fipronil* and *fipronil-sulfone* (MB46136), expressed as *fipronil*.

Definition of the residue for dietary risk assessment for plant and animal commodities: sum of *fipronil*, *fipronil-sulfone* (MB46136), *fipronil-sulfide* (MB45950) and *fipronil-desulfinyl* (MB46513), expressed as *fipronil*.

In deciding whether the residue for compliance monitoring is regarded as fat-soluble, the Meeting noted mean residues at the highest dose level in the lactating cow transfer study according to the compliance residue definition were 0.04 mg/kg in muscle and 0.53 mg/kg in fat while residues in a separate study on milk were 14× higher in milk fat compared to whole milk. In the laying hen transfer study, residues in the highest dose group were 0.012 mg/kg in muscle and 0.196 mg/kg in skin+fat while in a separate study on eggs, residues in whites were 0.02 mg/kg and in yolks 0.39 mg/kg.

The Meeting considers the residue should be classified as fat-soluble.

Results of supervised residue trials on crops

Supervised trials were available for the use of fipronil on a broad range of crops with product labels available from countries in Europe, North America, Central America, South America, Asia and Oceania.

No trials were available for head cabbage and flowerhead brassicas and the Meeting decided to withdraw its previous recommendations for these commodities.

The residue concentrations in the evaluation tables are expressed in terms of the individual compounds and not as fipronil equivalents. To convert to fipronil equivalents, correction for differences in molecular weight need to be made. The factors are: MB45950 1.04 (437.1/421.1); MB46136 0.965 (437.1/453.1); MB46513 1.1 (437.1/389.02); RPA200766 0.96 (437.1/455.08). The values reported below are as fipronil equivalents according to the respective residue definition (compliance or risk assessment). The LOQs of the individual compounds are not corrected for molecular weight, that is the correction for molecular weight is only applied when the residue is > LOQ. Examples are shown below:

As reported, mg/kg				Value for deriving total residue, mg/kg				Total, mg/kg
Fipronil	MB45950	MB46136	MB46513	Fipronil	MB45950	MB46136	MB46513	
< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.004
< 0.001	< 0.001	0.02	< 0.001	0.001	0.001	0.0193	0.001	0.0223
0.02	0.01	0.03	0.01	0.02	0.0104	0.02895	0.011	0.07035

In evaluating the crop residue data, a range of values are required to be derived. Estimates are made for residues of:

- fipronil and fipronil-sulfone (MB46136) for estimation of maximum residue levels and livestock dietary burden
- total residues (fipronil+MB46136+MB45950+MB46513) for estimation of STMR and HR values
- fipronil+MB46136+MB45950 for estimation of risk assessment livestock dietary burdens
- MB46513 for estimation of risk assessment livestock dietary burdens

Bananas

The critical GAP for fipronil on banana in Australia is application to the butt of a stool at 0.15 g ai/stool or to soil /litter at 8 g ai/100 m² (=0.8 kg ai/ha), maximum number of sprays not specified, and a PHI not required when used as directed. No trials matched cGAP in Australia. However, the Meeting noted residues were < LOQ (0.002 mg/kg) for fipronil and each metabolite in all trials. The application rates to soil ranged from 0.25 to 1.6 kg ai/ha with PHIs of 0 to 295 days.

The Meeting agreed that the use represents a nil residue situation and agreed to replace its previous recommendation (0.005 mg/kg) and recommend a maximum residue level of 0.004* mg/kg for the sum of fipronil and MB46136, and an STMR of 0 and HR of 0 mg/kg for total residues.

Onions

The Meeting received supervised residue trials on onion. In New Zealand cGAP is foliar application at 4 × 0.04 kg ai/ha with a minimum interval between sprays of 7 days and a PHI of 3 days. No trials matching cGAP in New Zealand were available to the Meeting.

In Bolivia, critical GAP for fipronil on onion is seed treatment at 0.25 kg ai/ha and a PHI not required. In fifteen trials approximating GAP (0.1875–0.3125 kg ai/ha) residues were (n = 15): 0.004, 0.0042, 0.0042, 0.0043, 0.0046, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.018 mg/kg for fipronil and MB46136 (as fipronil).

The highest individual analytical result was 0.023 mg/kg (as fipronil).

Total residues were: 0.008, 0.0082, 0.0082, 0.0083, 0.0086, 0.0146, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02, 0.021, 0.028 mg/kg (as fipronil). The highest individual analytical result for total residues was 0.033 mg/kg (as fipronil).

The Meeting estimated a maximum residue level of 0.03 mg/kg, an STMR of 0.02 mg/kg and an HR of 0.033 mg/kg for fipronil in onion bulb, dry.

Fruiting vegetables, other than cucurbits (tomato, egg plant, chili)

The Meeting received supervised residue trials on tomato. In Armenia, Georgia and the Ukraine cGAP is soil application at sowing/planting at up to 0.2 kg ai/ha with a PHI of 30 days. The Meeting considered the practical interval between sowing and harvest to be longer than 30 days (typically 4–5 months). In eight trials approximating cGAP (0.15–0.27 kg ai/ha) residues at harvest 93–117 days after application were: < 0.004 (8) mg/kg for fipronil and MB46136 and < 0.008 mg/kg (7) for total residues.

The Meeting estimated a maximum residue level of 0.01(*) mg/kg, an STMR of 0.008 mg/kg and an HR of 0.008 mg/kg for fipronil in tomatoes.

The Meeting received supervised residue trials on egg plant. In the Philippines cGAP is foliar application at 0.05 kg ai/ha with a PHI of 7 days. No trials matched GAP.

The Meeting received supervised residue trials on chili. In Indonesia cGAP is foliar application at 0.01 kg ai/hL with a PHI of 14 days. No trials matched GAP.

Legume vegetables - Beans with pods

The Meeting received supervised residue trials on green beans. In Brazil, critical GAP for fipronil on beans is seed application at 0.1 kg ai/100 kg seed. In six trials approximating GAP (0.072–0.12 kg ai/100 kg seed) residues were: < 0.004 (5), 0.0059 mg/kg (as fipronil).

Total residues were: < 0.008 (5), 0.0099 mg/kg (as fipronil).

The Meeting estimated a maximum residue level of 0.01 mg/kg, an STMR of 0.008 mg/kg and an HR of 0.0099 mg/kg for fipronil in beans with pods

Pulses

Beans (dry)

The Meeting received supervised residue trials on beans dry. In Myanmar, cGAP is seed treatment at 0.125 kg ai/100 kg seed. No trials matched cGAP of Myanmar.

In Honduras, cGAP is seed treatment at 0.05 kg ai/100 kg seed. In available trials application was at 0.039–0.057 kg ai/100 kg seed.

Residues approximating GAP ($\pm 25\%$ application rate) were (n = 12): < 0.001, < 0.001, < 0.001, < 0.001, < 0.001, < 0.001, 0.0014, 0.0021, < 0.004, < 0.004, < 0.004 mg/kg.

Total residues were (n = 12): < 0.002, < 0.002, < 0.002, < 0.002, < 0.002, < 0.002, < 0.002, 0.0024, 0.0031, < 0.008, < 0.008, < 0.008 mg/kg.

The Meeting estimated a maximum residue level of 0.01 mg/kg, and an STMR of 0.002 mg/kg for fipronil in the subgroup beans (dry) except soya beans.

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens. The median residue for the sum of fipronil and fipronil-sulfone (MB46136) in beans (dry) except soya beans was 0.001 mg/kg.

Residues of fipronil-desulfinyl for use in estimation of livestock dietary burden were (n = 12): < 0.0005, < 0.0005, < 0.0005, < 0.0005, < 0.0005, < 0.0005, < 0.0005, < 0.0005, < 0.0005, < 0.002, < 0.002, < 0.002 mg/kg. The Meeting estimated a median residue of 0.0005 mg/kg.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide for estimation of livestock dietary burdens were (n = 12): < 0.0015, < 0.0015, < 0.0015, < 0.0015, < 0.0015, < 0.0015, < 0.0015, 0.0019, 0.0026, < 0.006, < 0.006, < 0.006 mg/kg. The Meeting estimated a median residue of 0.0015 mg/kg.

Soya beans (dry)

The Meeting received supervised residue trials on soya bean. In Indonesia, critical GAP for fipronil on soya bean is seed treatment at 0.125 kg ai/100 kg seed followed by foliar application at 0.01 kg ai/hL with a PHI of 14 days. No trials matched cGAP in Indonesia.

In Brazil cGAP is seed treatment at 0.1 kg ai/100 kg seed with PHI not required. In three trials approximating cGAP (0.08 kg ai/100 kg seed) residues were: < 0.004 (3) mg/kg for fipronil and MB46136. Total residues were: < 0.008 (3) mg/kg.

In three trials at 0.048 kg ai/100 kg seed, residues were: 0.001, 0.00135 and 0.00143 mg/kg with total residues 0.0015, 0.0018 and 0.0019 mg/kg.

In three trials at 0.05 kg ai/100 kg seed residues were: < 0.001 (3) mg/kg with total residues < 0.0015 (3) mg/kg).

In three trials at 0.057 kg ai/100 kg seed, residues were: < 0.001 (2), 0.00131 mg/kg with total residues < 0.0015 (2), 0.00181 mg/kg.

The Meeting agreed to use proportionality to scale residues using factors of 1.25, 2.083, 2 and 1.75 for trials conducted at 0.08, 0.048, 0.05 and 0.057 kg ai/100 kg seed.

Scaled residues were (n = 12): < 0.00175, < 0.00175, < 0.002, < 0.002, < 0.002, < 0.00208, 0.00230, 0.00281, 0.00298, < 0.005, < 0.005, < 0.005 mg/kg.

Total residues were (n = 12): < 0.00351, < 0.00351, < 0.004, < 0.004, < 0.004, < 0.00406, 0.00417, 0.00578, 0.00877, < 0.01, < 0.01, < 0.01 mg/kg.

The Meeting estimated a maximum residue level of 0.01 mg/kg and an STMR of 0.00411 mg/kg for fipronil in soya bean (dry).

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens. The median residue for the sum of fipronil and fipronil-sulfone (MB46136) in soya beans (dry) was 0.00219 mg/kg.

Scaled residues of fipronil-desulfinyl for use in estimation of livestock dietary burdens were (n = 12): < 0.00088, < 0.00088, < 0.00088, < 0.001, < 0.001, < 0.001, < 0.00104, 0.00156, < 0.0025, < 0.0025, < 0.0025, 0.00438 mg/kg. The Meeting estimated a median residue of 0.00102 mg/kg. Scaled residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide for estimation of livestock dietary burdens were (n = 12): 0.00263, 0.00263, 0.003, 0.003, 0.003, 0.00313, 0.00330, 0.00385, 0.00402, 0.0075, 0.0075, 0.0075 mg/kg. The Meeting estimated a median residue of 0.003215 mg/kg.

Root and tuber vegetables

Potato

The Meeting received supervised residue trials on potato. In Indonesia, critical GAP for fipronil on potato is application to soil 7 days after planting at 0.075 kg ai/ha followed by foliar application at 0.005 kg ai/hL with a PHI of 14 days. No trials matched GAP.

In Brazil, critical GAP for fipronil on potatoes is in-furrow application at planting at 0.225 kg ai/ha followed by application at 0.225 kg ai/ha at hilling for a total of 0.45 kg ai/ha and a PHI not required.

In six trials matching cGAP, residues were: 0.001, 0.00248, 0.00323, 0.00399, 0.00866, 0.0262 mg/kg.

Highest individual analytical result 0.0268 mg/kg.

Total residues were: 0.002, 0.00378, 0.00487, 0.00499, 0.0115, 0.0287 mg/kg.

Highest individual analytical result 0.0296 mg/kg.

The Meeting estimated a maximum residue level of 0.05 mg/kg, an STMR of 0.00493 mg/kg and an HR of 0.0296 mg/kg for fipronil in potatoes to replace its previous recommendation (0.02 mg/kg).

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens. The median residue for the sum of fipronil and fipronil-sulfone (MB46136) in potato (culls) was 0.00361 mg/kg.

Residues of fipronil-desulfinyl for use in estimation of livestock dietary burden were (n = 6): < 0.0005, < 0.0005, < 0.0005, < 0.0005, < 0.0005, 0.00094 mg/kg. The Meeting estimated a median residue of 0.0005 mg/kg and a highest residue of 0.0094 mg/kg for potato culls.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide for estimation of livestock dietary burdens were (n = 6): 0.0015, 0.00328, 0.00437, 0.00449, 0.01043, 0.02816 mg/kg. The Meeting estimated a median residue of 0.00443 mg/kg and highest residue of 0.02816 mg/kg.

Sugar beet

The Meeting received supervised residue trials on sugar beet. In Armenia, Georgia and the Ukraine, critical GAP for fipronil on sugar beet is soil application at planting at 0.2 kg ai/ha with a PHI not required. In thirteen trials matching cGAP residues at harvest were: < 0.00197, 0.00197, 0.00197, 0.00197, 0.00197, 0.00197, 0.002, 0.00206, < 0.004, < 0.004, < 0.004, < 0.004 and < 0.004 mg/kg.

Total residues were (n = 13): 0.00296, 0.00296, 0.00296, 0.00296, 0.00296, 0.00296, 0.003, 0.00306, 0.006, 0.006, 0.006, 0.006, 0.006 mg/kg (the sum of fipronil, MB46136, MB45950 as fipronil; soil application with no residues of MB46513 detected).

The Meeting estimated a maximum residue level of 0.01 mg/kg and an STMR of 0.003 mg/kg for fipronil in sugar beet root to replace its previous recommendation (0.2 mg/kg).

Cereal grains

Barley

The Meeting received supervised residue trials on barley. In the Russian Federation, critical GAP for fipronil on barley is a single foliar application at 0.016 kg ai/ha with a PHI of 30 days. No trials matched GAP in the Russian Federation.

In Brazil, critical GAP for fipronil on barley is seed treatment at 0.1 kg ai/100 kg seed and a PHI of not required.

In one trial matching cGAP in Brazil, residues in grain at harvest were: < 0.004 mg/kg with total residues < 0.008 mg/kg.

In five trials conducted at exaggerated rates (1 × 0.15 kg ai/100 kg seed) residues at harvest were: < 0.004 (5) mg/kg for the sum of fipronil and MB46136. Total residues were < 0.008 (5) mg/kg.

Using proportionality (SF 0.67) residues in trials at cGAP are (n = 6): < 0.0027 (5), < 0.004 mg/kg and for total residues < 0.00536 (5) and < 0.008 mg/kg (the sum of fipronil, MB46136, MB45950 as fipronil; seed treatment with no residues of MB46513 expected).

The Meeting noted residues were all <LOQ and estimated a maximum residue level of 0.004 (*) mg/kg and an STMR of 0.00536 mg/kg for fipronil in barley and agreed to extrapolate to the subgroup of barley withdrawing its previous recommendations (0.002 (*) mg/kg) for barley and oats.

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens. The median residue for the sum of fipronil and fipronil-sulfone (MB46136) in grain in the subgroup of barley was 0.0027 mg/kg.

Scaled residues of fipronil-desulfinyl for use in estimation of livestock dietary burden were (n = 6): < 0.00134 (5), < 0.002 mg/kg. The Meeting estimated a median residue of 0.00134 mg/kg for grain in the subgroup of barley.

Scaled residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide for estimation of livestock dietary burdens were (n = 6): 0.00402 (5), 0.006 mg/kg. The Meeting estimated a median residue of 0.00402 mg/kg for grain in the subgroup of barley.

Maize

The Meeting received supervised residue trials on maize. In the Philippines, critical GAP for fipronil on maize is seed treatment at 0.0625 kg ai/100 kg seed followed by foliar application at 0.039 kg ai/ha, application is into the whorl at 30–35 days post-emergence, with a PHI of 7 days (in practice longer as

Total residues were (n = 12): 0.00206, 0.00279, 0.00287, 0.00311, 0.00380, 0.00386, 0.00436, 0.01251, 0.02005, 0.03666, 0.04656, 0.29997 mg/kg for the sum of fipronil, MB46136, MB45950 and MB46513.

The Meeting estimated a maximum residue level of 0.4 mg/kg and an STMR of 0.00411 mg/kg for fipronil in the subgroup of rice and withdrew its previous recommendation (0.01 mg/kg).

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens for maximum residue level estimation. The median residue for the sum of fipronil and fipronil-sulfone (MB46136) in rice (paddy rice) was 0.001445 mg/kg.

Residues of fipronil-desulfinyl (MB46513) in paddy rice grain were (n = 12): 0.0005, 0.0011, 0.0011, 0.0013, 0.0017, 0.0021, 0.0021, 0.00398, 0.0046, 0.0062, 0.0088, 0.0115 mg/kg. The Meeting estimated a median residue of 0.0021 mg/kg for fipronil-desulfinyl in paddy rice grain for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) in paddy rice grain were (n = 12): 0.0015, 0.00150, 0.00155, 0.00163, 0.00165, 0.00189, 0.00200, 0.00554, 0.01017, 0.03149, 0.04209, 0.28705 mg/kg. The Meeting estimated a median residue of 0.00195 mg/kg for the sum of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) in paddy rice grain for use in the estimation of livestock dietary burdens.

In fourteen trials conducted according to cGAP residues in husked rice (brown rice) were: 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.00102, 0.00133, 0.0025, 0.00946, 0.04139, 0.33987 mg/kg for the sum fipronil and MB46136.

Total residues were (n = 14): 0.002, 0.002, 0.002, 0.002, 0.00206, 0.00208, 0.00229, 0.00230, 0.00249, 0.00307, 0.00626, 0.01513, 0.04367, 0.36010 mg/kg for the sum of fipronil, MB46136, MB45950 and MB46513.

The Meeting estimated a maximum residue level of 0.4 mg/kg and an STMR of 0.00230 mg/kg for fipronil in husked rice (brown rice).

In fifteen trials conducted according to cGAP residues in polished rice were: 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.00398, 0.03569, 0.05766, 0.08979 mg/kg for the sum fipronil and MB46136.

Total residues were (n = 15): 0.002, 0.002, 0.002, 0.002, 0.002, 0.002, 0.002, 0.002, 0.00206, 0.0027, 0.00275, 0.00639, 0.03799, 0.06280, 0.09115 mg/kg for the sum of fipronil, MB46136, MB45950 and MB46513.

The Meeting estimated a maximum residue level of 0.15 mg/kg and an STMR of 0.002 mg/kg for fipronil in polished rice.

Sorghum

The Meeting received supervised residue trials on sorghum. In Australia, critical GAP for fipronil on sorghum is seed treatment at 0.075 kg ai/100 kg seed followed by a single foliar/soil spray of locust bands at 0.00125 kg ai/ha with a PHI of 14 days. No trials matched cGAP in Australia.

In Mexico and Paraguay, cGAP is seed treatment at 0.062 kg ai/100 kg seed with PHI not required. Two trials matched cGAP with residues: 0.004 (2) mg/kg and total residues 0.006 (2) mg/kg. The Meeting agreed there were insufficient trials matching cGAP to estimate a maximum residue level for sorghum grain.

Wheat

The Meeting received supervised residue trials on wheat. In Brazil, critical GAP for fipronil on wheat is ground application prior to crop emergence at 1 × 0.004 kg ai/ha and a PHI of not required. No trials matched cGAP of Brazil.

In Paraguay, critical GAP for fipronil on wheat is seed treatment at 0.045 kg ai/100 kg seed and a PHI of not required. In eight trials approximating cGAP (0.03375–0.05625 kg ai/100 kg seed) residues at harvest were: < 0.001 (3), < 0.004 (5) mg/kg. In five of the trials MB46513 was not measured, however this photo-degradate is not expected to be present at levels above LOQ following use as a seed treatment and residues were assumed to be present at <LOQ for estimation of total residues.

Total residues were: < 0.002 (3), < 0.008 (5) mg/kg.

The Meeting estimated a maximum residue level of 0.004 (*) mg/kg, an STMR of 0.008 mg/kg for fipronil in wheat and agreed to extrapolate the recommendations to rye and triticale to make a recommendation for the subgroup of wheat, withdrawing its previous recommendations for these commodities (0.002 (*) mg/kg).

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens. The median residue for the sum of fipronil and fipronil-sulfone (MB46136) in the subgroup of wheat was 0.004 mg/kg.

Residues of fipronil-desulfinyl (MB46513) in wheat grain were (n = 3): < 0.0005 (3) mg/kg. The Meeting estimated a median residue of 0.0005 mg/kg for fipronil-desulfinyl in the subgroup of wheat for use of estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in wheat grain were (n = 8): < 0.0015 (3), < 0.006 (5) mg/kg. The Meeting estimated a median residue of 0.006 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in the subgroup of wheat for use in the estimation of livestock dietary burdens.

Sugar cane

The Meeting received supervised residue trials on sugar cane. In Indonesia, critical GAP for fipronil on sugar cane is application at 0.15 kg ai/ha when plants are 2–3 months old followed by foliar applications at 0.025 kg ai/hL with a PHI of 14 days. No trials matched the GAP of Indonesia.

In Brazil, cGAP is a single in-furrow application at 0.54 kg ai/ha with a PHI not required. In nine trials approximating cGAP (0.405–0.675 kg ai/ha) residues at harvest were: < 0.001, 0.00143, 0.00175, 0.00204, 0.00204, 0.00218, 0.00324, 0.00382, 0.00633 mg/kg.

Highest individual analytical sample 0.00715 mg/kg.

Total residues were: < 0.002, 0.00243, 0.00275, 0.00304, 0.00304, 0.00318, 0.00424, 0.00482, 0.00733 mg/kg.

Highest individual analytical sample 0.00815 mg/kg.

The Meeting estimated a maximum residue level of 0.01 mg/kg, an STMR of 0.00304 mg/kg and an HR of 0.00815 mg/kg for fipronil in sugar cane.

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens. The median and highest residues for the sum of fipronil and fipronil-sulfone (MB46136) in sugar cane were 0.00204 and 0.00715 mg/kg, respectively.

Residues of fipronil-desulfinyl for use in estimation of livestock dietary burdens were (n = 9): < 0.0005 (9) mg/kg. The Meeting estimated a median residue of 0.0005 mg/kg.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide for estimation of livestock dietary burdens were (n = 9): 0.0015, 0.00193, 0.00225, 0.00254, 0.00254, 0.00268, 0.00374, 0.00432, 0.00683 mg/kg. The Meeting estimated a median residue of 0.00245 mg/kg.

*Oilseed**Cotton*

The Meeting received supervised residue trials on cotton. In Australia, cGAP is foliar application at 0.025 kg ai/ha with the number of sprays and interval not specified and with a PHI of 28 days. No trials matched cGAP in Australia.

In Brazil, cGAP is seed treatment at 0.25 kg ai/100 kg seed followed by further applications at or close to planting at 2×0.004 kg ai/ha with PHI not required. No trials matched cGAP of Brazil.

In Mexico, cGAP is seed treatment at 0.062 kg ai/100 kg seed followed by foliar application at 2×0.048 kg ai/ha with an interval between sprays of 7 days and a PHI of 45 days. No trials matched cGAP in Mexico.

In Paraguay, cGAP is seed treatment at 0.1 kg ai/100 kg seed with PHI not required.

In eight trials approximating this cGAP (0.075–0.125 kg ai/100 kg seed) residues at harvest were: 0.00098, < 0.001, < 0.001, < 0.001, < 0.001, < 0.001, < 0.001, 0.0023 mg/kg.

Total residues were: < 0.0012, < 0.0012, < 0.002, < 0.002, < 0.002, < 0.002, 0.00283, 0.0033 mg/kg.

The Meeting estimated a maximum residue level of 0.01 mg/kg and an STMR of 0.002 mg/kg for fipronil in cotton seed.

The Meeting also estimated a range of values for use in estimation of livestock dietary burdens. The median residue for the sum of fipronil and fipronil-sulfone (MB46136) in cotton seed was 0.001 mg/kg.

Residues of fipronil-desulfinyl (MB46513) in cotton seed were (n = 8): < 0.0001 (2), < 0.0005 (5), < 0.002 mg/kg. The Meeting estimated a median residue of 0.0005 mg/kg for fipronil-desulfinyl in cotton seed for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in cotton seed were (n = 8): 0.0011, 0.0011, 0.0015, 0.0015, 0.0015, 0.0015, 0.0015, 0.0028 mg/kg. The Meeting estimated a median residue of 0.0015 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in cotton seed for use in the estimation of livestock dietary burdens.

Sunflower

The Meeting received supervised residue trials on sunflower. In Armenia, Georgia and the Ukraine, cGAP is soil application at planting at 0.2 kg ai/ha with a PHI of 30 days (interval between application and harvest will be longer, about 80–120 days). In six trials on sunflower matching cGAP residues at harvest were: < 0.004 (6) mg/kg.

Total residues were: < 0.008 (6) mg/kg.

The Meeting estimated a maximum residue level of 0.004 (*) mg/kg and an STMR of 0.008 mg/kg for fipronil in the subgroup of sunflower seeds and withdrew its previous recommendation (0.002 (*) mg/kg).

Basil

In Thailand cGAP for application to basil is two foliar sprays at 7-day intervals and at a rate of 0.005 kg ai/hL with a 7-day PHI. Five independent trials were made available to the Meeting. In three trials, residues of fipronil-sulfide (MB45950) were not measured. From the trials where fipronil-sulfide was analysed, the sum of fipronil and MB46136 residues in basil were (n = 2): 0.103 and 0.30 mg/kg. From the trials where fipronil-sulfide was not analysed, the fipronil and MB46136 residues in basil were (n = 3): 0.033, 0.12 and 0.36 mg/kg.

The total residues (four compounds) in basil were (n = 2): 0.23 and 0.42 mg/kg. The sum of residues of three compounds in basil were (n = 3): 0.080, 0.23 and 0.57 mg/kg. Residues of fipronil-sulfide did not contribute significantly to the total residue (> 90% of total residues covered by sum of the three compounds). The Meeting estimated an STMR and HR using all available data.

The total residues in basil were (n = 5): 0.080, 0.23, 0.23, 0.42 and 0.57 mg/kg. The Meeting estimated a maximum residue level of 0.8 mg/kg, an STMR of 0.23 mg/kg and an HR of 0.57 mg/kg for basil to replace its previous recommendations.

Residues in animal feeds

Bean forage (green)

The Meeting received supervised residue trials on beans. In Brazil, critical GAP for fipronil on beans is seed application at 0.1 kg ai/100 kg seed. Residues in forage in one trial at a lower rate (0.049 kg ai/100 kg seed) were 0.0348 mg/kg for fipronil and MB46136 and for total residues 0.0388 mg/kg. In another two trials at exaggerated rates (0.14–0.15 kg ai/100 kg seed) residues were 0.0175 and 0.006 mg/kg with total residues 0.0195 and 0.01 mg/kg.

Scaling residues for application rate resulted in residues of 0.004, 0.0125 and 0.0710 mg/kg with total residues 0.007, 0.0139 and 0.0791 mg/kg.

The number of trials was insufficient to estimate a maximum residue level.

In Indonesia, critical GAP for fipronil on soybean is seed treatment at 0.125 kg ai/100 kg seed followed by foliar application at 0.01 kg ai/hL with a PHI of 14 days. No trials matched cGAP in Indonesia. None of the trials matched critical GAP in Indonesia.

Barley straw and fodder dry

The Meeting received supervised residue trials on barley. In the Russian Federation, critical GAP for fipronil on barley is a single foliar application at 0.016 kg ai/ha with a PHI of 30 days. No trials matched cGAP of the Russian Federation.

In Brazil, critical GAP for fipronil on barley is seed treatment at 0.1 kg ai/100 kg seed and a PHI of not required. In four trials conducted at 0.1 kg ai/100 kg seed residues in straw were: < 0.02, 0.0245, 0.0292 and 0.0332 mg/kg for the sum of fipronil and MB46136. Total residues were < 0.04, 0.0445, 0.0493 and 0.0532 mg/kg.

Residues in two trials conducted at an exaggerated rate of 0.15 kg ai/100 kg seed were: < 0.02 (2) mg/kg with total residues < 0.04 (2) mg/kg.

Using proportionality to scale the residues in the trials at the exaggerated rate are < 0.013 (2) mg/kg and for total residues < 0.027 (2) mg/kg.

Using the combined data set residues are: < 0.013 (2), < 0.02, 0.0245, 0.02923 and 0.0332 mg/kg.

After adjusting for a typical dry matter content (88% DM), the Meeting estimated a maximum residue level of 0.07 mg/kg, a median residue of 0.02225 mg/kg (as received) and highest residue of 0.0332 mg/kg (as received) for fipronil+MB46136 in barley straw. The Meeting agreed to extrapolate the results for barley straw to oats straw.

Residues of fipronil-desulfinyl (MB46513) were not measured in barley straw.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in barley straw were (n = 6): 0.02, 0.02, 0.03, 0.0345, 0.0393, 0.0432 mg/kg. The Meeting estimated a median residue of 0.03225 mg/kg and highest residue of 0.0432 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in barley and oat straw for use in the estimation of livestock dietary burdens.

Maize forage, fodder (stover) and silage

The Meeting received supervised residue trials on maize. In the Philippines critical GAP for fipronil on maize is seed treatment at 0.0625 kg ai/100 kg seed followed by foliar application at 0.039 kg ai/ha, application is into the whorl at 30–35 days post-emergence, with a PHI of 7 days (in practice longer as interval between 30–35 days post-emergence and harvest is about 130 days). No trials matched cGAP of the Philippines.

In Indonesia, critical GAP for fipronil on maize is seed treatment at 0.125 kg ai/100 kg seed followed by a single foliar application at 0.03 kg ai/ha at the time cobs have formed (PHI not specified) or at 0.0025 kg ai/hL with the number of sprays and interval not specified and with a PHI of 14 days. No trials matched cGAP of Indonesia.

In Mexico cGAP is seed application at 1 kg ai/100 kg seed or in-furrow application at 0.19 kg ai/ha at sowing. No trials matched cGAP of Mexico.

In Armenia, Georgia and the Ukraine, cGAP is application to soil at the time of planting at 0.2 kg ai/ha with a PHI of 30 days, in practice harvest would be 100-200 days after planting.

In thirteen trials conducted matching cGAP residues in silage were: < 0.004 (11), < 0.01 (2) mg/kg.

The Meeting estimated a median residue of 0.004 mg/kg and highest residue of 0.01 mg/kg for fipronil+MB46136 in maize silage.

Residues of fipronil-desulfinyl (MB46513) in maize silage were (n = 12): < 0.002 (10), < 0.005 (2) mg/kg. The Meeting estimated a median residue of 0.002 mg/kg and highest residues of 0.005 mg/kg for fipronil-desulfinyl in maize silage for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in maize silage were (n = 13): < 0.002 (11), < 0.005 (2) mg/kg. The Meeting estimated a median residue of 0.002 mg/kg and a highest residue of 0.005 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in maize silage for use in the estimation of livestock dietary burdens.

In seven trials conducted matching cGAP residues in forage (BBCH83-85) were: < 0.01 (7) mg/kg.

The Meeting estimated a median residue of 0.01 mg/kg (as received) and highest residue of 0.01 mg/kg (as received) for fipronil+MB46136 in maize forage.

No data was available for maize fodder (stover) and the Meeting withdrew its previous recommendation of 0.1 mg/kg (dw) for maize fodder (dry).

Residues of fipronil-desulfinyl (MB46513) in maize forage were (n = 7): < 0.005 mg/kg. The Meeting estimated a median and highest residue of 0.005 mg/kg for fipronil-desulfinyl in maize forage for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in maize forage were (n = 7): < 0.015 (6), 0.0175 mg/kg. The Meeting estimated a median residue of 0.015 mg/kg and highest residue of 0.0175 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in maize forage for use in the estimation of livestock dietary burdens.

As residues in forage and silage were all <LOQ, the Meeting agreed to use the lowest of the forage and silage values for estimation of livestock dietary burdens.

Rice straw and fodder dry

The Meeting received supervised residue trials on rice. In the Philippines, critical GAP for fipronil on rice is one application at 0.051 kg ai/ha 30-45 days after planting or 40-45 days after seeding followed by foliar application at 0.033 kg ai/ha with the number of sprays and interval not specified and with a PHI of 7 days. No trials matched cGAP of the Philippines.

In Malaysia, critical GAP for fipronil on rice is seed treatment at 0.1 kg ai/100 kg seed followed by foliar application at 2×0.031 kg ai/ha with a minimum interval between sprays of 7 days, and a PHI of 45 days.

In fourteen trials approximating cGAP residues in rice straw were (controls < 10%): 0.01062, 0.03133, 0.04622, 0.04983, 0.05076, 0.07827, 0.08085, 0.09832, 0.10600, 0.12282, 0.13293, 0.20641, 0.26400, 0.33400 mg/kg (as received) for fipronil and MB46136.

Assuming 90% DM, the Meeting estimated a maximum residue level of 0.6 mg/kg (dry weight) to replace its previous recommendation (0.2 mg/kg dw), as well as a median residue of 0.089585 mg/kg (as received) and highest residue of 0.3340 mg/kg (as received) for fipronil+MB46136 in rice straw.

Residues of fipronil-desulfinyl (MB46513) in rice straw were (n = 14): 0.01236, 0.03595, 0.04157, 0.04269, 0.04606, 0.05618, 0.05730, 0.05955, 0.07865, 0.07977, 0.08651, 0.12359, 0.20223, 0.31459 mg/kg. The Meeting estimated a median residue of 0.05843 mg/kg and highest residues of 0.31459 mg/kg for fipronil-desulfinyl in rice straw for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in rice straw were (n = 14): 0.01685, 0.04005, 0.05513, 0.06802, 0.06841, 0.08201, 0.08666, 0.12219, 0.12883, 0.14850, 0.14981, 0.22613, 0.26836, 0.34749 mg/kg. The Meeting estimated a median residue of 0.10443 mg/kg and highest residue of 0.34749 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in rice straw for use in the estimation of livestock dietary burdens.

Rice bran

Residues in rice bran were (n = 10): 0.00138, 0.0015, 0.0021, 0.00226, 0.0024, 0.00406, 0.00885, 0.02763, 0.55315, 1.2425 mg/kg for fipronil and MB46136.

Total residues in rice bran were (n = 10): 0.00346, 0.00586, 0.00593, 0.00715, 0.00797, 0.01321, 0.02395, 0.04424, 0.58945, 1.2687 mg/kg for total residues.

The Meeting estimated a maximum residue level of 2 mg/kg and median residues of 0.00323 mg/kg for fipronil+MB46136 and 0.01059 mg/kg for total residues in rice bran.

Residues of fipronil-desulfinyl (MB46513) in rice bran were (n = 10): 0.00146, 0.00303, 0.00326, 0.00438, 0.00607, 0.00865, 0.01461, 0.01573, 0.02022, 0.03371 mg/kg. The Meeting estimated a median residue of 0.00736 mg/kg for fipronil-desulfinyl in rice bran for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in rice bran were (n = 10): 0.00190, 0.002, 0.0026, 0.00276, 0.0029, 0.00456, 0.00935, 0.02851, 0.55575, 1.2485 mg/kg. The Meeting estimated a median residue of 0.00373 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in rice bran for use in the estimation of livestock dietary burdens.

Rice hulls

Residues in rice hulls were (n = 15): 0.00178, 0.00178, 0.00264, 0.00277, 0.00297, 0.00344, 0.00362, 0.00667, 0.01960, 0.02761, 0.04833, 0.09037, 0.17222, 1.0199, 1.1463 mg/kg (as received) for fipronil and MB46136.

The Meeting estimated a maximum residue level of 2 mg/kg and a median residue of 0.00667 mg/kg for fipronil+MB46136 in rice hulls.

Residues of fipronil-desulfinyl (MB46513) in rice hulls were (n = 15): 0.00225, 0.00427, 0.00483, 0.00539, 0.00573, 0.00618, 0.00629, 0.00910, 0.01348, 0.014615, 0.02359, 0.02809, 0.03146, 0.04382, 0.06067 mg/kg. The Meeting estimated a median residue of 0.00910 mg/kg for fipronil-desulfinyl in rice hulls for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in rice hulls were (n = 15): 0.00228, 0.0023, 0.00314, 0.00327, 0.00347, 0.00394, 0.00412, 0.00725, 0.02034, 0.02840, 0.04989,

0.09203, 0.17284, 1.0262, 1.1528 mg/kg. The Meeting estimated a median residue of 0.00725 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in rice hulls for use in the estimation of livestock dietary burdens.

Sorghum forage (green) and stover (Sorghum silage)

The Meeting received supervised residue trials on sorghum. In Australia, critical GAP for fipronil on sorghum is seed treatment at 0.075 kg ai/100 kg seed followed by a single foliar/soil spray of locust bands at 0.00125 kg ai/ha with a PHI of 14 days. Do not harvest, graze or cut for stock food for 14 days after application. No trials matched cGAP in Australia.

In Mexico and Paraguay, cGAP is seed treatment at 0.062 kg ai/100 kg seed with PHI not required.

In two trials approximating cGAP (0.0465–0.0775 kg ai/100 kg seed) residues in forage were: < 0.004 (2) mg/kg.

Total residues in forage were < 0.0048(2) mg/kg.

In trials at the same sites but with application at 2× the maximum rate, residues in forage were also < 0.004 (2) mg/kg with total residues < 0.008 (2) mg/kg.

In two trials approximating cGAP (0.0465–0.0775 kg ai/100 kg seed) residues in straw at harvest were: < 0.004 (2) mg/kg.

Total residues were < 0.0048(2) mg/kg.

In trials at the same sites but with application at 2× the maximum rate, residues were also < 0.004 (2) mg/kg with total residues < 0.008 (2) mg/kg.

The Meeting agreed the number of trials was insufficient to estimate maximum residue levels, median and highest residues for sorghum livestock feed commodities.

Wheat straw and fodder dry (hay, straw, silage)

The Meeting received supervised residue trials on wheat. In Brazil, critical GAP for fipronil on wheat is ground application prior to crop emergence at 1 × 0.004 kg ai/ha and a PHI of not required. No trials matched cGAP of Brazil.

In Paraguay, critical GAP for fipronil on wheat is as a seed treatment at 0.045 kg ai/100 kg seed and a PHI of not required. In five trials approximating cGAP (0.03375–0.05625 kg ai/100 kg seed) residues in straw at harvest were: < 0.02 (3), 0.0205, 0.0241 mg/kg (as received).

MB46513 was not measured but based on metabolism studies, and use as a seed treatment, residues of the photo-degradate are assumed to be <LOQ.

Using a typical dry matter content of 88%, the Meeting estimated a maximum residue level of 0.05 mg/kg, a median residue of 0.02 mg/kg (as received) and highest residue of 0.0241 mg/kg (as received) for fipronil+MB46136 in wheat straw. The Meeting agreed to extrapolate the recommendations to rye and triticale.

Residues of fipronil-desulfinyl (MB46513) in wheat straw were not measured.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in wheat straw were (n = 5): 0.03, 0.03, 0.03, 0.0305, 0.03408 mg/kg. The Meeting estimated a median residue of 0.03 mg/kg and highest residue of 0.03408 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in wheat, rye and triticale straw for use in the estimation of livestock dietary burdens.

Cotton gin trash

The Meeting received supervised residue trials on cotton. In Australia, cGAP is foliar application at 0.025 kg ai/ha with the number of sprays and interval not specified and with a PHI of 28 days. Do not graze or cut for stock food. No trials matched cGAP of Australia.

In Brazil, cGAP is soil application prior to emergence at 4 kg ai/ha and two applications at or close to planting at 2×0.004 kg ai/ha with PHI not required. No trials matched cGAP in Brazil.

In Mexico, cGAP is seed treatment at 0.062 kg ai/100 kg seed followed by foliar application at 2×0.048 kg ai/ha with an interval between sprays of 7 days and a PHI of 45 days. No trials matched cGAP of Mexico.

In Paraguay, cGAP is seed treatment at 0.1 kg ai/100 kg seed with PHI not required. No trials matched cGAP of Paraguay.

A scaling factor for residue trials could not be located that would allow both the seed treatment and foliar applications to match cGAP.

Sugar beet tops (Sugar beet fodder)

The Meeting received supervised residue trials on sugar beet. In Armenia, Georgia and the Ukraine, critical GAP for fipronil on sugar beet is application to soil at the time of planting at 0.2 kg ai/ha with a PHI not required. In fifteen trials matching cGAP, residues in leaves and tops were: 0.00196, < 0.002, 0.00297, 0.00297, 0.00297, 0.00297, 0.00297, < 0.004, < 0.004, < 0.004, 0.00432, 0.00731, 0.02, 0.02 mg/kg.

The Meeting estimated a median residue of 0.00297 mg/kg and a highest residue of 0.02 mg/kg for fipronil+MB46136 in sugar beet tops.

Residues of fipronil-desulfinyl (MB46513) in sugar beet tops were (n = 21): < 0.0005 (6), < 0.001 (8), < 0.002 (5), < 0.01 (2) mg/kg. The Meeting estimated a median residue of 0.001 mg/kg and a highest residue of 0.01 mg/kg for fipronil-desulfinyl in sugar beet tops for use in the estimation of livestock dietary burdens.

Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide in sugar beet tops were (n = 21): 0.0015, 0.00189, 0.00216, 0.00293, 0.00296, 0.003, 0.00303, 0.00380, 0.00396, 0.00396, 0.00396, 0.00396, 0.00396, 0.00396, 0.006, 0.006, 0.006, 0.00632, 0.00931, 0.03, 0.03 mg/kg. The Meeting estimated a median residue of 0.00396 mg/kg and highest residue of 0.03 mg/kg for the sum of fipronil, fipronil-sulfone and fipronil-sulfide in sugar beet tops for use in the estimation of livestock dietary burdens.

Residues in rotational crops

Fipronil and metabolites are moderately persistent to persistent in the environment and may contribute to residues in follow/rotational crops through uptake from soil.

In assessing the potential uptake of residues, the Meeting considered the maximal season rate to be 0.4 kg ai/ha for potato, the highest application rate for the uses evaluated by the Meeting.

Field rotational crop studies were used to derive estimates of residues in various commodities if the field were treated at the maximal seasonal rate. In combining residues to derive the various inputs required, individual compound residues were scaled to the maximum seasonal rate. The Meeting noted that residues above the LOQ in rotational crops were infrequent and agreed that residues of fipronil and individual metabolites <LOQ would be assumed to be zero in calculating the sums to avoid overestimating the contribution from uptake in rotational crops.

The commodity groups studied in the field crop rotational studies were:

Leafy vegetables (leaf lettuce, collard greens, mustard greens)

Root and tuber (radish, sugar beet)

Legumes (green pea, cowpea, soya beans, Wando pea)

Cereals (maize, sorghum, wheat)

No residues were detected in cereal grains, legume grains and legume straw.

Leafy vegetables (leaf lettuce, kale (collard) greens, mustard greens, radish tops, sugar beet tops)

Residues of fipronil and fipronil-sulfone (MB46136) (n = 10): 0, 0, 0, 0, 0, 0, 0, 0, 0.00193, 0.01135 mg/kg.

Total residues (n = 10): 0, 0, 0, 0, 0, 0, 0, 0.00193, 0.01652, 0.02919 mg/kg.

Residues of fipronil-desulfinyl (n = 10): 0, 0, 0, 0, 0, 0, 0, 0, 0.01652, 0.01784 mg/kg.

Residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) (n = 10): 0, 0, 0, 0, 0, 0, 0, 0, 0.00193, 0.01135 mg/kg.

The Meeting estimated a maximum residue level of 0.01 mg/kg for leafy vegetables, an STMR and HR of 0 and 0.02919 mg/kg respectively, median and highest residues of fipronil-desulfinyl of 0 and 0.01784 mg/kg, median and highest residues for the sum of fipronil and fipronil-sulfone and also the sum of fipronil, fipronil-sulfone and fipronil-sulfide of 0 and 0.01135 mg/kg.

Root and tubers

Residues of fipronil and fipronil-sulfone (MB46136): 0, 0, 0, 0, 0.00193, 0.00212 mg/kg.

Total residues, 0, 0, 0, 0, 0.00193, 0.00212 mg/kg.

Residues of fipronil-desulfinyl: 0 (6) mg/kg.

Residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950): 0, 0, 0, 0, 0.00193, 0.00212 mg/kg.

The Meeting estimated a maximum residue level of 0.002 mg/kg for group 16 root and tuber vegetables (except potato and sugar beet), an STMR and HR for total residues of 0 and 0.00212 mg/kg; median and highest residues of fipronil-desulfinyl of 0 and 0 mg/kg; median and highest residues for the sum of fipronil and fipronil-sulfone, and also the sum of fipronil, fipronil-sulfone and fipronil-sulfide of 0 and 0.00212 mg/kg.

Legume forages

Residues of fipronil and fipronil-sulfone (MB46136) (n = 4): 0 (4) mg/kg.

Residues of fipronil-desulfinyl (n = 4): 0, 0, 0.00727, 0.01190 mg/kg.

Residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) (n = 4): 0 (4) mg/kg.

The Meeting estimated median and highest residues of fipronil-desulfinyl of 0.00364 and 0.01190 mg/kg, median and highest residues for the sum of fipronil and fipronil-sulfone, and also the sum of fipronil, fipronil-sulfone and fipronil-sulfide of 0 and 0 mg/kg for legume forages (bean, cowpea, crown vetch, Lespedeza, pea, peanut, soya bean, trefoil, vetch).

Cereal forage

Residues of fipronil and fipronil-sulfone (MB46136) (n = 9): 0 (9) mg/kg.

Residues of fipronil-desulfinyl (n = 9): 0, 0, 0, 0, 0, 0, 0, 0.009253, 0.01322 mg/kg.

Residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) (n = 9): 0 (9) mg/kg.

The Meeting estimated median and highest residues of fipronil-desulfinyl of 0 and 0.01322 mg/kg, median and highest residues for the sum of fipronil and fipronil-sulfone, and also the sum of fipronil, fipronil-sulfone and fipronil-sulfide of 0 and 0 mg/kg for cereal forage except maize forage.

Cereal straw

Residues of fipronil and fipronil-sulfone (MB46136) (n = 11): 0, 0, 0.01109, 0.01475, 0.01646, 0.02117, 0.02270, 0.02329, 0.02383, 0.02724, 0.03112 mg/kg.

Residues of fipronil-desulfinyl (n = 11): 0, 0, 0, 0, 0.01322, 0.02379, 0.02842, 0.03172, 0.04230, 0.04362 mg/kg.

Residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) (n = 11): 0, 0, 0.01109, 0.01475, 0.01646, 0.02117, 0.02270, 0.02329, 0.02383, 0.02724, 0.03112 mg/kg.

The Meeting estimated a maximum residue level of 0.03 mg/kg, median and highest residues of fipronil-desulfinyl of 0.01322 and 0.04362 mg/kg, median and highest residues for the sum of fipronil and fipronil-sulfone, and also the sum of fipronil, fipronil-sulfone and fipronil-sulfide of 0.02117 and 0.03112 mg/kg for cereal straw and fodder dry (except for barley, oats, rice, rye, triticale and wheat).

Fate of residues during processing

The Meeting received information on the fate of fipronil residues during processing in soya beans, potatoes, sugar beet (no residues RAC), maize (no residues RAC), rice, sugarcane, cottonseed and sunflower (no residues RAC).

The hydrolytic stability of fipronil, MB46136 and MB46153 was studied in sterile aqueous buffers at pH 4 at 90 °C for 20 minutes, pH 5 at 100 °C for 60 minutes and pH 6 at 120 °C for 20 minutes to simulate commercial processing practices (pasteurization, baking/brewing/boiling and sterilization).

Fipronil was not degraded during the simulation of pasteurization (pH 4, 90 °C) and baking/brewing/boiling (pH 5, 100 °C) with only minor degradation occurring during sterilization (pH 6, 120 °C). Under sterilisation conditions fipronil accounted for 91% AR at the end of the experiment with fipronil-amide (RPA200766) accounting for 5.5% AR.

The effect of processing of fipronil-sulfone was also studied. Fipronil-sulfone (MB46136) was also not degraded during the simulation of pasteurization (pH 4, 90 °C) and baking/brewing/boiling (pH 5, 100 °C) with only minor degradation occurring during sterilization (pH 6, 120 °C). Fipronil-sulfone (MB46136) accounted for ≥ 88% AR at the end of the experiment with RPA105320 present at 8.4% AR.

In a study of fipronil-desulfinyl (MB46513), the compound was stable during the simulation of pasteurization (pH 4, 90 °C) with limited degradation to RPA105048 under baking/brewing/boiling (pH 5, 100 °C). Significant degradation occurred during sterilization (pH 6, 120 °C) with fipronil-desulfinyl (MB46513) accounting for 53.9% AR at the end of the experiment. MB46400 represented 10.5% AR and RPA105048 25.9% AR.

Table 1 Processing factors and median and highest residue values for the sum of fipronil and fipronil-sulfone (MB46136) – used for estimation of maximum residue levels including livestock dietary burdens for livestock commodity maximum residue levels

Processed commodity	Raw commodity [median residue]	Individual processing factors	Median or best estimate processing factor	Median residue-P (mg/kg)	Highest residue-P (mg/kg)
Soya bean hulls	0.00219	5.3, 6.8, 8.9	6.8	0.01489	
Potato peel	0.00361	1.0, 1.0, 1.4, 3.5, 8.2	1.4	0.00505	

Processed commodity	Raw commodity [median residue]	Individual processing factors	Median or best estimate processing factor	Median residue-P (mg/kg)	Highest residue-P (mg/kg)
Rice hulls	0.00327	1.3, 2.2, 3.8	2.2	See residue trials	
Rice bran		2.1, 2.2, 2.3	2.2	See residue trials	
Sugarcane bagasse	0.00204	0.011, 0.014, 0.65, 0.87, 1.0, 1.4, 1.5, 1.6	0.935	0.0019	
Molasses		0.011, 0.012	0.0115	0.00002	
Cotton seed meal	0.001	0.17	0.17	0.00017	
Cotton seed hulls		0.18	0.18	0.00018	

Table 2 Processing factors and STMR-P and HR-P residue values for total residues – used for STMR and HR estimation

Processed commodity	Raw commodity [STMR/HR]	Individual processing factors	Median or best estimate processing factor	STMR-P = $STMR_{RAC} \times PF$ (mg/kg)	HR-P = $HR_{RAC} \times PF$ (mg/kg)
Soya bean crude oil	0.00411	3.0, 4.4, 4.5	4.4	0.01808	
Washed potato	0.00493/ 0.0296	0.43, 0.44, 0.55, 0.87	0.495	0.00244	0.01465
Peeled potato		0.19, 0.22, 0.42, 0.43	0.32	0.00158	0.00947
Cooked peeled potato		0.18, 0.18, 0.31, 0.43	0.245	0.00121	0.00725
Microwave/unpeeled potato		0.34, 0.62, 0.73, 0.81	0.675	0.00333	0.01998
French fries		0.18, 0.20, 0.37, 0.39, 0.47	0.37	0.00182	0.01095
Flakes		0.45	0.45	0.00222	
Brown rice	0.00411 (rice grain = with husk)	0.45, 0.45, 0.46	0.45	See residue trials	
Polished rice		0.12, 0.13, 0.13	0.13	See residue trials	
Rice flour		0.10, 0.13, 0.14	0.13	0.00053	
Cooked (polished) rice		0.03, 0.04, 0.04	0.04	0.00016	
Steamed (polished) rice		0.03, 0.03, 0.04	0.03	0.00012	
Sake		0.01, 0.02, 0.03	0.02	0.00008	
Sugarcane juice	0.00304	0.022, 0.022, 0.3, 0.4, 0.8, 1.0, 1.0, 1.0	0.6	0.00182	
Sugarcane molasses		0.022, 0.022	0.022	0.00007	
Cane sugar		0.022, 0.022	0.022	0.00007	

Processed commodity	Raw commodity [STMR/HR]	Individual processing factors	Median or best estimate processing factor	STMR-P = $STMR_{RAC} \times PF$ (mg/kg)	HR-P = $HR_{RAC} \times PF$ (mg/kg)
Cotton seed crude oil	0.002	0.3, 0.4	0.4	0.0008	
Cotton seed refined oil		0.3	0.3	0.0006	

Using the estimated maximum residue level of 0.01 mg/kg for soya beans and applying the processing factors of 6.0 for hulls and 4.4 for crude oil, the Meeting estimated maximum residue levels of 0.06 mg/kg for soybean hulls and 0.05 mg/kg for soybean crude oil.

Table 3 Processing factors and median-P residue values for fipronil-desulfinyl (MB46513) and for the sum of fipronil, fipronil-sulfone (MB46316) and fipronil-sulfide (MB45950) residues – used as inputs for livestock dietary burden calculations

Processed commodity	Raw commodity Median (mg equiv/kg)	Individual processing factors	Median or best estimate processing factor	median-P = median_RAC × PF (mg/kg)
Fipronil-desulfinyl				
Soya bean hulls	0.00102	4.9, 6.0, 7.0	6.0	0.00612
Potato peels	0.0005	3.2	3.2	0.0016
Sugar cane bagasse	0.0005	0.4, 0.4, 0.44, 1.0	0.42	0.00021
Sugar cane molasses		0.4, 0.4	0.4	0.0002
Cotton seed meal	0.0005	0.001	0.001	0.0000005
Cotton seed hulls		0.15	0.15	0.00008
Sum of fipronil, fipronil-sulfone and fipronil-sulfide				
Soya bean hulls	0.003215	4.2, 6.5, 8.2	6.5	0.02090
Potato peels	0.00443	1.0, 1.0, 1.4, 3.6, 7.7	1.4	0.00620
Sugar cane bagasse	0.00245	0.02, 0.02, 0.88, 1.3, 1.4	0.88	0.00216
Sugar cane molasses		0.02, 0.02	0.02	0.00005
Cotton seed meal	0.0015	0.22	0.22	0.00033
Cotton seed hulls		0.23	0.23	0.00035

Residues in animal commodities

Farm animal feeding studies

The Meeting received a number of studies on the transfer of fipronil and metabolites to tissues and milk of dairy cows following oral dosing.

In the first study, groups of lactating cows were dosed once daily with fipronil at the equivalent of 0.04, 0.13, and 0.43 ppm in the feed for 35 days with sacrifice 7 hours after the last dose.

It was not clear that residues in milk had plateaued by the end of the dosing period. Residues of fipronil and fipronil-sulfide (MB45950) were <LOQ (0.01 mg/kg) in all milk samples. Maximum residues of fipronil-sulfone were < 0.01, 0.018 and 0.052 mg/kg for the three dose levels, respectively. In tissues, fipronil-sulfide (MB45950) was <LOQ in all samples. Fipronil was only present at levels above LOQ in fat of the high dose group at levels with a mean level of 0.033 mg/kg (0.026–0.042 mg/kg). Mean residues

of fipronil-sulfone (MB46136) were <LOQ in muscle and kidney of the low dose group but detected in all other tissues with highest levels in fat. Mean residues in milk and tissues showed a linear relationship with dose.

In the second study, lactating cows were dosed orally once daily with fipronil for 20 days at the equivalent of 1 ppm in the feed. Only fipronil-sulfone (MB46136) was detected at above LOQ in whole milk. Residues in milk appeared to reach a plateau at between 14 and 20 days of dosing. Once dosing stopped, residues declined with a DT_{50} of 5.2 days. Milk fat was isolated from the day 20 whole milk sample (0.036 mg/kg fipronil-sulfone) and contained the majority of the residue (0.035 mg/kg fipronil, 0.0058 mg/kg MB45950, 0.51 mg/kg MB46136).

In a separate study, lactating cows were dosed orally once daily with fipronil for 50 days at the equivalent of 0.004 and 0.04 ppm in the feed. Residues in milk reached a plateau by 35 to 50 days of dosing. In the 0.04 ppm dose group, residues of fipronil in milk were < 0.0005 mg/kg from day 0 to day 30 and at day 50. At days 35 and 40, mean residues of fipronil were 0.0007 and 0.0006 mg/kg, respectively. In day 45 cream, mean fipronil levels were 0.0056 mg/kg, whereas in skim milk levels of fipronil were < 0.0005 mg/kg.

Residues of fipronil-sulfone (MB46136) in milk samples were up to 0.0064 mg/kg for the 0.04 ppm dose group. In day 45 cream, the mean MB46136 level was 0.057 mg/kg, whereas in skim milk residues of MB46136 were < 0.0005 mg/kg.

Mean fipronil residues in fat were 0.0006 and 0.0051 mg/kg for the 0.004 and 0.04 ppm feed levels, respectively, and < 0.0005 mg/kg in liver, kidney and in muscle at both dose levels.

Mean fipronil-sulfone (MB46136) residues in fat were 0.0062 and 0.055 mg/kg for the 0.004 and 0.04 ppm feed levels, in liver 0.0017 and 0.014 mg/kg, in kidney 0.0006 and 0.0041 mg/kg and in muscle < 0.0005 and 0.0037 mg/kg.

The transfer of the photo-degradata fipronyl-desulfinyl (MB46513) was also studied. Lactating cows were dosed once daily with fipronil-desulfinyl (MB46513) at the equivalent of 0.025, 0.076, 0.31 and 1.03 ppm in the diet for 35 days with sacrifice 14 hours after the last dose.

Residue concentrations of fipronil-desulfinyl (MB46513) in whole milk ranged from <LOQ (0.002 mg/kg) to 0.084 mg/kg with a plateau in fipronil-desulfinyl (MB46513) levels attained by 15–20 days. Residues of fipronil-desulfinyl (MB46513) in milk were mainly associated with the milk fat. Milk fat isolated from the day 35 milk sample from the 1.03 ppm dose level accounted for > 88% of the total amount of fipronil-desulfinyl (MB46513) in whole milk while on a residue level basis milk fat levels (mean 0.97 mg/kg) were 15.9× higher than in whole milk (mean 0.061 mg/kg).

Mean residues of fipronil-desulfinyl (MB46513) in tissues were highest in fat at 0.040, 0.102, 0.363 and 0.973 mg/kg for the 0.025, 0.076, 0.3 and 1.03 ppm dose levels, respectively, with corresponding levels for liver of 0.036, 0.087, 0.273 and 0.563 mg/kg; kidney 0.005, 0.012, 0.038 and 0.102 mg/kg; and muscle 0.0023, 0.0037, 0.015 and 0.033 mg/kg.

A laying hens transfer study for fipronil was made available to the Meeting. Hens were dosed orally daily for 42 days at the equivalent of 0.01, 0.031 and 0.103 ppm in the diet with sacrifice 3 hours after the last dose. Samples were analysed for fipronil, fipronil-sulfide (MB45950) and fipronil-sulfone (MB46136) residues. The only compound detected above LOQ (0.01 mg/kg) in eggs and tissues was fipronil-sulfone (MB46136). Residues in eggs appeared to reach a plateau after 3–4 weeks of dosing with mean levels (day 25–41) of 0.013, 0.036 and 0.116 mg/kg for fipronil-sulfone (MB46136) for the 0.01, 0.031 and 0.103 ppm dose levels, respectively. Levels in skin with fat were 0.013, 0.054 and 0.191 mg/kg for the same dose levels, in liver < 0.01, 0.020 and 0.069 mg/kg and in muscle < 0.01, < 0.01 and 0.012 mg/kg.

In a separate study, residues in egg white and egg yolk were investigated following dosing of laying hens with fipronil-sulfone (MB46136) at 0.05 ppm in the diet for 24 days and 2 ppm in the diet for 31 days. The decline on ceasing dosing suggests a DT_{50} of the order 12 days. For the 2 ppm dose group,

levels of fipronil-sulfone (MB46136) attained 82% of the predicted plateau levels by the last day of dosing (by day 31). Only low levels of fipronil-sulfone (MB46136) were detected in eggs from the 0.05 ppm dose group (0.08–0.11 mg/kg in yolks and 0.01–0.04 mg/kg in egg whites for days 20–30). For the 2 ppm dose group levels in egg yolks for days 20–32 were 2.24–5.05 mg/kg and in egg whites 0.10–0.14 mg/kg.

The interval between last dosing and sacrifice in the lactating cow and laying hen transfer studies were shorter than commercial practice, typically 24 hours for cattle and 8 hours for hens. However, as the DT_{50} for residues in lactating cows and laying hens is greater than 8 days, the impact of the sacrifice time on predicted residues will be small and the Meeting agreed the studies could be used for estimation of maximum residue levels.

Farm animal dietary burden

Those commodities and their input values used in estimating livestock dietary burdens are listed in the Recommendations Tables.

Dietary burdens were calculated for beef cattle, dairy cattle, broilers and laying poultry based on feed items evaluated by the JMPR by the current Meeting. The dietary burdens, estimated using the most recent version of the OECD livestock dietary burden calculator, are presented in Annex 6 and summarised below.

Separate feeding studies were carried out in cattle with fipronil and the photodegradation product fipronil-desulfinyl to determine residues. For laying hens, a feeding study was carried out with fipronil, and a metabolism study was available for fipronil-desulfinyl. Detectable amounts of fipronil-desulfinyl were found in some feed commodities with highest residues in rice straw. The livestock dietary burden was therefore calculated separately for:

- the sum of fipronil and fipronil-sulfone (MB46136);
- the sum of residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) in feed; and
- fipronil-desulfinyl (MB46513) in feed.

Residues in animal commodities were estimated for the sum of fipronil and fipronil-sulfone (MB46136) for use in estimating maximum residue levels.

Residues in animal commodities for dietary risk assessment were estimated for fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950) and separately for fipronil-desulfinyl (MB46513) and the results added to provide an estimate of total residues (fipronil, fipronil-sulfone, fipronil-sulfide and separately for fipronil-desulfinyl).

Foliar use of fipronil on rice is not approved in Australia. As Australia does not allow the importation of fodders due to biosecurity concerns, rice straw was excluded from the dietary burden calculation for Australia.

Table 4 Estimated maximum and mean dietary burdens of farm animals (fipronil+fipronil-sulfone (MB46136))

	Animal dietary burden: fipronil+MB46136, ppm of dry matter diet							
	US-Canada		EU		Australia		Japan	
	max	mean	max	mean	max	mean	max	mean
Beef cattle	0.0619	0.0229	0.1151	0.0374	0.0672	0.0259	0.2126 ①	0.0633
Dairy cattle	0.0353	0.0150	0.1040	0.0313	0.0637	0.0213	0.1085 ②	0.0377
Broilers	0.0081	0.0081	0.0234	0.0116	0.0081	0.0081	0.0083	0.0083
Layers	0.0081	0.0081	0.0280 ③	0.0118	0.0081	0.0081	0.0073	0.0073

- ❶ Highest maximum beef or dairy cattle dietary burden suitable for MRL estimates for mammalian tissues
- ❷ Highest maximum dairy cattle dietary burden suitable for MRL estimates for mammalian milk
- ❸ Highest maximum poultry dietary burden suitable for MRL estimates for poultry tissues and eggs.

Table 5 Estimated maximum and mean dietary burdens of farm animals (sum of fipronil+fipronil-sulfone (MB46136)+fipronil-sulfide (MB45950))

	Animal dietary burden: fipronil+MB46136+MB45950, ppm of dry matter diet							
	US-Canada		EU		Australia		Japan	
	max	mean	max	mean	max	mean	max	mean
Beef cattle	0.0657	0.0285	0.1276	0.0463	0.0579	0.0370	0.2226❶	0.0740❸
Dairy cattle	0.0323	0.0166	0.1186	0.0394	0.0654	0.0299	0.1120❷	0.0437❹
Broilers	0.0108	0.0108	0.0266	0.0147	0.0099	0.0099	0.0103	0.0103
Layers	0.0108	0.0108	0.0338❺	0.0153❻	0.0090	0.0090	0.0092	0.0092

- ❶ Highest maximum beef or dairy cattle dietary burden suitable for HR estimates for mammalian tissues
- ❷ Highest maximum dairy cattle dietary burden suitable for HR estimates for mammalian milk
- ❸ Highest mean beef or dairy cattle dietary burden suitable for STMR estimates for mammalian tissues.
- ❹ Highest mean dairy cattle dietary burden suitable for STMR estimates for milk.
- ❺ Highest maximum poultry dietary burden suitable for HR estimates for poultry tissues and eggs.
- ❻ Highest mean poultry dietary burden suitable for STMR estimates for poultry tissues and eggs.

Table 6 Estimated maximum and mean dietary burdens of farm animals (fipronil-desulfinyl MB46513)

	Animal dietary burden: fipronil-desulfinyl, ppm of dry matter diet							
	US-Canada		EU		Australia		Japan	
	max	mean	max	mean	max	mean	max	mean
Beef cattle	0.0254	0.0071	0.0921	0.0184	0.0531	0.0140	0.1955❶	0.0390❸
Dairy cattle	0.0489	0.0111	0.0772	0.0143	0.0800	0.0148	0.0978❷	0.0213❹
Broilers	0.0035	0.0035	0.0088	0.0043	0.0038	0.0038	0.0035	0.0035
Layers	0.0035	0.0035	0.0195❺	0.0047❻	0.0038	0.0038	0.0042	0.0042

- ❶ Highest maximum beef or dairy cattle dietary burden suitable for HR estimates for mammalian tissues
- ❷ Highest maximum dairy cattle dietary burden suitable for HR estimates for mammalian milk
- ❸ Highest mean beef or dairy cattle dietary burden suitable for STMR estimates for mammalian tissues.
- ❹ Highest mean dairy cattle dietary burden suitable for STMR estimates for milk.
- ❺ Highest maximum poultry dietary burden suitable for HR estimates for poultry tissues and eggs.
- ❻ Highest mean poultry dietary burden suitable for STMR estimates for poultry tissues and eggs.

Animal commodity maximum residue levels

Cattle

The calculations used to estimate highest total residues for use in estimating maximum residue levels, STMR and HR values are shown below.

Table 7 Residues of fipronil and fipronil-sulfone (MB46136)

	Feed Level (ppm) for milk residues	Fipronil+MB46136 residues (mg eq/kg) in milk	Feed Level (ppm) for tissue residues	Fipronil+MB46136 residues (mg eq/kg)			
				Muscle	Liver	Kidney	Fat
Highest residue for maximum residue level estimation (beef or dairy cattle)							
Feeding Study	0.13	0.02736	0.13 0.43	0.02 0.037	0.06885 0.16435	0.024 0.044	0.22030 0.56872
Dietary burden and estimate of highest residue	0.1085	0.02284	0.2126	0.02468	0.09514	0.02951	0.31623

Table 8 Residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950)

	Feed Level (ppm) for milk residues	Total residues (mg eq/kg) in milk	Feed Level (ppm) for tissue residues	Fipronil+MB46136+MB45950 residues (mg eq/kg)			
				Muscle	Liver	Kidney	Fat
HR Determination (beef or dairy cattle)							
Feeding Study	0.13	0.03736	0.13 0.43	0.03 0.047	0.07885 0.17435	0.034 0.054	0.24033 0.58872
Dietary burden and estimate of highest residue	0.1120	0.03219	0.2226	0.03525	0.10833	0.04017	0.34787
STMR Determination (beef or dairy cattle)							
Feeding Study	0.04	0.00554	0.04 0.13 0.43	0.00457 0.047	0.01644 0.06695	0.00496 0.034	0.06460 0.19014
Dietary burden and estimate of highest residue	0.0437	0.00605	0.0740	0.00827	0.03552	0.01593	0.11203

Table 9 Residues of fipronil-desulfinyl (MB46513)

	Feed Level (ppm) for milk residues	Total residues (mg/kg) in milk	Feed Level (ppm) for tissue residues	MB46513 (mg/kg)			
				Muscle	Liver	Kidney	Fat
HR Determination (beef or dairy cattle)							
Feeding Study	0.076 0.3	0.008 0.02658	0.076 0.3	0.005 0.019	0.098 0.28	0.013 0.044	0.12 0.41
Dietary burden and estimate of highest residue	0.0978	0.00981	0.1955	0.01247	0.19509	0.02954	0.27471
STMR Determination (beef or dairy cattle)							
Feeding Study	0.025	0.00308	0.025 0.076	0.00233 0.00367	0.0357 0.087	0.005 0.0117	0.0403 0.1017
Dietary burden and estimate of highest residue	0.0213	0.00263	0.0390	0.002700	0.04978	0.00684	0.05716

The Meeting estimated the following maximum residue levels: milk 0.03 mg/kg; meat (mammalian except marine mammals) 0.03 mg/kg, mammalian fat (except milk fat) 0.4 mg/kg and edible offal 0.1 mg/kg. The Meeting estimated the following HRs: mammalian meat 0.04926 mg/kg; mammalian fat 0.65651 mg/kg; mammalian offal (liver 0.32752 mg/kg) and milk 0.04321 mg/kg and STMRs: mammalian meat 0.00850 mg/kg; mammalian fat 0.17625 mg/kg; mammalian offal (liver 0.09145 mg/kg) and milk 0.00845 mg/kg.

Residues of the sum of fipronil and fipronil-sulfone were 14× higher in milk fat compared to whole milk. Residues of the sum of fipronil, fipronil-sulfone and fipronil-sulfide were 13× higher in milk fat compared to whole milk. Residues of fipronil-desulfinyl were 16× higher in milk fat compared to whole milk. Using this information, the Meeting estimated a maximum residue level of 0.3 mg/kg for milk fat ($0.02284 \text{ mg/kg} \times 14 = 0.32 \text{ mg/kg}$) and HR and STMR values of 0.59 ($0.03219 \text{ mg/kg} \times 13 + 0.00981 \times 16 \times 1.1$) and 0.12 ($0.00605 \text{ mg/kg} \times 13 + 0.00263 \times 16 \times 1.1$) mg/kg, respectively.

The Meeting withdrew its previous recommendations for cattle commodities.

Poultry

Table 10 Residues of fipronil and fipronil-sulfone (MB46136)

	Feed Level (ppm) for egg residues	Fipronil+MB46136 residues (mg eq/kg) in egg	Feed Level (ppm) for tissue residues	Fipronil+MB46136 residues (mg eq/kg)			
				Muscle	Liver	Kidney	Fat
Highest residue for maximum residue level estimation (layers or broilers)							
Feeding Study	0.01	0.02254	0.01	0.02351	0.01965		0.02351
	0.031	0.04473	0.031		0.02929		0.06788
	0.103						
Dietary burden and estimate of highest residue	0.0280	0.04156	0.0280	0.00639	0.02792		0.06154

Table 11 Residues of fipronil, fipronil-sulfone (MB46136) and fipronil-sulfide (MB45950)

	Feed Level (ppm) for egg residues	Total residues (mg eq/kg) in egg	Feed Level (ppm) for tissue residues	Total residues (mg eq/kg)			
				Muscle	Liver	Kidney	Fat
HR Determination (layers or broilers)							
Feeding Study	0.031	0.05473	0.031		0.03929		0.07788
	0.103	0.13190	0.103	0.03351	0.08849		0.22066
Dietary burden and estimate of highest residue	0.0338	0.05773	0.0338	0.01100	0.04121		0.08343
STMR Determination (layers or broilers)							
Feeding Study	0.01	0.03061	0.01		0.02965		0.03286
	0.031	0.04766	0.031		0.03897		0.07242
			0.103	0.03158			
Dietary burden and estimate of highest residue	0.0153	0.03491	0.0153	0.00469	0.032		0.04285

Table 12 Residues of fipronil-desulfinyl (MB46513)

	Feed Level (ppm) for egg residues	Total residues (mg eq/kg) in egg	Feed Level (ppm) for tissue residues	MB46513 (mg equiv/kg)			
				Muscle	Liver	Kidney	Fat
HR Determination (layers or broilers)							
Feeding Study	10A	1.886	10A	0.354	0.564		8.8018
Dietary burden and estimate of highest residue	0.0195	0.00368	0.0195	0.00069	0.0011		0.01716
STMR Determination (layers or broilers)							
Feeding Study	10A	1.886	10A	0.354	0.564		8.8018
Dietary burden and estimate of highest residue	0.0047	0.00089	0.0047	0.00017	0.00027		0.00414

^A laying hen metabolism study

The Meeting estimated the following maximum residue levels for poultry commodities: eggs 0.04 mg/kg; poultry meat 0.007 mg/kg, poultry fat 0.07 mg/kg and poultry edible offal 0.03 mg/kg to replace its previous recommendations. The Meeting estimated the following HRs: poultry meat 0.01169 mg/kg; poultry fat 0.10060 mg/kg; poultry edible offal (liver 0.04231 mg/kg) and eggs 0.06141 mg/kg and STMRs: poultry meat 0.00486 mg/kg; poultry fat 0.04698 mg/kg; poultry edible offal (liver 0.03227 mg/kg) and eggs 0.03580 mg/kg.

RECOMMENDATIONS

On the basis of the data obtained from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI and IESTI assessments.

Definition of the residue for compliance with the MRL for plant and animal commodities: sum of *fipronil* and *5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfonylpyrazole* (MB46136) expressed in terms of *fipronil*.

Definition of the residue for dietary risk assessment for plant and animal commodities: sum of *fipronil* and *5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfonylpyrazole* (MB46136), *5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylthiopyrazole* (MB45950) and *5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylpyrazole* (MB46513) expressed in terms of *fipronil*.

The residue is fat-soluble.

DIETARY RISK ASSESSMENT

Long-term dietary exposure

The ADI for fipronil is 0–0.0002 mg/kg bw. The International Estimated Daily Intakes (IEDIs) for fipronil were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs ranged from 30–170% of the maximum ADI.

Shorter than lifetime and high consumer long-term dietary exposure

The Global Estimates of Chronic Dietary Exposure (GECDEs) for fipronil were estimated for male and female, adults and children.

The long elimination half-lives of fipronil and fipronil sulfone (the major chemical form in the body) means that intermittent and repeated exposures to fipronil dietary residues over any duration should not exceed the maximum ADI. The mean dietary exposure for each population subgroup all exceeded the maximum ADI. Shorter-than-lifetime estimated dietary exposures for each age group considered also exceeded the maximum ADI for fipronil, with the infants and toddlers having the highest range from 159–1014% of the maximum ADI. For this age group milks including infant formula and other dairy products were the main food contributor, but for other age groups a range of foods were major contributors to the high exposures (> 50%), for example milks and dairy products, pig fat, beef, beef offal, yeast leavened pastries, basil and semolina.

The Meeting concluded that results using national food consumption data confirmed the finding from the IEDI calculation that long-term dietary exposure to residues of fipronil from uses considered by the Meeting may present a public health concern.

Acute dietary exposure

The ARfD for fipronil is 0.03 mg/kg bw. The International Estimate of Short-Term Intakes (IESTIs) for fipronil were calculated for the food commodities and their processed commodities for which HRs/HR-Ps or STMRs/STMR-Ps were estimated by the present Meeting and for which consumption data were available. The results are shown in Annex 4 of the 2021 JMPR Report.

The IESTIs varied from 0–20% of the ARfD for children and 0–10% of the ARfD for the general population. The Meeting concluded that acute dietary exposure to residues of fipronil from uses considered by the present Meeting is unlikely to present a public health concern.

Threshold of toxicological concern (TTC) consideration for metabolites

The Meeting agreed metabolites MB45897, MB46400, fipronil-carboxamide (RPA105048), fipronil-sulfone-amide (RPA105320), RPA108058 and fipronil-acid (RPA200761) could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

The current Meeting estimated dietary exposures for metabolite MB45897 of 0.0176 µg/kg bw per day, MB46400 of 0.0000067 µg/kg bw per day, fipronil-carboxamide (RPA105048) of 0.0199 µg/kg bw per day, fipronil-sulfone-amide (RPA105320) of 0.0252 µg/kg bw per day, RPA108058 of 0.0196 µg/kg bw per day and fipronil-acid (RPA200761) of 0.0642 µg/kg bw per day.

The Meeting concluded that the estimated dietary exposure to residues of MB45897, MB46400, fipronil-carboxamide (RPA105048), fipronil-sulfone-amide (RPA105320), RPA108058 and fipronil-acid (RPA200761) from uses considered by the JMPR is below the TTC for Cramer Class III compounds and is unlikely to present a public health concern. Should further uses be considered in the future, these conclusions may need to be re-evaluated.

5.5 Flutianil (319)

TOXICOLOGY

Flutianil is the ISO-approved common name for [(Z)-2-[2-fluoro-5-(trifluoromethyl)phenylthio]-2-[3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile (IUPAC), with the Chemical Abstracts Service number 958647-10-4.

Flutianil (OK-5203) is a novel fungicide belongs to the thiazolidine chemical class exhibiting both fungicidal and fungistatic contact action. Flutianil is a systemic fungicide for use in controlling *Rhizoctonia solani* (black scurf) and some other basidiomycetes in rice, turf, potato, vegetables and peanuts. Its proposed mode of action (MOA) is the inhibition of haustorium formation in the fungus, thus inhibiting sporulation and spore isolation and preventing disease expansion. This MOA is different from existing fungicides. Flutianil is classified by the Fungicide Resistance Action Committee (FRAC) as having an unknown MOA, placing it in group U13. Flutianil has not previously been evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) and was reviewed by the present Meeting at the request of the Codex Committee on Pesticide Residues (CCPR).

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with current test guidelines, unless otherwise specified. No additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Biochemical aspects

The absorption, distribution and excretion of flutianil was investigated in rats following the administration of flutianil uniformly labelled in either the methoxyphenyl ring or the fluorotolyl ring, as a single oral low dose (10 mg/kg bw), a single oral high dose (1000 mg/kg bw) or 14 unlabelled daily oral low doses followed by a radioactive low dose. Based on the sum of the radioactivity in urine, bile, cage wash plus debris and the residual carcass, approximately 20% of the oral dose was absorbed at 10 mg/kg bw [¹⁴C]-flutianil. There was no significant sex difference following administration of either radiolabel. Urinary excretion of radioactivity from methoxyphenyl-labelled flutianil was slightly lower (ca. 10%) following repeat dosing than from a single dose (ca. 20%), but this might have been due to variation across studies. At 1000 mg/kg bw, based on urinary excretion data (bile not measured), absorption was approximately 1%.

At the low dose (10 mg/kg bw), maximum plasma concentrations were achieved within 13 hours following administration of fluorotolyl-labelled flutianil and within three hours following administration of methoxyphenyl-labelled flutianil. At the high dose level, concentrations were generally below the limit of detection. In contrast to plasma, maximum whole blood concentrations were achieved within eight hours following administration of fluorotolyl-labelled flutianil and within five hours following administration of methoxyphenyl-labelled flutianil at the low dose level. Peak blood levels of radioactivity were approximately four-fold higher than peak plasma levels. At the high dose, maximum blood concentrations were achieved within four hours following administration of fluorotolyl-labelled flutianil but could not be determined following administration of methoxyphenyl-labelled flutianil. There were no significant sex differences in plasma or blood kinetics of radioactivity following administration of either radiolabelled form.

Flutianil was widely distributed throughout the tissues following oral administration and was rapidly eliminated at 10 mg/kg bw. Generally, radioactivity seemed to be associated with the organs of metabolism and excretion, and fatty tissues. Liver, after fat, was the tissue containing the highest level of radioactivity at all time points. Tissue residues were not proportional to dose, and with the exception of residues in the gastrointestinal tract, an increase in dose levels from 10 to 1000 mg/kg bw resulted in an increase in tissue residues of much less than 100-fold. Thus, absorption was saturated at the high dose level. At comparable times after dosing (24 and 48 hours) tissue concentrations of radioactivity

were generally greater in rats that received fluorotolyl-labelled flutianil than in rats given methoxyphenyl-labelled flutianil. At the low dose, clearance of radioactivity from the tissues was relatively rapid although slightly greater following administration of methoxyphenyl-labelled flutianil. At the high dose many tissues and blood exhibited a relatively slow clearance, with significant levels remaining after 48 hours.

Repeated administration was less extensively investigated than single doses, but there was no evidence it resulted in a major effect on the absorption, distribution, or excretion of flutianil.

Excretion was extensive; over 90% in 48 hours. Faecal elimination was the major route of excretion and accounted for 70–95% of the administered dose (AD), varying with label position and dose level.

Samples of urine, faeces and bile from the studies described above were analysed for metabolites. In urine, no parent flutianil was detected in any group dosed, irrespective of cannulation status or label position. The major peak (called either Met 6 or M11, which are structural isomers) from the urine of rats dosed with fluorotolyl-labelled flutianil was identified as a mercapturate conjugate of a hydroxylated methylsulphoxy trifluoromethyl ring structure, and this accounted for up to 5.5% of AD following a single dose, and up to 11% of AD following repeated low dose administration. Several other minor metabolites were identified which resulted from cleavage of the parent compound between the methoxyphenyl and fluorotolyl ring structures. The major component in the faeces was the parent compound (94–98% of the faecal elimination) and two minor, tentative metabolites, OC 53429 and OC 53982, resulting from sequential defluorination of the parent compound, without cleavage of the two ring structures. The majority of metabolites remained unidentified. The metabolic profile of flutianil was independent of sex, single or repeated administration and of dose level.

Toxicological data

The acute oral median lethal dose (LD₅₀) of flutianil was >5000 mg/kg bw and the dermal LD₅₀ was >5000 mg/kg bw. The inhalation median lethal concentration (LC₅₀) of flutianil was >5.17 mg/L. Flutianil was not irritating to skin but transiently irritating to eyes in rabbits and was not skin sensitizing in the guinea pig maximization test.

In repeated-dose toxicity studies on mice and rats, the main effects were adaptive changes in the liver. In male rats, hyaline droplets in the proximal tubular cells increased by treatment were positive for α₂u-globulin and were therefore concluded not to be relevant to humans. In dog studies, no treatment-related effects were observed.

In a 90-day dietary toxicity study in mice, flutianil was administered at dietary concentrations of 0, 1000, 3000 or 10 000 ppm (equal to 0, 138, 409 and 1387 mg/kg bw per day for males, 0, 159, 481 and 1555 mg/kg bw per day for females). The NOAEL was 10 000 ppm (equal to 1387 mg/kg bw per day), the highest dose tested.

In a 90-day dietary toxicity study in rats, flutianil was administered at dietary concentrations of 0, 20, 200, 2000 or 20 000 ppm (equal to 0, 1.22, 12.5, 122 and 1271 mg/kg bw per day for males, 0, 1.46, 14.3, 149 and 1500 mg/kg bw per day for females). The NOAEL was 20 000 ppm (equal to 1271 mg/kg bw per day), the highest dose tested.

In a 90-day oral toxicity study flutianil was administered to beagle dogs by capsule at dose levels of 0, 30, 300 and 1000 mg/kg bw per day. The NOAEL was 1000 mg/kg bw per day, the highest tested dose.

In a one-year oral toxicity study flutianil was administered to dogs via capsule at dose levels of 0, 30, 300 or 1000 mg/kg bw per day. The NOAEL was 1000 mg/kg bw per day, the highest dose tested.

In a 78-week carcinogenicity study in mice flutianil was administered in the diet at concentrations of 0, 1000, 3000 or 10 000 ppm (equal to 0, 106, 321 and 1084 mg/kg bw per day for males, 0, 105, 316 and 1063 mg/kg bw per day for females). The NOAEL for toxicity was 10 000 ppm (equal to

1063 mg/kg bw per day), the highest dose tested. The NOAEL for carcinogenicity was 10 000 ppm (equal to 1063 mg/kg bw per day), the highest dose tested.

In a 104-week combined chronic toxicity and carcinogenicity study in rats, flutianil was administered in the diet to males at concentrations of 0, 60, 600, 2000 or 6000 ppm (equal to 0, 2.45, 35.2, 81.9, and 249 mg/kg bw per day), and at 0, 60, 2000, 6000 and 20 000 ppm (equal to 0, 3.15, 111, 334, and 1130 mg/kg bw per day) to females. The NOAEL for toxicity was 2000 ppm (equal to 81.9 mg/kg bw per day), based upon a slight increase in liver foci of cellular alteration in males (a similar increase was seen at 52 weeks in both eosinophilic and basophilic foci) at 6000 ppm (equal to 249 mg/kg bw per day). The NOAEL for carcinogenicity was 2000 ppm (equal to 81.9 mg/kg bw per day), based upon an equivocally increased incidence of pancreatic islet cell adenomas at 6000 ppm in males (equal to 249 mg/kg bw per day).

The Meeting concluded that flutianil was carcinogenic in male rats but not in mice or female rats.

Flutianil was tested for genotoxicity in an adequate range of in vitro and in vivo assays. No evidence of genotoxicity was found.

The Meeting concluded that flutianil is unlikely to be genotoxic.

In view of the lack of genotoxicity, the absence of carcinogenicity in mice and female rats and the fact that only benign pancreatic tumours were observed at the highest dose tested in male rats, the Meeting concluded that flutianil is unlikely to pose a carcinogenic risk to humans from the diet.

In a two-generation reproductive toxicity study in rats, flutianil was administered at dietary concentrations of 0, 20, 200, 2000 or 20 000 ppm (equal to 0, 9.3, 91 and 963 mg/kg bw per day in males, 0, 16.6, 171 and 1754 mg/kg bw per day in females). The NOAEL for parental effects was 2000 ppm (equal to 171 mg/kg bw per day), based upon increased absolute and relative thyroid weight in females at 20 000 ppm. The reproductive NOAEL was 2000 ppm (equal to 91 mg/kg bw per day in males, 171 mg/kg bw per day in females), based upon a slightly reduced number of implantation sites and the number of pups delivered by the F₁ parental generation at 20 000 ppm. The offspring NOAEL was 20 000 ppm (equal to 963 mg/kg bw per day), the highest dose tested.

In a developmental toxicity study in rats, flutianil was administered by gavage at dose levels of 0, 100, 333 and 1000 mg/kg bw per day. The maternal and fetal NOAEL was 1000 mg/kg bw per day, the highest dose tested.

In a developmental toxicity study in rabbits, flutianil was administered by gavage at dose levels of 0, 100, 300 and 1000 mg/kg bw per day. The maternal and fetal NOAEL was 1000 mg/kg bw per day, the highest dose tested.

The Meeting concluded that flutianil is not teratogenic.

No evidence of neurotoxicity was reported in routine toxicological studies with flutianil.

The Meeting concluded that flutianil is unlikely to be neurotoxic.

In an immunotoxicity study (functional assessment of the humoral T-lymphocyte-dependent antigenic immune response, TDAR) in male rats, flutianil was administered at dietary concentrations of 0, 3250, 6500 or 13 000 ppm (equal to 0, 304, 616 and 1251 mg/kg bw per day) for 28 days. The NOAEL for systemic toxicity was 13 000 ppm (equal to 1251 mg/kg bw per day), the highest dose tested. No specific indications of immunotoxicity were noted, therefore the NOAEL for immunotoxicity was 13 000 ppm (equal to 1251 mg/kg bw per day), the highest dose tested.

The Meeting concluded that flutianil is not immunotoxic.

Toxicological data on metabolites and/or degradates

Metabolites OC 56635, OC 53276, OC 53279, OC 56574, OC 56634, Bz5 and Bz6 were identified in residues of plant and animal origin, none of them were identified, or identified at relevant amounts, in rat metabolism studies.

For metabolites OC 53279, OC 56574, OC 56634, Bz5 and Bz6, identified in plant and animal commodities, QSAR assessments for negative genotoxicity predictions were provided. Genotoxicity studies (in vitro and in vivo) were conducted with metabolites OC 56635 and OC 53276 and were negative.

An acute toxicity study in the rat with OC 63421, the sodium salt of OC 56635, revealed an LD₅₀ of >5000 mg/kg bw: an equivalent study with OC 56635 identified an LD₅₀ of between 300 and 2000 mg/kg bw. A 28-day rat toxicity study with OC 63421, the sodium salt of OC 56635, revealed a NOAEL of 16 000 ppm (equal to 1380 mg/kg bw per day), based upon increased food consumption and reduced food conversion efficiency in both sexes, as well as slight effects on some haematological parameters observed at 50 000 ppm (equal to 4740 mg/kg bw per day).

As regards the metabolite OC 63421 (sodium salt of OC 56635), the Meeting concluded that the ADI of its parent flutianil is sufficiently protective, since the NOAEL in the 28-day rat study with OC 63421 was >1000 mg/kg bw per day. This also applies to the acid form of OC 63421, the metabolite OC 56635.

For metabolites OC 53276, OC 53279 and OC 56574, the Meeting concluded that the ADI for flutianil is considered sufficiently protective, based upon their close structural similarity to flutianil and absence of functional groups likely to confer higher biological activity.

Based on the available data, the Meeting concluded that the metabolites, OC 56634, Bz5 and Bz6 could be evaluated using the threshold of toxicological concern (TTC) approach based on Cramer class III, with a threshold of 1.5 µg/kg bw per day.

Microbiological data

Flutianil has a novel mode of action against powdery mildew, which is classified under the FRAC code U13. No information is available on mechanism and type of any antimicrobial action on the microbiome of the human gastrointestinal tract, antimicrobial spectrum of activity or antimicrobial resistance mechanisms and genetics.

Human data

No information was provided on the health of workers involved in the manufacture or use of flutianil. No information on accidental or intentional poisoning in humans was available.

The Meeting concluded that the existing database on flutianil was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting established an ADI of 0–0.8 mg/kg bw, for flutianil based on the NOAEL of 82 mg/kg bw per day in the two-year chronic toxicity and carcinogenicity study in rats, and using a safety factor of 100. The margin between the upper bound ADI and the LOAEL for an equivocal increase in pancreatic tumours in male rats was 300.

The Meeting concluded that it was not necessary to establish an ARfD for flutianil in view of its low acute oral toxicity and the absence of developmental toxicity likely to be elicited by a single dose and any other toxicological effects that would be likely to be elicited by a single dose.

A toxicological monograph was prepared.

Levels relevant to risk assessment of flutianil

Species	Study	Effect	NOAEL	LOAEL
Mouse	90-day study of toxicity ^a	Toxicity	10 000 ppm, equal to 1387 mg/kg bw per day ^c	-
	Two-year study of toxicity and carcinogenicity ^a	Toxicity	10 000 ppm, equal to 1063 mg/kg bw per day ^c	-
		Carcinogenicity	10 000 ppm, equal to 1063 mg/kg bw per day ^c	-
Rat	90-day study of toxicity ^a	Toxicity	20 000 ppm, equal to 1271 mg/kg bw per day ^c	-
	Two-year study of toxicity and carcinogenicity ^a	Toxicity	2000 ppm, equal to 81.9 mg/kg bw per day	6000 ppm, equal to 249 mg/kg bw per day
		Carcinogenicity	2000 ppm, equal to 81.9 mg/kg bw per day	6000 ppm, equal to 249 mg/kg bw per day
	Two-generation study of reproductive toxicity ^a	Reproductive toxicity	2000 ppm, equal to 91 mg/kg bw per day	20 000 ppm, equal to 963 mg/kg bw per day
		Parental toxicity	2000 ppm, equal to 171 mg/kg bw per day	20 000 ppm, equal to 963 mg/kg bw per day
		Offspring toxicity	20 000 ppm, equal to 963 mg/kg bw per day ^c	-
	Developmental toxicity study ^b	Maternal toxicity	1000 mg/kg bw per day ^c	-
		Embryo and fetal toxicity	1000 mg/kg bw per day ^c	-
Rabbit	Developmental toxicity study ^b	Maternal toxicity	1000 mg/kg bw per day ^c	-
		Embryo and fetal toxicity	1000 mg/kg bw per day ^c	-
Dog	28-day, thirteen-week and one-year studies of toxicity ^{d,e}	Toxicity	1000 mg/kg bw per day ^c	-

Metabolite OC 63421 (sodium 2-fluoro-5-(trifluoromethyl)benzenesulfonate)

Rat	Four-week study of toxicity ^a	Toxicity	16 000 ppm, equal to 1380 mg/kg bw per day	50 000 ppm, equal to 4740 mg/kg bw per day
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^a Dietary administration^b Gavage administration^c Highest dose tested^d Two or more studies combined^e Capsule administration

Acceptable daily intake (ADI) applies to flutianil, OC 56635 / OC 63421, OC 53276, OC 53279 and OC 56574, expressed as flutianil

0–0.8 mg/kg bw

Acute reference dose (ARfD)

Not necessary

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Critical end-points for setting guidance values for exposure to flutianil**Absorption, distribution, excretion and metabolism in mammals**

Rate and extent of oral absorption	Moderately rapid ($T_{\max}$ 3–8 h). Poorly absorbed <20% (based on bile and urine at 10 mg/kg bw)
Dermal absorption (in vitro)	8% for a 50 g/L formulation; 23% for a 0.02 g/L dilution
Distribution	Widely distributed
Potential for accumulation	Low at 10 mg/kg bw
Rate and extent of excretion	Extensively excreted, >90% in 48 h at 10 mg/kg bw, in bile, urine and predominantly faeces (80%)
Metabolism in animals	Limited (10% of the administered dose) Cleavage at sulfur bridge, oxidation and conjugation
Toxicologically significant compounds in animals and plants	Flutianil and OC 56635, OC 63421, OC 53276, OC 53279 and OC 56574

Acute toxicity

Rat, LD ₅₀ , oral	>5000 mg/kg bw
Rat, LD ₅₀ , dermal	>5000 mg/kg bw
Rat, LC ₅₀ , inhalation	>5.17 mg/L
Rabbit, dermal irritation	Not irritating
Rabbit, ocular irritation	Transiently irritating
Guinea pig, dermal sensitization	Not sensitizing (Magnussen & Kligman)
Mouse, dermal sensitization	No data

Short-term studies of toxicity

Target/critical effect	Thyroid hypertrophy, inhalation route
Lowest relevant oral NOAEL	1000 mg/kg bw per day, highest dose tested (dog)
Lowest relevant dermal NOAEL	1000 mg/kg bw per day, highest dose tested (rat)
Lowest relevant inhalation NOAEC	0.1 mg/L (rat)

Long-term studies of toxicity and carcinogenicity

Target/critical effect	Liver foci and pancreatic adenoma (male rat)
Lowest relevant NOAEL	81.9 mg/kg bw per day (rat)
Carcinogenicity	Carcinogenic in male rats ^a

Genotoxicity

Unlikely to be genotoxic

Reproductive toxicity

Target/critical effect	Increased thyroid weight in females; reduced implantation rates and pup numbers
Lowest relevant parental NOAEL	171 mg/kg bw per day (rat)
Lowest relevant offspring NOAEL	963 mg/kg bw per day, highest dose tested (rat)

Lowest relevant reproductive NOAEL	91 mg/kg bw per day (rat)
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Developmental toxicity

Target/critical effect	None
Lowest relevant maternal NOAEL	1000 mg/kg bw per day, highest dose tested (rat/rabbit)
Lowest relevant embryo/fetal NOAEL	1000 mg/kg bw per day, highest dose tested (rat/rabbit)

Neurotoxicity

Acute neurotoxicity NOAEL	No study
Subchronic neurotoxicity NOAEL	No study
Developmental neurotoxicity NOAEL	No study

Immunotoxicity

1251 mg/kg bw per day, highest dose tested (rat)

Studies on toxicologically relevant metabolites

OC 56635	Oral LD ₅₀ 300–2000 mg/kg bw (rat) Unlikely to be genotoxic (negative in vitro & in vivo)
OC 63421	Oral LD ₅₀ > 2000 mg/kg bw (rat) Unlikely to be genotoxic (salt of OC 56635) 28-day NOAEL 1380 mg/kg bw per day, reduced body weight and haematology changes
OC 56634	Unlikely to be genotoxic (QSAR)
OC 53276	Unlikely to be genotoxic (negative in vitro and in vivo)
OC 53279	Unlikely to be genotoxic (QSAR)
OC 56574	Unlikely to be genotoxic (QSAR)
Bz5	Unlikely to be genotoxic (QSAR)
Bz6	Unlikely to be genotoxic (QSAR)
	No data submitted

Microbiological data

No data submitted.

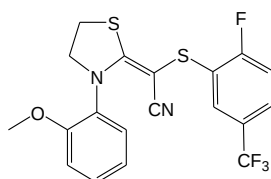
Human data^a Unlikely to pose a carcinogenic risk to humans via exposure from the diet**Summary**

	Value	Study	Safety factor
ADI	0–0.8 mg/kg bw ^a	Two-year study of toxicity and carcinogenicity (rat)	100
ARfD	Unnecessary		

^a Applies to flutianil, OC 56635/OC 63421, OC 53276, OC 53279 and OC 56574, expressed as flutianil**RESIDUE AND ANALYTICAL ASPECTS**

Flutianil, (2Z)-2-[2-fluoro-5-(trifluoromethyl)phenyl]sufanyl-2-[3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile (IUPAC name), is a new systemic thiazolidine fungicide used for control of powdery

mildew on a broad range of crops. Its fungicidal mode of action is inhibition of haustorium formation and inhibition of spore formation/isolation.



Flutianil

Flutianil has been registered in a number of countries for use on various crops, mostly tree fruits and fruiting vegetables. Flutianil was scheduled at the Fifty-first Session of the CCPR in 2019 for toxicological and residue evaluation by the 2020 JMPR as a new compound, which was postponed to the 2021 JMPR. No specification has been established for flutianil by the Joint FAO/WHO Meeting on Pesticide Specifications.

The Meeting received information on identity, chemical and physical properties, plant and animal metabolism, environmental fate, residue analysis, storage stability, use pattern, supervised trials on apple, cherries, grapes, strawberry and cucurbits, and processing studies on apple and grape.

The following abbreviated names were used for the metabolites referred to in the appraisal of flutianil.

Table 1 List of compounds appearing in this appraisal

Compound Name/Code (MW, g/mol)	IUPAC name	Structure
Flutianil (426.45)	(2Z)-2-[[2-fluoro-5-(trifluoromethyl) phenyl]sulfanyl]-2-[3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile	
OC 42121 (412.42)	(2Z)-{[2-fluoro-5-(trifluoromethyl) phenyl]sulfanyl}[3-(2-hydroxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile	
OC 53274 (428.42)	(2Z)-[2-fluoro-5-(trifluoromethyl) phenyl-sulfinyl]-2-[3-(2-hydroxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile	
OC 53276 (442.45)	(2Z)-{[2-fluoro-5-(trifluoromethyl) phenyl]sulfinyl}[3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile	
OC 53277 (458.45)	(2Z)-{[2-fluoro-5-(trifluoromethyl) phenyl]sulfonyl}[3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile	

Compound Name/Code (MW, g/mol)	IUPAC name	Structure
OC 53279 (442.45)	(2Z)-{[2-fluoro-5-(trifluoromethyl) phenyl]sulfanyl}[4-hydroxy-3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile	
OC 53429 (372.48)	(2Z)-{[2-fluoro-5-methyl phenyl]sulfanyl}[3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene]acetonitrile	
OC 56574 (442.45)	(2Z)-{[2-fluoro-5-(trifluoromethyl) phenyl]sulfanyl}[3-(2-methoxyphenyl)-1-oxido-1,3-thiazolidin-2-ylidene] acetonitrile	
OC 56631 (123.15)	2-methoxyaniline	
OC 56634 (390.31)	Bis[2-fluoro-5-(trifluoromethyl) phenyl]disulfide	
OC 56635 (244.16)	2-fluoro-5-(trifluoromethyl) benzenesulfonic acid	
OC 59291 (209.26)	3-(2-methoxyphenyl)-1,3-thiazolidin-2-one	
AP5A in photolysis (232.30)	[3-(2-methoxyphenyl)-1,3-thiazolidin-2-ylidene] acetonitrile	
Bz5	Not completely identified: S-[2-(methanesulfinyl)-4-(trifluoromethyl)phenyl] cysteine + oxygen	
Bz6	Not completely identified. S-[2-fluoro-5-(trifluoromethyl)phenyl] hydrogen carbonothioate + oxygen + glucose	

Based on the information on physical and chemical properties, flutianil is not volatile and is more soluble in organic solvents than water with a Log P_{ow} of around 3. Flutianil is hydrolytically stable. Aqueous photolysis is likely to be a major degradation pathway of flutianil in the environment. Photodegradation produces polar compounds and the major photodegradation product is OC 56635.

Plant metabolism

The Meeting received information on the fate of flutianil in apple, grape, cucumber and lettuce after three to five foliar spray applications. For the studies, flutianil labelled uniformly with ^{14}C in either of the two phenyl rings: in 2-methoxyphenyl group (abbreviated as MP-label hereafter); or 2-fluoro-5-(trifluoromethyl)phenyl group (abbreviated as TFBZ-label hereafter) were used. In the metabolism studies, total radioactive residues (TRR) are expressed in mg flutianil equivalents/kg.

Apple

MP- or TFBZ-labelled flutianil was applied to apple plants, grown outdoors, three times as a foliar spray, 12 days apart, each at a target rate of 0.075 kg ai/ha. Fruit and foliage samples were collected 1–35 days and 14–35 days, respectively, after the last application.

After applications of either of the labelled flutianil, the TRR in the fruit decreased from 0.15–0.19 mg eq/kg at 1 DALA to 0.077–0.078 mg eq/kg at 14 DALA. However, after 14 DALA, it did not change significantly following the application of MP-labelled flutianil while, following the application of TFBZ-labelled flutianil, continuously decreased to 0.046 mg eq/kg at 21 DALA and further to 0.027 mg eq/kg at 35 DALA.

The TRR in foliage analysed 14–35 DALA were much higher following the application of MP-labelled flutianil than TFBZ-labelled flutianil. The TRR were similar at 14 and 21 DALA and then showed some decline from 21 DALA to 35 DALA (from the application with MP-label, 8.5 mg eq/kg at 14 DALA, 8.2 mg eq/kg at 21 DALA, and 6.7 mg eq/kg at 35 DALA; and from the application with TFBZ-label, 4.9 mg eq/kg at 14 DALA, 5.1 mg eq/kg at 21 DALA and 4.5 mg eq/kg at 35 DALA).

Samples of fruits and foliage were washed with acetonitrile and then extracted sequentially with methanol, methanol/water (1/1, v/v), water, 0.1 mol/L HCl, 0.1 mol/L NaOH and acetone. Most of the radioactive residue was recovered in acetonitrile surface washes following the applications of the radiolabelled flutianil. The sum of radioactivity recovered in the surface washes and methanol extracts accounted for 88–98% TRR. Ratio of radioactivity in the acetonitrile surface washes gradually decreased over time: in the fruit, 80–89% TRR at 1 DALA to 71–74% TRR at 35 DALA; and in the foliage, 85–87% TRR at 14 DALA to 71% TRR at 35 DALA. The percent TRR in the methanol extracts increased with the time: in the fruit extracts, 7.5–16% TRR at 1 DALA, to 13–24% TRR at 21–35 DALA; and in the foliage extracts, 9.1% TRR at 14 DALA and 15–20% TRR at 30–35 DALA. The sum of radioactivity in all other extracts accounted for < 10% TRR in the fruit and foliage and unextracted radioactivity accounted for < 4% TRR.

Regardless of the position of ^{14}C , the predominant radioactive residue in the acetonitrile surface washes and methanol extracts was flutianil in both the fruit and foliage samples. In the washes and methanol extracts of fruits, flutianil accounted for 69–87% TRR (0.021–0.16 mg/kg) between 1 DALA and 35 DALA. In the foliage washes and methanol extracts, flutianil accounted for 50–74% TRR (2.4–6.3 mg/kg) between 14 DALA and 35 DALA.

Four metabolites were identified from the washes and methanol extracts of fruit samples: OC 53279 (hydroxylated carbon on thiazolidine ring), OC 56574 (oxidized sulfur on the thiazolidine ring), OC 53276 (oxidized sulfur attached to fluoro-trifluoromethyl phenyl ring), and OC 56635 (detected only from the application of TFBZ-labelled flutianil, after its cleavage producing 2-fluoro-5-(trifluoromethyl)benzenesulfonic acid; and is also a major photodegradation product). From the fruit samples, the above individually accounted for $\leq 2.6\%$ TRR (≤ 0.005 mg eq/kg).

From the foliage samples, in addition to the four above, one metabolite, OC 56634 (detected only from the application of TFBZ-labelled flutianil, consisting of two 2-fluoro-5-(trifluoromethyl)phenyl rings

connected by S-S bond) was identified ($\leq 1.2\%$ TRR). Except for OC 56635 in the 35 DALA foliage sample, the metabolites individually accounted for $\leq 4.1\%$ TRR, while they were found up to 0.22 mg eq/kg. OC 56635 was found in the methanol extracts and increased from 2.3% TRR (0.11 mg eq/kg) at 14 DALA to 15% TRR (0.67 mg eq/kg) at 35 DALA. After hydrolysis of the methanol/water extract of the foliage samples from the MP-labelled flutianil in 2 mol/L HCl at 60 °C overnight and partitioning into dichloromethane, the presence of additional metabolites was shown in the TLC analysis: OC 42121 and/or OC 56631 ($\leq 0.5\%$ TRR) and OC 53277 ($\leq 0.2\%$ TRR), which may be present in foliage as conjugates.

In the fruit, a total of 71–88% TRR and in the foliage, 64–78% TRR were identified in the acetonitrile surface washes and methanol extracts. Unidentified residues were found in the surface washes and methanol extracts, but no peak exceeded 5.4% TRR individually. A small proportion of radioactive residues remained unextracted (up to 3.7% TRR).

Grape

Outdoor grown grapevines received four foliar sprays of MP- or TFBZ-labelled flutianil, 14 days apart, each application at a target rate of 0.040 kg ai/ha. Fruit and foliage samples were collected 1–45 days after the final application.

After applications of radiolabelled flutianil, the TRR in the fruit declined from 0.30–0.36 mg eq/kg at 1 DALA to 0.15–0.22 mg eq/kg at 21 DALA and 0.17–0.23 mg eq/kg at 45 DALA. The TRR in the foliage ranged 2.7–5.4 mg eq/kg during the study period.

Samples of fruits and foliage were washed with acetonitrile and then extracted in the same manner as the apple samples. The sum of radioactivity recovered in the surface washes and methanol extracts accounted for 96–100% TRR. Most of the radioactivity was recovered in acetonitrile surface washes of the fruit and foliage throughout the study period: in the fruit, 87–94% TRR, and in the foliage, 83–92% TRR. The percentage of TRR in the methanol extracts show some increasing trend: in the fruit, 5.1–5.3% TRR at 1 DALA to 9.1–11% TRR at 45 DALA; and in the foliage, 4.5–5.6% TRR at 1 DALA increasing to 10–11% TRR at 21 DALA and then decreasing to 8.1–8.8% TRR. In other extracts, radioactivity was low: in the fruit, $< 1\%$ TRR or not detected, and in the foliage, $\leq 2.2\%$ TRR. Small proportions of radioactivity remained unextracted (up to 4.2% TRR).

Most of the radioactivity in the surface washes and methanol extracts was flutianil (89–96% TRR in the fruit; and 89–94% TRR in the foliage). No metabolites were identified from the acetonitrile washes.

From the methanol extracts of the fruit, two metabolites, OC 53276 and OC 56574, were identified but they accounted for at the maximum 0.3% TRR (< 0.001 mg eq/kg). One metabolite in the polar fraction of the methanol extracts of the fruit from the TFBZ-labelled flutianil treatment co-chromatographed with OC 56635 in TLC but was not quantified. From the methanol extracts of the foliage, two metabolites OC 53276 and OC 56574 from both treatments and OC 56635 from the TFBZ-labelled treatment only were identified. Neither OC 53276 nor OC 56574 accounted for more than 0.4% TRR or 0.018 mg eq/kg. OC 56635 accounted for up to 3.5% TRR and 0.19 mg eq/kg.

The metabolism of flutianil in grape vines is limited.

Cucumber

Greenhouse grown cucumber plants received four foliar sprays of MP- or TFBZ-labelled flutianil applied 13 or 14 days apart, each at a nominal rate of 0.060 kg ai/ha. Fruit and foliage samples were collected 1–15 DALA and 3–15 DALA, respectively.

After applications of radiolabelled flutianil, the TRR in the fruits and foliage declined over time: in the fruit with low TRR, from 0.012–0.026 mg eq/kg at 1 DALA to 0.001–0.004 mg eq/kg at 15 DALA; in the foliage with much higher TRR, from 2.1–3.2 mg eq/kg at 3 DALA to 1.1–1.3 mg eq/kg at 15 DALA.

Samples of fruits and foliage were washed with acetonitrile, and then extracted in the similar manner as with the apple and grape samples. Most of the radioactivity was recovered in acetonitrile

washes and methanol extracts (total of 94–100% TRR for the fruit and the foliage). The TRR in the surface washes decreased significantly: in the fruit, from 82–84% TRR (up to 0.022 mg eq/kg) at 1 DALA to 61–68% TRR to 23–35% TRR at 15 DALA (< 0.01 mg eq/kg); and in the foliage, 75–85% TRR (1.6–2.8 mg eq/kg) at 3 DALA to 61–65% (0.69–0.86 mg eq/kg) at 15 DALA.

The percentage TRR in the methanol extracts increased in fruit, 16–17% TRR at 1 DALA to 59–74% TRR at 15 DALA (all < 0.01 mg eq/kg); and in the foliage, 13–23% TRR to 33–36% TRR. Throughout the study period, 0.3–6.4% TRR in the fruit and 0.5–1.1% TRR in the foliage remained unextracted after neutral, acid and base extractions.

Parent flutianil was the predominant residue in the washes and methanol extracts of fruit and foliage samples: in the fruit, 75–95% TRR at 1 and 3 DALA and 35–47% TRR at 15 DALA; and in the foliage, 93–95% TRR at 3 and 15 DALA. No other metabolites were identified and quantified. TLC analysis of the surface washes and methanol extracts of the foliage showed trace amounts of OC 53279, OC 53276 and OC 56574 but they were below the limit of detection of HPLC.

The metabolism of flutianil in cucumber is limited. The PHI for the GAP in the USA for the squash/cucumber subgroup (9B) is 0 days after the spray has dried. The above results may indicate that at the PHI of 0 days, the ratio of flutianil would be even higher with limited metabolism.

Lettuce

MP- or TFBZ-labelled flutianil was applied to lettuce grown in a greenhouse as a foliar spray five times, 7 days apart, each at a nominal rate of 0.045 kg ai/ha. Lettuce samples were collected 7 DALA and each head was split into head and wrapper leaves.

Following applications of radiolabelled flutianil, the TRR in the lettuce head samples (0.026–0.050 mg eq/kg) were significantly lower than those in the wrapper leaf samples (1.9–2.1 mg eq/kg).

Samples of heads and wrapper leaves were washed with acetonitrile, and then extracted with methanol. Most of the radioactivity (98–100% TRR) was recovered in the acetonitrile washes and methanol extracts. In the head samples, 73–74% TRR was recovered in the washes and 25–26% TRR from the methanol extracts; and in the wrapper leaf samples, 63–79% TRR from the washes and 21–35% TRR from the methanol extracts. Residues remaining unextracted were at or less than 0.8% TRR in both the head and wrapper leaf samples.

Parent flutianil was the predominant residue in the washes and methanol extracts: in the head samples, 84–89% TRR; and in the wrapper leaf samples, 89% TRR. From the methanol extracts of the head samples, three metabolites, OC 53279, OC 56574 and OC 53276, were identified at minor levels ($\leq$ 0.001 mg eq/kg and $\leq$ 2.5% TRR). From the methanol extracts of wrapper leaf samples, in addition to the above three metabolites, OC 56635 was identified. They accounted for $\leq$ 1.0% TRR and $\leq$ 0.015 mg eq/kg.

The metabolism of flutianil in lettuce is also limited.

Summary of plant metabolism

When flutianil was applied as a foliar spray to apple, grape, cucumber and lettuce plants, residue levels were much higher in leaves than fruits, or wrapper leaves of lettuce than in the head. Most radioactive residues were on the surface of fruits or leaves. As flutianil is susceptible to photolysis, some differences in the flutianil concentrations were anticipated between outdoor- and greenhouse-grown crops. However, in the metabolism studies, no significant differences were observed between outdoor and greenhouse studies.

Flutianil was the predominant residue, accounting for 35–96% TRR (fruit, foliage). From the fruit samples and lettuce, up to four metabolites, OC 53279 (apple and lettuce), OC 56574 (apple, grape and lettuce), OC 53276 (apple, grape and lettuce), and OC 56635 (apple, grape and lettuce) were identified and quantified. No metabolites accounted for more than 0.01 mg eq/kg in edible parts. They were not detected or only found at very low levels in the rat metabolism. From apple foliage samples, OC 56634

was also identified ($\leq 1\%$ TRR) and after acid hydrolysis, OC 42121 and/or OC 56631 and OC 53277 ($\leq 0.5\%$ TRR) were extracted and identified. Only OC 56635 was found above 10% TRR, up to 15% TRR (0.67 mg eq/kg), in the 35 DALA apple foliage but in edible parts at < 0.01 mg eq/kg. They were not detected in the rat metabolism.

The metabolism of flutianil in plants is limited and may be through multiple routes. Oxidation of sulfur groups in the molecule formed OC 53276 and OC 56574 and hydroxylation to form OC 53279. From these metabolites and the parent, OC 56635 could occur by cleavage of the link between two phenyl rings.

Animal metabolism

Metabolism studies on laboratory animals were reviewed in the framework of the toxicological evaluation by the current JMPR.

In a goat metabolism study, either MP-labelled or TFBZ-labelled flutianil in capsules were administered to one lactating goat orally using a balling gun at approximately 24 h intervals for five consecutive days at a nominal rate of 10 ppm in the diet on a dry weight basis. Goats were sacrificed 12 h (MP-label) or 6 h (TFBZ-label) after the last dose. Milk was collected twice daily and fractionated.

The TRR in whole milk was low and variable for 5 days (0.008–0.010 mg eq/kg) following the MP-label dose but following the TFBZ-label dose it continued to increase and reached a maximum concentration of 0.020 mg eq/kg on day 5. Fractionation of whole milk indicated that the TRR in milk fat was higher than that in skimmed milk.

Following the doses of MP-labelled flutianil, radioactivity in tissues and milk accounted for 0.57% of the cumulative administered dose (AD). Of the AD, 9.4% and 62% was excreted in the urine and faeces, and 18% and 0.1% dose was recovered in the gastro-intestinal tract and cage wash. Total recovered radioactivity was 90% of the AD.

Following the doses of TFBZ-labelled flutianil, radioactivity in tissues and milk accounted for 1.0% of the AD. Of the AD, 7.9% and 52% was excreted in the urine and faeces, and 21% and 0.3% was recovered in the gastro-intestinal tract and cage wash. Total recovered radioactivity was 83% of the AD.

Among edible tissues, TRR was the highest in the liver (0.37–0.39 mg eq/kg) and kidney (0.061–0.069 mg eq/kg). TRR were much lower in the fat (0.009–0.061 mg eq/kg) and muscle (0.003–0.021 mg eq/kg) samples.

Identification of extracts of edible tissues and milk revealed differences in the metabolites between the treatment with MP-labelled flutianil or TFBZ-labelled flutianil. Flutianil was found in all the tissues, except in the muscle (not subject to identification), from the MP-label administration and was the predominant residue in the fat (84% TRR, 0.011 mg/kg), whole milk (33% TRR, 0.003 mg/kg) and milk fat (60% TRR, 0.019 mg/kg). In liver and kidney, flutianil accounted for 5.6% (0.022 mg/kg) and 8.2% TRR (0.006 mg/kg), respectively. Flutianil was also detected in fat (9.9% TRR, 0.006 mg/kg) and milk fat (12% TRR 0.007 mg/kg) from the TFBZ-label treatment. Flutianil was not detected in other tissues from the TFBZ-label treatment.

Numerous peaks, other than flutianil, were observed in the extracts of the liver, kidney and milk samples.

From the MP-labelled flutianil administration, 1, 3 and 2 peaks, respectively, in the liver, kidney and whole milk samples accounted individually for 12–18% TRR but their concentrations were ≤ 0.047 mg eq/kg.

From the TFBZ-labelled flutianil administration, 1 peak coded as Bz11 accounted for 82, 66 and 79% TRR, respectively in the muscle, whole milk and milk fat samples but at levels ≤ 0.044 mg eq/kg. In the fat sample, Bz11 accounted for 71% TRR (0.042 mg eq/kg), and additionally Bz10 was found at 14% TRR (0.008 mg eq/kg). In whole milk Bz1 accounted for 22% TRR and 0.003 mg eq/kg. In the liver and kidney, Bz5 accounted for 25% (0.094 mg eq/kg) and 59% (0.036 mg eq/kg) TRR, respectively, and Bz6

accounted for 16% TRR (0.058 mg eq/kg) in the liver. Bz5 and Bz6 were partially identified as the S-[2-(methanesulfinyl)-4-(trifluoromethyl)phenyl]cysteine conjugate with one additional oxygen atom (position unknown), and the glucose conjugate of S-[2-fluoro-5-(trifluoromethyl)phenyl]-hydrogen carbonothioate with one additional oxygen atom (positions of oxygen or glucose unknown). Attempts were made to identify 4 other peaks, including Bz11, as well as Bz5 and Bz6, but were unsuccessful.

Animal metabolism summary

In goat, flutianil was extensively metabolized into numerous compounds. Flutianil was detected in the fat (84% TRR), kidney (8.2% TRR), liver (5.6% TRR), whole milk (33% TRR) and milk fat (60% TRR) from the MP-labelled flutianil administration; and the fat (9.9% TRR) and milkfat (12% TRR) from the TFBZ-labelled flutianil administration. Among many peaks in the extracts of these edible tissues and milk, only Bz5 and Bz6 found in kidney and liver, were partly identified and no other detected peaks were identified. Bz5 and Bz6 were not detected in rats.

No information on the metabolism of flutianil in laying hens was provided to the Meeting.

Environmental fate

The Meeting received information on environmental fate of flutianil: hydrolytic degradation and photodegradation in aqueous solutions, photodegradation on the soil surface, aerobic and anaerobic degradation in soils, field dissipation, and confined crop rotation.

Hydrolysis

Hydrolysis of MP-labelled flutianil was investigated in sterilized aqueous buffer solutions at pH 4, 7 and 9 at 50 °C for 5 days. After 5 days of incubation, 94–99% of the applied radioactivity remained and the concentrations of flutianil did not change. In conclusion, flutianil was hydrolytically stable. Hydrolysis is not expected to be a significant route of degradation at environmental pH.

Photolysis in buffer and natural water

After irradiation equivalent to UK/US summer sunlight for 30 days in sterilized pH 7 buffer at 25 °C, no flutianil was detected, while 87–95% of flutianil remained after 30 days in the dark. After irradiation of 30 days, about 60% of the applied TFBZ-labelled flutianil was detected as OC 56635.

The DT₅₀ was calculated to be 1.0 day for both MP- and TFBZ-labelled flutianil in pH 7 buffer. In natural water, the DT₅₀ was calculated to be 1.1–1.2 days. OC 56635 was found to be much more persistent than flutianil or other degradates and its DT₅₀ was calculated to be 62–72 days.

Irradiation was regarded to be a significant factor contributing to environmental degradation of flutianil.

Photodegradation in soil

Photodegradation of MP-labelled and TFBZ-labelled flutianil, at a rate equivalent to a field application rate of 0.040 kg ai/ha, was studied on soil surfaces irradiated with artificial sunlight. DT₅₀ of irradiated flutianil was calculated to be 92 days while in the dark DT₅₀ was 556 days. This confirms that photodegradation on the soil surface also contributes to environmental degradation of flutianil.

After irradiation of the MP-labelled flutianil for 45 days, OC 53276 and OC 56574 were identified but at a maximum 2.8% AR. Total of volatiles and carbon dioxide accounted for 10% AR. After irradiation of the TFBZ-labelled flutianil for 37 days, OC 56635 (11% AR), OC 53276 (3.3% AR) and OC 56574 (1.7% AR) were identified. Total of volatiles and carbon dioxide accounted for 3.3% AR.

Aerobic degradation in soil

The aerobic degradation of MP-labelled and TFBZ-labelled flutianil in four microbiologically viable soils was studied in the dark at 20 °C for 365 days at a dose equivalent to a field application rate of 0.227 kg ai/ha.

DT₅₀ values of flutianil were calculated to be in a range of 1100 to 2900 days in four soils. The results indicate that flutianil was stable in the aerobic soils.

Four minor degradates were identified: OC 56635, OC 53276, OC 56574 and OC 53279. At the end of incubation, they accounted for, at a maximum, 3.9% AR (OC 56635) or 3.4% AR (OC 56574). Others accounted for less than 2.5% AR at the end of incubation or earlier. Carbon dioxide accounted for a maximum of 1.2% AR at the end of incubation. Organic volatile compounds were not detected in any of the soils throughout the study.

Anaerobic degradation in soils

The anaerobic degradation of MP-labelled and TFBZ-labelled flutianil in four microbiologically viable soils as used in aerobic degradation study was studied in the dark at 20 °C for 365 days under the anaerobic condition at a dose equivalent to a field application rate of 0.227 kg ai/ha. After 30 days of incubation in the dark, the soils were flooded with water under a nitrogen atmosphere and incubated for additional 335 days.

Calculated DT₅₀ values flutianil ranged from 2100 to > 13000 days.

Four minor degradates were identified: OC 56635, OC 53276, OC 56574 and OC 53279. They accounted for at a maximum 2.6% AR (OC 56635, 365 days) or 2.2% AR (OC 56574, 246 days). Others accounted for less than 1.5% AR at the end of incubation or earlier. Carbon dioxide was never found above 1% AR. Organic volatile compounds were not detected in any of the soils throughout the study.

Field dissipation

Field dissipation of flutianil after four applications made 7 days apart each at a target rate of 0.045 kg ai/ha to bare soil. Samples were collected for analysis of flutianil and degradates up to 450–488 days after the last application.

Flutianil degraded continuously over time and was not quantified, i.e., < 0.01 mg/kg, at 181–271 DALA. When quantified, flutianil was present only in the top layer of soil (0–7.6 cm depth). DT₅₀ values of flutianil were calculated to be in a range of 55–400 days in four sites (soils). The variability may be due to the difference in the level of photolysis on the surface of soils and viability of microorganisms in the soils.

Degradation products, OC 53276, OC 56574 and OC 56635, which were also identified in the soil metabolism studies, were identified in most of the samples. Flutianil, OC 53276 and OC 56574 were detected only in the top layer soil (0–7.6 cm) and not detected in deeper soils. OC 56635 was detected in different layers of soil depending on the soils and sampling intervals. In one soil at 271 DALA, OC 56635 was detected only in the depth of 76–91 cm, while in another soil, it was not quantified at any sampling intervals. The cause of this variability of occurrence and movement of OC 56635 is unknown.

Residues in succeeding or rotational crops

A confined crop rotation study was conducted. Either MP-labelled or TFBZ-labelled flutianil was applied to bare soil at a target rate of 0.224 kg ai/ha (equivalent to the maximum annual rate of 0.224 kg ai/ha specified on the label). At plant back intervals (PBI) of 30, 120 or 365 days, radish, spinach and sorghum were planted.

The TRR in plant samples grown in the soil treated with TFBZ-labelled flutianil were much greater than those from the soil treated with MP-labelled flutianil. Among the samples from the TFBZ-labelled flutianil treatment, radish leaves contained significantly higher TRR than the corresponding roots; and the

TRR in the vegetative portions of sorghum increased with maturity. The TRR in the grain samples were less than one tenth of the TRR in the corresponding stover samples.

Identification was attempted for samples containing significant radioactivity. Among them, flutianil was found only in immature radish root samples (0.006 mg/kg, 120 PBI following the MP-label treatment; and 0.001 mg/kg, 365 PBI following the TFBZ-label treatment). Although flutianil is persistent in soil, its uptake by succeeding crops after one application at the maximum annual rate was limited, while OC 56635 is persistent in soil and taken up by succeeding crops at significant levels.

Three metabolites (OC 56635, OC 53276 and OC 56574) were found. Among them, from the TFBZ-label treatment, OC 56635 was the predominant residue in all crop samples subject to identification, at or higher than 83% TRR and at significant levels from all PBIs (immature radish roots, 0.019–0.17 mg eq/kg; immature radish leaves, 0.43–0.79 mg eq/kg; mature radish roots, 0.026–0.043 mg eq/kg; mature radish leaves, 0.16–0.32 mg eq/kg; immature spinach, 0.094–0.34 mg eq/kg; mature spinach, 0.14–0.41 mg eq/kg; sorghum grains, 0.036–0.059 mg eq/kg; and sorghum stover, 0.46–0.84 mg eq/kg). The other two metabolites were found only in immature radish root samples (0.003–0.005 mg eq/kg, 120-day PBI, MP-label; and 0.0005 mg eq/kg, 365 PBI, TFBZ-label).

Radioactive residues remaining in the PES were low at ≤ 0.02 mg eq/kg.

The Meeting noted that flutianil is persistent in soil (calculated $DT_{50} > 2000$ days in aerobic soil and about 400 days in the field dissipation study) and flutianil, as well as its soil degradates, may accumulate significantly in soil after multiple years of flutianil use on primary crops. The label indicates that any crops may be planted immediately after the last application. As a result, residues of flutianil or its degradates, in particular OC 56635, may be taken up by succeeding crops. The confined rotational crop study provided only used one application at the maximum annual application rate on the label.

Summary of environmental fate

Flutianil is hydrolytically stable in aqueous solutions. It is readily photolysed in pH 7.0 buffer and natural water with half-lives of 1.0 day and 1.2 days, respectively, with OC 56635 as a major photodegradation product. Flutianil is photodegraded on the surface of soils with a half-life of 92 days. OC 56635 occurred slightly above 10% of applied radioactivity. As minor metabolites, OC 53276 and OC 56574 were identified. In the environment, aqueous photolysis is likely to be a major degradation pathway of flutianil followed by photolysis on the surface of soils.

Flutianil and its minor degradates OC 56635, OC 53276, OC 56574 and OC 53279 are persistent in soil.

Residues of flutianil are not expected in rotational crops above 0.01 mg/kg. Degradate OC 56635 is predominant in rotational crops at significant levels, in particular in leafy parts of crops (up to 0.41 mg eq/kg in mature spinach, and up to 0.84 mg eq/kg in sorghum stover).

The Meeting concluded that the information available on rotational crops does not allow estimation of OC 56635 in succeeding crops after use of flutianil for multiple years.

Methods of analysis

Analytical methods for the determination of residues of flutianil in data development and enforcement for apple and its processed commodities, cherry, grape and its processed commodities, strawberry, cucumber, summer squash and cantaloupe, and OC 56635 in grape and its processed products were provided to the Meeting.

In general, the methods employed extraction by homogenization with acetonitrile/water (4:1, v/v) or acetone and partitioning with hexane for analysis of flutianil. After clean-up with silica gel or without clean-up, the hexane phase was analysed by GC-MSD or GC-ECD. The GC-MSD method was validated for apple and its processed commodities, cherry, strawberry, cucumber, summer squash and cantaloupe to be suitable for data development to determine flutianil with an LOQ of 0.01 mg/kg (0.1 mg/kg for wet apple

pomace) and mean recoveries (83–107%) and repeatability ($RSD \leq 6.2\%$) within the acceptable range. The GC-ECD method was validated for apple, grape, strawberry and cucumber to be suitable for data development with an LOQ of 0.01 mg/kg (0.1 mg/kg for dried grape) and mean recoveries (86–103%) and $RSD (\leq 9.9\%)$ in the acceptable range. These methods were subjected to independent laboratory validations on grape matrices and cucumber. For the dried grape, the mean recoveries were low (25–45%) at the fortification levels tested (0.1 and 1 mg/kg). For other matrices, the mean recoveries (78–101%) and repeatability ($RSD \leq 16\%$) were within the acceptable range.

An LC-MS/MS method for determining OC 56635 in grape, grape juice and dried grape was also provided, with an LOQ of 0.01 mg/kg (0.1 mg/kg for dried grape) with mean recoveries (94–102%) and repeatability ($RSD \leq 13\%$) in the acceptable range. This method was also validated in independent laboratory and the mean recoveries were within the acceptable range (76–109%).

The GC-ECD method was radio-validated by using the [^{14}C]-flutianil treated apple and cucumber samples obtained in the respective metabolism studies. The extraction efficiency was $> 90\%$ and accountability was $> 80\%$.

The DFG S19 Method with some modification was validated (including ILV) to be suitable for multi-residue analysis with LOQ of 0.01 mg/kg for flutianil in high water (cucumber), high acid (grape), dry (dry bean, wheat grain) and high oil (sunflower seed) matrices. This method did not include the determination of OC 56635. The DFG S19 method was also validated (without ILV) to be suitable for multi-residue analysis with LOQ of 0.01 mg/kg for flutianil in animal matrices.

Stability of residues in stored analytical samples

The stability of flutianil during frozen storage at around $-20\text{ }^{\circ}\text{C}$ was investigated in the high water content category (apple and its processed commodities, cherry, cucumber, summer squash and cantaloupe) and high acid content category (grapes and its processed commodities and strawberry). The stability of OC 56635 was also investigated in grape and its processed commodities. The fortification levels were in general 0.1 mg/kg, except it was 1.0 mg/kg in dried grape, and 2.0 mg/kg in strawberry.

The Meeting considered that flutianil was stable for at least the following time periods (longest periods tested) in the commodities.

High water content category (including processed commodities): 18 months

High acid content category (including processed commodities): 15 months

OC 56635 was found to be stable in frozen state for at least 6 months (longest periods tested) in high acid content category (grape and its processed commodities).

Definition of the residue

Plant commodities

For enforcement, the predominant residue was parent flutianil: 35–96% TRR in edible parts of the crops tested (apple, grape, cucumber and lettuce) and was observed in all field trial and processing studies.

Suitable analytical methods using GC (ECD or MSD) are available for various plant commodities to determine flutianil.

The Meeting considered that flutianil was a suitable marker for enforcement purposes.

In deciding which compounds should be included in the residue definition for dietary risk assessment, the Meeting noted that four metabolites, OC 53276, OC 53279, OC 56574 and OC 56635, were identified in the extracts of edible portions in the plant metabolism studies, and OC 56634 in apple leaves, as minor metabolites. In the fruit samples of apple, grape, and cucumber and in head lettuce, these metabolites were $< 10\%$ TRR and either not detected or $< 0.01\text{ mg eq/kg}$ after adjusting to the critical GAP rate in the USA. From the plant metabolism studies, these metabolites would not be detected at or above

0.01 mg/kg in edible portions after the use of flutianil on fruit trees and fruiting vegetables as primary crops. However, OC 56635 was analysed in the residue trials on grape (0–28 DALA) and found to be present occasionally above 0.01 mg eq/kg. The calculated ratios of OC 56635 (expressed in flutianil equivalents) to flutianil at the specific PHI for crops specified in the GAP were larger than the ratios in the plant metabolism studies and indicated that sometimes OC 56635 (expressed in flutianil equivalents) accounted for more than 10% of flutianil.

In the confined crop rotation study, following the application of TFBZ-labelled flutianil to bare soil at a rate equivalent to the maximum annual rate in the US GAP, OC 56635 occurred as a predominant residue ($\geq 83\%$ TRR, 0.016–0.84 mg eq/kg) in all the crops analysed, in particular, foliage of crops. Flutianil, OC 53276 or OC 56574 were detected only in immature radish roots (flutianil ≤ 0.006 mg/kg, and the others ≤ 0.0005 mg eq/kg).

The Meeting evaluated, for toxicology, five metabolites (OC 56635, OC 53276, OC53279, OC56574, OC 56634) not detected or found only at very low levels in the rat metabolism, based on available information. As regards OC 56635, the Meeting concluded that the ADI of flutianil is sufficiently protective for the sodium salt of OC 56635 and this conclusion also applies to the free acid form of OC 56635. For metabolites OC 53276, OC 53279 and OC 56574, the ADI is considered sufficiently protective based upon their close structural similarity to flutianil and absence of functional groups of higher reactivity. The Meeting also concluded OC 56634 would not be genotoxic. As it was detected only in the foliage samples of apple (up to 1% TRR), OC 56634 was not considered further.

The Meeting therefore concluded that for dietary risk assessment in plant commodities, the residue definition should be the sum of flutianil and 2-fluoro-5-(trifluoromethyl) benzenesulfonic acid (OC 56635), expressed as flutianil.

Animal commodities

In the goat metabolism study, flutianil was detected in fat (84 or 9.9% TRR), kidney (0 or 8.2% TRR), liver (0 or 5.6% TRR) and milk (33 or 0% TRR in whole milk and 60 or 12% TRR in milk fat). Flutianil was not detected in muscle.

Flutianil was metabolized into numerous compounds, a number of which exceeded 10% TRR and 0.01 mg eq/kg but only Bz5 and Bz6 found in kidney and liver from the administration of TFBZ-labelled flutianil were partially identified. They were not detected in rats.

Flutianil is the only component identified in the goat metabolism study and it can serve as a marker residue. A multi-residue method (LC-MS/MS) was available for the determination of flutianil in animal matrices.

Flutianil is more soluble in organic solvents than water with a Log P_{ow} of 3. Flutianil was not detected in muscle (< 0.001 mg/kg) but was found in fat at low concentrations, 0.008 or 0.011 mg/kg. A ratio of flutianil in milk fat obtained by centrifugation and whole milk was 0.019 mg/kg/0.003 mg/kg = 6.3. The Meeting concluded that flutianil is fat-soluble.

In deciding which compounds should be included in the residue definition for dietary risk assessment, it should be noted that flutianil was metabolized into numerous components. Although a number of these metabolites (components) accounted for $> 10\%$ TRR (e.g., Bz11 up to 82% TRR and 0.01 mg/kg in muscle and Bz5 up to 25% TRR and 0.09 mg eq/kg in liver), none of them were fully identified.

There would be a small dietary burden of flutianil for cattle (up to 0.060 ppm for cattle), significantly smaller compared to the 10 ppm dose used in the goat metabolism study. However, no information was available on the animal metabolism of OC 56635 present as a predominant residue in those succeeding crops that can be fed to animals.

No metabolism data were provided on laying hens dosed with flutianil or OC 56635. The dietary burden of flutianil calculated for poultry was extremely low.

The Meeting considered that it was not possible to determine the residue definition for dietary risk assessment for animal commodities.

Conclusion

Based on the above, the Meeting recommended the following residue definitions.

Definition of the residue for compliance with the MRL for plant and animal commodities: Flutianil

Definition of the residue for dietary risk assessment for plant commodities: Sum of flutianil and 2-fluoro-5-(trifluoromethyl)benzenesulfonic acid (OC 56635), expressed as flutianil

The residue is fat-soluble.

The Meeting noted that it would be necessary to revisit the residue definition for plant commodities, if uses were expanded beyond fruiting vegetables and leafy vegetables, and/or new data on metabolism became available for a new category of crops.

Results of supervised residue trials on crops

The Meeting received supervised trial data for flutianil on apple, cherry, grape, strawberry, cucumber, summer squash and cantaloupe.

The Meeting considered that potential accumulation of flutianil in soil after its use over multiple years and uptake of residues into rotational crops may significantly contribute to the residue of flutianil and OC 56635 in food and feed commodities. The Meeting decided that residues found in annual crops in the residue trials, following treatment as primary crops (cucumber, summer squash and melon), may significantly underestimate the potential residues, which impacts the estimation of maximum residue levels and STMRs. This applies to strawberry which is a perennial plant but rotated every 1–5 years. For these crops, maximum residue levels and STMRs may be estimated by a future Meeting, should additional information addressing the contribution of the residue uptake from soil become available.

For estimating an STMR value for apple (PHI 14 days), the Meeting calculated the ratios of OC 56635 (expressed in flutianil equivalents) to flutianil reported at 14 DALA in the supervised trials on grapes, which were in the range of 0.022 to 0.93 with a median of 0.33. The Meeting decided to use the conversion factor ($0.33 + 1 = 1.33$) for apple. For estimating an STMR value for cherries (PHI 3 days), the Meeting calculated a ratio of OC 56635 to flutianil from the grape metabolism study at 3 DALA, which was 0.08 (no data on apple fruit at 3 DALA in the apple metabolism study). Therefore, a conversion factor of $1 + 0.08 = 1.08$ was used for cherries. For estimating an STMR for grapes, the Meeting used the sum of reported flutianil and OC 56635, expressed in flutianil equivalents, from the residue trials. When summing up, if OC 56635 was reported as < LOQ, it was assumed to be 0 mg/kg.

Pome fruits

Apple

GAP in the USA for apples allows four foliar applications (RTI of 7 days), each at a rate of 0.045 kg ai/ha, and a PHI of 14 days, using a concentrated or diluted spray.

Twelve supervised trials were conducted on apple in the USA according to GAP.

The use of either a diluted or concentrated spray did not significantly affect the residue concentrations and the highest residue level was selected from each trial. Flutianil residues from independent trials matching the critical GAP in the USA in rank order were (n = 10): 0.025, 0.026, 0.030, 0.030, 0.031, 0.038, 0.043, 0.044, 0.052 and 0.080 mg/kg.

The Meeting estimated a maximum residue level of 0.15 mg/kg for apple. An STMR was estimated to be $0.035 \times 1.33 = 0.047$ mg/kg.

Stone fruits

Cherry

GAP in the USA for the “cherry subgroup” allows four foliar applications (RTI of 7 days), each at a rate of 0.045 kg ai/ha, and a PHI of 3 days, using a concentrated or diluted spray.

Eight independent supervised trials were conducted on sweet and sour cherries in the USA according to the GAP rate, using either a diluted or concentrated spray, with or without adjuvant.

In four trials, samples were taken at 3 DALA. Flutianil residues from independent trials approximating the critical GAP in the USA were in rank order ($n = 4$): 0.070, 0.12, 0.20 and 0.24 mg/kg. In the other four trials, cherry samples were not collected at 3 DALA but either at 2 DALA or 4 DALA. According to the two decline trials, only slight decline was observed between 2 DALA and 4 DALA. The Meeting decided to use these four trials. In these trials, flutianil residues were in rank order ($n = 4$): 0.081 (2 DALA), 0.090 (2 DALA), 0.092 (4 DALA) and 0.16 mg/kg (2 DALA). The combined data were ($n = 8$): 0.070, 0.081, 0.090, 0.092, 0.12, 0.16, 0.20 and 0.24 mg/kg.

As the GAP in the USA was for the cherry subgroup and sweet and sour cherries are the representative commodities of the Codex subgroup of cherries, the Meeting estimated a maximum residue level of 0.4 mg/kg for the subgroup of cherries. An STMR was estimated to be $0.106 \times 1.08 = 0.11$ mg/kg.

Berries and other small fruits

Subgroup of small fruit vine climbing

Grapes

GAP in the USA for the “small fruit, vine climbing subgroup (except fuzzy kiwifruit)” allows four foliar applications (RTI of 7 days), each at a rate of 0.045 kg ai/ha, and a PHI of 14 days, using concentrated or diluted spray.

Nine independent supervised trials were conducted on grapes in the USA approximating the GAP rate (five applications instead of four applications) with either a diluted spray or a concentrated spray, and with adjuvants.

Flutianil residues from independent trials approximating the critical GAP in the USA were in rank order ($n = 9$): 0.012, 0.019, 0.047, 0.051, 0.051, 0.056, 0.077, 0.11 and 0.50 mg/kg.

The GAP in the USA covers the subgroup of “small fruit, vine climbing” and as grape is the representative commodity of this Codex subgroup, the Meeting estimated a maximum residue level of 0.7 mg/kg for the subgroup of small fruit vine climbing.

The calculated sum of residues (flutianil and OC 56635, expressed as flutianil) from nine independent trials were: 0.012, 0.029, 0.047, 0.068, 0.075, 0.077, 0.108, 0.109 and 0.495 mg/kg. The Meeting estimated an STMR of 0.075 mg/kg.

Subgroup of low growing berries

Strawberry

The GAP in the USA for the “berry, low growing subgroup” allows five foliar applications (RTI 7 days), each at 0.045 kg ai/ha, with a PHI of 0 days.

Nine trials were conducted on strawberry in the USA according to GAP.

Flutianil residues from trials matching (8) or approximating (1) the critical GAP in the USA were in rank order (n = 9): 0.025, 0.030, 0.032, 0.048, 0.062, 0.076, 0.098, 0.098 and 0.16 mg/kg.

Fruiting vegetables, Cucurbits

GAP in the USA for the squash/cucumber subgroup and melon subgroup allows five applications (RTI 7 days), each at 0.045 kg ai/ha, and a PHI of 0 days. While a maximum of 2 seasons are allowed in one year, a maximum total annual rate is set at 0.224 kg ai/ha.

Subgroup of fruiting vegetables, Cucurbits - cucumbers and summer squashes

Cucumber and summer squash

Eight field trials on cucumber and six field trials on summer squash were conducted in the USA according to GAP.

Flutianil residues on cucumber were in rank order (n = 8): < 0.01 (6), 0.012 and 0.012 mg/kg.

Flutianil residues on summer squash in ranked order were (n = 6): < 0.01 (4), 0.013 and 0.020 mg/kg.

The combined data set is in rank order (n = 14): < 0.01 (10), 0.012, 0.012, 0.013 and 0.020 mg/kg.

Subgroup of fruiting vegetables, Cucurbits - melons, pumpkins and winter squashes

Melon

Seven supervised trials were conducted on cantaloupe.

Flutianil residues in the trials matching the critical GAP in the USA were in rank order (n = 7): < 0.01 (2), 0.014, 0.014, 0.016, 0.018 and 0.042 mg/kg.

Fate of residues during processing

High temperature hydrolysis

The hydrolysis of MP-labelled and TFBZ-labelled flutianil was studied in sterile buffered aqueous solutions under the conditions simulating pasteurization (pH 4, 90 °C, 20 min), baking/brewing/boiling (pH 5, 100 °C, 60 min), and sterilization (pH 6, 120 °C, 20 min).

Flutianil was stable under the conditions representing pasteurization, baking/brewing/boiling and sterilization, with 99–103% of the initial concentrations recovered at the end of incubation.

Processing

The Meeting received information on processing of apple to juice and wet pomace, and grape to juice and dried grape. For the processing studies, samples of apple and grape were obtained from the supervised trials with the applications at 5-fold the critical GAP rate and flutianil was analysed. Processing factors (flutianil) for apple to its processed commodities and grape to its processed commodities are summarized below. Using the best estimates of processing factors and the STMR values for apple and grape, the STMR-P values were calculated for processed commodities of apple and grape, assuming that OC56635 reacts in a similar manner to flutianil, and included in the following table.

Table 2 Processing factors of flutianil for apple to its processed commodities and grape to its processed commodities

Processed commodity	Individual processing factor	Best estimate	Median residue of flutianil	STMR/STMR-P
Apple			0.035	0.047

Processed commodity	Individual processing factor	Best estimate	Median residue of flutianil	STMR/STMR-P
Juice	< 0.091, 0.11, 0.16	0.11 (median)		0.005 ^a
Wet pomace	3.2, 3.3, 14	3.3 (median)	0.012	
Grape				0.075
Juice	0.71	0.71		0.05 ^a
Dried grape	1.2	1.2		0.09 ^a

^a Tentatively calculated assuming that OC 56635 reacts in a similar manner as flutianil in processing.

The median residue for apple wet pomace was calculated for animal dietary burden calculation. Since the processing factor from grape to dried grape is 1.2, indicating only slight concentration, the Meeting considered that the maximum residue level for grape would cover residues in dried grape.

Residues in animal commodities

No livestock feeding studies were provided to the current Meeting. Among all the commodities for which information on supervised residue trials and processing studies were submitted, apple pomace can be fed to cattle. Some rotational crops and their parts can be fed to livestock. Using the STMR-P of 0.12 mg/kg for apple wet pomace and highest levels in rotational crops, an animal dietary burden of flutianil was calculated. The maximum and mean burden for beef cattle was 0.060 ppm and for dairy cattle was 0.030 ppm.

The goat metabolism study was conducted at a daily dose of 10 ppm. Comparing the calculated burden with the dose in the goat metabolism study, flutianil or any metabolites are not expected to be found, individually or in combination, above 0.01 mg/kg. Without appropriate data on the accumulation of flutianil or its degradates in soil after multiple years of use of flutianil, it was not possible to calculate a dietary burden for OC 56635 and there is no information about the metabolism of OC 56635 in animals.

No metabolism studies or feeding studies were provided on laying hens. Among all the commodities for which information on supervised residue trials or processing studies were submitted, grape pomace can be fed to turkey in the Australian ration. However, grape pomace was not analysed.

The Meeting concluded that it was not possible to estimate maximum residue levels or STMRs for animal commodities.

RECOMMENDATIONS

On the basis of the data from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI assessment.

Definition of the residue for compliance with the MRL for plant and animal commodities: *Flutianil*.

Definition of the residue for dietary risk assessment for plant commodities: *sum of flutianil and 2-fluoro-5-(trifluoromethyl)benzenesulfonic acid (OC 56635), expressed as flutianil*.

The residue is fat-soluble.

FUTURE WORK OR INFORMATION*Desirable information:*

- Long-term soil accumulation of flutianil and OC 56635 and their uptake by succeeding crops in the field, addressing soil residues taking into account the estimated soil plateau concentrations after application over multiple years
- Metabolism of OC 56635 in livestock.

DIETARY RISK ASSESSMENT***Long-term dietary exposure***

The current Meeting established an ADI of 0–0.8 mg/kg bw for flutianil. The International Estimated Daily Intakes (IEDIs) for flutianil were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the current JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs was 0% of the maximum ADI. The Meeting concluded that long-term dietary exposure to residues of flutianil from uses considered by the JMPR is unlikely to present a public health concern.

Acute dietary exposure

The current Meeting decided that an ARfD for flutianil was unnecessary. The Meeting therefore concluded that the acute dietary exposure to residues of flutianil from the uses considered is unlikely to present a public health concern.

Threshold of toxicological concern (TTC) consideration for metabolites

The metabolites Bz5 and Bz6 were not detected in rat and could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

Bz5

Bz5 was detected in muscle at 0.001 mg eq/kg, in kidney at 0.036 mg eq/kg, in liver at 0.094 mg eq/kg and in whole milk at 0.01 mg eq/kg following the administration of 10 ppm flutianil to a goat. Although the calculated dietary burden of flutianil was less than 1/100 of this dose level, no information was available on the metabolism of OC 56635 contained in feed. Bz5 and OC 56635 both have 5-trifluoromethyl phenyl moiety in common and conversion from OC 56635 to Bz5 cannot be excluded.

The Meeting considered that it was not possible to complete an exposure estimation for Bz5. The Meeting noted that this issue is not relevant for the permanent crops currently being considered. If rotated crops are considered in future this issue will need to be revisited.

Bz6

Bz6 was detected in muscle at 0.001 mg eq/kg, in kidney at 0.005 mg eq/kg, and in liver at 0.058 mg eq/kg following the administration of 10 ppm flutianil to a goat. Although the calculated dietary burden of flutianil was less than 1/100 of this dose level, no information was available on the metabolism of OC 56635 contained in feed. Bz6 and OC 56635 both have 2-fluoro-5-trifluoromethyl phenyl moiety in common and conversion from OC 56635 to Bz6 cannot be excluded.

The Meeting considered that it was not possible to complete an exposure estimation for Bz6. The Meeting noted that this issue is not relevant for the permanent crops currently being considered. If rotated crops are considered in future this issue will need to be revisited.

5.6 Guazatine (114)

Background

Guazatine is the ISO common name for a mixture of the reaction products from polyamines, comprising mainly octamethylenediamine, iminodi(octamethylene)diamine, octamethylene bis(iminoctamethylene)diamine, and carbamonitrile. The Chemical Abstracts Service (CAS) number for guazatine acetates is 115044-19-4. Guazatine is a fungicide having multi-site contact activity.

Guazatine was previously evaluated by the FAO/WHO Joint Meeting on Pesticide Residues (JMPR) in 1978 and 1997. In 1978 an acceptable daily intake (ADI) of 0–0.03 mg/kg body weight (bw) was established based on a no-observed-adverse-effect level (NOAEL) of 3 mg/kg bw per day in a two-year dog study. In 1997 the Meeting considered that it could not establish an ADI for guazatine owing to the inadequate information on its composition, and concerns about the production of rare malignant tumours in mice. The ADI for guazatine from 1978 was therefore withdrawn. The 1997 Meeting identified the following studies that would provide information necessary for continued evaluation of guazatine:

1. Data on the levels of individual components in batches of guazatine from recent production runs
2. Investigation of the mechanism of tumour production in mice
3. Clarification of the extent of absorption, excretion and metabolism of all components of guazatine
4. Clarification as to whether the stated doses used in the studies of toxicity were expressed as free-base or the triacetate.

Comment by JMPR

Guazatine was considered for re-evaluation at the present Meeting under the periodic review programme (awaiting confirmation) of the Codex Committee on Pesticide Residues (CCPR). A literature review identified no relevant information on the toxicology of guazatine. An initial screen of the submitted dossier indicated that it was insufficient to enable a toxicological evaluation. The new notifier had been unable to gain access to the original study reports, and relied on a summary document prepared for an EU evaluation. In addition, of the four points identified by the 1997 Meeting as required for further evaluation of guazatine, the only additional information presented was on the composition of guazatine batches produced between 1990 and 2007.

The Meeting concluded that the submitted dossier for guazatine was inadequate to address the concerns identified by the 1997 JMPR Meeting and For this reason, it was not possible for the Meeting to conclude the toxicological re-evaluation of guazatine.

5.7 Isoprothiolane (299)

TOXICOLOGY

Isoprothiolane is the ISO-approved name for diisopropyl 1,3-dithiolan-2-ylidenemalonate (IUPAC), with the Chemical Abstract Service number 50512-35-1. Isoprothiolane is a systemic fungicide with protective and curative action which is used on rice crops. In bananas it is used to control black sigatoka (*Mycosphaerella fijiensis*). Isoprothiolane inhibits the penetration and elongation of infecting hyphae by inhibiting the formation of infecting peg or cellulase secretion.

Isoprothiolane was previously evaluated by the FAO/WHO Joint Meeting on Pesticide Residues (JMPR) in 2017, when an ADI of 0–0.1 mg/kg body weight (bw) was established, based on a NOAEL of 10.9 mg/kg bw per day identified in a two-year study of toxicity and carcinogenicity in rats, using a safety factor of 100. This ADI was also applicable to the isoprothiolane metabolites M-2, M-3 and M-5. The Meeting in 2017 concluded that it was unnecessary to establish an acute reference dose (ARfD).

Isoprothiolane is being evaluated by the current Meeting in support of the FAO panel review of isoprothiolane for additional maximum residue limits (MRLs) for bananas. The new information on isoprothiolane included: 28-day and 90-day oral studies of toxicity in mice, a developmental toxicity study in rabbits, in vitro mammalian cell gene mutation assay, a bacterial reverse mutation assay and in vivo micronucleus assay on isoprothiolane. Also, with respect to its metabolites: acute toxicity studies and genotoxicity studies on metabolites 4-hydroxy-isoprothiolane and isoprothiolane monosulfoxide, and QSAR predictions for 4-hydroxy-isoprothiolane and isoprothiolane monosulfoxide.

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with relevant national or international test guidelines unless otherwise specified. No additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Biochemical aspects

No new data available.

Toxicological data

In a 28-day study of toxicity in mice, isoprothiolane was administered at dietary concentrations of 0, 500, 1500 or 5000 ppm (equal to 0, 92.2, 266 and 870 mg/kg bw per day for males; 0, 107, 347, and 1042 mg/kg bw per day for females). The NOAEL was 1500 ppm (equal to 266 mg/kg bw per day) based on decreased absolute and relative heart weight in females seen at the LOAEL of 5000 ppm (equal to 870 mg/kg bw per day) in the absence of histopathological examination.

In a 90-day study of toxicity in mice, isoprothiolane was administered at dietary concentrations of 0, 200, 1000, or 5000 ppm (equal to 0, 32.3, 168, and 793 mg/kg bw per day for males, 0, 40.1, 197, and 958 mg/kg bw per day for females). The NOAEL was 1000 ppm (equal to 168 mg/kg bw per day) based on changes in haematology, clinical chemistry, liver weight increases and histopathological findings in the liver seen at the LOAEL of 5000 ppm (equal to 793 mg/kg bw per day).

Isoprothiolane was negative for genotoxicity in an additional bacterial reverse mutation assay, gene mutation in mouse lymphoma assay and an additional in vivo mouse micronucleus assay.

The Meeting confirmed the conclusion of the 2017 Meeting that isoprothiolane is unlikely to be genotoxic in vivo.

In an additional developmental toxicity study in rabbits, isoprothiolane was administered by gavage at dose levels of 0, 30, 100 or 300 mg/kg bw per day. The NOAEL for maternal toxicity was 100 mg/kg bw per day based on lower feed consumption and body weight gains, body weight losses, and decreased defaecation observed at the LOAEL of 300 mg/kg bw per day. The NOAEL for embryo/fetal

toxicity was 300 mg/kg bw per day, the highest dose tested. Isoprothiolane was not teratogenic in rabbits in this study.

The Meeting confirmed the conclusion of the 2017 Meeting that isoprothiolane is not teratogenic.

Toxicological data on metabolites and/or degradates

M-3 (4-hydroxy-isoprothiolane)

The acute oral median lethal dose (LD₅₀) for metabolite M-3 was 300–2000 mg/kg bw for female rats. M-3 was tested for genotoxicity in bacterial reverse mutation and in vitro human lymphocytes assays in the presence and absence of metabolic activation. It gave a negative response in both assays. The Meeting confirmed the conclusion of the 2107 Meeting that it was unlikely that the metabolite M-3 or its conjugates would be of greater toxicity than the parent, isoprothiolane.

M-1 (monosulfoxide-isoprothiolane)

The acute oral LD₅₀ for metabolite M-1 was 300–2000 mg/kg bw for female rats. M-1 was tested for genotoxicity in bacterial reverse mutation and in vitro human lymphocytes assays in the presence and absence metabolic activation. It gave a negative response in both assays.

The JMPR 2017 Meeting concluded that:

"...these metabolites (M-3, M-5, M-2) are not of greater toxicological concern than the parent and considered that they would be covered by the acceptable daily intake (ADI) established for isoprothiolane."

The new studies on M-3 support the previous conclusion from the JMPR 2017 Meeting.

Metabolite M-1 was not detected in the rat metabolism study. No repeat-dose studies were available for metabolite M-1. It was negative for mutagenicity, therefore dietary exposure to M-1 should be compared to the TTC value for Cramer class III, that is 1.5 µg/kg bw per day.

Microbiological data

No data for antimicrobial activity or impact on the human gut microbiome was available.

Human data

No new data available.

The Meeting concluded that the existing database on isoprothiolane was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting re-affirmed an ADI of 0–0.1 mg/kg bw on the basis of a NOAEL of 300 ppm (equal to 10.9 mg/kg bw per day) from a two-year study of toxicity and carcinogenicity in rats, based on an increase in blood urea nitrogen (BUN) in females, and an increase in the relative weight of liver and kidneys in both sexes at 3000 ppm (equal to 115 mg/kg bw per day). A safety factor of 100 was applied. Although the NOAEL of 50 ppm (equal to 3.4 mg/kg bw per day), in the 90-day oral toxicity study in rats was lower, the LOAEL in that study was based on marginal effects seen at 300 ppm (equal to 20.9 mg/kg bw per day). The Meeting therefore concluded that the NOAEL of the two-year combined toxicity/carcinogenicity study was the more appropriate value on which to establish the ADI.

The Meeting re-affirmed that it was unnecessary to establish an ARfD for isoprothiolane in view of its low acute oral toxicity and absence of developmental toxicity or any other toxicological effects likely to be elicited by a single dose. The Meeting concluded that the lower feed consumption and body weight losses seen in the early phase of dosing in the developmental toxicity study in rabbits were not an

appropriate basis for an ARfD because the decrease in body weight was small in relation to the rabbits' overall body weights.

The reaffirmed ADI of 0–0.1 mg/kg bw can be applied to metabolites M-2, M-3 and M-5.

A toxicological monograph addendum was prepared.

Addendum to levels relevant to risk assessment of isoprothiolane

Species	Study	Effect	NOAEL	LOEL
Mouse	90-day study of toxicity ^a	Toxicity	1000 ppm, equal to 168 mg/kg bw per day	5000 ppm, equal to 793 mg/kg bw per day ^c
Rabbit	Developmental toxicity study ^b	Maternal toxicity	100 mg/kg bw per day	300 mg/kg bw per day ^c
		Embryo and fetal toxicity	100 mg/kg bw per day	300 mg/kg bw per day ^c

^a Dietary administration

^b Gavage administration

^c Highest dose tested

Acceptable daily intake (ADI) applies to isoprothiolane, M-2, M-3 and M-5 expressed as isoprothiolane.

0–0.1 mg/kg bw

Acute reference dose (ARfD) applies to isoprothiolane, M-2, M-3 and M-5 expressed as isoprothiolane.

Not necessary

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Addendum to critical end-points for setting guidance values for exposure to isoprothiolane

Short-term studies of toxicity

Target/critical effect	Liver, red blood cells, haemoglobin, haematocrit, alanine amino transferase and total protein
Lowest relevant oral NOAEL	168 mg/kg bw per day (mouse)

Genotoxicity

Unlikely to be genotoxic

Developmental toxicity

Target/critical effect	Decreased in body weight gain and feed consumption
Lowest relevant maternal NOAEL	100 mg/kg bw per day (rabbit)
Lowest relevant embryo/fetal NOAEL	100 mg/kg bw per day (rabbit)

Studies on toxicologically relevant metabolites

4-hydroxy isoprothiolane (M-3)	Acute oral LD ₅₀ : 300–2000 mg/kg bw (rat) Not genotoxic (Ames and human lymphocytes in vitro)
Isoprothiolane monosulfoxide (M-1)	Acute oral LD ₅₀ : 300–2000 mg/kg bw (rat) Not genotoxic (Ames and human lymphocytes in vitro)

Summary

	Value	Study	Safety factor
ADI	0–0.1 mg/kg bw ^a	Two-year study of toxicity and carcinogenicity (rat)	100
ARfD	Not necessary		

^a Applies to parent, metabolites M-2, M-3 and M-5

5.8 Mefentrifluconazole (320)

TOXICOLOGY

Mefentrifluconazole is the International Standards Organization (ISO)-approved common name for 2*RS*-2-[4-(4-chlorophenoxy)- α,α,α -trifluoro-*o*-tolyl]-1-(1*H*-1,2,4-triazol-1-yl)propan-2-ol (IUPAC), Chemical Abstracts Service number 1417782-03-6. Chemically, mefentrifluconazole is a triazole compound. It is a racemic mixture of two (*R*- and *S*-) enantiomers. The substance is a novel fungicide that is used to control fungal diseases in various crops including cereals, oilseeds, fruits or vegetables. Its fungicidal mode of action (MOA) is by blocking ergosterol biosynthesis due to inhibition of the P450 sterol demethylase (CYP51) resulting in growth inhibition and cell membrane disruption.

Mefentrifluconazole has not previously been evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) and was reviewed by the present Meeting at the request of the Codex Committee on Pesticide Residues (CCPR). All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with current guidelines, unless otherwise specified. No additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Biochemical aspects

Mefentrifluconazole was subject to extensive absorption, distribution, metabolism, excretion (ADME) investigations. Most of the experiments had been performed with test substance that was radiolabelled with ^{14}C in the chlorophenyl ring or in the triazole moiety, but some data were also available for trifluoromethyl ring-labelled material. The metabolic profile observed was not significantly affected by sex, repeated administration nor the position of the radiolabel.

Following oral administration of a single dose, absorption was rapid (T_{max} of radioactivity 0.5–1.2 h) and extensive (80% or higher) at 5 mg/kg bw, and somewhat less extensive (50–70%) at 180 mg/kg bw, suggesting some saturation of absorption. The applied radioactivity was widely distributed throughout the body. Total organ and tissue residues after seven days accounted for less than 1% of the applied dose, with highest concentrations found in the gut, in stomach and gut contents, in liver and kidneys.

In all experimental groups, the major part of excretion took place within 72 hours. Elimination was complete seven days after a single low dose and accounted for more than 90% after a single high dose given alone or after repeated (15 days) low dose administration of unlabelled compound. Most of the radioactivity was excreted via the faeces (60–90%, depending on dose and radiolabel) whereas urinary elimination was less important (5–15%). In bile duct-cannulated rats, 40–60% of radioactivity was found in the bile, demonstrating that this was in fact the major excretion route. There was limited evidence of enterohepatic circulation.

Plasma kinetics of mefentrifluconazole in the mouse were similar to those in the rat.

Metabolism in the rat was extensive and complex with a total of 68 identified metabolites resulting from phase I and phase II reactions. Main metabolic pathways comprised hydroxylation, methylation and cleavage of the ether group or of the triazole ring from the parent molecule, often followed by conjugations. Most metabolites occurred at comparatively low concentrations and only very few of them, like M750F015, M750F016 plus M750F017 in the faeces, accounted for more than 30% of the dose. In urine, the most abundant metabolite was 1,2,4-triazole (M750F001) with a maximum abundance of 10.5%. In bile, the five main metabolites were all glucuronides which had been formed subsequent to hydroxylation. Amongst individual experiments and groups the percentage of the remaining parent compound varied very greatly. In faeces, there was a 1:1 ratio of the two isomers of mefentrifluconazole, whereas a shift towards the *R*-enantiomer was observed in methanolic liver and kidney extracts.

A comparative study in human, rat and mouse hepatocytes revealed a completely different metabolic pattern in vitro. In rat hepatocytes, biotransformation of the parent compound was complete after three hours of incubation but there was only one metabolite that was most likely identical to one of those found in rat bile. The same metabolite was detected in the experiment with human hepatocytes, but accounted for only 19–21% of the applied radioactivity, suggesting more delayed metabolism. In mouse hepatocytes, no metabolism at all was observed since parent mefentrifluconazole remained unchanged. Based on this limited information, no reliable conclusions regarding similarities and differences in human and animal metabolism could be drawn.

Toxicological data

The acute oral median lethal dose (LD_{50}) in rats was >2000 mg/kg bw and the dermal LD_{50} >5000 mg/kg bw. The inhalation median lethal concentration (LC_{50}) was above 5.3 mg/L. The compound was not irritating to the skin or to the eyes of rabbits. The test substance was devoid of a phototoxic potential in vitro. However, mefentrifluconazole was identified a skin sensitizer in a Guinea pig maximization test.

Oral (feeding) short-term toxicity studies were performed in mice (28-day and 90-day) and rats (28-day and 90-day). Dogs were administered mefentrifluconazole in capsules for ca. 30 or for 90 days, as well as in a one-year study. In these studies, the liver was the common target organ in all the species. In the dog and in the rat, body weight and its gain were also affected. Clinical signs of toxicity were observed, but only at higher dose levels in dogs. In the mouse, there was some evidence of higher mortality in high-dose animals under isoflurane anaesthesia during blood sampling. In the short-term studies, the rat and the dog appeared less sensitive than the mouse when NOAELs and LOAELs were compared.

In a 28-day range-finding study in mice, the dietary dose levels were 0, 30, 100, 300 and 1000 ppm (equal to 0, 4.8, 15.5, 47.9 and 128 mg/kg bw per day in males, 0, 5.8, 18.5, 51 and 145 mg/kg bw per day in females). The NOAEL was 30 ppm (equal to 5.8 mg/kg bw per day) based on a marked increase in liver weight with associated alterations in clinical chemistry parameters in females at 100 ppm (equal to 18.5 mg/kg bw per day).

In a 90-day study in mice, dietary doses of 0, 10, 50, 250, and 750 ppm (equal to mean daily intakes of 0, 2, 11, 58 and 174 mg/kg bw in males, 0, 3, 15, 67 and 211 mg/kg bw in females) were administered. The NOAEL was 50 ppm (equal to 11 mg/kg bw per day), based on increased liver weight with associated histopathology such as cytoplasmic alterations (hyaline vacuoles) and single cell necrosis at 250 ppm (equal to 58 mg/kg bw per day).

In a 28-day study in rats, dietary dose levels of 0, 500, 1500 and 4000 ppm (equal to 0, 47, 135, and 388 mg/kg bw per day in males, 0, 47, 138, and 334 mg/kg bw per day in females) were fed to the animals. The NOAEL of 1500 ppm (135 mg/kg bw per day) was based on an increase in relative liver weight in females, and associated clinical chemistry findings and reductions in body weight and its gain at 4000 ppm (equal to 334 mg/kg bw per day) in both sexes.

In a 90-day study in rats, dietary doses of 0, 400, 1200, or 3600 ppm (equal to 0, 27, 76, and 256 mg/kg bw per day in males, 0, 30, 91, and 314 mg/kg bw per day in females) were administered. The NOAEL was 1200 ppm (equal to 76 mg/kg bw per day), based on weak evidence of liver toxicity (increased liver weight with associated clinical chemistry findings) and reductions in terminal body weight and body weight gain at 3600 ppm (equal to 256 mg/kg bw per day).

In a four-week range-finding study in dogs, the LOAEL was 125 mg/kg bw per day, the lowest dose tested, based on clinical signs, reduced body weight and indications of liver toxicity.

In a 90-day oral toxicity study in dogs, the dose levels were 0, 15, 90, or 180 mg/kg bw per day. The NOAEL was 180 mg/kg bw per day, the highest dose tested. Minor treatment-related effects on body and liver weight and liver function at lower dose levels were not considered adverse.

In a one-year oral toxicity study in dogs, doses of 0, 10, 30, or 150 mg/kg bw per day were administered. The NOAEL was 30 mg/kg bw per day, based on a lower body weight and body weight gain at 150 mg/kg bw per day, supported by limited evidence of liver toxicity in both sexes.

In the 18-month feeding study in mice, mefentrifluconazole was administered at dietary doses of 0, 20, 50 or 200 ppm (equal to 0, 3.5, 9.1 and 36 mg/kg bw per day) to males and of 0, 20, 50, or 250 ppm (equal to 0, 4.9, 12.6, and 61.5 mg/kg bw per day) to females. The NOAEL was 20 ppm (equal to 3.5 mg/kg bw per day) based on liver weight increase in both sexes, a higher frequency of macrovesicular fatty change in males and an increase in severity of fatty degeneration in male and female mice at 50 ppm (equal to 9.1 mg/kg bw per day). Increased follicular cell hyperplasia of thyroids was noted in high-dose animals which apparently progressed to thyroid follicular cell adenomas in a few animals in both sexes. Mechanistic studies conducted in liver *in vivo* and *in vitro* (see below) indicated that mefentrifluconazole, like other conazoles, could activate the CAR receptor. However, the evidence was against any induction of thyroxine-glucuronosyl transferase. In addition, there were a number of species and tissue differences that were not consistent with a MOA mediated by CAR alone for either the liver or the thyroid effects. Hence, the Meeting was unable to dismiss the potential human relevance of the thyroid tumours observed in the mouse and the NOAEL for carcinogenicity was 50 ppm (equal to 9.1 mg/kg bw per day).

In the two-year study in rats, mefentrifluconazole was administered at dietary concentrations of 0, 100, 600 or 3600 ppm (equal to 0, 4, 25, and 163 mg/kg bw per day in males, 0, 6, 38, and 302 mg/kg bw per day in females). The NOAEL for chronic toxicity was 600 ppm (equal to 25 mg/kg bw per day) based on lower body weight, higher relative liver weights and histopathological liver findings at 3600 ppm (equal to 163 mg/kg bw per day). There was no evidence of substance-related carcinogenicity in the rat. Accordingly, the NOAEL for carcinogenicity was 3600 ppm (equal to 163 mg/kg bw per day), the highest dose tested.

The meeting concluded that there was limited evidence that mefentrifluconazole is carcinogenic in mice but no evidence for its carcinogenicity in rats.

Mefentrifluconazole was tested for genotoxicity in an adequate range of studies *in vitro* and *in vivo*, all of which were negative.

The meeting concluded that mefentrifluconazole is unlikely to be genotoxic.

In view of the lack of genotoxicity, the absence of carcinogenicity in the rat and since the higher incidence of benign thyroid tumours in the mouse was confined to the highest dose levels, the Meeting concluded that mefentrifluconazole is unlikely to pose a carcinogenic risk to humans via exposure from the diet.

In a two-generation study, mefentrifluconazole was administered to rats at dietary concentrations that were weekly adjusted to keep constant dose levels of 0, 25, 75, or 200 mg/kg bw per day over the whole course of the study. The NOAEL for parental effects was 25 mg/kg bw per day, based on increased liver weight and associated clinical chemistry findings at 75 mg/kg bw per day. A (conservative) reproductive toxicity NOAEL of 75 mg/kg bw per day was based on lower fertility amongst the high-dose (200 mg/kg bw per day) F₁ females, and perinatal litter losses in both generations, suggesting a possible adverse effect on certain reproductive parameters that may be related to parental toxicity. The offspring NOAEL was 75 mg/kg bw per day, based on a lower pup body weight gain and a slight developmental delay at 200 mg/kg bw per day.

In a developmental study in rats, mefentrifluconazole was administered by oral gavage at dose levels of 0, 50, 150, or 400 mg/kg bw per day. The maternal NOAEL was 150 mg/kg bw per day based on reduced body weight gain and food consumption at 400 mg/kg bw per day. In addition, mean placental weight was higher in this group. The highest dose level of 400 mg/kg bw per day was considered the developmental NOAEL. There was no increase in malformations.

In a range-finding study in female rabbits (non-pregnant), mefentrifluconazole appeared more toxic than in other species since all animals receiving doses of 50 mg/kg bw per day or greater died or

had to be killed for humane reasons. A single death was also noted after repeated administration of 25 mg/kg bw per day. Even though these fatalities cannot be considered acute effects, clinical signs were observed after only very few doses. These signs included a marked reduction in food intake, missing defaecation and steady weight loss. Based on the mainly gastrointestinal nature of these signs and the pathological findings in the decedents, this apparently treatment-related effect was described as "mucoid enteropathy". On the other hand, pregnant does in a subsequent range-finding study tolerated administration of up to 20 mg/kg bw per day without any adverse effects, which tends to suggest a steep dose-response curve. In the main developmental study in rabbits, mefentrifluconazole was administered by oral gavage at doses of 0, 5, 15, and 25 mg/kg bw per day. Since no adverse effects on the does or the fetuses were observed, 25 mg/kg bw per day, the highest dose tested, is both the maternal and the developmental NOAEL.

The Meeting concluded that mefentrifluconazole is not teratogenic.

In an acute neurotoxicity study in rats in which gavage doses of 0, 200, 600, and 2000 mg/kg bw were administered, a transient reduction in body weight gain, unsteady gait or reduced forelimb grip strength were observed at 2000 mg/kg bw. Accordingly, the overall NOAEL was 600 mg/kg bw. Since these effects had disappeared within seven days after dosing and since there were no neuropathological findings, they were not considered specific neurotoxic effects. A separate neurotoxicity study with repeated administration was not submitted, but no concern for neurotoxicity was identified from the available studies.

The Meeting concluded that mefentrifluconazole is unlikely to be neurotoxic.

An immunotoxicity study was not submitted but no concern was identified from the available studies.

The Meeting concluded that mefentrifluconazole is unlikely to be immunotoxic.

A number of mechanistic studies were conducted to further investigate the effects on the endocrine system and on the liver.

The few in vitro studies on the presumed endocrine disrupting potential of mefentrifluconazole were all on inhibition of human recombinant aromatase. They revealed a very weak inhibition of this enzyme by the parent compound ($IC_{50} = 0.6-0.9 \mu M$) whereas the metabolite M750F022 was inactive. The aromatase-inhibiting potential of mefentrifluconazole was by orders of magnitudes smaller than that of other compounds from the triazole group.

A number of studies on liver enzyme induction and cell proliferation were conducted in vivo in rats and in both wild type and PXR/CAR knock-out strains of mice, as well as in vitro. Some induction of a variety of CYP enzymes was observed. It seems that the liver effects of mefentrifluconazole were mainly CAR receptor-mediated, though PXR might be also involved, in contrast to the AH and PPAR receptors. It could also be shown that liver effects in rodents develop very early after the commencement of treatment. A final conclusion on the relevance of this to humans could not be drawn.

Toxicological data on metabolites and/or degradates

M750F022

Toxicological studies are only available for the metabolite M750F022 that was detected in livestock matrices but not in rat metabolism. This metabolite was devoid of genotoxic potential in an adequate range of in vitro tests. It is of low acute oral toxicity with an LD_{50} in the rat >2000 mg/kg bw.

In a 28-day feeding study in mice, the metabolite caused similar liver effects to the parent compound, but only at higher dose levels.

The Meeting concluded that M750F022 is of equal or lower toxicity than the parent compound and that the reference doses as established for mefentrifluconazole are applicable.

Other metabolites

For the metabolites M750F015, M750F016, M750F017, M750F038, and M750F039, negative *in silico* predictions for Ames mutagenicity and chromosome aberrations (clastogenicity and aneugenicity) were made in two independent, but complimentary, QSAR models. The same negative predictions were made for parent mefentrifluconazole and the metabolite M750F022, this being in line with the negative results obtained with both in experimental studies.

M750F015, M750F016 and M750F017 were considered structurally similar to the parent, and are present in rat metabolism (>10%, taking into account downstream metabolites). M750F019 was considered structurally similar to M750F016 and M750F017, and several other metabolites (M750F038, M750F043, M750F068) were considered structurally similar to metabolite M750F022.

Therefore, the Meeting concluded that the reference doses for mefentrifluconazole are applicable also to M750F015, M750F016, M750F017, M750F019, M750F038, M750F043, and M750F068, in case such a risk assessment should prove to be needed.

No genotoxicity concern was raised in the case of the metabolites M750F023, M750F024, and M750F025 due to their similarity to M750F022. However, it was noted that for these two metabolites the additional fatty acid side chain may cause differences in kinetics compared to M750F022, and therefore the Meeting concluded to use the TTC approach for non-genotoxic compounds of Cramer class III, that is 1.5 µg/kg bw per day. The same reasoning was adopted towards M750F072 which may be cleaved in the gastrointestinal tract to give the (presumably non-genotoxic) metabolite M750F039, but for which no further data was available. No conclusion on metabolite M750F034 could be drawn.

Microbiological data

No data for the impact of mefentrifluconazole on the human gut microbiome was available.

Human data

From observations in manufacturing personnel, no adverse health effects have been reported. As expected for a new compound, no information on poisoning incidents or other human exposure is available.

The Meeting concluded that the existing database on mefentrifluconazole was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting established an ADI of 0–0.04 mg/kg bw on the basis of the NOAEL of 3.5 mg/kg bw per day in the long-term study in mice. A safety factor of 100 was applied. The margin from this ADI to the dose level at which an increased incidence of thyroid tumours in mice was observed, was 900.

The Meeting established an ARfD of 0.3 mg/kg bw, based on the NOAEL of 25 mg/kg bw per day for maternal and developmental toxicity in the main developmental toxicity study in the rabbit. A safety factor of 100 was applied.

A toxicological monograph was prepared.

Levels relevant to risk assessment of mefentrifluconazole

Species	Study	Effect	NOAEL	LOAEL
Mouse	78-week study of toxicity and carcinogenicity ^a	Toxicity	20 ppm, equal to 3.5 mg/kg bw per day	50 ppm, equal to 9.1 mg/kg bw per day
		Carcinogenicity	50 ppm, equal to 9.1 mg/kg bw per day	200 ppm, equal to 36 mg/kg bw per day

Species	Study	Effect	NOAEL	LOAEL
Rat	Acute neurotoxicity study ^b	Neurotoxicity	600 mg/kg bw	2000 mg/kg bw
	Two-year studies of toxicity and carcinogenicity ^a	Toxicity	600 ppm, equal to 25 mg/kg bw per day	3600 ppm, equal to 163 mg/kg bw per day
		Carcinogenicity	3600 ppm, equal to 163 mg/kg bw per day ^c	-
	Two-generation study of reproductive toxicity ^a	Reproductive toxicity	75 mg/kg bw per day	200 mg/kg bw per day
		Parental toxicity	25 mg/kg bw per day	75 mg/kg bw per day
		Offspring toxicity	75 mg/kg bw per day	200 mg/kg bw per day
	Developmental toxicity study ^b	Maternal toxicity	150 mg/kg bw per day	400 mg/kg bw per day
		Embryo and fetal toxicity	400 mg/kg bw per day ^c	-
Rabbit	Developmental toxicity study ^b	Maternal toxicity	25 mg/kg bw per day ^c	-
		Embryo and fetal toxicity	25 mg/kg bw per day ^c	-
Dog	One-year study ^d	Toxicity	30 mg/kg bw per day	150 mg/kg bw per day
Metabolite M750F022				
Mouse	Four-week study of toxicity ^a	Toxicity	87 ppm, equal to 20 mg/kg bw per day	872 ppm, equal to 180 mg/kg bw per day

^a Dietary administration^b Gavage administration^c Highest dose tested^d Capsule administration

Acceptable daily intake (ADI) applies to mefentrifluconazole and its metabolites M750F015, M750F016, M750F017, M750F019, M750F022, M750F038, M750F043, and M750F068, expressed as mefentrifluconazole:

0–0.04 mg/kg bw

Acute reference dose (ARfD) applies to mefentrifluconazole and its metabolites M750F015, M750F016, M750F017, M750F019, M750F022, M750F038, M750F043, and M750F068, expressed as mefentrifluconazole:

0.3 mg/kg bw

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other observational studies of human exposure.

Critical end-points for setting guidance values for exposure to mefentrifluconazole

Absorption, distribution, excretion and metabolism in mammals

Rate and extent of oral absorption

Rapid ($T_{\max}$ 0.5–1.2 h at low dose of 5 mg/kg bw);
≥80% at low dose, 50–70% at high dose of 180 mg/kg bw

Dermal absorption

No data

Distribution	Widely distributed; highest residues in liver, gut and stomach and their contents
Potential for accumulation	Not observed
Rate and extent of excretion	Nearly complete within 3 days, mainly via faeces ($\geq 80\%$); biliary excretion accounting for elimination of most of absorbed dose (40–60%), urine less important (5–15%)
Metabolism in animals	Extensive with up to 68 metabolites occurring in the rat in the different matrices; mainly by hydroxylation, methylation and cleavage of ether bond or triazole ring, followed by conjugations
Toxicologically significant compounds in animals and plants	Parent compound and livestock metabolite M750F022 (others may be identified by FAO evaluation in 2022)
Acute toxicity	
Rat, LD ₅₀ , oral	>2000 mg/kg bw
Rat, LD ₅₀ , dermal	>5000 mg/kg bw
Rat, LC ₅₀ , inhalation	>5.3 mg/L
Rabbit, dermal irritation	Not irritating
Rabbit, ocular irritation	Not irritating
Dermal sensitization	Sensitizing (Magnusson & Kligman test)
Short-term studies of toxicity	
Target/critical effect	Liver (organ weight increase, histopathological lesions, clinical chemistry alterations) in all species; body weight loss in rats and dogs
Lowest relevant oral NOAEL	11 mg/kg bw per day (mouse)
Lowest relevant dermal NOAEL	1000 mg/kg bw per day (rat)
Lowest relevant inhalation NOAEC	No data
Long-term studies of toxicity and carcinogenicity	
Target/critical effect	Liver (histopathology, clinical chemistry organ weight) and reduced body weight in rodents; in addition thyroid (histopathology) only in mice
Lowest relevant NOAEL	3.5 mg/kg bw per day (mouse)
Carcinogenicity	Not carcinogenic in rats, carcinogenic in mice (thyroid adenoma) ^a
Genotoxicity	No evidence of genotoxicity in vitro or in vivo
Reproductive toxicity	
Target/critical effect	Reproductive toxicity: possible impact on female fertility and increase in perinatal losses Offspring toxicity: low pup weight and developmental delay Parental toxicity: increased liver weight, clinical chemistry
Lowest relevant parental NOAEL	25 mg/kg bw per day
Lowest relevant offspring NOAEL	75 mg/kg bw per day
Lowest relevant reproductive NOAEL	75 mg/kg bw per day

Developmental toxicity	
Target/critical effect	Maternal: decreased food intake and body weight loss, mortality and gastrointestinal signs in range-findings studies in rabbits Developmental: none
Lowest relevant maternal NOAEL	25 mg/kg bw per day, highest dose tested in main study (rabbit)
Lowest relevant embryo/fetal NOAEL	25 mg/kg bw per day, highest dose tested in main study (rabbit)
Neurotoxicity	
Acute neurotoxicity NOAEL	No evidence of a specific neurotoxic potential up to 2000 mg/kg bw
Subchronic neurotoxicity NOAEL	No data, but no evidence from routine studies
Developmental neurotoxicity NOAEL	No data
Phototoxicity	
No data, no concern from routine studies	
Mechanistic studies	
Negative in vitro; Early occurring liver effects (more pronounced in mice) most likely CAR-mediated; high cytotoxicity in vitro confirmed; low potential for aromatase inhibition in vitro (in particular as compared to other triazole fungicides)	
Studies on toxicologically relevant metabolites	
M750F022 Acute oral toxicity (rat): LD ₅₀ >2000 mg/kg bw; 28-day feeding study in mice: NOAEL 20 mg/kg bw per day, liver and possible kidney identified as target organs; Genotoxicity: negative in vitro (experiment and QSARs); Aromatase inhibition: much lower effect than with parent M750F015, M750F016, M750F017, M750F038, M750F039 Negative in silico predictions for genotoxicity	
Human data	
Not available for this new compound	

^a Unlikely to pose a carcinogenic risk to humans via exposure from the diet

Summary

	Value	Study	Safety factor
ADI	0.04 mg/kg bw ^a	18-month carcinogenicity study, mouse	100
ARfD	0.3 mg/kg bw ^a	Developmental toxicity, rabbit	100

^a Applies to mefentrifluconazole and its metabolites M750F015, M750F016, M750F017, M750F019, M750F022, M750F038, M750F043, and M750F068, expressed as mefentrifluconazole

5.9 Metalaxyl (138)

TOXICOLOGY

Metalaxyl is a 1:1 mixture of (*R*)-2-[(2,6-dimethylphenyl)methoxyacetylamino]propionic acid methyl ester (*R*-enantiomer) and (*S*)-2-[(2,6-dimethylphenyl)methoxyacetylamino]propionic acid methyl ester (*S*-enantiomer). Metalaxyl-M is the ISO common name for methyl *N*-(2,6-dimethylphenyl)-*N*-(methoxyacetyl)alaninate (IUPAC). Metalaxyl-M is a single enantiomer (*R* enantiomer) of the racemic mixture metalaxyl. The two compounds are fungicides used in agriculture, horticulture, and forestry, and they act by inhibiting mycelial growth and spore formation. They are of the phenylamide class of pesticides and are thought to act by inhibition of RNA polymerase I.

The toxicity of metalaxyl and of metalaxyl-M was evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) in 1982. Investigations with metalaxyl-M were confined to studies of its absorption, distribution, metabolism and excretion, acute and short-term toxicity, mutagenicity and developmental toxicity, and were designed to establish whether there are qualitative or quantitative differences between the toxicological properties of metalaxyl-M and metalaxyl. None of the studies revealed any unexpected effects of metalaxyl-M, and the quantitative dose–effect relationships found with the racemate and the *R*-enantiomer were similar. Therefore, the 2002 Meeting concluded that the database on metalaxyl could be used for the toxicological evaluation of metalaxyl-M. The JMPR Meeting in 2002 established a group ADI for metalaxyl and metalaxyl-M (alone or in combination) of 0–0.08 mg/kg body weight (bw) per day on the basis of a NOAEL of 8 mg/kg bw per day in a two-year toxicity study in dogs with metalaxyl, and applying a safety factor of 100. The 2002 Meeting concluded that it was not necessary to establish an acute reference dose (ARfD).

Metalaxyl and metalaxyl-M were reviewed by the present Meeting under the periodic review programme of the Codex Committee on Pesticide Residues (CCPR). The only study submitted since the previous JMPR evaluation of metalaxyl and metalaxyl-M was a study of the absorption and distribution of metalaxyl-M following 14-day, repeat-dose administration.

The evaluation of the biochemical and toxicological aspects of metalaxyl and metalaxyl-M was based on previous JMPR evaluations; they have been re-evaluated and updated in this report to meet current scientific standards.

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with relevant national or international test guidelines, unless otherwise specified. A search of the open literature did not reveal any relevant publications that would have an impact on the evaluation.

Biochemical aspects

Studies of the biokinetics and metabolism of both metalaxyl and metalaxyl-M have been performed. Absorption, distribution and excretion of the two compounds were similar, and both were rapidly absorbed and eliminated after oral administration with no potential for accumulation observed. In rats, maximum blood concentrations of radioactivity were detected 0.5–1 hour after administration for both metalaxyl and metalaxyl-M, irrespective of dose, except that the maximum concentrations for females in the higher dose group of metalaxyl occurred at 4 h. The decline in radioactivity was biphasic, with half-lives of 1.5 h and 125 h in males, and 3 h and 96 h in females for the first and second phases, respectively. Rats eliminated 90–100% of the total administered dose of either substance within 72 hours, with the majority eliminated within 24 hours. The extent of urinary excretion of radioactivity was about 1.5 times higher in females than in males, whereas the faecal elimination extent was about 1.5-fold to 2-fold higher in males than in females. The similarity in excretion pattern of radioactivity after oral and intravenous administration of metalaxyl indicates that the compound was well absorbed. In a study with bile duct-cannulated rats, elimination of metalaxyl in the bile was substantial. Approximately 66–71% of the administered dose (AD) was excreted in bile within 24 hours in males and females at the low dose of

2 mg/kg bw and in males at the high dose of 80 mg/kg bw. At this dose, excretion was 54.5% in females. Based on the biliary excretion study, oral absorption of metalaxyl is estimated to be greater than 90% within 24 hours, except in high-dose females where it is 70%. The concentration of residues of both compounds in organs and tissues was about 0.01 ppm at seven days after the low dose. The pattern of distribution was similar at the higher dose but approximately 100-fold greater.

The total residues remaining after seven days at the low dose represented 0.16–0.55% of the AD for both metalaxyl and metalaxyl-M in males and females. The total residues remaining after seven days at the high dose amounted to 0.17–0.43% of AD for metalaxyl-M and metalaxyl in males and females. The terminal half-life for tissue depletion was variable, with the shortest estimate in the plasma and longest in the lungs, 1.0 day and 37.4 days respectively. The terminal half-lives for brain, heart, kidneys, liver, muscle, spleen and whole blood were 17.11, 24.11, 18.44, 18.35, 10.06, 24.12 and 7.41 days, respectively.

Metalaxyl and metalaxyl-M were both extensively metabolized, showing a similar pattern of metabolites irrespective of sex or administered dose. Metabolism involved hydrolysis of side-chains and oxidation of the phenyl ring. Most of the phase I metabolites were excreted as conjugates with glucuronic acid and sulfate. Treatment with metalaxyl resulted in modest induction of hepatic and renal P450 enzymes and some other drug-metabolizing enzymes.

Toxicological data

Median lethal dose (LD₅₀) values in rats treated orally with metalaxyl-M were 953 and 375 mg/kg bw for males and females, respectively. The LD₅₀ value in rats after dermal application was >2000 mg/kg bw. Median lethal concentration (LC₅₀) values in rats treated by inhalation for four hours was >2.3 mg/L, the highest achievable concentration. Metalaxyl was non-irritating to the skin of rabbits but severely irritating to the eyes of rabbits. It was negative for sensitization in Guinea pigs in Magnusson & Kligman and Buehler tests.

Metalaxyl-M and metalaxyl showed similar toxicological properties. A comparative 28-day study in rats given metalaxyl-M and metalaxyl by gavage confirmed the toxicological equivalence of the *R*-enantiomer and the racemate, as both the nature of their effects and their dose-effect relationships were similar. In a comparative toxicity study in rats via gavage for 28 days, the NOAEL was 300 mg/kg bw per day, the highest dose tested for metalaxyl and metalaxyl-M.

In studies with mice, rats and dogs treated orally, both substances had low toxicity, and treatment was well tolerated, even at relatively high doses. Dog, where red blood cells were the target, was the most sensitive species. Increased liver weights were observed in rats and dogs, but in this case the effect was considered not to be adverse.

In a 28-day toxicity study in rats via gavage, metalaxyl at escalating doses up to 600 mg/kg bw per day produced liver effects (increased absolute and relative liver weights) which were not considered adverse.

In a 90-day toxicity study with rats, metalaxyl-M was administered via the diet at concentrations of 0, 25, 50, 250, 625 or 1250 ppm (equal to 0, 1.7, 3.5, 17, 45 and 91 mg/kg bw per day for males, 0, 1.9, 3.7, 18, 49 and 95 mg/kg bw per day for females). The NOAEL was 1250 ppm (equal to 91 mg/kg bw per day) the highest dose tested.

In a separate 90-day toxicity study in rats, metalaxyl was administered via the diet at a concentration of 0, 50, 250 or 1250 ppm (equal to 0, 3.2, 16 and 79 mg/kg bw per day for males, 0, 3.5, 18 and 86 mg/kg bw per day for females). The NOAEL was 1250 ppm (equal to 79 mg/kg bw per day) the highest dose tested. Liver cell hypertrophy observed at 1250 ppm was considered not to be adverse.

The toxicity of metalaxyl-M in dogs was investigated in a 13-week dietary administration study. Metalaxyl was tested in a six-month study with administration via the diet and in a two-year study with administration via capsules.

In a 90-day toxicity study in dogs, metalaxyl-M was administered via the diet at a concentration of 0, 50, 125, 250 or 1250 ppm (equal to 0, 1.6, 4.1, 7.3 and 39 mg/kg bw per day for males, 0, 1.6, 4.3, 7.9 and 40 mg/kg bw per day for females). The NOAEL was 1250 ppm (equal to 39 mg/kg bw per day) the highest dose tested, as the increased alkaline phosphatase activity and increased liver weights seen at 1250 ppm were not considered adverse.

In a six-month toxicity study in dogs, metalaxyl was administered via the diet at a concentration of 0, 50, 250 or 1000 ppm (equal to 0, 1.6, 7.8 and 31 mg/kg bw per day for males, 0, 1.7, 7.4 and 32 mg/kg bw per day for females). The NOAEL was 250 ppm (equal to 7.8 mg/kg bw per day), on the basis of decreases in red blood cell (RBC) count, haematocrit and haemoglobin in males (marginal but consistent throughout the study and also observed in a two-year toxicity study in dogs) at 1000 ppm (equal to 31 mg/kg bw per day).

In a two-year toxicity study in dogs, metalaxyl was administered in gelatine capsules at a dose of 0, 0.8, 8 or 80 mg/kg bw per day. The NOAEL was 8 mg/kg bw per day based on RBC toxicity at the LOAEL of 80 mg/kg bw per day.

In a study of carcinogenicity in mice, metalaxyl was administered via diet for 104 weeks at a concentration of 0, 50, 250 or 1250 ppm (equal to 0, 4.5, 22.7 and 119 mg/kg bw per day for males, 0, 5.0, 21.5 and 132 mg/kg bw per day for females). The NOAEL for systemic toxicity was 1250 ppm (equal to 119 mg/kg bw per day) the highest dose tested. The NOAEL for carcinogenicity was 1250 ppm (equal to 119 mg/kg bw per day) the highest dose tested.

In a study of chronic toxicity and carcinogenicity in rats, metalaxyl was administered for 105 weeks via the diet at a concentration of 0, 50, 250 or 1250 ppm (equal to 0, 1.7, 8.7 and 43 mg/kg bw per day for males, 0, 2, 10 and 55 mg/kg bw per day for females). The NOAEL for systemic toxicity was 1250 ppm (equal to 43 mg/kg bw per day) the highest concentration tested; the mild liver changes seen at this concentration were considered not to be adverse. The NOAEL for carcinogenicity was 1250 ppm (equal to 43 mg/kg bw per day) the highest dose tested.

Since technical grade metalaxyl contains approximately 50% of the *R*-enantiomer, the results also apply to metalaxyl-M.

The Meeting concluded that metalaxyl and metalaxyl-M are not carcinogenic in mice or rats.

Metalaxyl was tested for genotoxicity in an adequate range of in vitro and in vivo assays. No evidence of genotoxicity was found.

The Meeting concluded that metalaxyl and metalaxyl-M are unlikely to be genotoxic.

In view of the lack of carcinogenicity in mice and rats, and that both compounds are unlikely to be genotoxic, the Meeting concluded that metalaxyl and metalaxyl-M are unlikely to pose a carcinogenic risk to humans.

In a three-generation study of reproductive toxicity in rats, metalaxyl was administered via the diet at concentrations of 0, 50, 250 or 1250 ppm. This resulted in intakes of 4.1 (3.2–4.7), 21 (17–23) and 96 (84–100) mg/kg bw per day for males, and for females 4.6 (3.6–5.1), 24 (20–27) and 150 (100–240) mg/kg bw per day. The NOAEL for parental systemic toxicity, reproductive toxicity and offspring toxicity was 1250 ppm (equal to 96 mg/kg bw per day) the highest dose tested.

In a developmental toxicity study in rats, metalaxyl-M was administered once daily by gavage at 0, 10, 50 or 250 mg/kg bw per day on gestation days (GDs) 6–15. The NOAEL for maternal toxicity was 50 mg/kg bw per day, on the basis of a reduction in body weight gain and feed consumption at the LOAEL of 250 mg/kg bw per day. The NOAEL for embryo/fetal toxicity was 250 mg/kg bw per day, the highest dose tested.

In a separate developmental toxicity study in rats, metalaxyl was administered once daily by gavage at 0, 50, 250 or (initially) 575 mg/kg bw per day on GDs 6–15. Owing to maternal deaths, the highest dose was reduced to 500 and then to 400 mg/kg bw per day after two days of treatment.

The NOAEL for maternal toxicity was 50 mg/kg bw per day, on the basis of a reduction in body weight gain at higher doses and clinical signs (including convulsions), the majority of which occurred at the beginning of the treatment period (GD 6 and/or GD 7) at 250 mg/kg bw per day. The NOAEL for developmental toxicity was 400 mg/kg bw per day, the highest dose tested.

In a developmental toxicity study in rabbits, metalaxyl was administered once daily by gavage at 0, 30, 150 or 300 mg/kg bw per day on GDs 7–19. The NOAEL for maternal toxicity was 150 mg/kg bw per day, on the basis of reductions in body weight gain and food consumption at the LOAEL of 300 mg/kg bw per day. The NOAEL for embryo/fetal toxicity was 300 mg/kg bw per day, the highest dose tested.

The Meeting concluded that metalaxyl and metalaxyl-M are not teratogenic.

No neurotoxicity studies were available, however no evidence of neurotoxicity or neuropathology were observed in any of the systemic toxicity studies.

The Meeting concluded that metalaxyl and metalaxyl-M are unlikely to be neurotoxic.

No evidence of direct immunotoxic effects was observed in the available toxicity studies.

The Meeting concluded that metalaxyl and metalaxyl-M are unlikely to be immunotoxic.

Toxicological data on metabolites and/or degradates

The metabolism of metalaxyl and metalaxyl-M is similar in animals and plants. There appear to be three major plant metabolites of metalaxyl:

- N-(2-6-dimethylphenyl)-N-(methoxyacetyl)alanine (M-1; CGA62826),
- N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine (M-6; CGA107955) and
- N-(2 - carboxy-6-methylphenyl)-N-methoxyacetyl)alanine (M-12; CGA108906)

Their acute toxicity was studied after oral administration. These metabolites showed little toxicity, with LD₅₀ values >2000 mg/kg bw.

NOAELs in 28-day studies were 1000 mg/kg bw per day, the highest doses tested for CGA108906 and CGA62826.

Metabolites CGA108906 and CGA62826 had no mutagenic potential in vitro.

Metabolites CGA108906 (M-12) and CGA62826 (M-1; found in rat metabolism study) are expected to be no more toxic than the parent compound based on available toxicity studies.

The Meeting concluded that the toxicity of plants and animal metabolites M-1, M-3, M-6, M-7, M-8 and M-12 would be covered by the acceptable daily intake (ADI) and acute reference dose (ARfD) established for metalaxyl.

Microbiological data

No data were available.

Human data

No cases of adverse effects were reported in personnel involved in the production and formulation of metalaxyl or metalaxyl-M or in the field use of these products.

The Meeting concluded that the existing database was adequate to characterize the potential hazards of metalaxyl and metalaxyl-M to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting re-affirmed the group ADI of 0–0.08 mg/kg bw for metalaxyl and metalaxyl-M (alone or in combination) established by the 2002 Meeting on the basis of the NOAEL of 8 mg/kg bw per day in the

two-year toxicity study in dogs, based on red blood cell toxicity at the LOAEL of 80 mg/kg bw per day, and using a safety factor of 100.

The Meeting established a group ARfD of 0.5 mg/kg bw for metalaxyl and metalaxyl-M (alone or in combination) on the basis of the NOAEL of 50 mg/kg bw per day observed in the developmental toxicity study in rats with metalaxyl, based on ataxia, tremors, hypoactivity and hypothermia, and using a safety factor of 100. In addition, clinical signs of toxicity were observed consistently in acute oral toxicity studies and after the first dose in a number of repeat-dose toxicity studies, with a LOAEL of around 200 mg/kg bw. The ARfD applies to all age groups.

A toxicological monograph was prepared.

Levels relevant to risk assessment of metalaxyl and metalaxyl-M

Species	Study	Effect	NOAEL	LOAEL
Mouse	104-week study of toxicity and carcinogenicity ^a	Toxicity	1250 ppm, equal to 119 mg/kg bw per day ^d	-
		Carcinogenicity	1250 ppm, equal to 119 mg/kg bw per day ^d	-
Rat	Two-year study of toxicity and carcinogenicity ^a	Toxicity	1250 ppm, equal to 43 mg/kg bw per day ^d	-
		Carcinogenicity	1250 ppm, equal to 43 mg/kg bw per day ^d	-
	Three-generation study of reproductive toxicity ^a	Reproductive toxicity	1250 ppm, equal to 96 mg/kg bw per day ^d	-
		Parental toxicity	1250 ppm, equal to 96 mg/kg bw per day ^d	-
		Offspring toxicity	1250 ppm, equal to 96 mg/kg bw per day ^d	-
	Developmental toxicity study ^b	Maternal toxicity	50 mg/kg bw per day	250 mg/kg bw per day
		Embryo and fetal toxicity	250 mg/kg bw per day ^d	-
Rabbit	Developmental toxicity study ^b	Maternal toxicity	150 mg/kg bw per day	300 mg/kg bw per day
		Embryo and fetal toxicity	300 mg/kg bw per day ^d	-
Dog	Six-month studies of toxicity ^a	Toxicity	250 ppm, equal to 7.8 mg/kg per day	1000 ppm, equal to 31 mg/kg bw per day
	Two-year studies of toxicity ^c	Toxicity	8 mg/kg bw per day	80 mg/kg bw per day ^d
Metabolite CGA 62826 (M-1)				
Rat	Four-week study of toxicity ^b	Toxicity	1000 mg/kg bw per day ^d	-
Metabolite CGA 108906 (M-12)				
Rat	Four-week study of toxicity ^b	Toxicity	1000 mg/kg bw per day ^d	-

^a Dietary administration

^b Gavage administration

^c Capsule administration

^d Highest dose tested

Acceptable daily intake (ADI); group ADI for metalaxyl and metalaxyl-M (alone or in combination) and for M-1, M-3, M-6, M-7, M-8 and M-12.

0–0.08 mg/kg bw

Acute reference dose (ARfD); group ARfD for metalaxyl and metalaxyl-M (alone or in combination) and for M-1, M-3, M-6, M-7, M-8 and M-12.

0.5 mg/kg bw

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Critical end-points for setting guidance values for exposure to metalaxyl and metalaxyl-M

Absorption, distribution, excretion and metabolism in mammals

Rate and extent of oral absorption	About 70% excreted within 24 h; >90% oral absorption based on biliary excretion study
Dermal absorption	No data provided
Distribution	Widely distributed
Potential for accumulation	None
Rate and extent of excretion	95–100% after 72 hours; slightly higher urinary excretion for females
Metabolism in animals	Extensive metabolism via hydrolysis of the methyl ester and methyl ether groups; oxidation of the 2-(6)-methyl group; oxidation of the phenyl ring; <i>N</i> -dealkylation. Conjugation to glucuronides and sulfates
Toxicologically significant compounds in animals and plants	Parent compound

Acute toxicity

Rat, LD ₅₀ , oral	375 mg/kg bw
Rat, LD ₅₀ , dermal	>2000 mg/kg bw
Rat, LC ₅₀ , inhalation	>2.3 mg/L
Rabbit, dermal irritation	Not irritating
Rabbit, ocular irritation	Severely irritating
Guinea pig, dermal sensitization	Not sensitizing (Magnusson & Kligman and Buehler)

Short-term studies of toxicity

Target/critical effect	Red blood cells
Lowest relevant oral NOAEL	8 mg/kg bw per day (six-month dietary study and two-year capsule study)
Lowest relevant dermal NOAEL	1000 mg/kg bw per day (28-day rat and rabbit, the highest dose tested)
Lowest relevant inhalation NOAEC	No data

Long-term studies of toxicity and carcinogenicity

Target/critical effect	None
Lowest relevant NOAEL	34 mg/kg bw per day; the highest dose tested (rat)

Carcinogenicity	Not carcinogenic in mice and rats ^a
Genotoxicity	No evidence of genotoxicity ^a
Reproductive toxicity	
Target/critical effect	No reproductive effects were observed
Lowest relevant parental NOAEL	96 mg/kg bw per day; the highest dose tested (rat)
Lowest relevant offspring NOAEL	96 mg/kg bw per day; the highest dose tested (rat)
Lowest relevant reproductive NOAEL	96 mg/kg bw per day; the highest dose tested (rat)
Developmental toxicity	
Target/critical effect	Not teratogenic in rats and rabbits (clinical signs, reduced body weight gains and mortality)
Lowest relevant maternal NOAEL	50 mg/kg bw per day (rat)
Lowest relevant embryo/fetal NOAEL	250 mg/kg bw per day; the highest dose tested (rat)
Neurotoxicity	
Acute neurotoxicity NOAEL	No data
Subchronic neurotoxicity NOAEL	No data
Developmental neurotoxicity NOAEL	No data
Immunotoxicity	No data
Mechanism studies	Weak inducer liver, kidney and lungs
Studies on toxicologically relevant metabolites	
CGA 108906 (M-12)	
Acute oral LD ₅₀	>2000 mg/kg bw (rat)
28-day toxicity	1000 mg/kg bw per day; the highest dose tested (rat)
Genotoxicity	Negative (in vitro)
CGA 62828 (M-1)	
Acute oral LD ₅₀	>2000 mg/kg bw (rat)
28-day toxicity	1000 mg/kg bw per day; the highest dose tested (rat)
Genotoxicity	Negative (in vitro)
CGA 107955 (M-6)	
Acute oral LD ₅₀	>3000 mg/kg bw (rat)
Microbial data	Not applicable
Human data	No adverse effects on health of manufacturing personnel

^a Unlikely to pose a carcinogenic risk to humans via exposure from the diet.

Summary

	Value	Study	Safety factor
ADI ^a	0–0.08 mg/kg bw	Two-year study of toxicity in dogs	100
ARfD ^a	0.5 mg/kg bw	Developmental toxicity study in rats	100

^a Applies to metalaxyl and metalaxyl-M (alone or in combination, and to M-1, M-3, M-6, M-7, M-8 and M-12).

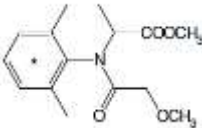
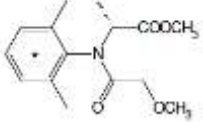
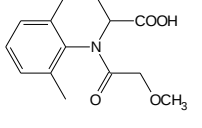
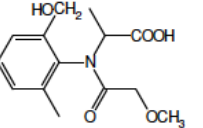
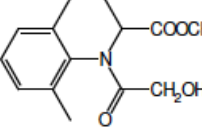
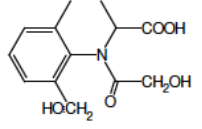
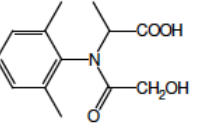
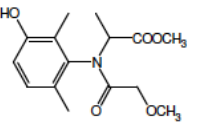
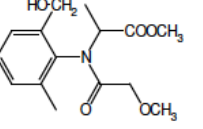
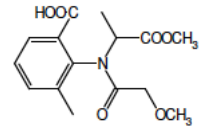
RESIDUE AND ANALYTICAL ASPECTS

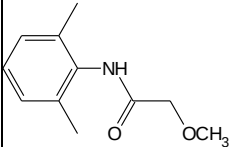
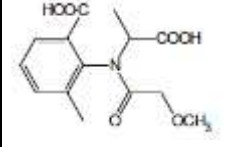
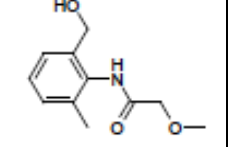
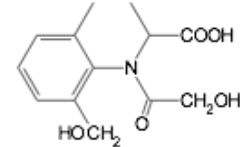
Metalaxyl and metalaxyl-M are systemic fungicides. Metalaxyl is a racemic mixture of R- and S-enantiomers, where the R-enantiomer is the biologically active form, and metalaxyl-M consists of a minimum of 97% of the R-enantiomer.

Metalaxyl and metalaxyl-M were first evaluated by the JMPR for residues in 1982 and 2004, respectively, and were evaluated together for toxicology in 2002. In 2008, CCPR agreed to retain all existing CXLs for metalaxyl, and to retain all the draft MRLs for metalaxyl-M, at Step 7, awaiting the JMPR periodic review of metalaxyl for residues and toxicology. The Forty-ninth Session of the CCPR scheduled metalaxyl for periodic review evaluation and metalaxyl-M for evaluation of new uses by the 2018 JMPR, which were postponed to the 2021 JMPR.

Available studies on metalaxyl included physical and chemical properties, metabolism, environmental fate and field rotation studies, analytical methods, GAP information, residue trials and monitoring data in pepper, black, white. Available studies on metalaxyl-M included animal and plant metabolism, rotational crop studies, analytical methods, GAP information and residue trials. Residue trials on metalaxyl-M submitted to the 2004 JMPR that were conducted according to the current GAP, and other relevant information were also considered by the current Meeting.

Table 1 The labelled parent compounds used in the studies and the main metabolites found in plants, livestock and soil are shown below.

 ^{*14}C Metalaxyl	 ^{*14}C Metalaxyl-M	 Metabolite 1 <i>N</i>-(2,6-dimethylphenyl)-<i>N</i>-(methoxyacetyl) alanine	 Metabolite 2 <i>N</i>-(2-hydroxymethyl-6-methylphenyl)-<i>N</i>-(methoxyacetyl)alanine	 Metabolite 3 <i>N</i>-(2,6-dimethylphenyl)-<i>N</i>-(hydroxyacetyl)alanine methyl ester
 Metabolite P1/P2 <i>N</i>-(2-hydroxymethyl)-6-methylphenyl]-<i>N</i>-(hydroxyacetyl)alanine	 Metabolite 6 <i>N</i>-(2,6-dimethylphenyl)-<i>N</i>-(hydroxyacetyl)alanine	 Metabolite 7 <i>N</i>-(2,6-dimethyl-5-hydroxyphenyl)-<i>N</i>-(methoxyacetyl)alanine methyl ester	 Metabolite 8 <i>N</i>-(2-hydroxymethyl-6-methylphenyl)-<i>N</i>-(methoxyacetyl)alanine methyl ester (2 isomers)	 Metabolite 9 <i>N</i>-(2-carboxy-6-methylphenyl)-<i>N</i>-(methoxyacetyl)alanine methyl ester

	 Metabolite 10 N-methoxyacetyl-2,6-dimethyl-aniline	 Metabolite 12 N-(2-carboxy-6-methylphenyl)-N-(methoxyacetyl)alanine	 SYN546555	 Metabolite P1/P2 N-(2-hydroxymethyl)-6-methylphenyl]-N-(hydroxyacetyl)alanine
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Plant metabolism

Metabolism studies were conducted with ^{14}C -metalaxyl or ^{14}C -metalaxyl-M on grape (2 studies), lettuce (2 studies) and potato (3 studies).

Field grown grapevines received 7 applications of [^{14}C] metalaxyl at 0.05 kg ai/hL, grape samples were collected 52 days after the last application (DALA) and processed to juice and presscake. Residues in juice samples were partitioned into hexane, dichloromethane (DCM) and ethyl acetate (EthAc). Residues in presscake and leaves were first extracted with methanol/water and partitioned into DCM and EthAc; leaf extracts were also submitted to clean-up with silica gel or anion exchange column before analysis. Each extract was analysed separately by TLC, GC-FID and/or GC-MS.

Over 90% of the radioactivity was extracted from grapes and leaves. TRRs were 0.9 mg eq/kg in juice, 1.7 mg eq/kg in presscake and 19.8 mg eq/kg in leaves. Metalaxyl accounted for 64.1% TRR in grapes (56.3% in presscake) and 22% TRR in leaves. The water extracts submitted to cellulase hydrolysis contained less than 10% TRR. Metabolite 8 (M8, free + conjugated), was the major metabolite, accounting for 20.4% TRR in whole grape (mostly in presscake) and 55.5% TRR in the leaves. Other metabolites included M7 (up to 13% in leaves), M1 and M6 (up to 6% TRR in the leaves). As the current GAP PHI for metalaxyl and/or metalaxyl-M in grapes is 7 or 20 days, a lower proportion of metabolites compared to the parent is expected when the products are used according to GAP.

In a glasshouse study conducted in lettuce, plants were treated with ^{14}C -metalaxyl at 2×0.25 kg ai/ha and samples harvested 14 DALA. Lettuce samples were extracted with methanol/water, partitioned with hexane and DCM, and the water-soluble residues treated with cellulase. The TRR were 5.5 mg eq/kg, of which 74% TRR were recovered in methanol/water (40.2% partitioned in organic solvent). Parent metalaxyl accounted for 18.6% TRR. M8 was the major metabolite found (25.1% TRR, mainly in the conjugated form), followed by M6 (10.1% free and conjugated). Another 6 metabolites were present at levels < 10% TRR (free and conjugated).

In a field study, lettuce was treated with either ^{14}C -metalaxyl-M or ^{14}C -metalaxyl at 3×0.2 kg ai/ha and samples harvested 0, 14 and 21 DALA. Samples were extracted with methanol/water, partitioned with hexane and DCM, and the water phases treated with glucosidase. At least 96% of the radioactivity was extracted by organic solvents.

At 14 DALA (relevant for the registered use), TRR were 2.4 and 1.8 mg eq/kg, respectively. Metabolite patterns were qualitatively similar for the 2 substances, with metabolites present mainly as conjugates. The amount of metalaxyl declined with time, and at 21 DALA accounted for 6.6 and 10.2% TRR after treatment with metalaxyl-M and metalaxyl, respectively. M8 (mostly conjugated) accounted for 57.1 and 34.7% TRR at 14 DALA, and 61.3 and 38.8% TRR at 21 DALA, in samples treated with metalaxyl-M and metalaxyl, respectively. Metabolite 1 accounted for 2.4–2.1% TRR (free) at 21 DALA.

Potato plants were treated in the field with ^{14}C -metalaxyl at 3×1.2 kg ai/ha (28 day re-treatment interval) and harvested 28 DALA. Leaves and tubers were extracted with methanol/water, the aqueous phase partitioned with methylene chloride and ethyl acetate at pH 6 and 3, and the water-soluble residues treated with cellulase.

TRR in tubers decreased from 2.47 mg eq/kg at day 0 to 0.14 mg eq/kg at 28 DALA, 22.7% of which was found in the organic phase, 32.1% in the aqueous phase and 38.6% not extracted. TRR in the potato stalks dropped from 84.1 mg eq/kg at day 0 to 7 mg eq/kg at 28 DALA (54% in the organic phase). Metalaxyl accounted for 8.5% TRR and M8 was the major metabolite (40% TRR) in stalks.

In another field study, potato received foliar treatment with ^{14}C -metalaxyl at 5×0.2 kg ai/ha, samples were harvested 47 DALA, extracted as described above and TLC and GC-MS used for characterization and identification of the residues.

The TRR in potato leaves was 2.35 mg eq/kg, with 77% being extracted in the organic fraction. Metalaxyl accounted for 2.7% TRR in the leaves, with M9 and M1 the major components (42.9 and 13.5% TRR, respectively). Residues in the tubers were 0.02 mg eq/kg, comprising mostly polar compounds, but were at concentrations too low to characterize.

In a third field study, potato plants were treated six times with ^{14}C -metalaxyl at 0.426 or 1.28 kg ai/ha (GAP rate is $3\text{--}4 \times 0.2$ kg ai/ha), with mature potatoes harvested 7 DALA and samples extracted and analysed as described previously.

In the leaves (TRR of 31.9 mg eq/kg), metalaxyl accounted for 2.2% TRR and M8 (free and conjugated) for 50.6% TRR. TRR in the tubers of plants treated at the highest rate were 0.5 mg eq/kg, with metalaxyl accounting for 57% TRR, and M8 for 4.1% TRR (free and conjugated). Metabolites M7, M1 and M9, M12 and M6 were also detected (< 3% TRR each).

Enantioselective metabolism of metalaxyl in plants

In the lettuce metabolism study with metalaxyl-M, the enantiomeric S/R ratio was stable over the whole time of the experiment while in plants sprayed with metalaxyl the S-enantiomer was always slightly dominating over the R-enantiomer. In a study conducted with tomato and cucumber treated with metalaxyl in the greenhouse and harvested at 28 DALA, the (-)-R-enantiomer of the parent was preferentially degraded. Formation of the metalaxyl acid enantiomers was also shown to be enantioselective, with the R-metalaxyl acid formed at a faster rate. In another study also conducted with metalaxyl in the greenhouse, the R-enantiomer dissipated faster (half-life of 2.8 days) than S-enantiomer (half-life of 4.1 days) in cucumber, while a preferential degradation of the S-form was found in spinach and pak-choi under field and greenhouse conditions. No stereoselective degradation was observed in cabbage.

In summary, metabolism of metalaxyl in plants involves mainly hydroxylation and oxidation of the ring methyl groups, to yield M8 and M9, hydroxylation of the phenyl ring to yield M7, hydrolysis of the methyl ester to yield M1, and conjugation, mainly with glucose. Parent compound was found in all crops, being the major residue in grape and potato tubers; M8 (free and conjugated) was the major component of the residues in lettuce (up to 71.9% TRR) and potato leaves (58.8% TRR), mainly in conjugated form. The enantiomeric ratio varied among the crops, being stable for lettuce treated with metalaxyl-M and cabbage treated with metalaxyl, the S-enantiomer slightly dominating the residues in lettuce, tomato and cucumber treated with metalaxyl or degrading faster in spinach and pakchoi.

Rotational crops

In a confined study, potatoes were treated at 6×0.45 kg ai/ha of ^{14}C -metalaxyl, and the potatoes harvested at 14 DALA. Over 80% of the radioactivity was extracted from the soil (14 DALA) with organic solvent or water, with TRR levels of 1.82 mg eq/kg at 7.5 cm layer and 0.05 mg eq/kg at 22.5 cm. Lettuce, oat, soya beans, maize and sugar beet were planted in this soil at ~250 DALA, and wheat at 28 DALA. About 65% of the radioactivity was extracted from wheat straw, with TRR of 0.56 mg eq/kg at 51 weeks after the last treatment made to potato plants; only 24% of the radioactivity was extracted from wheat grain (water phase), with TRR of 0.11 mg eq/kg. Only 35% of the radioactivity was extracted from oat straw, comprising mostly polar compounds, with TRR of 0.21 mg eq/kg; residues in oat grain were 0.09 mg eq/kg. About 66% of the residues were extracted from lettuce, with a TRR of 0.11 mg eq/kg at 51 weeks after treatment. Residues in maize samples (stalks, cob and grain) and in tops and roots of sugar

beet samples ranged from 0.02 to 0.06 mg eq/kg (extractability not reported). The organic solvent used in the extractions was not specified in the study.

In another study, rotational crops were planted after tobacco grown in soil treated at 3.36 kg ai/ha ¹⁴C-metalaxyl. Tobacco plants were removed, and lettuce, wheat, soya beans and sugar beet were planted 232 DALA. TRR in soil (232 DALA) ranged from 1 mg eq/kg (0–7.5 cm) to 0.6 mg eq/ha (15–20 cm). TRR in the plants ranged from 0.275 to 1.1 mg eq/kg in sugar beet, soya beans, wheat grains and lettuce, 3.61 mg eq/kg in beans stalk, and 7.2–7.8 mg eq/kg in wheat stalk and hulls. Crop samples were extracted with methanol/water, followed by enzyme and acid hydrolysis. Metalaxyl represented a minor component of the residues (0.1% TRR of the organic extract in wheat stalks, 3.3% TRR in sugar beet root, and 15.5% TRR in immature lettuce). Various metabolites were identified, with M8 the major metabolite found in wheat stalks (32.5% of the extracted residues, 25.7% TRR, mainly as glucose conjugate), and M1 the major metabolite in sugar beet root (20% of the extracted residues and 12.5% TRR).

In a third confined study, ¹⁴C-metalaxyl-M was applied to soil at 0.62 kg ai/ha and lettuce, radish and wheat planted at 30, 120 or 270 days plant back interval (PBI). Samples were extracted with methanol/water and acid hydrolysis and enzyme treatment were used to release conjugates. At 30 days PBI, TRR ranged from 0.03 mg eq/kg in radish root to 3.2 mg eq/kg in wheat forage.

Metalaxyl levels were higher at the 120 day PBI, accounting for > 10% TRR only in radish leaves; it was not detected in wheat grain at any PBI. M8 was the major compound in the crops, mainly in conjugated form (up to 21.1% TRR in mature lettuce at 120 days PBI), in radish leaves (18.8% TRR at 120 days PBI, free and conjugated), and in wheat forage, hay and straw at 30 days PBI (40, 21 and 12% TRR, mainly free). M5 was also a major metabolite in radish leaves and wheat forage, hay and straw at levels ranging from 13 to 45% TRR (in hay, 20% conjugated), at the 120 day PBI. Metabolite SYN546555, a breakdown product of M8, was a major metabolite in hay (40% TRR, 19% conjugated). At 30 days PBI, the concentration ratio of M8/metalaxyl-M was 10 for forage and 30 for straw. For wheat hay, this ratio could only be estimated at 120 days PBI (0.07).

Two field rotational studies were conducted in Europe in 2011–2013. Soil was treated with metalaxyl-M at 0.6 kg ai/ha (France, Germany, Spain and the UK) or with metalaxyl at 0.8 kg ai/ha (Spain and the UK), which are comparable to the seasonal rate recommended for most crops. Crop samples were extracted with methanol or acetonitrile, clean-up with SPE and analyzed by LC-MS/MS. In the metalaxyl-M study, crops were planted one month (spinach), two months (wheat or barley) or one year (carrots) later. Residues were < 0.01 mg/kg in most samples, and highest residues were found in cereal whole plants (up to 0.12 mg/kg at the 30 day PBI) and straw (up to 0.06 mg/kg). In the study conducted with metalaxyl, wheat, spinach and turnip were planted after 30, 70, 120 and 365 days PBI and harvested at different growth stages. Metalaxyl residues in spinach leaves were always < 0.01 mg/kg and were detected only in turnip whole plants and wheat forage up to 0.07 mg/kg at PBIs of 30 to 120 days. No metalaxyl residues were found in wheat grain.

In summary, with the exception of SYN546555, all the residues found in soil and in rotational crops were found in primary crops, indicating a similar degradation/metabolic pathway. The rotational studies suggest that residues of parent metalaxyl at levels higher than 0.01 mg/kg are unlikely to occur in rotational crops, taking into account the maximum annual application rate of 0.4 kg ai/ha from the uses considered by the current Meeting. Various metabolites were found in rotational crops, mainly M8.

Expected residues (metalaxyl + M8) in rotation crops (cereal forage and hay)

An M8/metalaxyl-M ratio of 10 and 30 were found in the confined rotational crop study (30 day PBI) for forage and straw, respectively.

The highest application rate for metalaxyl or metalaxyl-M is 2×0.75 kg ai/ha (cabbage in South Korea), which gives a GAP rate adjustment related to the field rotational study of 2.5.

Highest residues of metalaxyl in whole wheat or barley plants in the four field trials were 0.04, 0.05, 0.09 and 0.12 mg/kg.

Highest residues of metalaxyl in wheat or barley straw in the four field trials were < 0.01, 0.02, 0.03 and 0.06 mg/kg.

Considering the M8/metalaxyl ratios, the GAP rate adjustment, and the residues found in the field rotational study, the Meeting estimated median and highest residues for metalaxyl (sum of enantiomers) and M8 (free and conjugated; sum of enantiomers), expressed as metalaxyl of 0.192 and 3.3 mg/kg for cereal forage, respectively, and median and highest residues of 1.94 and 4.65 mg/kg for cereal straw, respectively.

Animal metabolism

Rat

Metalaxyl and metalaxyl-M were evaluated together by the WHO Core Assessment Group of the 2021 JMPR. Four main metabolic pathways were identified: (i) hydrolytic cleavage of the carboxyl methyl ester group, (ii) hydrolytic (or oxidative) cleavage of the methyl ether moiety, (iii) oxidation of the toluene methyl side-chain to the benzylic alcohol derivative and (iv) oxidation of the phenyl moiety to form phenols. Secondary biotransformation pathways are N-dealkylation at the 2-aniline propionic acid bond, oxidation of benzylic alcohol to the benzoic acid derivative and conjugation of metabolites with glucuronic acid. Metabolism led to at least 33 metabolites, from which nine were identified in urine: M1, M3, M4, M5, M6, M7, M8, M9 and M10.

Lactating goats

In one study conducted with goats dosed for 10 days at 7 ppm in the feed (0.14 mg/kg bw per day), and animals sacrificed 24 hours later, 0.1% of total administered radioactivity (TAR) was found in tissues (mainly liver) and < 0.01% in milk. Most of the radioactivity was found in urine (93.9% TAR), and urinary radioactivity characterization after enzyme treatment showed that metabolites were mostly in conjugated form.

In another study, the goats were dosed for 4 days with 77 ppm in the feed (3.9 mg/kg bw per day), and animals sacrificed 6 hours later. Less than 0.2% TAR was found in milk, liver, kidney and muscle, and < 0.01% TAR in fat. From 25 to 98% of TRR was extracted from milk and tissues with acetonitrile, and further treated with glucuronidase to hydrolyze conjugates. TRR in milk from goat 2 was 0.09 mg eq/kg, comprising mainly M3 (70% TRR; 65% in fatty acid conjugated form). TRR was 1.37 mg eq/kg in liver, 1.1 mg eq/kg in kidney, 0.14 mg eq/kg in muscle, and 0.25 mg eq/kg in fat. M6 accounted for 13.9 to 31.6% TRR in tissues, M8 contributed with 34% TRR in kidney and from 10.8 to 13.6% TRR in muscle and fat, and M7 was major only in tenderloin muscle (16.9% TRR). M1 and M5 were present in tissues up to 11% TRR. Metalaxyl was not detected in milk or tissues.

Laying hens

In the study conducted with hens, the birds received daily oral doses of ¹⁴C-metalaxyl for 4–5 consecutive days, at 100 or 80 ppm in feed and the animals sacrificed 6 hours later. Tissue and egg samples were extracted with methanol, water, dichloromethane, ethyl acetate, acetonitrile, and/or hexane; enzymes (protease, glucuronidase and/or collagenase) and/or base hydrolysis were used to release conjugates. About 0.9% of TAR was recovered in the edible tissues and 0.1% in the eggs. From 69 (liver) to 95.9% (fat) of the radioactivity were extracted. TRRs ranged was 0.157 mg eq/kg in egg white and 0.289 mg eq/kg in egg yolk (mean of 0.21 mg eq/kg in whole egg), 0.525 mg eq/kg in liver (after enzyme treatment), 1.5 mg eq/kg in kidney, up to 0.674 mg eq/kg in muscle and 0.25 mg eq/kg in peritoneal fat. Metalaxyl accounted for 5.6% TRR in eggs, 1.3% TRR in liver, and was not found in muscle and fat. Metabolites were found in eggs and tissues in free and conjugated forms, but the absolute contribution of each was not always clear. Metabolite P1 and its stereoisomer P2 were major in eggs (16.2 and 13% TRR, respectively), muscle (37.3 and 9.1% TRR). M1 accounted for 40.3% TRR in fat and 10.8% TRR in egg. M3 was found mainly as sulfate conjugate (up to 30% TRR in muscle). In liver, M6 and M12 accounted for 17 and 12%

TRR, respectively. Another ten metabolites were found at < 10% TRR. An unknown metabolite P3 accounted for up to 15% TRR in egg white.

In summary, metalaxyl is extensively metabolized in goats and hens, through ester hydrolysis, hydroxylation, oxidation and demethylation of the parent compound, and conjugation. This pathway is similar to what was observed in rat, and to the metabolism in plants discussed earlier, except for metabolites P1/P2, which were only found in hens, and for sugar conjugates found in plants.

Metalaxyl was not detected in goat and was a minor component of the residues in hens (up to 8.2% in egg white, not detected in muscle or fat). Major metabolites (> 10% TRR) are M3 (mainly conjugated), M6, M8, M7 and M1, in addition to M12 and P1/P2 in hens.

Environmental fate

Soil

Two loam soil photolysis studies conducted with ^{14}C -metalaxyl (at 10 mg/kg) indicated fast degradation under natural and artificial lights, with calculated DT_{50} values of 7 and 15.9 days, respectively. No significant degradation was found in dark controls.

Three studies conducted under aerobic laboratory conditions showed that the degradation rate of ^{14}C -metalaxyl varies according to the soil type, temperature, moisture conditions and concentration of the product, with the DT_{50} varying from 7 to 43 days. M1 is the major degradation product, which reached a maximum of 31–34% of the recovered activity (RR) after 56–84 days, which decreased to < 15% RR after 253 days. In a study conducted with M1, the compound degraded to M10, with a DT_{50} of 102 days.

Seven field dissipation studies were conducted with metalaxyl-M at ~1 kg ai/ha in France, Italy, Spain and Switzerland. The residues occurred mostly in the top 10 cm of soil, with M1 at levels below the parent compound, but in some cases, residues in the 10–30 cm soil horizon were present (from day 7 to 21; 0.011 to 0.087 mg/kg) with no residues of the parent detected (< 0.01 mg/kg). Metalaxyl-M residues in treated soil disappeared with DT_{50} ranging from 5 to 21 days.

The enantiomeric ratios of parent and M1 in soil from treated lettuce was shown to depend on the product applied. While R-isomers of both compounds were relatively stable after metalaxyl-M treatment, the R-isomer of the parent decreased significantly while the R-isomer of M1 became predominant after metalaxyl treatment. In a laboratory study retrieved from the literature, the enantiomers were sorbed to the same extent on different soils treated with metalaxyl, but the column leachates had higher S-enantiomer levels, with R-enantiomer degrading slower in the fine-textured soil. Another study showed that, although sorption was not enantioselective in all 3 soils (pH 7.7–8.3), degradation was higher for the R-isomer in unamended and olive-mill waste (OMWc) amended soil than in the biochar (BC) amended soil. BC indirectly affected the enantioselective behavior of metalaxyl in soil by reducing its bioavailability through sorption to a greater extent than OMWc. A third published study showed that the enantioselectivity of metalaxyl degradation depends on soil pH, with lower degradation of R-enantiomer in soils with pH < 4.

In summary, metalaxyl (and metalaxyl-M) are non-persistent in soil under field conditions (DT_{50} up to 21 days), with M1 the main degradate, which was shown to be persistent (DT_{50} = 102 days). The extent and enantioselectivity of degradation depend on which compound the soil was exposed to (higher enantioselective change for metalaxyl), the type of soil, pH and whether the soil was amended.

Methods of analysis

Plant matrices

The methods to analyse residues of metalaxyl in plant matrices were mostly based on acetonitrile extraction and LC-MS/MS determination. LC-MS/MS methods for analysis of metalaxyl residues in crop treated with metalaxyl-M (same m/z transitions) involved extraction with methanol and SPE clean-up,

extraction with acetonitrile or QuEChERS extraction/clean-up. In GC-MS methods, samples were either extracted with acidified acetone or acetonitrile and residues partitioned into dichloromethane. As chiral columns for enantiomeric separation were not used in any of the above methods, metalaxyl residues are measured as the sum of the enantiomers. The methods were validated at LOQs of 0.005 mg/kg (cacao products) or 0.01 mg/kg.

Animal commodities

An LC-MS/MS QuEChERS method was validated to analyse metalaxyl-M in animal commodities, with an LOQ of 0.01 mg/kg.

A common moiety method is available to analyze metalaxyl and metabolites in animal commodities (method AG-576). Samples are extracted with acetonitrile, acetonitrile/water or hexane (poultry fat samples), the extract is cleaned up by hexane partitioning, except for poultry fat samples which are partitioned with ACN. The acetonitrile phase is evaporated, water is added and refluxed with methane sulfonic acid to convert the metabolites to 2,6-dimethylaniline, alkalized and steam distilled. The hexane phase containing 2,6 dimethylaniline is subject to clean-up in silica gel before 2,6-dimethylaniline is analyzed by GC-NPD.

In one study, where the samples were fortified with metalaxyl, the common moiety method was validated at LOQs of 0.05 mg/kg in tissues and eggs and 0.01 mg/kg for milk. In a second study, samples were fortified with M1 and M8 at levels of 0.05 (n = 2), 0.1 (n = 2) and 0.5 (n = 1) mg/kg. Recoveries were < 60% for both metabolites in eggs and for M8 in beef liver, and < 11% for both metabolites in poultry skin/attached fat. The method was validated for both metabolites at the LOQ of 0.05 mg/kg for poultry muscle (recoveries 65 to 116%) and for M1 at 0.10 mg/kg for beef liver (recoveries 72–78%).

More recently, the common moiety method was developed for the analysis of metalaxyl-M residues in animal commodities by LC-MS/MS. Fat samples are first extracted with ethyl acetate/cyclohexane and the fat is separated by freezing-out. Samples are extracted with water/acetonitrile or acetonitrile (egg), the extract partitioned with hexane, the acetonitrile phase evaporated, the remainder heated under reflux in methane sulfonic acid, the extract alkalized and 2,6-dimethylaniline analyzed by LC-MS/MS. The method was validated for metalaxyl-M at LOQs of 0.01 mg/kg for milk, eggs, meat and fat and 0.05 mg/kg for liver and kidney.

The common moiety method AG-576 was validated for metalaxyl, and partially validated for M1 and M8, but no other metabolite found in animal commodities was included in the validation procedure. The Meeting agreed that this method is not appropriate to analyze residues other than parent compound in animal commodities.

Stability of residues in stored analytical samples

Storage stability studies conducted with lettuce and potato fortified with metalaxyl at 5 mg/kg and 0.2 mg/kg, respectively, and stored at -20 to -15 °C showed residues of metalaxyl were stable for at least 6 months (lettuce) or 12 months (potato). Fresh ginseng and processed product samples were spiked with metalaxyl-M at 0.3 or 0.6 mg/kg, respectively, and stored at about -20 °C. Over 80% of the residues remained after 40 to 80 days of storage. Studies on plant samples (oranges, potatoes, rape seed, tomatoes and wheat) conducted with metalaxyl-M were evaluated by the 2004 JMPR and showed that metalaxyl residues were stable in frozen storage for at least 24 months. The demonstrated sample storage stability covers the sample storage periods used in the supervised residue trials considered by the current Meeting.

One study was conducted with beef muscle and liver, milk and egg samples fortified with metalaxyl-M, M1 and M8, stored up to 24 months in the freezer and analysed by the common moiety method AG-576. Metalaxyl-M was stable for the whole study period. Method recovery for M1 in eggs was < 60%, but the residues were stable for at least 2 years compared to the initial measured level. M1 was also stable for at least 2 years in liver, muscle and milk. Procedural recoveries for M8 were lower than

60% in all matrices, but the measured residues were stable for up to 1 year in milk, up to 1.5 years in liver and up to 3 months in muscle.

Definition of the residue

Metalaxyl is a racemic mixture of an R-enantiomer and an S-enantiomer, and metalaxyl-M is 97% R-enantiomer.

Metalaxyl is the main identified component of the residue in grapes and potato tuber resulting from the use of metalaxyl, accounting for 64.1% (68 DALA) and 57% TRR (7 DALA), respectively. M8 was the major metabolite, accounting for 20.4% TRR in grapes (ratio M8/metalaxyl=0.32) and 4.1% TRR in potato tuber (ratio M8/metalaxyl = 0.07).

In lettuce, metalaxyl accounted for up to 19% TRR (metalaxyl treatment at 14 DALA). M8 was the main metabolite accounting for up to 61.3% TRR (mainly in conjugated form) in plants treated with metalaxyl-M (21 DALA). The ratios M8/metalaxyl (14 or 21 DALA) ranged from 1.8 to 10.2.

Available analytical methods determine the sum of metalaxyl enantiomers and were used in the supervised trials. The Meeting agreed that the parent metalaxyl is an appropriate marker for enforcement of metalaxyl and metalaxyl-M uses according to GAP.

Studies conducted with metalaxyl or metalaxyl-M evaluated by the current Meeting indicated that all major plant metabolites also occur in rats and have been investigated in toxicological studies. Based on the available toxicological data for M12 and that the identified plant metabolites M1, M3, M6, M7 and M8 were also observed in the rat metabolism, the Meeting concluded that the toxicity of these metabolites are covered by the ADI and ARfD for metalaxyl and metalaxyl-M.

M8 is a major metabolite in some plant commodities, and the parent plus M8 (free and conjugated) covers the majority of the residues expected in plant commodities. Although no toxicological studies were conducted with M8, its toxicity is expected to be similar to the parent compound. The residue trials do not include measurement of the metabolites.

The Meeting agreed that:

The residue definition for metalaxyl *and* metalaxyl-M for compliance with the MRL in plant commodities is: metalaxyl (sum of enantiomers).

The residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in plant commodities is: metalaxyl (sum of enantiomers) and N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (M8; free and conjugated; sum of enantiomers), expressed as metalaxyl.

Based on the metabolism studies the Meeting estimated conservative conversion factors of 11 for leafy crops (based on lettuce) and 1.3 for fruits and non-leafy vegetables (based on grapes) to account for residues of metabolite M8 (free and conjugated). The factors can be applied to the parent residues in order to estimate residues of parent + M8 (free and conjugated) for estimating STMR and HR values. No factor was estimated for bulb, root and tuber crops as M8 (free and conjugated) represented less than 10% of residues of the parent in potato tubers.

Metabolism studies conducted in goat and hens with metalaxyl showed that metalaxyl is rapidly metabolized to various compounds, in free and conjugated forms. Metalaxyl was only detected in hens, contributing to up to 5.6% TRR (eggs). M3 (70% TRR; 65% in conjugated form) was the major residue in milk and in hen muscle (30% TRR), M6 accounted for 13.9 to 31.6% TRR in tissues, M8 contributed with 34% TRR in kidney and from 10.8 to 13.6% TRR in muscle and fat, and M7 and M1 were major metabolites in goat muscle (16.9 and 10.8% TRR, respectively) and hen fat (40% TRR).

No metabolism study conducted with metalaxyl-M in farm animals was provided to the Meeting, but the results are not expected to be different from the studies conducted with metalaxyl, which include

metalaxyl-M. Analytical methods for metalaxyl residues in animal commodities are mainly based on a common moiety method (2,6-dimethylaniline), which were not sufficiently validated for the metabolites.

In farm feeding studies on goats and hens conducted with metalaxyl, residues were measured by the common moiety method.

The log P_{OW} for metalaxyl is 1.6, and animal metabolism studies confirm that metalaxyl is not fat-soluble.

The Meeting agreed that:

The residue definition for metalaxyl and metalaxyl-M for compliance with the MRL in animal commodities is: the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.

The residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in animal commodities is: the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M1 (N-(2,6-dimethylphenyl)-N-(methoxyacetyl) alanine), M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester), M6 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine), M7 (N-(2,6-dimethyl-5-hydroxyphenyl)-N-(methoxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.

The residue is not fat-soluble.

As no validated method is available that includes the determination of metabolites, the Meeting agreed that it could not estimate maximum residue levels for metalaxyl in animal commodities.

Results of supervised residue trials on crops

The Meeting considered supervised residue trials conducted with metalaxyl and metalaxyl-M provided to the current Meeting and residue trials conducted with metalaxyl-M provided to the 2004 JMPR that matched the current GAP information.

The only difference between metalaxyl and metalaxyl-M is the ratio of the enantiomers, and analytical methods determine residues of metalaxyl (sum of enantiomers) in plants treated with either metalaxyl or metalaxyl-M.

The following approach was considered for the estimations:

- When data for a given crop was submitted for metalaxyl and metalaxyl-M, the residue data according to GAP that led to the highest maximum residue level was used for the estimations.
- Regardless of which product was applied in the residue trials, the estimations were made for metalaxyl.
- The source of the residue data (metalaxyl or metalaxyl-M) is shown below for each crop and in the recommendations Table.

Table 2 A summary of the residue data considered by the current Meeting

Crop	Submitted to this Meeting	Submitted to the 2004 JMPR*
Orange	Metalaxyl	
Apples		Metalaxyl-M
Grapes	Metalaxyl	Metalaxyl-M
Strawberry	Metalaxyl	
Onion	Metalaxyl	Metalaxyl-M
Broccoli	Metalaxyl and metalaxyl-M	
Brussels sprouts	Metalaxyl and metalaxyl-M	
Cabbage	Metalaxyl and metalaxyl-M	
Cauliflower	Metalaxyl and metalaxyl-M	
Melon	Metalaxyl-M	
Pepper	Metalaxyl	
Tomato	Metalaxyl	Metalaxyl-M
Lettuce	Metalaxyl	Metalaxyl-M
Spinach		Metalaxyl-M
Carrot	Metalaxyl-M	
Potato	Metalaxyl	Metalaxyl-M
Sunflower		Metalaxyl-M
Cacao	Metalaxyl-M	
Ginseng	Metalaxyl-M	
Wasabi	Metalaxyl-M	
Pepper, Black	Metalaxyl-M	
Pepper, black, white	Monitoring data	

* Only trials conducted with metalaxyl-M and in accordance with the current GAP

No residue data or GAP information was available for metalaxyl on asparagus, avocado, citrus fruit (other than oranges), cereal grains, cotton seed, cucumber, gherkin, hops (dry), peas shelled (succulent seeds), peanut, pome fruits (other than apples), raspberries (red, black), summer squash, sugar beet, soya bean (dry), watermelon and winter squash. Hence, the Meeting withdrew its previous recommendations for these commodities and they are not discussed further in this document.

Based on the approach outlined above, the Meeting withdrew all previous recommendations for metalaxyl-M and made new estimations in terms of metalaxyl, when trials according to GAP were available. The estimated values for metalaxyl are intended to cover uses of metalaxyl-M.

For dietary risk assessment, factors of 11 for leafy crops and 1.3 for fruits and non-leafy vegetables, estimated from the metabolism studies on lettuce and grape, respectively, were applied to STMR and HR values found in the crops to account for the presence of metabolite M8 (free and conjugated). No factor is needed for bulb, root, and tuber crops or for crops when no residues are expected to be found.

Orange, sweet, sour (Subgroup 001C)

Metalaxyl is registered for use in orange orchards in Spain at a critical GAP of 3×0.2 kg ai/ha foliar applications (14-day interval) with a 14-day PHI. This registration is extrapolated to the other crops in the Subgroup of oranges (Regulation EC 62/2018, product group code for oranges: 0110020).

Seven independent trials conducted in orange orchards in Greece, Italy and Spain approximating this GAP gave residues in the whole fruit of 0.03, 0.10 (2), 0.26, 0.28, 0.29 and 0.30 mg/kg. Residues in the pulp were <0.01 (6) and 0.02 mg/kg. Although the application intervals in these trials (17 days) were longer than what was specified on the label, data show that residues in oranges decline slowly or do not change within 7 days. Therefore, the longer application interval is unlikely to significantly affect the final residues.

No trials conducted with metalaxyl-M were provided to the current Meeting and trials submitted to the 2004 JMPR (post-harvest application) are for a GAP that is no longer supported. Metalaxyl-M is registered in Italy for soil application at the plant base of orange trees, but it is unlikely that residues in orange fruit from the use of metalaxyl-M on soil will exceed those found after foliar application of metalaxyl.

Based on the data from foliar application of metalaxyl, the Meeting estimated a maximum residue level of 0.7 mg/kg for metalaxyl for Orange, sweet, sour (Subgroup 001C).

A factor of 1.3 was applied to estimate an STMR of 0.013 mg/kg and an HR of 0.026 mg/kg for metalaxyl for Orange, sweet, sour.

The factor of 1.3 was also applied to the residues in whole orange to estimate median and highest residues of 0.338 and 0.39 mg/kg in orange fruit for use in livestock dietary burden calculation and estimation of residues in relevant processed commodities.

Apples and pears

No GAP information or supervised trials with metalaxyl on apples or pears was provided to the current Meeting.

Metalaxyl-M is registered in Spain for soil application in apple and pear orchards at 2×0.96 g ai/tree (182 days interval) with a 15-day PHI. Assuming 3000 trees/ha, the GAP rate corresponds to 3 kg ai/ha. Six trials provided to the 2004 JMPR were conducted with metalaxyl-M in apple orchards in France, Italy and Spain using 2 soil treatments at 0.78 (one trial), 1.9 (one trial) or 10 (four trials) kg ai/ha (31 to 158 days application interval) gave residues at 14 days DAT of < 0.02 mg/kg (6). The data indicated that no metalaxyl residues are expected in apples when metalaxyl-M is applied at GAP to soil.

The Meeting confirms its previous recommendation of a maximum residue level of 0.02(*) mg/kg for metalaxyl in apples and extrapolated this estimation to pears.

The Meeting also estimated an STMR of 0 mg/kg and an HR of 0 mg/kg for metalaxyl in apples and pears.

As no residues are expected in apples and pears after soil application, no factor to account for M8 needs to be applied.

Grapes

The critical GAP for metalaxyl on grapes in Europe is 4×0.2 kg ai/ha foliar applications at a 10-day interval with a 21-day PHI (Romania and Macedonia). Sixteen trials conducted in Czech Republic, France, Germany, Hungary, Italy and Spain according to this GAP gave residues of: 0.08, 0.09, 0.10 (2), 0.11 (2), 0.13, 0.15, 0.16 (2), 0.21, 0.22 (2), 0.25, 0.27 and 0.52 mg/kg.

The GAP for metalaxyl-M in Australia is 4×0.11 kg ai/ha at a 7 to 21 day interval with a 7-day PHI. Six trials conducted in Australia within GAP rate (5 applications) submitted to the 2004 JMPR gave

residues of: 0.03, 0.04, 0.06, 0.11, 0.34 and 0.68 mg/kg. Four trials conducted in Italy within this GAP (3–5 applications) gave residues of: 0.07, 0.17, 0.26, and 0.60 mg/kg. Ten trials conducted with metalaxyl-M according to the Australian GAP belong to the same residue population and can be combined as: 0.03, 0.04, 0.06, 0.07, 0.11, 0.17, 0.26, 0.34, 0.60 and 0.68 mg/kg.

Based on residue data from metalaxyl-M trials, the Meeting estimated a maximum residue level of 1.5 mg/kg, for metalaxyl in grapes.

This estimation replaced the previous recommendation of 1 mg/kg for metalaxyl in grapes.

For dietary risk assessment, a factor of 1.3 was applied to account for M8 and the Meeting estimated an STMR of 0.182mg/kg and an HR of 0.884 mg/kg.

Strawberry

The GAP for metalaxyl in strawberry in Spain is 1×0.38 kg ai/ha (drip irrigation, greenhouse) with a 14-day PHI. No trials conducted according to this GAP were provided.

Metalaxyl-M is registered in Italy and Spain (soil application) but no trials conducted according to this GAP were provided.

As no trials were conducted according to either the metalaxyl or the metalaxyl-M GAPs, the Meeting was unable to estimate a maximum residue level for metalaxyl in strawberry.

Onion, bulb

Metalaxyl is registered in onion in North Macedonia at a foliar rate of 3×0.024 kg ai/hL (10–14 day interval) and a 21-day PHI. No trials according to this GAP were provided to the Meeting.

The Brazilian GAP for metalaxyl-M in onion is 4×0.1 kg ai/ha (10-day interval) and a 7-day PHI. Seven trials conducted with metalaxyl-M approximating this GAP (3–4 applications) were evaluated by the 2004 JMPR, 3 in Brazil, 2 in Italy and 2 in Switzerland. Residues were ($n = 7$): < 0.02 (6) and 0.02 mg/kg.

Based on the residue trials conducted with metalaxyl-M, the Meeting estimated a maximum residue level of 0.03 mg/kg for metalaxyl in onion, bulb.

This estimation replaced the previous recommendation of a maximum residue level of 2 mg/kg for metalaxyl in onion, bulb.

No factor needs to be applied to onion residues and the Meeting estimated an STMR of 0.02 mg/kg and an HR of 0.02 mg/kg for metalaxyl in onion, bulb.

Flowerhead brassicas

The critical GAP for metalaxyl in broccoli and cauliflower in Spain is 3×0.2 kg ai/ha (10-day interval) and a 14-day PHI.

In four trials conducted on broccoli in Greece, Italy, Portugal and Spain according to the Spanish GAP, residues were: < 0.01 , 0.01, 0.09 and 0.11 mg/kg.

Residues in cauliflower in four trials conducted in Italy and Spain matching the Spanish GAP were: < 0.01 , 0.02, 0.03 and 0.04 mg/kg.

As the median of the two populations are within the 5-fold range, the Meeting agreed that the datasets could be combined to give a population for flowerhead brassica of ($n = 8$): < 0.01 (2), 0.01, 0.02, 0.03, 0.04, 0.09 and 0.11 mg/kg.

In Italy, the critical GAP for metalaxyl-M in broccoli is 3×0.1 kg ai/ha (10–14 day interval) and a 20-day PHI. In four trials conducted on broccoli in Spain according to the Italian GAP, residues were ($n=4$): < 0.02 (4) mg/kg.

Australian GAP for metalaxyl-M in Brassica vegetables is 2×0.1 kg ai/ha (7–14 day interval) and a 14-day PHI. In 16 trials conducted on cauliflower in France, Italy and Switzerland according to the Australian GAP, residues were: < 0.01 (3), < 0.02 (11) and 0.02 (2) mg/kg.

Based on the residue data from trials conducted with metalaxyl on broccoli and cauliflower the Meeting estimated a maximum residue level of 0.2 mg/kg for metalaxyl in flowerhead brassicas.

The Meeting withdrew its previous recommendations of maximum residue levels of 0.5 mg/kg for metalaxyl on broccoli and cauliflower.

For dietary risk assessment, a factor of 11 was applied to account for M8 and the Meeting estimated an STMR of 0.275 mg/kg and an HR of 1.21 mg/kg for metalaxyl in flowerhead brassicas.

Brussel sprouts

The critical GAP for metalaxyl in Brussels sprouts in Spain is 3×0.2 kg ai/ha (10-day interval) and a 14-day PHI. Residues from trials conducted in Greece, Italy and Spain according to the Spanish GAP were (n = 4): 0.03 (2), 0.05 and 0.07 mg/kg.

Australian GAP for metalaxyl-M in Brassica vegetables is 2×0.1 kg ai/ha (7–14 day interval) and a 14-day PHI. No trials were conducted according to this GAP with metalaxyl-M in Brussels sprout.

Based on the residue data from trials conducted with metalaxyl, the Meeting estimated a maximum residue level of 0.15 mg/kg for metalaxyl in Brussels sprouts.

This estimation replaced the previous recommendation of maximum residue level of 0.2 mg/kg for metalaxyl on Brussels sprouts.

For dietary risk assessment, a factor of 11 was applied to account for M8 and the Meeting estimated an STMR of 0.44 mg/kg and an HR of 0.77 mg/kg for metalaxyl for in Brussels sprouts.

Cabbage, head

The critical GAP for metalaxyl in cabbage in Spain is 3×0.2 kg ai/ha (10-day interval) and a 14-day PHI. In four trials conducted in cabbage according to the Spanish GAP, residues were: < 0.01 and 0.02 (3) mg/kg.

In Italy, the critical GAP for metalaxyl-M in cabbage is 3×0.1 kg ai/ha (10–14 day interval) and a 20-day PHI. In five trials conducted with cabbage in Belgium, France and Italy according to the Italian GAP, residues were: < 0.02 (4) and 0.04 mg/kg.

Based on the residue data from trials conducted with metalaxyl-M, the Meeting estimated a maximum residue level of 0.08 mg/kg for metalaxyl in cabbage, head.

This estimation replaced the previous recommendation of maximum residue level of 0.5 mg/kg for metalaxyl on cabbage, head.

For dietary risk assessment, a factor of 11 was applied to account for M8 and the Meeting estimated an STMR of 0.22 mg/kg and an HR of 0.44 mg/kg for metalaxyl for in cabbage, head.

Melons, except watermelon

Metalaxyl is registered in North Macedonia in melon at 3×0.024 kg ai/hL (10-day interval) and a 21-day PHI. No trials conducted with metalaxyl were provided to the Meeting.

The GAP for metalaxyl-M in melons in Italy is 3×0.1 kg ai/ha (10–14 day interval) with a 3-day PHI.

In 16 greenhouse or polytunnel trials conducted in France, Germany, the Netherlands and Spain according to the Italian GAP, residues were: < 0.01(5), < 0.02 (7), 0.02 (2), 0.04, 0.10 mg/kg in the fruit

and < 0.01 (9), < 0.02 (5) mg/kg in the pulp. Residues found in four field trials were < 0.01 (3) and 0.03 mg/kg in the fruit and < 0.01 (4) in the pulp.

Based on residue data from greenhouse/polytunnel trials conducted with metalaxyl-M, the Meeting estimated a maximum residue level of 0.15 mg/kg for metalaxyl in melons, except watermelon.

This estimation replaced the previous recommendation of 0.2 mg/kg for metalaxyl in melons, except watermelon.

For dietary risk assessment, a factor of 1.3 was applied to account for M8 and the Meeting estimated an STMR of 0.013 mg/kg and an HR of 0.026 mg/kg for metalaxyl in melons, except watermelon.

Peppers

The GAP for metalaxyl in greenhouse peppers in Spain is 2×0.38 kg ai/ha (drip irrigation; 14-day interval; greenhouse) and a 14-day PHI. No trials according to this GAP were provided to the Meeting.

The GAP for metalaxyl-M in peppers in Italy involves 3 applications: 1×0.46 kg/ha (pre-planting) followed by 2×0.92 kg ai/ha (30-day interval) and a 15-day PHI. No trials were provided to the current Meeting and those provided to the 2004 JMPR were not conducted according to this GAP.

The Meeting withdrew its previous recommendation of a maximum residue level of 1 mg/kg for metalaxyl on peppers.

Tomatoes (subgroup 12A)

The GAP for metalaxyl on tomatoes in Romania and Spain is 4×0.2 kg ai/ha (10-day interval, greenhouse or field) and a 21-day PHI) or 3×0.2 kg ai/ha (10-day interval, greenhouse or field) and a 14-day PHI.

In 23 field or greenhouse trials conducted in Czech Republic, France, Germany, Greece, Hungary, Italy, Poland, Portugal and Spain at 4×0.2 kg ai/ha and a 21-day PHI (including cherry tomato), residues were: < 0.01 (22) and 0.01 mg/kg. Samples collected at a 14-day PHI, within the Spanish GAP, gave residues of (n = 23): <0.01 (17), 0.01 (2), 0.02 (3) and 0.04 mg/kg.

The critical GAP for metalaxyl-M in tomato is 4×0.14 kg ai/ha (10-day interval) and a 3-day PHI (Morocco). Sixteen trials conducted with metalaxyl-M in 1997/1998 in France, Spain and Switzerland according to this GAP were provided to the 2004 JMPR. Residues in 12 greenhouse trials were: < 0.02 (2), 0.02, 0.03, 0.04 (2), 0.05 (2), 0.08, 0.09, 0.12 and 0.18 mg/kg and residues in the field were <0.02 (3) and 0.02 mg/kg.

Based on the greenhouse trials conducted with metalaxyl-M, the Meeting estimated a maximum residue level of 0.3 mg/kg for metalaxyl in tomatoes.

For dietary risk assessment, a factor of 1.3 was applied to account for M8 and the Meeting estimated an STMR of 0.058 mg/kg and an HR of 0.234 mg/kg for metalaxyl in the Subgroup of tomatoes.

The Meeting withdrew its previous recommendation of a maximum residue level of 0.5 mg/kg for metalaxyl in tomato.

Lettuce

Metalaxyl is registered in lettuce in Spain at 3×0.2 kg ai/ha (10-day interval) and a 14-day PHI. In 11 field trials conducted in France, Greece, Italy and Spain in leafy lettuce according to this GAP, residues were: 0.05, 0.07, 0.08, 0.12 (2), 0.13, 0.15, 0.17, 0.36, 0.71 and 0.74 mg/kg.

The critical GAP for metalaxyl-M in Europe for lettuce is 3×0.1 kg ai/ha (10-day interval), with a 14-day PHI in the greenhouse (Italy). In one trial on head lettuce conducted in Germany according to this GAP and provided to the 2004 JMPR, residues were 0.07 mg/kg.

Based on the data from trials conducted with metalaxyl, the Meeting estimated a maximum residue level of 1.5 mg/kg, for metalaxyl in lettuce, leaf.

For dietary risk assessment, a factor of 11 was applied to account for M8 and the Meeting estimated an STMR of 1.43 mg/kg and an HR of 8.14 mg/kg for metalaxyl in lettuce, leaf.

The Meeting agreed that there are not enough trials conducted according to GAP to estimate a maximum residue level for head lettuce.

The Meeting withdrew its previous recommendation for a maximum residue level of 2 mg/kg for metalaxyl in head lettuce.

Spinach

Metalaxyl is registered for use on spinach in Spain at 3×0.2 kg ai/ha with a 14-day PHI, but no residue trials were provided to the Meeting.

The GAP for the use of metalaxyl-M on spinach in Chile involves 3×0.1 kg ai/ha foliar applications (7-day interval) and a 10-day PHI. Four trials conducted in France and Switzerland according to GAP provided to the 2004 JMPR gave residues of < 0.02 (4) mg/kg in the leaves and the whole plant. Furthermore, samples from trials conducted at 3×0.14 kg ai/ha or harvested at 7 DALA gave the same results, which supports the conclusion that it is unlikely that residues in spinach will be higher or equal to the LOQ.

Based on the data on metalaxyl-M, the Meeting estimated a maximum residue level of 0.02* mg/kg for metalaxyl in spinach.

For dietary risk assessment, a factor of 11 was applied to account for M8 and the Meeting estimated an STMR of 0.22 mg/kg and an HR of 0.22 mg/kg for metalaxyl in spinach.

The Meeting withdrew its previous recommendation of a maximum residue level of 2 mg/kg for metalaxyl in spinach.

Wasabi, tops

No GAP or residue trials for metalaxyl in wasabi were provided to the Meeting.

The GAP for use of metalaxyl-M on wasabi in the USA is 6×0.8 kg ai/ha with a 7-day PHI (protected). In three trials conducted in the USA according to GAP residues were: 1.82, 2.04 and 2.24 mg/kg in the top.

As not enough trials were conducted according to GAP, the Meeting could not estimate a maximum residue level for wasabi tops.

Carrot

No GAP information or residue trials for the use of metalaxyl on carrot were provided to the Meeting.

The GAP for metalaxyl-M on carrot in France is one soil application *plus* one foliar application at 0.23 kg ai/ha, and a PHI of 28 days. In one trial conducted in Switzerland at GAP residues were: < 0.02 mg/kg. In three trials, one foliar application at an exaggerated rate of 0.62 kg ai/ha gave the same results, which indicated that no residues are expected in carrot when metalaxyl-M is applied according to GAP.

Based on the data on metalaxyl-M, the Meeting estimated a maximum residue level of 0.02* mg/kg for metalaxyl in carrot.

The Meeting withdrew its previous recommendation of a maximum residue level of 0.05 mg/kg for metalaxyl in carrot.

For dietary risk assessment, a factor to account for M8 is unnecessary and the Meeting estimated an STMR of 0.02 and an HR of 0.02 mg/kg for metalaxyl in carrot.

Potato

The critical GAP for metalaxyl in potato in Romania and Spain is 4×0.2 kg ai/ha (10-day interval) and a 21-day PHI. Sixteen trials conducted in France, Germany, Hungary, Italy, Poland, Portugal and Spain according to GAP gave residues of: < 0.01 (12), 0.01 (3) and 0.02 mg/kg.

The critical GAP for metalaxyl-M on potato in Brazil is 4×0.1 kg ai/ha (10-day interval) and a 7-day PHI. Trials conducted with metalaxyl-M in potato in 1995–1997 were provided to the 2004 JMPR. Trials conducted in Brazil (3) and Europe (9; Germany, Switzerland and the UK) within this GAP gave residues of: < 0.02 mg/kg (12). Three other trials conducted at twice the GAP rate gave the same results.

Based on the data from trials conducted with metalaxyl at a 21-day PHI, the Meeting estimated a maximum residue level of 0.02 mg/kg, for metalaxyl in potato.

The Meeting replaced its previous recommendation of a maximum residue level of 0.05(*) mg/kg for metalaxyl in potato.

For dietary risk assessment, a factor to account for M8 is unnecessary and the Meeting estimated an STMR of 0.01 mg/kg and HR of 0.02 mg/kg for metalaxyl in potato.

Sunflower seed

No GAP information or residue trials for the use of metalaxyl on sunflower were provided to the Meeting.

Metalaxyl-M is registered in Italy as a seed treatment at 0.1 kg ai/100 kg seed. Eight trials conducted with metalaxyl-M in France and Spain were evaluated by the 2004 JMPR, giving residues of: < 0.01 (2) and < 0.02 (6) mg/kg at 125 to 151 days after sowing.

Based on the data from trials conducted with metalaxyl-M, the Meeting estimated a maximum residue level of 0.01(*) mg/kg for metalaxyl in sunflower seed.

The Meeting replaced its previous recommendation of a maximum residue level of 0.05(*) mg/kg for metalaxyl in sunflower seed.

As no residues are expected after seed treatment, the Meeting estimated an STMR of 0 mg/kg for metalaxyl in sunflower seed.

Cacao beans

No GAP information or residue trials for the use of metalaxyl on cacao were provided to the Meeting.

Metalaxyl-M is registered for use in cacao in Cameroon at 6×0.072 kg ai/ha (21-day interval) and a 14-day PHI. Eight trials were conducted according to this GAP rate, but the application interval ranged from 12 to 17 days.

As no trials were conducted according to GAP, the Meeting could not estimate a maximum residue level for metalaxyl in cacao beans.

Ginseng

No GAP information or residue trials for the use of metalaxyl on ginseng were provided to the Meeting.

Metalaxyl-M is registered in ginseng in South Korea at 3×0.08 kg ai/ha (10-day interval) and a 14-day PHI. In nine trials conducted in South Korea according to GAP, residues were < 0.03 (9) mg/kg.

Based on the data from trials conducted with metalaxyl-M, the Meeting estimated a maximum residue level of 0.03* mg/kg for metalaxyl in ginseng.

For dietary risk assessment, a factor to account for M8 is unnecessary and the Meeting estimated an STMR of 0.03 mg/kg and an HR of 0.03 mg/kg for metalaxyl in ginseng.

Wasabi, stems

No GAP or residue trials for the use of metalaxyl on wasabi were provided to the Meeting.

The GAP for metalaxyl-M in the USA is 6×0.8 kg ai/ha with a 7-day PHI (protected). In three trials conducted in the USA according to GAP residues were: 0.259, 0.273 and 1.32 mg/kg in the stems.

As not enough trials were conducted according to GAP, the Meeting could not estimate a maximum residue level for wasabi stems.

Pepper, black, white (peppercorn)

No GAP information or residue trials for the use of metalaxyl on pepper, black, white were provided to the Meeting.

The GAP for metalaxyl-M in Vietnam is 4×0.12 kg ai/ha (7-day interval) and a 14-day PHI. Four residue trials conducted in Vietnam according to GAP gave residues of: 0.23, 0.305, 0.395 and 0.86 mg/kg.

Based on the data from trials conducted with metalaxyl-M, the Meeting estimated a maximum residue level of 2 mg/kg for metalaxyl in pepper, black, white.

For dietary risk assessment, a factor of 1.3 was applied to account for M8 and the Meeting estimated an STMR of 0.455 mg/kg for metalaxyl in pepper, black, white.

Monitoring data (2011 to 2015) on metalaxyl residues in dry pepper in 531 samples produced in Brazil, Ecuador, India, Indonesia, Malaysia, Sri Lanka and Vietnam were provided.

A total of 398 samples (75%) contained detectable residues ranging from 0.005 to 0.54 mg/kg, with a median of 0.06 mg/kg. The upper 95% confidence limit of the detected residues is 0.32 mg/kg.

The Meeting noted that none of the results of the monitoring data exceeded the estimated maximum residue level of 2 mg/kg for metalaxyl in pepper, black, white.

Spices, seeds

No GAP information or residue trials for the use of metalaxyl or metalaxyl-M on spices were provided to the Meeting

Monitoring data on metalaxyl for anise seeds and fennel seeds were provided to the 2004 JMPR. A total of 1306 samples were analyzed, from which 10 gave residue levels above the LOQ (from 0.2 to 3.2 mg/kg, median of 0.43 mg/kg).

According to the current procedure (2015 JMPR) no estimation can be made because less than 59 samples had detectable residues.

The Meeting withdrew its previous recommendation of maximum residue level of 5 mg/kg for metalaxyl in spices, seeds.

Residues in processed commodities

Based on the estimated maximum residue levels, STMRs and median residues, processing information on oranges, grapes and cacao beans and the processing factors (PF), the estimations as outlined in Table 3 could be made.

Table 3 Estimation of processing factors, STMRs and median residues used for estimation of maximum residue levels

Commodity	Mean PF	STMR-P, mg/kg	Maximum residue level, mg/kg
<i>Orange: maximum residue level = 0.7 mg/kg; median = 0.338 mg/kg, STMR = 0.013</i>			
Juice, pasteurised	0.06, n=4	0.020	-
Marmalade	0.39, n=4	0.132	-
Oil	9.0, n=4	3.04	7
Pomace, wet	1.1, n=4	-	-
Pomace, dry	4.1, n=4	Median: 1.38 ^a	-
<i>Grape: STMR = 0.182 mg/kg (metalaxyl and metalaxyl-M)</i>			
Juice	0.4, n=7	0.073	
Wine	0.76, n=21	0.138	
<i>Cacao beans:</i>			
Cocoa beans	1.31, n=2		
Cocoa powder	1.05, n=2		
Cocoa butter	2.0, n=2		
Chocolate	1.43, n=2		

^a applied to citrus dry pomace for dietary burden calculations

Feeding studies

The Meeting received the results of studies of feeding metalaxyl to lactating dairy cows and laying hens. Samples were analysed by the 2,6-dimethylaniline common moiety method, which measured metalaxyl and all metabolites.

Two lactating dairy cows were dosed daily with metalaxyl, equivalent to 75 ppm in their diet, and slaughtered for tissue collection on days 21 and 28 (23.5 hours after dosing). Residues were 0.12 and 0.11 mg/kg in kidney, 0.12 and 0.14 mg/kg in liver, < 0.05 and 0.06 mg/kg in tenderloin muscle, and 0.07 and 0.08 mg/kg in round muscle. No residues were found in any fat sample (< 0.05 mg/kg) and the levels in milk were 0.02 mg/kg for all cows at all days (n = 6).

Groups of 15 laying hens were dosed daily with metalaxyl at levels equivalent to 10, 30 and 100 ppm in the feed, and birds were slaughtered on days 7, 14, 21 and 28 days after the first dose. At 10 ppm level, residues were < 0.05 (3) and 0.06 mg/kg in muscle, < 0.05 (4) mg/kg in skin + attached fat and in fat peritoneal and < 0.05 (2), 0.05 and 0.10 mg/kg in liver. No residues were found in eggs (< 0.05 mg/kg).

The levels reached 0.10 mg/kg in muscle, fat and liver at 30 ppm and 0.40 mg/kg in skin + attached fat at 100 ppm. No residues were found in eggs (< 0.05 mg/kg) at any dose (n = 18).

Dietary burden

The Meeting estimated the dietary burden of metalaxyl/metalaxyl M in farm animals using the most recent version of the OECD livestock dietary burden calculator; the STMR, STMR-Ps, median or highest residue levels estimated by the current Meeting (citrus pomace dry, potato, cabbage and carrots); and the residues of metalaxyl+M8 expected in rotational crops (cereal forage and straw).

Table 4 Livestock dietary burden for metalaxyl/metalaxyl-M, (metalaxyl (sum of enantiomers) and M8 (free and conjugated; sum of enantiomers), expressed as metalaxyl) ppm of dry matter diet

Commodity	US-Canada		EU		Australia		Japan	
	Max	Mean	Max	Mean	Max	Mean	Max	Mean
Beef cattle	1.59	0.513	10.1	1.31	13.2 ^①	2.26	2.84	1.19
Dairy cattle	11.0	0.966	10.6	1.76	12.3 ^②	2.26 ^{③ ④}	4.68	0.672
Poultry - broiler	-	-	0.17	0.017	-	-	-	-
Poultry - layer	-	-	2.51 ^⑤	0.448 ^⑥	-	-	-	-

① Highest maximum beef or dairy cattle dietary burden suitable for maximum residue level estimated for mammalian tissues

② Highest maximum dairy cattle dietary burden suitable for maximum residue level estimated for mammalian milk

③ Highest mean beef or dairy cattle dietary burden suitable for STMR estimated for mammalian tissues.

④ Highest mean dairy cattle dietary burden suitable for STMR estimated for milk.

⑤ Highest maximum poultry dietary burden suitable for maximum residue level estimated for poultry tissues and eggs.

⑥ Highest mean poultry dietary burden suitable for STMR estimated for poultry tissues and eggs.

Animal commodity maximum residue level

As no suitable method is available to determine the residues expected to be found in animal commodities, the Meeting could not estimate maximum residue levels for metalaxyl in animal commodities.

RECOMMENDATIONS

On the basis of the data obtained from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI and IESTI assessments.

Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL for plant commodities: *metalaxyl (sum of enantiomers)*.

Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in plant commodities: *metalaxyl (sum of enantiomers) and N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (M8; free and conjugated; sum of enantiomers), expressed as metalaxyl*.

Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL in animal commodities is: *the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl*.

Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in animal commodities is: *the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M1 (N-(2,6-dimethylphenyl)-N-(methoxyacetyl) alanine), M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester), M6 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine), M7 (N-(2,6-dimethyl- 5-hydroxyphenyl)-N-(methoxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl*.

The residue is not fat-soluble.

DIETARY RISK ASSESSMENT

Long-term dietary exposure

The ADI for metalaxyl and metalaxyl-M (alone or in combination) is 0–0.08 mg/kg bw. The International Estimated Daily Intakes (IEDIs) were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the current JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs ranged from 0 to 1% of the maximum ADI. The Meeting concluded that long-term dietary exposure to residues of metalaxyl residues from the uses considered the JMPR is unlikely to present a public health concern.

Acute dietary exposure

The ARfD for metalaxyl and metalaxyl-M (alone or in combination) is 0.5 mg/kg bw. The International Estimate of Short Term Intakes (IESTIs) for metalaxyl were calculated for the food commodities and their processed commodities for which HRs/HR-Ps or STMRs/STMR-Ps were estimated by the current Meeting and for which consumption data were available. The results are shown in Annex 4 of the 2021 JMPR Report.

The IESTIs varied from 0–100% of the ARfD for children and 0–30% for the general population. The Meeting concluded that acute dietary exposure to residues of metalaxyl from uses considered by the current Meeting is unlikely to present a public health concern.

5.10 Metalaxyl-M (212)

RESIDUE AND ANALYTICAL ASPECTS

Metalaxyl and metalaxyl-M are closely related systemic fungicides with registered uses in a variety of crops around the world. Metalaxyl is a racemic mixture of R- and S-enantiomers, where the R-enantiomer is the biologically active form, and metalaxyl-M consists of a minimum of 97% of the R-enantiomer.

The current Meeting evaluated metalaxyl and metalaxyl-M together for toxicology and established a single ADI and ARfD that apply to metalaxyl and metalaxyl-M, either alone or in combination.

Metabolism studies on plants and animals and environmental fate studies were provided for metalaxyl and/or metalaxyl-M and the conclusions applied to both. The Meeting agreed to prepare a single residues evaluation for metalaxyl and metalaxyl-M. The information and studies relating to both metalaxyl and metalaxyl-M including the proposed residue definitions, maximum residue levels, STMRs, HRs and dietary risk assessment are reported in the appraisal for metalaxyl (See Section 5.9).

As the available analytical methods do not use enantiomeric selective columns and the residues are determined as metalaxyl (sum of R- and S- enantiomers), the Meeting agreed to withdraw all previous recommendations made in terms of metalaxyl-M and to make new recommendations in terms of metalaxyl for apple, grapes, onion, bulb, potato, spinach, and sunflower seed. Due to the lack of residue data according to GAP, no recommendation could be made for cacao beans, head lettuce, sweet peppers and tomato.

RECOMMENDATIONS

Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL for plant commodities: *metalaxyl (sum of enantiomers)*.

Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in plant commodities: *metalaxyl (sum of enantiomers) and N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (M8; free and conjugated; sum of enantiomers), expressed as metalaxyl.*

Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL in animal commodities: *the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.*

Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in animal commodities: *the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M1 (N-(2,6-dimethylphenyl)-N-(methoxyacetyl) alanine), M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester), M6 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine), M7 (N-(2,6-dimethyl- 5-hydroxyphenyl)-N-(methoxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.*

The residue is not fat-soluble.

5.11 Pendimethalin (292)

RESIDUE AND ANALYTICAL ASPECTS

Pendimethalin is the ISO-approved common name for N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine (IUPAC), with the CAS number 040487-42-1. Pendimethalin is a selective herbicide belonging to the dinitroaniline chemical class. It is a selective herbicide used to control most annual grasses and certain broadleaf weeds in various crops, such as fruits, vegetables, cereals, pulses, oilseeds, root crops and ornamentals. Its primary mode of action is to prevent plant cell division and elongation in susceptible species by inhibition of microtubule formation. Pendimethalin was evaluated for the first time by the 2016 JMPR where an ADI of 0–0.1 mg/kg bw and an ARfD of 1 mg/kg bw were established. It was also evaluated for residues by the 2019 Extra JMPR for additional uses.

The residue definition for compliance with the MRL and dietary risk assessment for plant and animal commodities is parent *Pendimethalin*.

The residue is fat-soluble.

In addition, previous Meetings assessed metabolites M455H025, M455H029 and M455H030 under the threshold of toxicological concern (TTC) approach.

Pendimethalin was scheduled at the Fifty-first Session of the CCPR for evaluation of additional uses by the 2020 JMPR, which was postponed to the 2021 JMPR. The current Meeting received new information on use patterns for pendimethalin in grapes, leek, broccoli, cauliflower, tomato, pepper, beans/peas (dry), soya bean (dry), celeriac, wheat, rice, maize (corn), sugarcane, sunflower, parsley (leaves), sage and related salvia species supported by additional analytical methods and recovery data, storage stability studies, supervised field trials and studies simulating typical processing conditions.

Methods of analysis

The Meeting received several analytical methods for pendimethalin in plant matrices. The basic principle employs liquid extraction by homogenisation with various solvent combinations. Clean-up is normally achieved by Florisil or C₁₈ solid-phase extraction (SPE) or/and gel permeation chromatography (GPC). Residues are determined by LC-MS/MS, GC-ECD, GC-MS or GC-MS/MS. The Meeting concluded that the presented methods were sufficiently validated and are suitable for measuring pendimethalin with an LOQ of 0.01 or 0.02 or 0.05 mg/kg depending on the plant matrix. In several methods the metabolite M455H025 was also validated with the same respective LOQs.

Stability of residues in stored analytical samples

The storage stability of pendimethalin was investigated by the 2016 JMPR and it was concluded that in plant matrices with high water, high starch and high acid content pendimethalin is stable for at least 24 months. In soya bean and almond nutmeat, which are representative commodities for high oil content matrices, stability was achieved for up to 18 and 12 months, respectively. Additional storage stability studies for dried peas (high protein content), carrots (high starch), potatoes (high starch) and wheat forage (high water) were provided to the current meeting. The Meeting concluded that pendimethalin is stable in dried peas, carrots, potatoes and wheat forage for at least 2 years (24 months) when stored frozen at -18 °C, confirming the conclusion of the 2016 JMPR.

Results of supervised residue trials on crops

The Meeting received new GAP information and supervised residue trial data involving applications of pendimethalin on grapes, leek, broccoli, cauliflower, tomato, pepper, beans/peas (dry), soya bean (dry), celeriac, wheat, rice, maize (corn), sugarcane, sunflower, parsley (leaves), sage and related salvia species.

Grapes

The cGAP for grapes is in the USA with a maximum rate of 1 × 6.7 kg ai/ha (band or broadcast soil application, pre-emergence of weeds) and a PHI of 21 days.

Supervised field trials conducted in the USA, matching the cGAP with one application at BBCH 79–87, were provided. In grapes, residues of pendimethalin were (n = 9): < 0.05 (9) mg/kg.

The Meeting estimated a maximum residue level of 0.05(*) mg/kg, an STMR of 0.05 mg/kg and an HR of 0.05 mg/kg for grapes.

Leek

The cGAP for leek is in Canada with a maximum rate of 1 × 3 kg ai/ha (after transplanting leeks and before weed emergence, BBCH growth stage not indicated on the label) and a PHI of 30 days. Supervised field trials on leek matching or approximating the Canadian GAP were not available.

In Hungary, pendimethalin is registered at a maximum rate of 1 × 1.6 kg ai/ha (directly after transplanting, BBCH growth stage not mentioned on the label) with a PHI of not less than 60 days.

Supervised field trials performed in Europe at a maximum rate of 1 × 1.6 kg ai/ha (BBCH 13–15) with PHIs of not less than 60 days, approximating the Hungarian GAP, were provided. In leeks, residues of pendimethalin were (n = 22): < 0.01 (6), < 0.02 (9), < 0.05 (5), 0.082, and 0.23 mg/kg.

The Meeting estimated a maximum residue level of 0.3 mg/kg, an STMR of 0.02 mg/kg and an HR of 0.23 mg/kg for leeks.

Flowerhead Brassica vegetables (Broccoli, Cauliflower)

The cGAP for broccoli and cauliflower (the only commodities in the Subgroup of Flowerhead Brassica vegetables) is in Germany with a maximum rate of 1 × 1.6 kg ai/ha (pre-planting) and a PHI determined by growth stage at application.

Supervised field trials performed in Belgium, France, Germany, Italy, the Netherlands and Spain on broccoli (two trials) and cauliflower (eight trials), matching the European GAP were provided. In flower heads, residues of pendimethalin were (n = 10): < 0.01 (3) and < 0.05 (7) mg/kg.

The Meeting noted that the more contemporary trials used a new method with an LOQ of 0.01 mg/kg and decided to estimate a maximum residue level of 0.01(*) mg/kg.

The Meeting noted that at higher rates (2.64 kg ai/ha) in the field trials residues were also below the LOQ and decided to recommend an STMR of 0 mg/kg and an HR of 0 mg/kg for the Subgroup of Flowerhead Brassica vegetables.

Fruiting vegetables, other than cucurbits

The cGAP for pendimethalin is in the USA with a maximum rate of 1 × 1.65 kg ai/ha (BBCH not specified) and a PHI of 21 days for tomatoes and 70 days for all other fruiting vegetables.

Supervised field trials conducted in the USA on tomatoes according to the critical GAP were provided. In tomato fruits, residues of pendimethalin were (n = 11): < 0.05 (11) mg/kg.

Supervised field trials conducted in the USA on sweet peppers were provided. Pendimethalin was applied once, pre-transplant, at maximum rates of 1 × 1.46–2.35 kg ai/ha with PHIs 36–90 days. Since in all cases residues were < LOQ, the trials were considered to sufficiently support the US GAP. Residues of pendimethalin in sweet peppers were (n = 8): < 0.05 (8) mg/kg.

The Meeting estimated maximum residue levels of 0.05(*) mg/kg, STMRs of 0.05 mg/kg and HRs of 0.05 mg/kg for both tomatoes and peppers. Noting the absence of residues in these representative crops the Meeting agreed to extrapolate the recommendation to the whole group of fruiting vegetables, other than cucurbits.

Based on the early soil application and that all residues in sweet peppers were below the LOQ the Meeting decided not to estimate a maximum residue level for dried chili peppers.

Soya bean (dry)

The cGAP for soya bean is in the USA with a ground application (pre-planting) at a maximum rate of 1×2.13 kg ai/ha and a PHI of 85 days.

Supervised field trials approximating the US GAP were provided. In seeds, residues of pendimethalin were ($n = 5$): < 0.05 (5) mg/kg. The number of available trials was not sufficient to estimate a maximum residue level according to the US GAP.

In addition, pendimethalin is registered in Hungary as a ground application up to BBCH 08 at a maximum rate of 1×1.4 kg ai/ha with a PHI determined by growth stage at application.

Supervised field trials conducted in France, Italy, Spain, approximating the Hungarian GAP were provided. In seeds, residues of pendimethalin were ($n = 7$): < 0.05 (7) mg/kg.

Noting that residues in the two US trials at a more critical GAP were also below the LOQ the Meeting estimated a maximum residue level of $0.05^{(*)}$ mg/kg and an STMR of 0.05 mg/kg for soya bean (dry) based on the European data set.

Celeriac

The cGAP for celeriac is in Germany with a ground application up to BBCH 13 at a maximum rate of 1×1.6 kg ai/ha and a PHI determined by growth stage at application.

In supervised field trials conducted in Germany, only one trial (< 0.02 mg/kg) matched the German GAP. The Meeting was unable to estimate a maximum residue level for celeriac due to an insufficient number of trials matching the critical GAP.

Wheat

The cGAP for wheat is in Belgium with a pre or post emergence application up to BBCH 30, at a maximum rate of 1×1 kg ai/ha and the PHI determined by growth stage at application.

In four trials (conducted in Austria, Germany, Netherlands, and the UK) matching the Belgian GAP (post emergence application at BBCH 30) residues of pendimethalin in grain were ($n = 4$): < 0.01 (4) mg/kg. In two additional trials in Italy and Spain at a slightly higher rate of 1×1.6 kg ai/ha residues of pendimethalin in grain were: ($n = 2$): < 0.01 (2) mg/kg.

The Meeting noted that in 12 US trials conducted at an earlier growth stage (BBCH 22) but at a higher rate (1×1.6 kg ai/ha) residues were: < 0.05 (12) mg/kg.

The Meeting agreed the data were sufficient to estimate a maximum residue level of $0.01^{(*)}$ mg/kg and an STMR of 0.01 mg/kg for wheat.

Rice

The cGAP for rice is in Hungary with a post emergent ground application up to BBCH 12 at a maximum rate of 1×1.6 kg ai/ha and a PHI determined by growth stage at application. Supervised field trials (in upland and paddy rice) approximating the European GAP were provided. The residues of pendimethalin in rice grain were ($n = 7$): < 0.005 (1), < 0.01 (4), and < 0.05 (2) mg/kg.

Pendimethalin is also registered in Brazil as a pre-emergent ground application in rice at maximum rates of 1×1.8 kg ai/ha. Supervised field trials (upland and paddy rice) approximating the Brazilian GAP were provided. In seeds, residues of pendimethalin were ($n = 2$): < 0.05 (2) mg/kg.

The Meeting noted that in 2 US trials conducted at BBCH 11–12 with exaggerated rates (2.2–4.6 kg ai/ha) residues were: < 0.01 (2) mg/kg.

The Meeting estimated a maximum residue level of 0.01(*) mg/kg and an STMR of 0 mg/kg for rice.

Maize

The cGAP for maize is in the USA with a post-emergent ground application at BBCH 18 with a rate of 1 × 2.1 kg ai/ha. The PHI is determined by growth stage at application.

In supervised field trials, conducted in France, Germany, Italy and Spain, at BBCH 15–16 approximating the US GAP residues of pendimethalin were (n = 12): < 0.05 (12) mg/kg.

The Meeting estimated a maximum residue level of 0.05(*) mg/kg and an STMR of 0.05 mg/kg for maize.

Sugarcane

The cGAP for sugarcane is in the USA with a pre-emergent or “layby under the canopy” ground application at maximum rates of 1 × 4.5 kg ai/ha and a PHI determined by growth stage at application.

In supervised field trials approximating the US GAP with early post emergent application or pre-canopy closure (directed at the soil between rows) residues of pendimethalin in stems were (n = 4): < 0.05 (4) mg/kg.

In addition, sugarcane is registered in Brazil as a pre-emergent application at a maximum rate of 1 × 2.5 kg ai/ha with a PHI determined by growth stage at application. Supervised field trials according to the Brazilian GAP were provided. In stems, residues of pendimethalin were (n = 5): < 0.01 (3) and < 0.05 (2) mg/kg.

The Meeting noted that the more contemporary trials used a new method with an LOQ of 0.01 mg/kg and decided to estimate a maximum residue level of 0.01(*) mg/kg.

Noting that residues in a US trial conducted at an exaggerated rate (of 6.7 kg ai/ha) were also below the LOQ the Meeting estimated an STMR of 0 mg/kg for sugarcane.

Sunflower

The cGAP for sunflower is in the USA with a ground or aerial pre-plant application at a maximum rate of 1 × 1.84 kg ai/ha and a PHI of 60 days. Supervised field trials matching or approximating the US GAP were not provided.

In addition, pendimethalin is registered in the UK as a pre or post sowing application (before crop and weed emergence) at a maximum rate of 1 × 1.3 kg ai/ha with a PHI determined by growth stage at application. In supervised field trials approximating the UK GAP residues of pendimethalin in seeds were (n = 9): < 0.05 (9) mg/kg. The Meeting also noted that in 9 trials treatments were conducted at higher rates (1.98–2.64 kg ai/ha) and residues of pendimethalin in seeds were < 0.05 (9) mg/kg.

The Meeting estimated a maximum residue level of 0.05(*) mg/kg and an STMR of 0.05 mg/kg for sunflower seed.

Parsley

The cGAP for parsley is in Germany as a ground application after planting (BBCH 12–15) at a maximum rate of 1 × 1.6 kg ai/ha with a PHI of 28 days.

In supervised field trials, performed in France, Germany and the Netherlands, approximating the German GAP residues of pendimethalin in leaves/aerial plant parts were (n = 10): 0.065, 0.066, 0.12, 0.28, 0.29, 0.32, 0.45, 0.6, 0.61 and 0.76 mg/kg.

The Meeting estimated a maximum residue level of 1.5 mg/kg, an STMR of 0.305 mg/kg and an HR of 0.76 mg/kg for parsley.

Sage

The cGAP for sage is in France as a ground application, after planting (BBCH stage not specified), at a maximum rate of 1×1.32 kg ai/ha with a PHI of 42 days.

As no trials matched GAP, the Meeting was unable to estimate a maximum residue level for sage.

Animal feedstuffs

Wheat forage and hay

The cGAP for wheat is in the USA as a pre or post emergence application up to BBCH 37 at a maximum rate of 1×1.6 kg ai/ha with a PHI of 11 days for forage and 28 days for hay.

The Meeting received no trials matching the cGAP for forage and one trial in hay. As a result, the Meeting was unable to estimate a maximum residue level in wheat hay.

Wheat straw

The cGAP for wheat is in the USA as a pre or post emergence application up to BBCH 37 at a maximum rate of 1×1.6 kg ai/ha.

Twelve supervised trials conducted in the USA and matching the US GAP at a PHI of 54–279 days were provided. Residues of pendimethalin were ($n = 12$): < 0.05 (10), 0.052 and 0.17 mg/kg.

The Meeting estimated a maximum residue level of 0.3 mg/kg, a median residue of 0.05 mg/kg and a highest residue of 0.17 mg/kg for wheat straw.

Rice hay

The cGAP for rice is in Hungary as a pre or post emergence ground application up to BBCH 12 at a maximum rate of 1×1.6 kg ai/ha. The PHI is determined by growth stage at application.

The Meeting received no trials matching the cGAP for hay. Therefore, the Meeting was unable to estimate a maximum residue level in rice hay.

Rice straw

The cGAP for rice is in Hungary as a pre or post emergence ground application up to BBCH 12 at a maximum rate of 1×1.6 kg ai/ha. The PHI is determined by the growth stage at application.

Seven supervised trials conducted in Hungary, Italy and Spain, matching the cGAP at a PHI of 54–279 days were provided. Residues of pendimethalin were ($n = 7$): < 0.01 (4), < 0.02 and < 0.05 (2) mg/kg.

The Meeting noted that the more contemporary trials used a new method with an LOQ of 0.01 mg/kg and decided to estimate a maximum residue level of 0.01(*) mg/kg.

Noting that residues in the Hungarian and Spanish trials conducted at exaggerated rates (of 2.31 and 2.24 kg ai/ha) were also below the LOQ the Meeting recommended median and highest residues of 0 mg/kg for rice straw.

Maize forage

The cGAP for maize forage is in the USA as a pre or post emergence application up to BBCH 18 at a maximum rate of 1×2.1 kg ai/ha with a PHI of 21 days.

Supervised field trials matching the US GAP were provided. In early forage (21 days after application), residues of pendimethalin were ($n = 2$): < 0.05 (2) mg/kg.

In addition, pendimethalin is registered in Hungary as a pre or post emergence application up to BBCH 14 at a maximum rate of 1×1.6 kg ai/ha with a 90-day PHI for maize forage.

In seven supervised field trials conducted in France, Germany, Italy and Spain approximating the Hungarian GAP residues of pendimethalin in maize forage were: < 0.05 (7) mg/kg.

The Meeting estimated median and highest residues of 0.05 mg/kg for maize forage, as received.

Maize fodder (stover)

The cGAP for maize is in the USA as a post-emergent ground application at BBCH 18 with a rate of 1 × 2.1 kg ai/ha. The PHI is determined by growth stage at application.

In six supervised field trials, conducted in France, Germany, Spain and the USA, matching the US GAP, residues of pendimethalin in maize stover were (n = 6): < 0.05 (6) mg/kg.

The Meeting estimated a maximum residue level of 0.05(*) mg/kg, a median residue of 0.05 mg/kg and a highest residue of 0.05 mg/kg for maize stover.

Fate of residues during processing

The Meeting received new processing studies on wheat, rice and sugar cane. Processed commodities from wheat (e.g. bran, flour, wholemeal bread), rice (e.g. bran, hulls) and sugar cane (e.g. raw sugar, molasses) were derived using simulated commercial practices. However, no processing factors could be calculated since residues in both the raw and processed commodities were below the LOQ (0.05 mg/kg).

Residues in animal commodities

Farm animal feeding studies

No additional information on transfer of residues to livestock was provided to the current Meeting. Please refer to the 2016 JMPR Report and Evaluation⁵.

Estimated maximum and mean dietary burdens of livestock and animal commodities maximum residue levels

Dietary burdens were calculated for beef cattle, dairy cattle, broilers and laying poultry based on feed items evaluated by the Meeting. The dietary burdens, estimated using the most recent version of the OECD livestock dietary burden calculator, are presented in Annex 6 and summarised below. In the 2016 JMPR the calculations were made according to the animal diets listed in Appendix IX of the 2016 edition of the FAO manual. Results of the estimated maximum and mean dietary burdens are summarised in Table 1.

Table 1 Estimated maximum and mean dietary burdens of farm animals

Animal dietary burden, pendimethalin residues, ppm of dry matter diet (DM)					
		US-Canada	EU	Australia	Japan
Beef cattle	max	176	586	0.85	468
	mean	79	263	0.23	210
Dairy cattle	max	527	703	1.34	820 ❶
	mean	237	316	0.29	368 ❷
Poultry Broiler	max	0.057	0.37	0.057	0.04
	mean	0.057	0.1	0.057	0.04
Poultry Layer	max	0.057	0.46 ❸	0.057	0.045
	mean	0.057	0.12 ❹	0.057	0.05

❶ Suitable for estimation of maximum residue levels in meat and milk.

❷ Suitable for estimation of median residue levels in meat and milk

❸ Suitable for estimation of maximum residue levels in poultry meat and eggs

⁵ Pesticide residues in food – 2016. Report of the Joint Meeting of the FAO Panel of Experts on Pesticide Residues in Food and the Environment and the WHO Core Assessment Group on Pesticide Residues. FAO Plant Production and Protection Paper, 229, 2016. and Pesticide residues in food – 2016 evaluations. Part I. Residues. FAO Plant Production and Protection Paper, 231, 2017

④ Suitable for estimation of median residue levels in poultry meat and eggs

The pendimethalin dietary burden reached a maximum level of 586 ppm of dry matter diet in beef cattle, 820 ppm diet in dairy cattle and 0.46 mg/kg diet in poultry. The mean dietary burdens were 263 mg/kg in beef cattle, 368 mg/kg diet in dairy cattle and 0.12 mg/kg in poultry. These results are similar to the previous livestock dietary burden calculations performed by the 2016 JMPR (highest maximum dietary burden was 590 ppm of dry matter diet in beef cattle, 820 ppm diet in dairy cattle and 0.41 mg/kg diet in poultry). The meeting confirmed its previous recommendations for animal commodities.

RECOMMENDATIONS

On the basis of the data from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI and IESTI assessment.

The residue definition for compliance with the MRL and dietary risk assessment for plant and animal commodities is parent *Pendimethalin*.

The residue is fat-soluble.

DIETARY RISK ASSESSMENT

Long-term dietary exposure

The ADI for pendimethalin is 0–0.1 mg/kg bw. The International Estimated Daily Intakes (IEDIs) for pendimethalin were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs were 1% of the maximum ADI. The Meeting concluded that long-term dietary exposure to residues of pendimethalin from uses considered by the JMPR is unlikely to present a public health concern.

Acute dietary exposure

The ARfD of pendimethalin is 1 mg/kg bw. The International Estimate of Short-Term Intakes (IESTIs) for pendimethalin were calculated for the food commodities and their processed commodities for which HRs/HR-Ps or STMRs/STMR-Ps were estimated by the present Meeting and for which consumption data were available. The results are shown in Annex 4 of the 2021 JMPR Report.

The IESTIs were up to 1% of the ARfD for the general population to up to 1% of the ARfD for children. The Meeting concluded that acute dietary exposure to residues of pendimethalin from uses considered by the present Meeting is unlikely to present a public health concern.

Assessment of metabolites M455H025, M455H029 and M455H030 using the threshold of toxicological concern (TTC) approach

The metabolite M455H025 could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

The current Meeting noted that the 2016 JMPR estimated a dietary exposure of 1.3 µg/kg bw per day for M455H025, based on rotational crops taking into account a maximum annual rate of 6.7 kg/ha. The uses considered by the current Meeting do not involve higher annual rates. The Meeting confirmed its previous exposure estimation for M455H025 and concluded that the estimated dietary exposure from uses considered by the JMPR is below the TTC for Cramer Class III compounds and is unlikely to present a public health concern. Should further uses be considered in the future, these conclusions may need to be re-evaluated.

The metabolite M455H029 could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

The current Meeting noted that the 2016 JMPR estimated a dietary exposure of 0.3 µg/kg bw per day for M455H029, based on its occurrence exclusively in ruminant liver and kidney. The uses considered by the current Meeting do not result in higher dietary feed burdens than estimated by the 2016 JMPR. The Meeting confirmed its previous exposure estimation for M455H029 and concluded that the estimated dietary exposure from uses considered by the JMPR is below the TTC for Cramer Class III compounds and is unlikely to present a public health concern. Should further uses be considered in the future, these conclusions may need to be re-evaluated.

The metabolite M455H030 could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

The present Meeting noted that the 2016 JMPR estimated a dietary exposure of 0.1 µg/kg bw per day for M455H030, exclusively based on rotational radish roots taking into account a maximum annual rate of 6.7 kg/ha. The uses considered by the current Meeting do not involve higher annual rates. The Meeting confirmed its previous exposure estimation for M455H030 and concluded that the estimated dietary exposure from uses considered by the JMPR is below the TTC for Cramer Class III compounds and is unlikely to present a public health concern. Should further uses be considered in the future, these conclusions may need to be re-evaluated.

5.12 Pyrasulfotole (321)

TOXICOLOGY

Pyrasulfotole is the ISO-approved common name for (5-hydroxy-1,3-dimethylpyrazol-4-yl)(α,α,α -trifluoro-2-mesyl-*p*-tolyl)methanone (IUPAC), Chemical Abstracts Service number 365400-11-9.

The primary mode of action (MOA) of pyrasulfotole is as an inhibitor of the enzyme 4-hydroxyphenylpyruvate dioxygenase (4-HPPD) in susceptible plants, thereby disrupting the synthesis of carotenoids that are produced by plants to protect against oxidative and photolytic damage.

Pyrasulfotole has not previously been evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) and was reviewed by the present Meeting at the request of the Codex Committee on Pesticide Residues (CCPR).

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with current guidelines, unless otherwise specified. No additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Overall, the Meeting considered that the database was adequate for the risk assessment.

Biochemical aspects

The toxicokinetics and metabolism of pyrasulfotole have been investigated in the rat, following oral dosing. Although only a preliminary study, the toxicokinetics of pyrasulfotole were investigated following a single oral dose via gavage of 100 mg/kg bw [^{14}C]pyrasulfotole administered to male and female rats. Excretion via urine was greater than via the faeces for both sexes. The C_{max} value for radioactivity was higher for males (75.48 $\mu\text{g eq/g}$) than for females (56.84 $\mu\text{g eq/g}$). The absorption of the radioactivity was very rapid with mean T_{max} interpolated values of 0.5 hours for both sexes. The elimination half-life values based upon the terminal data points also indicated a slight sex difference and the values obtained were found to be around four hours for males and around six hours for females. Pyrasulfotole was found not to be extensively metabolized in the rat, with only five radioactive fractions in urine and four radioactive fractions in faecal extracts.

In the main study, absorption, distribution, metabolism and excretion (ADME) were studied in male rats following single dosing with 10 mg/kg bw [^{14}C]pyrasulfotole, labelled either at the phenyl or pyrazole positions. At time of termination (48 to 52 hours post dosing) oral absorption was 74–76% based on urine, tissues and carcass. Distribution showed limited amounts of pyrasulfotole in tissues, as less than 2% of the administered dose (AD) remained in the carcass and tissues. Highest residue levels were found in the liver and kidney. Excretion following a single oral or intravenous dose at 10 mg/kg body weight (bw) was rapid; 96–111% of the AD excreted within 24 hours. Irrespective of route or label, the majority of the radioactivity was excreted in the urine (73–91%), with less being excreted in the faeces (8–32%). Much of the administered dose was rapidly excreted in the urine within six hours of dosing (57–84%). Regarding metabolism, 87–95% of AD was excreted unchanged as pyrasulfotole. Minor metabolites in urine and faeces were hydroxymethyl pyrasulfotole (up to 2.3% found), desmethyl pyrasulfotole (up to 8.1% found), and AE B197555 (up to 1.4% found). The major metabolic pathway occurred via *N*-demethylation of pyrasulfotole.

Toxicological data

The acute oral median lethal dose (LD_{50}) of pyrasulfotole was >5000 mg/kg bw. No data on acute dermal or inhalation toxicity, skin or eye irritation or skin sensitization were provided.

In repeated-dose toxicity studies on mice, rats and dogs, the main effects were on urinary tract, bladder, kidney, and eyes.

In an exploratory 14-day oral gavage study in mice, pyrasulfotole was administered at doses of 0, 100, 300 or 1000 mg/kg bw per day. In this study, no NOAEL could be identified as kidney effects (multifocal cortical tubuloe epithelial degeneration and/or multifocal basophilic tubules) were seen at all dose levels tested.

In a 28-day dietary toxicity study in mice, pyrasulfotole was administered at dietary concentrations of 0, 200, 1000 or 5000 ppm (equal to 0, 35.8, 192, and 961 mg/kg bw per day in males, 0, 45.0, 233, and 1082 mg/kg bw per day in females). The NOAEL was 1000 ppm (equal to 192 mg/kg bw per day) based on gritty content in the urinary bladder and histopathological findings in the bladder seen in males only, at 5000 ppm (equal to 961 mg/kg bw per day).

In a 90-day dietary toxicity study in mice, pyrasulfotole was administered at dietary concentrations of 0, 100, 750, 1500 or 3000 ppm (equal to 0, 16.5, 124, 259, and 500 mg/kg bw per day for males, 0, 19.7, 152, 326, and 617 mg/kg bw per day for females). The NOAEL was 3000 ppm (equal to 500 mg/kg bw per day), the highest dose tested.

In a 14-day exploratory dietary study in rats, pyrasulfotole was administered at dietary concentrations of 0, 400, 2000 or 7500 ppm (equal to 0, 27.4, 135 and 497 mg/kg bw per day for males, 0, 32.0, 157, and 547 mg/kg bw per day for females). The NOAEL was 2000 ppm (equal to 135 mg/kg bw per day) based on decreased body weight gain seen in males at 7500 ppm (equal to 497 mg/kg bw per day).

In a 90-day toxicity study in rats, pyrasulfotole was administered at dietary concentrations of 0, 2, 30, 1000, 7000 or 12 000 ppm (equal to 0.0, 0.13, 1.96, 66, 454, and 830 mg/kg bw per day for males, 0, 0.15, 2.32, 77, 537, and 956 mg/kg bw per day for females). The NOAEL was 1000 ppm (equal to 66 mg/kg bw per day) based on mortality, clinical signs, increased liver and kidney weight, effects on the eyes, macroscopic findings in the bladder and ureter, and histopathological findings in kidney, bladder and ureter at 7000 ppm (equal to 454 mg/kg bw per day).

In a 28-day toxicity study in dogs, pyrasulfotole was administered at dietary concentrations of 0, 5000, 13 000 or 26 000 ppm (equal to 0, 174, 469, and 860 mg/kg bw per day for males, 0, 171, 440, and 782 mg/kg bw per day for females). No NOAEL could be identified in this study; the low dose of 5000 ppm (equal to 171 mg/kg bw per day) was the LOAEL based on clinical chemistry findings.

In a 90-day toxicity study in dogs, pyrasulfotole was administered at dietary concentrations of 0, 100, 5000 or 1000 ppm (equal to 0, 3, 17, and 40 mg/kg bw per day for males, 0, 3, 17, and 33 mg/kg bw per day for females). The NOAEL was 1000 ppm (equal to 33 mg/kg bw per day), the highest dose tested.

In a one-year toxicity study in dogs, pyrasulfotole was administered at dietary concentrations of 0, 250, 1000 or 3000 ppm (equal to 0, 7, 34, and 101 mg/kg bw per day for males, 0, 9, 33, and 93 mg/kg bw per day for females). The NOAEL was 250 ppm in males (equal to 7 mg/kg bw per day) based on increased incidence and severity of tubular dilatation in the kidneys accompanied by increased kidney weight seen at 1000 ppm (equal to 34 mg/kg bw per day).

In a 78-week chronic toxicity/carcinogenicity study in mice, pyrasulfotole was administered at dietary concentrations of 0, 100, 1000 or 4000 ppm (equal to 0, 13.6, 137, and 560 mg/kg bw per day for males, 0, 16.7, 168, and 713 mg/kg bw per day for females). A NOAEL for systemic toxicity could not be identified. The lowest dose of 100 ppm (equal to 13.6 mg/kg bw per day) was the LOAEL based on gall bladder stones seen at all dose levels tested. Regarding carcinogenicity, the NOAEL was 1000 ppm (equal to 137 mg/kg bw per day) based on urinary tract carcinoma and papilloma observed at 4000 ppm (equal to 560 mg/kg bw per day). These tumours were seen in the presence of urinary stones/calculi.

In a two-year chronic toxicity/carcinogenicity study in rats, pyrasulfotole was administered at dietary concentrations of 0, 25, 250, 1000 or 2500 ppm (equal to 0, 1.0, 10, 41, and 104 mg/kg bw per day for males, 0, 1.4, 14, 57, and 140 mg/kg bw per day for females). The NOAEL for systemic toxicity was 25 ppm (equal to 1.0 mg/kg bw per day) based on the effect on the eyes and increased plasma cholesterol at 250 ppm (equal to 10 mg/kg bw per day). In males one squamous cell papilloma and one squamous

cell carcinoma were found at the high dose of 2500 ppm (equal to 104 mg/kg bw per day), and these were considered to be treatment-related. The findings were considered to be related to a chronic stimulation of the cornea, and thus caused by a nongenotoxic mechanism.

The Meeting concluded that pyrasulfotole is carcinogenic in mice and rats.

Pyrasulfotole was tested for genotoxicity in an adequate range of in vitro and in vivo assays. No evidence of genotoxicity was found.

The Meeting concluded that pyrasulfotole is unlikely to be genotoxic.

In view of the lack of genotoxicity, the fact that only urinary tract tumours were observed in mice at the highest dose tested in the presence of urinary stones/calculi, and that the corneal effects observed in rats at the highest dose tested are most likely related to pyrasulfotole's pesticidal mode of action to which rats are more sensitive, the Meeting concluded that pyrasulfotole is unlikely to pose a carcinogenic risk to humans from the diet.

In a two-generation reproductive toxicity study in rats, pyrasulfotole was administered at dietary concentrations of 0, 30, 300 or 3000 ppm (equal to 0, 2.5–3.7, 26–34 and 272–354 mg/kg bw per day for males, 0, 2.0–4.2, 22–39 and 229–393 mg/kg bw per day for females). No NOAEL for parental toxicity could be identified. In males at all dose levels tested, thyroid weight was increased and histopathological changes were seen (pigment deposition and colloid alteration), although the toxicological significance of these changes was considered equivocal. Therefore, the parental LOAEL was 30 ppm in males (equal to 2.5 mg/kg bw per day). The reproductive NOAEL was 3000 ppm (equal to 272 mg/kg bw per day) the highest dose tested. The offspring NOAEL was 30 ppm (equal to 2.0 mg/kg bw per day) based on a delay in balano-preputial separation in F₁ weanlings seen at 300 ppm (equal to 22 mg/kg bw per day).

In a developmental toxicity study in rats, pyrasulfotole was administered by gavage at dose levels of 0, 10, 100 or 300 mg/kg bw per day. The maternal NOAEL was 300 mg/kg bw per day, the highest dose tested. The embryo/fetal NOAEL was 10 mg/kg bw per day based on delayed ossification and reduced fetal weights observed at 100 mg/kg bw per day.

In a developmental toxicity study in rabbits, pyrasulfotole was administered by gavage at dose levels of 0, 10, 75 or 250 mg/kg bw per day. The maternal NOAEL was 75 mg/kg bw per day based on decreased food consumption and body weight gain observed at 250 mg/kg bw per day. The embryo/fetal NOAEL was 10 mg/kg bw per day, based on altered ossification patterns and minor skeletal variants observed at 75 mg/kg bw per day.

The Meeting concluded that pyrasulfotole is not teratogenic.

In an acute neurotoxicity study in rats, pyrasulfotole was administered by gavage at dose levels of 0, 200, 500 or 2000 mg/kg bw. The NOAEL was 500 mg/kg bw based on decreased motor and locomotor activity seen at 2000 mg/kg bw.

In a 90-day neurotoxicity study in rats, pyrasulfotole was administered at dietary concentrations of 0, 500, 2500 or 5000 ppm (equal to 0, 32.3, 166 and 345 mg/kg bw per day for males, 0, 41.9, 206 and 416 mg/kg bw per day for females). The systemic and neurotoxicity NOAELs were both 5000 ppm (equal to 345 mg/kg bw per day) the highest dose tested.

In a developmental neurotoxicity study in rats, pyrasulfotole was administered at dietary concentrations of 0, 45, 450 or 4500 ppm (equal to averaged values of 0, 3.8, 37.1 and 354 mg/kg bw per day). The NOAEL for maternal toxicity was 45 ppm (equal to 3.8 mg/kg bw per day) based on a decreased fertility index and ocular opacity observed at 450 ppm (equal to 37.1 mg/kg bw per day). The embryo/fetal NOAEL was 45 ppm (equal to 3.8 mg/kg bw per day) based on decreased body weight, delayed preputial separation, decreased cerebrum and/or cerebellum length and decreased cerebellum thickness observed at 450 ppm (equal to 37.1 mg/kg bw per day).

The Meeting concluded that pyrasulfotole showed potential developmental neurotoxicity in rat pups at the highest dose level tested, however, these effects were observed in the presence of maternal toxicity.

No evidence of immunotoxicity was reported in routine toxicological studies with pyrasulfotole.

The Meeting concluded that pyrasulfotole is unlikely to be immunotoxic.

Toxicological data on metabolites and/or degradates

Metabolite RPA 203328

RPA 203328 (MTFM-BA; AE B197555) is a minor rat metabolite and a plant metabolite. The acute oral LD₅₀ of this metabolite was >5000 mg/kg bw.

In a 28-day dietary toxicity study in rats, RPA 203328 was administered at dietary concentrations of 0, 150, 500, 5000 or 15 000 ppm (equal to 0, 11.1, 37.6, 377 and 1118 mg/kg bw per day for males, 0, 12.7, 42.7, 421 and 1269 mg/kg bw per day for females). The NOAEL was 15 000 ppm (equal to 1118 mg/kg bw per day) the highest dose tested.

In a 90-day dietary toxicity study in rats, RPA 203328 was administered at dietary concentrations of 0, 1200, 4800 or 12 000 ppm (equal to 0, 73.2, 306 and 769 mg/kg bw per day for males, 0, 93.1, 371 and 952 mg/kg bw per day for females). The NOAEL was 12 000 ppm (equal to 769 mg/kg bw per day) the highest dose tested.

Metabolite RPA 203328 was tested for genotoxicity in an adequate range of in vitro and in vivo assays. No evidence of genotoxicity was found.

In a developmental toxicity study in rats RPA 203328 was administered by gavage at dose levels of 0, 75, 250 or 750 mg/kg bw per day. The maternal NOAEL and embryo/fetal NOAEL were 750 mg/kg bw per day, the highest dose tested.

The Meeting concluded that metabolite RPA 203328 is not of any toxicological concern.

Desmethyl pyrasulfotole (AE 1073910)

Desmethyl pyrasulfotole is a metabolite that was found in the rat ADME study at levels up to 8.1%. Taking into account the oral absorption of pyrasulfotole at about 75%, total level for this metabolite in the rat will be around 10% of the absorbed dose. No specific toxicity studies with this metabolite were available. Considering the amount found in the rat and its structural similarity to pyrasulfotole, it can be assumed that desmethyl pyrasulfotole is of similar toxicity to the parent compound. The reference values of pyrasulfotole can therefore be applied.

Pyrasulfotole-desmethyl-O-glucoside

Pyrasulfotole-desmethyl-O-glucoside was not found as such in the rat ADME study and no specific toxicity studies are available. However, considering this metabolite is the conjugate of desmethyl-pyrasulfotole, the same conclusion applies. The reference values of pyrasulfotole can be applied.

Hydroxymethyl-pyrasulfotole (AE 0317309)

Hydroxymethyl pyrasulfotole is a minor metabolite in the rat (up to 2.3% found). No specific toxicity studies with this metabolite were available. Considering the structural similarity to the parent compound pyrasulfotole, the reference values of pyrasulfotole can be applied.

Microbiological data

In the case of pyrasulfotole no information was available concerning type of antimicrobial action and its mechanism, effects on the microbiome of the human gastrointestinal tract, antimicrobial spectrum of activity or antimicrobial resistance mechanisms and genetics.

Human data

In reports on manufacturing plant personnel, no adverse health effects were noted.

No information on accidental or intentional poisoning in humans was available.

The Meeting concluded that the existing database on pyrasulfotole was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting established an ADI of 0–0.01 mg/kg bw, based on the NOAEL of 1 mg/kg bw per day in the two-year study in rats and using a safety factor of 100. The margin between the upper bound ADI and the LOAEL for urinary tract carcinoma and papilloma in mice is 56 000. It is noted that in the two-generation rat study a LOAEL of 2.5 mg/kg bw per day (lowest dose tested) was identified. However, the effects seen at this LOAEL were considered of equivocal toxicological significance, therefore the Meeting decided to base the ADI on the NOAEL from the two-year rat study.

The Meeting concluded that it was not necessary to establish an ARfD for pyrasulfotole in view of its low acute oral toxicity and the absence of developmental toxicity or any other toxicological effects that would be likely to be elicited by a single dose.

A toxicological monograph was prepared.

Levels relevant to risk assessment of pyrasulfotole

Species	Study	Effect	NOAEL	LOAEL
Mouse	78-week study of toxicity and carcinogenicity ^a	Toxicity	-	100 ppm, equal to 13.6 mg/kg bw per day ^d
		Carcinogenicity	1000 ppm, equal to 137 mg/kg bw per day	4000 ppm, equal to 560 mg/kg bw per day
Rat	Acute neurotoxicity study ^b	Neurotoxicity	500 mg/kg bw	2000 mg/kg bw
	Two-year studies of toxicity and carcinogenicity ^{a, d}	Toxicity	25 ppm, equal to 1.0 mg/kg bw per day	250 ppm, equal to 10 mg/kg bw per day
		Carcinogenicity	1000 ppm, equal to 41 mg/kg bw per day	2500 ppm, equal to 104 mg/kg bw per day
	Two-generation study of reproductive toxicity ^a	Reproductive toxicity	3000 ppm, equal to 272 mg/kg bw per day ^c	-
		Parental toxicity	-	30 ppm, equal to 2.5 mg/kg bw per day ^d
		Offspring toxicity	30 ppm, equal to 2.0 mg/kg bw per day	300 ppm, equal to 22 mg/kg bw per day
	Developmental toxicity study ^b	Maternal toxicity	300 mg/kg bw per day ^c	-
		Embryo and fetal toxicity	10 mg/kg bw per day	100 mg/kg bw per day
	Developmental neurotoxicity study ^a	Maternal toxicity	45 ppm, equal to 3.8 mg/kg bw per day	450 ppm, equal to 37.1 mg/kg bw per day

Species	Study	Effect	NOAEL	LOAEL
		Embryo and fetal toxicity	45 ppm, equal to 3.8 mg/kg bw per day	450 ppm, equal to 37.1 mg/kg bw per day
Rabbit	Developmental toxicity study ^b	Maternal toxicity	75 mg/kg bw per day	250 mg/kg bw per day
		Embryo and fetal toxicity	10 mg/kg bw per day	75 mg/kg bw per day
Dog	One-year studies of toxicity ^a	Toxicity	250 ppm, equal to 7 mg/kg bw per day	1000 ppm, equal to 34 mg/kg bw per day

Metabolite MTFM-BA (RPA 203328, AE B197555)

Rat	Four-week study of toxicity ^a	Toxicity	15 000 ppm, equal to 1118 mg/kg bw per day ^c	-
	Ninety-day study of toxicity ^a	Toxicity	12 000 ppm, equal to 769 mg/kg bw per day ^c	-
	Developmental toxicity study ^b	Maternal toxicity	750 mg/kg bw per day ^c	-
		Embryo and fetal toxicity	750 mg/kg bw per day ^c	-

^a Dietary administration

^b Gavage administration

^c Highest dose tested

^d Lowest dose tested

Acceptable daily intake (ADI) applies to pyrasulfotole, desmethyl pyrasulfotole (AE 1073910) and pyrasulfotole-desmethyl-O-glucoside, expressed as pyrasulfotole

0–0.01 mg/kg bw

Acute reference dose (ARfD) applies to pyrasulfotole, desmethyl pyrasulfotole (AE 1073910) and pyrasulfotole-desmethyl-O-glucoside, expressed as pyrasulfotole

Not necessary

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Critical end-points for setting guidance values for exposure to pyrasulfotole

Absorption, distribution, excretion and metabolism in mammals

Rate and extent of oral absorption	74–76% at 10 mg/kg bw within 48–52 hours based on urine, tissues and carcass
Dermal absorption	No data
Distribution	Limited; less than 2% remaining in carcass and tissues; highest concentrations in liver and kidney
Potential for accumulation	No evidence of accumulation
Rate and extent of excretion	Rapid excretion (96–111% excreted within 24 hours), predominantly in urine (73–91%), less in faeces (8–32%)

Metabolism in animals	Limited, as 87–95% was excreted as unchanged pyrasulfotole; hydroxymethyl pyrasulfotole, desmethyl pyrasulfotole and AE B197555 were minor metabolites in urine and faeces
Toxicologically significant compounds in animals and plants	Pyrasulfotole
Acute toxicity	
Rat, LD ₅₀ , oral	>5000 mg/kg bw
Rat, LD ₅₀ , dermal	No data
Rat, LC ₅₀ , inhalation	No data
Rabbit, dermal irritation	No data
Rabbit, ocular irritation	No data
Mouse, dermal sensitization	No data
Guinea pig, dermal sensitization	No data
Short-term studies of toxicity	
Target/critical effect	Urinary bladder (mouse); kidney (rat, dog); eyes (rat); urinary tract (rat, dog)
Lowest relevant oral NOAEL	7 mg/kg bw per day (dog)
Lowest relevant dermal NOAEL	No data.
Lowest relevant inhalation NOAEC	No data.
Long-term studies of toxicity and carcinogenicity	
Target/critical effect	Gall bladder stones (mouse); eyes, cholesterol (rat)
Lowest relevant NOAEL	1.0 mg/kg bw per day (rat)
Carcinogenicity	Carcinogenic in mice (urinary tract carcinoma and papilloma) and rats (corneal carcinoma and papilloma) ^a
Genotoxicity	
Unlikely to be genotoxic ^a	
Reproductive toxicity	
Target/critical effect	Parental: thyroid Offspring: delay in balanopreputial separation
Lowest relevant parental NOAEL	<2.5 mg/kg bw per day (rat)
Lowest relevant offspring NOAEL	2.0 mg/kg bw per day (rat)
Lowest relevant reproductive NOAEL	272 mg/kg bw per day (rat)
Developmental toxicity	
Target/critical effect	Delayed ossification, fetal weight (rat); food consumption, body weight gain, ossification and skeletal variations (rabbit)
Lowest relevant maternal NOAEL	75 mg/kg bw per day (rabbit)
Lowest relevant embryo/fetal NOAEL	10 mg/kg bw per day (rat, rabbit)
Neurotoxicity	
Acute neurotoxicity NOAEL	500 mg/kg bw (rat)
Subchronic neurotoxicity NOAEL	345 mg/kg bw per day, highest dose tested (rat)
Developmental neurotoxicity NOAEL	3.8 mg/kg bw per day (rat)

Immunotoxicity

No data

Studies on toxicologically relevant metabolites

MTFM-BA (RPA 203328, AE B197555)	Acute oral LD ₅₀ : >5000 mg/kg bw (rat) 28-day NOAEL: 1118 mg/kg bw per day, highest dose tested (rat) 90-day NOAEL: 769 mg/kg bw per day, highest dose tested (rat) Developmental toxicity: maternal and embryo/fetal NOAEL of 750 mg/kg bw per day, highest dose tested (rat) Not genotoxic (Ames, in vitro chromosome aberration, in vitro mammalian cell gene mutation, micronucleus in vivo)
Microbiological data	No data
Human data	No clinical cases or poisoning incidents had been recorded

^a Unlikely to pose a carcinogenic risk to humans via exposure from the diet**Summary**

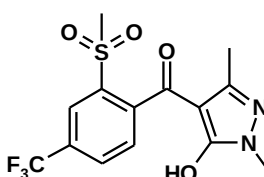
	Value	Study	Safety factor
ADI	0–0.01 mg/kg bw ^a	Two-year study of toxicity and carcinogenicity (rat)	100
ARfD	Unnecessary	-	-

^a Applies to pyrasulfotole, desmethyl-pyrasulfotole (AE 1073910) and pyrasulfotole-desmethyl-O-glucoside, expressed as pyrasulfotole**RESIDUE AND ANALYTICAL ASPECTS**

Pyrasulfotole is a pyrazole herbicide which acts as an inhibitor of the enzyme 4-hydroxyphenylpyruvate dioxygenase (HPPD) and consequently blocks the pathway of phenylquinone biosynthesis in plants, thus reducing photosynthetic yield. It is registered for selective post-emergence control of dicotyledonous weeds in cereals including wheat, barley, oats, rye, and triticale.

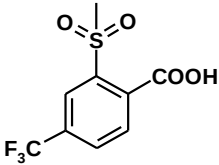
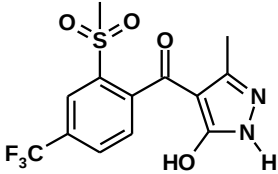
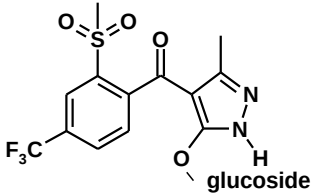
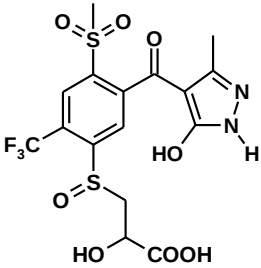
Pyrasulfotole was scheduled at the Fifty-first Session of the CCPR for evaluation as a new compound by the 2020 JMPR, which was postponed to the 2021 JMPR. The Meeting received information on identity, physical and chemical properties, animal and plant metabolism, rotational crops, environmental fate, analytical methods, GAP information, storage stability, processing, supervised residue trials and farm animal feeding.

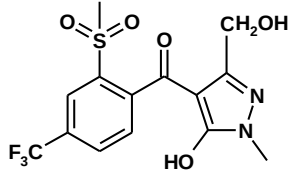
The IUPAC name for pyrasulfotole is (5-hydroxy-1,3-dimethyl-1H-pyrazol-4-yl)[2-(methylsulfonyl)-4-(trifluoromethyl)phenyl]methanone.



The following abbreviations are used for the major metabolites discussed below:

Table 1 Metabolites referred to in the appraisal

Code	Name and Matrix	Structure
MTFM-BA	2-mesyl-4-trifluoromethylbenzoic acid Wheat, Soil	
Desmethyl-pyrasulfotole	(5-hydroxy-3-methylpyrazol-4-yl)(2-mesyl-4-trifluoromethylphenyl)methanone Goat, Hen	
Pyrasulfotole-desmethyl-O-glucoside	Conjugates of desmethyl-pyrasulfotole Wheat	
Pyrasulfotole-sulfinyl-lactate	Conjugates of desmethyl-pyrasulfotole Wheat	

Code	Name and Matrix	Structure
Hydroxymethyl-pyrasulfotole	Methanone, (5-hydroxy-3-hydroxy-methyl-1-methyl-1H-pyrazol-4-yl)[2-(methylsulfonyl)-4-(trifluoromethyl)-phenyl] Goat	

Physical and chemical properties

Pyrasulfotole has a high solubility in water and is not volatile. Pyrasulfotole was shown to be hydrolytically and photolytically stable.

Plant metabolism

The Meeting received plant metabolism studies for pyrasulfotole after foliar application on spring wheat with pyrasulfotole labelled at [Phenyl-U-¹⁴C] and [Pyrazole-3-¹⁴C].

Wheat

Spring wheat grown outdoors received a single foliar spray of [¹⁴C]-Pyrasulfotole applied at a rate of 0.10 kg ai/ha (early tillering). Samples were taken at 27–28 days after treatment (DAT) (forage), 49–50 DAT (hay) and 89–90 DAT (grain and straw). The TRRs in treated samples were 0.44–0.47 mg eq/kg in forage, 0.06–0.18 mg eq/kg in hay, 0.38–0.55 mg eq/kg in straw, and 0.03–0.30 mg eq/kg in grain. Total extractability with acetonitrile/water (4:1, v/v) and Soxhlet (acetonitrile /water (4:1, v/v)) was ≥ 89% TRR in samples treated with [Phenyl-U-¹⁴C]-pyrasulfotole. Extractability of [Pyrazole-3-¹⁴C]-pyrasulfotole samples decreased in the order forage (87%), hay (65%), straw (59%) and grain (33%). The post-extraction solids of forage, straw and grain were submitted to enzymatic hydrolysis followed by either acidic or alkaline hydrolysis, releasing 5–47% TRR. Unextracted residues after enzymatic/acidic hydrolysis amounted to 7.2% TRR (forage), 25% TRR (straw) and 20% TRR (grain). Unextracted residues after enzymatic/alkaline hydrolysis amounted to 1.8% TRR (forage) and 4.1% TRR (straw).

Pyrasulfotole was not detected in either the [Phenyl-U-¹⁴C] or [Pyrazole-3-¹⁴C] study. For the [Phenyl-U-¹⁴C] study, MTFM-BA was the predominant metabolite in the hay, straw and grain (61–90% TRR, 0.11–0.36 mg eq/kg) and a major residue in the forage (26% TRR, 0.12 mg eq/kg). The O-glucoside of desmethyl-pyrasulfotole was the predominant metabolite in forage (34% TRR, 0.15 mg eq/kg) with lesser amounts in the hay (10% TRR, 0.02 mg eq/kg) and straw (5.1% TRR, 0.03 mg eq/kg) but not detected in grain. Polar constituents were also characterized in the extracts none of which exceeded 5.9% TRR.

For the [Pyrazole-3-¹⁴C] study, pyrasulfotole-desmethyl-O-glucoside was the predominant metabolite in forage (43% TRR, 0.20 mg eq/kg), hay (25% TRR, 0.02 mg eq/kg) and straw (22% TRR, 0.08 mg eq/kg) but only small amount was found in grain (0.7% TRR, < 0.001 mg eq/kg). Very polar fraction p1, classified based on the chromatographic behaviour on reversed-phase HPLC, was a major part of the residue in hay (21% TRR, 0.01 mg eq/kg), straw (13% TRR, 0.05 mg eq/kg) and grain (11% TRR, 0.003 mg eq/kg). Several polar fractions were detected, none of them exceeding 7.7% TRR in forage, hay, straw and grain.

For the [Phenyl-U-¹⁴C] study in the presence and absence of a safener, to investigate the influence of the safener on the metabolism, total extractability was ≥ 92% TRR in all samples extracted with acetonitrile/water (4:1, v/v) and microwave (acetonitrile /water (1:1, v/v) and acetonitrile/0.1 mol/L NaOH

(1:1, v/v)). The major metabolite in all commodities consisted of MTFM-BA. Pyrasulfotole-desmethyl-O-glucoside was a major metabolite in forage, hay and straw, but it was not detected in grain.

The parent compound was a major component of the residues in forage, but was readily degraded (hay and straw) and not detected in grains. In the order forage – hay – straw, MTFM-BA (13–17% TRR to 27–33% TRR) and pyrasulfotole-sulfinyl-lactate (3.8–7.8% TRR to 5.7–9.6% TRR) increased, while pyrasulfotole-desmethyl-O-glucoside (30–44% TRR to 20–28% TRR) decreased. In grain, only pyrasulfotole-benzoic acid was present and represented 98% TRR (0.15–0.23 mg eq/kg). Several unidentified fractions and regions were detected, none of them exceeding 4.4% TRR in forage, hay and straw.

Conclusions

Parent pyrasulfotole is rapidly metabolised in wheat. The potential metabolic pathway in wheat is demethylation to desmethyl-pyrasulfotole, although not detected in wheat parts, followed by glucosylation to pyrasulfotole-desmethyl-O-glucoside, or conjugation with glutathione to pyrasulfotole-sulfinyl-lactate. Pyrasulfotole-desmethyl-O-glucoside was a major metabolite in forage, hay and straw, but not detected in grain.

Another pathway is cleavage of the molecule leaving MTFM-BA and the very polar fraction (p1). MTFM-BA was a major metabolite in forage, hay and straw. Minor portions of the residue were attributed to the parent compound and pyrasulfotole-sulfinyl-lactate. In grain, nearly all identified residues consisted of MTFM-BA.

Animal metabolism

The Meeting received animal metabolism studies on rats, lactating goats and laying hens where animals were dosed with [^{14}C]-pyrasulfotole. The metabolism and distribution of pyrasulfotole in farm animals were investigated using the [Phenyl- ^{14}C] and the [Pyrazole-3- ^{14}C]-pyrasulfotole.

Rats

The metabolism of pyrasulfotole in rats was reviewed in the framework of the toxicological evaluation by the WHO Core Assessment Group of the 2021 JMPR.

Lactating goats

Lactating goats received daily oral dosing of [^{14}C]-pyrasulfotole for 3 consecutive days at 0.93 mg/kg bw per day (equivalent to 51 ppm in the diet) for [Phenyl- ^{14}C]-pyrasulfotole and at 1.2 mg/kg bw per day (equivalent to 28 ppm in the diet) for [Pyrazole-3- ^{14}C]-pyrasulfotole. The goats were sacrificed 23 hours after the last dose.

Total radioactive residues (TRR) were highest in the liver (1.5–1.7 mg eq/kg) and kidney (0.27–0.53 mg eq/kg), followed by muscle (0.007–0.011 mg eq/kg) and fat (0.008–0.010 mg eq/kg). The concentration of radioactivity in the milk reached a plateau of about 0.02–0.04 mg eq/kg by day 1.

In the [Phenyl- ^{14}C]-pyrasulfotole study, liver was extracted with acetonitrile/water (4:1, v/v) and acetonitrile. Kidney was extracted with acetonitrile and the remaining extraction solids were refluxed and extracted with methanol/water (4:1, v/v). Muscle was extracted with acetonitrile. Fat was extracted with hexane. Milk was extracted with hexane and methanol. Most of the radioactive residues in milk and tissues were extracted (92–100% TRR).

In the [Pyrazole-3- ^{14}C]-pyrasulfotole study, liver was extracted with methanol and the remaining extraction solids were refluxed and extracted with methanol/water (4:1, v/v). Kidney was extracted with methanol. Muscle was extracted with acetonitrile. Milk and fat were extracted with hexane. The post-extraction solids of fat were extracted with acetonitrile. Most of the radioactive residues in milk and tissues were extracted (70–97% TRR).

Pyrasulfotole was the predominant residue in liver (93–96% TRR, 1.4–1.6 mg/kg), kidney (92–100% TRR, 0.25–0.53 mg/kg), muscle (80% TRR, 0.008 mg/kg) and milk on Day 3 (39–83% TRR, 0.014–0.017 mg/kg).

Desmethyl-pyrasulfotole was found as a major metabolite in milk (12% TRR, 0.002 mg eq/kg) and as a minor metabolite in liver (1.4% TRR, 0.025 mg eq/kg). The ratio of desmethyl-pyrasulfotole residues to pyrasulfotole residues was 14% in milk and 1.5% in liver.

Hydroxymethyl-pyrasulfotole was identified as a minor metabolite in milk (4.4% TRR, 0.001 mg eq/kg) and muscle (8.3% TRR, 0.001 mg eq/kg).

Laying hens

Laying hens received daily oral dosing of [^{14}C]-pyrasulfotole for 14 consecutive days at a rate equivalent to 8.6 ppm in the diet (0.82 mg/kg bw per day) for [Phenyl- ^{14}C]-pyrasulfotole and at 11 ppm in the diet (0.81 mg/kg bw per day) for [Pyrazole-3- ^{14}C]-pyrasulfotole. The hens were sacrificed 30 minutes after the last dose, while commercial practice is considered to be 6–8 hours.

TRRs in tissues were highest in liver (1.3–1.6 mg eq/kg). Residue levels in muscle and fat were 0.02–0.07 mg eq/kg. Residue levels in eggs were ≤ 0.004 mg eq/kg (maximum at day 12). Radioactivity levels in eggs reached a plateau of 0.002–0.003 mg eq/kg by day 6–8.

Liver, muscle and egg were extracted with acetonitrile/water (4:1, v/v) and acetonitrile. The post-extraction solids of liver were refluxed and further extracted with acetonitrile/water (4:1, v/v). Fat was extracted with acetonitrile/water (4:1, v/v), acetonitrile, and hexane. Good extractability was achieved for all tissue samples (97–100% TRR). The residue in eggs was extracted at 47–84% TRR.

As expected for birds slaughtered so close to the last dose, pyrasulfotole was the predominant component in liver (93–95% TRR, 1.2–1.5 mg/kg), muscle (93–95% TRR, 0.02–0.04 mg/kg) and fat (97% TRR, 0.06 mg/kg).

Desmethyl-pyrasulfotole was identified as a minor metabolite in liver (4.8–6.5% TRR, 0.06–0.10 mg eq/kg), muscle (2.2% TRR, ≤ 0.001 mg eq/kg) and fat (1.8% TRR, 0.001 mg eq/kg). No other metabolites were identified.

Conclusions

The metabolic pathway of pyrasulfotole in goat and hen is consistent with that in the rat where very little biotransformation occurred. N-Demethylation to desmethyl-pyrasulfotole was observed in all metabolism studies. Oxidation of parent pyrasulfotole to hydroxymethyl-pyrasulfotole was observed in rat and lactating goat (the [Phenyl- ^{14}C]-pyrasulfotole labelled) metabolism studies at very low levels.

Environmental fate

The Meeting received aerobic soil metabolism, soil photolysis, soil degradation, aqueous hydrolysis and aqueous photolysis studies for pyrasulfotole.

In the aerobic soil metabolism studies, pyrasulfotole was moderately persistent in a variety of soils with a DT_{50} of 11–72 days (first order kinetics) at 20–25 °C. MTFM-BA was the only identified metabolite (< 12% applied radioactivity (AR)) but did not accumulate. Pyrasulfotole was substantially mineralized yielding $^{14}\text{CO}_2$ (34–41% AR). The other major residue of the applied radioactivity was unextracted soil residues (31–35% AR). Pyrasulfotole and its metabolite are not persistent in soil. The Meeting noted that the benzoic acid metabolite (MTFM-BA) is common with isoxaflutole where the observed DT_{50} was up to 1000 days. No clear evidence for this variation could be identified.

In the soil photolysis studies, pyrasulfotole was stable in irradiated soils (81% AR) and in non-irradiated soils (92% AR) for 9 days. Photolysis in soil is unlikely to be a major route of environmental degradation.

In the soil degradation studies, the DT_{50s} of pyrasulfotole and MTFM-BA were 5–31 days and 5–87 days, respectively.

In the aqueous hydrolysis study, pyrasulfotole was hydrolytically stable at pH 5, 7 and 9 at 25 °C. Hydrolysis is unlikely to be a major route of environmental degradation.

In the aqueous photolysis study, pyrasulfotole was photolytically stable at 25 °C for 30 days. Photolysis in water is unlikely to be a major route of environmental degradation.

Pyrasulfotole was moderately degraded in the environmental fate studies (soil), and the breakdown product MTFM-BA was also moderately degraded to form unextracted residue and CO₂. Pyrasulfotole is not persistent in soil.

Rotational crop metabolism

The Meeting received a confined rotational crop study with ¹⁴C-labeled pyrasulfotole and a field rotational crop study.

In a confined rotational crop study, rotational crops (Swiss chard, turnip and wheat) were planted in soil that had been treated at 0.082 kg ai/ha with [Phenyl-U-¹⁴C] or [Pyrazole-3-¹⁴C]-pyrasulfotole and then aged for approximately 120 days and wheat was also planted 301 days after treatment (DAT). Shorter intervals were not investigated due to phytotoxicity to follow crops.

In Swiss chard, turnip tops and turnip roots planted at 122 DAT, the radioactive residues were < 0.01 mg eq/kg (0.002–0.008 mg eq/kg).

In 120 DAT and 301 DAT wheat commodities (forage, hay, straw and grain), MTFM-BA was the only metabolite detected greater than 0.01 mg eq/kg (27–91% TRR, 0.004–0.078 mg eq/kg). Parent pyrasulfotole was ≤ 0.001 mg/kg (≤ 2% TRR) in all of the analysed extracts.

In the confined rotational crop study, MTFM-BA was the only metabolite occurring at > 0.01 mg/kg. Environmental fate studies suggest that this metabolite is formed in the soil, taken up by plant roots, and then distributed through the plant.

In a field rotational crop study, a pyrasulfotole SE formulation was applied to wheat at 0.049–0.051 kg ai/ha and after the mature wheat crop was harvested (fate of remaining plant material unspecified) maize and soya bean were planted at 114–123 days plant back interval. The total pyrasulfotole residue (pyrasulfotole + MTFM-BA + desmethyl-pyrasulfotole, expressed as pyrasulfotole) was less than the LOQ (0.01 mg/kg for each analyte) for maize forage, maize grain, maize fodder, soya bean forage, soya bean hay and soya bean seed with exception of one trial in soya bean hay (total residue 0.014 mg/kg).

Residues related to pyrasulfotole are not expected to be significant in rotational crops.

Methods of analysis

The Meeting received information on analytical methods for pyrasulfotole and its metabolites in plant and animal matrices.

Plant matrices

In the methods (AI-001-P04-01 and 01019) for determination of pyrasulfotole, desmethyl-pyrasulfotole (free and conjugated) and pyrasulfotole-benzoic acid in plant matrices, samples were extracted with acetonitrile/water/conc. HCl (30/15/3 v/v/v or 20/10/3 v/v/v) at 60 °C for 30 min. After SPE clean-up, residues were determined by LC-MS/MS. The methods of analysis were validated with an LOQ of 0.005–0.01 mg/kg expressed as pyrasulfotole equivalents for each analyte.

The study to determine the efficiency of the analytical residue method, AI-001-P04-01, in extracting pyrasulfotole derived residues from aged radioactive plant matrices showed that pyrasulfotole, pyrasulfotole-desmethyl-O-glucoside and MTFM-BA residues in plant matrices were effectively extracted

by the method AI-001-P04-01. Pyrasulfotole-desmethyl-O-glucoside and other conjugates of desmethyl-pyrasulfotole were detected as desmethyl-pyrasulfotole due to the hydrolytic extraction of this method.

Method 01019 was adapted from method AI-0001-P04-01. Since the extraction conditions used in method 01019 are similar to those used in AI-001-P04-01, the extraction efficiency of method 01019 is demonstrated for forage but not for grain and hay. The Meeting concluded that the results from the metabolism study support suitable extraction efficiency in hay. In addition, the changes are the use of matrix-matched standards instead of stable isotopic internal standards and two different MS/MS transitions (instead of only one) to monitor each analyte. The Meeting noted that method 01019 is suitable as an enforcement method for the monitoring of residues of pyrasulfotole, desmethyl-pyrasulfotole (free and conjugated) and MTFM-BA in the plant matrices considered by the meeting.

In the other method (ATM-0018) for determination of pyrasulfotole, desmethyl-pyrasulfotole and MTFM-BA in plant matrices, samples were extracted with methanol/0.1% formic acid. The extract was filtered and analysed by LC-MS/MS using internal standardisation. This method was validated with an LOQ of 0.02 mg/kg for each analyte.

Animal matrices

In the methods for determination of pyrasulfotole, desmethyl-pyrasulfotole and MTFM-BA in animal matrices, samples were extracted with acetonitrile/water or acidified acetonitrile/water at 60 °C for 30 min. After SPE clean-up, residues were determined by LC-MS/MS. The methods of analysis were validated with an LOQ of 0.005–0.01 mg/kg.

The Meeting concluded that the presented methods were sufficiently validated and are suitable to measure pyrasulfotole and its metabolites, desmethyl-pyrasulfotole (free and conjugated) and MTFM-BA, in plant and animal commodities.

Stability of pesticide residues in stored analytical samples

The Meeting received information on the storage stability of pyrasulfotole and its metabolites in plant commodities (soya bean seed, wheat grain, wheat forage and wheat hay).

Storage stability studies showed that pyrasulfotole and MTFM-BA were stable for at least 11 months at -10 °C in crop commodities representative of the high water (wheat forage), high oil (soya bean) and high starch (wheat grain) commodity groups, and dry feed.

Storage stability studies showed that desmethyl-pyrasulfotole was stable for at least 11 months at ≤-10 °C in crop commodities representative of the high oil (soya bean) and high starch (wheat grain) commodity groups, but it was gradually degraded to 75% within 3 months in the high water (wheat forage) commodity group and dry feed.

The Meeting noted that the residue data on desmethyl-pyrasulfotole in high-water commodities and dry feed can be used if the storage intervals were 3 months or less.

The Meeting agreed that the demonstrated storage stability for pyrasulfotole and MTFM-BA on various representative plant commodities covered the residue sample storage intervals used in the field trials considered by the current Meeting.

Definition of the residue

Plant commodities

In plant metabolism studies on pyrasulfotole in wheat, pyrasulfotole was readily metabolised and not detected in grains.

MTFM-BA (24–98% TRR) was found in all wheat commodities (forage, hay, straw and grains). Glucose conjugates of desmethyl-pyrasulfotole (pyrasulfotole desmethyl-O-glucoside: 4.5–44% TRR) were identified as metabolites in wheat forage, hay and straw.

In rotational crops (wheat forage, hay, straw and grain), MTFM-BA (27–91% TRR), was the only metabolite detected greater than 0.01 mg/kg.

In field trials desmethyl-pyrasulfotole was the predominant residue detected. The method hydrolysed conjugates and therefore represents the sum of free and conjugated forms.

MTFM-BA is the predominant metabolite and is relevant for consideration in the residue definition for enforcement for plant commodities. However, MTFM-BA (2-mesyl-4-trifluoromethylbenzoic acid) can also arise in plant commodities as a result of treatment with isoxaflutole. For this reason, MTFM-BA is not a suitable analyte for enforcing MRLs for pyrasulfotole residues.

In the analytical method for monitoring, method 01019, samples are extracted with acidified acetonitrile/water to hydrolyse pyrasulfotole desmethyl-O-glucoside to desmethyl-pyrasulfotole. This method also hydrolyses a minor metabolite pyrasulfotole-sulfinyl-lactate, which is a conjugate of desmethyl-pyrasulfotole with glutathione, to desmethyl-pyrasulfotole.

The Meeting decided that the suitable analytes for enforcement purposes in cereal commodities are parent pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated).

In deciding which compounds should be included in the residue definition for dietary risk assessment, the Meeting considered the toxicological properties of the candidates MTFM-BA and desmethyl-pyrasulfotole (free and conjugated).

MTFM-BA is a minor metabolite in rat. Assessment of toxicological data on MTFM-BA demonstrated that this metabolite is not of any toxicological concern. From a toxicological point of view it is not necessary to include MTFM-BA in the residue definition.

The free desmethyl-pyrasulfotole is considered to be covered by toxicity on parent pyrasulfotole since it was found (up to 8.1% TRR) in the rat metabolism study. Conjugates of desmethyl-pyrasulfotole are considered relevant for dietary exposure since the free form can be released during the metabolic process in humans.

The Meeting decided that the suitable analytes for dietary risk assessment are pyrasulfotole, desmethyl-pyrasulfotole and its conjugates.

Animal commodities

The Meeting noted that predominant exposure to animals in feed would be MTFM-BA, however it is a common metabolite to isoxaflutole and is therefore not a suitable marker for compliance.

In animal metabolism studies with pyrasulfotole, pyrasulfotole was the predominant component of the residue (80–100% TRR) in liver, kidney, muscle, egg and hen fat, and the major residue in milk (39–83% TRR). Desmethyl-pyrasulfotole was a major metabolite identified in milk (12% TRR, 0.002 mg eq/kg). Hydroxymethyl-pyrasulfotole was a minor metabolite in milk and goat muscle (4.4–8.3% TRR, 0.001 mg eq/kg).

The Meeting decided to define the residue for compliance with the MRL for animal commodities as sum of pyrasulfotole and desmethyl-pyrasulfotole, expressed as pyrasulfotole.

In deciding which compounds should be included in the residue definition for dietary risk assessment, the Meeting noted that MTFM-BA was of no toxicological concern. The Meeting considered the likely occurrence and toxicological properties of the hydroxymethyl-pyrasulfotole and desmethyl-pyrasulfotole.

The Meeting concluded that the toxicity of hydroxymethyl-pyrasulfotole was covered by the parent compound. As it was found at very low levels in the metabolism study involving exaggerated rates, the Meeting agreed not to consider it further.

The free desmethyl-pyrasulfotole was a major metabolite identified in milk (12% TRR, 0.002 mg eq/kg) and also found at low levels in liver. As it is considered to be covered by toxicity on

parent pyrasulfotole the Meeting considered it should be included in the residue definition for dietary exposure.

The Meeting decided to define the residue for dietary risk assessment for animal commodities as the sum of pyrasulfotole and desmethyl-pyrasulfotole, expressed as pyrasulfotole.

Animal metabolism and feeding studies show that residues of pyrasulfotole and desmethyl-pyrasulfotole in muscle and fat, and in cream and skim milk, are similar. The Meeting considered that the residue is not fat-soluble.

The Meeting recommended the following residue definitions for pyrasulfotole:

Definition of the residue for compliance with the MRL and dietary risk assessment for plant commodities: *Sum of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated), expressed as pyrasulfotole.*

Definition of the residue for compliance with the MRL and dietary risk assessment for animal commodities: *Sum of pyrasulfotole and desmethyl-pyrasulfotole, expressed as pyrasulfotole*

The residue is not fat-soluble.

Results of supervised residue trials on crops

Supervised trials were available for the use of pyrasulfotole on wheat, barley, oats and sorghum.

Product labels were available from Australia, Canada and the USA.

Analytical methods used in trials adopted a hydrolytic extraction. Conjugates of desmethyl-pyrasulfotole were detected as desmethyl-pyrasulfotole.

Total residues for estimation of maximum residue levels in plant commodities are calculated by summing up the concentrations of pyrasulfotole and desmethyl-pyrasulfotole (expressed as pyrasulfotole equivalents). The method of calculation is illustrated below.

Example of the method for calculation of total residues

Pyrasulfotole	Desmethyl-pyrasulfotole	Total
<0.01	<0.01	<0.02
<0.01	0.025	0.035

Cereal grains

Wheat, similar grains, and pseudocereals without husks

The critical GAP for triticale and wheat in the USA allows one spray application of 0.05 kg ai/ha with a PHI of 50 days. The critical GAP for rye in the USA allows one spray application of 0.04 kg ai/ha with a PHI of 60 days.

Data were available from supervised trials on wheat in Canada and the USA.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in wheat grain from independent trials in Canada and the USA matching the US GAP for triticale and wheat were (n = 30): < 0.02 (30) mg/kg.

Based on the residues in wheat grains from trials in Canada and the USA, the Meeting estimated a maximum residue level of 0.02 (*) mg/kg and an STMR value of 0.02 mg/kg for pyrasulfotole in wheat, and the Meeting agreed to extrapolate this to triticale. As the GAP for rye has a slightly lower rate and longer PHI, and residues according to the GAP for wheat are below LOQ, the Meeting also agreed to extrapolate the maximum residue level to rye.

*Barley, similar grains, and pseudocereals with husks**Barley*

The critical GAP for barley in the USA allows one spray application of 0.05 kg ai/ha with a PHI of 45 days.

Data were available from supervised trials on barley in Canada and the USA.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in barley grain from independent trials in Canada and the USA matching the US GAP were (n = 39): < 0.02 (37), 0.023 and 0.029 mg/kg.

Based on the residues in barley grains from trials in Canada and the USA, the Meeting estimated a maximum residue level of 0.03 mg/kg and an STMR value of 0.02 mg/kg for pyrasulfotole in barley.

Oats

The critical GAP for oats in the USA allows one spray application of 0.05 kg ai/ha with a PHI of 45 days.

Data were available from supervised trials on oats in Canada and the USA.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in oat grain from independent trials in Canada and the USA matching the US GAP were (n = 24): < 0.02 (13), 0.020, 0.021 (2), 0.022, 0.025, 0.027, 0.030, 0.031, 0.033, 0.043 and 0.11 mg/kg.

Based on the residues in oat grain from trials in Canada and the USA, the Meeting estimated a maximum residue level of 0.15 mg/kg and an STMR value of 0.02 mg/kg for pyrasulfotole in oats.

*Sorghum Grain and Millet**Sorghum Grain*

Data were available from supervised trials on sorghum in the USA.

The critical GAP for sorghum in the USA allows two spray applications of 0.043 kg ai/ha with a maximum seasonal rate of 0.087 kg ai/ha and a PHI of 60 days.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in sorghum grain from independent trials in the USA matching the US GAP were (n = 12): < 0.02 (2), 0.035, 0.057, 0.061, 0.072, 0.11, 0.12, 0.17, 0.21 (2) and 0.32 mg/kg.

Based on the residues in sorghum grain from trials in the USA, the Meeting estimated a maximum residue level of 0.5 mg/kg and an STMR value of 0.091 mg/kg for pyrasulfotole in sorghum grain.

Residues in animal feeds

Due to the instability of residues of desmethyl-pyrasulfotole in stored samples of high-water commodities and dry feed, trials were only used when the sample storage intervals were less than 3 months.

Forage of barley, oat, rye, triticale and wheat

The critical GAP for forage of barley, oat, triticale and wheat in the USA allows one spray application of 0.05 kg ai/ha with a PHI of 25 days. The critical GAP for rye forage in the USA allows one spray application of 0.04 kg ai/ha with a PHI of 25 days.

Oat forage

Data were available from supervised trials on oat forage in Canada and the USA.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in oat forage from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA, and matching GAP were ($n = 2$): 0.050 and 0.11 mg/kg (as received basis).

Wheat forage

Data were available from supervised trials on wheat forage in Canada and the USA.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in wheat forage from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA, and matching GAP were ($n = 15$): < 0.02 (5), 0.022, 0.038, 0.051, 0.063, 0.090, 0.11, 0.12, 0.16, 0.18 and 0.19 mg/kg (as received basis).

The Meeting agreed that residues on forage of oat and wheat can be combined and extrapolated to forage from barley, rye and triticale.

The combined total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in forage of oat and wheat were in rank order ($n = 17$): < 0.02 (5), 0.022, 0.038, 0.050, 0.051, 0.063, 0.090, 0.11 (2), 0.12, 0.16, 0.18 and 0.19 mg/kg (as received basis).

Based on the residues in forage of oat and wheat from trials in Canada and the USA, the Meeting estimated a median residue value of 0.051 mg/kg and a highest residue value of 0.19 mg/kg for pyrasulfotole in forage of barley, oat, rye, triticale and wheat (as received basis).

Sorghum forage

The critical GAP for sorghum forage in the USA allows two spray applications of 0.043 kg ai/ha and a PHI of 7 days.

Data were available from supervised trials on sorghum forage in the USA.

All samples were stored for > 3 months prior to analysis and the Meeting could not estimate median and highest residue values for pyrasulfotole in sorghum forage.

Straw and fodder (dry) of barley, oat, rye, triticale and wheat

Hay or fodder dry of barley, oat, triticale and wheat

The critical GAP for hay of barley, oat, triticale and wheat in the USA allows one spray application of 0.05 kg ai/ha with a PHI of 25 days.

Data were available from supervised trials on hay of barley, oat and wheat in Canada and the USA.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in barley hay from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA, and matching GAP were ($n = 2$): 0.21 and 0.22 mg/kg on a dry weight basis.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in oat hay from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA, and matching GAP were ($n = 10$): 0.084, 0.17, 0.18, 0.19, 0.20, 0.25, 0.26, 0.40, 0.44 and 0.50 mg/kg on a dry weight basis.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in wheat hay from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA, and matching GAP were ($n = 8$): 0.088, 0.13, 0.16, 0.21 (2), 0.26, 0.37 and 0.43 mg/kg on a dry weight basis.

The Meeting recognized that the GAPs were same and the residue populations were similar for hay of barley, oat and wheat. Therefore, the Meeting decided to combine the data from hay of barley, oat and wheat.

The combined total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in hay of barley, oat and wheat were in rank order ($n = 20$): 0.084, 0.088, 0.13, 0.16, 0.17, 0.18, 0.19, 0.20, 0.21 (3), 0.22, 0.25, 0.26 (2), 0.37, 0.40, 0.43, 0.44 and 0.50 mg/kg on a dry weight basis.

Straw of barley, oat, rye, triticale and wheat

The critical GAP for straw of barley, oat, rye, triticale and wheat in the USA allows one spray application of 0.05 kg ai/ha with a PHI of 45 (barley and oat), 50 (triticale and wheat) and 60 days (rye).

Data were available from supervised trials on straw of barley, oat and wheat in Canada and the USA.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in barley straw from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA matching GAP were ($n = 2$): 0.067 and 0.098 mg/kg on a dry weight basis.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in oat straw from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA matching GAP were ($n = 16$): 0.075, 0.078, 0.079, 0.090, 0.12, 0.15 (2), 0.16 (3), 0.17 (3), 0.21, 0.32 and 0.36 mg/kg on a dry weight basis.

Total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in wheat straw from independent trials with sample storage durations ≤ 3 months conducted in Canada and the USA matching GAP were ($n = 20$): < 0.02 , 0.028, 0.054 (2), 0.058, 0.064, 0.068, 0.073, 0.081, 0.088, 0.094 (2), 0.10, 0.11, 0.12, 0.13, 0.14 (2), 0.23 and 0.38 mg/kg on a dry weight basis.

The Meeting recognized that the GAPs and the residue populations were similar for straw of barley, oat and wheat. Therefore, the Meeting decided to combine the data from straw of barley, oat and wheat.

The combined total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) in straw of barley, oat and wheat were in rank order ($n = 38$): < 0.02 , 0.028, 0.054 (2), 0.058, 0.064, 0.067, 0.068, 0.073, 0.075, 0.078, 0.079, 0.081, 0.088, 0.090, 0.094 (2), 0.098, 0.10, 0.11, 0.12 (2), 0.13, 0.14 (2), 0.15 (2), 0.16 (3), 0.17 (3), 0.21, 0.23, 0.32, 0.36 and 0.38 mg/kg on a dry weight basis.

As the dataset from hay of barley, oat and wheat led to the highest maximum residue level the Meeting decided to use this dataset to estimate a maximum residue level for straw and fodder, dry, of barley, oat and wheat.

The Meeting agreed that residues on straw and fodder of barley, oat and wheat may be extrapolated to straw and fodder from rye and triticale.

Based on the residues in hay of barley, oat and wheat from trials in Canada and the USA, the Meeting estimated a maximum residue level of 0.8 mg/kg for pyrasulfotole in straw and fodder, dry, of barley, oat, rye, triticale and wheat on a dry weight basis.

The Meeting estimated a median residue value of 0.21 mg/kg and a highest residue value of 0.50 mg/kg for pyrasulfotole in hay of barley, oat, rye, triticale and wheat, and a median residue value of 0.105 mg/kg and a highest residue value of 0.38 mg/kg for pyrasulfotole in straw of barley, oat, rye, triticale and wheat.

Sorghum straw and fodder, dry

The critical GAP for sorghum stover in the USA allows two spray applications of 0.043 kg ai/ha with a maximum seasonal rate of 0.087 kg ai/ha and a PHI of 60 days.

Data were available from supervised trials on sorghum in the USA.

All samples were stored for > 3 months prior to analysis and the Meeting could not estimate a maximum residue level for pyrasulfotole in sorghum stover.

Fate of residues during processing

Residues in processed commodities

The Meeting received information on the fate of total residues of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated) during processing in wheat. All data relevant for an estimation of maximum residue levels in processed commodities or for dietary exposure calculations are summarised in the following table.

Table 2 Processing factors and STMR-P/HR-P

Raw commodity [STMR]	Processed commodity	Processing factor	STMR-P = STMR _{RAC} × PF (mg/kg)
Wheat grains [0.02 mg/kg]	Bran	1.4	0.028
	Flour	<0.71	<0.014
	Germ	0.79	0.016
	Middlings (milled by-product)	<0.71	<0.014
	Shorts (milled by-product)	<0.71	<0.014
	Aspirated grain fractions	22	0.44

Using the estimated maximum residue level of 0.02 (*) mg/kg for wheat and applying the processing factor of 1.4 for wheat bran the Meeting estimated a maximum residue level of 0.03 mg/kg for pyrasulfotole in wheat bran.

Residues in animal commodities

Farm animal feeding studies

The Meeting received lactating dairy cow and laying hen feeding studies, which provided information on likely residues resulting in animal commodities, milk and eggs from pyrasulfotole (cows) and MTFM-BA (hens) residues in the animal diet.

Lactating dairy cows

Holstein dairy cows were dosed with pyrasulfotole for 29 days at the equivalent of 3.0, 9.0 and 30 ppm in the diet. Pyrasulfotole residues were determined in milk and tissues (liver, kidney, muscle and fat).

Whole milk contained no pyrasulfotole residues (< 0.005 mg/kg) at the 3.0 and 9.0 ppm feeding levels. Pyrasulfotole residues in whole milk achieved a plateau concentration of 0.009–0.013 mg/kg at the 30 ppm feeding level.

Pyrasulfotole residues were below the LOQ (< 0.01 mg/kg) in muscle and fat at all feeding levels, and were detected in liver (0.69–1.9 mg/kg) and kidney (0.12–0.42 mg/kg) at all feeding levels.

Laying hens

Laying hens were dosed with MTFM-BA for 29 days at the equivalent of 0.4, 1.2 and 4.0 ppm in the diet. MTFM-BA residues were determined in eggs, liver, muscle and fat.

Residues of MTFM-BA were below the LOQ (< 0.01 mg/kg) in eggs, muscle and fat at all feeding levels, and in liver at the 0.4 and 1.2 ppm feeding levels. Residues of MTFM-BA in liver were detected at 0.010–0.021 mg/kg at the 4.0 ppm feeding level.

Farm animal dietary burden

Dietary burdens were calculated for beef cattle, dairy cattle, broilers and laying poultry based on feed items evaluated by the JMPR.

Potential feed items were barley, oat, rye, triticale and wheat (forage, hay, straw and grain). For poultry, the items were only grains of those crops.

Some processed and forage commodities do not appear in the Recommendations Table (because no maximum residue level is needed), but they are used in estimating livestock dietary burdens. Those commodities are included in the list below. The input was based on the intake of pyrasulfotole and desmethyl-pyrasulfotole (free and conjugated).

Table 3 Potential feed items

Commodity	Median residue (-P) (mg/kg)	Highest residue (-P) (mg/kg)
Rye grain, Triticale grain, Wheat grain	0.02	
Wheat aspirated grain fractions	0.44	
Wheat milled by-product	<0.014	
Barley grain, Oat grain	0.02	
Hay of barley, oat, triticale, and wheat	0.21 (dry weight)	0.50 (dry weight)
Straw of barley, oat, rye, triticale, and wheat	0.105 (dry weight)	0.38 (dry weight)
Forage of barley, oat, rye, triticale and wheat	0.051 (as received)	0.19 (as received)

The dietary burdens, estimated using the most recent version of the OECD livestock burden calculator, are presented in Annex 6 and summarised below.

Table 4 Estimated maximum and mean dietary burdens of farm animals

	Animal dietary burden: Pyrasulfotole, ppm of dry matter diet							
	US-Canada		EU		Australia		Japan	
	max	mean	max	mean	max	mean	max	mean
Beef cattle	0.12	0.073	0.23	0.075	0.76 ^①	0.21 ^⑤	0.021	0.021
Dairy cattle	0.23	0.077	0.23	0.073	0.71 ^②	0.21 ^④	0.048	0.027
Poultry – broiler	0.021	0.020	0.019	0.019	0.023	0.023	0.003	0.003
Poultry – layer	0.021	0.021	0.096 ^{⑤ ⑦}	0.041 ^{⑥ ⑧}	0.016	0.016	0.005	0.005

① Highest maximum beef or dairy cattle dietary burden suitable for MRL estimates for mammalian tissues

② Highest maximum dairy cattle dietary burden suitable for MRL estimates for mammalian milk

③ Highest mean beef or dairy cattle dietary burden suitable for STMR estimates for mammalian tissues.

④ Highest mean dairy cattle dietary burden suitable for STMR estimates for milk.

⑤ Highest maximum poultry dietary burden suitable for MRL estimates for poultry tissues.

⑥ Highest mean poultry dietary burden suitable for STMR estimates for poultry tissues.

⑦ Highest maximum poultry dietary burden suitable for MRL estimates for poultry eggs.

⑧ Highest mean poultry dietary burden suitable for STMR estimates for poultry eggs.

Animal commodity maximum residue levels

Cattle

Residues in tissues and milk at the expected dietary burden for dairy and beef cattle are shown in the Table below.

Table 5 Maximum residue level and STMR in mammalian animal commodities

	Feed Level (ppm) for milk residues	Pyrasulfotole residues (mg eq/kg) in milk	Feed Level (ppm) for tissue residues	Pyrasulfotole residues (mg eq/kg)			
				Muscle	Liver	Kidney	Fat
MRL Determination (beef or dairy cattle)							
Feeding Study	3.0	<0.005	3.0	<0.01	1.2	0.22	<0.01
Dietary burden and estimate of highest residue	0.71	<0.005	0.76	<0.01	0.30	0.056	<0.01
STMR Determination (beef or dairy cattle)							
Feeding Study	3.0	<0.005	3.0	<0.01	1.2	0.20	<0.01
Dietary burden and estimate of highest residue	0.21	<0.005	0.21	<0.01	0.084	0.014	<0.01

In the lactating goat metabolism studies, desmethyl-pyrasulfotole was detected only in milk (14% of parent) and liver (1.6% of parent). Therefore, desmethyl-pyrasulfotole residues do not contribute significantly to the estimation of maximum residue levels for animal commodities.

Based on pyrasulfotole residues in milk and cattle tissues, the Meeting estimated maximum residue levels of 0.01 (*) mg/kg in milk, 0.02 (*) mg/kg in mammalian meat (in the fat) and mammalian fat, and 0.5 mg/kg in mammalian edible offal.

Based on the mean estimated pyrasulfotole residues in milk, muscle and fat, the Meeting estimated STMR values of 0.01 mg/kg in milk, 0.02 mg/kg in mammalian meat and mammalian fat.

Based on the mean estimated pyrasulfotole residues in liver and kidney, the Meeting estimated an STMR value of 0.084 mg/kg in mammalian edible offal.

Poultry

Residues in tissues and eggs at the expected dietary burden for laying hen are shown in the Table below.

The results of the laying hen metabolism studies dosed with parent pyrasulfotole were used to estimate maximum residue levels in poultry commodities because pyrasulfotole and desmethyl-pyrasulfotole were the residues included in the residue definition for animal commodities.

Table 6 Maximum residue level and STMR in poultry commodities

	Feed Level (ppm) for egg residues	Total residues (mg eq/kg) in egg	Feed Level (ppm) for tissue residues	Total residues* (mg eq/kg)		
				Muscle	Liver	Fat
MRL Determination (poultry broiler or layer)						
Metabolism Study	8.6	<0.02	8.6	0.037	1.6	0.065
Dietary burden and estimate of highest residue	0.096	<0.02	0.096	<0.02	0.02	<0.02
STMR Determination (poultry broiler or layer)						
Metabolism Study	8.6	<0.02	8.6	0.037	1.6	0.065
Dietary burden and estimate of highest residue	0.041	<0.02	0.041	<0.02	<0.02	<0.02

* Total residue: sum of pyrasulfotole and desmethyl-pyrasulfotole expressed as pyrasulfotole

Based on total residues of pyraulfotole and desmethyl-pyraulfotole expressed as pyraulfotole in eggs and poultry tissues, the Meeting estimated a maximum residue level of 0.02 (*) mg/kg in eggs, poultry meat and poultry fat, and 0.05 mg/kg in poultry, edible offal of.

Based on the mean estimated total residues of pyraulfotole and desmethyl-pyraulfotole expressed as pyraulfotole in eggs, muscle, liver and fat, the Meeting estimated STMR values of 0 mg/kg in eggs, 0.02 mg/kg in poultry meat, 0.02 mg/kg in poultry, edible offal of and 0.02 mg/kg in poultry fat.

RECOMMENDATIONS

On the basis of the data from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI assessment.

Definition of the residue for compliance with the MRL and estimation of dietary exposure for plant commodities: *Sum of pyraulfotole and desmethyl-pyraulfotole and its conjugates, expressed as pyraulfotole.*

Definition of the residue for compliance with the MRL and estimation of dietary exposure for animal commodities: *Sum of pyraulfotole and desmethyl-pyraulfotole, expressed as pyraulfotole.*

The residue is not fat-soluble.

DIETARY RISK ASSESSMENT

Long-term dietary exposure

The ADI for pyraulfotole is 0–0.01 mg/kg bw. The International Estimated Daily Intakes (IEDIs) for pyraulfotole were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs ranged from 1–2% of the maximum ADI. The Meeting concluded that long-term dietary exposure to residues of pyraulfotole from uses considered by the JMPR is unlikely to present a public health concern.

Acute dietary exposure

The current Meeting decided that an ARfD for pyraulfotole was unnecessary. The Meeting therefore concluded that the acute dietary exposure to residues of pyraulfotole from the uses considered is unlikely to present a public health concern.

5.13 Pyraziflumid (322)

TOXICOLOGY

Pyraziflumid is the International Organization for Standardization (ISO)-approved common name for *N*-(3',4'-difluorobiphenyl-2-yl)-3-(trifluoromethyl)pyrazine-2-carboxamide (IUPAC), which has the Chemical Abstracts Service number 942515-63-1.

Pyraziflumid is a pyrazine-biphenyl type carboxamide SDHI (succinate dehydrogenase inhibitor) fungicide, for use on fruits such as pome, stone and citrus, as well as persimmon and grape. Its mode of action (MOA) is suppression of spore germination, mycelium elongation and sporogenesis by inhibiting the activity of mitochondrial electron transport chain complex II (succinate dehydrogenase complex) in filamentous fungi.

Pyraziflumid has not previously been evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) and was reviewed by the present Meeting at the request of the Codex Committee on Pesticide Residues (CCPR).

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with the relevant national or international test guidelines, unless otherwise specified. No additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Biochemical aspects

Toxicokinetic studies were performed with pyraziflumid with a ^{14}C label in the pyrazinyl, aniline or difluorophenyl rings. Following the administration of a single oral dose of 1 or 100 mg/kg body weight (bw) of [^{14}C]pyraziflumid to rats, absorption was relatively rapid, with a peak concentration of radioactivity in plasma (depending on the position of the radiolabel) at the low dose after 3–12 hours, and the high dose after 9–24 hours. Excretion was relatively rapid, with more than 84–92% of the radioactivity excreted within 72 hours following a 1 mg/kg bw dose, indicating that pyraziflumid has a low potential for accumulation. At a dose of 1 mg/kg bw urinary and faecal excretion in males was 9–13% and 84–88%, respectively after 168 hours. In studies with pyrazinyl-labelled pyraziflumid up to 5% of radiolabel was excreted in expired air. Studies in bile duct-cannulated male rats indicated that the majority of an administered dose of [^{14}C]pyraziflumid at 1 mg/kg bw was excreted in the bile, 83–85%, and that at least 90% of the administered dose was absorbed. There were no major differences in excretion or metabolism due to dose or sex. Tissue distribution was widespread, but levels in tissues were low. Highest levels were found in the gastrointestinal tract, liver, fat, kidneys, thyroid and adrenals.

Up to 10 metabolites of pyraziflumid were identified in metabolism studies. Pyraziflumid is metabolized mainly via hydroxylation of the pyrazinyl and aniline rings followed by conjugation with glucuronic acid. There is some limited cleavage of the molecular backbone of pyraziflumid, yielding pyraziflumid-oxamic acid, pyraziflumid-acid and pyraziflumid-amide. The major metabolites were 4-hydroxy-pyraziflumid and 3-hydroxy-pyraziflumid, which are excreted unchanged or as their glucuronide conjugates. No parent compound was found in urine or bile. No marked sex- or dose-related differences in metabolism were seen. Studies in vitro with hepatic liver microsomes indicated that metabolism by humans was qualitatively similar to that in other mammalian species.

Toxicological data

The acute oral median lethal dose (LD_{50}) in rats was greater than 2000 mg/kg bw, the acute dermal LD_{50} was greater than 2000 mg/kg bw and the acute inhalation median lethal concentration (LC_{50}) was greater than 2.1 mg/L. Pyraziflumid was not irritating to the skin of rabbits or to the eyes of rabbits. Pyraziflumid was not skin sensitizing in a local lymph node assay (LLNA) in mice.

In repeat-dose oral toxicity studies with pyraziflumid in mice, rats and dogs the main targets for toxicity were the liver and thyroid.

In a 28-day range-finding study in mice using dietary pyraziflumid concentrations of 0, 300, 1000, 3000 or 10 000 ppm (equal to 0, 49, 164, 520 and 1744 mg/kg bw per day for males, 0, 66, 195, 759 and 2475 mg/kg bw per day for females), the NOAEL was 3000 ppm (equal to 520 mg/kg bw per day), based on increased relative liver weight in males and changes in blood biochemistry in both sexes at 10 000 ppm (equal to 1744 mg/kg bw per day).

In a 90-day study in mice using dietary pyraziflumid concentrations of 0, 800, 3000 or 8000 ppm (equal to 0, 119, 433 and 1183 mg/kg bw per day for males, 0, 146, 514 and 1456 mg/kg bw per day for females), the NOAEL was 800 ppm (equal to 119 mg/kg bw per day), based on reductions in cholesterol and increases in liver weight in males, and hepatocyte hypertrophy and increased periportal hepatocyte fat in both sexes at 3000 ppm (equal to 433 mg/kg bw per day).

A 13-week study in rats was performed using dietary pyraziflumid concentrations in males of 0, 100, 500 or 5000 ppm (equal to 0, 7.1, 36 and 435 mg/kg bw/day), this highest dose being reduced to 2000 ppm (151 mg/kg bw per day) from week 9 due to excessive toxicity. Females received 0, 100, 500 or 2000 ppm (equal to 0, 8.6, 42 and 172 mg/kg bw per day) throughout the study period. The NOAEL was 100 ppm (equal to 7.1 mg/kg bw per day), based on changes in clinical signs, clinical chemistry and effects on liver and thyroid in both sexes, and effects on kidney in females at 500 ppm (equal to 36 mg/kg bw per day).

A 90-day study in dogs was performed using dietary pyraziflumid concentrations of 0, 200, 1000 and 10 000 ppm. In males the high dose was reduced to 5000 ppm from week 3 due to severe toxicity in these animals. The dietary doses were equal to 0, 6.0, 29 and 167 mg/kg bw per day for males and 0, 6.2, 31 and 320 mg/kg bw per day for females. The NOAEL was 1000 ppm (equal to 29 mg/kg bw per day), based on effects on body weight and feed consumption in males, and several signs of liver toxicity in both sexes at 10 000↓5000 ppm (equal to 167 mg/kg bw per day).

In a one-year oral toxicity study in dogs, pyraziflumid was administered in the diet at dose levels of 0, 200, 1000 or 5000 ppm. Due to severe toxicity the treatment of the high-dose animals was discontinued from day 9 for males and day 4 for females. After a withdrawal and recovery period, treatment of the high dose group recommenced at 2000 ppm. The dietary doses were equal to 0, 5.4, 28 and 51 mg/kg bw per day for males, 0, 5.5, 28, and 48 mg/kg bw per day for females. The NOAEL was 1000 ppm (equal to 28 mg/kg bw per day) based on mortality, clinical signs, body weight loss and reduced feed consumption in males, and on clinical biochemistry and histopathological signs of liver and kidney toxicity in both sexes at 5000↓2000 ppm, equal to 48 mg/kg bw per day.

The overall NOAEL for the 90-day and one-year dog studies was 1000 ppm (equal to 29 mg/kg bw per day), and the overall LOAEL was 5000↓2000ppm, equal to 48 mg/kg bw per day.

In an 18-month carcinogenicity study in mice using dietary pyraziflumid concentrations of 0, 200, 2000 or 8000 ppm (equal to 0, 21, 227 and 905 mg/kg bw per day for males, 0, 25, 251 and 1030 mg/kg bw per day for females) the NOAEL was 200 ppm (equal to 21 mg/kg bw per day) based on histopathological signs of liver toxicity observed in both sexes, accompanied by slight reductions in body weight gain in males, particularly at the first week of treatment at 2000 ppm (equal to 227 mg/kg bw per day). There were no treatment-related neoplastic lesions.

In a two-year combined toxicity and carcinogenicity study in rats using dietary pyraziflumid concentrations of 0, 50, 100, 300 and 1000 ppm (equal to 0.0, 2.2, 4.3, 13 and 46 mg/kg bw per day for males, 0.0, 2.9, 5.7, 18 and 66 mg/kg bw per day for females) the NOAEL was 50 ppm (equal to 2.2 mg/kg bw per day) based on macroscopic and histopathological signs of liver and thyroid toxicity observed in both sexes, and indications of minor kidney toxicity in females at 100 ppm (equal to 4.3 mg/kg bw per day). Thyroid adenomas and carcinomas in both sexes and hepatocellular adenomas in females were observed. The NOAEL for neoplastic changes was 300 ppm (equal to 13 mg/kg bw per

day) based on increased incidences of thyroid tumours in males and hepatocellular adenomas in females at 1000 ppm (equal to 46 mg/kg bw per day).

A 1–4 week study was conducted to elucidate a possible mechanism for the increase in hepatocellular adenomas that had been observed (particularly in female rats) after dietary treatment with pyraziflumid at 1000 ppm in a two-year study. Pyraziflumid at 1000 ppm (equal to 70 mg/kg bw per day) caused hepatic enlargement associated with induction of the liver enzymes of the CYP sub-family CYP2B and to a lesser extent CYP1A. The activation of the nuclear receptors AhR, CAR, PXR, PPAR α and PPAR γ , was not investigated. The data are consistent with the induction of liver adenomas by pyraziflumid through activation of CAR, although it was noted that no hepatocyte proliferation was found after pyraziflumid treatment at doses of 1000 and 2000 ppm for 1–4 weeks.

Several mechanistic studies were performed to investigate the MOA for the increased incidence of thyroid tumours observed at doses of 300 ppm and higher in rats. In repeat-dose studies, pyraziflumid, at dietary doses of 300 ppm (equal to 17 mg/kg bw per day) or higher, caused hepatocellular and thyroid follicular cell hypertrophy and an increase in total microsomal P450 levels in liver. Induction of CYP2B activity, and to a lesser extent CYP1A activity, was observed at doses of 1000 ppm (lowest dose tested) and higher. In a seven-day dietary study and a 15-week dietary study, the activity of T4-UDP-GT was also increased at doses of 300 ppm and higher, whereas T3-UDP-GT activity was unchanged. The Meeting noted that thyroid-stimulating hormone (TSH) plasma levels were increased at doses of 1000 ppm and higher but not at 300 ppm, and that plasma levels of thyroxine (T4) and triiodothyronine (T3) were not affected by treatment at doses up to 2000 ppm. An *in vitro* study with thyroid microsomes found no direct inhibition of thyroid peroxidase (TPO) by pyraziflumid. The data suggest that pyraziflumid may cause an enhanced hepatic clearance of thyroid hormones by induction of T4-UDP-GT and a compensatory stimulation of the thyroid by TSH. However, the lowest dose at which an increased incidence of thyroid tumours was observed did not affect plasma TSH levels, nor the levels of T4 and T3. Thus, the mechanistic studies were not sufficient to elucidate unequivocally the MOA of thyroid activation in rats.

The Meeting concluded that pyraziflumid is carcinogenic in rats, but not in mice.

Pyraziflumid was tested for genotoxicity in an adequate range of *in vitro* and *in vivo* assays. No evidence of induction of gene mutation or structural chromosomal aberrations was found. Pyraziflumid induced polyploidy *in vitro* in Chinese hamster lungs cells, a cell line that is known to be susceptible to induction of polyploidy by various factors acting via non-specific mechanisms.

Although it cannot be unequivocally concluded that the parent compound does not have the ability to induce polyploidy *in vivo*, as no specific test was conducted, the negative finding obtained in the *in vivo* micronucleus test in mice can be considered reliable indirect evidence of the absence of polyploidy induction.

In addition, the induction of polyploidy is a threshold phenomenon since it is not related to direct interaction with the DNA. Taking all these considerations into account, the observation of polyploidy in the *in vitro* test was considered to be of low concern.

The Meeting concluded that pyraziflumid is unlikely to be genotoxic.

In view of the lack of genotoxicity, the absence of carcinogenicity in mice, and the fact that increases in liver and thyroid tumours in rats were observed only at a high dose by non-genotoxic mode of action, the Meeting concluded that pyraziflumid is unlikely to pose a carcinogenic risk to humans from the diet.

In a two-generation reproductive toxicity study in rats administered pyraziflumid in the diet at a dose of 0, 50, 100, 300 or 1000 ppm (equal to 0, 2.8, 5.6, 17 and 57 mg/kg bw per day for F₀ males; 0, 3.6, 7.1, 21 and 72 mg/kg bw per day for F₁ males; 0, 3.5, 7.0, 21 and 70 mg/kg bw per day for F₀ pre-mating females) the NOAEL for parental toxicity was 100 ppm (equal to 5.3 mg/kg bw per day) based on adverse effects on liver and thyroid, as indicated by changes in organ weights and histopathology at 300 ppm (equal to 16 mg/kg bw per day).

The NOAEL for offspring toxicity was 300 ppm (equal to 16 mg/kg bw per day) based on reduced body weight gain at 1000 ppm (equal to 57 mg/kg bw per day). The NOAEL for reproductive toxicity was 1000 ppm (equal to 57 mg/kg bw per day) the highest dose tested.

In a developmental toxicity study in rats using gavage doses of pyraziflumid at 0, 20, 100 or 500 mg/kg bw per day, the NOAEL for maternal toxicity was 20 mg/kg bw per day based on reduced body weight gain and feed intake at 100 mg/kg bw per day. The NOAEL for embryo and fetal toxicity was 500 mg/kg bw per day, the highest dose tested.

In a developmental toxicity study in rabbits administered pyraziflumid by gavage at a dose of 0, 10, 30 or 100 mg/kg bw per day from gestation day (GD) 6 to GD 27, the NOAEL for maternal toxicity was 30 mg/kg bw per day, based on reduced body weight gain, feed consumption and a slightly increased incidence of abortion at 100 mg/kg bw per day. The NOAEL for embryo and fetal toxicity was 100 mg/kg bw per day, the highest dose tested.

The Meeting concluded that pyraziflumid is not teratogenic.

In an acute neurotoxicity study in which rats were administered pyraziflumid by gavage at a dose of 0, 500, 1000 or 2000 mg/kg bw and then observed for 14 days, the NOAEL for neurotoxicity was 2000 mg/kg bw. The LOAEL for general toxicity was 500 mg/kg bw, the lowest dose tested, based on a transient decrease in total and ambulatory motor activity observed eight hours after treatment at this dose, in the absence of any neuropathological changes.

Although there were no indications of neuropathological effects due to pyraziflumid, the Meeting concluded that pyraziflumid may cause transient, acute neurobehavioural effects at high doses.

No studies on immunotoxicity were available. In the available toxicity studies on pyraziflumid no indications of an immunotoxic potential were seen.

The Meeting concluded that pyraziflumid is unlikely to be immunotoxic.

Special studies in rats revealed no effect of pyraziflumid at doses up to 2000 mg/kg bw on the cardiovascular and respiratory systems or on small intestinal transport.

Toxicological data on metabolites and/or degradates

No studies were available on the toxicity of pyraziflumid metabolites. The major metabolites identified in livestock were pyraziflumid-4'-OH (BC-01) and its glucuronide conjugate, and pyraziflumid-amide (BC-10). In the rat at least 90% of pyraziflumid was absorbed and subsequently its metabolites were largely excreted in the bile (83–85%). Pyraziflumid-4'-OH (BC-01) and its glucuronide are the major metabolites in bile (approximately 6% and 48% of the administered dose, respectively). Of the administered radiolabel 6–13% was excreted in urine, of which a major part was pyraziflumid-4'-OH and its glucuronide (2–7% of the administered dose, or 29–58% of total excretion in urine). Thus it is concluded that the toxicity of pyraziflumid-4'-OH and its glucuronide is covered by that of pyraziflumid.

The metabolite pyraziflumid-amide (BC-10) only occurs at low levels in urine (up to 2% of excreted radioactivity). QSAR analysis (OECD QSAR Toolbox) indicated that pyraziflumid-amide may bind to DNA and have the ability to induce chromosomal aberrations. As no genotoxicity studies were available for this metabolite the threshold of toxicological concern (TTC) for genotoxicity can be applied for the assessment of chronic toxicity, that is 0.0025 µg/kg bw per day.

Microbiological data

No data for the impact of pyraziflumid on the human gut microbiome was available.

Human data

As this is a new compound no data for the effects of the pyraziflumid in humans were available.

The Meeting concluded that the existing database on pyraziflumid was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting established an acceptable daily intake (ADI) of 0–0.02 mg/kg bw for pyraziflumid on the basis of a NOAEL of 2.2 mg/kg bw per day in a two-year rat study, based on macroscopic and histopathological signs of liver and thyroid toxicity observed in both sexes, and indications of minor kidney toxicity in females, observed at 4.3 mg/kg bw per day. A safety factor of 100 was used. The upper bound of the ADI gives a margin of 2300 relative to the LOAEL for the observed thyroid tumours and hepatocellular adenomas in rats.

The Meeting established an acute reference dose (ARfD) of 2 mg/kg bw for pyraziflumid on the basis of a LOAEL of 500 mg/kg bw, based on a reduction in locomotor activity in an acute neurotoxicity study in rats. A safety factor of 300 was used, which includes an additional factor of three for the use of a LOAEL instead of a NOAEL.

A toxicological monograph was prepared.

Levels relevant to risk assessment of pyraziflumid

Species	Study	Effect	NOAEL	LOAEL
Mouse	18-month study of carcinogenicity ^a	Toxicity	200 ppm, equal to 21 mg/kg bw per day	2000 ppm, equal to 227 mg/kg bw per day
		Carcinogenicity	8000 ppm, equal to 905 mg/kg bw per day ^c	–
Rat	Acute neurotoxicity study ^b	Neurotoxicity	–	–
		Toxicity	–	500 mg/kg bw ^e
	Two-year study of toxicity and carcinogenicity ^a	Toxicity	50 ppm, equal to 2.2 mg/kg bw per day	100 ppm, equal to 4.3 mg/kg bw per day
		Carcinogenicity	300 ppm, equal to 13 mg/kg bw per day	1000 ppm, equal to 46 mg/kg bw per day
	Two-generation study of reproductive toxicity ^a	Reproductive toxicity	1000 ppm, equal to 57 mg/kg bw per day ^c	–
		Parental toxicity	100 ppm, equal to 5.3 mg/kg bw per day	300 ppm, equal to 16 mg/kg bw per day
		Offspring toxicity	300 ppm, equal to 16 mg/kg bw per day	1000 ppm, equal to 57 mg/kg bw per day
	Developmental toxicity study ^b	Maternal toxicity	20 mg/kg bw per day	100 mg/kg bw per day
		Embryo and fetal toxicity	500 mg/kg bw per day ^c	–
Rabbit	Developmental toxicity study ^b	Maternal toxicity	30 mg/kg bw per day	100 mg/kg bw per day
		Embryo and fetal toxicity	100 mg/kg bw per day ^c	–
Dog	13-week and one-year studies of toxicity ^d	Toxicity	1000 ppm, equal to 29 mg/kg bw per day	5000↓2000 ppm, equal to 48 mg/kg bw per day

^a Dietary administration

^b Gavage administration

^c Highest dose tested

^d Two or more studies combined

^e Lowest dose tested

Acceptable daily intake (ADI) applies to pyraziflumid, pyraziflumid-4'-OH (BC-01) and its glucuronide conjugate, expressed as pyraziflumid

0–0.02 mg/kg bw

Acute reference dose (ARfD) applies to pyraziflumid, pyraziflumid-4'-OH (BC-01) and its glucuronide conjugate, expressed as pyraziflumid

2.0 mg/kg bw

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Critical end-points for setting guidance values for exposure to [compound]

Absorption, distribution, excretion and metabolism in mammals

Rate and extent of oral absorption	Relatively rapid ($T_{\max}$ 3–12 hours at 1 mg/kg bw, 9–24 hours at 100 mg/kg bw) and almost complete (>90%) in rats
Dermal absorption	No data
Distribution	Widely distributed, highest concentrations found in gastrointestinal tract, liver, fat, kidneys, thyroid and adrenals
Potential for accumulation	Low
Rate and extent of excretion	Relatively rapid; 84–92% in 72 hours
Metabolism in animals	Extensively metabolized; major metabolites are 4-hydroxy-pyraziflumid, 3-hydroxy-pyraziflumid and their glucuronides.
Toxicologically significant compounds in animals and plants	Pyraziflumid

Acute toxicity

Rat, LD ₅₀ , oral	>2000 mg/kg bw
Rat, LD ₅₀ , dermal	>2000 mg/kg bw
Rat, LC ₅₀ , inhalation	>2.1 mg/L
Rabbit, dermal irritation	Not irritating
Rabbit, ocular irritation	Not irritating
Mouse, dermal sensitization	Not sensitizing (LLNA)
Guinea pig, dermal sensitization	No data

Short-term studies of toxicity

Target/critical effect	Liver, thyroid
Lowest relevant oral NOAEL	7.1 mg/kg bw per day (rat)
Lowest relevant dermal NOAEL	1000 mg/kg bw per day, highest dose tested (rat)
Lowest relevant inhalation NOAEC	No data

Long-term studies of toxicity and carcinogenicity

Target/critical effect	Liver (rat and mouse), thyroid and kidney (rat)
Lowest relevant NOAEL	2.2 mg/kg bw per day (rat)

Carcinogenicity	Carcinogenic in rats (thyroid and liver tumours), but not in mice ^a
Genotoxicity	Unlikely to be genotoxic
Reproductive toxicity	
Target/critical effect	No reproductive effects, liver and thyroid effect (parental), pup body weight gain
Lowest relevant parental NOAEL	5.3 mg/kg bw per day (rat)
Lowest relevant offspring NOAEL	16 mg/kg bw per day (rat)
Lowest relevant reproductive NOAEL	57 mg/kg bw per day, highest dose tested (rat)
Developmental toxicity	
Target/critical effect	No developmental toxicity, body weight gain and feed consumption (rat, rabbit), slight effect on abortions (rabbit)
Lowest relevant maternal NOAEL	20 mg/kg bw per day (rat)
Lowest relevant embryo/fetal NOAEL	100 mg/kg bw per day, highest dose tested (rabbit)
Neurotoxicity	
Acute neurotoxicity NOAEL	2000 mg/kg bw (LOAEL general toxicity: 500 mg/kg bw, lowest dose tested) (rat)
Subchronic neurotoxicity NOAEL	No data
Developmental neurotoxicity NOAEL	No data
Immunotoxicity	No data
Studies on toxicologically relevant metabolites	
	No data
Microbiological data	No data is available for the impact on the human gut microbiome.
Human data	No data

^a Unlikely to pose a carcinogenic risk to humans via exposure from the diet

Summary

	Value	Study	Safety factor
ADI	0–0.02 mg/kg bw ^a	Two-year study of toxicity and carcinogenicity (rat)	100
ARfD	2.0 mg/kg bw ^a	Acute neurotoxicity study (rat)	300

^a Applies to pyraziflumid, pyraziflumid-4'-OH (BC-01) and its glucuronide conjugate, expressed as pyraziflumid

RESIDUE AND ANALYTICAL ASPECTS

Pyraziflumid is a broad-spectrum fungicide active against *Sclerotinia sclerotiorum* and used on vegetables, fruits and turf. It is a pyrazine-biphenyl fungicide of the carboxamide class for suppression of spore germination, mycelium elongation and sporogenesis. Pyraziflumid acts by inhibition of activity of

mitochondria electron transport chain complex II (succinate dehydrogenase complex) of pathogenic filamentous fungi.

It was scheduled by the Fifty-first Session of the CCPR for evaluation as a new compound by the 2020 JMPR, which was postponed to the 2021 JMPR.

The Meeting received information on identity, physicochemical properties, metabolism and environmental fate, methods of residue analysis, freezer storage stability, registered use patterns, supervised residue trials and fate of residues in processing.

GAP information was available for the use of pyraziflumid as foliar treatments on apples, pears and grapes in Japan.

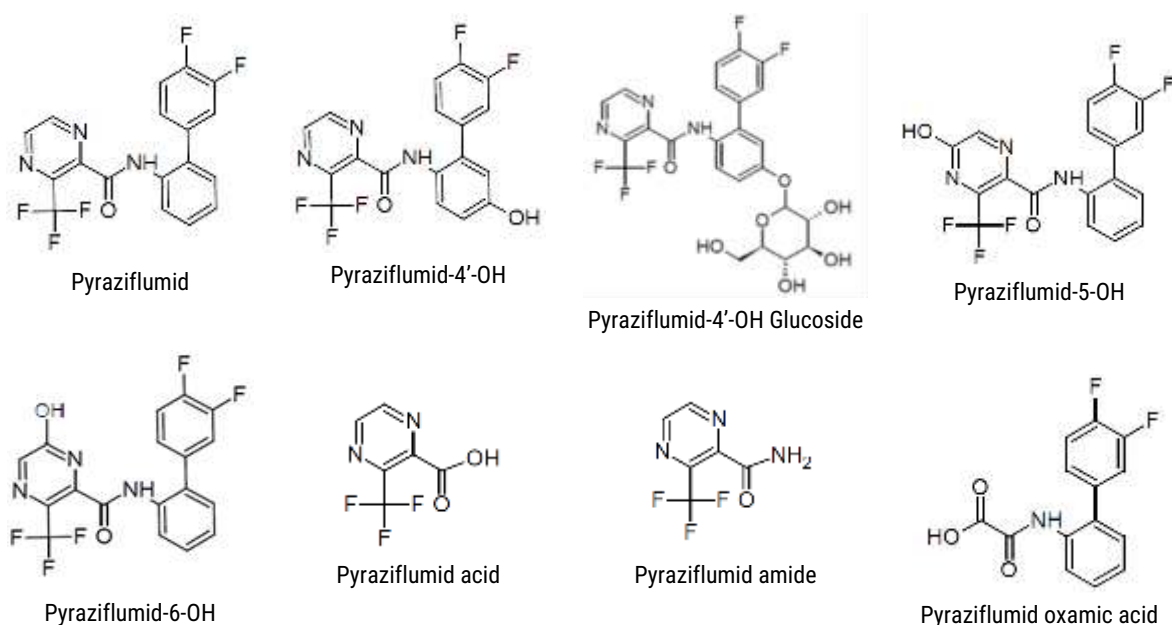


Figure 1 Pyraziflumid and metabolites/degradates

Physical and chemical properties

Pyraziflumid is not volatile. It generally has a higher solubility in organic solvents in comparison to water. The n-octanol water partition coefficient $\log P_{ow}$ is 3.5 at 25 °C, suggesting that the parent has the potential to partition into fat. Pyraziflumid is stable to hydrolysis and photolysis.

Plant metabolism

The Meeting received plant metabolism studies on tomatoes, lettuce and paddy rice following foliar applications of [14 C]-pyraziflumid.

In the metabolism study conducted on greenhouse grown tomatoes, three foliar applications of [14 C]-pyraziflumid were applied at 0.3 kg ai/ha at intervals of 7 days. Treated mature fruit and leaf samples were collected at 0, 1, 7 and 14 days after the final application. Acetonitrile and water extraction of tomatoes resulted in extraction efficiencies of > 98%.

Pyraziflumid was the only radioactive component identified, representing greater than 98% of the TRR in fruit.

In the metabolism study conducted on greenhouse grown lettuces, three foliar applications of [14 C]-pyraziflumid were applied at 0.3 kg ai/ha at intervals of 7 days. Treated lettuce heads were collected at 0, 7 and 14 days after the final application. Acetonitrile and water extraction of lettuce resulted in extraction efficiencies of > 99%.

Pyraziflumid was the major radioactive component identified, representing greater than 99% of the TRR in lettuce head and outer leaves. Residues were highest on the outer leaves.

In the metabolism study conducted on paddy rice, three foliar applications of [¹⁴C]-pyraziflumid were applied at 0.1 kg ai/ha at intervals of 7 days. Panicle and foliage were collected at the milk stage, while mature grain, hull and straw samples were collected at the ripe stage. Acetonitrile and water extraction of rice, resulted in extraction efficiencies of > 94% in rice grain, > 88% in hulls and > 91% in straw.

Pyraziflumid was the major radioactive component identified, representing greater than 95% of the TRR in grain, greater than 79% of the TRR in hulls and greater than 77% of the TRR in straw. Pyraziflumid-4'-OH was a minor component in all matrices accounting for up to < 5% of the TRR in grain and hulls and 5.5% of the TRR in straw. In addition, small amounts of pyraziflumid-4'-OH glucoside were characterized by treatment of enzymatic and acidic hydrolysis (up to 7.4% TRR in hulls and up to 12% TRR in straw).

The Meeting noted that the metabolism studies were conducted under greenhouse grown conditions however as pyraziflumid is not susceptible to photolysis this is not expected to influence the expected residue in crops grown under field conditions.

In summary, the metabolism of pyraziflumid is comparable in all crops investigated. Most of the radioactivity was recovered, with the majority (generally > 75%) of this being identified as parent pyraziflumid. No metabolites were observed at > 10% TRR in plant matrices except pyraziflumid-4'-OH glucoside which was observed in rice straw at 12% TRR. The principal metabolic pathway of pyraziflumid in plants is considered to be the formation of metabolite pyraziflumid-4'-OH by hydroxylation of the phenyl ring and subsequent glucosidation.

Animal metabolism

The Meeting received animal metabolism studies on rats, lactating goats and laying hens in which animals were dosed with [¹⁴C]-pyraziflumid.

Evaluation of the metabolism studies in rodents was carried out by the WHO Core Assessment Group.

Lactating goats were orally dosed via gelatine capsule at a nominal concentration of 15 ppm in feed for 5 consecutive days and sacrificed 6 hours after the last dose.

The majority of the administered dose (AD) was excreted with faeces accounting for 53% AD, while excretion via urine accounted for between 15–19% AD. The total recovery of radioactivity was low in all tissues and milk, representing between 0.2–0.5% AD in tissues and < 0.1–0.3% AD in whole milk.

In tissues, the majority of radioactivity was extracted with acetonitrile: hexane and then acetonitrile ranging from 38% TRR (loin muscle) to 128% TRR (subcutaneous fat) via the aniline labelled experiment and 49% TRR (liver and loin muscle) to 98% TRR (omental fat) via the pyrazinyl experiment. Much lower levels were extracted via aqueous extractions. The PES required further acid, base and protease hydrolysis extractions for loin muscle, liver and kidney. In the pyrazinyl labelled experiment, total hydrolysis released up to 21%, 23% and 29% TRR in loin muscle, liver and kidney, respectively, with the majority at 14% TRR via protease hydrolysis in loin muscle, 9% TRR via acid or base hydrolysis in liver and 12% TRR via protease or acid hydrolysis in kidney. In milk the majority of the radioactivity was extracted with acetonitrile: hexane, then acetonitrile and acetonitrile/water (pyrazinyl labelled milk aqueous only) solvents. This represented between 47–66% TRR in skim milk and 18–31% TRR in milk fat.

In goat muscle, pyraziflumid was the major component, representing 16–55% TRR (0.003–0.015 mg/kg). Pyraziflumid-4'-OH was also observed at low levels 0.002–0.003 mg eq/kg (3–15% TRR).

In goat fat, pyraziflumid was the major component, representing 64–107% TRR (0.043–0.084 mg/kg). Pyraziflumid-4'-OH was also observed in omental fat at 20% TRR (0.017 mg eq/kg) via the

aniline labelled experiment but was only observed as a minor metabolite in most other fats in both labelled experiments at up to 6% TRR.

In goat liver, pyraziflumid-4'-OH was the major metabolite at up to 25–50% TRR (0.187–0.188 mg eq/kg). In goat kidney, pyraziflumid-4'-OH, pyraziflumid amide and the pyraziflumid-4'-OH glucuronide were all significant metabolites at up to 14% (0.023 mg eq/kg), 11% (0.018 mg eq/kg) and 36% TRR (0.023 mg eq/kg), respectively. Parent pyraziflumid was observed at < 5% TRR in liver and kidney.

In milk fat, pyraziflumid was the major component, representing 13–26% TRR (0.004–0.010 mg/kg). In skim milk, pyraziflumid-4'-OH glucuronide was observed at low levels 0.003 mg eq/kg (22% TRR).

Laying hens were orally dosed at a nominal dose of 10 ppm in the diet for 10 consecutive days and sacrificed 6 hours after the last dose.

The majority of the administered dose was eliminated in the excreta, accounting for 86–87% AD. The total recovery of radioactivity was low in all tissues and eggs, representing between < 0.1–0.1% AD.

In tissues, the majority of radioactivity was recovered in the acetonitrile: hexane and then acetonitrile extracts, ranging from 66–69% TRR (breast muscle), 74–88% TRR (thigh muscle), 53–58% TRR (liver), 95–105% TRR (abdominal fat) and 91–97% TRR (subcutaneous fat) across both label experiments. In eggs, the majority of the radioactivity was recovered in the sum of the acetonitrile: hexane and acetonitrile extracts, ranging from 69–82% TRR (egg white) and 58–61% TRR (egg yolk) across both labelled experiments.

In hen muscle, pyraziflumid was observed at 15–38% TRR (0.001–0.006 mg/kg) and pyraziflumid amide observed at 41–46% TRR (0.010–0.013 mg eq/kg). Pyraziflumid-4'-OH was also observed at low levels up to 0.003 mg eq/kg (20% TRR).

In hen fat, pyraziflumid was the major component at up to 84% TRR (0.087 mg/kg). In hen liver, pyraziflumid-4'-OH was the major metabolite, representing up to 39% TRR (0.182 mg eq/kg).

In egg white, pyraziflumid-4'-OH glucuronide and pyraziflumid amide were observed at up to 42% TRR (0.003 mg eq/kg) and 35% TRR (0.009 mg eq/kg), respectively. In egg yolk, pyraziflumid and pyraziflumid-4'-OH were major components at 9–19% TRR (0.001–0.027 mg/kg) and 4–15% TRR (0.001–0.021 mg eq/kg), respectively.

In summary, the metabolism of pyraziflumid is comparable in the goat and hen. Pyraziflumid was mainly metabolised via hydroxylation of the aniline ring followed by conjugation by glucuronic acid. In the pyrazinyl ring, pyraziflumid amide and the corresponding acid, pyraziflumid acid, were also observed after cleavage of the amide bond. Other minor metabolites were observed.

Environmental fate

The Meeting considered that pyraziflumid is stable to aqueous photolysis ($DT_{50} > 1000$ days), soil photolysis (DT_{50} 135–304 days) and is persistent in soils under aerobic conditions ($DT_{50} > 1$ year).

No rotational crop studies were made available to the Meeting. However, the uses under consideration are all for permanent crops. Uptake of residues by rotational crops cannot be excluded and should uses expand to non-permanent crops, rotational crop studies will be required to estimate the potential for carry over of residues.

Methods of analysis

Analytical methods have been reported and validated for the analysis of pyraziflumid in plant matrices.

The enforcement and data generation method GPL-MTH-096 for pyraziflumid and pyraziflumid-4'-OH involved a number of extractions with acetonitrile: 0.1 M hydrochloric acid in water. The extracts were combined and an aliquot of the combined extract was concentrated and acidified with 8M hydrochloric acid in water and incubated at 50 °C for 16 hours or overnight to hydrolyse the pyraziflumid-

4'-OH conjugate to pyraziflumid-4'-OH. After incubation, water was added followed by partitioning twice in ethyl acetate. An aliquot of the combined organic layer was brought to a known volume with ethyl acetate and evaporated to dryness, reconstituted in acetonitrile: water (50:50, v/v) and analysed by HPLC-MS/MS. The LOQs for pyraziflumid and pyraziflumid-4'-OH were demonstrated to be 0.01 mg/kg for each analyte in apple (whole fruit), pear (whole fruit), grapes and apple juice (fresh).

The Meeting concluded that a suitable analytical method GPL-MTH-096 to measure pyraziflumid and free and conjugated 4'-OH-pyraziflumid in plant commodities, including apples, pears and grapes, is available and suitable for both data generation and monitoring compliance with MRLs. The QuEChERS method was found to be suitable for analysis of parent pyraziflumid and pyraziflumid-4'-OH with acceptable recoveries in apples (72–108%), grapes (71–99%), tomato (99–110%) and wheat grain (72–99%) by GC-MS, with an LOQ of 0.01 mg/kg. The method by LC-MS/MS had satisfactory recoveries for parent pyraziflumid only for grapes (74–119%).

The Meeting noted that there is no available method for pyraziflumid or its metabolites in animal commodities.

Stability of residues in stored analytical samples

A storage stability study showed that residues of pyraziflumid and pyraziflumid-4'-OH were stable for at least 24 months in apple (whole fruit) and apple juice (fresh) and for at least 12 months in grapes. In storage stability studies conducted concurrently with the supervised residues trials residues of pyraziflumid and pyraziflumid-4'-OH were observed to be stable for at least 5, 6 and 7 months in apples, pears and grapes, respectively. These durations covered the longest period of storage for all samples obtained from supervised residue trials.

Definition of the residue

Plant commodities

In the plant metabolism studies involving foliar applications to tomatoes, lettuce and paddy rice, pyraziflumid was the predominant residue accounting for 99–100% TRR in tomato fruit, lettuce heads and lettuce leaves at 0 to 14 DALA and 96, 80 and 77% TRR in rice grain, hulls and straw, respectively, at 28 DALA. The only metabolite observed at > 10% was pyraziflumid-4'-OH glucoside at 12% in rice straw. In the supervised residue trials on apple, pear and grapes, pyraziflumid was detected in all trials at 0.06–0.73 mg/kg (apples), 0.04–0.48 mg/kg (pears) and 0.06–1.21 mg/kg (grapes). Pyraziflumid-4'-OH (free and conjugated) was not observed above 0.01 mg/kg (LOQ) in any of the apple, pear or grape trials.

Method GPL-MTH-096 has been validated for the analysis of pyraziflumid (and pyraziflumid-4'-OH) in plant commodities.

The Meeting noted that in residue trials in apple, pear, and grape which monitored pyraziflumid and its free and conjugated 4'-OH metabolite, only the parent compound was observed at quantifiable levels. The hydrolysis steps were considered unnecessary for the crops considered due to the LOQ levels of pyraziflumid-4'-OH observed in apples, grapes and pears. The Meeting agreed that for the purposes of enforcement and dietary exposure, the residue definition for plant commodities is *pyraziflumid*.

The definition of the residue for compliance with the MRL for plant commodities: *pyraziflumid*.

The definition of the residue for dietary risk assessment for plant commodities: *pyraziflumid*.

Animal commodities

Pyraziflumid was the major component in goat muscle (up to 55% TRR), goat fat (up to 107% TRR) and milk fat (up to 26% TRR), hen muscle (up to 38% TRR), hen fat (up to 84% TRR) and egg yolk (up to 19% TRR).

Pyraziflumid-4'-OH was observed as a significant metabolite in goat muscle (up to 15% TRR), goat fat (up to 20% TRR) and goat kidney (up to 14% TRR), hen muscle (up to 20% TRR), egg yolk (up to 15% TRR) and was the major component in goat liver (up to 50% TRR) and hen liver (up to 39% TRR).

Pyraziflumid-4'-OH glucuronide was observed in goat kidney (up to 36% TRR) and at trace levels in skim milk (up to 22% TRR) and egg white (up to 42% TRR).

Pyraziflumid amide was observed in goat kidney (up to 11% TRR), hen muscle (up to 46% TRR) and at trace levels in egg white (up to 35% TRR).

No validated analytical method is available for the measurement of pyraziflumid and its metabolites in animal commodities. Furthermore, no animal feeding studies were available to the Meeting.

When considering which components should be included in the enforcement definition the Meeting concluded that the conjugated pyraziflumid-4'-OH did not contribute significantly in most animal commodities compared to pyraziflumid and its free pyraziflumid-4'-OH metabolite except in goat kidney, skim milk and egg white where sufficient amounts of parent and the free metabolite were observed to allow enforcement. The Meeting decided that an appropriate residue definition for animal commodities for compliance with the MRL is pyraziflumid and its pyraziflumid-4'-OH metabolite (free). Based on the proportions of pyraziflumid and pyraziflumid-4'-OH observed in fat and egg yolk compared to muscle and egg white the residue is considered to be fat-soluble.

When considering which components should be included in the residue definition for dietary risk assessment, the Meeting concluded that the toxicity of the metabolites pyraziflumid-4'-OH (free and conjugated) is covered by parent pyraziflumid but were observed at significant levels in animal commodities and should be included in the definition. For pyraziflumid amide, the Meeting noted that QSAR analysis identified that pyraziflumid amide is potentially genotoxic, however no specific genotoxicity data were available for the metabolite and the TTC approach for genotoxicity should be followed⁶. The toxicological threshold for genotoxicity (0.15 µg/person) was applied. The estimated exposure based on animal commodities from the goat metabolism study, adjusted to the estimated livestock dietary burden, resulted in a maximum long-term exposure of 0.02 µg/person, below the applicable threshold of toxicological concern. The Meeting concluded that pyraziflumid amide was unlikely to present a dietary exposure concern from the uses evaluated by the current Meeting. Should further uses be considered in the future, these conclusions may need to be re-evaluated.

For dietary exposure, the appropriate residue definition is pyraziflumid and its pyraziflumid-4'-OH metabolite (free and conjugated), expressed as pyraziflumid.

The definition of the residue for compliance with the MRL for animal commodities: pyraziflumid and its pyraziflumid-4'-OH metabolite (free), expressed as pyraziflumid.

The definition of the residue for dietary risk assessment for animal commodities: pyraziflumid and its pyraziflumid-4'-OH metabolite (free and conjugated), expressed as pyraziflumid.

The residue is fat-soluble.

Results of supervised residue trials on crops

The Meeting received residue trials on apples, pears and grapes.

Apples and Pears

In Japan, the critical GAP for apples and pears is 2 foliar applications of pyraziflumid at a 2000-fold dilution (7.5 g ai/100 L) in conjunction with a harvest withholding period of 1 day.

⁶ See Toxicology section for further details.

Sixteen residue trials (8× apples and 8× pears) conducted in Japan involved 2 applications at a 2000-fold dilution (7.5 g ai/100 L with spray volumes of 4000–5000 L/ha) and matched the cGAP.

In eight trials conducted on apples, pyraziflumid residues in apple (whole fruit) were 0.23, 0.24, 0.28, 0.30, 0.33, 0.36, 0.46 and 0.73 mg/kg.

In eight trials conducted on pears, pyraziflumid residues in pear (whole fruit) were 0.27, 0.34, 0.36, 0.39, 0.40, 0.43, 0.46 and 0.48 mg/kg.

The residue data sets for apples and pears were similar (Mann-Whitney U test) and could be combined for mutual support. The combined residue dataset was 0.23, 0.24, 0.27, 0.28, 0.30, 0.33, 0.34, 0.36, 0.36, 0.39, 0.40, 0.43, 0.46, 0.46, 0.48 and 0.73 mg/kg. The Meeting noted that the cGAPs were the same for apples, pears and persimmons and decided to extrapolate the recommendation to Japanese persimmons.

The Meeting estimated a maximum residue level of 1.5 mg/kg, an STMR of 0.36 mg/kg and an HR of 0.73 mg/kg for pyraziflumid in apples, Japanese persimmons and pears.

Grapes

In Japan, the critical GAP for grapes is 2 foliar applications of pyraziflumid at a 2000-fold dilution (7.5 g ai/100 L) in conjunction with a harvest withholding period of 7 days.

Nine residue trials conducted in Japan involved 2 applications at a 2000-fold dilution (7.5 g ai/100 L with spray volumes of 3100–3530 L/ha (233–265 g ai/ha)) and matched the cGAP.

Pyraziflumid residues were (n = 9): 0.41, 0.48, 0.50, 0.51, 0.57, 0.92, 0.97, 0.98 and 0.98 mg/kg.

The Meeting estimated a maximum residue level of 3 mg/kg, an STMR of 0.57 mg/kg and an HR of 0.98 mg/kg for pyraziflumid in grapes.

Fate of residues during processing

In a high-temperature hydrolysis study, pyraziflumid was stable under conditions simulating pasteurisation (pH 4 at 90 °C for 20 minutes), baking, brewing and boiling (pH 5 at 100 °C for 60 minutes) and sterilisation (pH 6 and autoclaving at 120 °C for 20 minutes).

The processing of apples to apple juice (fresh) and apple wet pomace, and grapes to grape juice (fresh) and dried grapes (raisins) was also investigated. The studies showed that pyraziflumid did not concentrate in apple or grape juice (PF = 0.1) but did concentrate in apple wet pomace (PF = 2.5) and dried grapes (PF = 2.0).

Residues of pyraziflumid-4'-OH (free and conjugated) were < 0.01 mg eq/kg in both the apple and grape processing studies.

Table 1 Processing factors, for pyraziflumid, used for dietary risk assessment and livestock dietary burden

Raw commodity	Processed commodity	Processing factor	STMR-P (Median-P) = STMR _{RAC} × PF (mg/kg)	HR-P = HR _{RAC} × PF (mg/kg)
Apples	Apple juice	0.1	0.36 × 0.1 = 0.036	-
	Apple pomace	2.5	0.36 × 2.5 = 0.90	-
Grapes	Dried grapes	2.0	0.57 × 2.0 = 1.14	0.98 × 2.0 = 1.96
	Grape juice	0.1	0.57 × 0.1 = 0.057	-

Table 2 Processing factors, for pyraziflumid, used for maximum residue level estimation.

Raw commodity	Processed commodity	Processing factor	Maximum residue level $\text{RAC} \times \text{PF}$ (mg/kg)	Maximum residue level (mg/kg)
Grapes	Dried grapes	2.0	$3 \times 2.0 = 6$	6

Based on the STMR in apples at 0.36 mg/kg and the processing factor (2.5) for wet pomace (noting that residues of pyraziflumid-4'-OH (free and conjugated) were below LOQ in the apple residue trials and processing study), the Meeting estimated a Median-P of 0.90 mg/kg in apple wet pomace.

Based on the estimated maximum residue level of 3 mg/kg, the HR and STMR in grapes at 0.98 mg/kg and 0.57 mg/kg, respectively, and the processing factor (2.0) for dried grapes, the Meeting estimated a maximum residue level in dried grapes (DF 0269) of 6 mg/kg, an HR-P of 1.96 mg/kg and an STMR-P of 1.14 mg/kg.

Residues in animal commodities

Based on the uses considered, only apple wet pomace is considered to be a significant animal feed for ruminants. Apple wet pomace is not a significant animal feed for poultry.

Estimation of livestock dietary burden

The Meeting estimated the dietary burden of pyraziflumid in cattle using the most recent version of the OECD livestock dietary burden calculator.

The only animal feed considered by the Meeting was apple wet pomace (Median-P = 0.9 mg/kg). The maximum dietary burdens for beef cattle and dairy cattle were estimated to be 0.45 ppm in the feed DW and 0.225 ppm in the feed DW, respectively.

A feeding study was not available.

The Meeting used the residue levels of pyraziflumid and its pyraziflumid-4'-OH metabolite (free and conjugated), expressed as pyraziflumid from the goat metabolism study to estimate expected residue levels for cattle commodities. In the metabolism study, a lactating goat was dosed with [¹⁴C]-pyraziflumid at up to 15.7–16.7 ppm in the diet/day on a dry weight basis. Highest residues of pyraziflumid and its pyraziflumid-4'-OH metabolite (free and conjugated) were 0.005 mg eq/kg in skim milk, 0.011 mg eq/kg in milk fat, 0.005 mg eq/kg in whole milk (calculated based on 4% fat in goat milk), 0.019 mg eq/kg in muscle, 0.087 mg eq/kg in fat, 0.058 mg eq/kg in kidney, and 0.222 mg eq/kg in liver. When scaled to the expected dietary burdens, the highest anticipated residues in milk and tissues were: < 0.0001 mg/kg in whole milk (calculated), 0.0001 mg/kg in milk fat, 0.0005 mg/kg in muscle, 0.002 mg/kg in fat and 0.005 mg/kg in mammalian edible offal.

The Meeting noted that no analytical method for the measurement of pyraziflumid and its metabolites in animal commodities was available, therefore no maximum residue levels were estimated for animal commodities.

RECOMMENDATIONS

On the basis of the data provided, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI and IESTI assessment.

The definition of the residue for compliance with the MRL for plant commodities: *pyraziflumid*.

The definition of the residue for dietary risk assessment for plant commodities: *pyraziflumid*.

The definition of the residue for compliance with the MRL for animal commodities: *pyraziflumid and its pyraziflumid-4'-OH metabolite (free), expressed as pyraziflumid*.

The definition of the residue for dietary risk assessment for animal commodities: *pyraziflumid and its pyraziflumid-4'-OH metabolite (free and conjugated), expressed as pyraziflumid*.

The residue is fat-soluble.

FUTURE WORK

Desirable information:

An analytical method allowing the enforcement of pyraziflumid residues in animal commodities.

If, in the future, the Meeting is asked to consider additional uses involving potentially rotated crops, information on rotational crop metabolism and residues in field rotational crops will become necessary. These studies need to address soil residues taking into account the estimated soil plateau concentrations after subsequent applications over multiple years. In addition, information on long-term soil accumulation may become necessary.

DIETARY RISK ASSESSMENT

Long-term dietary exposure

The ADI for pyraziflumid is 0–0.02 mg/kg bw. The International Estimated Daily Intakes (IEDIs) for pyraziflumid were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs ranged from 0–8% of the maximum ADI. The Meeting concluded that long-term dietary exposure to residues of pyraziflumid from uses considered by the JMPR is unlikely to present a public health concern.

Acute dietary exposure

The ARfD for pyraziflumid is 2 mg/kg bw. The International Estimate of Short-Term Intakes (IESTIs) for pyraziflumid were calculated for the food commodities and their processed commodities for which HRs/HR-Ps or STMRs/STMR-Ps were estimated by the current Meeting and for which consumption data were available. The results are shown in Annex 4 of the 2021 JMPR Report.

The IESTIs varied from 0–3% of the ARfD for children and 0–2% for the general population. The Meeting concluded that acute dietary exposure to residues of pyraziflumid from uses considered by the current Meeting is unlikely to present a public health concern.

5.14 Spiropidion (323)

TOXICOLOGY

Spiropidion is the common name approved by the International Organization for Standardization for 3-(4-chloro-2,6-dimethylphenyl)-8-methoxy-1-methyl-2-oxo-1,8-diazaspiro[4.5]dec-3-en-4-yl ethyl carbonate (IUPAC), with the Chemical Abstracts Service (CAS) number 1229023-00-0. It is a pro-insecticide incorporating a novel tetramic acid derivative. The pesticidal mode of action (MOA) is by inhibiting the enzyme acetyl-CoA carboxylase.

Spiropidion has not previously been evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) and was reviewed by the present Meeting at the request of the Codex Committee on Pesticide Residues (CCPR).

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with relevant national or international test guidelines, unless otherwise specified. The Meeting noted that in short term toxicity studies in rodents and in the 90-day dog study a number of control blood samples subject to bioanalysis were contaminated, either with the parent compound spiropidion (SYN546330) and/or the enol-metabolite SYN547305, and that the reason for the contamination was not clearly identified. However, the Meeting concluded that the contamination did not invalidate the results and interpretation of the study's toxicological findings. No additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Biochemical aspects

In disposition studies rats were administered [^{14}C]spiropidion labelled at either the phenyl or spirodecanone ring as a single oral dose of 5 or 250 mg/kg body weight (bw), or as a single intravenous dose of 1 mg/kg bw of [^{14}C]spiropidion labelled at the phenyl ring. Excretion was >95% (urine, faeces and cage wash) of the administered dose (AD), of which more than 90% was excreted within 48 hours of oral dosing, or 24 hours when dosing was intravenous. There was no significant sex-, dose- or radiolabel-related differences in either the rates or routes of excretion. The major route of excretion was through the urine, accounting for approximately 60% and 50% of the AD in animals given an oral dose of 5 mg/kg bw or 250 mg/kg bw respectively, or 60% of the AD when an intravenous dose of 1 mg/kg bw was administered. In animals given an oral dose of 5 mg/kg bw, an oral dose of 250 mg/kg bw or an intravenous dose of 1 mg/kg bw, faecal excretion accounted for approximately 30%, 40% and 30% of the AD respectively.

In bile duct-cannulated rats, excreted radioactivity in urine, faeces and cage wash was >96% of the AD following either a single dose of 5 or 250 mg/kg bw. Elimination via urine was comparable to that of intact animals. Following a single low dose of 5 mg/kg bw the majority of the dose was excreted in the urine (65–66%) in the first 48 hours, in faeces (14–15%), and via biliary excretion (6–11%). Following a single oral administration of 250 mg/kg bw the majority of the dose was excreted in the urine (45–64%) in the first 48 hours. Elimination of spiropidion via the faeces accounted for 19–30% of the dose and biliary excretion for 13–14% of the dose. The oral absorption was 72–77% in male and female rats, based on urinary, biliary excretion, cage wash and carcass.

Following intravenous administration of 1 mg/kg bw, blood concentrations of radioactivity steadily declined to 72 hours post dose. The systemic exposure (represented by the area under the concentration–time curve, $\text{AUC}_{(0-t)}$) was comparable irrespective of sex and the estimated half-life ($t_{1/2}$) of total radioactivity was longer in males (42 hours) than females (22 hours). In animals given a single oral dose of 5 or 250 mg/kg bw, peak whole blood and plasma concentrations (C_{max}) were observed at 1–2 h or 1–4 h, respectively. Overall total systemic exposure was comparable between whole blood and plasma with the same dose levels and radiolabel position. Systemic exposure to total radioactivity increased in a broadly proportional manner with dose. The estimated $t_{1/2}$ of total radioactivity in whole blood was longer

in males (65 hours) than females (44 hours) administered 5 mg/kg bw, while no sex differences were observed in animals given 250 mg/kg bw.

The pharmacokinetics of non-radiolabelled spiropidion were investigated in the rat following repeated oral exposure at dose levels of 3, 30 and 300 mg/kg bw per day over seven days, or after a single intravenous administration at a dose level of 1 mg/kg bw. Spiropidion blood concentrations were generally very low (<10 ng/mL or below the limit of quantification). The low and variable blood concentrations of spiropidion are due to its rapid hydrolysis to metabolite SYN547305 after both oral and intravenous administration. This is supported by the concentrations of metabolite SYN547305 observed after intravenous and oral administration of spiropidion. There was no evidence supporting accumulation.

Radioactivity was widely distributed, with the highest concentrations observed in the gastrointestinal tract and its contents, followed by liver and kidney, consistent with the excretion profile of spiropidion. Radioactivity in all other tissues was lower or equal to 0.1% of the AD. The tissue depletion half-life of radioactivity in the majority of tissues was either similar, or shorter, than in plasma. However, longer estimates were obtained in whole blood. At termination (96 h post dose), total tissue and carcass residues accounted for equal or less than 2.5% of the administered dose.

Metabolism occurred rapidly in both intact and bile duct-cannulated rats. Unchanged spiropidion was found in faeces at 2.7% and 7% respectively for intact or cannulated animals given 250 mg/kg bw. The observed metabolites were qualitatively and quantitatively similar in urine, faeces, bile and plasma irrespective of dose. Spiropidion was extensively metabolized in rat via ester hydrolysis and subsequent loss of the methoxy moiety in the piperidine ring. No cleavage of the cyclic moieties (phenyl, spirodecanone or piperidine) was observed. In rat excreta, only SYN548430 and an oxidised product of SYN548430, accounted for >10% of the AD. The major components (>10% of total radioactivity AUC) identified in plasma were SYN547305 and SYN548430. In an in vitro study, metabolism of spiropidion in human liver microsomes was extensive and qualitatively comparable to that in rat microsomes, irrespective of the radiolabel position.

Toxicological data

In rats, the acute oral median lethal dose (LD₅₀) was greater than 2000 mg/kg bw, the dermal LD₅₀ was greater than 5000 mg/kg bw and the acute inhalation lethal median concentration (LC₅₀) was greater than 1.12 mg/L. Spiropidion was not irritating to the skin of rabbits but was mildly irritating to the eyes of rabbits. Spiropidion was sensitizing to the skin of guinea pigs and not phototoxic in vitro.

The short-term toxicity of spiropidion was orally tested in mice, rats and dogs, and its long-term toxicity and carcinogenicity tested in mice and rats. The most common effects in rodent studies were on body weights along with reductions in cholesterol and triglycerides. In dogs, critical effects consisted of severe clinical signs indicative of systemic neurotoxicity, which warranted unscheduled termination. No necropsy or microscopic treatment-related effects were reported that were associated with those observations.

In a 90-day toxicity study in mice in which spiropidion was administered at a dietary concentration of 0, 250, 700 or 1500 ppm (equal to 0, 35.2, 105 and 236 mg/kg bw per day for males, 0, 44.1, 115 and 252 mg/kg bw per day for females) the NOAEL was 700 ppm (equal to 105 mg/kg bw per day) based on reduced body weight in females and reduced body weight gain, reduced food utilization, and changes in clinical biochemistry parameters in both sexes at 1500 ppm (equal to 236 mg/kg bw per day).

In a 90-day rat toxicity study in which spiropidion was administered at dietary concentrations of 0, 100, 500 or 2500 ppm for males, the high dose being 1500 ppm in the case of females (equal to 0, 6.2, 31.5 and 159 mg/kg bw per day in males, 0, 7.0, 36.1 and 110 mg/kg bw per day in females) the NOAEL was 100 ppm (equal to 6.2 mg/kg bw per day) based on reduced cholesterol in both sexes and triglycerides in females at 500 ppm (equal to 31.5 mg/kg bw per day).

In a 28-day dog study in which spiropidion was administered via gelatine capsules at dose levels of 0, 10, 30 or 65†100 mg/kg bw per day, the NOAEL was 30 mg/kg bw per day, based on clinical signs

and mortality at 65–100 mg/kg bw per day. Severe clinical signs were observed after one or two doses of 100 mg/kg bw per day, and after 2–4 doses of 65 mg/kg bw per day.

In a 90-day study in dogs in which spiropidion was administered via gelatine capsules at dose levels of 0, 5, 15 or 30 mg/kg bw per day, the NOAEL was 15 mg/kg bw per day, based on mortality and clinical signs in females at 30 mg/kg bw per day.

In a one-year dog study in which spiropidion was administered via gelatine capsules at dose levels of 0, 1, 3, 10 or 30 mg/kg bw per day, the NOAEL was 10 mg/kg bw per day, based on mortality and clinical signs in two animals at 30 mg/kg bw per day.

Combining the 90-day and one-year studies, the Meeting identified an overall NOAEL of 15 mg/kg bw per day.

In a carcinogenicity study in mice, spiropidion was administered at dietary concentrations of 0, 50, 250 or 500 ppm (equal to 0, 6.4, 31.8 and 65.4 mg/kg bw per day for males, 0, 7.0, 36.8 and 72.8 mg/kg bw per day for females). The NOAEL was 50 ppm (equal to 6.4 mg/kg bw per day) based on lower body weight and reduced body weight gain at 250 ppm (equal to 31.8 mg/kg bw per day). No treatment-related increases in tumour incidence were observed in this study.

In a two-year toxicity and carcinogenicity study in rats, spiropidion was administered at dietary concentrations of 0, 50, 100 or 500 ppm in males, the high dose being 300 ppm for females (equal to 0, 2.4, 4.7 and 24.0 mg/kg bw per day for males, 0, 3.1, 6.1 and 18.7 mg/kg bw per day for females). The NOAEL for non-neoplastic lesions was 100 ppm (equal to 4.7 mg/kg bw per day), based on increased incidence of luminal dilatation and inflammation in the bile duct observed in males at 500 ppm (equal to 24.0 mg/kg bw per day). The NOAEL for carcinogenicity was 50 ppm (equal to 2.4 mg/kg bw per day) based on an equivocal increase in testicular interstitial cell adenomas at 100 ppm (equal to 4.7 mg/kg bw per day). Equivocal increases in malignant ependymomas in males were seen at the highest dose level.

The Meeting concluded that spiropidion is not carcinogenic in mice and female rats, and produced equivocal carcinogenic effects in male rats.

Spiropidion was tested for genotoxicity in an adequate range of in vitro and in vivo assays. No evidence of genotoxicity was found.

The Meeting concluded that spiropidion is unlikely to be genotoxic.

In view of the lack of genotoxicity, the absence of carcinogenicity in mice and the fact that only an equivocal increase in testicular interstitial cell adenomas and ependymomas were seen in male rats, the Meeting concluded that spiropidion is unlikely to pose a carcinogenic risk to humans from the diet.

In a two generation reproductive toxicity study in rats spiropidion was administered at dietary concentrations of 50, 100 or 500 ppm for males, the high dose for females being 300 ppm (equal to 0, 3, 6 and 31 mg/kg bw per day for males, 0, 4, 8 and 24 mg/kg bw per day, for females, pre-pairing). The NOAEL for parental toxicity was 100 ppm (equal to 7.8 mg/kg bw per day), based on increased incidence of thyroid follicular hypertrophy in females of both generations at 300 ppm (equal to 23 mg/kg bw per day). The NOAEL for offspring toxicity was 300 ppm (equal to 23 mg/kg bw per day), the highest dose tested. The NOAEL for reproductive toxicity was 300 ppm (equal to 23 mg/kg bw per day) the highest dose tested.

In a developmental toxicity study in rats given spiropidion by gavage at dose levels of 0, 10, 30 or 100 mg/kg bw per day from gestation day (GD) 6 to GD 19, the NOAEL for maternal toxicity was 30 mg/kg bw per day, based on body weight loss, reduced body weight gain and food consumption at 100 mg/kg bw per day. The NOAEL for embryo/fetal toxicity was 100 mg/kg bw per day, the highest dose tested.

In a developmental toxicity study in rabbits given spiropidion once daily by gavage at dose levels of 0, 10, 30 or 60 mg/kg bw per day from GD 6 to GD 27, the NOAEL for maternal toxicity was 10 mg/kg bw per day based on body weight stasis in the first six days of treatment and decreased body weight gain

thereafter at 30 mg/kg bw per day. The NOAEL for embryo/fetal toxicity was 30 mg/kg bw per day, based on increased incidence of skeletal effects (unossified pelvic girdle and incomplete xyphoid cartilage) at 60 mg/kg bw per day.

The Meeting concluded that spiropidion is not teratogenic.

In an acute neurotoxicity study, spiropidion was given to rats by gavage at a dose of 0, 50, 150 or 500 mg/kg bw. There were no histopathological changes in the nervous system. The NOAEL for neurotoxicity was 150 mg/kg bw based on clinical signs (convulsions) in females at 500 mg/kg bw. The NOAEL for systemic toxicity was 150 mg/kg bw based on body weight loss in males at 500 mg/kg bw.

Although there were no indications of neuropathological effects due to spiropidion, the Meeting concluded that spiropidion may cause, acute neurobehavioral effects at high doses.

No evidence of immunotoxicity was reported in routine toxicological studies with spiropidion.

The Meeting concluded that spiropidion is unlikely to be immunotoxic.

An ex vivo and an in vitro study were conducted to assess treatment-related thyroid effects observed in the short term and reproductive toxicity studies in rats. Results indicated that spiropidion is an inducer of hepatic UDP-glucuronosyltransferase (UGT) activity in both male and female rats, and that it does not inhibit rat thyroid peroxidase activity in vitro.

Microbiological data

No information on the potential effects of spiropidion on the microbiome of the human gastrointestinal tract was available.

Toxicological data on metabolites and/or degradates

The Meeting noted that the ethyl carbonate group of spiropidion (SYN546330) is rapidly removed by hydrolysis to leave the metabolite spiropidion-enol (SYN547305). It is very likely this metabolite that is responsible for spiropidion's toxicity, as substantial exposure to spiropidion-enol was demonstrated in several toxicity studies.

Spiropidion-enol (SYN547305) free and conjugate

Spiropidion-enol is a metabolite found in the rat (< 1% in urine), in plants (tomato, potato, cotton), and other animals (goat, hen, eggs).

No specific toxicological data are available. However, toxicity studies with the parent include exposure to SYN547305. The parent molecule, spiropidion, is a pro-insecticide which is hydrolysed to SYN547305; it is very likely that SYN547305 is responsible for the insecticidal effect. SYN547305 is the precursor of all the other metabolites identified in rats. It was found in plasma in the rat biotransformation study and exposure was demonstrated in several toxicological studies. The Meeting concluded that the toxicity of spiropidion-enol (SYN547305) is covered by that of the parent compound.

Spiropidion-enol (SYN547305) dehydrogenated

SYN547305 is a metabolite found in plants (tomato).

No specific toxicological data are available. Based on structural considerations the Meeting concluded that it is unlikely to be genotoxic. For chronic toxicity, the TTC approach can be applied using Cramer class III, 1.5 µg/kg bw per day.

Metabolite SYN548430

SYN548430 is a major metabolite in the rat (>30% in urine), in plants (tomato, potato, cotton) and animals (goat, hen, milk, eggs).

As SYN548430 is a major rat metabolite, the Meeting concluded that it would be covered by toxicological studies on the parent.

Metabolite SYN547435

SYN547435 is a metabolite in the rat (<1% in urine) and in plants (tomato, potato).

No specific toxicological data are available. A chemical read-across approach examined the structural similarities of SYN547435, spiropidion (SYN546330), SYN547305 and SYN548430 via the appropriate chemistry parameters, and an OECD (Q)SAR Toolbox analysis (v. 4.3) for all toxicity endpoints of SYN547435. On the basis of this analysis, the Meeting concluded that metabolite SYN547435 would be covered by toxicological studies on the parent.

Metabolites SYN550820, SYN550839, SYN548033 and SYN549098 (free and conjugate)

SYN550820, SYN550839, SYN548033 and SYN549098 (free and conjugate) are metabolites of plants (cotton, gin trash or rotational crops).

No specific toxicological data are available. The Meeting noted that these metabolites are structurally different from the parent compound. According to the results of evaluations made using Derek Nexus (v.6.0.1), VEGA platform (CAESAR mutagenicity models, v.2.1.13) and the OECD (Q)SAR Toolbox (v.4.3), no alert for genotoxicity was identified. For chronic toxicity, the TTC approach can be applied using Cramer class III, 1.5 µg/kg bw per day.

Metabolite SYN548939

SYN548939 is a metabolite in animals (goat, poultry).

No specific toxicological data are available. The Meeting noted that metabolite is structurally similar to the parent compound and is of no genotoxicological concern based on the results of Derek Nexus (v.6.0.1), VEGA platform (CAESAR mutagenicity models, v.2.1.13) and OECD (Q)SAR Toolbox (v.4.3) evaluations.

The Meeting concluded that metabolite SYN548939 is not of greater toxicity than spiropidion and would be covered by toxicological studies on the parent.

Human data

In reports on manufacturing plant personnel, no adverse health effects were noted. No information on accidental or intentional poisoning in humans is available.

The Meeting concluded that the existing database on spiropidion was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting established an ADI of 0–0.02 mg/kg bw, based on a NOAEL of 2.4 mg/kg bw per day for an equivocal increase in testicular interstitial cell adenomas at 100 ppm (equal to 4.7 mg/kg bw per day) in the two-year rat study, and using a safety factor of 100.

The Meeting established an ARfD of 0.3 mg/kg bw based on a NOAEL of 30 mg/kg bw for mortality and clinical signs in a 28-day dog study, using a safety factor of 100. This NOAEL is supported by the NOAEL of 30 mg/kg bw for an initial body weight loss (GDs 6–9) in dams in the rat developmental study.

A toxicological monograph was prepared.

Levels relevant to risk assessment of spiropidion

Species	Study	Effect	NOAEL	LOAEL
Mouse	80-week carcinogenicity ^a	Toxicity	50 ppm, equal to 6.4 mg/kg bw per day	250 ppm, equal to 31.8 mg/kg bw per day
		Carcinogenicity	500 ppm, equal to 65.4 mg/kg bw per day ^b	-
Rat	Two-year toxicity and carcinogenicity ^a	Toxicity	100 ppm, equal to 4.7 mg/kg bw per day	500 ppm, equal to 24.0 mg/kg bw per day
		Carcinogenicity	50 ppm, equal to 2.4 mg/kg bw per day	100 ppm, equal to 4.7 mg/kg bw per day
	Two-generation reproductive toxicity ^a	Reproduction/fertility	300 ppm, equal to 23 mg/kg bw per day ^b	-
		Parental toxicity	100 ppm, equal to 7.8 mg/kg bw per day	300 ppm, equal to 23 mg/kg bw per day
		Offspring toxicity	300 ppm, equal to 23 mg/kg bw per day ^b	-
	Developmental toxicity study ^c	Maternal toxicity	30 mg/kg bw per day	100 mg/kg bw per day
		Fetal toxicity	100 mg/kg bw per day ^b	-
	Acute oral neurotoxicity study ^c	Neurotoxicity	150 mg/kg bw	500 mg/kg bw
Rabbit	Developmental toxicity study ^c	Maternal toxicity	10 mg/kg bw per day	30 mg/kg bw per day
		Fetal toxicity	30 mg/kg bw per day	60 mg/kg bw per day
Dog	28 day toxicity ^d	Toxicity	30 mg/kg bw per day	65/100 mg/kg bw per day
	13-week and one-year studies of toxicity ^{d, e}	Toxicity	15 mg/kg bw per day	30 mg/kg bw per day

^a Dietary administration^b Highest dose tested^c Gavage administration^d Gelatine capsule administration^e Two studies combined

Acceptable daily intake (ADI) for spiropidion, spiropidion-enol (SYN547305 free and conjugate), SYN548430, SYN547435 and SYN548939, expressed as spiropidion:

0–0.02 mg/kg bw

Acute reference dose (ARfD) for spiropidion, spiropidion-enol (SYN547305 free and conjugate), SYN548430, SYN547435 and SYN548939, expressed as spiropidion:

0.3 mg/kg bw

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Critical end-points for setting guidance values for exposure to spiropidion**Absorption, distribution, excretion, and metabolism in mammals**

Rate and extent of oral absorption	Oral absorption of 72–77% in male and female rats, based on urinary, biliary excretion, cage wash and carcass (excluding gastrointestinal content) within 72 hours The majority of a 5 mg/kg bw oral dose was systemically available, based on the urinary excretion ratio following oral and intravenous administration
Distribution	Widely distributed, with highest concentrations of radioactivity observed in the GI tract, liver and kidney
Rate and extent of excretion	>94 % of the administered dose was excreted within 48 h with excretion complete by 168 h; the major route of excretion was via urine (65%), followed by faeces (~15%) and bile (~10%) and <0.7% of dose recovered in the carcass
Potential for accumulation	No evidence of accumulation
Metabolism in mammals	SYN546330 was extensively metabolized in rat via ester hydrolysis to form SYN547305 and subsequent loss of the methoxy moiety in the piperidine ring to form SYN548430. SYN548430 and an oxidised moiety of SYN548430, individually accounted for >10% of the administered dose in excreta. The major components (>10% of total radioactivity AUC) identified in plasma were SYN547305 and SYN548430

Acute toxicity

Rat LD ₅₀ oral	>2000 mg/kg bw
Rat LD ₅₀ dermal	>5000 mg/kg bw
Rat LC ₅₀ inhalation (nose-only exposure)	>1.12 mg/L
Rabbit, skin irritation	Non-irritating
Rabbit, eye irritation	Mildly irritating
Mouse LLNA, skin sensitization	Sensitizing (EC ₃ value: 0.13%)
In vitro 3T3 NRU phototoxicity test	Non-phototoxic

Short-term studies of toxicity

Target/critical effect	Reduced cholesterol and triglycerides
Lowest relevant oral NOAEL	6.2 mg/kg bw per day (rat)
Lowest relevant dermal NOAEL	300 mg/kg bw per day (rat)

Long-term studies of toxicity and carcinogenicity

Target/critical effect	Body weight (mouse), equivocal increase in Leydig cell adenomas and in malignant ependymomas (rat)
Lowest relevant oral NOAEL	2.4 mg/kg bw per day (male rats)
Carcinogenicity	Not carcinogenic in mice; equivocal increase in Leydig cell adenomas and in malignant ependymomas (rat) ^a

GenotoxicityUnlikely to be genotoxic^a**Reproductive toxicity**

Target/critical effect	Parental: thyroid follicular hypertrophy
Lowest relevant parental NOAEL	7.8 mg/kg bw per day (rat)
Lowest relevant offspring NOAEL	23 mg/kg bw per day, highest dose tested (rat)

Lowest relevant reproductive NOAEL	23 mg/kg bw per day, highest dose tested (rat)
Developmental toxicity	
Target/critical effect	Parental: decreased body weight gain (rabbit); body weight loss and food consumption (rat)
Lowest relevant maternal NOAEL	10 mg/kg bw per day (rabbit)
Lowest relevant developmental NOAEL	30 mg/kg bw per day (rabbit)
Neurotoxicity	
Acute neurotoxicity NOAEL	150 mg/kg bw (rat)
Subchronic neurotoxicity NOAEL	Not available
Developmental neurotoxicity NOAEL	Not available
Immunotoxicity	
Mechanism studies	
UDP-glucuronosyltransferase induction, rat liver samples, diet: SYN546330 is an inducer of hepatic UGT activity Rat thyroid peroxidase activity, microsomal preparation: SYN546330 and SYN547305 are not inhibitors of rat thyroid peroxidase activity in vitro	
Studies on toxicologically relevant metabolites or impurities	
SYN547305	See rat thyroid peroxidase activity above
SYN550820, SYN550839, SYN548033, SYN548939, SYN549098 (free and conjugate)	Unlikely to be genotoxic (QSAR)
SYN547435	Structurally similar to spiropidion and rat metabolites for all toxicity end-points (read-across and QSAR)
Human data	Limited information as this is a new substance

^a Unlikely to pose a carcinogenic risk to humans via exposure from the diet

Summary

	Value	Study	Safety factor
ADI ^a	0–0.02 mg/kg bw	Two-year study (rat)	100
ARfD ^a	0.3 mg/kg bw	28-day study (dog)	100

^a Applies to spiropidion, spiropidion-enol (SYN547305 free and conjugate), SYN548430, SYN547435 and SYN548939, expressed as spiropidion

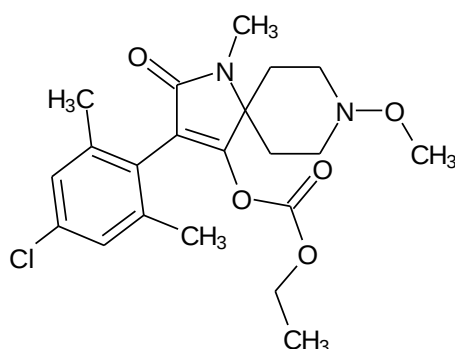
RESIDUE AND ANALYTICAL ASPECTS

Spiropidion is a systemic keto-enol acid insecticide for the control of sucking insects. It acts by inhibiting acetyl-CoA carboxylase and thus interfering with lipid metabolism and inhibiting juvenile growth and adult reproduction.

Spiropidion was scheduled at the Fifty-first Session of the CCPR for toxicology and residue evaluation as a new compound by the 2019 JMPR, which was postponed to the 2021 JMPR.

The Meeting received information on the identity, physical and chemical properties, metabolism of spiropidion in livestock, plants and rotational crops, environmental fate studies, methods of residue analysis, freezer storage stability, GAP information, supervised residue trials (fruiting vegetables, soya bean and potato), processing studies and a livestock feeding study (lactating cow).

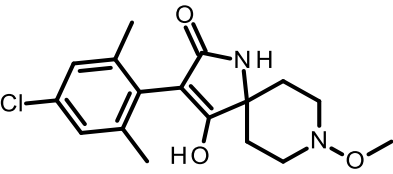
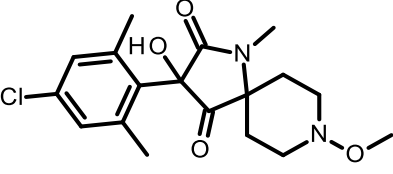
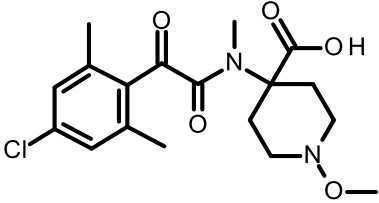
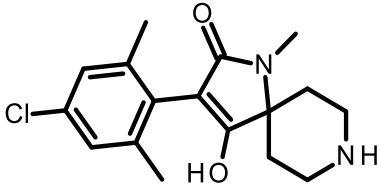
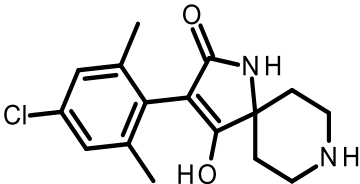
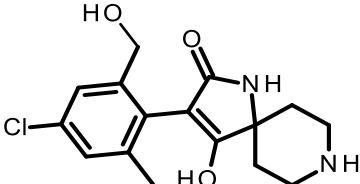
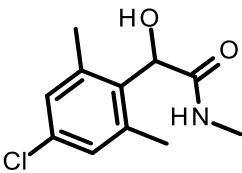
Spiropidion is 3-(4-chloro-2,6-dimethylphenyl)-8-methoxy-1-methyl-2-oxo-1,8-diazaspiro[4.5]dec-3-en-4-yl ethyl carbonate.



The following abbreviations are used for the major metabolites discussed below:

Table 1 Major spiropidion metabolites identified in plant and animal matrices

Compound code (other names)	Description	Chemical structure
Spiropidion SYN546330	[3-(4-chloro-2,6-dimethyl-phenyl)-8-methoxy-1-methyl-2-oxo-1,8-diazaspiro[4.5]dec-3-en-4-yl] ethyl carbonate MW 422.9 g/mol	
Spiropidion-enol SYN547305 (CSCD710959)	3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-8-methoxy-1-methyl-1,8-diazaspiro[4.5]dec-3-en-2-one MW 350.9 g/mol	

Compound code (other names)	Description	Chemical structure
SYN547435	3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-8-methoxy-1,8-diazaspiro[4.5]dec-3-en-2-one MW 336.8 g/mol	
SYN548033	3-(4-chloro-2,6-dimethyl-phenyl)-3-hydroxy-8-methoxy-1-methyl-1,8-diazaspiro[4.5]decane-2,4-dione MW 366.8 g/mol	
SYN548388	4-[[2-(4-chloro-2,6-dimethyl-phenyl)-2-oxo-acetyl]-methyl-amino]-1-methoxy-piperidine-4-carboxylic acid MW 382.8 g/mol	
SYN548430	3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-1-methyl-1,8-diazaspiro[4.5]dec-3-en-2-one MW 320.8 g/mol	
SYN548939	3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-1,8-diazaspiro[4.5]dec-3-en-2-one MW 306.8 g/mol	
SYN548939-OH U5 (Proposed structure)	3-[4-chloro-2-(hydroxymethyl)-6-methyl-phenyl]-4-hydroxy-1,8-diazaspiro[4.5]dec-3-en-2-one MW 322.8 g/mol	
SYN549098	2-(4-chloro-2,6-dimethyl-phenyl)-2-hydroxy-N-methyl-acetamide MW 227.7 g/mol	

Compound code (other names)	Description	Chemical structure
SYN549098-OH (Proposed structure)	2-[4-chloro-2-(hydroxymethyl)-6-methyl-phenyl]-2-hydroxy-N-methyl-acetamide MW 243.7 g/mol	
SYN549937	3-(4-chloro-2,6-dimethyl-phenyl)-3-hydroxy-1-methyl-1,8-diazaspiro[4.5]decane-2,4-dione MW 336.8 g/mol	
SYN550820	4-[[2-(4-chloro-2,6-dimethyl-phenyl)-2-hydroxy-acetyl]amino]-1-methoxy-piperidine-4-carboxylic acid MW 370.8 g/mol	
SYN550821	3-(4-chloro-2,6-dimethyl-phenyl)-1-methyl-4-oxa-1,9-diazaspiro[5.5]undecane-2,5-dione MW 336.8 g/mol	
SYN550839	4-[[2-(4-chloro-2,6-dimethyl-phenyl)-2-hydroxy-acetyl]-methyl-amino]-1-methoxy-piperidine-4-carboxylic acid MW 384.5 g/mol	

Physical and chemical properties

Spiropidion is not volatile ($< 5.0 \times 10^{-6}$ Pa at 25 °C), relatively insoluble in water (46 mg/L) but more soluble in organic solvents (up to 500 g/L in dichloromethane). It is rapidly hydrolysed, especially under alkaline conditions, with DT_{50} values ranging from 0.1 days (pH 9) to about 11 days (pH 4). Photolysis is not expected to be a significant route of degradation. The octanol/water partition coefficient $\log P_{ow}$ of 3.2 for spiropidion suggests a potential to partition into fat.

Plant metabolism

The Meeting received studies on the metabolism of spiropidion after foliar applications to tomato, potato and cotton, using spiropidion radiolabelled (^{14}C) in the phenyl ring or in the spiro-5 positions.

In all studies, in order to minimise potential spiropidion hydrolysis during sample preparation, frozen samples were homogenised with dry ice to a fine powder and stored frozen at $-20\text{ }^{\circ}\text{C}$ until extracted with aqueous acetonitrile and analysed within 12–15 days after extraction. Repeat analysis of representative extracts following 5–27 months frozen storage showed that the nature of the residue was stable throughout the analytical period.

Tomato

Tomato plants grown in a glasshouse in a sandy loam soil were treated three times with 273–302 g ai/ha [^{14}C]-label spiropidion, at growth stages BBCH 16, BBCH 61 and BBCH 85. Mature fruit samples taken 1 day (1DAT3) and 21 days after the third treatment (21DAT3) were surface-washed with acetonitrile.

TRRs in mature fruit were 0.013–0.017 mg eq/kg in the 1DAT3 fruit and 0.072–0.18 mg/kg in 21DAT3 fruit.

Extractability in aqueous acetonitrile was good, with unextracted radioactivity not exceeding 11% TRR in any sample. Surface washes from the fruit sampled one day after the third treatment (1DAT3) contained about 29–34% of the radioactivity and the 21DAT3 fruit surface washes contained 41–59% of the radioactivity. TRRs in the 21DAT3 foliage were 0.6 and 2.3 mg eq/kg for [phenyl- ^{14}C]- and [spiro-5- ^{14}C]- spiropidion treated experiments, respectively.

Parent spiropidion was the predominant residue in fruit and foliage, accounting for 28–32% TRR (0.037–0.055 mg/kg) in 1DAT3 fruit, 38–55% TRR in the 21DAT3 fruit and 35–44% TRR (0.21–1.0 mg/kg) in foliage.

Significant metabolites in fruit were spiropidion-enol (SYN547305) and SYN548430. The -enol metabolite accounted for about 9% TRR in the 1DAT3 fruit, increasing to 17–26% TRR (0.019–0.031 mg eq/kg) in the 21DAT3 fruit. Metabolite SYN548430 was found at up to 19% TRR (0.028 mg eq/kg) in the 1DAT3 fruit and up to 15% TRR (0.018 mg eq/kg) in the 21DAT3 fruit.

A minor component, identified as the equivalent of dehydrogenated spiropidion-enol made up about 10% TRR in fruit (≤ 0.01 mg eq/kg) and SYN547435 accounted for up to 7% of the fruit TRR. Unidentified components in fruit across all harvest points, accounted for 4.0–14% (0.003–0.018 mg eq/kg), none of which individually accounted for more than 2.8% TRR and 0.005 mg eq/kg.

Significant metabolites in foliage were SYN548430, accounting for 7.2–14% TRR (0.042–0.32 mg eq/kg) and spiropidion-enol, making up about 10% TRR (0.058–0.24 mg eq/kg).

The dehydrogenated spiropidion-enol component accounted for about 2.7% TRR (0.015–0.064 mg eq/kg) and metabolite SYN548033 accounted for 0.4–1.6% TRR (0.009–0.01 mg eq/kg) in foliage.

Other identified metabolites measured in foliage were SYN548338 (0.3–0.4% TRR) and unidentified components accounted for a total of 9.7–23% TRR (0.135–0.22 mg eq/kg), with no individual component present at more than 6.8% TRR and 0.043 mg eq/kg.

Potato

Potato plants, grown in a glasshouse in a sandy loam soil were treated three times with phenyl- or spiro-labelled spiropidion (SC) as a foliar spray at rates of 192–200 g ai/ha, applied at growth stages BBCH 16, BBCH 42 and BBCH 48. Total radioactive residues (TRRs) in immature (23DAT2) tubers were 0.03–0.031 mg eq/kg and 0.064–0.065 mg eq/kg in mature (7DAT3) tubers. TRRs in immature foliage were 2.4–2.5 mg eq/kg and 14–21 mg eq/kg in mature foliage (7DAT3).

Extractability in aqueous acetonitrile was good. Extracted radioactivity was greater than 80% TRR in all commodities, with unextracted residues being highest in mature tubers (15.7–20.3% TRR but 0.01–0.013 mg eq/kg).

Parent spiropidion was not found in tubers and was the primary residue in foliage, making up 79–83% TRR (1.97–2.0 mg/kg) in immature (23DAT2) foliage and 59–68% TRR (9.5–12.7 mg/kg) in mature (7DAT3) foliage.

Spiropidion-enol was the predominant residue in tubers, accounting for 27–33% TRR (up to 0.021 mg eq/kg) in mature tubers and was the only significant metabolite in foliage (11–18% TRR, 1.5–3.9 mg eq/kg in mature foliage).

Metabolite SYN547435 was a significant metabolite in tubers, accounting for 22–25% TRR (about 0.015 mg eq/kg in mature tubers) but was a minor residue in foliage (1.3% TRR, 0.033 mg eq/kg).

SYN548430 was identified as a minor metabolite, accounting for up to 7.8% TRR (0.005 mg eq/kg) in mature tubers and both SYN548430 and SYN548388 were each present in mature foliage at < 0.6% TRR (0.11 mg eq/kg).

SYN548033 was also a minor component in foliage (up to 1.9% TRR, 0.42 mg eq/kg) in mature foliage and tentatively characterised in tubers at up to 1.4% TRR (< 0.001 mg eq/kg).

In tubers, unidentified components accounted for less than 10% TRR (0.005 mg eq/kg) with no single residue greater than 4.3% TRR (0.002 mg eq/kg). In foliage the unidentified components accounted for up to 15% TRR (3.2 mg eq/kg) with no single component present at more than 2.5% TRR (0.54 mg eq/kg).

Cotton

Cotton plants grown in a glasshouse in a sandy loam soil were treated three times with 149–156 g ai/ha [¹⁴C]-label spiropidion at growth stages BBCH 16, BBCH 60 and BBCH 87 and samples of immature foliage (27DAT2) and mature foliage, seeds and gin trash (cotton burs, leaves, stems and immature seeds) (14DAT3) were collected. TRRs in immature foliage (27DAT2) were 0.61–1.7 mg eq/kg and were 2.7–3.4 mg eq/kg in the 14DAT3 gin trash. In mature seeds (14DAT3), TRRs were 0.033–0.035 mg eq/kg.

Extractability was good, with initial aqueous acetonitrile extractions recovering 88–97% TRR (immature foliage), 69–75% TRR (seeds) and 93% TRR (trash). More aggressive enzyme, alkaline and acidic extraction techniques were able to extract up to a further 10% TRR from the immature foliage and mature seeds PES. Unextracted radioactivity was 6.7% TRR in trash, did not exceed 12% TRR in foliage, but was higher (25–31% TRR) in seeds.

Parent spiropidion was not found above the LOQ of 0.001 mg/kg in cotton seed (0.3–0.6% TRR), but was the major residue in cotton foliage, making up 9.6–35% TRR (0.059–0.59 mg/kg) in immature foliage and 21–29% TRR (0.56–0.96 mg/kg) in mature gin trash.

Spiropidion-enol was detected in all samples, 18–23% TRR (0.006–0.008 mg eq/kg) in seed, 7.9–8.7% TRR (0.048–0.15 mg eq/kg) in immature foliage and 13–14% TRR (0.34–0.48 mg eq/kg) in gin trash.

SYN550820 was not found in cotton seed, but accounted for about 8–12% TRR in both immature foliage (0.072–0.13 mg eq/kg) and gin trash (0.29–0.32 mg eq/kg).

SYN550839 was also not found in cotton seed, but found in immature foliage (3.6–7.3% TRR) and gin trash (7.2–9.8% TRR).

SYN548430 was a minor metabolite in all samples, accounting for up to about 3% TRR in immature foliage (up to 0.05 mg eq/kg), gin trash (up to 0.53 mg eq/kg) and seed (0.001 mg eq/kg).

Conclusions

The metabolic pathways in tomato, potato and cotton are similar and the distribution of the characterised metabolites was similar across both label experiments, indicating that cleavage of the parent molecule does not occur.

The principal biotransformations involve hydrolysis of the spiropidion ethyl carbonate side chain to form the -enol metabolite (SYN547305) which can then be dehydrogenated or is demethylated to SYN547435; undergoes N-methoxide reduction to SYN548430; ring-opening to SYN550839 or hydroxylation to SYN548033. Subsequently, SYN548033 can undergo ring opening to form SYN548388 and SYN550820 can be formed by demethylation of SYN550839.

Metabolites SYN548033, SYN548388, SYN550820, SYN550839 and the dehydrogenated spiropidion-enol were not observed in laboratory animal metabolism studies.

Rotational crop metabolism

Confined rotational crop studies

In a confined rotational crop study with wheat, lettuce and turnip grown as rotational crops, bare sandy loam soil was treated once with [¹⁴C]-spiropidion (phenyl- and spiro-labels) at a rate equivalent to 0.59 kg ai/ha (approximating the maximum seasonal GAP rate) and crops were sown 30, 120 and 275 days after treatment.

Radioactive residues in rotational crops did not exceed 0.085 mg eq/kg in immature lettuce, 0.031 mg eq/kg in mature lettuce, 0.027 mg eq/kg in wheat grain or 0.01 mg eq/kg in turnip roots. Higher residues were present in feed commodities, up to 0.64 mg eq/kg in the 120-day PBI (plant back interval) wheat straw, up to 0.34 mg eq/kg in the 30-day PBI wheat hay, 0.21 mg eq/kg in the 30-day PBI wheat forage and 0.053 mg eq/kg in the 30-day PBI turnip foliage. In most cases, the phenyl-labelled TRRs were significantly higher than the spiro-labelled TRRs, particularly at the longer rotations.

Spiropidion metabolism in rotational crops was extensive with intact molecules being more abundant at earlier PBIs but declining at later PBIs with a corresponding increase in phenyl label specific (cleavage) products.

Spiropidion and its -enol hydrolysis product were not found in any samples. At the 30-day PBI, the principal metabolites making up more than 10% TRR were SYN548033, SYN550821 and SYN549937, these declining at later PBIs. The phenyl-specific SYN549098 (free, conjugated or hydroxylated) was also found above 10% TRR at all PBIs

In food commodities at all rotational intervals, radioactivity in turnip roots was < 0.01 mg eq/kg (and not investigated further) and no identified metabolites were found above 0.002 mg eq/kg in wheat grain. In lettuce, at the 30-day PBI, SYN548033 was the predominant residue (27–54% TRR), found at up to 0.026 mg eq/kg in immature lettuce but < 0.01 mg eq/kg in mature lettuce. In later lettuce rotations, SYN548033 was either not detected or < 0.01 mg eq/kg.

Total SYN549098 residues (free and conjugated) were significant components of the residue in lettuce from the later PBIs, making up 54–56% TRR, up to 0.048 mg eq/kg in the 120-day immature lettuce and 55% TRR (0.012 mg eq/kg) in mature lettuce.

While the SYN548939-OH enantiomers made up 12–20% TRR in immature lettuce from the later PBIs, residue levels were all < 0.01 mg eq/kg and were not more than 0.001 mg eq/kg in mature lettuce.

In animal feed commodities, SYN548033 was the predominant residue in 30-day rotation wheat forage (27% TRR, up to 0.057 mg eq/kg) but was either < 0.01 mg eq/kg or not detected in later rotations. In other feed commodities (turnip tops, wheat hay and straw), highest residues of SYN548033 were in the 30-day PBIs (< 0.02 mg eq/kg) except in wheat straw, where spiro-labelled residues in the 120-day PBI reached 0.05 mg eq/kg.

SYN549937 and SYN550821 were only present above 10% TRR in the 30-day rotation wheat forage, at levels of up to 0.037 mg eq/kg (18–21% TRR) and 0.013 mg eq/kg (5–17% TRR), respectively.

At later PBIs, the phenyl-specific SYN549098 (free and hydroxylated/conjugated) were the principal identified metabolites present at more than 0.01 mg eq/kg. Combined residues of free and conjugated SYN549098 accounted for maximum absolute residue levels of 0.016 mg eq/kg in turnip tops (275-day rotation). In wheat forage, hay and straw, total residues (i.e. the sum of free plus conjugated/hydroxylated SYN549098) accounted for maximum absolute residue levels of 0.038 mg eq/kg in wheat forage (275-day PBI), 0.11 mg eq/kg in wheat straw (120-day PBI) and 0.076 mg eq/kg in wheat hay (275-day PBI).

Unidentified components accounted for up to 28–45% TRR in the different crops from the three rotations, with no single component found above 0.024 mg eq/kg (wheat straw from the 120-day rotation).

Further characterisation of the unextracted residues in wheat hay and straw demonstrated incorporation into fractions; characteristic of lignin (15% TRR, < 0.1 mg eq/kg), hemicellulose (12% TRR, 0.054 mg eq/kg) and cellulose (6.7% TRR, 0.017 mg eq/kg).

Major metabolic pathways included cleavage of the ethyl carbonate side chain and hydroxylation to form SYN548033 or with N-methoxide reduction to form SYN548430. Oxidation of SYN548430 to SYN548939-OH or SYN549937. Expansion of the SYN549937 pyrrolidine ring to form SYN550821 and subsequent ring-opening to form SYN550838. Loss of the SYN550838 spiro-ring forms the phenyl specific metabolite SYN549098 and its -glucose and malonyl-glucose conjugates.

Conclusions

The Meeting concluded that in rotational crops, significant residues related to spiropidion are generally not anticipated above 0.01 mg/kg in food commodities (turnip roots, mature lettuce, wheat grain) with the highest residue being 0.012 mg eq/kg for free and conjugated SYN549098 in mature lettuce. In feed commodities (turnip tops, wheat forage, hay and straw), residues are generally not expected above 0.1 mg eq/kg, with the highest residues being 0.11 mg eq/kg for free and conjugated/hydroxylated SYN549098 in 120-day PBI wheat straw and 0.057 mg eq/kg for SYN548033 in wheat forage.

Animal metabolism

The Meeting received animal metabolism studies on rats, lactating goats and laying hens where animals were dosed with spiropidion radiolabelled in the phenyl-U ring or the spirodecanone-5 ring.

Rats

The metabolism of spiropidion in rats was reviewed in the framework of the toxicological evaluation by the WHO Core Assessment Group of the 2021 JMPR. In addition to parent, metabolites identified in the rat metabolism study included spiropidion-enol (SYN547305), SYN548430 and SYN547435.

Lactating goats

Two lactating goats were dosed orally with [¹⁴C]-spiropidion once a day (after morning milking) for eight days at dose rates equivalent to 37.3 ppm in the diet (phenyl-label) or 31 ppm in the diet (spiro-label). Milk was collected twice a day, urine and faeces were collected once a day and liver, kidney, muscle, fat and GI tract contents were collected 12 hours after the last dose,

The total recovery of the administered radioactivity was 92–95.5%, mostly from urine (51–59% AR) and faeces (16–29% AR). Only low levels (0.3–0.6% AR) were recovered in milk and tissue samples. Total radioactive residues were highest in kidney (up to 2.5 mg eq/kg) and liver (up to 1.6 mg eq/kg) and much lower in muscle (up to 0.082 mg eq/kg) and fat (up to 0.014 mg eq/kg). Residues in milk reached a plateau concentration of about 0.017 mg eq/kg after 3 days (spiro-label) and 0.033 mg eq/kg after 6 days (phenyl-label).

Extractability was good. Acetonitrile extractions were able to release 100% TRR in fat, more than 97% TRR in muscle and kidney, 91–94% TRR in milk (acetonitrile:hexane) and 89–92% TRR in liver.

Additional ^{14}C was released from liver PES using more aggressive extraction (protease and sodium decyl sulphate solubilisation) with a further 15.6% TRR in total released, made up of at least 13 polar components, each making up not more than 2.4% TRR (0.038 mg eq/kg).

Spiropidion was not detected in milk or any tissue, with the only identified metabolites being spiropidion-enol and its glucuronide conjugate, SYN548430 and SYN548939.

Spiropidion-enol was the predominant residue in muscle (47–57% TRR) and also a major component of the residue in kidney (37% TRR) and fat (30–53% TRR). In liver, residues were 16–37% TRR and its glucuronide conjugate made up a further 8–10% TRR. Total (free and conjugated) spiropidion-enol residues made up 27–45% TRR in liver, and in milk spiropidion-enol was a minor residue (1.9–7.3% TRR). Absolute residue levels were highest in kidney (up to 0.95 mg eq/kg) and liver (up to 0.45 mg eq/kg) and were lower in muscle (up to 0.044 mg eq/kg) and were < 0.01 mg eq/kg in fat and 0.001 mg eq/kg in milk.

SYN548430 was the major metabolite in milk (77–81% TRR), kidney (40–50% TRR) and liver (34–47% TRR) and was also a significant component of the residue in muscle (27–39% TRR) and fat (20–43% TRR but < 0.01 mg eq/kg). Absolute residue levels were highest in kidney (up to 1.3 mg eq/kg) and liver (up to 0.79 mg eq/kg) and were lower in muscle (up to 0.028 mg eq/kg) and milk (up to 0.027 mg eq/kg) and were < 0.01 mg eq/kg in fat.

SYN548939 was a minor residue, found only in liver (up to 10% TRR, 0.17 mg eq/kg) and kidney (up to 3.5% TRR, 0.089 mg eq/kg). Unidentified components did not exceed 18.5% TRR in fat, 6.7% TRR in other tissues or 3.7% TRR in milk, with no individual unidentified component present above 10% TRR or more than 0.035 mg eq/kg.

Laying hens

Laying hens were dosed orally with [^{14}C]-spiropidion in gelatin capsules once a day for 14 days at dose rates equivalent to 16.4 ppm in the diet (phenyl-label) and 14.5 ppm in the diet (spiro-label). Eggs were collected twice a day and separated into yolks and egg white, excreta was collected daily and liver, muscle and fat samples were collected 12 hours after the last dosing. The majority (91.5–94%) of the administered radioactivity was found in excreta (84–88% AR) and in the cage wash (6.2–7.7% AR). Less than 0.1% AR was recovered from any single tissue or egg sample. Total radioactive residues were highest in liver (up to 0.23 mg eq/kg) and much lower in eggs (< 0.04 mg eq/kg) and < 0.02 mg eq/kg in other tissues. After 6–7 days, the mean radioactive residues in eggs reached plateau concentrations of 0.032–0.036 mg eq/kg (0.033–0.038 mg eq/kg in yolks and 0.025–0.031 mg eq/kg in egg whites).

Extractability was good. Acetonitrile extractions were able to release 98–100% TRR in egg white, 86% TRR in muscle, 70–72% TRR in skin & fat and 64–67% TRR in liver.

Additional ^{14}C was released using more aggressive extraction of the liver PES (protease hydrolysis and sodium dodecyl sulphate solubilisation) with a total of up to 49% TRR released, made up to at least 10 polar components, each accounting for not more than 5.6% TRR (0.011 mg eq/kg).

Spiropidion was extensively metabolised in eggs and hen tissues, and not detected in any samples. The predominant metabolites are spiropidion-enol (SYN547305), SYN548430 and SYN548939 (only in liver).

Spiropidion-enol (SYN547305) was the predominant residue in egg white (65–73% TRR), muscle (39–42% TRR), skin plus fat (25–34% TRR) and liver (22–23% TRR) and was present in egg yolk at up to 10% TRR. Absolute residue levels were highest in liver (0.056 mg eq/kg) and egg white (0.02 mg eq/kg) but were 0.003–0.007 mg eq/kg in muscle, skin and fat, and egg yolk. Calculated maximum residues in whole eggs (70:30 white:yolk) were 0.015 mg/kg.

SYN548430 was the predominant residue in egg yolk (14–19% TRR) and a major component of the residue in muscle (19–21% TRR), skin plus fat (13–14% TRR), liver (8–13% TRR) and egg white (5.9–9.6% TRR). Highest absolute levels were in liver (0.032 mg eq/kg) with lower levels (0.001–0.007 mg eq/kg) in eggs and other tissues. Calculated maximum residues in whole eggs were 0.004 mg/kg.

SYN548939, a minor metabolite, was only detected in liver, at up to 6.5% TRR (0.016 mg eq/kg) and unidentified components did not exceed 35% TRR in skin and fat, 23.5% TRR in egg yolk, 17–19% TRR in egg white and liver and 7.2% TRR in muscle, with no individual unidentified component present above 13% TRR or more than 0.013 mg eq/kg.

Conclusions

Spiropidion was mostly excreted and extensively metabolised in goats and hens, with not more than 0.6% AR remaining in eggs, milk and tissues. Parent compound was not found in any of the livestock commodities, and none of the characterised metabolites indicated cleavage of the parent structure.

The metabolic pathways in poultry and ruminants (hen and goat) were similar, primarily involving rapid hydrolysis to spiropidion-enol (SYN547305) followed by N-demethoxylation to produce SYN548430, this being subsequently demethylated to form SYN548939 (not observed in the laboratory animal metabolism). This pathway (hydrolysis, N-demethoxylation and N-demethylation) is consistent with that observed in the spiropidion rat metabolism study.

Environmental fate

The Meeting received information on the hydrolytic stability, photochemical and aerobic soil degradation of spiropidion.

Hydrolysis

Spiropidion was prone to hydrolysis at all pHs, more rapidly at higher pHs and temperatures. Calculated hydrolytic degradation half-lives (SFO kinetics model) ranged from 1 hour (pH 9 and 35 °C) to 16.5 days (pH 7 and 15 °C). The major degradate was spiropidion-enol, with levels still increasing at the end of the 30-day study period and unidentified components did not exceed 4.5% TRR.

Photochemical degradation

In a soil photolysis study, spiropidion was steadily degraded in irradiated dry soil (64–67.5% AR remaining at the end of the 15-day study period), with degradation in moist soils being more rapid, with less than 0.5% AR remaining after 10 days.

The major degradate was spiropidion-enol, making up 22–34% AR in dry soils after 15 days irradiation and accounting for up to 57% AR in moist soils after 1 day. Other degradates present at up to about 10% AR (in moist soils) were SYN552045 and SYN548037, with other identified or characterised degradates not present above 8% AR in any sample.

Spiropidion degradation half-lives (SFO) in dry soil were about 30 days and were much shorter (0.6 days) in moist soils. Degradation rates in irradiated and dark control samples were similar, and the Meeting concluded that photolysis is not a major degradation route.

Aerobic soil metabolism

Spiropidion degraded quickly in soil under aerobic conditions, with half-lives in five soils ranging from 0.5–1.3 days. The predominant degradation product was spiropidion-enol, reaching up to 20–65% AR after 1–3 days and subsequently decreasing to 5–24% AR after 7–10 days and < 3% AR after 120 days.

Other degradates present at more than 5% AR at one or more sampling points were SYN552257 (up to 14% AR), SYN550839 (up to 6.6% AR), SYN548430 (up to 5.6% AR), SYN548033 (up to 5.3% AR) and

the phenyl-specific degradates SYN549098 (up to 7.4% AR) and SYN549979 (up to 5.5% AR). No individual unidentified component accounted for more than 5.9% AR.

Conclusions

Spiropidion is readily hydrolysed in aqueous solutions, more rapidly at higher temperatures and increasing pHs. The calculated half-life at pH 7 and 15 °C was 16.5 days. Soil degradation under aerobic conditions is also rapid, with spiropidion half-lives of 0.5–1.3 days. Photolysis is not considered to be a major degradation route.

Methods of analysis

The meeting received analytical method descriptions and validation data for spiropidion, spiropidion-enol, SYN548430 and SYN547435 in commodities of plant origin and of spiropidion, spiropidion-enol, SYN548430 and SYN548939 in livestock commodities. An LOQ of 0.01 mg/kg was validated for all metabolites in all matrices.

For plant commodities, validated methods for the analysis of spiropidion, spiropidion-enol, SYN548435, SYN550820 and SYN550839 involve dry ice or cryogenic milling, homogenisation in acetonitrile:water and after one or two centrifugation steps, analysis by LC-MS/MS. For SYN548430 and for SYN550820 and SYN550839 in dry commodities, additional steps included suspension in formic acid or acetic acid (to achieve a pH of 3–4) and SPE clean-up

The QuEChERS extraction method and LC-MS/MS analysis was validated for the measurement of spiropidion, spiropidion-enol, SYN547435 and SYN548430 in representative plant commodities except for SYN548430 in soya bean extracts, where recovery rates were not acceptable.

For animal commodities, a QuEChERS extraction method and LC-MS/MS analysis was validated for the measurement of spiropidion, spiropidion-enol, SYN548430 and SYN548939 in milk, eggs and bovine liver, muscle and fat.

A method was also validated for the analysis of spiropidion-enol, SYN548430 and SYN548939 in animal commodities, where samples were generally homogenised in acetonitrile:water and after one or two centrifugation steps, aliquots of the supernatant were adjusted to pH 3–4 with formic acid and analysed by LC-MS/MS.

For analysis of free and conjugated spiropidion-enol in liver, the validated method involves homogenisation in acetonitrile:water, alkaline (NaOH) incubation (2 hours at 60 °C), and adjustment to pH 3–4 prior to LC-MS/MS analysis.

Residues in cool-stored (4 °C) final extracts were generally stable for at least 5–14 days in plant commodity extracts and at least 37 days in animal commodity samples. Exceptions were SYN548430 (citrus after 5 days), SYN550820 (dry beans after 13 days) and SYN550839 (tomato, potato, wheat grain and dry beans after 8–13 days), where recovery rates were outside the acceptable (70–120%) range.

Conclusions

The Meeting concluded that the methods used in the supervised residue trials were validated with an LOQ of 0.01 mg/kg for all analytes, and are suitable to measure residues of spiropidion and its metabolites in plant and animal commodities provided sample preparation and analysis is completed within 5–14 days. The Meeting also noted that the extract storage periods in the trials were within the acceptable storage stability intervals.

Stability of pesticide residues in stored analytical samples

The Meeting received information on the stability of residues of spiropidion, spiropidion-enol, SYN547435, SYN548430, SYN550820 and SYN550839 in a range of representative plant commodities with high water content, high oil content, high acid content, high starch content, high protein content and low moisture

content. Studies were also received on the stability of spiropidion, spiropidion-enol, SYN548430 and SYN548939 in milk, eggs and bovine tissues.

Spiropidion is not stable in frozen stored crop commodities, degrading to spiropidion-enol. However, the sum of spiropidion plus spiropidion-enol (expressed as parent) residues and individual residues of spiropidion-enol are stable in a range of representative commodities with high water content (apple), high oil content (cotton seed), high acid content (orange), high starch content (wheat/barley grain, potato), high protein content (dried beans) and low moisture content (wheat/barley straws) when stored at or below -18 °C for at least 24 months (8 months for cotton seed).

In frozen homogenised soya bean samples, residues of spiropidion-enol are not stable, and while spiropidion residues are also not stable in frozen whole soya beans, the combined residues of spiropidion plus -enol in whole soya beans are stable in frozen storage for at least 15 months.

Residues of metabolites SYN547435 and SYN548430 in the above commodities and in homogenised soya beans are stable for at least 9 months (15 months in whole soya beans) in samples stored at or below -18 °C.

Residues of SYN550820 and SYN550839 in the above commodities are stable for at least 6 months (up to 3 months in oranges, 15 months in whole soya beans) in samples stored at or below -18 °C.

In livestock commodities, spiropidion is also not stable in frozen samples, but the combined residues of spiropidion plus spiropidion-enol (expressed as parent), as well as individual residues of the -enol, SYN548430 and SYN548939 are stable for at least 1 month in milk, eggs and bovine muscle, fat, liver and kidney when stored at or below -18 °C.

Conclusions

The Meeting agreed that the combined residues of spiropidion and its -enol metabolite (SYN547305) were stable for at least 8 months in cotton seed (high oil content), 15 months in whole soya bean and at least 24 months in other representative plant commodities. SYN547435, SYN548430, SYN550820 and SYN550839 were also stable for at least 6 months in most representative commodities (3 months in orange – high acid content). In animal commodities, the combined residues of spiropidion and its -enol metabolite were stable for at least 1 month and metabolites SYN548430 and SYN548939 were also stable for at least 1 month.

These storage intervals cover the residue sample storage intervals used in the field trials considered by the current Meeting.

Definition of the residue

Plant commodities

The metabolism of spiropidion was similar in the three plant metabolism studies provided on crops representative of root and tuber vegetables (potatoes), fruiting vegetables (tomato) and pulses/oilseeds (cotton).

Spiropidion was the major residue in tomato fruit (28–55% TRR) and in foliage of potato (59–83% TRR), tomato (35–44% TRR) and cotton (9.6–35% TRR) but was not a significant residue in cotton seed (0.3–0.6% TRR) and not detected in potato tubers.

Spiropidion-enol (SYN547305) was the major residue in potato tubers (27–36% TRR) and cotton seed (18–23% TRR) and a major residue in tomato fruit (8.5–26% TRR) and tomato, potato and cotton foliage and cotton gin trash (9.9–18% TRR). In addition, hydrolysis studies simulating industrial processing conditions indicated significant hydrolysis of spiropidion to spiropidion-enol.

Metabolite SYN547435 was a significant residue in potato tubers (22–25% TRR) and a minor component of the TRR in tomato fruit and potato foliage.

Metabolite SYN548430 was a significant residue in both tomato fruit (10–19% TRR) and foliage (7.2–14% TRR) but a minor residue in potato tubers (< 8% TRR) and cotton matrices ($\leq$ 3.3% TRR).

The dehydrogenated spiropidion-enol metabolite made up about 10% TRR in fruit ($\leq$ 0.01 mg/kg) and metabolites SYN550839 and SYN550820 accounted for up to 10–12% TRR respectively in immature cotton foliage and gin trash.

In rotational crops, residues of spiropidion and metabolites in food commodities (mature lettuce, turnip, wheat grain) were all < 0.01 mg eq/kg except in mature lettuce (120-day PBI) where total SYN549098 residues (free and conjugated) made up 55% TRR (0.012 mg/kg).

For MRL-compliance, spiropidion and spiropidion-enol are the most significant residues in all commodities investigated and validated analytical methods are available for their determination. The Meeting noted that since spiropidion was readily hydrolysed to the -enol in treated crops and also in stored frozen analytical samples, a suitable residue definition for MRL-compliance would be the sum of spiropidion and spiropidion-enol, expressed as spiropidion.

For risk assessment, in deciding which metabolites should be included in the residue definition for plant commodities, the Meeting considered the likely occurrence and toxicological relevance of the compounds present at more than 10% of total identified residues in the metabolism studies and more than 0.01 mg eq/kg. Compounds considered were spiropidion-enol, SYN547435, SYN548430, SYN550820, SYN 550839 and dehydrogenated spiropidion-enol.

Spiropidion-enol was a major residue in most plant matrices both in plant metabolism and supervised field trial studies. It is readily formed by the hydrolysis of spiropidion during industrial processing and can also be formed in stored samples. It was also found in the rat metabolism study and is covered by the HBGVs for spiropidion. Validated analytical methods are available for its determination and the Meeting considered that spiropidion-enol should be included in the residue definition for risk assessment.

In the metabolism studies, SYN547435 was a significant residue in potato tubers (up to 25% TRR, 0.016 mg eq/kg) and a minor component (up to 7% TRR, 0.012 mg eq/kg) of the residue in tomato fruit. It was also found at levels above the LOQ (0.01 mg/kg) in a number of the processing studies and supervised residue trials on fruiting vegetables, potatoes and soya beans. It was also found in the rat metabolism study and is covered by the HBGVs for spiropidion. Validated analytical methods are available for its determination and the Meeting considered that SYN547435 should be included in the residue definition for risk assessment.

Metabolite SYN548430 was a significant residue in tomato fruit (10–19% TRR, up to 0.028 mg eq/kg) and was also found at levels above the LOQ in a number of the processing studies and in field trials on fruiting vegetables, potatoes and soya beans. It was also found in the rat metabolism study and is covered by the HBGVs for spiropidion. Validated analytical methods are available for its determination and the Meeting considered that SYN548430 should be included in the residue definition for risk assessment.

SYN550820, SYN550839, SYN549098 and the dehydrogenated spiropidion-enol were not found in the rat metabolism study.

Metabolites SYN550820 and SYN550839 were only found in immature cotton foliage and gin trash (up to 12% TRR) in the metabolism studies. Residues were < 0.01 mg/kg in all of the processing studies and supervised residue trials.

The dehydrogenated spiropidion-enol was a minor residue in tomato fruit (up to 0.01 mg eq/kg) and in rotational crops. The phenyl-specific SYN549098 (free and conjugated) accounted for 26–55% TRR (up to 0.012 mg eq/kg) in mature lettuce from the later crop rotations.

The Meeting concluded that metabolites SYN550820, SYN550839, SYN549098 and the dehydrogenated spiropidion-enol can be assessed against the TTC for a Cramer Class III compound and should not be included in the residue definition for risk assessment.

The Meeting agreed that for plant commodities, the residue definition for risk assessment should be the sum of spiropidion, spiropidion-enol, SYN547435 and SYN548430, expressed as spiropidion.

Animal commodities

The metabolic pathways in poultry and ruminants (hen and goat) were similar, primarily involving hydrolysis to spiropidion-enol followed by N-demethoxylation to produce SYN548430 and subsequent demethylation to form SYN548939. This pathway is generally consistent with that observed in the spiropidion rat metabolism study although SYN548939 was not found in the rat metabolism study.

Spiropidion was not found in any livestock commodities, with the predominant metabolites in milk, egg and tissues being spiropidion-enol and SYN548430.

Spiropidion-enol was the major residue in muscle (up to 57% TRR; 0.044 mg eq/kg), eggs (calculated as up to 47% TRR; 0.015 mg eq/kg), kidney (37% TRR; up to 0.95 mg eq/kg), fat (up to 53% TRR; 0.007 mg eq/kg) and poultry liver (up to 23% TRR; 0.056 mg eq/kg). It also accounted for up to 45% TRR (0.45 mg eq/kg) in goat liver (including 8–10% TRR as the glucuronide conjugate released by alkaline incubation) and was a minor residue in milk (up to 7.3% TRR; 0.001 mg eq/kg).

Metabolite SYN548430 was the major residue in milk (up to 81% TRR; 0.027 mg eq/kg) and a significant residue in eggs (up to 12% TRR; 0.004 mg eq/kg), poultry tissues (8–21% TRR; up to 0.32 mg eq/kg in liver) and all goat tissues (20–50% TRR; up to 1.3 mg eq/kg in kidney).

Metabolite SYN548939 was a minor residue in liver and kidney, up to 10% TRR (0.17 mg eq/kg) in goat liver, up to 6.5% TRR (0.016 mg eq/kg) in poultry liver and up to 3.5% TRR; 0.09 mg eq/kg) in kidney.

For MRL-compliance, since spiropidion residues are rapidly hydrolysed in animal commodities and are also unstable in frozen stored analytical samples, the Meeting considered that spiropidion was not a suitable marker for compliance.

Spiropidion-enol was the major or a significant residue in the animal metabolism studies and the cattle feeding study, present at concentrations above 0.01 mg/kg in all commodities except milk and poultry muscle and skin+fat.

For poultry commodities, since TRRs in poultry muscle and skin+fat, from hens dosed with 14.5–16.4 ppm spiropidion were < 0.02 mg/kg, the Meeting concluded that no spiropidion-related residues are expected in muscle, skin or fat from poultry exposed to residues at the maximum dietary burden.

For milk, although the goat metabolism study would indicate that SYN548430 might be a suitable marker residue, no spiropidion-related residues were found in milk from any dose group in the cattle feeding study (up to 31 ppm) and the Meeting concluded that no spiropidion-related residues are expected in milk from cows exposed to residues at the maximum dietary burden.

Validated analytical methods are available for measuring spiropidion-enol in animal commodities and the Meeting considered that a suitable residue definition for MRL-compliance would be spiropidion-enol (expressed as spiropidion).

In deciding whether the residue for MRL-compliance should be regarded as fat-soluble, the Meeting noted that in the animal metabolism studies and the dairy cow feeding study, residues of spiropidion-enol did not appear to preferentially partition into fat or egg yolk and the Meeting considered that the residue is not fat-soluble.

For risk assessment, in deciding which metabolites in addition to spiropidion-enol (free) should be included in the residue definition the Meeting considered the likely occurrence and toxicological relevance of the compounds present at more than 10% of total identified residues in the metabolism

studies and more than 0.01 mg eq/kg. Compounds considered were the glucuronide conjugate of spiropidion-enol, SYN548430 and SYN548939.

In the goat metabolism study, residues of the glucuronide conjugate of spiropidion-enol were found in liver at 7.9–10% TRR (up to 0.17 mg eq/kg) and in the cattle feeding study, residues ranged from 0.03 mg/kg (3.1 ppm dose group) to 0.35 mg/kg (31 ppm dose group).

Because of the potential cleavage to release the free spiropidion-enol, the Meeting considered that conjugated spiropidion-enol should be included in the residue definition for dietary risk assessment.

SYN548430 was a significant residue in the goat metabolism study (30–37 ppm dose rates), with highest residues of 1.3 mg eq/kg (kidney), 0.78 mg eq/kg (liver), 0.028 mg eq/kg (muscle), 0.027 mg eq/kg (milk) and residues were < 0.01 mg eq/kg in fat. In most cases, the maximum levels were close to or higher than those for spiropidion-enol.

However, in the cattle feeding study, SYN548430 residues were all < 0.01 mg/kg except in liver and kidney from the higher dose groups. Residues in milk and tissues from the low dose group (approximating the calculated livestock dietary burden) were < 0.01 mg/kg.

In the poultry metabolism study (14.5–16 ppm dose rates), residues of SYN548430 were < 0.01 mg eq/kg except in liver (0.032 mg eq/kg). In most cases, the maximum levels were about half those of spiropidion-enol.

The HBGVs for spiropidion also cover SYN548430. Although it was a significant residue in the livestock metabolism studies, SYN548430 was only found in liver and kidney from the higher dose groups in the cattle feeding study (and then only at levels less than 10% of the spiropidion-enol concentrations) and in the poultry metabolism study, residues were only found above 0.01 mg eq/kg in liver. The Meeting considered that SYN548430 need not be included in the residue definition for risk assessment.

Metabolite SYN548939 was only found in liver (up to 0.17 mg eq/kg) and kidney (up to 0.09 mg eq/kg) in the goat metabolism study, with maximum levels being about 40% of the spiropidion-enol concentrations in liver and < 10% in kidney. However, SYN548939 was only detected at levels up to 0.03 mg/kg in cattle liver from the medium and high dose group animals in the dairy cow feeding study (i.e. < 10% of the spiropidion-enol concentrations).

In the poultry metabolism study (14.5–16 ppm dose rates), SYN548939 residues were only found in liver (up to 0.016 mg eq/kg), at levels < 30% of the spiropidion-enol concentrations and the Meeting concluded that SYN548939 residues are not expected in commodities from poultry exposed to residues at the anticipated maximum dietary burden.

Although SYN548939 was not found in the rat metabolism study, based on the structural similarity to the parent and using a read-across approach, the Meeting noted that the HBGVs for spiropidion also cover SYN548939. Since only low concentrations are expected in food commodities of animal origin, and then only in liver following medium-high exposure levels, the Meeting considered that SYN548939 need not be included in the residue definition for risk assessment.

The Meeting agreed that for animal commodities, a suitable residue definition for risk assessment is spiropidion-enol (free and conjugated), expressed as spiropidion.

Conclusions

Based on the above, The Meeting recommended the following residue definitions for spiropidion.

Definition of the residue for compliance with the MRL for plant commodities: *Sum of spiropidion and spiropidion-enol, expressed as spiropidion.*

Definition of the residue for dietary risk assessment for plant commodities: Sum of spiropidion, spiropidion-enol, SYN547435 and SYN548430, expressed as spiropidion.

Definition of the residue for compliance with the MRL for animal commodities: *Spiropidion-enol*, expressed as *spiropidion*.

Definition of the residue for dietary risk assessment for animal commodities: *Spiropidion-enol* (free and conjugated), expressed as *spiropidion*.

The residue is not fat-soluble.

Results of supervised residue trials on crops

Supervised trials were available for the use of spiropidion on cucurbits, other fruiting vegetables, soya beans and potatoes. Product labels were available from Guatemala and Paraguay.

For estimating maximum residue limits and for dietary risk assessment, the Meeting converted the relevant metabolite concentrations in each sample to parent equivalents (based on their molecular weight ratios) and added them to the spiropidion concentrations.

Sum of residues = spiropidion + (spiropidion-enol $\times$ 1.205) mg/kg (for maximum residue level estimation)

Total residues = spiropidion + (spiropidion-enol $\times$ 1.205) + (SYN547435 $\times$ 1.256) + (SYN548430 $\times$ 1.318) mg/kg (for dietary risk assessment)

While in the Evaluation, calculations have been made using the residue values as provided (generally to 3 decimal places), the values used for the estimation of maximum residue levels (underlined) have been rounded to two significant digits in this Appraisal.

For acute dietary exposure estimation, the highest individual total residue values from the trials have been used to derive the highest residues.

Fruiting vegetables, Cucurbits

Cucumber

The GAP for spiropidion on cucumbers in Guatemala is for up to 3 foliar applications of 180 g ai/ha from first appearance of pests, with a minimum retreatment interval of 7 days and a PHI of 1 day.

In nine independent trials on outdoor cucumbers, conducted in the USA and matching this GAP, the sum of residues (for maximum residue level estimation) were: 0.12, 0.16, 0.17, 0.20, 0.26, 0.30, 0.36, 0.36 and 0.44 mg/kg.

For risk assessment, total residues (for dietary exposure) were (n = 9): 0.16, 0.20, 0.23, 0.23, 0.34, 0.46, 0.54, 0.55 and 0.65 mg/kg and the highest individual value was 0.7 mg/kg.

The Meeting estimated a maximum residue level of 0.8 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and an STMR of 0.34 mg/kg and an HR of 0.7 mg/kg for total spiropidion residues in Cucumber.

Summer squash

While the Meeting received results of supervised residue trials conducted in the USA on summer squash, since the Guatemala label only refers to *Cucurbita maxima* (i.e. winter squash varieties), no GAP information was available to support a maximum residue level estimation for summer squash.

Melons

The GAP for spiropidion on melons in Guatemala is for up to 3 foliar applications of 180 g ai/ha from first appearance of pests, with a minimum retreatment interval of 7 days and a PHI of 1 day.

In nine independent trials on outdoor melons, conducted in USA and matching this GAP, the sum of residues (for maximum residue level estimation) were: 0.12, 0.14, 0.16, 0.18, 0.20, 0.29, 0.36, 0.46 and 0.52 mg/kg.

For risk assessment, total residues in these trials were (n = 9): 0.19, 0.21, 0.24, 0.24, 0.25, 0.40, 0.44, 0.71 and 0.90 mg/kg. The highest individual residue was 0.91 mg/kg.

The Meeting estimated a maximum residue level of 0.9 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and an STMR of 0.25 mg/kg and an HR of 0.91 mg/kg for total spiropidion residues in Melons (except Watermelon).

Pumpkins, Winter squash, Watermelon

The GAP for spiropidion on pumpkins, winter squash and watermelon in Guatemala is for up to 3 foliar applications of 180 g ai/ha from first appearance of pests, with a minimum retreatment interval of 7 days and a PHI of 1 day.

Noting that melons is a representative crop for the sub-group of melons, pumpkins and winter squash, the Meeting agreed to extrapolate the outdoor melon data (matching the above GAP for watermelon, winter squash and pumpkins) to estimate a maximum residue level for pumpkins, winter squash and watermelon.

The Meeting estimated a maximum residue level of 0.9 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and an STMR of 0.25 mg/kg and an HR of 0.91 mg/kg for total spiropidion residues in pumpkins, in winter squash and in watermelon.

Fruiting vegetables, other than Cucurbits

Tomatoes

The GAP for spiropidion on tomatoes in Guatemala is for up to 3 foliar applications of 180 g ai/ha from first appearance of pests, with a minimum re-treatment interval of 7 days and a PHI of 1 day.

In 16 independent trials on outdoor tomatoes, including cherry tomatoes, conducted in North America and matching this GAP, the sum of residues (for maximum residue level estimation) were: 0.044, 0.067, 0.075, 0.10, 0.13, 0.14, 0.16, 0.18, 0.21, 0.21, 0.22, 0.28, 0.34, 0.35, 0.35 and 0.5 mg/kg. The median residue, for livestock dietary burden estimation, was 0.195 mg/kg.

For risk assessment, total residues in these trials were (n = 16): 0.07, 0.093, 0.1, 0.13, 0.16, 0.17, 0.18, 0.24, 0.25, 0.25, 0.29, 0.39, 0.39, 0.40, 0.41 and 0.67 mg/kg and the highest individual value was 0.7 mg/kg.

The Meeting estimated a maximum residue level of 0.8 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and an STMR of 0.245 mg/kg and an HR of 0.7 mg/kg for total spiropidion residues in tomato.

Pepper and pepper-like commodities (sub-group)

The GAP for spiropidion on peppers in Guatemala is for up to 3 foliar applications of 180 g ai/ha from first appearance of pests, with a minimum retreatment interval of 7 days and a PHI of 1 day.

In nine independent trials on outdoor sweet peppers, conducted in USA and matching this GAP, the sum of residues (for maximum residue level estimation) were: 0.11, 0.13, 0.13, 0.29, 0.33, 0.33, 0.37, 0.38 and 0.49 mg/kg.

In ten independent trials on outdoor chili peppers, conducted in North America, matching the GAP in Guatemala for peppers, the sum of residues (for maximum residue level estimation) were: 0.073, 0.083, 0.14, 0.18, 0.26, 0.26, 0.32, 0.34, 0.49 and 0.69 mg/kg.

Noting that the median residues for sweet peppers and chili peppers were similar, considering that the datasets were similar (Mann-Whitney test) and that the GAP covered all commodities in the peppers sub-group except martynia, okra and roselle, the Meeting agreed to combine the residue datasets to estimate a group maximum residue level.

The combined data set matching the Guatemala GAP is (n = 19): 0.073, 0.083, 0.11, 0.13, 0.13, 0.14, 0.18, 0.26, 0.26, 0.29, 0.32, 0.33, 0.33, 0.34, 0.37, 0.38, 0.49, 0.49 and 0.69 mg/kg.

For risk assessment (fresh peppers), total residues in the combined sweet pepper and chili pepper trials were (n = 19): 0.11, 0.14, 0.16, 0.17, 0.17, 0.27, 0.34, 0.34, 0.41, 0.49, 0.53, 0.54, 0.57, 0.58, 0.65, 0.71, 0.98, 1.0 and 1.1 mg/kg and the highest individual value was 1.2 mg/kg.

For risk assessment (dried chili peppers), total residues in fresh chili peppers were (n = 10): 0.11, 0.14, 0.27, 0.34, 0.34, 0.49, 0.54, 0.58, 0.65 and 0.98 mg/kg and the highest individual value was 1 mg/kg.

The Meeting estimated a maximum residue level of 1 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and an STMR of 0.49 mg/kg and an HR of 1.2 mg/kg for total spiropidion residues in the sub-group of pepper and pepper-like commodities (except martynia, okra and roselle).

Using the default concentration factor of 7 for deriving residues in dried chili pepper from data on fresh chili peppers, the Meeting estimated a maximum residue level of 7 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and an STMR of 2.905 mg/kg and an HR of 7 mg/kg for total spiropidion residues in dried chili pepper.

Eggplants

While the Meeting received results of supervised residue trials conducted in the USA on eggplants, since the Guatemala label only refers to tomatoes and peppers, no GAP information was available to support a maximum residue level estimation for eggplants.

Pulses

Soya bean (dry)

The GAP for spiropidion on soya beans in Paraguay is for up to 3 foliar applications of 120 g ai/ha from first appearance of pests, with a minimum retreatment interval of 14 days and a PHI of 14 days.

In 20 independent trials on soya beans, conducted in the USA and matching this GAP, the sum of residues (for maximum residue level estimation) were: 0.022, 0.17, 0.22, 0.23, 0.24, 0.30, 0.32, 0.35, 0.35, 0.41, 0.42, 0.46, 0.57, 0.68, 0.74, 0.87, 0.99, 1.1, 1.2 and 2.1 mg/kg. The median residue, for livestock dietary burden estimation, was 0.415 mg/kg.

For risk assessment, total residues in these trials were (n = 20): 0.048, 0.20, 0.26, 0.26, 0.27, 0.34, 0.36, 0.37, 0.43, 0.48, 0.50, 0.57, 0.65, 0.74, 0.81, 0.94, 1.1, 1.3, 1.4 and 2.4 mg/kg.

The Meeting estimated a maximum residue level of 3 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and an STMR of 0.49 mg/kg for total spiropidion residues in Soya bean (dry).

Root and tuber vegetables

Potato

The GAP for spiropidion on potatoes in Guatemala is for up to 3 foliar applications of 180 g ai/ha from first appearance of pests, with a minimum retreatment interval of 7 days and a PHI of 7 days.

In 25 independent trials on potatoes, conducted in North America and matching this GAP, the sum of residues (for maximum residue level estimation) were: < 0.022, 0.055, 0.071, 0.09, 0.099, 0.11,

0.16, 0.19, 0.19, 0.21, 0.23, 0.23, 0.24, 0.25, 0.26, 0.29, 0.29, 0.43, 0.55, 0.73, 0.74, 0.76, 0.78, 0.84 and 0.90 mg/kg. For livestock dietary burden estimation, the median residue was 0.24 mg/kg and the highest residue was 0.9 mg/kg.

For risk assessment, total residues in these trials were ($n = 25$): < 0.048, 0.081, 0.097, 0.12, 0.12, 0.13, 0.18, 0.22, 0.23, 0.24, 0.25, 0.27, 0.28, 0.29, 0.30, 0.32, 0.33, 0.47, 0.61, 0.8, 0.8, 0.82, 0.84, 0.89 and 0.95 mg/kg and the highest individual value was 0.98 mg/kg.

The Meeting estimated a maximum residue level of 1.5 mg/kg for spiropidion (sum of spiropidion and spiropidion-enol, expressed as spiropidion) and a STMR of 0.28 mg/kg and an HR of 0.98 mg/kg for total spiropidion residues in Potato.

Fate of residues during processing

The Meeting received information on the hydrolysis of spiropidion under simulated processing conditions and the effects of processing on residues of spiropidion in tomatoes, soya beans and potatoes.

High temperature hydrolysis

In studies on the hydrolytic stability of aqueous solutions of radiolabelled spiropidion and metabolites spiropidion-enol, SYN547435 and SYN548430, samples were incubated under conditions simulating pasteurisation (20 minutes at 90 °C in pH 4 buffer), baking/brewing/boiling (60 minutes at 100 °C in pH 5 buffer) and sterilisation (20 minutes at 120 °C in pH 6 buffer).

The total recovery of radioactivity in both the control and test samples ranged from 94% to 105% AR, indicating that there were no significant losses of radioactivity during the experimental procedures.

Under conditions simulating pasteurisation, spiropidion residues decreased to 85% AR, with a corresponding increase in the SYN547305 concentration to 9.9% AR.

Under conditions simulating baking/brewing/boiling, spiropidion residues decreased to 80% AR with spiropidion-enol concentrations increasing to 16% AR.

Under conditions simulating sterilisation, spiropidion residues decreased to 14% AR with the - enol concentrations increasing to 85% AR.

Unidentified components were present at very low levels, accounting for not more than 5% AR, with no single component present at more than 4.4% AR.

Metabolites spiropidion-enol, SYN547435 and SYN548430 made up more than 97% AR in all samples from all incubations, with unidentified components totalling not more than 2.8% AR.

Spiropidion is not hydrolytically stable under conditions representative of pasteurisation, baking/brewing/boiling and sterilisation, being hydrolyzed to spiropidion-enol, particularly under simulated sterilisation conditions. Spiropidion-enol, SYN547435 and SYN548430 were shown to be stable to high temperature hydrolysis.

Residues in processed commodities

In processing studies on tomatoes, soya beans and potatoes, residues of spiropidion, spiropidion-enol, SYN547435, SYN548430, SYN550820 and SYN550839 were measured in a range of processed commodities.

No residues of SYN550820 or SYN550839 were detected in any RAC or processed commodity samples.

In food commodities, residues increased in soya flour, tomato paste, tomato puree, dried tomatoes and in raw peeled potato, baked whole potatoes, potato fries (with peel), potato flakes and potato protein. In non-food commodities, residues concentrated in soya meal, pollard and aspirated grain fractions, in wet tomato peel and dried pomace and in potato process waste and dried pulp.

Processing factors were calculated for the sum of spiropidion plus its –enol metabolite (for maximum residue level estimation and for calculating livestock dietary burdens) and for total residues (for risk assessment). If residues in the RAC were below the LOQ, no processing factor was derived. In case of residues below the LOQ in the processed product, the numeric value of the LOQ was used for the calculation.

Where residues concentrated in the processed food commodities, maximum residue levels were estimated using the estimated maximum residue levels for the raw commodities and applying the calculated mean processing factors.

Table 2 Estimated spiropidion maximum residue levels for processed commodities

Commodity	Processing factors		Maximum Residue Level (mg/kg) ^a
	Calculated Processing Factors ^b	Best Estimate	
Tomato			MRL=0.8
Puree	0.78, 1.3	1.1	
Paste	1.5, 2.5	2.0	1.6
Fruit (dried)	4.3, 12	8.1	6.5
Soya bean			MRL=3.0
Meal	1.4, 1.6	1.5	4.5
Soy flour	1.5, 1.6	1.6	4.8
Potato			MRL=1.5
Flakes	2.8, 3.0	2.9	4.4

^a Sum of spiropidion + (spiropidion-enol×1.205), as parent

^b Each value represents a separate study where residues were above the LOQ in the RAC. The factors are the ratios of the residue in the processed item divided by the residue in the Raw Agricultural Commodity

The Meeting estimated maximum residue levels of 1.5 mg/kg for tomato puree (based on the higher processing factor for tomato paste); 7 mg/kg for dried tomatoes; 5 mg/kg for soya bean flour, 5 mg/kg for soya bean meal and 5 mg/kg for potato flakes.

For processed food commodities, STMR-Ps and HR-Ps (where relevant) were calculated using the STMRs for the raw commodities and applying the calculated mean processing factors.

Table 3 Calculated spiropidion STMR-Ps and HR-Ps for processed food commodities

RAC	Processing factors		Total residue ^a	
	Calculated Processing factors ^b	Best Estimate	STMR-P (mg/kg)	HR-P (mg/kg)
Tomato			STMR=0.245	HR=0.7
Juice	0.66, 0.89	0.78	0.19	-
Canned, peeled fruit	0.61, < 0.64	0.62	0.15	0.44
Paste	1.3, 2.5	1.9	0.46	-
Puree	0.85, 1.3	1.1	0.27	-
Fruit (dried)	3.2, 10	6.8	1.7	4.8
Soya bean			STMR=0.49	
Oil (refined)	< 0.016, < 0.026	< 0.021	< 0.01	
Soy flour	1.6, 1.6	1.6	0.79	
Soy milk	0.073, 0.087	0.08	0.039	
Tofu	0.088, 0.12	0.1	0.051	
Soy sauce	0.039, 0.043	0.041	0.02	
Miso	0.19, 0.21	0.2	0.098	
Potato			STMR=0.28	HR=0.98

RAC	Processing factors		Total residue ^a	
	Calculated Processing factors ^b	Best Estimate	STMR-P (mg/kg)	HR-P (mg/kg)
Peeled tubers	1.3, 1.4	1.35	0.38	1.3
Baked (with peel)	1.1, 2.9	2.0	0.56	2.0
Fries (peeled)	0.55, < 0.87	0.71	0.2	-
Crisps	0.76, 0.94	0.84	0.24	-
Flakes	2.0, 2.9	2.4	0.67	-
Starch	0.29, < 0.87	0.58	0.16	-

^a Total residue: spiropidion + (spiropidion-enol×1.205) + (SYN547435×1.256) + (SYN548430×1.318), as parent

^b Each value represents a separate study where residues were above the LOQ in the RAC. The factors are the ratios of the residue in the processed item divided by the residue in the Raw Agricultural Commodity

For processed feed commodities, median residues and highest residues (where relevant) were calculated using the median sum of residues (spiropidion plus spiropidion-enol) for the raw commodities and applying the calculated mean processing factors for the sum of residues.

Table 4 Calculated spiropidion median-Ps and highest-Ps for processed feed commodities

RAC	Processing factors		Sum of residues ^a	
	Calculated Processing factors ^b	Best Estimate	Median-P	Highest-P
Tomato			median=0.195	
Pomace	12, 31	21	4.1	-
Soya bean			median=0.415	
Meal	1.4, 1.6	1.5	0.62	
Hulls	0.63, 0.71	0.67	0.28	
Aspirated Grain Fraction	0.57, 3.4	2.0	0.83	
Potato			median=0.24	highest=0.9
Potato culls	-	-	0.24	0.9
Waste	0.88, 1.9	1.4	0.34	-
Pulp (dried)	1.5, 1.6	1.5	0.36	-

^a Sum of residues, as parent = spiropidion+(SYN547305×1.205)

^b Each value represents a separate study where residues were above the LOQ in the RAC. The factors are the ratios of the residue in the processed item divided by the residue in the Raw Agricultural Commodity

Residues in animal commodities

Farm animal feeding studies

The Meeting received the results of a feeding study where dairy cows were dosed with spiropidion and residues of spiropidion and metabolites spiropidion-enol (SYN547305), SYN548430 and SYN548939 were measured in milk and edible tissues.

Lactating dairy cows were dosed orally with spiropidion in gelatine capsules each morning (after milking) for 29 consecutive days. Average equivalent 'in the feed' dose rates were 3.07 ppm (0.1 mg/kg bw), 9.21 ppm (0.3 mg/kg bw) and 30.8 ppm (0.93 mg/kg bw). Milk was collected twice daily and the animals were sacrificed within 24 hours after the last dose.

Samples were stored frozen for up to 21 days before analysis for spiropidion, spiropidion-enol, SYN548430 and SYN548439 using method GRM069.14A except for liver, where method GRM069.15A (with an additional hydrolysis step) was also used to measure free and conjugated spiropidion-enol. The validated LOQs for each analyte was 0.01 mg/kg and mean procedural recovery rates in samples fortified with mixed standards of spiropidion, spiropidion-enol, SYN548430, and SYN548939 ranged from 81–119%

for the combined residues of spiropidion and its -enol and from 74–100% for metabolites SYN548430 and 548939.

Spiropidion residues were below the LOQ of 0.01 mg/kg in all the milk and tissue samples from all dose groups. Residues of spiropidion-enol were not detected in any of the milk samples but were found in all tissues, highest in kidney and in liver (where residues of conjugated spiropidion-enol were also measured). In the medium and high dose groups, SYN548430 residues were only found in kidney and liver samples and SYN548939 residues were only found in liver.

For maximum residue level estimation and dietary risk assessment, a molecular weight correction factor of 1.205 was applied to the residue values for spiropidion-enol (free and conjugated) to express them as spiropidion equivalents.

The calculated maximum and mean residues in tissues and milk from animals dosed with 3.07 ppm spiropidion in the diet are summarized in the table below.

Table 5 Residues of spiropidion and metabolites in tissues from lactating cows dosed with the equivalent of 3.07 ppm spiropidion daily for 28 days

Matrix	Residues of spiropidion and metabolites in tissues (mg/kg)				
	Spiropidion	Spiropidion-enol	SYN548430	SYN548939	Spiropidion-enol parent equivalents ^a
Mean residues					
Muscle	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012
Liver	< 0.01	0.09 (0.11) ^b	< 0.01	< 0.01	0.11 (0.13) ²
Kidney	< 0.01	0.15	< 0.01	< 0.01	0.18
Fat (perineal)	< 0.01	0.02	< 0.01	< 0.01	0.024
Fat (mesenteric)	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012
Fat (subcutaneous)	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012
Milk	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012
Maximum residues					
Muscle	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012
Liver	< 0.01	0.1 (0.12) ^b	< 0.01	< 0.01	0.12 (0.14) ²
Kidney	< 0.01	0.19	< 0.01	< 0.01	0.23
Fat (perineal)	< 0.01	0.02	< 0.01	< 0.01	0.024
Fat (mesenteric)	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012
Fat (subcutaneous)	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012
Milk	< 0.01	< 0.01	< 0.01	< 0.01	< 0.012

^a Spiropidion-enol parent equivalents = (Spiropidion-enol × 1.205) mg/kg

^b Calculated free+conjugated residues, based on a mean free:total ratio 1.2 (Individual ratios of 0.7, 1.33 and 1.55 in the three dose groups – 3.07, 9.21 and 30.8 ppm, respectively)

Farm animal dietary burden

Dietary burden calculations for beef cattle, dairy cattle and poultry are provided below.

The feed commodities used in estimating livestock dietary burdens are listed below, and do not appear in the Recommendations Table unless a maximum residue level is recommended. The input values used in the calculations are for the sum of spiropidion and spiropidion-enol (expressed as spiropidion).

Table 6 Residues in feed commodities used in estimating livestock dietary burdens

Codex classification	Commodity	Median residue (-P) (mg/kg) ^a	Highest residue (-P) (mg/kg) ^a
	Tomato pomace (dry)	4.1	
	Soya bean seed	0.415	
	Soya bean meal	0.62	
	Soya bean hulls	0.28	
	Soya bean aspirated grain fraction	0.83	
	Potato waste	0.34	
	Potato pulp (dried)	0.36	
	Potato culls	0.24	0.9

^a Sum of residues = (spiropidion) + (spiropidion-enol×1.205) mg/kg

Dietary burdens were calculated for beef cattle, dairy cattle, broilers and laying poultry based on feed items evaluated by the current JMPR. The dietary burdens, estimated using the most recent version of the OECD livestock dietary burden calculator, are presented in Annex 6 and summarised below.

Table 7 Estimated maximum and mean dietary burdens of farm animals

Animal dietary burden: Sum of spiropidion and spiropidion-enol (SYN547305) residues (as parent), ppm dry matter diet								
	US-Canada		EU		Australia		Japan	
	max	mean	max	mean	max	mean	max	mean
Beef cattle	2.30	1.31	2.67 ^①	1.675 ^⑤	1.04	0.71	0.51	0.51
Dairy cattle	0.85	0.52	2.44 ^②	1.45 ^④	1.08	0.75	0.45	0.45
Poultry – broiler	0.26	0.26	0.895	0.565	0.24	0.24	0.24	0.24
Poultry – layer	0.26	0.26	0.75	0.42	0.24	0.24	0.2	0.2

① Highest maximum beef or dairy cattle dietary burden suitable for MRL estimates for mammalian tissues

② Highest maximum dairy cattle dietary burden suitable for MRL estimates for mammalian milk

③ Highest mean beef or dairy cattle dietary burden suitable for STMR estimates for mammalian tissues.

④ Highest mean dairy cattle dietary burden suitable for STMR estimates for milk.

Animal commodity maximum residue levels

Cattle

The calculations used to estimate maximum residue levels, STMR and HR values for cattle matrices are shown below.

Table 8 Estimated residues for maximum residue levels, HRs and STMRs for spiropidion in cattle commodities

Spiropidion feeding study	Feed level (ppm) for milk residues	Residues (mg/kg) in milk	Feed level (ppm) for tissue residues	Residues (mg/kg)			
				Muscle	Liver	Kidney	Fat
Maximum residue level beef or dairy cattle (spiropidion equivalents)							
Feeding study	3.07	< 0.012	3.07	< 0.012	0.12	0.23	0.024
Dietary burden and highest residue	2.44	< 0.0095	2.67	< 0.01	0.1	0.2	0.021
HR beef or dairy cattle (spiropidion equivalents)							
Feeding study			3.07	< 0.012	0.14	0.23	0.024

Spiropidion feeding study	Feed level (ppm) for milk residues	Residues (mg/kg) in milk	Feed level (ppm) for tissue residues	Residues (mg/kg)			
				Muscle	Liver	Kidney	Fat
Dietary burden and residue estimate			2.67	< 0.01	0.12	0.2	0.021
STMR beef or dairy cattle (spiropidion equivalents)							
Feeding study	3.07	< 0.012	3.07	< 0.012	0.13	0.18	0.024
Dietary burden and residue estimate	1.45	< 0.0057	1.675	< 0.0065	0.071	0.098	0.013

Based on these calculated spiropidion-enol residues (expressed as spiropidion equivalents) in milk and cattle tissues, the Meeting estimated maximum residue levels of 0.012(*) mg/kg in milk and mammalian meat, 0.025 mg/kg in mammalian fat and 0.2 mg/kg in mammalian offal.

Based on the highest spiropidion-enol (free and conjugated) residues (expressed as spiropidion equivalents) in cattle tissues, the Meeting estimated HR values of 0.01 mg/kg in mammalian muscle, 0.021 mg/kg in mammalian fat and 0.2 mg/kg in mammalian offal (based on residues in kidney).

Based on the mean spiropidion-enol (free and conjugated) residues (expressed as spiropidion equivalents) in milk and cattle tissues, the Meeting estimated STMR values of 0.0057 mg/kg in milk, 0.0065 mg/kg in mammalian muscle, 0.013 mg/kg in mammalian fat and 0.098 mg/kg in mammalian edible offal (based on residues in kidney).

Poultry

A poultry feeding study was not available and the Meeting used the laying hen metabolism study conducted at feeding levels of 14.5–16.4 ppm to evaluate residue levels in poultry.

In this study, spiropidion-enol residues (expressed as spiropidion equivalents) were up to 0.007 mg eq/kg in muscle, up to 0.006 mg eq/kg in skin+fat, up to 0.015 mg eq/kg (calculated) in eggs and up to 0.056 mg eq/kg in liver.

When these maximum values are scaled to the highest poultry dietary burden (0.895 ppm in broilers), a scaling factor of 0.059×, the calculated residues are 0.00041 mg/kg (muscle), 0.00035 mg/kg (skin+fat), 0.00089 mg/kg (eggs) and 0.0033 mg/kg (liver).

The Meeting estimated maximum residue levels of 0.012(*) mg/kg for spiropidion-enol (expressed as spiropidion) in eggs, poultry meat, poultry offal and poultry fat.

For dietary risk assessment, the Meeting estimated HRs and STMRs of 0.00041 mg/kg in poultry muscle, 0.00035 mg/kg in poultry fat, 0.0033 mg/kg in poultry edible offal (based on residues in liver) and 0.00089 mg/kg in eggs.

RECOMMENDATIONS

On the basis of the data obtained from supervised trials, the Meeting concluded that the residue levels listed in Annex 1 are suitable for establishing maximum residue limits and for IEDI and IESTI assessments.

Definition of the residue for compliance with the MRL for plant commodities: the sum of spiropidion and spiropidion-enol (SYN547305) expressed as spiropidion

Definition of the residue for dietary risk assessment for plant commodities: the sum of spiropidion, spiropidion-enol (SYN547305), 3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-8-methoxy-1,8-diazaspiro[4.5]dec-3-en-2-one (SYN547435) and 3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-1-methyl-1,8-diazaspiro[4.5]dec-3-en-2-one (SYN548430), expressed as spiropidion.

Definition of the residue for compliance with the MRL for animal commodities: spiropidion-enol (SYN547305) expressed as spiropidion.

Definition of the residue for dietary risk assessment for animal commodities: free and conjugated spiropidion-enol (SYN547305) expressed as spiropidion.

The residue is not fat-soluble.

DIETARY RISK ASSESSMENT

Long-term dietary exposure

The ADI for Spiropidion is 0–0.02 mg/kg bw. The International Estimated Daily Intakes (IEDIs) for spiropidion were estimated for the 17 GEMS/Food Consumption Cluster Diets using the STMR or STMR-P values estimated by the JMPR. The results are shown in Annex 3 of the 2021 JMPR Report.

The IEDIs ranged from 0–10% of the maximum ADI. The Meeting concluded that long-term dietary exposure to residues of spiropidion from uses considered by the JMPR is unlikely to present a public health concern.

Acute dietary exposure

The ARfD for spiropidion is 0.3 mg/kg bw. The International Estimate of Short Term Intakes (IESTIs) for spiropidion were calculated for the food commodities and their processed commodities for which HRs/HR-Ps or STMRs/STMR-Ps were estimated by the present Meeting and for which consumption data were available. The results are shown in Annex 4 of the 2021 JMPR Report.

The IESTIs varied from 0–60% of the ARfD for children and 0–40% of the ARfD for the general population. The Meeting concluded that acute dietary exposure to residues of spiropidion from uses considered by the present Meeting is unlikely to present a public health concern.

Threshold of toxicological concern (TTC) consideration for metabolites

The metabolite SYN550820 could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

While residues of SYN550820 were < 0.01 mg/kg in all the fresh and processed commodities considered by the Meeting, in the cotton metabolism study, SYN550820 residues accounted for up to 12% TRR in cotton foliage and in gin trash, at concentrations of up to 0.13 mg eq/kg and 0.32 mg eq/kg, respectively.

The Meeting agreed that while the dietary exposure to residues of SYN550820 from uses considered by the current Meeting need not be estimated (since no leafy crops were involved), should further uses be considered in the future, the TTC approach for a Cramer Class III compound could be used to estimate dietary exposure.

The metabolite SYN550839 could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

While residues of SYN550839 were < 0.01 mg/kg in all the fresh and processed commodities considered by the Meeting, in the cotton metabolism study, SYN550830 residues accounted for up to 10% TRR in cotton foliage and in gin trash, at concentrations of up to 0.06 mg eq/kg and 0.26 mg eq/kg, respectively.

The Meeting agreed that while the dietary exposure to residues of SYN550839 from uses considered by the current Meeting need not be estimated (since no leafy crops were involved), should further uses be considered in the future, the TTC approach for a Cramer Class III compound could be used to estimate dietary exposure.

The metabolite identified as dehydrogenated spiropidion-enol could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

Residues of dehydrogenated spiropidion-enol were not measured in the supervised field trials but in the tomato metabolism study, residues of dehydrogenated spiropidion-enol accounted for up to 10% TRR (0.01 mg eq/kg) in fruit.

The Meeting concluded that low concentrations of the dehydrogenated spiropidion-enol might be expected in fruiting vegetables treated with spiropidion.

The Meeting estimated a dietary exposure for the dehydrogenated spiropidion-enol metabolite of 0.07 µg/kg bw per day.

The Meeting concluded that the estimated dietary exposure to residues of dehydrogenated spiropidion-enol from uses considered by the JMPR is below the TTC for Cramer Class III compounds and is unlikely to present a public health concern. Should further uses be considered in the future, these conclusions may need to be re-evaluated.

The metabolite SYN549098 (free and conjugated) could be assessed using the TTC approach (Cramer Class III threshold of 1.5 µg/kg bw per day).

In the confined rotational crop study, residues of free and conjugated SYN549098 were a significant component of the TRR in mature lettuce from the longer plant-back intervals, accounting for up to 55% TRR and 0.012 mg eq/kg.

The Meeting concluded that low concentrations of free and conjugated SYN549098 might be expected in leafy vegetables grown in rotation with spiropidion-treated crops.

The Meeting estimated a dietary exposure for metabolite SYN549098 (free and conjugated) of 0.025 µg/kg bw per day.

The Meeting concluded that the estimated dietary exposure to residues of SYN549098 (free and conjugated) from uses considered by the JMPR is below the TTC for Cramer Class III compounds and is unlikely to present a public health concern. Should further uses be considered in the future, these conclusions may need to be re-evaluated.

5.15 Tetraniliprole (324)

TOXICOLOGY

Tetraniliprole is the ISO-approved common name for 1-(3-chloropyridin-2-yl)-*N*-[4-cyano-2-methyl-6-(methylcarbamoyl)phenyl]-3-[[5-(trifluoromethyl)-2H-tetrazol-2-yl]methyl]-1H-pyrazole-5-carboxamide (IUPAC), with the CAS number 1229654-66-3.

Tetraniliprole is an anthranilic diamide-class insecticide. The proposed pesticidal mode of action (MOA) for tetraniliprole is by activation of ryanodine receptor channels, leading to internal calcium store depletion that impairs regulation of muscle contraction. Mammalian ryanodine receptors are substantially less sensitive to the effects of anthranilic diamides than insect ryanodine receptors.

Tetraniliprole has not previously been evaluated by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR) and was reviewed by the present Meeting at the request of the Codex Commission for Pesticide Residues (CCPR).

All critical studies contained statements of compliance with good laboratory practice (GLP) and were conducted in accordance with relevant national or international test guidelines, unless otherwise specified. No additional information from a literature search was identified that complemented the toxicological information submitted for the current assessment.

Biochemical aspects

Absorption, distribution, metabolism, excretion (ADME) studies were conducted in rats, using tetraniliprole labelled with ^{14}C at the pyrazole-carboxamide, pyridinyl-2, phenyl-carbamoyl or tetrazolyl positions. The absorption of tetraniliprole at a low dose of 2 mg/kg body weight (bw) was rapid ($T_{\max} < 1-7$ hours), approximately 30 to 46% (based on bile, urine and carcass), and radioactivity was widely distributed to the tissues. The values for the area under the concentration–time curve for the interval 0 to 168 hours (AUC_{0-168}) in females were approximately double those for males: 1.21 g $\times$ hours/g in males and 2.36 g $\times$ hours/g in females. Absorption at 20 and 200 mg/kg bw, the mid- and high-dose levels, was very limited.

Elimination was rapid and nearly complete by 72 hours post dose. The half-life of elimination was slightly longer (less than two-fold) in males than in females at the low-dose level. Very low to non-detectable levels of radioactivity at the mid and high doses meant that the half-life could not be calculated for comparison. Elimination was predominantly via the faeces, bile being the secondary route with up to 39% of the administered radioactivity recovered in the bile. Radioactivity recovered in the bile and faeces accounted for 96–98% of the administered dose (AD). In proportional terms the highest levels of radioactivity recovered in the urine were at the low dose, with 7% of the AD based on the results from the pyridinyl labelled groups. In the mid- and high-dose groups, urinary excretion was negligible, suggesting a decrease in absorption with increased dose. There were no changes following repeat-dose treatment in males.

Seventy-two hours following dosing, trace amounts of radioactivity were found in the tissues, with higher concentrations in females than males. The highest concentrations were found in the liver, followed by the skin, with measurable amounts in the kidneys and in the ovaries and uterus of females. Relative residual concentrations were higher in the low-dose groups than in the high-dose groups.

A quantitative whole-body autoradiography study was performed and this confirmed the results of the main study. Absorption was rapid, though limited, with small amounts of radioactivity distributed to the liver and kidney and some distribution to the glandular organs and fatty tissues. There were no substantial sex-related differences.

Unchanged tetraniliprole was found in both the urine and faeces, though not the bile, and comprised 51–70% of the AD in the low-dose groups and over 90% of the AD in the mid- and high-dose groups. Hydroxylation was favoured in the males, however the metabolite profiles were essentially similar

in males and females. Other major metabolites comprised 1–9% of the AD and included the tetraniliprole-deshydrochloro-dihydrate, dihydroxy and hydroxy-*N*-methyl metabolites. Metabolites making up 1–4% of the AD included the benzyl alcohol glucuronide, hydroxypyridyl glucuronide, deschloro-desmethylamide-dihydroxy, despyridyl, benzylalcohol, pyridinyl-pyrazole-5-carboxylic acid, and hydroxypyridine derivatives of tetraniliprole. The remaining identified metabolites accounted for most of metabolites, and were present in quantities equal to or less than 2.5% of the administered dose.

The proposed metabolic pathway in rats involves hydroxylation in several positions including the pyridinyl moiety, the *N*-methyl moiety, and the methyl group of the phenyl moiety. Conjugation with glucuronic acid takes place following hydroxylation. Intramolecular condensation (cyclization) of the parent molecule also occurred, yielding quinazolinone compounds, one of which was identified as BCS-CQ63359 (tetraniliprole-*N*-methyl-quinazolinone), a primary plasma metabolite. Also noted was cleavage of the phenyl moiety yielding an amide, followed by oxidation or methylation and further cleavage reactions involving the pyridine and tetrazole rings.

There were no significant differences in tissue distribution, biokinetics or metabolite profile due to the label position.

In several toxicity studies, the parent compound was measured in plasma. Concentrations were higher in females than in males. At the highest dose tested in the two-year rat study, parental plasma concentrations were 1.7 mg/L, which is equivalent to around 3 µM.

A series of *in vitro* liver microsome assays on [pyrazole-carboxamide-¹⁴C]tetraniliprole, [pyridinyl-2-¹⁴C]tetraniliprole and [tetrazolyl-4-¹⁴C]tetraniliprole at concentrations of 10 µM indicated that there were no metabolites unique to humans detected. Metabolism in the rat and the male dog microsome assays were the lowest with 91–100% of the administered parent compound found in rat assays and 91–97% of the administered parent compound found in the male dog assays. Unchanged parent was detected at 88–94% of the starting concentrations in the case of the male rabbit microsomes. In the mouse microsome study, unchanged parent compound represented 73–96% of the relative percentage, and in the human assays unchanged parent amounted to 74–87% of the starting concentration. Of the five metabolites identified and labelled as BCS-1 through BCS-5, only BCS-03 occurred at relative percentages greater than 5%. Metabolite BCS-3 occurred at relative percentages between 13% and 25% in the mouse microsome assays, and at 11% to 21% in human microsome assays, while relative percentages were in the range of 1–7% in rats, male dogs and male rabbits. Metabolite BCS-4 was found in mouse, male rabbit and human microsome assays at between 0.5% and 2% and BCS-2 was found in the male rabbit and human microsome assays only. There were no substantial differences due to labelling position.

Toxicological data

In rats, tetraniliprole had an acute oral and dermal median lethal dose (LD₅₀) greater than 2000 mg/kg body weight (bw) and an acute inhalation median lethal concentration (LC₅₀) greater than 5.01 mg/L. Tetraniliprole was minimally irritating to the skin and transiently irritating to the eyes of rabbits. Tetraniliprole was a dermal sensitizer in the local lymph node assay (LLNA) in mice.

The rat was the species most sensitive to tetraniliprole, with decreased body weights and effects on the ovaries, uterus and sexual development in females occurring at or above the limit dose.

In a 28-day toxicity study, mice received tetraniliprole in the diet at a concentration of 0, 600, 3000 or 6000 ppm (equal to 0, 100, 523 and 1010 mg/kg bw per day for males, 0, 113, 576 and 1159 mg/kg bw per day for females). The NOAEL was 6000 ppm (equal to 1010 mg/kg bw per day), the highest dose tested.

In a 90-day toxicity study, mice received tetraniliprole in the diet at a concentration of 0, 900, 2700 or 6000 ppm (equal to 0, 145, 426 or 973 mg/kg bw per day for males, 0, 180, 544 or 1224 mg/kg bw per day for females). The NOAEL was 6000 ppm (equal to 973 mg/kg bw per day), the highest dose tested.

In a 28-day toxicity study, rats received tetraniliprole dermally at a concentration of 0, 100, 300 or 1000 mg/kg bw per day for six hours per day. The NOAEL for dermal toxicity was 1000 mg/kg bw per day, the highest dose tested.

In a 28-day toxicity study, rats received tetraniliprole in the diet at a concentration of 0, 500, 2000 or 8000 ppm (equal to 0, 38, 148 and 599 mg/kg bw per day for males, 0, 43, 171 or 700 mg/kg bw per day for females). The NOAEL was 8000 ppm (equal to 599 mg/kg bw per day), the highest dose tested.

In a 90-day toxicity study, rats received tetraniliprole in the diet at a concentration of 0, 900, 3000 or 10 000 ppm (equal to 0, 55.5, 178 and 608 mg/kg bw per day for males, 0, 65.7, 213 and 723 mg/kg bw per day for females). The NOAEL was 10 000 ppm (equal to 608 mg/kg bw per day), the highest dose tested.

In a 90-day toxicity study, dogs received tetraniliprole in the diet at a concentration of 0, 800, 3200 or 12 800 ppm (equal to 0, 25.6, 126 or 440 mg/kg bw per day for males, 0, 29.9, 138 or 485 mg/kg bw per day for females). The NOAEL was 12 800 ppm (equal to 440 mg/kg bw per day), the highest dose tested.

In a one-year toxicity study, dogs received tetraniliprole in the diet at a concentration of 0, 650, 2900 or 12 800 ppm (equal to 0, 19.8, 91.2 and 440 mg/kg bw per day for males, 0, 18.3, 88.8 or 408 mg/kg bw per day for females). The NOAEL was 12 800 ppm (equal to 408 mg/kg bw per day), the highest dose tested.

In a 19-month toxicity and carcinogenicity study, mice received tetraniliprole in the diet at a concentration of 0, 260, 1300 or 6500 ppm (equal to 0, 32.9, 166 and 825 mg/kg bw per day for males, 0, 43.1, 225 and 1073 mg/kg bw per day for females). The NOAEL was 6500 ppm (equal to 825 mg/kg bw per day), the highest dose tested. No tumours were observed that were considered to be related to treatment with tetraniliprole.

In a two-year chronic toxicity and carcinogenicity study, rats received tetraniliprole in the diet at a concentration of 0, 900, 4000 or 18 000 ppm (equal to 0, 35.3, 159 and 741 mg/kg bw per day for males; 0, 51.2, 221, and 1052 mg/kg bw per day for females). The NOAEL was 4000 ppm (equal to 221 mg/kg bw per day) in females, based on decreased body weight, increased diffuse squamous cell hyperplasia in the cervix and vagina and increased severity of corpora lutea depletion at the LOAEL of 18 000 ppm (equal to 1052 mg/kg bw per day). No tumours were observed that were considered to be related to treatment with tetraniliprole.

The Meeting concluded that tetraniliprole is not carcinogenic in mice or rats.

Tetraniliprole was tested for genotoxicity in an adequate range of in vitro and in vivo assays. No evidence of genotoxicity was found.

The Meeting concluded that tetraniliprole is unlikely to be genotoxic.

In view of the lack of genotoxicity and the absence of carcinogenicity in mice and rats, the Meeting concluded that tetraniliprole is unlikely to pose a carcinogenic risk to humans.

In a two-generation reproductive toxicity study, rats received a dietary concentration of tetraniliprole of 0, 300, 600, 2700 or 12 000 ppm (equal to 0, 22, 44, 196 and 896 mg/kg bw per day in males, 0, 25, 41, 224 and 1032 mg/kg bw per day in females). In the lactation phase, dietary concentrations in females were decreased to 0, 150, 300, 1350 and 6000 ppm (equal to 0, 23, 47, 211 and 890 mg/kg bw per day). The NOAEL for parental toxicity was 2700 ppm (equal to 196 mg/kg bw per day), based on decreased body weights in F₁ animals at the LOAEL of 12 000 ppm (equal to 896 mg/kg bw per day). The NOAEL for reproductive toxicity was 12 000 ppm (equal to 896 mg/kg bw per day), the highest dose tested. The NOAEL for offspring toxicity was 2700 ppm (equal to 196 mg/kg bw per day), based on decreased pup body weights resulting in delayed completion of vaginal opening at the LOAEL of 12 000 ppm (equal to 896 mg/kg bw per day).

In a developmental toxicity study, female rats received a gavage dose of tetraniliprole at concentrations of 0, 62.5, 250 or 1000 mg/kg bw per day mixed in 0.5% aqueous sodium methylcellulose 400, from days 6–20 of gestation. The NOAEL for maternal toxicity was 1000 mg/kg bw per day, the highest dose tested. The NOAEL for embryo/fetal toxicity was 250 mg/kg bw per day, based on lack of ossification in the fifth or sixth sternbrae at the LOAEL of 1000 mg/kg bw per day.

In a developmental toxicity study, predated rabbits were given tetraniliprole in 0.5% aqueous sodium methylcellulose 400 at a concentration of 0, 62.5, 250 or 1000 mg/kg bw per day from days 6–28 of gestation. The NOAELs for maternal and embryo/fetal toxicity were both 1000 mg/kg bw per day, the highest dose tested.

In an uterotrophic assay, groups of six immature (19-day-old) female Sprague Dawley rats were given tetraniliprole in 0.5% aqueous methylcellulose 400 at a concentration of 0, 100, 400 or 800 mg/kg bw per day for three days. There were no effects on vaginal opening or uterine weight. The reliability of the study is unclear due to the lack of a positive control group in the antiestrogenic assay and the small group numbers.

In a steroidogenesis assay, tetraniliprole and its main mammalian metabolite BCS-CO63359, were evaluated for their effects on progesterone, testosterone, estradiol and cortisol secretion using the H295R cell line. Concentrations tested were 0.1, 0.3, 1, 3, 10, 12 and 15 μ M for BCS-CL73507 and 0.1, 0.3, 1, 2, 4, 8, and 12 μ M for BCS-CQ63359, all presented in 0.1% dimethyl sulfoxide (DMSO). There were clear increases in estradiol and cortisol levels with the tetraniliprole treatments at and above 3 μ M. There were slight increases in progesterone and testosterone levels at the same doses, but these did not show a dose–response. In the case of the metabolite, increases occurred in estradiol and cortisol at 1 μ M, and testosterone was increased at 8 μ M. There were no apparent effects on progesterone. Increases seen in any of the hormones were much lower when treated with tetraniliprole and its metabolite than with the positive control.

The Meeting concluded that tetraniliprole is not teratogenic.

No specific studies on neurotoxicity were submitted. No evidence of neurotoxicity was reported in routine toxicological studies with tetraniliprole.

The Meeting concluded that tetraniliprole was unlikely to be neurotoxic.

No specific studies on immunotoxicity were submitted. No evidence of immunotoxicity was reported in routine toxicological studies with tetraniliprole.

The Meeting concluded that tetraniliprole was unlikely to be immunotoxic.

Toxicological data on metabolites and/or degradates

BCS-CR74541 (tetraniliprole-carboxylic acid)

The acute oral LD₅₀ of BCS-CR74541, a soil metabolite, was greater than 2000 mg/kg bw and it tested negative in reverse mutation, gene mutation and chromosomal aberration assays. In a 28-day dietary toxicity study in rats, the NOAEL was 10 000 ppm (equal to 775 mg/kg bw per day), the highest dose tested. BCS-CR74541 was considered to be covered by the toxicity of the parent compound.

BCS-CU81055 (tetraniliprole-desmethyl-amide-carboxylic acid)

The acute oral LD₅₀ of BCS-CU81055, another soil metabolite, was greater than 2000 mg/kg bw, and it tested negative in mutation, gene mutation and chromosomal aberration assays. In a 28-day dietary toxicity study in rats, the NOAEL was 10 000 ppm (equal to 768 mg/kg bw per day), the highest dose tested. BCS-CU81055 was considered to be covered by the toxicity of the parent compound.

BCS-CT30673 (tetraniliprole-N-methyl-quinazolinone-carboxylic acid),

BCS-CT30673, a third soil metabolite, tested negative in mutation, gene mutation and chromosomal aberration assays. BCS-CT30673 was considered to be covered by the toxicity of the parent compound.

BCS-CU81056 (tetraniliprole-quinazolinone-carboxylic acid)

BCS-CU81056, a fourth soil metabolite, tested negative in mutation, gene mutation and chromosomal aberration assays. BCS-CU81056 was considered to be covered by the toxicity of the parent compound.

BCS-CQ63359 (tetraniliprole-N-methyl-quinazolinone amide)

BCS-CQ63359 was the major plasma metabolite; it gave rise to no alerts for genotoxicity following QSAR analysis and is of similar structure to the parent. BCS-CQ63359 was considered to be covered by the toxicity of the parent compound.

BCS-CZ91631 (tetraniliprole-benzyl alcohol)

BCS-CZ91631 produced no alerts for genotoxicity following QSAR analysis and is of similar structure to the parent. BCS-CZ91631 was considered to be covered by the toxicity of the parent compound.

Tetraniliprole-despyridyl-N-methyl-quinazolinone

Tetraniliprole-despyridyl-N-methyl-quinazolinone gave rise to alerts for genotoxicity and is not covered by the toxicity of the parent. It is a Cramer class III compound and a TTC of 0.002 5 µg/kg bw per day was recommended.

Microbiological data

No data for antimicrobial activity or impact of tetraniliprole on the human gut microbiome are available.

Human data

In reports on manufacturing plant personnel, no adverse health effects had been noted.

The Meeting concluded that the existing database on tetraniliprole was adequate to characterize the potential hazards to the general population, including fetuses, infants and children.

Toxicological evaluation

The Meeting established an ADI for tetraniliprole of 0–2 mg/kg bw per day, from the two-year dietary toxicity study in rats and the two-generation reproductive toxicity study in rats, on the basis of the NOAEL of 221 mg/kg bw per day in the two-year rat study, and 196 mg/kg bw per day for offspring in the two-generation rat reproductive toxicity study. Findings at the LOAEL consisted of decreased body weight and changes to cervix, vagina and ovaries in the two-year study, and decreased body weight resulting in delayed vaginal opening in the two-generation study. A safety factor of 100 was applied.

The Meeting concluded that it was not necessary to establish an ARfD for tetraniliprole in view of its low acute oral toxicity and the absence of developmental toxicity and any other toxicological effects that would be likely to be elicited by a single dose.

A toxicological monograph was prepared.

Levels relevant to risk assessment of tetraniliprole

Species	Study	Effect	NOAEL	LOAEL
Mouse	28-day oral toxicity study ^a	Toxicity	6000 ppm equal to 1010 mg/kg bw per day	-

Species	Study	Effect	NOAEL	LOAEL
Rat	90-day oral toxicity study ^a	Toxicity	6000 ppm equal to 973 mg/kg bw per day ^b	-
	Eighteen-month study of carcinogenicity ^a	Toxicity	6500 ppm equal to 825 mg/kg bw per day ^b	-
	28-day oral toxicity study ^a	Toxicity	8000 ppm equal to 599 mg/kg bw per day ^b	-
	90-day oral toxicity study ^a	Toxicity	10 000 ppm equal to 608 mg/kg bw per day ^b	-
	Two-year study of toxicity and oncogenicity ^a	Toxicity	4000 ppm equal to 221 mg/kg bw per day (females)	1052 mg/kg bw per day (females)
	Two-generation study of reproductive toxicity ^a	Reproductive toxicity	12 000 ppm equal to 896 mg/kg bw per day ^b	-
		Parental toxicity	12 000 ppm equal to 896 mg/kg bw per day ^b	-
		Offspring toxicity	2700 ppm equal to 196 mg/kg bw per day	12 000 ppm equal to 896 mg/kg bw per day
	Developmental toxicity study ^c	Maternal toxicity	1000 mg/kg bw per day ^b	-
		Embryo and fetal toxicity	250 mg/kg bw per day	1000 mg/kg bw per day
Rabbit	Developmental toxicity study ^c	Maternal toxicity	1000 mg/kg bw per day ^b	-
		Embryo and fetal toxicity	1000 mg/kg bw per day ^b	-
Dog	90-day study of toxicity ^a	Toxicity	12 800 ppm, equal to 440 mg/kg bw per day ^b	-
	One-year study of toxicity ^a	Toxicity	12 800 ppm, equal to 408 mg/kg bw per day ^b	-
Metabolite BCS-CU81055				
Rat	28-day study of toxicity ^a	Toxicity	10 000 ppm equal to 755 mg/kg bw per day ^b	-
Metabolite BCS-CU81055				
Rat	28-day study of toxicity ^a	Toxicity	10 000 ppm equal to 768 mg/kg bw per day ^b	-

^a Dietary administration^b Highest dose tested.^c Gavage administration^d Two or more studies combined^e Lowest dose tested.

Acceptable daily intake (ADI), applies to tetraniliprole, BCS-CQ63359 and BCS-CZ91631 expressed as tetraniliprole

0–2 mg/kg bw

Acute reference dose (ARfD), applies to tetraniliprole, BCS-CQ63359 and BCS-CZ91631 expressed as tetraniliprole

Not necessary

Information that would be useful for the continued evaluation of the compound

Results from epidemiological, occupational health and other such observational studies of human exposure.

Critical end-points for setting guidance values for exposure to tetraniliprole**Absorption, distribution, excretion and metabolism in mammals**

Rate and extent of oral absorption	Rapid: $T_{\max}$ = 1 hour for all doses, except 20 mg/kg bw in females, where $T_{\max}$ = 7 hours Oral absorption: 46% (male), 30% (female), based on bile, urine and carcass (rat)
Dermal absorption	No data
Distribution	Extensive; highest concentrations in liver and kidneys
Potential for accumulation	No evidence of accumulation
Rate and extent of excretion	Relatively rapid; (complete within 48–72 hours in rats, mainly in faeces)
Metabolism in animals	Major metabolic reactions: demethylation of the <i>N</i> -methyl group and deschlorodination in the pyridine ring, two separate conjugations of tetraniliprole (after deschlorodination) with glutathione occurred, followed by degradation of one of the glutathione groups to the mercapto-alcohol (rat)
Toxicologically significant compounds in animals and plants	Tetraniliprole, BCS-CQ63359, BCS-CZ91631

Acute toxicity

Rat, LD ₅₀ , oral	>2000 mg/kg bw
Rat, LD ₅₀ , dermal	>2000 mg/kg bw
Rat, LC ₅₀ , inhalation	>5.01 mg/L
Rabbit, dermal irritation	Minimally irritating
Rabbit, ocular irritation	Minimally irritating
Mouse, dermal sensitization	Sensitizer (LLNA)

Short-term studies of toxicity

Target/critical effect	Decreased body weight/body weight gain (dog)
Lowest relevant oral NOAEL	408 mg/kg bw per day, the highest dose tested (dog)
Lowest relevant dermal NOAEL	1000 mg/kg bw, the highest dose tested (rat)
Lowest relevant inhalation NOAEC	No data.

Long-term studies of toxicity and carcinogenicity

Target/critical effect	Decreased body weight, prolapsed vagina, squamous cell hyperplasia of cervix/vagina, increased severity of corpora lutea depletion (rat)
Lowest relevant NOAEL	221 mg/kg bw per day (female rat)
Carcinogenicity	No evidence of carcinogenicity ^a

Genotoxicity

Unlikely to be genotoxic^a

Reproductive toxicity

Target/critical effect	Decreased pup body weight resulting in delayed vaginal opening
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Lowest relevant parental NOAEL	896 mg/kg bw per day (rat)
Lowest relevant offspring NOAEL	196 mg/kg bw per day (rat)
Lowest relevant reproductive NOAEL	896 mg/kg bw per day, the highest dose tested (rat)

Developmental toxicity

Target/critical effect	Unossified fifth or sixth sternbrae
Lowest relevant maternal NOAEL	1000 mg/kg bw per day, the highest dose tested (rat)
Lowest relevant embryo/fetal NOAEL	250 mg/kg bw per day (rat)

Neurotoxicity

Acute neurotoxicity NOAEL	No data
Subchronic neurotoxicity NOAEL	No data
Developmental neurotoxicity NOAEL	No data

Immunotoxicity

	No data
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Studies on toxicologically relevant metabolites

BCS-CR74541	Acute oral LD ₅₀ : > 2000 mg/kg bw (rat) 28-day NOAEL: 775 mg/kg bw per day, the highest dose tested (rat) Not genotoxic: mutation, gene mutation and chromosomal aberration
BCS-CU81055	Acute oral LD ₅₀ : > 2000 mg/kg bw (rat) 28-day NOAEL: 768 mg/kg bw per day, the highest dose tested (rat) Not genotoxic: mutation, gene mutation and chromosomal aberration
BCS-CT30673	Not genotoxic: mutation, gene mutation and chromosomal aberration
BCS-CU81056	Not genotoxic: mutation, gene mutation and chromosomal aberration
Microbiological data	No data
Human data	No clinical cases or poisoning incidents have been recorded

^a Unlikely to pose a carcinogenic risk to humans via exposure from the diet

Summary

	Value	Study	Safety factor
ADI	0–2 mg/kg bw ^a	Two-year toxicity (rat) and two-generation reproductive toxicity (rat)	100
ARfD	Not necessary		

^a Applies to tetraniliprole, BCS-CQ63359, and BCS-CZ91631 expressed as tetraniliprole

6. Corrigenda

Pesticide Residues in Food 2019. Report of the Joint Meeting of the FAO Panel of Experts on Pesticide Residues in Food and the Environment and the WHO Core Assessment Group on Pesticide Residues. FAO Plant Production and Protection Paper, 229, 2019

Changes are shown in bold

Metconazole (313)

Annex 1 Page 415

Definition of the residue for dietary risk assessment for animal commodities: *Sum of metconazole (cis and trans-isomer) and metabolites (1SR,2SR,5RS)-5-(4-chlorobenzyl)-2-(hydroxymethyl)-2-methyl-1-(1H-1,2,4-triazol-1-ylmethyl)cyclopentanol (M1; free and conjugated) and (1RS,2SR,3RS)-3-(4-chlorobenzyl)-2-hydroxy-1-methyl-2-(1H-1,2,4-triazol-1-ylmethyl)cyclopentanecarboxylic acid (M12; free and conjugated), expressed as metconazole.*

Annex 1: Acceptable daily intakes, acute reference doses, recommended maximum residue levels, supervised trials median residue values and other values recorded by the 2021 JMPR Meeting.

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P	HR or HR-P
			New	Previous	mg/kg	mg/kg
Afidopyropen (312) ADI: 0–0.08 mg/kg bw ARfD: 0.2 mg/kg bw (for women of child-bearing age) ARfD: 0.3 mg/kg bw (for general population)	Definition of the residue for compliance with the MRL for plant commodities: <i>Afidopyropen</i> Definition of the residue for dietary risk assessment for plant commodities: <i>the sum of afidopyropen + dimer of [(3R,6R,6aR,12S,12bR)-3-[(cyclopropanecarbonyl)oxy]-6,12-dihydroxy-4,6a,12b-trimethyl-11-oxo-9-(pyridin-3-yl)-1,3,4,4a,5,6,6a,12,12a,12b-decahydro-2H,11H-naphtho[2,1-b]pyrano[3,4-e]pyran-4-yl)methyl rac-cyclopropanecarboxylate (M007)</i> Definition of the residue for compliance with the MRL for animal commodities: <i>Afidopyropen</i> Definition of the residue for dietary risk assessment for animal commodities, excluding liver: <i>Afidopyropen + (3S,4R,4aR,6S, 6aS, 12R,12aS,12bS)-3,6,12-trihydroxy-4-(hydroxymethyl)-4,6a, 12b-trimethyl-9-(pyridin-3-yl)-1, 3,4,4a,5,6,6a,12, 12a,12b-decahydro-2H,11H-benzo- [f] pyrano[4,3-b]chromen-11-one (M001) + Cyclopropane carboxylic acid (CPCA/M061) and (2R)-3-carboxy-2- [(cyclopropylcarbonyl)oxy]- N, N, N-trimethylpropan-1- aminium chloride (CPCA-carnitine conjugate/M060), expressed as afidopyropen.</i> Definition of the residue for dietary risk assessment for animal commodities, liver: <i>Afidopyropen + (3S,4R,4aR,6S, 6aS, 12R,12aS,12bS)-3,6,12-trihydroxy-4-(hydroxymethyl)-4,6a, 12b-trimethyl-9-(pyridin-3-yl)-1, 3,4,4a,5,6,6a,12, 12a,12b-decahydro-2H,11H-benzo- [f] pyrano[4,3-b]chromen-11-one (M001) + Cyclopropane carboxylic acid (CPCA/M061) and (2R)-3-carboxy-2- [(cyclopropylcarbonyl)oxy]- N, N, N-trimethylpropan-1- aminium chloride (CPCA-carnitine conjugate/M060) + [(3S,4R,4aR,6S,6aS,12R,12aS,12bS)-3-(cyclopropylcarbonyl)oxy]-6,12-dihydroxy-4,6a,12b-trimethyl-9-(1-oxidopyridin-3-yl)-11-oxo-1,3,4,4a,5,6,6a,12,12a,12b-decahydro-2H, 11H-benzo[f]pyrano[4,3-b]chromen-4-yl)methyl cyclopropane-carboxylate (M017), expressed as afidopyropen.</i> <i>The residue is not fat-soluble.</i>					
Fenpyroximate (193) ^b ADI: 0–0.005 mg/kg bw ARfD: 0.005 mg/kg bw	FC 0001	Citrus Fruit, Group of	W	0.6		
	FC 0002	Lemons and Limes (including Citron), Subgroup of	1	-	0.37 (RAC)	0.59 (RAC)
					0.085 (flesh)	0.14 (flesh)
	FC 0003	Mandarins (including Mandarin-like hybrids), Subgroup of	1 ^a	-	0.37 (RAC)	0.59 (RAC)
					0.085 (flesh)	0.14 (flesh)
	FC 0004	Oranges, sweet, sour (including orange- like hybrids), Subgroup of	0.7 ^a	-	0.225 (RAC)	0.48 (RAC)
					0.052 (flesh)	0.11 (flesh)
	FC 0005	Pummelo and Grapefruits (including Shaddock-like hybrids), Subgroup of	0.5	-	0.19 (RAC)	0.32 (RAC)
					0.044 (flesh)	0.074 (flesh)
	FS 0014	Plums (including fresh Prunes), Subgroup of	0.05	0.8	0.025 (RAC)	0.040 (RAC)
	FB 0272	Raspberries, Red, Black	W	0.2		
	FB 2005	Cane berries, Subgroup of	3 ^a	-	0.84	1.4

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P	HR or HR-P
			New	Previous	mg/kg	mg/kg
	Alternative GAP data were available for plums, so the 2017 JMPR exceedances noted for FS 0014 Plums and dried plum no longer exist.					
	In addition, the current Meeting also concluded, based on the revised ARfD, that the estimated acute dietary exposure to residues of fenpyroximate for the consumption of commodities FP 0226 Apple, FP 0230 Pear, FS 0240 Apricot, VC 0046 Melons (except watermelon), VO 2045 Subgroup of Tomatoes, VO 2046 Subgroup of Eggplants, VP 2060 Subgroup of Beans with pods as previously considered by the 2017 and 2018 JMPRs may present a public health concern.					
	Definition of the residue for compliance with the MRL for plant commodities: fenpyroximate					
	Definition of the residue for dietary risk assessment for plant commodities and for dietary burden calculations: Sum of parent fenpyroximate and tert-butyl (Z)- α -(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneamino-oxy)-p-toluate (its Z-isomer M-1), expressed as fenpyroximate.					
	Definition of the residue for compliance with the MRL for animal commodities: Sum of fenpyroximate and (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl) methyleneaminooxymethyl]benzoic acid (M-3), expressed as fenpyroximate.					
	Definition of the residue for dietary risk assessment for animal commodities: Sum of fenpyroximate, 2-hydroxymethyl-2-propyl (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)-methyleneaminooxymethyl]benzoate (Fen-OH), 2-hydroxy-2-methylpropyl (E)- α -(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneamino-oxy)-p-toluate (R-UL-1) and (E)-4-[(1,3-dimethyl-5-phenoxy-pyrazol-4-yl)methyleneaminooxymethyl]benzoic acid (M-3), expressed as fenpyroximate.					
	The residue is fat-soluble.					
Fipronil (202)** a ADI: 0–0.0002 mg/kg bw ARfD: 0.03 mg/kg bw	FI 0327	Banana	0.004 *	0.005	0	0
	GC 0640	Barley	W	0.002*		
	GC 2087	Barley, similar grains, and pseudocereals with husks, Subgroup of	0.004 *		0.00536	
	AS 0640	Barley, straw and fodder dry Barley, hay and/or straw #	0.07 dw			
	HH 0722	Basil, leaves	0.8	1.5	0.23	0.57
	VD 2065	Dry beans, Subgroup of (except soya beans)	0.01		0.002	
	VB 0041	Cabbage, head	W	0.02		
	MO 1280	Cattle, kidney	W	0.02		
	MO 1281	Cattle, liver	W	0.1		
	MM 0812	Cattle meat	W	0.5 (fat)		
	ML 0812	Cattle milk	W	0.02		
	SO 0691	Cottonseed	0.01		0.002	
	PE 0112	Eggs	0.04	0.02	0.0358	0.06141
	VB 0042	Flowerhead Brassicas, Subgroup of	W	0.02		
	VL 0053	Leafy vegetables, Group of	0.01 ^b		0	0.02919
	VP 2060	Beans with pods, Subgroup of	0.01		0.008	0.0099
	GC 0645	Maize	W	0.01		
	GC 2091	Maize cereals, Subgroup of	0.01		0.004	
	AS 0645	Maize fodder (dry)	W	0.1 dw		
	GC 0647	Oats	W	0.002*		
	AS0647	Oat straw and fodder dry	0.07 dw			
	AS 3554 #	Oat, hay and/or straw #				
	VA 0385	Onion, bulb	0.03		0.02	0.033

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P mg/kg	HR or HR-P mg/kg
			New	Previous		
	VR 0589	Potato	0.05	0.02	0.00493	0.0296
	PF 0111	Poultry fats	0.07	0.02	0.04698	0.1006
	PM 0110	Poultry meat	0.007	0.01*	0.00486 muscle 0.04698 fat	0.01169m uscle 0.1006 fat
	PO 0111	Poultry, Edible offal of	0.03	0.02	0.03227 Liver	0.04231 Liver
	GC 0649	Rice	W	0.01		
	GC 2088	Rice cereals, Subgroup of	0.4		0.00411	
	CM 0649	Rice, husked	0.4		0.0023	
	CM 1205	Rice, polished	0.15		0.002	
	CM 1206	Rice bran, unprocessed	2		0.00323	
	CM 1207	Rice hulls	2			
	AS 0649	Rice straw and fodder, Dry <i>Rice, hay and/or straw</i> #	0.6 dw	0.2 dw		
	VR 0075	Root and Tuber vegetables, Group of (except potato and sugar beet)	0.002 ^b		0	0.00212
	GC 0650	Rye	W	0.002*		
	AS 0650	Rye straw and fodder dry	0.05 dw			
	AS 3555#	<i>Rye, hay and/or straw</i> #				
	VD 0541	Soya bean (dry)	0.01		0.00411	
	AB 0541	Soya bean hulls	0.06			
	AL 3538 #					
	AS 0081#	Straw and fodder (dry) of cereal grains (except of barley, oats, rice, rye, triticale and wheat)	0.03 ^b dw			
	VR 0596	Sugar beet	0.01	0.2	0.003	
	GS 0659	Sugar cane	0.01		0.00304	0.00815
	SO 0702	Sunflower seed	W	0.002*		
	SO 2091	Sunflower seeds, Subgroup of	0.004 *		0.008	
	VO 2045	Tomato, Subgroup of	0.01 *		0.008	0.008
	GC 0653	Triticale	W	0.002*		
	AS 0653	Triticale straw and fodder dry <i>Triticale, hay and/or straw</i> #	0.05 dw			
	GC 0654	Wheat	W	0.002*		
	GC2086	Wheat, similar grains, and pseudocereals with husks, Subgroup of	0.004 *		0.008	
	AS 0654	Wheat straw and fodder dry <i>Wheat, hay and/or straw</i> #	0.05 dw			
	MO 0105	Edible offal (Mammalian)	0.1		0.09145 Liver	0.32752 Liver
	MF 0100	Mammalian fats (except milk fats)	0.4		0.17625	0.65651
	MM 0095	Meat (from mammals other than marine mammals)	0.03		0.0085 muscle 0.17625 fat	0.04926 muscle 0.65651 fat
	ML 0106	Milks	0.03		0.00845	0.04321
	FM 0183	Milk fats	0.3		0.12	0.59
	OC 0541	Soya bean oil, crude	0.05		0.01808	
		Potato washed			0.00244	0.01465
		Potato peeled			0.00158	0.00947
		Potato, cooked peeled			0.00121	0.00725
		Potato, microwave unpeeled			0.00333	0.01998
		Potato, French fries			0.00182	0.01095

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P mg/kg	HR or HR-P mg/kg
			New	Previous		
		Potato flakes			0.00222	
	DV 0589 [#]	Potato, flakes/granules				
	CF 3513	Rice, flour			0.00053	
		Rice, polished cooked			0.00016	
		Rice polished steamed			0.00012	
		Sake			0.00008	
		Sugarcane juice			0.00182	
	DM 0659	Sugar cane molasses			0.00007	
	DM3524	Sugar cane, sugar refined			0.00007	
	OC 0691	Cotton seed oil, crude			0.0008	
	OR 0691	Cotton seed oil, edible			0.0006	
	^a On the basis of the information provided to the JMPR it was concluded that the estimated long-term dietary exposure to residues of fipronil may present a public health concern. ^b residues resulting from rotational cropping [#] New codes and/or commodity names as agreed by CCPR 52 and proposed for adoption by CAC 43; Definition of the residue for compliance with the MRL for plant and animal commodities: Sum of fipronil and 5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfonylpyrazole (MB46136) expressed in terms of fipronil. Definition of the residue for dietary risk assessment for plant and animal commodities: Sum of fipronil and 5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylsulfonylpyrazole (MB46136), 5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylthiopyrazole (MB45950) and 5-amino-3-cyano-1-(2,6-dichloro-4-trifluoromethylphenyl)-4-trifluoromethylpyrazole (MB46513) expressed in terms of fipronil. The residue is fat-soluble.					
Fluensulfone (265) ADI: 0–0.01 mg/kg bw ARfD: 0.3 mg/kg bw	FP 0009	Pome fruits, Group of (except Persimmon, Japanese)	0.3	0.2	0	0
	Definition of the residue for compliance with the MRL for plant commodities: the sum of fluensulfone and 3,4,4-trifluorobut-3-ene-1-sulfonic acid (BSA), expressed as fluensulfone equivalents. Definition of the residue for compliance with the MRL for animal commodities: fluensulfone Definition of the residue for dietary risk assessment for plant and animal commodities: fluensulfone The residue is fat-soluble.					
Flutianil (319)* ADI: 0–0.8 mg/kg bw ARfD: Unnecessary	FP 0226	Apple	0.15	-	0.047	
	FS 0013	Cherries, Subgroup of	0.4	-	0.11	
	FB 2008	Small fruit vine climbing, Subgroup of	0.7	-	0.075	
	JF 0226	Apple, juice			0.005	
	JF 0269	Grape, juice			0.05	
	DF 0269	Grape, dried (=Currants, Raisins and Sultanas)			0.09	

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P	HR or HR-P
			New	Previous	mg/kg	mg/kg
	Definition of the residue for compliance with the MRL for plant and animal commodities: <i>Flutianil</i>					
	Definition of the residue for dietary risk assessment for plant commodities: <i>Sum of flutianil and 2-fluoro-5-(trifluoromethyl)benzenesulfonic acid (OC 56635), expressed as flutianil</i>					
	<i>The residue is fat-soluble.</i>					
Isoprothiolane (299) ADI: 0–0.1 mg/kg bw ARfD: Unnecessary						
Mefentrifluconazole (320)* ADI: 0–0.04 mg/kg bw ARfD: 0.3 mg/kg bw						
Metalaxyl (138)**^a ADI: 0–0.08 mg/kg bw ^b ARfD: 0.5 mg/kg bw ^b	FP 0226	Apple	0.02* (MM)		0	0
	VS 0621	Asparagus	W	0.05*		
	FI 0326	Avocado	W	0.2		
	VB 0400	Broccoli	W	0.5		
	VB 0402	Brussels sprouts	0.15 (M)	0.2	0.44	0.77
	VB 0041	Cabbages, Head	0.08 (MM)	0.5	0.22	0.44
	SB 0715	Cacao bean	W	0.2		
	VR 0577	Carrot	0.02* (MM)	0.05*	0.02	0.02
	VB 0404	Cauliflower	W	0.5		
	GC 0080	Cereal grains	W	0.05		
	FC 0001	Citrus fruits, Group of	W	5		
	SO 0691	Cottonseed	W	0.05*		
	VC 0424	Cucumber	W	0.5		
	VB 0042	Flowered brassicas, Subgroup of	0.2 (M)		0.275	1.21
	VC 0425	Gherkin	W	0.5		
	VR 0604	Ginseng	0.03* (MM)		0.03	0.03
	FB 0269	Grapes	1.5 (MM)	1	0.182	0.884
	JF 0269	Grape, juice			0.073	
		Grape wine			0.138	
	DH 1100	Hops, dry	W	10		
	VL 0482	Lettuce, head	W	2		
	VL 0483	Lettuce, leaf	1.5 (M)		1.43	8.14
	VC 0046	Melons, except Watermelon	0.15 (MM)	0.2	0.013	0.026
	VA 0385	Onion, Bulb	0.03 (MM)	2	0.02	0.02
	FC 0004	Oranges, Sweet, Sour, Subgroup of	0.7 (M)		0.013 (flesh) 0.338 (RAC)	0.026 (flesh) 0.39 (RAC)
	JF 0004	Orange juice			0.016	
		Orange marmalade			0.132	
	OR 0004	Orange oil, edible	7		3.04	

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P mg/kg	HR or HR-P mg/kg
			New	Previous		
	SO 0697	Peanut	W	0.1		
	FP 0230	Pear	0.02* (MM)		0	0
	VP 0064	Peas, shelled (succulent seeds)	W	0.05*		
	VO 0051	Peppers	W	1		
	HS 0790	Pepper, black, White, pink, green	2 (MM)		0.455	
	VO 0444	Peppers Chili, dried	W	10		
	FP 0009	Pome fruits	W	1		
	VR 0589	Potato	0.02 (M)	0.05*	0.01	0.02
	FB 0272	Raspberries, red, black	W	0.2		
	VD 0451	Soya bean (dry)	W	0.05*		
	HS 0190	Spices, seeds	W	5 (Mt)		
	VL 0502	Spinach	0.02* (MM)	2	0.22	0.22
	VC 0431	Squash, summer	W	0.2		
	VR 0596	Sugar beet	W	0.05*		
	SO 0702	Sunflower seed	0.01* (MM)	0.05*	0	
	VO 0448	Tomato	W	0.5		
	VO 2045	Tomatoes, Subgroup of	0.3 (MM)		0.058	0.234
	VC 0432	Watermelon	W	0.2		
	VC 0433	Winter squash	W	0.2		
	RAC: Raw Agricultural Commodity					
	^a Residue data that was the basis for the estimation: metalaxyl (M), metalaxyl-M (MM) or monitoring (Mt)					
	^b Applies to metalaxyl and metalaxyl-M (alone or in combination)					
	Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL for plant commodities: <i>metalaxyl (sum of enantiomers).</i>					
	Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in plant commodities: <i>metalaxyl (sum of enantiomers) and N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (M8; free and conjugated; sum of enantiomers), expressed as metalaxyl.</i>					
	Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL in animal commodities is: <i>the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.</i>					
	Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in animal commodities is: <i>the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M1 (N-(2,6-dimethylphenyl)-N-(methoxyacetyl) alanine), M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester), M6 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine), M7 (N-(2,6-dimethyl- 5-hydroxyphenyl)-N-(methoxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.</i>					
	<i>The residue is not fat-soluble.</i>					
Metalaxyl-M (212) ADI: 0–0.08 mg/kg bw ^b ARfD: 0.5 mg/kg bw ^b	FP 0226	Apple	W	0.02 *		
	SB 0715	Cacao beans	W	0.02		
	FB 0269	Grapes	W	1		
	VL 0482	Lettuce, head	W	0.5		
	VA 0385	Onion, bulb	W	0.03		
	VO 0445	Peppers, sweet (including pimento or pimienta)	W	0.5		
	VR 0589	Potato	W	0.02 *		

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P	HR or HR-P
			New	Previous	mg/kg	mg/kg
	VL 0502	Spinach	W	0.1		
	SO 0702	Sunflower seed	W	0.02 *		
	VO 0448	Tomato	W	0.2		
	^b Applies to metalaxyl and metalaxyl-M (alone or in combination)					
	Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL for plant commodities: <i>metalaxyl (sum of enantiomers)</i> .					
	Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in plant commodities: <i>metalaxyl (sum of enantiomers) and N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (M8; free and conjugated; sum of enantiomers), expressed as metalaxyl.</i>					
	Residue definition for metalaxyl and metalaxyl-M for compliance with the MRL in animal commodities is: <i>the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.</i>					
	Residue definition for metalaxyl and metalaxyl-M for dietary risk assessment in animal commodities is: <i>the sum of metalaxyl (sum of enantiomers) and metabolites (free + conjugated) M1 (N-(2,6-dimethylphenyl)-N-(methoxyacetyl) alanine), M3 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine methyl ester), M6 (N-(2,6-dimethylphenyl)-N-(hydroxyacetyl)alanine), M7 (N-(2,6-dimethyl- 5-hydroxyphenyl)-N-(methoxyacetyl)alanine methyl ester) and M8 (N-(2-hydroxymethyl-6-methylphenyl)-N-(methoxyacetyl)alanine methyl ester (sum of enantiomers), expressed as metalaxyl.</i>					
	<i>The residue is not fat-soluble.</i>					
Metconazole (313) ADI: 0–0.04 mg/kg bw ARfD: 0.04 mg/kg bw Triazole alanine and Triazole acetic ADI: 0–1 mg/kg bw ARfD: Unnecessary 1,2,4-triazole ADI: 0–0.2 mg/kg bw ARfD: 0.3 mg/kg bw	GC 0654	Wheat	0.15		0.035	
	GC 0653	Triticale	0.15		0.035	
	CM 0654	Wheat bran, unprocessed	0.3		0.067	
	CF 1212	Wheat, wholemeal			0.026	
	CF 1211	Wheat, flour			0.008	
	CF 1210	Wheat, germ			0.035	
	CP 1211	White bread			0.021	
	Definition of the residue for compliance with the MRL for plant and animal commodities: <i>Metconazole (sum of cis and trans isomer)</i> .					
	Definition of the residue for dietary risk assessment for plant commodities: <i>Metconazole (sum of cis and trans isomer)</i> .					
	Definition of the residue for dietary risk assessment for animal commodities: <i>Sum of metconazole (cis and trans-isomer) and metabolites (1SR,2SR,5RS)-5-(4-chlorobenzyl)-2-(hydroxymethyl)-2-methyl-1-(1H-1,2,4-triazol-1-ylmethyl)cyclopentanol (M1; free and conjugated) and (1RS,2SR,3RS)-3-(4-chlorobenzyl)-2-hydroxy-1-methyl-2-(1H-1,2,4-triazol-1-ylmethyl)cyclopentanecarboxylic acid (M12; free and conjugated), expressed as metconazole.</i>					

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P	HR or HR-P
			New	Previous	mg/kg	mg/kg
	The residue is not fat-soluble					
Pendimethalin (292) ADI: 0–0.1 mg/kg bw ARfD: 1 mg/kg bw	FB 0269	Grapes	0.05*	-	0.05	0.05
	VA 0384	Leek	0.3	-	0.02	0.23
	VB 0042	Flowerhead Brassicas, Subgroup of	0.01*	-	0	0
	VO 0050	Fruiting vegetables, other than Cucurbits, Group of	0.05*	-	0.05	0.05
	VD 0541	Soya bean (dry)	0.05*	-	0.05	-
	GC 0654	Wheat	0.01*	-	0.01	-
	GC 0649	Rice	0.01*	-	0	-
	GC 0645	Maize	0.05*	-	0.05	-
	GS 0659	Sugar cane	0.01*	-	0	-
	SO 0702	Sunflower Seed	0.05*	-	0.05	-
	HH 0740	Parsley, leaves	1.5	-	0.305	0.76
	AS 0654	Wheat straw and fodder dry Wheat, hay and/or straw #	0.3	-	0.05	0.17
	AS 0649	Rice straw and fodder, dry Rice, hay and/or straw #	0.01*	-	0	0
	AS 0645	Maize fodder (dry)	0.05*	-	0.05	0.05
	AS 3552 #	Maize, hay and/or straw #	0.05*	-	0.05	0.05
	# New codes and/or commodity names as agreed by CCPR 52 and proposed for adoption by CAC43					
	Definition of the residue for compliance with the MRL and dietary risk assessment for plant and animal commodities: Pendimethalin.					
	The residue is fat-soluble.					
Pyrasulfotole (321)* ADI: 0–0.01 mg/kg bw ARfD: Unnecessary	GC 0640	Barley	0.03	-	0.02	-
	AS 0640	Barley straw and fodder, dry	0.8 (dw)	-	0.21 ^a Hay (dw)	0.50 ^b Hay (dw)
		Barley, hay and/or straw #	-	-	0.105 ^a Straw (dw)	0.38 ^b Straw (dw)
	MO 0105	Edible offal (Mammalian)	0.5	-	0.084	-
	PE 0112	Eggs	0.02*	-	0	-
	MF 0100	Mammalian fats (except milk fats)	0.02*	-	0.02	-
	MM 0095	Meat (from mammals other than marine mammals)	0.02*	-	0.02 muscle 0.02 fat	-
	ML 0106	Milks	0.01*	-	0.01	-
	GC 0647	Oats	0.15	-	0.02	-
	AS 0647	Oat straw and fodder, dry	-	-	0.21 ^a Hay (dw)	0.50 ^b Hay (dw)
	AS 3554 #	Oat, hay and/or straw #	0.8 (dw)	-	0.105 ^a Straw (dw)	0.38 ^b Straw (dw)

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P mg/kg	HR or HR-P mg/kg
			New	Previous		
	PM 0110	Poultry meat	0.02*		0.02 muscle 0.02 fat	
	PO 0111	Poultry, Edible offal of	0.05		0.02	
	PF 0111	Poultry fats	0.02*		0.02	
	GC 0650	Rye	0.02*		0.02	
	AS 0650	Rye, straw and fodder, dry	0.8 (dw)		0.105 ^a	0.38 ^b
		Rice, hay and/or straw [#]			Straw (dw)	Straw (dw)
	GC 0651	Sorghum Grain	0.5		0.091	
	GC 0653	Triticale	0.02*		0.02	
	AS 0653	Triticale, straw and fodder, dry	0.8 (dw)		0.21 ^a Hay (dw)	0.50 ^b Hay (dw)
		Triticale, hay and/or straw [#]			0.105 ^a Straw (dw)	0.38 ^b Straw (dw)
	CM 0654	Wheat bran, unprocessed	0.03		0.028	
	GC 0654	Wheat	0.02*		0.02	
	AS 0654	Wheat straw and fodder, dry	0.8 (dw)		0.21 ^a Hay (dw)	0.50 ^b Hay (dw)
		Wheat, hay and/or straw [#]			0.105 ^a Straw (dw)	0.38 ^b Straw (dw)
	CF 1211	Wheat, flour			0.02	
	CF 1210	Wheat germ			0.016	
# New codes and/or commodity names as agreed by CCPR 52 and proposed for adoption by CAC43						
^a Median						
^b Highest						
Definition of the residue for compliance with the MRL and for dietary risk assessment for plant commodities: Sum of pyrasulfotole and desmethyl-pyrasulfotole and its conjugates, expressed as pyrasulfotole.						
Definition of the residue for compliance with the MRL and for dietary risk assessment for animal commodities: Sum of pyrasulfotole and desmethyl-pyrasulfotole, expressed as pyrasulfotole.						
The residue is not fat-soluble.						
Pyrasulfotole (322)* ADI: 0–0.02 mg/kg bw ARfD: 2 mg/kg bw	FP 0226	Apple	1.5		0.36	0.73
	JF 0226	Apple juice			0.036	-
	DF 0269	Grape, dried (=Currants, Raisins and Sultanas)	6		1.14	1.96
	MO 0105	Edible offal (mammalian) ^a			0.005	0.005
	FB 0269	Grapes	3		0.57	0.98
	JF 0269	Grape juice			0.057	-
	MF 0100	Mammalian fats (except milk fats) ^a			0.002	0.002
	MM 0095	Meat (from mammals other than marine mammals) ^a			0.0005 (muscle)	0.0005 (muscle)
					0.002 (fat)	0.002 (fat)
	ML 0106	Milks ^a			0.0001	-
	FP 0230	Pear	1.5		0.36	0.73
	FP 0307	Persimmon, Japanese	1.5		0.36	0.73

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P	HR or HR-P
			New	Previous	mg/kg	mg/kg
	^a No maximum residue level recommendation due to the absence of an enforcement method					
	Definition of the residue for compliance with the MRL for plant commodities: <i>pyraziflumid</i> .					
	Definition of the residue for dietary risk assessment for plant commodities: <i>pyraziflumid</i> .					
	Definition of the residue for compliance with the MRL for animal commodities: <i>Pyraziflumid and its pyraziflumid-4'-OH metabolite (free), expressed as pyraziflumid</i> .					
	Definition of the residue for dietary risk assessment for animal commodities: <i>pyraziflumid and its pyraziflumid-4'-OH metabolite (free and conjugated), expressed as pyraziflumid</i> .					
	<i>The residue is fat-soluble.</i>					
Spiropidion (323)* ADI: 0–0.02 mg/kg bw ARfD: 0.3 mg/kg bw	VC 0424	Cucumber	0.8		0.34	0.7
	VC 0046	Melons (except watermelon)	0.9		0.25	0.91
	VC 0429	Pumpkins	0.9		0.25	0.91
	VC 0432	Watermelon	0.9		0.25	0.91
	VC 0433	Winter squash	0.9		0.25	0.91
	VO 0448	Tomato	0.8		0.245	0.7
	VO 0051	Peppers, Subgroup of (except martynia, okra, roselle)	1		0.49	1.2
	HS 0444	Peppers, Chili, dried	7		2.905	7
	VD 0541	Soya bean (dry)	3		0.49	
	VW 0448	Tomato paste	1.5		0.46 paste	
	DM 0448#	Tomato, puree#			0.27 puree	
	VR 0589	Potato	1.5		0.28	0.98
	DV 0448	Tomato (dried)	7		1.7	4.8
		Soya flour	5		0.79	
	AB 1265	Soya bean meal	5		0.62	
	AL 3539 #					
		Potato, flakes	5		0.67	
	DV 0589#	Potato, flakes/granules#				
	MO 0105	Edible offal (mammalian)	0.2		0.098 kidney	0.2 kidney
	MF 0100	Mammalian fats (except milk fats)	0.025		0.013	0.021
	MM 0095	Meat (from mammals other than marine mammals)	0.012 *		0.0065 muscle 0.013 fat	0.01 muscle 0.021 fat
	ML 0106	Milks	0.012 *		0.0057	
	PE 0112	Eggs	0.012 *		0.00089	0.00089
	PM 0110	Poultry meat	0.012 *		0.00041 muscle 0.00035 fat	0.00041m uscle 0.00035 fat
	PO 0111	Poultry, edible offal of	0.012 *		0.0033	0.0033
	PF 0111	Poultry fat	0.012 *		0.00035	0.00035
	JF 0448	Tomato juice			0.19	
		Canned tomatoes (peeled)			0.15	0.44
OR 0541	Soya bean oil, refined			0.01		
	Soya milk			0.039		

Pesticide (Codex reference number)	CCN	Commodity	Recommended Maximum residue level (mg/kg)		STMR or STMR-P mg/kg	HR or HR-P mg/kg
			New	Previous		
		Tofu			0.051	
		Soy sauce			0.02	
		Miso			0.098	
		Potato (peeled)			0.37	1.3
		Potato crisps			0.23	
		Potato (baked, with peel)			0.55	2
		Potato fries (without peel)			0.2	
		Potato starch			0.16	
# New codes and/or commodity names as agreed by CCPR 52 and proposed for adoption by CAC 43;						
Definition of the residue for compliance with the MRL for plant commodities: <i>the sum of spiropidion and spiropidion-enol (SYN547305) expressed as spiropidion</i>						
Definition of the residue for dietary risk assessment for plant commodities: <i>the sum of spiropidion, spiropidion-enol (SYN547305), 3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-8-methoxy-1,8-diazaspiro[4.5]dec-3-en-2-one (SYN547435) and 3-(4-chloro-2,6-dimethyl-phenyl)-4-hydroxy-1-methyl-1,8-diazaspiro[4.5]dec-3-en-2-one (SYN548430), expressed as spiropidion.</i>						
Definition of the residue for compliance with the MRL for animal commodities: <i>spiropidion-enol (SYN547305) expressed as spiropidion.</i>						
Definition of the residue for dietary risk assessment for animal commodities: <i>free and conjugated spiropidion-enol (SYN547305) expressed as spiropidion.</i>						
<i>The residue is not fat-soluble.</i>						
Tetraniliprole (324)*						
ADI: 0–2 mg/kg bw ARfD: Unnecessary						

Annex 2: Index of reports and evaluations of pesticides by the JMPR

Numbers in parentheses after the names of pesticides are Codex classification numbers. The abbreviations used are:

T, evaluation of toxicology

R, evaluation of residue and analytical aspects

E, evaluation of effects on the environment

Abamectin (177)	1992 (T,R), 1994 (T,R), 1995 (T), 1997 (T,R), 2000 (R), 2015 (R), 2017 (T), 2018 (R)
Acephate (095)	1976 (T,R), 1979 (R), 1981 (R), 1982 (T), 1984 (T,R), 1987 (T), 1988 (T), 1990 (T,R), 1991 (corr. to 1990 R evaluation), 1994 (R), 1996 (R), 2002 (T), 2003 (R), 2004 (corr. to 2003 report), 2005 (T), 2006 (R), 2011 (R)
Acetamiprid (246)	2011 (T, R), 2012 (R), 2015 (R), 2017 (R), 2021 (R)
Acetochlor (280)	2015 (T, R), 2019 (T, R)
Acibenzolar-S-methyl (288)	2016 (T, R)
Acrylonitrile	1965 (T, R)
Afidopyropen (312)	2019 (T, R), 2021 (R)
Aldicarb (117)	1979 (T, R), 1982 (T,R), 1985 (R), 1988 (R), 1990 (R), 1991 (corr. to 1990 evaluation), 1992 (T), 1993 (R), 1994 (R), 1996 (R), 2001 (R), 2002 (R), 2006 (R)
Aldrin (001)	1965 (T), 1966 (T, R), 1967 (R), 1974 (R), 1975 (R), 1977 (T), 1990 (R), 1992 (R)
Allethrin	1965 (T, R)
Ametoctradin (253)	2012 (T, R)
Aminocarb (134)	1978 (T, R), 1979 (T, R)
Aminocyclopyrachlor (272)	2014 (T, R)
Aminomethylphosphonic acid (AMPA, 198)	1997 (T, R)
Aminopyralid (220)	2006 (T, R), 2007 (T,R)
Amitraz (122)	1980 (T, R), 1983 (R), 1984 (T,R), 1985 (R), 1986 (R), 1989 (R), 1990 (T,R), 1991 (R & corr. to 1990 R evaluation), 1998 (T)
Amitrole (079)	1974 (T, R), 1977 (T), 1993 (T, R), 1997 (T), 1998 (R)
Anilazine (163)	1989 (T, R), 1992 (R)
Atrazine	2007 (T)
Azinphos-ethyl (068)	1973 (T, R), 1983 (R)
Azinphos-methyl (002)	1965 (T), 1968 (T, R), 1972 (R), 1973 (T), 1974 (R), 1991 (T, R), 1992 (corr. to 1991 report), 1993 (R), 1995 (R), 2007 (T)
Azocyclotin (129)	1979 (R), 1981 (T), 1982 (R), 1983 (R), 1985 (R), 1989 (T,R), 1991 (R), 1994 (T), 2005 (T,R)
Azoxystrobin (229)	2008 (T,R), 2011 (R), 2012 (R), 2013 (R), 2017 (R), 2019 (R)
Benalaxyl (155)	1986 (R), 1987 (T), 1988 (R), 1992 (R), 1993 (R), 2005 (T), 2009 (R)
Bendiocarb (137)	1982 (T,R), 1984 (T,R), 1989 (R), 1990 (R)
Benomyl (069)	1973 (T,R), 1975 (T,R), 1978 (T,R), 1983 (T,R), 1988 (R), 1990 (R), 1994 (R), 1995 (T,E), 1998 (R)

Bentazone (172)	1991 (T,R), 1992 (corr. to 1991 report, Annex I), 1994 (R), 1995 (R), 1998 (T,R), 1999 (corr. to 1998 report), 2004 (T), 2012 (T), 2013 (R), 2016 (T), 2018 (R)
Benzovindiflupyr (261)	2013 (T), 2014 (R), 2016 (R), 2019 (R)
BHC (technical-grade)	1965 (T), 1968 (T,R), 1973 (T,R) (see also Lindane)
Bicyclopyrone (295)	2017 (T, R)
Bifenazate (219)	2006 (T,R), 2008 (R), 2010 (R)
Bifenthrin (178)	1992 (T,R), 1995 (R), 1996 (R), 1997 (R), 2009 (T), 2010 (R), 2015 (R), 2019 (R)
Binapacryl (003)	1969 (T,R), 1974 (R), 1982 (T), 1984 (R), 1985 (T,R)
Bioresmethrin (093)	1975 (R), 1976 (T,R), 1991 (T,R)
Biphenyl	See Diphenyl
Bitertanol (144)	1983 (T), 1984 (R), 1986 (R), 1987 (T), 1988 (R), 1989 (R), 1991 (R), 1998 (T), 1999 (R), 2002 (R)
Bixafen (262)	2013 (T,R), 2016 (R), 2021 (R)
Boscalid (221)	2006 (T,R), 2008 (R), 2010 (R), 2019 (T, R)
Bromide ion (047)	1968 (R), 1969 (T, R), 1971 (R), 1979 (R), 1981 (R), 1983 (R), 1988 (T, R), 1989 (R), 1992 (R)
Bromomethane (052)	1965 (T, R), 1966 (T,R), 1967 (R), 1968 (T,R), 1971 (R), 1979 (R), 1985 (R), 1992 (R)
Bromophos (004)	1972 (T,R), 1975 (R), 1977 (T,R), 1982 (R), 1984 (R), 1985 (R)
Bromophos-ethyl (005)	1972 (T,R), 1975 (T,R), 1977 (R)
Bromopropylate (070)	1973 (T,R), 1993 (T,R)
Butocarboxim (139)	1983 (R), 1984 (T), 1985 (T), 1986 (R)
Buprofezin (173)	1991 (T,R), 1995 (R), 1996 (corr. to 1995 report.), 1999 (R), 2008 (T,R), 2009 (R), 2012 (R), 2014 (R), 2016 (R), 2019 (T, R)
sec-Butylamine (089)	1975 (T,R), 1977 (R), 1978 (T,R), 1979 (R), 1980 (R), 1981 (T), 1984 (T,R: withdrawal of temporary ADI, but no evaluation)
Cadusafos (174)	1991 (T,R), 1992 (R), 1992 (R), 2009 (R), 2010 (R)
Campheclor (071)	1968 (T,R), 1973 (T,R)
Captafol (006)	1969 (T,R), 1973 (T,R), 1974 (R), 1976 (R), 1977 (T,R), 1982 (T), 1985 (T,R), 1986 (corr. to 1985 report), 1990 (R), 1999 (ARfD)
Captan (007)	1965 (T), 1969 (T,R), 1973 (T), 1974 (R), 1977 (T,R), 1978 (T,R), 1980 (R), 1982 (T), 1984 (T,R), 1986 (R), 1987 (R and corr. to 1986 R evaluation), 1990 (T,R), 1991 (corr. to 1990 R evaluation), 1994 (R), 1995 (T), 1997 (R), 2000 (R), 2004 (T), 2007 (T), 2017 (R)
Carbaryl (008)	1965 (T), 1966 (T,R), 1967 (T,R), 1968 (R), 1969 (T,R), 1970 (R), 1973 (T,R), 1975 (R), 1976 (R), 1977 (R), 1979 (R), 1984 (R), 1996 (T), 2001 (T), 2002 (R), 2007 (R)
Carbendazim (072)	1973 (T,R), 1976 (R), 1977 (T), 1978 (R), 1983 (T,R), 1985 (T,R), 1987 (R), 1988 (R), 1990 (R), 1994 (R), 1995 (T,E), 1998 (T,R), 2003 (R), 2005 (T), 2012 (R)
Carbofuran (096)	1976 (T,R), 1979 (T,R), 1980 (T), 1982 (T), 1991 (R), 1993 (R), 1996 (T), 1997 (R), 1999 (corr. to 1997 report), 2002 (T,R), 2003 (R) (See also carbosulfan), 2004 (R), 2008 (T), 2009 (R)
Carbon disulfide (009)	1965 (T,R), 1967 (R), 1968 (R), 1971 (R), 1985 (R)

Carbon tetrachloride (010)	1965 (T,R), 1967 (R), 1968 (T,R), 1971 (R), 1979 (R), 1985 (R)
Carbophenothion (011)	1972 (T,R), 1976 (T,R), 1977 (T,R), 1979 (T,R), 1980 (T,R), 1983 (R)
Carbosulfan (145)	1984 (T,R), 1986 (T), 1991 (R), 1992 (corr. to 1991 report), 1993 (R), 1997 (R), 1999 (R), 2002 (R), 2003 (T,R), 2004 (R, corr. to 2003 report)
Cartap (097)	1976 (T,R), 1978 (T,R), 1995 (T,R)
Chinomethionat (080)	1968 (T,R) (as oxythioquinox), 1974 (T,R), 1977 (T,R), 1981 (T,R), 1983 (R), 1984 (T,R), 1987 (T)
Chlorantraniliprole (230)	2008 (T,R), 2010 (R), 2013 (R), 2014 (R), 2016 (R), 2019 (R)
Chlorbenside	1965 (T)
Chlordane (012)	1965 (T), 1967 (T,R), 1969 (R), 1970 (T,R), 1972 (R), 1974 (R), 1977 (T,R), 1982 (T), 1984 (T,R), 1986 (T)
Chlordimeform (013)	1971 (T,R), 1975 (T,R), 1977 (T), 1978 (T,R), 1979 (T), 1980 (T), 1985 (T), 1986 (R), 1987 (T)
Chlorfenapyr (254)	2013 (T), 2018 (T,R)
Chlorfenson	1965 (T)
Chlorfenvinphos (014)	1971 (T,R), 1984 (R), 1994 (T), 1996 (R)
Chlormequat (015)	1970 (T,R), 1972 (T,R), 1976 (R), 1985 (R), 1994 (T,R), 1997 (T), 1999 (ARfD), 2000 (R), 2017 (T, R)
Chlorobenzilate (016)	1965 (T), 1968 (T,R), 1972 (R), 1975 (R), 1977 (R), 1980 (T)
Chloropicrin	1965 (T,R)
Chloropropylate	1968 (T,R), 1972 (R)
Chlorothalonil (081)	1974 (T,R), 1977 (T,R), 1978 (R), 1979 (T,R), 1981 (T,R), 1983 (T,R), 1984 (corr. to 1983 report and T evaluation), 1985 (T,R), 1987 (T), 1988 (R), 1990 (T,R), 1991 (corr. to 1990 evaluation), 1992 (T), 1993 (R), 1997 (R), 2009 (T), 2010 (R), 2012 (R), 2015 (R), 2019 (T, R)
Chlorpropham (201)	1965 (T), 2000 (T), 2001 (R), 2005 (T), 2008 (R)
Chlorpyrifos (017)	1972 (T,R), 1974 (R), 1975 (R), 1977 (T,R), 1981 (R), 1982 (T,R), 1983 (R), 1989 (R), 1995 (R), 1999 (T), 2000 (R), 2004 (R), 2006 (R), 2021 (R)
Chlorpyrifos-methyl (090)	1975 (T,R), 1976 (R, Annex I only), 1979 (R), 1990 (R), 1991 (T,R), 1992 (T and corr. to 1991 report), 1993 (R), 1994 (R), 2001 (T), 2009 (R)
Chlorthion	1965 (T)
Clethodim (187)	1994 (T,R), 1997 (R), 1999 (R), 2002 (R), 2019 (T, R)
Clofentezine (156)	1986 (T,R), 1987 (R), 1989 (R), 1990 (R), 1992 (R), 2005 (T), 2007 (R), 2021 (R)
Clothianidin (238)	2010 (T,R), 2011 (R), 2014 (R), 2021 (R)
Coumaphos (018)	1968 (T,R), 1972 (R), 1975 (R), 1978 (R), 1980 (T,R), 1983 (R), 1987 (T), 1990 (T,R)
Crufomate (019)	1968 (T,R), 1972 (R)
Cyanophenfos (091)	1975 (T,R), 1978 (T: ADI extended, but no evaluation), 1980 (T), 1982 (R), 1983 (T)
Cyantraniliprole (263)	2013 (T,R), 2015 (R), 2018 (R)
Cyazofamid (281)	2015 (T, R), 2018 (R)
Cyclaniliprole (296)	2017 (T, R), 2019 (R)

Cycloxydim (179)	1992 (T,R), 1993 (R), 2009 (T), 2012 (R)
Cyflumetofen (273)	2014 (T,R)
Cyfluthrin (157)	1986 (R), 1987 (T and corr. to 1986 report), 1989 (R), 1990 (R), 1992 (R), 2006 (T), 2007 (R)
Cyhalothrin (including lambda-cyhalothrin(146)	1984 (T,R), 1986 (R), 1988 (R), 2007 (T), 2008 (R), 2015 (R), 2018 (T), 2019 (R)
Cyhexatin (067)	1970 (T,R), 1973 (T,R), 1974 (R), 1975 (R), 1977 (T), 1978 (T,R), 1980 (T), 1981 (T), 1982 (R), 1983 (R), 1985 (R), 1988 (T), 1989 (T), 1991 (T,R), 1992 (R), 1994 (T), 2005 (T,R)
Cypermethrin (118)	1979 (T,R), 1981 (T,R), 1982 (R), 1983 (R), 1984 (R), 1985 (R), 1986 (R), 1987 (corr. to 1986 evaluation), 1988 (R), 1990 (R), 2006 (T), 2008 (R), 2009 (R), 2011 (R), 2019 (R), 2021 (R)
Cyproconazole (239)	2010 (T,R), 2013 (R)
Cyprodinil (207)	2003 (T,R), 2004 (corr. to 2003 report), 2013 (R), 2015 (R), 2017 (R), 2018 (R), 2019 (T, R), 2021 (R)
Cyromazine (169)	1990 (T,R), 1991 (corr. to 1990 R evaluation), 1992 (R), 2006 (T), 2007 (R), 2012 (R)
2,4-D (020)	1970 (T,R), 1971 (T,R), 1974 (T,R), 1975 (T,R), 1980 (R), 1985 (R), 1986 (R), 1987 (corr. to 1986 report, Annex I), 1996 (T), 1997 (E), 1998 (R), 2001 (R), 2017 (R)
Daminozide (104)	1977 (T,R), 1983 (T), 1989 (T,R), 1991 (T)
DDT (021)	1965 (T), 1966 (T,R), 1967 (T,R), 1968 (T,R), 1969 (T,R), 1978 (R), 1979 (T), 1980 (T), 1983 (T), 1984 (T), 1993 (R), 1994 (R), 1996 (R)
Deltamethrin (135)	1980 (T,R), 1981 (T,R), 1982 (T,R), 1984 (R), 1985 (R), 1986 (R), 1987 (R), 1988 (R), 1990 (R), 1992 (R), 2000 (T), 2002 (R), 2016 (R)
Demeton (092)	1965 (T), 1967 (R), 1975 (R), 1982 (T)
Demeton-S-methyl (073)	1973 (T,R), 1979 (R), 1982 (T), 1984 (T,R), 1989 (T,R), 1992 (R), 1998 (R)
Demeton-S-methylsulfon (164)	1973 (T,R), 1982 (T), 1984 (T,R), 1989 (T,R), 1992 (R)
Dialifos (098)	1976 (T,R), 1982 (T), 1985 (R)
Diazinon (022)	1965 (T), 1966 (T), 1967 (R), 1968 (T,R), 1970 (T,R), 1975 (R), 1979 (R), 1993 (T,R), 1994 (R), 1996 (R), 1999 (R), 2001 (T), 2006 (T,R), 2016 (T)
1,2-Dibromoethane (023)	1965 (T,R), 1966 (T,R), 1967 (R), 1968 (R), 1971 (R), 1979 (R), 1985 (R)
Dicamba (240)	2010 (T,R), 2011 (R), 2012 (R), 2013 (R), 2019 (T, R)
Dichlobenil (274)	2014 (T,R)
Dicloran (083)	2003 (R)
Dichlorfluanid (082)	1969 (T,R), 1974 (T,R), 1977 (T,R), 1979 (T,R), 1981 (R), 1982 (R), 1983 (T,R), 1985 (R)
1,2-Dichloroethane (024)	1965 (T,R), 1967 (R), 1971 (R), 1979 (R), 1985 (R)
Dichlorvos (025)	1965 (T,R), 1966 (T,R), 1967 (T,R), 1969 (R), 1970 (T,R), 1974 (R), 1977 (T), 1993 (T,R), 2011 (T), 2012 (R)
Dicloran (083)	1974 (T,R), 1977 (T,R), 1998 (T,R)
Dicofol (026)	1968 (T,R), 1970 (R), 1974 (R), 1992 (T,R), 1994 (R), 2011 (T), 2012 (R)

Dieldrin (001)	1965 (T), 1966 (T,R), 1967 (T,R), 1968 (R), 1969 (R), 1970 (T,R), 1974 (R), 1975 (R), 1977 (T), 1990 (R), 1992 (R)
Difenoconazole (224)	2007 (T,R), 2010 (R), 2013 (R), 2015 (R), 2017 (R), 2021 (R)
Diflubenzuron (130)	1981 (T,R), 1983 (R), 1984 (T,R), 1985 (T,R), 1988 (R), 2001 (T), 2002 (R), 2011 (R)
Dimethenamid-P (214)	2005 (T,R)
Dimethipin (151)	1985 (T,R), 1987 (T,R), 1988 (T,R), 1999 (T), 2001 (R), 2004 (T)
Dimethoate (027)	1965 (T), 1966 (T), 1967 (T,R), 1970 (R), 1973 (R in evaluation of formothion), 1977 (R), 1978 (R), 1983 (R), 1984 (T,R), 1986 (R), 1987 (T,R), 1988 (R), 1990 (R), 1991 (corr. to 1990 evaluation), 1994 (R), 1996 (T), 1998 (R), 2003 (T,R), 2004 (corr. to 2003 report), 2006 (R), 2008 (R), 2019 (T, R)
Dimethomorph (225)	2007 (T,R), 2014 (R), 2016 (R)
Dimethrin	1965 (T)
Dinocap (087)	1969 (T,R), 1974 (T,R), 1989 (T,R), 1992 (R), 1998 (R), 1999 (R), 2000 (T), 2001 (R)
Dinotefuran (255)	2012 (T,R)
Dioxathion (028)	1968 (T,R), 1972 (R)
Diphenyl (029)	1966 (T,R), 1967 (T)
Diphenylamine (030)	1969 (T,R), 1976 (T,R), 1979 (R), 1982 (T), 1984 (T,R), 1998 (T), 2001 (R), 2003 (R), 2008 (R)
Diquat (031)	1970 (T,R), 1972 (T,R), 1976 (R), 1977 (T,R), 1978 (R), 1994 (R), 2013 (T,R), 2018 (R)
Disulfoton (074)	1973 (T,R), 1975 (T,R), 1979 (R), 1981 (R), 1984 (R), 1991 (T,R), 1992 (corr. to 1991 report, Annex I), 1994 (R), 1996 (T), 1998 (R), 2006 (R)
Dithianon (180)	1992 (T,R), 1995 (R), 1996 (corr. to 1995 report), 2010 (T), 2013 (T,R)
Dithiocarbamates (105)	1965 (T), 1967 (T,R), 1970 (T,R), 1983 (R propineb, thiram), 1984 (R propineb), 1985 (R), 1987 (T thiram), 1988 (R thiram), 1990 (R), 1991 (corr. to 1990 evaluation), 1992 (T thiram), 1993 (T,R), 1995 (R), 1996 (T,R ferbam, ziram; R thiram), 2004 (R), 2012 (R), 2014 (R)
4,6-Dinitro- <i>ortho</i> -cresol (DNOC)	1965 (T)
Dodine (084)	1974 (T,R), 1976 (T,R), 1977 (R), 2000 (T), 2003 (R), 2004 (corr. to 2003 report)
Edifenphos (099)	1976 (T,R), 1979 (T,R), 1981 (T,R)
Emamectin benzoate (247)	2011 (T,R), 2014 (R)
Endosulfan (032)	1965 (T), 1967 (T,R), 1968 (T,R), 1971 (R), 1974 (R), 1975 (R), 1982 (T), 1985 (T,R), 1989 (T,R), 1993 (R), 1998 (T), 2006 (R), 2010 (R)
Endrin (033)	1965 (T), 1970 (T,R), 1974 (R), 1975 (R), 1990 (R), 1992 (R)
Esfenvalerate (204)	2002 (T,R)
Ethephon (106)	1977 (T,R), 1978 (T,R), 1983 (R), 1985 (R), 1993 (T), 1994 (R), 1995 (T), 1997 (T), 2002 (T), 2015 (T, R)
Ethiofencarb (107)	1977 (T,R), 1978 (R), 1981 (R), 1982 (T,R), 1983 (R)

Ethion (034)	1968 (T,R), 1969 (R), 1970 (R), 1972 (T,R), 1975 (R), 1982 (T), 1983 (R), 1985 (T), 1986 (T), 1989 (T), 1990 (T), 1994 (R), 2021 (T, R)
Ethiprole (304)	2018 (T, R), 2021 (R)
Ethoprophos (149)	1983 (T), 1984 (R), 1987 (T), 1999 (T), 2004 (R)
Ethoxyquin (035)	1969 (T,R), 1998 (T), 1999 (R), 2005 (T), 2008 (R)
Ethylene dibromide	See 1,2-Dibromoethane
Ethylene dichloride	See 1,2-Dichloroethane
Ethylene oxide	1965 (T,R), 1968 (T,R), 1971 (R)
Ethylenethiourea (ETU) (108)	1974 (R), 1977 (T,R), 1986 (T,R), 1987 (R), 1988 (T,R), 1990 (R), 1993 (T,R)
Etofenprox (184)	1993 (T,R), 2011 (T,R)
Etiozazole (241)	2010 (T,R), 2011 (R)
Etrinfos (123)	1980 (T,R), 1982 (T,R), 1986 (T,R), 1987 (R), 1988 (R), 1989 (R), 1990 (R)
Famoxadone (208)	2003 (T,R)
Fenamidone (264)	2013 (T), 2014 (T,R)
Fenamiphos (085)	1974 (T,R), 1977 (R), 1978 (R), 1980 (R), 1985 (T), 1987 (T), 1997 (T), 1999 (R), 2002 (T), 2006 (R)
Fenarimol (192)	1995 (T,R,E), 1996 (R and corr. to 1995 report)
Fenazaquin (297)	2017 (T, R), 2019 (R)
Fenbuconazole (197)	1997 (T,R), 2009 (R), 2012 (T), 2013 (R), 2021 (R)
Fenbutatin oxide (109)	1977 (T,R), 1979 (R), 1992 (T), 1993 (R)
Fenchlorfos (036)	1968 (T,R), 1972 (R), 1983 (R)
Fenhexamid (215)	2005 (T,R), 2021 (R)
Fenitrothion (037)	1969 (T,R), 1974 (T,R), 1976 (R), 1977 (T,R), 1979 (R), 1982 (T), 1983 (R), 1984 (T,R), 1986 (T,R), 1987 (R and corr. to 1986 R evaluation), 1988 (T), 1989 (R), 2000 (T), 2003 (R), 2004 (R, corr. to 2003 report), 2007 (T,R)
Fenpicoxamid (305)	2018 (T,R), 2021 (T, R)
Fenpropathrin (185)	1993 (T,R), 2006 (R), 2012 (T), 2014 (R)
Fenpropimorph (188)	1994 (T), 1995 (R), 1999 (R), 2001 (T), 2004 (T), 2016 (T), 2017 (T, R)
Fenpyrazamine (298)	2017 (R, T)
Fenpyroximate (193)	1995 (T,R), 1996 (corr. to 1995 report), 1999 (R), 2004 (T), 2007 (T), 2010 (R), 2013 (R), 2017 (T, R), 2018 (R), 2021 (T, R)
Fensulfothion (038)	1972 (T,R), 1982 (T), 1983 (R)
Fenthion (039)	1971 (T,R), 1975 (T,R), 1977 (R), 1978 (T,R), 1979 (T), 1980 (T), 1983 (R), 1989 (R), 1995 (T,R,E), 1996 (corr. to 1995 report), 1997 (T), 2000 (R)
Fentin compounds (040)	1965 (T), 1970 (T,R), 1972 (R), 1986 (R), 1991 (T,R), 1993 (R), 1994 (R)
Fenvalerate (119)	1979 (T,R), 1981 (T,R), 1982 (T), 1984 (T,R), 1985 (R), 1986 (T,R), 1987 (R and corr. to 1986 report), 1988 (R), 1990 (R), 1991 (corr. to 1990 R evaluation), 2012 (T,R)
Ferbam	See Dithiocarbamates, 1965 (T), 1967 (T,R), 1996 (T,R)
Fipronil (202)	1997 (T), 2000 (T), 2001 (R), 2016 (R), 2021 (T, R)
Fipronil-desulfinyl	1997 (T), 2021 (T, R)
Flonicamid (282)	2015 (T,R), 2016 (R), 2017 (R), 2019 (R)
Fluazifop-P-butyl	2016 (T,R), 2019 (R)
Flubendiamide (242)	2010 (T,R)

Flucythrinate (152)	1985 (T,R), 1987 (R), 1988 (R), 1989 (R), 1990 (R), 1993 (R)
Fludioxonil (211)	2004 (T,R), 2006 (R), 2010 (R), 2012 (R), 2013 (R), 2018 (R)
Fluensulfone (265)	2013 (T), 2014 (T,R), 2016 (T,R), 2017 (R), 2019 (R), 2021 (R)
Flufenoxuron (275)	2014 (T,R)
Flumethrin (195)	1996 (T,R)
Fluazinam (306)	2018 (T,R)
Fluopicolide (235)	2009 (T,R), 2014 (R)
Fluopyram (243)	2010 (T,R), 2012 (R), 2014 (R), 2015 (R), 2017 (R), 2021 (R)
Flupyradifurone (285)	2015 (T), 2016 (R), 2017 (R), 2019 (R)
Flusilazole (165)	1989 (T,R), 1990 (R), 1991 (R), 1993 (R), 1995 (T), 2007 (T, R)
Flutianil (319)	2021 (T, R)
Flutolanil (205)	2002 (T,R), 2013 (R)
Flutriafol (248)	2011 (T,R), 2015 (R)
Fluxapyroxad (256)	2012 (T,R), 2015 (R), 2018 (T,R)
Folpet (041)	1969 (T,R), 1973 (T), 1974 (R), 1982 (T), 1984 (T,R), 1986 (T), 1987 (R), 1990 (T,R), 1991 (corr. to 1990 R evaluation), 1993 (T,R), 1994 (R), 1995 (T), 1997 (R), 1998 (R), 1999 (R), 2002 (T), 2004 (T), 2007 (T)
Formothion (042)	1969 (T,R), 1972 (R), 1973 (T,R), 1978 (R), 1998 (R)
Fosetyl Aluminium (302)	2017 (T, R), 2019 (R)
Glufosinate-ammonium (175)	1991 (T,R), 1992 (corr. to 1991 report, Annex I), 1994 (R), 1998 (R), 1999 (T,R), 2012 (T,R), 2014 (R)
Glyphosate (158)	1986 (T,R), 1987 (R and corr. to 1986 report), 1988 (R), 1994 (R), 1997 (T,R), 2004 (T), 2005 (R), 2011 (T,R), 2013 (R), 2016 (T), 2019 (R)
Guazatine (114)	1978 (T,R), 1980 (R), 1997 (T,R)
Haloxypop (194)	1995 (T,R), 1996 (R and corr. to 1995 report), 2001 (R), 2006 (T), 2009 (R)
Heptachlor (043)	1965 (T), 1966 (T,R), 1967 (R), 1968 (R), 1969 (R), 1970 (T,R), 1974 (R), 1975 (R), 1977 (R), 1987 (R), 1991 (T,R), 1992 (corr. to 1991 report, Annex I), 1993 (R), 1994 (R)
Hexachlorobenzene (044)	1969 (T,R), 1973 (T,R), 1974 (T,R), 1978 (T), 1985 (R)
Hexaconazole (170)	1990 (T,R), 1991 (R and corr. to 1990 R evaluation), 1993 (R)
Hexythiazox (176)	1991 (T,R), 1994 (R), 1998 (R), 2008 (T), 2009 (R)
Hydrogen cyanide (045)	1965 (T,R)
Hydrogen phosphide (046)	1965 (T,R), 1966 (T,R), 1967 (R), 1969 (R), 1971 (R)
Imazalil (110)	1977 (T,R), 1980 (T,R), 1984 (T,R), 1985 (T,R), 1986 (T), 1988 (R), 1989 (R), 1991 (T), 1994 (R), 2000 (T), 2001 (T), 2005 (T), 2018 (T,R), 2021 (R)
Imazamox (276)	2014 (T,R), 2017 (R)
Imazapic (266)	2013 (T,R), 2015 (R)
Imazapyr (267)	2013 (T,R), 2015 (R), 2017 (R)
Imazethapyr (289)	2016 (T,R)
Imidacloprid (206)	2001 (T), 2002 (R), 2006 (R), 2008 (R), 2012 (R), 2015 (R), 2017 (R)

Indoxacarb (216)	2005 (T,R), 2007 (R), 2009 (R), 2012 (R), 2013 (R)
Iprodione (111)	1977 (T,R), 1980 (R), 1992 (T), 1994 (R), 1995 (T), 2001 (R)
Isofenphos (131)	1981 (T,R), 1982 (T,R), 1984 (R), 1985 (R), 1986 (T,R), 1988 (R), 1992 (R)
Isofetamid (290)	2016 (T,R), 2018 (R)
Isoprothiolane (299)	2017 (T, R), 2021 (R), 2021 (T)
Isopyrazam (249)	2011 (T,R), 2017 (R)
Isoxaflutole (268)	2013 (T,R), 2021 (R)
Kresoxim-methyl (199)	1998 (T,R), 2001 (R), 2018 (T,R), 2019 (R)
Lead arsenate	1965 (T), 1968 (T,R)
Leptophos (088)	1974 (T,R), 1975 (T,R), 1978 (T,R)
Lindane (048)	1965 (T), 1966 (T,R), 1967 (R), 1968 (R), 1969 (R), 1970 (T,R, published as Annex VI to 1971 evaluations), 1973 (T,R), 1974 (R), 1975 (R), 1977 (T,R), 1978 (R), 1979 (R), 1989 (T,R), 1997 (T), 2002 (T), 2003 (R), 2004 (corr. to 2003 report), 2015 (R)
Lufenuron (286)	2015 (T, R), 2018 (R)
Malathion (049)	1965 (T), 1966 (T,R), 1967 (corr. to 1966 R evaluation), 1968 (R), 1969 (R), 1970 (R), 1973 (R), 1975 (R), 1977 (R), 1984 (R), 1997 (T), 1999 (R), 2000 (R), 2003 (T), 2004 (R), 2005 (R), 2008 (R), 2013 (R), 2016 (T)
Maleic hydrazide (102)	1976 (T,R), 1977 (T,R), 1980 (T), 1984 (T,R), 1996 (T), 1998 (R)
Mancozeb (050)	1967 (T,R), 1970 (T,R), 1974 (R), 1977 (R), 1980 (T,R), 1993 (T,R)
Mandestrobin (307)	2018 (T,R), 2019 (R)
Mandipropamid (231)	2008 (T,R), 2013 (R), 2018 (R), 2021 (R)
Maneb	See Dithiocarbamates, 1965 (T), 1967 (T,R), 1987 (T), 1993 (T,R)
MCPA (257)	2012 (T,R)
Mecarbam (124)	1980 (T,R), 1983 (T,R), 1985 (T,R), 1986 (T,R), 1987 (R)
Mefentrifluconazole (320)	2021 (T)
Meptyldinocap (244)	2010 (T, R)
Mesotrione (277)	2014 (T, R), 2019 (T, R)
Metaflumizone (236)	2009 (T, R), 2019 (T, R)
Metalaxyl (138)	1982 (T, R), 1984 (R), 1985 (R), 1986 (R), 1987 (R), 1989 (R), 1990 (R), 1992 (R), 1995 (R), 2021 (T, R)
Metalaxyl –M (212)	2002 (T), 2004 (R), 2021 (R)
Metconazole (313)	2019 (T, R), 2021(R)
Methacrifos (125)	1980 (T,R), 1982 (T), 1986 (T), 1988 (T), 1990 (T,R), 1992 (R)
Methamidophos (100)	1976 (T,R), 1979 (R), 1981 (R), 1982 (T,R), 1984 (R), 1985 (T), 1989 (R), 1990 (T,R), 1994 (R), 1996 (R), 1997 (R), 2002 (T), 2003 (R), 2004 (R, corr. to 2003 report)
Methidathion (051)	1972 (T,R), 1975 (T,R), 1979 (R), 1992 (T,R), 1994 (R), 1997 (T)
Methiocarb (132)	1981 (T,R), 1983 (T,R), 1984 (T), 1985 (T), 1986 (R), 1987 (T,R), 1988 (R), 1998 (T), 1999 (R), 2005 (R)
Methomyl (094)	1975 (R), 1976 (R), 1977 (R), 1978 (R), 1986 (T,R), 1987 (R), 1988 (R), 1989 (T,R), 1990 (R), 1991 (R), 2001 (T,R), 2004 (R), 2008 (R)

Methoprene (147)	1984 (T,R), 1986 (R), 1987 (T and corr. to 1986 report), 1988 (R), 1989 (R), 2001 (T), 2005 (R), 2016 (R), 2019 (R), 2021 (R)
Methoxychlor	1965 (T), 1977 (T)
Methoxyfenozide (209)	2003 (T,R), 2004 (corr. to 2003 report), 2006 (R), 2009 (R), 2012 (R), 2021 (R)
Methyl bromide (052)	See Bromomethane
Metrafenone (278)	2014 (T,R), 2016 (R)
Metiram (186)	1993 (T), 1995 (R)
Mevinphos (053)	1965 (T), 1972 (T,R), 1996 (T), 1997 (E,R), 2000 (R)
MGK (264)	1967 (T,R)
Monocrotophos (054)	1972 (T,R), 1975 (T,R), 1991 (T,R), 1993 (T), 1994 (R)
Myclobutanil (181)	1992 (T,R), 1997 (R), 1998 (R), (2001 (R)), 2014 (T,R)
Nabam	See Dithiocarbamates, 1965 (T), 1976 (T,R)
Natamycin (300)	2017 (T, R)
Nitrofen (140)	1983 (T,R)
Norflurazon (308)	2018 (T,R)
Novaluron (217)	2005 (T,R), 2010 (R)
Omethoate (055)	1971 (T,R), 1975 (T,R), 1978 (T,R), 1979 (T), 1981 (T,R), 1984 (R), 1985 (T), 1986 (R), 1987 (R), 1988 (R), 1990 (R), 1998 (R)
Organomercury compounds	1965 (T), 1966 (T,R), 1967 (T,R)
Oxamyl (126)	1980 (T,R), 1983 (R), 1984 (T), 1985 (T,R), 1986 (R), 2002 (T,R), 2017 (T, R)
Oxathiapiprolin (291)	2016 (T,R), 2018 (R)
Oxydemeton-methyl (166)	1965 (T, as demeton-S-methyl sulfoxide), 1967 (T), 1968 (R), 1973 (T,R), 1982 (T), 1984 (T,R), 1989 (T,R), 1992 (R), 1998 (R), 1999 (corr. to 1992 report), 2002 (T), 2004 (R)
Oxythioquinox	See Chinomethionat
Paclobutrazol (161)	1988 (T,R), 1989 (R)
Paraquat (057)	1970 (T,R), 1972 (T,R), 1976 (T,R), 1978 (R), 1981 (R), 1982 (T), 1985 (T), 1986 (T), 2003 (T), 2004 (R), 2009 (R)
Parathion (058)	1965 (T), 1967 (T,R), 1969 (R), 1970 (R), 1984 (R), 1991 (R), 1995 (T,R), 1997 (R), 2000 (R)
Parathion-methyl (059)	1965 (T), 1968 (T, R), 1972 (R), 1975 (T,R), 1978 (T,R), 1979 (T), 1980 (T), 1982 (T), 1984 (T,R), 1991 (R), 1992 (R), 1994 (R), 1995 (T), 2000 (R), 2003 (R)
Penconazole (182)	1992 (T, R), 1995 (R), 2015 (T), 2016 (R)
Pendimethalin (292)	2016 (T, R), 2019 (R), 2021 (R)
Penthiopyrad (253)	2011 (T), 2012 (R), 2013 (R), 2019 (R)
Permethrin (120)	1979 (T, R), 1980 (R), 1981 (T,R), 1982 (R), 1983 (R), 1984 (R), 1985 (R), 1986 (T,R), 1987 (T), 1988 (R), 1989 (R), 1991 (R), 1992 (corr. to 1991 report), 1999 (T)
2-Phenylphenol (056)	1969 (T,R), 1975 (R), 1983 (T), 1985 (T,R), 1989 (T), 1990 (T,R), 1999 (T,R), 2002 (R)
Phenothrin (127)	1979 (R), 1980 (T,R), 1982 (T), 1984 (T), 1987 (R), 1988 (T,R)
Phenthoate (128)	1980 (T,R), 1981 (R), 1984 (T)

Phorate (112)	1977 (T,R), 1982 (T), 1983 (T), 1984 (R), 1985 (T), 1990 (R), 1991 (R), 1992 (R), 1993 (T), 1994 (T), 1996 (T), 2004 (T), 2005 (R), 2012 (R), 2014 (R)
Phosalone (060)	1972 (T,R), 1975 (R), 1976 (R), 1993 (T), 1994 (R), 1997 (T), 1999 (R), 2001 (T)
Phosmet (103)	1976 (R), 1977 (corr. to 1976 R evaluation), 1978 (T,R), 1979 (T,R), 1981 (R), 1984 (R), 1985 (R), 1986 (R), 1987 (R and corr. to 1986 R evaluation), 1988 (R), 1994 (T), 1997 (R), 1998 (T), 2002 (R), 2003 (R), 2007 (R)
Phosphine	See Hydrogen phosphide
Phosphamidon (061)	1965 (T), 1966 (T), 1968 (T,R), 1969 (R), 1972 (R), 1974 (R), 1982 (T), 1985 (T), 1986 (T)
Phosphonic acid (301)	2017 (T, R)
Phoxim (141)	1982 (T), 1983 (R), 1984 (T,R), 1986 (R), 1987 (R), 1988 (R)
Picoxystrobin (258)	2012 (T,R), 2013 (R), 2016 (R), 2017 (R), 2019 (R)
Pinoxaden (293)	2016 (T,R)
Piperonyl butoxide (062)	1965 (T,R), 1966 (T,R), 1967 (R), 1969 (R), 1972 (T,R), 1992 (T,R), 1995 (T), 2001 (R), 2002 (R)
Pirimicarb (101)	1976 (T,R), 1978 (T,R), 1979 (R), 1981 (T,R), 1982 (T), 1985 (R), 2004 (T), 2006 (R)
Pirimiphos-methyl (086)	1974 (T,R), 1976 (T,R), 1977 (R), 1979 (R), 1983 (R), 1985 (R), 1992 (T), 1994 (R), 2003 (R), 2004 (R, corr. to 2003 report), 2006 (T)
Prochloraz (142)	1983 (T,R), 1985 (R), 1987 (R), 1988 (R), 1989 (R), 1990 (R), 1991 (corr. to 1990 report, Annex I, and R evaluation), 1992 (R), 2001 (T), 2004 (R), 2009 (R)
Procymidone(136)	1981 (R), 1982 (T), 1989 (T,R), 1990 (R), 1991 (corr. to 1990 Annex I), 1993 (R), 1998 (R), 2007 (T)
Profenofos (171)	1990 (T,R), 1992 (R), 1994 (R), 1995 (R), 2007 (T), 2008 (R), 2011 (R), 2018 (R)
Propamocarb (148)	1984 (T,R), 1986 (T,R), 1987 (R), 2005 (T), 2006 (R), 2014 (R), 2018 (R)
Propargite (113)	1977 (T,R), 1978 (R), 1979 (R), 1980 (T,R), 1982 (T,R), 1999 (T), 2002 (R), 2006 (R)
Propham (183)	1965 (T), 1992 (T,R)
Propiconazole (160)	1987 (T, R), 1991 (R), 1994 (R), 2004 (T), 2006 (R), 2007 (R), 2013 (R), 2014 (R), 2015 (R), 2017 (R), 2018 (R)
Propineb	1977 (T, R), 1980 (T), 1983 (T), 1984 (R), 1985 (T, R), 1993 (T, R), 2004 (R)
Propoxur (075)	1973 (T, R), 1977 (R), 1981 (R), 1983 (R), 1989 (T), 1991 (R), 1996 (R)
Propylene oxide (250)	2011 (T, R), 2017 (T, R)
Propylenethiourea (PTU, 150)	1993 (T, R), 1994 (R), 1999 (T)
Prothioconazole (232)	2008 (T, R), 2009 (R), 2014 (R), 2017 (R), 2021 (R)
Pydiflumetofen (309)	2018 (T, R), 2019 (R), 2021 (R)
Pyflubumide (314)	2019 (T, R)
Pymetrozine (279)	2014 (T, R)
Pyraclostrobin (210)	2003 (T), 2004 (R), 2006 (R), 2011 (R), 2012 (R), 2014 (R), 2018 (T, R)
Pyrasulfotole (321)	2021 (T, R)

Pyraziflumid (322)	2021 (T, R)
Pyrazophos (153)	1985 (T, R), 1987 (R), 1992 (T,R), 1993 (R)
Pyrethrins (063)	1965 (T), 1966 (T, R), 1967 (R), 1968 (R), 1969 (R), 1970 (T), 1972 (T,R), 1974 (R), 1999 (T), 2000 (R), 2003 (T,R), 2005 (R)
Pyridate (315)	2019 (T)
Pyriproxyfen (316)	2019 (T, R)
Pyrimethanil (226)	2007 (T, R), 2013 (R)
Pyriofenone (310)	2018 (T, R), 2019 (R)
Pyriproxyfen (200)	1999 (R, T), 2000 (R), 2001 (T), 2018 (R), 2019 (R)
Quinclorac (287)	2015 (T, R), 2017 (R)
Quinoxifen (223)	2006 (T, R), 2021 (R)
Quintozene (064)	1969 (T, R), 1973 (T,R), 1974 (R), 1975 (T,R), 1976 (Annex I, corr. to 1975 R evaluation), 1977 (T,R), 1995 (T,R), 1998 (R)
Saflufenacil (251)	2011 (T, R), 2016 (R), 2017 (R)
Sedaxane (259)	2012 (T, R), 2014 (R)
Spices	2004 (R), 2005 (R), 2007 (R), 2010 (R), 2015 (R), 2019 (R)
Spinetoram (233)	2008 (T, R), 2012 (R), 2017 (R), 2021 (R)
Spinosad (203)	2001 (T, R), 2004 (R), 2008 (R), 2011 (R)
Spirodiclofen (237)	2009 (T, R)
Spiromesifen (294)	2016 (T, R), 2021 (R)
Spiropidion (323)	2021 (T, R)
Spirotetramat (234)	2008 (T, R), 2011 (R), 2012 (R), 2013 (R), 2015 (R), 2019 (R)
Sulfoxaflor (252)	2011 (T, R), 2013 (R), 2014 (R), 2016 (R), 2018 (R), 2021 (R)
Sulfuryl fluoride (218)	2005 (T, R)
2,4,5-T (121)	1970 (T,R), 1979 (T,R), 1981 (T)
Tebuconazole (189)	1994 (T, R), 1996 (corr. to Annex II of 1995 report), 1997 (R), 2008 (R), 2010 (T), 2011 (R), 2015 (R), 2017 (R), 2019 (R), 2021 (R)
Tebufenozide (196)	1996 (T, R), 1997 (R), 1999 (R), 2001 (T, R), 2003 (T)
Tecnazine (115)	1974 (T, R), 1978 (T, R), 1981 (R), 1983 (T), 1987 (R), 1989 (R), 1994 (T, R)
Teflubenzuron (190)	1994 (T), 1996 (R), 2016 (T, R)
Temephos	2006 (T)
Terbufos (167)	1989 (T, R), 1990 (T,R), 2003 (T), 2005 (R)
Tetraniliprole (324)	2021 (T)
Thiabendazole (065)	1970 (T, R), 1971 (R), 1972 (R), 1975 (R), 1977 (T,R), 1979 (R), 1981 (R), 1997 (R), 2000 (R), 2006 (T,R), 2019 (T, R)
Thiacloprid (223)	2006 (T, R)
Thiamethoxam (245)	2010 (T, R), 2011 (R), 2012 (R), 2014 (R), 2021 (R)
Thiodicarb (154)	1985 (T, R), 1986 (T), 1987 (R), 1988 (R), 2000 (T), 2001 (R)
Thiometon (076)	1969 (T, R), 1973 (T,R), 1976 (R), 1979 (T,R), 1988 (R)
Thiophanate-methyl (077)	1973 (T, R), 1975 (T, R), 1977 (T), 1978 (R), 1988 (R), 2002 (R), 1990 (R), 1994 (R), 1995 (T,E), 1998 (T,R), 2006 (T), 2017 (T)

Thiram (105)	See Dithiocarbamates, 1965 (T), 1967 (T, R), 1970 (T,R), 1974 (T), 1977 (T), 1983 (R), 1984 (R), 1985 (T,R), 1987 (T), 1988 (R), 1989 (R), 1992 (T), 1996 (R) 2018 (T, R)
Tioxazafen (211)	1994 (T, R), 1996 (corr. to Annex II of 1995 report), 2019 (T, R)
Tolclofos-methyl (191)	2013 (T), 2016 (R)
Tolfenpyrad (269)	1988 (T, R), 1990 (R), 1991 (corr. to 1990 report), 2002 (T, R), 2003 (R)
Toxaphene	See Camphechlor
Triadimefon (133)	1979 (R), 1981 (T, R), 1983 (T,R), 1984 (R), 1985 (T,R), 1986 (R), 1987 (R and corr. to 1986 R evaluation), 1988 (R), 1989 (R), 1992 (R), 1995 (R), 2004 (T), 2007 (R)
Triadimenol (168)	1989 (T, R), 1992 (R), 1995 (R), 2004 (T), 2007 (R), 2014 (R)
Triazolylalanine	1989 (T, R)
Triazophos (143)	1982 (T), 1983 (R), 1984 (corr. to 1983 report, Annex I), 1986 (T, R), 1990 (R), 1991 (T and corr. to 1990 R evaluation), 1992 (R), 1993 (T, R), 2002 (T), 2007 (R), 2010 (R), 2013 (R)
Trichlorfon (066)	1971 (T,R), 1975 (T,R), 1978 (T,R), 1987 (R)
Trichloronat	1971 (T,R)
Trichloroethylene	1968 (R)
Tricyclohexyltin hydroxide	See Cyhexatin
Trifloxystrobin (213)	2004 (T, R), 2012 (R), 2015 (R), 2017 (R), 2021 (R)
Triflumezopyrim (303)	2017 (T, R)
Triflumizole (270)	2013 (T, R)
Triforine (116)	1977 (T), 1978 (T, R), 1997 (T), 2004 (R), 2014 (T,R)
Trinexapac-ethyl (271)	2013 (T,R), 2021(T, R)
Triphenyltin compounds	See Fentin compounds
Vamidothion (078)	1973 (T, R), 1982 (T), 1985 (T, R), 1987 (R), 1988 (T), 1990 (R), 1992 (R)
Vinclozolin (159)	1986 (T, R), 1987 (R and corr. to 1986 report and R evaluation), 1988 (T, R), 1989 (R), 1990 (R), 1992 (R), 1995 (T)
Zineb (105)	See Dithiocarbamates, 1965 (T), 1967 (T, R), 1993 (T)
Ziram (105)	See Dithiocarbamates, 1965 (T), 1967 (T, R), 1996 (T,R)
Zoxamide (227)	2007 (T, R), 2009 (R)

Annex 3: International estimated daily intakes (IEDIs) of pesticide residues

FENPYROXIMATE (193)				International Estimated Daily Intake (IEDI)				ADI = 0–0.005 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
FC 0002	Subgroup of Lemons and limes, raw (excl kumquat commodities)	RAC	0.085	2.42	0.21	2.15	0.18	0.43	0.04	10.74	0.91	6.59	0.56	14.06	1.20
FC 0303	Kumquats, raw (incl juice)	RAC	0.37	2.36	0.87	0.27	0.10	3.19	1.18	14.44	5.34	1.66	0.61	1.71	0.63
-	Lemon, juice (single strength, incl. concentrated)	PP	0.037	0.01	0.00	0.01	0.00	0.11	0.00	0.09	0.00	0.18	0.01	0.17	0.01
FC 0003	Subgroup of Mandarins, raw	RAC	0.085	6.18	0.53	3.66	0.31	0.25	0.02	6.82	0.58	3.49	0.30	19.38	1.65
-	Subgroup of Mandarins, juice	PP	0.037	NC	-	NC	-	NC	-	NC	-	NC	-	NC	-
FC 0004	Subgroup of Oranges, sweet, sour, raw	RAC	0.052	20.66	1.07	5.23	0.27	11.90	0.62	37.90	1.97	21.16	1.10	56.46	2.94
JF 0004	Subgroup of Oranges, juice (single strength, incl. concentrated)	PP	0.022	1.27	0.03	2.20	0.05	0.09	0.00	11.81	0.26	0.46	0.01	1.69	0.04
FC 0005	Subgroup of Pummelo and grapefruits, raw	RAC	0.044	0.64	0.03	0.35	0.02	0.93	0.04	6.10	0.27	1.01	0.04	1.36	0.06
JF 0203	Grapefruits, juice (single strength, incl. concentrated)	PP	0.019	0.01	0.00	0.16	0.00	0.02	0.00	1.97	0.04	0.12	0.00	0.77	0.01
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.075	13.49	1.01	26.63	2.00	15.05	1.13	16.28	1.22	6.47	0.49	47.88	3.59
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.012	0.32	0.00	3.07	0.04	0.07	0.00	5.00	0.06	0.29	0.00	5.57	0.07
FP 0230	Pear, raw	RAC	0.078	2.16	0.17	6.24	0.49	0.05	0.00	4.07	0.32	1.16	0.09	5.34	0.42
FS 0013	Subgroup of Cherries, raw	RAC	0.585	0.92	0.54	9.15	5.35	0.01	0.01	0.61	0.36	0.06	0.04	6.64	3.88
FS 0014	Subgroup of Plums, raw	RAC	0.025	2.40	0.06	8.60	0.22	0.06	0.00	2.52	0.06	0.58	0.01	4.16	0.10
DF 0014	Plums, dried (prunes)	PP	0.05	0.09	0.00	0.06	0.00	0.01	0.00	0.18	0.01	0.04	0.00	0.06	0.00
FS 2001	Subgroup of peaches, raw (incl dried apricots)	RAC	0.17	8.01	1.36	5.87	1.00	0.18	0.03	8.19	1.39	1.64	0.28	22.46	3.82
FB 2005	Subgroup of Caneberries, raw	RAC	0.84	0.42	0.35	1.05	0.88	0.01	0.01	0.02	0.02	0.02	0.02	1.24	1.04
FB 2006	Subgroup of Bush berries, raw (including processed)	RAC	0.8	0.53	0.42	1.31	1.05	0.40	0.32	1.66	1.33	0.01	0.01	0.99	0.79
FB 0269	Grapes, raw (i.e. table grapes)	RAC	0.035	12.68	0.44	9.12	0.32	0.03	0.00	16.88	0.59	3.70	0.13	54.42	1.90
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	0.07	0.51	0.04	0.51	0.04	0.01	0.00	1.27	0.09	0.12	0.01	2.07	0.14
JF 0269	Grape juice (from wine grapes)	PP	0.005	0.14	0.00	0.29	0.00	0.05	0.00	0.30	0.00	0.24	0.00	0.05	0.00
-	Graps must (from wine-grapes)	PP	0.035	0.33	0.01	0.13	0.00	0.01	0.00	0.02	0.00	0.01	0.00	0.02	0.00
-	Grape wine (incl vermouths) (from wine-grapes)	PP	0.005	0.67	0.00	12.53	0.06	2.01	0.01	1.21	0.01	3.53	0.02	4.01	0.02
FB 0275	Strawberry, raw	RAC	0.06	0.70	0.04	2.01	0.12	0.04	0.00	1.36	0.08	0.37	0.02	2.53	0.15
FI 0326	Avocado, raw	RAC	0.05	0.13	0.01	0.03	0.00	2.05	0.10	2.54	0.13	2.34	0.12	0.12	0.01
VC 2039	Subgroup of Cucumbers and Squashes, raw	RAC	0.12	10.52	1.26	39.36	4.72	2.07	0.25	25.74	3.09	2.80	0.34	44.83	5.38
VC 0046	Melons, except watermelon, raw (Cantaloupe)	RAC	0.05	8.90	0.45	8.64	0.43	0.80	0.04	17.90	0.90	2.80	0.14	29.17	1.46

ADI = 0–0.005 mg/kg bw

Codex				Estimated daily intake (EDI)				ADI - 0.0005 mg/kg bw								
Commodity description		Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day								
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake	
VC 0432	Watermelon, raw	RAC	0.1	28.96	2.90	25.65	2.57	1.56	0.16	39.26	3.93	4.94	0.49	66.90	6.69	
VO 0448	Tomato, raw (incl paste, excl juice, excl canned)	RAC	0.1	51.07	5.11	80.96	8.10	16.96	1.70	99.83	9.98	26.09	2.61	212.26	21.23	
-	Tomato, canned (& peeled)	PP	0.04	0.20	0.01	0.31	0.01	0.02	0.00	1.11	0.04	0.11	0.00	1.50	0.06	
JF 0448	Tomato, juice (single strength, incl concentrated)	PP	0.064	0.29	0.02	0.29	0.02	0.01	0.00	0.38	0.02	0.05	0.00	0.14	0.01	
VO 0051	Subgroup of peppers, raw (incl dried sweet peppers, excl dried chilepeppers), excl okra	RAC	0.05	8.48	0.42	13.74	0.69	10.13	0.51	11.29	0.56	9.52	0.48	26.36	1.32	
VO 2046	Subgroup of eggplants	RAC	0.1	5.58	0.56	4.31	0.43	0.89	0.09	9.31	0.93	13.64	1.36	20.12	2.01	
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0.075	0.68	0.05	NC	-	NC	-	0.39	0.03	0.22	0.02	0.49	0.04	
VP 2062	Subgroup of succulent beans without pods (all commodities within this group)	RAC	0.1	5.07	0.51	1.02	0.10	0.49	0.05	1.78	0.18	1.19	0.12	8.57	0.86	
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0	59.74	0.00	316.14	0.00	9.78	0.00	60.26	0.00	54.12	0.00	119.82	0.00	
VS 2080	Subgroup of stems and petioles	RAC	0.845	3.11	2.63	5.52	4.66	3.42	2.89	8.29	7.01	0.02	0.02	4.00	3.38	
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl beer, incl germ, incl starch, excl flour, excl oil)	RAC	0.01	0.97	0.01	0.24	0.00	1.58	0.02	4.10	0.04	2.56	0.03	13.31	0.13	
CF 1255	Maize, flour (white flour and wholemeal flour)	PP	0.0037	22.72	0.08	35.61	0.13	87.27	0.32	34.92	0.13	46.71	0.17	49.12	0.18	
OR 0645	Maize oil	PP	0.0099	0.96	0.01	0.85	0.01	0.29	0.00	5.42	0.05	0.42	0.00	2.10	0.02	
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0	4.06	0.00	3.27	0.00	7.01	0.00	13.93	0.00	14.01	0.00	9.36	0.00	
SB 0716	Coffee bean raw (incl roasted, incl instant coffee, incl substitutes)	RAC	0.025	1.36	0.03	3.59	0.09	1.44	0.04	5.18	0.13	2.02	0.05	1.70	0.04	
DH 1100	Hops, dry	RAC	5.15	0.01	0.05	0.04	0.21	0.01	0.05	0.01	0.05	NC	-	0.01	0.05	
DT 1114	Tea, green or black, fermented and dried, (including concentrates)	RAC	1.4	2.28	3.19	1.98	2.77	0.46	0.64	2.43	3.40	1.29	1.81	3.04	4.26	
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.015	24.96	0.37	57.95	0.87	16.70	0.25	38.38	0.58	26.46	0.40	29.00	0.44	
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.063	6.24	0.39	14.49	0.91	4.18	0.26	9.60	0.60	6.62	0.42	7.25	0.46	
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.063	3.29	0.21	6.14	0.39	0.82	0.05	1.57	0.10	2.23	0.14	1.07	0.07	
MO 0105	Edible offal (mammalian), raw	RAC	0.4	4.79	1.92	9.68	3.87	2.97	1.19	5.49	2.20	3.84	1.54	5.03	2.01	
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.005	289.65	1.45	485.88	2.43	26.92	0.13	239.03	1.20	199.91	1.00	180.53	0.90	
Total intake (µg/person)=					28.8		45.9		12.2		50.2		15.0		73.5	
Bodyweight per region (kg bw) =					60		60		60		60		60		60	
ADI (µg/person)=					300		300		300		300		300		300	

Annex 3

FENPYROXIMATE (193)**International Estimated Daily Intake (IEDI)****ADI = 0–0.005 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
	%ADI=				9.6%		15.3%		4.1%		16.7%		5.0%		24.5%
	Rounded %ADI=				10%		20%		4%		20%		5%		20%

FENPYROXIMATE (193)**International Estimated Daily Intake (IEDI)****ADI = 0–0.005 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
FC 0002	Subgroup of Lemons and limes, raw (excl kumquat commodities)	RAC	0.085	3.78	0.32	8.84	0.75	0.92	0.08	6.71	0.57	4.09	0.35	4.57	0.39
FC 0303	Kumquats, raw (incl juice)	RAC	0.37	4.67	1.73	5.86	2.17	1.96	0.73	1.45	0.54	17.05	6.31	1.37	0.51
-	Lemon, juice (single strength, incl. concentrated)	PP	0.037	0.60	0.02	0.36	0.01	0.01	0.00	1.49	0.06	0.43	0.02	0.24	0.01
FC 0003	Subgroup of Mandarins, raw	RAC	0.085	12.34	1.05	14.99	1.27	16.08	1.37	10.76	0.91	9.94	0.84	NC	-
-	Subgroup of Mandarins, juice	PP	0.037	0.04	0.00	NC	-	0.01	0.00	0.01	0.00	NC	-	NC	-
FC 0004	Subgroup of Oranges, sweet, sour, raw	RAC	0.052	15.68	0.82	24.00	1.25	6.80	0.35	29.09	1.51	15.39	0.80	160.47	8.34
JF 0004	Subgroup of Oranges, juice (single strength, incl. concentrated)	PP	0.022	33.31	0.73	1.78	0.04	0.28	0.01	18.97	0.42	14.01	0.31	13.36	0.29
FC 0005	Subgroup of Pummelo and grapefruits, raw	RAC	0.044	2.19	0.10	1.24	0.05	0.60	0.03	3.44	0.15	4.60	0.20	299.96	13.20
JF 0203	Grapefruits, juice (single strength, incl. concentrated)	PP	0.019	2.89	0.05	1.61	0.03	0.02	0.00	1.15	0.02	7.39	0.14	33.07	0.63
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.075	41.14	3.09	56.49	4.24	26.64	2.00	31.58	2.37	51.94	3.90	3.05	0.23
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.012	14.88	0.18	11.98	0.14	0.15	0.00	9.98	0.12	30.32	0.36	3.47	0.04
FP 0230	Pear, raw	RAC	0.078	8.79	0.69	8.44	0.66	12.37	0.96	9.60	0.75	10.27	0.80	0.23	0.02
FS 0013	Subgroup of Cherries, raw	RAC	0.585	1.40	0.82	4.21	2.46	0.04	0.02	2.93	1.71	1.50	0.88	NC	-
FS 0014	Subgroup of Plums, raw	RAC	0.025	3.75	0.09	3.33	0.08	5.94	0.15	2.64	0.07	2.50	0.06	0.06	0.00
DF 0014	Plums, dried (prunes)	PP	0.05	0.61	0.03	0.35	0.02	0.05	0.00	0.35	0.02	0.49	0.02	0.13	0.01
FS 2001	Subgroup of peaches, raw (incl dried apricots)	RAC	0.17	13.03	2.22	16.29	2.77	8.29	1.41	12.95	2.20	5.35	0.91	0.04	0.01
FB 2005	Subgroup of Caneberries, raw	RAC	0.84	0.56	0.47	1.43	1.20	0.14	0.12	1.23	1.03	1.14	0.96	0.01	0.01
FB 2006	Subgroup of Bush berries, raw (including processed)	RAC	0.8	1.31	1.05	5.50	4.40	0.01	0.01	2.57	2.06	0.82	0.66	2.15	1.72
FB 0269	Grapes, raw (i.e. table grapes)	RAC	0.035	6.33	0.22	11.22	0.39	5.21	0.18	9.38	0.33	4.55	0.16	0.78	0.03
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	0.07	3.09	0.22	1.51	0.11	0.03	0.00	1.38	0.10	4.26	0.30	0.42	0.03
JF 0269	Grape juice (from wine grapes)	PP	0.005	0.56	0.00	1.96	0.01	0.02	0.00	2.24	0.01	2.27	0.01	0.34	0.00
-	Graps must (from wine-grapes)	PP	0.035	0.16	0.01	0.09	0.00	0.01	0.00	0.12	0.00	0.11	0.00	NC	-

Annex 3

FENPYROXIMATE (193)				International Estimated Daily Intake (IEDI)				ADI = 0–0.005 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
-	Grape wine (incl vermouths) (from wine-grapes)	PP	0.005	88.93	0.44	62.41	0.31	1.84	0.01	25.07	0.13	61.17	0.31	5.84	0.03
FB 0275	Strawberry, raw	RAC	0.06	4.49	0.27	5.66	0.34	0.02	0.00	6.63	0.40	5.75	0.35	0.05	0.00
FI 0326	Avocado, raw	RAC	0.05	2.65	0.13	0.87	0.04	0.46	0.02	1.64	0.08	1.30	0.07	0.96	0.05
VC 2039	Subgroup of Cucumbers and Squashes, raw	RAC	0.12	7.14	0.86	16.92	2.03	37.58	4.51	15.16	1.82	4.42	0.53	12.67	1.52
VC 0046	Melons, except watermelon, raw (Cantaloupe)	RAC	0.05	9.20	0.46	11.95	0.60	14.63	0.73	8.99	0.45	7.86	0.39	2.46	0.12
VC 0432	Watermelon, raw	RAC	0.1	4.60	0.46	9.82	0.98	68.50	6.85	13.19	1.32	1.99	0.20	14.56	1.46
VO 0448	Tomato, raw (incl paste, excl juice, excl canned)	RAC	0.1	51.98	5.20	64.09	6.41	35.52	3.55	79.82	7.98	42.65	4.27	10.96	1.10
-	Tomato, canned (& peeled)	PP	0.04	7.57	0.30	2.66	0.11	0.30	0.01	0.97	0.04	7.31	0.29	0.41	0.02
JF 0448	Tomato, juice (single strength, incl concentrated)	PP	0.064	0.80	0.05	0.07	0.00	0.05	0.00	0.61	0.04	0.40	0.03	0.08	0.01
VO 0051	Subgroup of peppers, raw (incl dried sweet peppers, excl dried chili peppers), excl okra	RAC	0.05	6.39	0.32	15.53	0.78	19.09	0.95	10.36	0.52	8.29	0.41	4.53	0.23
VO 2046	Subgroup of eggplants	RAC	0.1	1.01	0.10	1.69	0.17	21.37	2.14	3.00	0.30	1.40	0.14	NC	-
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0.075	5.07	0.38	0.83	0.06	0.17	0.01	3.70	0.28	NC	-	NC	-
VP 2062	Subgroup of succulent beans without pods (all commodities within this group)	RAC	0.1	2.42	0.24	6.09	0.61	4.33	0.43	2.09	0.21	18.99	1.90	0.17	0.02
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0	225.03	0.00	234.24	0.00	71.48	0.00	177.55	0.00	234.55	0.00	37.71	0.00
VS 2080	Subgroup of stems and petioles	RAC	0.845	9.31	7.87	8.57	7.24	NC	-	3.88	3.28	24.46	20.67	5.89	4.98
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl beer, incl germ, incl starch, excl flour, excl oil)	RAC	0.01	0.10	0.00	9.93	0.10	1.71	0.02	21.57	0.22	0.33	0.00	0.05	0.00
CF 1255	Maize, flour (white flour and wholemeal flour)	PP	0.0037	14.27	0.05	12.86	0.05	19.71	0.07	12.55	0.05	4.21	0.02	52.30	0.19
OR 0645	Maize oil	PP	0.0099	0.90	0.01	0.47	0.00	0.15	0.00	3.01	0.03	1.86	0.02	0.36	0.00
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0	8.52	0.00	8.94	0.00	15.09	0.00	9.60	0.00	14.57	0.00	26.26	0.00
SB 0716	Coffee bean raw (incl roasted, incl instant coffee, incl substitutes)	RAC	0.025	10.90	0.27	12.44	0.31	0.77	0.02	9.48	0.24	22.07	0.55	8.15	0.20
DH 1100	Hops, dry	RAC	5.15	NC	-	NC	-	0.02	0.10	0.02	0.10	NC	-	NC	-
DT 1114	Tea, green or black, fermented and dried, (including concentrates)	RAC	1.4	2.91	4.07	1.73	2.42	1.14	1.60	1.85	2.59	2.29	3.21	0.74	1.04
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.015	112.02	1.68	120.71	1.81	63.46	0.95	88.99	1.33	96.24	1.44	41.02	0.62
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.063	28.01	1.76	30.18	1.90	15.86	1.00	22.25	1.40	24.06	1.52	10.25	0.65
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.063	6.44	0.41	15.51	0.98	3.79	0.24	8.29	0.52	18.44	1.16	8.00	0.50
MO 0105	Edible offal (mammalian), raw	RAC	0.4	15.17	6.07	5.19	2.08	6.30	2.52	6.78	2.71	3.32	1.33	3.17	1.27

Annex 3

FENPYROXIMATE (193)**International Estimated Daily Intake (IEDI)****ADI = 0–0.005 mg/kg bw**

STMR				Diets as g/person/day				Intake as µg/person/day							
Codex Code	Commodity description	Expr as	mg/kg	G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.005	388.92	1.94	335.88	1.68	49.15	0.25	331.25	1.66	468.56	2.34	245.45	1.23
Total intake (µg/person)=				47.3		53.1		33.4		42.6		59.1		40.7	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (µg/person)=				300		300		275		300		300		300	
%ADI=				15.8%		17.7%		12.1%		14.2%		19.7%		13.6%	
Rounded %ADI=				20%		20%		10%		10%		20%		10%	

FENPYROXIMATE (193)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.005 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
FC 0002	Subgroup of Lemons and limes, raw (excl kumquat commodities)	RAC	0.085	0.61	0.05	0.73	0.06	4.01	0.34	0.01	0.00	NC	-
FC 0303	Kumquats, raw (incl juice)	RAC	0.37	18.35	6.79	0.23	0.09	1.78	0.66	0.08	0.03	3.35	1.24
-	Lemon, juice (single strength, incl. concentrated)	PP	0.037	0.01	0.00	0.01	0.00	0.16	0.01	0.01	0.00	NC	-
FC 0003	Subgroup of Mandarins, raw	RAC	0.085	0.16	0.01	0.27	0.02	9.06	0.77	0.01	0.00	0.02	0.00
-	Subgroup of Mandarins, juice	PP	0.037	0.01	0.00	NC	-	NC	-	NC	-	NC	-
FC 0004	Subgroup of Oranges, sweet, sour, raw	RAC	0.052	1.18	0.06	1.11	0.06	14.28	0.74	0.05	0.00	1.08	0.06
JF 0004	Subgroup of Oranges, juice (single strength, incl. concentrated)	PP	0.022	0.08	0.00	0.26	0.01	12.61	0.28	0.14	0.00	0.33	0.01
FC 0005	Subgroup of Pummelo and grapefruits, raw	RAC	0.044	0.63	0.03	0.01	0.00	1.58	0.07	0.01	0.00	NC	-
JF 0203	Grapefruits, juice (single strength, incl. concentrated)	PP	0.019	0.03	0.00	0.02	0.00	0.78	0.01	0.01	0.00	NC	-
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.075	66.67	5.00	2.06	0.15	55.83	4.19	188.29	14.12	1.38	0.10
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.012	0.03	0.00	0.10	0.00	7.19	0.09	0.03	0.00	NC	-
FP 0230	Pear, raw	RAC	0.078	0.07	0.01	0.14	0.01	9.45	0.74	0.01	0.00	0.14	0.01
FS 0013	Subgroup of Cherries, raw	RAC	0.585	0.01	0.01	0.01	0.01	5.96	3.49	0.01	0.01	NC	-
FS 0014	Subgroup of Plums, raw	RAC	0.025	0.07	0.00	0.01	0.00	15.56	0.39	0.01	0.00	NC	-
DF 0014	Plums, dried (prunes)	PP	0.05	0.01	0.00	0.01	0.00	0.37	0.02	0.01	0.00	NC	-
FS 2001	Subgroup of peaches, raw (incl dried apricots)	RAC	0.17	0.02	0.00	0.01	0.00	10.76	1.83	0.01	0.00	NC	-
FB 2005	Subgroup of Caneberries, raw	RAC	0.84	0.01	0.01	7.30	6.13	2.29	1.92	0.01	0.01	NC	-
FB 2006	Subgroup of Bush berries, raw (including processed)	RAC	0.8	0.82	0.66	4.05	3.24	5.94	4.75	0.43	0.34	2.66	2.13
FB 0269	Grapes, raw (i.e. table grapes)	RAC	0.035	0.14	0.00	0.36	0.01	15.22	0.53	0.01	0.00	0.09	0.00
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	0.07	0.01	0.00	0.13	0.01	1.06	0.07	0.01	0.00	0.03	0.00

Annex 3

FENPYROXIMATE (193)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.005 mg/kg bw					
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
JF 0269	Grape juice (from wine grapes)	PP	0.005	0.01	0.00	0.01	0.00	0.41	0.00	0.01	0.00	NC	-
-	Graps must (from wine-grapes)	PP	0.035	0.01	0.00	0.01	0.00	0.11	0.00	0.01	0.00	0.19	0.01
-	Grape wine (incl vermouths) (from wine-grapes)	PP	0.005	0.31	0.00	0.23	0.00	60.43	0.30	0.52	0.00	31.91	0.16
FB 0275	Strawberry, raw	RAC	0.06	0.01	0.00	0.01	0.00	3.35	0.20	0.01	0.00	0.01	0.00
FI 0326	Avocado, raw	RAC	0.05	1.12	0.06	0.01	0.00	0.84	0.04	0.01	0.00	6.60	0.33
VC 2039	Subgroup of Cucumbers and Squashes, raw	RAC	0.12	0.92	0.11	3.20	0.38	13.55	1.63	1.91	0.23	0.05	0.01
VC 0046	Melons, except watermelon, raw (Cantaloupe)	RAC	0.05	0.19	0.01	0.10	0.01	4.98	0.25	0.01	0.00	NC	-
VC 0432	Watermelon, raw	RAC	0.1	4.29	0.43	0.30	0.03	28.70	2.87	0.01	0.00	NC	-
VO 0448	Tomato, raw (incl paste, excl juice, excl canned)	RAC	0.1	15.33	1.53	5.65	0.57	67.23	6.72	1.88	0.19	12.48	1.25
-	Tomato, canned (& peeled)	PP	0.04	0.07	0.00	0.08	0.00	2.42	0.10	0.07	0.00	NC	-
JF 0448	Tomato, juice (single strength, incl concentrated)	PP	0.064	0.05	0.00	0.01	0.00	0.42	0.03	0.01	0.00	0.02	0.00
VO 0051	Subgroup of peppers, raw (incl dried sweet peppers, excl dried chilipeppers), excl okra	RAC	0.05	8.97	0.45	14.13	0.71	25.14	1.26	0.91	0.05	NC	-
VO 2046	Subgroup of eggplants	RAC	0.1	1.31	0.13	8.26	0.83	3.95	0.40	0.01	0.00	NC	-
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0.075	NC	-	NC	-	NC	-	NC	-	NC	-
VP 2062	Subgroup of succulent beans without pods (all commodities within this group)	RAC	0.1	0.37	0.04	3.14	0.31	4.88	0.49	0.01	0.00	NC	-
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0	23.96	0.00	13.56	0.00	213.41	0.00	104.35	0.00	8.56	0.00
VS 2080	Subgroup of stems and petioles	RAC	0.845	5.33	4.50	3.85	3.25	5.80	4.90	3.60	3.04	7.20	6.08
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl beer, incl germ, incl starch, excl flour, excl oil)	RAC	0.01	0.58	0.01	0.52	0.01	3.26	0.03	7.96	0.08	NC	-
CF 1255	Maize, flour (white flour and wholemeal flour)	PP	0.0037	94.34	0.35	8.09	0.03	28.03	0.10	55.94	0.21	28.07	0.10
OR 0645	Maize oil	PP	0.0099	0.33	0.00	0.07	0.00	0.81	0.01	0.01	0.00	NC	-
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0	4.39	0.00	135.53	0.00	6.11	0.00	0.72	0.00	317.74	0.00
SB 0716	Coffee bean raw (incl roasted, incl instant coffee, incl substitutes)	RAC	0.025	0.95	0.02	1.32	0.03	11.64	0.29	2.96	0.07	14.73	0.37
DH 1100	Hops, dry	RAC	5.15	NC	-	NC	-	0.04	0.21	NC	-	NC	-
DT 1114	Tea, green or black, fermented and dried, (including concentrates)	RAC	1.4	0.53	0.74	5.25	7.35	0.86	1.20	0.56	0.78	0.88	1.23
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.015	23.34	0.35	40.71	0.61	97.15	1.46	18.06	0.27	57.71	0.87
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.063	5.84	0.37	10.18	0.64	24.29	1.53	4.52	0.28	14.43	0.91
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.063	1.05	0.07	1.14	0.07	18.69	1.18	0.94	0.06	3.12	0.20

Annex 3

FENPYROXIMATE (193)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.005 mg/kg bw						
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person								
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake	
MO 0105	Edible offal (mammalian), raw	RAC	0.4	4.64	1.86	1.97	0.79	10.01	4.00	3.27	1.31	3.98	1.59	
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.005	108.75	0.54	70.31	0.35	436.11	2.18	61.55	0.31	79.09	0.40	
Total intake (µg/person)=					24.2		25.8		52.3		21.4		17.1	
Bodyweight per region (kg bw) =					60		60		60		60		60	
ADI (µg/person)=					300		300		300		300		300	
%ADI=					8.1%		8.6%		17.4%		7.1%		5.7%	
Rounded %ADI=					8%		9%		20%		7%		6%	

Annex 3

FIPRONIL (202)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.0002 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
FI 0327	Banana, raw (incl plantains) (incl dried)	RAC	0	5.23	0.00	6.94	0.00	99.45	0.00	32.47	0.00	48.30	0.00	24.70	0.00
VA 2031	Subgroup of bulb onions	RAC	0.02	31.65	0.63	43.28	0.87	3.68	0.07	38.48	0.77	20.46	0.41	47.29	0.95
VO 2045	Subgroup of tomatoes, raw (incl processed) (Lycopersicum spp. Only)	RAC	0.008	51.75	0.41	81.80	0.65	16.99	0.14	102.02	0.82	26.32	0.21	214.77	1.72
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0.008	0.68	0.01	NC	-	NC	-	0.39	0.00	0.22	0.00	0.49	0.00
VD 0071	Beans, dry, raw (Phaseolus spp)	RAC	0.002	2.39	0.00	1.61	0.00	10.47	0.02	1.84	0.00	12.90	0.03	7.44	0.01
VD 0541	Soya bean, dry, raw (incl paste, incl curd, incl oil, incl sauce)	RAC	0.00411	72.79	0.30	59.05	0.24	20.55	0.08	74.20	0.30	61.12	0.25	73.24	0.30
VR 0596	Sugar beet, raw (incl sugar)	RAC	0.003	0.13	0.00	NC	-	0.08	0.00	0.66	0.00	0.47	0.00	88.94	0.27
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0.00493	59.74	0.29	316.14	1.56	9.78	0.05	60.26	0.30	54.12	0.27	119.82	0.59
GC 2086	Subgroup of wheat, similar grains and pseudocereals without husks, raw (including processed)	RAC	0.008	381.29	3.05	360.94	2.89	38.45	0.31	282.01	2.26	173.32	1.39	436.22	3.49
GC 2087	Subgroup of barley, similar grains, and pseudocereals with husks, raw (including processed)	RAC	0.00536	19.96	0.11	38.62	0.21	5.13	0.03	4.81	0.03	10.80	0.06	4.44	0.02
CM 0649 (GC 0649)	Rice, husked, dry (incl flour, incl oil, incl beverages, incl starch, excl polished)	REP	0.0023	1.26	0.00	1.58	0.00	31.05	0.07	5.43	0.01	0.90	0.00	2.18	0.01
CM 1205	Rice polished, dry	PP	0.002	34.21	0.07	10.39	0.02	41.72	0.08	82.38	0.16	150.24	0.30	70.47	0.14
GC 2091	Subgroup of Maize Cereals	RAC	0.004	29.81	0.12	44.77	0.18	108.95	0.44	52.37	0.21	60.28	0.24	75.69	0.30
GS 0659	Sugar cane, raw	RAC	0.0000669	38.16	0.00	NC	-	12.58	0.00	0.34	0.00	17.79	0.00	42.78	0.00
-	Sugar cane, molasses	PP	0.0000669	NC	-	NC	-	NC	-	NC	-	0.01	0.00	NC	-
SO 0702	Sunflower seed, raw (incl oil)	RAC	0.008	7.40	0.06	35.86	0.29	1.15	0.01	8.76	0.07	5.45	0.04	13.62	0.11
OR 0691	Cotton seed oil, edible	PP	0.0006	3.22	0.00	1.54	0.00	1.01	0.00	0.74	0.00	1.12	0.00	2.93	0.00
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0085	24.96	0.21	57.95	0.49	16.70	0.14	38.38	0.33	26.46	0.22	29.00	0.25
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.17625	6.24	1.10	14.49	2.55	4.18	0.74	9.60	1.69	6.62	1.17	7.25	1.28
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.17625	3.29	0.58	6.14	1.08	0.82	0.14	1.57	0.28	2.23	0.39	1.07	0.19
MO 0105	Edible offal (mammalian), raw	RAC	0.09145	4.79	0.44	9.68	0.89	2.97	0.27	5.49	0.50	3.84	0.35	5.03	0.46
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.00845	289.65	2.45	485.88	4.11	26.92	0.23	239.03	2.02	199.91	1.69	180.53	1.53
PM 0110	Poultry meat, raw (incl prepared) - 90% as muscle	RAC	0.004856	13.17	0.06	26.78	0.13	7.24	0.04	116.71	0.57	22.54	0.11	32.09	0.16
PM 0110	Poultry meat, raw (incl prepared) - 10% as fat	RAC	0.046982	1.46	0.07	2.98	0.14	0.80	0.04	12.97	0.61	2.50	0.12	3.57	0.17

Annex 3

FIPRONIL (202)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.0002 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.046982	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.10	0.00	0.10	0.00
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.032265	0.12	0.00	0.12	0.00	0.11	0.00	5.37	0.17	0.24	0.01	0.10	0.00
PE 0112	Eggs, raw, (incl dried)	RAC	0.035799	7.84	0.28	23.08	0.83	2.88	0.10	14.89	0.53	9.81	0.35	14.83	0.53
Total intake (µg/person)=					10.3		17.1		3.0		11.6		7.6		12.5
Bodyweight per region (kg bw) =					60		60		60		60		60		60
ADI (µg/person)=					12		12		12		12		12		12
%ADI=					85.5%		142.8%		25.0%		97.0%		63.5%		104.0%
Rounded %ADI=					90%		140%		30%		100%		60%		100%

FIPRONIL (202)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.0002 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
FI 0327	Banana, raw (incl plantains) (incl dried)	RAC	0	25.76	0.00	23.65	0.00	23.83	0.00	24.37	0.00	19.43	0.00	101.55	0.00
VA 2031	Subgroup of bulb onions	RAC	0.02	20.67	0.41	31.32	0.63	37.52	0.75	35.08	0.70	11.77	0.24	13.74	0.27
VO 2045	Subgroup of tomatoes, raw (incl processed) (Lycopersicum spp. Only)	RAC	0.008	64.74	0.52	68.31	0.55	36.05	0.29	82.09	0.66	54.50	0.44	11.69	0.09
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0.008	5.07	0.04	0.83	0.01	0.17	0.00	3.70	0.03	NC	-	NC	-
VD 0071	Beans, dry, raw (Phaseolus spp)	RAC	0.002	1.51	0.00	1.50	0.00	1.90	0.00	5.11	0.01	1.36	0.00	23.43	0.05
VD 0541	Soya bean, dry, raw (incl paste, incl curd, incl oil, incl sauce)	RAC	0.00411	106.33	0.44	117.78	0.48	42.12	0.17	195.70	0.80	222.52	0.91	80.47	0.33
VR 0596	Sugar beet, raw (incl sugar)	RAC	0.003	0.01	0.00	NC	-	0.01	0.00	0.01	0.00	NC	-	NC	-
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0.00493	225.03	1.11	234.24	1.15	71.48	0.35	177.55	0.88	234.55	1.16	37.71	0.19
GC 2086	Subgroup of wheat, similar grains and pseudocereals without husks, raw (including processed)	RAC	0.008	256.28	2.05	280.29	2.24	134.94	1.08	241.61	1.93	217.88	1.74	167.40	1.34
GC 2087	Subgroup of barley, similar grains, and pseudocereals with husks, raw (including processed)	RAC	0.00536	43.68	0.23	60.49	0.32	9.72	0.05	40.47	0.22	49.83	0.27	18.90	0.10
CM 0649 (GC 0649)	Rice, husked, dry (incl flour, incl oil, incl beverages, incl starch, excl polished)	REP	0.0023	3.70	0.01	2.11	0.00	1.51	0.00	1.75	0.00	0.29	0.00	5.12	0.01
CM 1205	Rice polished, dry	PP	0.002	13.38	0.03	10.80	0.02	262.08	0.52	57.16	0.11	12.83	0.03	62.78	0.13
GC 2091	Subgroup of Maize Cereals	RAC	0.004	18.51	0.07	26.18	0.10	26.04	0.10	39.99	0.16	7.36	0.03	64.58	0.26

Annex 3

FIPRONIL (202)

International Estimated Daily Intake (IEDI)

ADI = 0 - 0.0002 mg/kg bw

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day		Intake as µg/person/day									
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
GS 0659	Sugar cane, raw	RAC	0.0000669	NC	-	NC	-	4.27	0.00	0.01	0.00	NC	-	3.24	0.00
-	Sugar cane, molasses	PP	0.0000669	NC	-	NC	-	0.08	0.00	NC	-	NC	-	NC	-
SO 0702	Sunflower seed, raw (incl oil)	RAC	0.008	23.40	0.19	29.33	0.23	1.24	0.01	13.85	0.11	6.48	0.05	6.91	0.06
OR 0691	Cotton seed oil, edible	PP	0.0006	1.68	0.00	0.66	0.00	1.13	0.00	1.18	0.00	0.89	0.00	0.37	0.00
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0085	112.02	0.95	120.71	1.03	63.46	0.54	88.99	0.76	96.24	0.82	41.02	0.35
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.17625	28.01	4.94	30.18	5.32	15.86	2.80	22.25	3.92	24.06	4.24	10.25	1.81
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.17625	6.44	1.14	15.51	2.73	3.79	0.67	8.29	1.46	18.44	3.25	8.00	1.41
MO 0105	Edible offal (mammalian), raw	RAC	0.09145	15.17	1.39	5.19	0.47	6.30	0.58	6.78	0.62	3.32	0.30	3.17	0.29
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.00845	388.92	3.29	335.88	2.84	49.15	0.42	331.25	2.80	468.56	3.96	245.45	2.07
PM 0110	Poultry meat, raw (incl prepared) - 90% as muscle	RAC	0.004856	66.38	0.32	48.47	0.24	21.58	0.10	78.41	0.38	48.04	0.23	76.01	0.37
PM 0110	Poultry meat, raw (incl prepared) - 10% as fat	RAC	0.046982	7.38	0.35	5.39	0.25	2.40	0.11	8.71	0.41	5.34	0.25	8.45	0.40
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.046982	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.71	0.03	NC	-
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.032265	0.33	0.01	0.72	0.02	0.27	0.01	0.35	0.01	0.80	0.03	NC	-
PE 0112	Eggs, raw, (incl dried)	RAC	0.035799	25.84	0.93	29.53	1.06	28.05	1.00	33.19	1.19	36.44	1.30	8.89	0.32
Total intake (µg/person)=				18.4		19.7		9.6		17.2		19.3		9.8	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (µg/person)=				12		12		11		12		12		12	
%ADI=				153.4%		164.3%		87.0%		143.1%		160.7%		82.0%	
Rounded %ADI=				150%		160%		90%		140%		160%		80%	

Annex 3

FIPRONIL (202)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.0002 mg/kg bw					
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
FI 0327	Banana, raw (incl plantains) (incl dried)	RAC	0	44.80	0.00	118.17	0.00	25.25	0.00	454.49	0.00	310.23	0.00
VA 2031	Subgroup of bulb onions	RAC	0.02	9.83	0.20	22.30	0.45	34.69	0.69	9.65	0.19	2.39	0.05
VO 2045	Subgroup of tomatoes, raw (incl processed) (Lycopersicum spp. Only)	RAC	0.008	15.50	0.12	5.78	0.05	71.52	0.57	2.00	0.02	12.50	0.10
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0.008	NC	-	NC	-	NC	-	NC	-	NC	-
VD 0071	Beans, dry, raw (Phaseolus spp)	RAC	0.002	7.11	0.01	2.33	0.00	3.76	0.01	44.70	0.09	3.27	0.01
VD 0541	Soya bean, dry, raw (incl paste, incl curd, incl oil, incl sauce)	RAC	0.00411	15.80	0.06	14.29	0.06	104.36	0.43	17.11	0.07	35.20	0.14
VR 0596	Sugar beet, raw (incl sugar)	RAC	0.003	3.93	0.01	1.68	0.01	NC	-	NC	-	36.12	0.11
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0.00493	23.96	0.12	13.56	0.07	213.41	1.05	104.35	0.51	8.56	0.04
GC 2086	Subgroup of wheat, similar grains and pseudocereals without husks, raw (including processed)	RAC	0.008	57.23	0.46	110.47	0.88	286.57	2.29	25.82	0.21	132.92	1.06
GC 2087	Subgroup of barley, similar grains, and pseudocereals with husks, raw (including processed)	RAC	0.00536	11.99	0.06	5.22	0.03	49.50	0.27	3.82	0.02	16.26	0.09
CM 0649 (GC 0649)	Rice, husked, dry (incl flour, incl oil, incl beverages, incl starch, excl polished)	REP	0.0023	13.58	0.03	4.29	0.01	2.17	0.00	0.01	0.00	8.84	0.02
CM 1205	Rice polished, dry	PP	0.002	30.20	0.06	218.34	0.44	12.77	0.03	15.24	0.03	51.35	0.10
GC 2091	Subgroup of Maize Cereals	RAC	0.004	116.66	0.47	10.52	0.04	38.46	0.15	76.60	0.31	34.44	0.14
GS 0659	Sugar cane, raw	RAC	0.0000669	5.62	0.00	50.91	0.00	NC	-	11.04	0.00	0.10	0.00
-	Sugar cane, molasses	PP	0.0000669	NC	-	NC	-	NC	-	NC	-	NC	-
SO 0702	Sunflower seed, raw (incl oil)	RAC	0.008	0.94	0.01	0.22	0.00	32.01	0.26	12.12	0.10	0.48	0.00
OR 0691	Cotton seed oil, edible	PP	0.0006	1.28	0.00	0.05	0.00	0.45	0.00	0.42	0.00	0.15	0.00
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0085	23.34	0.20	40.71	0.35	97.15	0.83	18.06	0.15	57.71	0.49
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.17625	5.84	1.03	10.18	1.79	24.29	4.28	4.52	0.80	14.43	2.54
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.17625	1.05	0.19	1.14	0.20	18.69	3.29	0.94	0.17	3.12	0.55
MO 0105	Edible offal (mammalian), raw	RAC	0.09145	4.64	0.42	1.97	0.18	10.01	0.92	3.27	0.30	3.98	0.36
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.00845	108.75	0.92	70.31	0.59	436.11	3.69	61.55	0.52	79.09	0.67
PM 0110	Poultry meat, raw (incl prepared) - 90% as muscle	RAC	0.004856	3.53	0.02	10.83	0.05	51.36	0.25	4.53	0.02	50.00	0.24
PM 0110	Poultry meat, raw (incl prepared) - 10% as fat	RAC	0.046982	0.39	0.02	1.20	0.06	5.71	0.27	0.50	0.02	5.56	0.26
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.046982	NC	-	NC	-	0.32	0.02	NC	-	NC	-
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.032265	0.10	0.00	0.70	0.02	0.97	0.03	0.10	0.00	NC	-

Annex 3

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FIPRONIL (202)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.0002 mg/kg bw						
Codex Code		Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
					G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
PE 0112	Eggs, raw, (incl dried)	RAC	0.035799	3.84	0.14	4.41	0.16	27.25	0.98	1.13	0.04	7.39	0.26	
Total intake (µg/person)=					4.6		5.4		20.3		3.6		7.2	
Bodyweight per region (kg bw) =					60		60		60		60		60	
ADI (µg/person)=					12		12		12		12		12	
%ADI=					37.9%		45.3%		169.1%		29.7%		60.4%	
Rounded %ADI=					40%		50%		170%		30%		60%	

Annex 3

FLUTIANIL (319)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.8 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day		Intake as µg/person/day									
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.047	13.49	0.63	26.63	1.25	15.05	0.71	16.28	0.77	6.47	0.30	47.88	2.25
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.005	0.32	0.00	3.07	0.02	0.07	0.00	5.00	0.03	0.29	0.00	5.57	0.03
FS 0013	Subgroup of Cherries, raw	RAC	0.11	0.92	0.10	9.15	1.01	0.01	0.00	0.61	0.07	0.06	0.01	6.64	0.73
FB 0269	Grapes, raw (incl must, incl wine, excl dried, excl juice)	RAC	0.075	13.94	1.05	26.46	1.98	2.79	0.21	18.58	1.39	8.54	0.64	59.95	4.50
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	0.09	0.51	0.05	0.51	0.05	0.01	0.00	1.27	0.11	0.12	0.01	2.07	0.19
JF 0269	Grape juice (from wine grapes)	PP	0.05	0.14	0.01	0.29	0.01	0.05	0.00	0.30	0.02	0.24	0.01	0.05	0.00
Total intake (µg/person)=				1.8		4.3		0.9		2.4		1.0		7.7	
Bodyweight per region (kg bw) =				60		60		60		60		60		60	
ADI (µg/person)=				48000		48000		48000		48000		48000		48000	
%ADI=				0.0%		0.0%		0.0%		0.0%		0.0%		0.0%	
Rounded %ADI=				0%		0%		0%		0%		0%		0%	

FLUTIANIL (319)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.8 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day		Intake as µg/person/day									
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.047	41.14	1.93	56.49	2.66	26.64	1.25	31.58	1.48	51.94	2.44	3.05	0.14
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.005	14.88	0.07	11.98	0.06	0.15	0.00	9.98	0.05	30.32	0.15	3.47	0.02
FS 0013	Subgroup of Cherries, raw	RAC	0.11	1.40	0.15	4.21	0.46	0.04	0.00	2.93	0.32	1.50	0.17	NC	-
FB 0269	Grapes, raw (incl must, incl wine, excl dried, excl juice)	RAC	0.075	128.64	9.65	97.04	7.28	7.74	0.58	43.94	3.30	88.68	6.65	8.80	0.66
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	0.09	3.09	0.28	1.51	0.14	0.03	0.00	1.38	0.12	4.26	0.38	0.42	0.04
JF 0269	Grape juice (from wine grapes)	PP	0.05	0.56	0.03	1.96	0.10	0.02	0.00	2.24	0.11	2.27	0.11	0.34	0.02
Total intake (µg/person)=				12.1		10.7		1.8		5.4		9.9		0.9	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (µg/person)=				48000		48000		44000		48000		48000		48000	
%ADI=				0.0%		0.0%		0.0%		0.0%		0.0%		0.0%	
Rounded %ADI=				0%		0%		0%		0%		0%		0%	

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FLUTIANIL (319)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.8 mg/kg bw						
Commodity description		Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person								
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake	
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.047	66.67	3.13	2.06	0.10	55.83	2.62	188.29	8.85	1.38	0.06	
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.005	0.03	0.00	0.10	0.00	7.19	0.04	0.03	0.00	NC	-	
FS 0013	Subgroup of Cherries, raw	RAC	0.11	0.01	0.00	0.01	0.00	5.96	0.66	0.01	0.00	NC	-	
FB 0269	Grapes, raw (incl must, incl wine, excl dried, excl juice)	RAC	0.075	0.57	0.04	0.69	0.05	98.34	7.38	0.73	0.05	44.12	3.31	
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	0.09	0.01	0.00	0.13	0.01	1.06	0.10	0.01	0.00	0.03	0.00	
JF 0269	Grape juice (from wine grapes)	PP	0.05	0.01	0.00	0.01	0.00	0.41	0.02	0.01	0.00	NC	-	
Total intake (µg/person)=					3.2		0.2		10.8		8.9		3.4	
Bodyweight per region (kg bw) =					60		60		60		60		60	
ADI (µg/person)=					48000		48000		48000		48000		48000	
%ADI=					0.0%		0.0%		0.0%		0.0%		0.0%	
Rounded %ADI=					0%		0%		0%		0%		0%	

Annex 3

METALAXYL (138)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.08 mg/kg bw								
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day								
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake	
FC 0004	Subgroup of Oranges, sweet, sour, raw	RAC	0.013	20.66	0.27	5.23	0.07	11.90	0.15	37.90	0.49	21.16	0.28	56.46	0.73	
JF 0004	Subgroup of Oranges, juice (single strength, incl. concentrated)	PP	0.02	1.27	0.03	2.20	0.04	0.09	0.00	11.81	0.24	0.46	0.01	1.69	0.03	
FP 0226	Apple, raw (incl juice, incl cider)	RAC	0	13.94	0.00	30.81	0.00	15.14	0.00	23.10	0.00	6.86	0.00	55.48	0.00	
FP 0230	Pear, raw	RAC	0	2.16	0.00	6.24	0.00	0.05	0.00	4.07	0.00	1.16	0.00	5.34	0.00	
FB 0269	Grapes, raw (incl must, incl dried, excl juice, excl wine)	RAC	0.182	15.15	2.76	11.38	2.07	0.05	0.01	22.18	4.04	4.19	0.76	63.07	11.48	
JF 0269	Grape juice (from wine grapes)	PP	0.073	0.14	0.01	0.29	0.02	0.05	0.00	0.30	0.02	0.24	0.02	0.05	0.00	
-	Grape wine (incl vermouths) (from wine-grapes)	PP	0.138	0.67	0.09	12.53	1.73	2.01	0.28	1.21	0.17	3.53	0.49	4.01	0.55	
VA 2031	Subgroup of bulb onions	RAC	0.02	31.65	0.63	43.28	0.87	3.68	0.07	38.48	0.77	20.46	0.41	47.29	0.95	
VB 0042	Subgroup of Flowerhead Brassica, raw	RAC	0.275	2.54	0.70	0.49	0.13	0.01	0.00	3.57	0.98	7.79	2.14	3.12	0.86	
VB 0402	Brussels sprouts, raw	RAC	0.44	0.63	0.28	6.41	2.82	0.13	0.06	1.03	0.45	NC	-	2.35	1.03	
VB 0041	Cabbages, head, raw	RAC	0.22	2.73	0.60	27.92	6.14	0.55	0.12	4.47	0.98	4.27	0.94	10.25	2.26	
VC 0046	Melons, except watermelon, raw (Cantaloupe)	RAC	0.013	8.90	0.12	8.64	0.11	0.80	0.01	17.90	0.23	2.80	0.04	29.17	0.38	
VO 2045	Subgroup of tomatoes, raw (incl processed) (Lycopersicum spp. Only)	RAC	0.058	51.75	3.00	81.80	4.74	16.99	0.99	102.02	5.92	26.32	1.53	214.77	12.46	
VL 0483	Lettuce, leaf, raw	RAC	1.43	0.53	0.76	0.36	0.51	0.16	0.23	6.21	8.88	1.90	2.72	6.05	8.65	
VL 0502	Spinach, raw	RAC	0.22	0.74	0.16	0.22	0.05	0.02	0.00	0.91	0.20	0.04	0.01	2.92	0.64	
VR 0577	Carrots, raw	RAC	0.02	9.51	0.19	30.78	0.62	0.37	0.01	8.75	0.18	2.80	0.06	6.10	0.12	
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0.01	59.74	0.60	316.14	3.16	9.78	0.10	60.26	0.60	54.12	0.54	119.82	1.20	
SO 2091	Subgroup of Sunflower seeds, raw (incl processed)	RAC	0	7.43	0.00	36.06	0.00	1.15	0.00	8.77	0.00	5.74	0.00	13.63	0.00	
HS 0790	Pepper (black, white, pink, green)	RAC	0.455	0.08	0.04	0.13	0.06	0.05	0.02	0.36	0.16	0.17	0.08	0.13	0.06	
Total intake (µg/person)=					10.2		23.2		2.1		24.3		10.0		41.4	
Bodyweight per region (kg bw) =					60		60		60		60		60		60	
ADI (µg/person)=					4800		4800		4800		4800		4800		4800	
%ADI=					0.2%		0.5%		0.0%		0.5%		0.2%		0.9%	
Rounded %ADI=					0%		0%		0%		1%		0%		1%	

Annex 3

METALAXYL (138)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.08 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
FC 0004	Subgroup of Oranges, sweet, sour, raw	RAC	0.013	15.68	0.20	24.00	0.31	6.80	0.09	29.09	0.38	15.39	0.20	160.47	2.09
JF 0004	Subgroup of Oranges, juice (single strength, incl. concentrated)	PP	0.02	33.31	0.67	1.78	0.04	0.28	0.01	18.97	0.38	14.01	0.28	13.36	0.27
FP 0226	Apple, raw (incl juice, incl cider)	RAC	0	61.44	0.00	72.81	0.00	26.84	0.00	45.18	0.00	93.28	0.00	7.78	0.00
FP 0230	Pear, raw	RAC	0	8.79	0.00	8.44	0.00	12.37	0.00	9.60	0.00	10.27	0.00	0.23	0.00
FB 0269	Grapes, raw (incl must, incl dried, excl juice, excl wine)	RAC	0.182	19.37	3.53	17.62	3.21	5.33	0.97	15.24	2.77	22.40	4.08	2.51	0.46
JF 0269	Grape juice (from wine grapes)	PP	0.073	0.56	0.04	1.96	0.14	0.02	0.00	2.24	0.16	2.27	0.17	0.34	0.02
-	Grape wine (incl vermouths) (from wine-grapes)	PP	0.138	88.93	12.27	62.41	8.61	1.84	0.25	25.07	3.46	61.17	8.44	5.84	0.81
VA 2031	Subgroup of bulb onions	RAC	0.02	20.67	0.41	31.32	0.63	37.52	0.75	35.08	0.70	11.77	0.24	13.74	0.27
VB 0042	Subgroup of Flowerhead Brassica, raw	RAC	0.275	9.50	2.61	6.77	1.86	NC	-	3.21	0.88	9.36	2.57	0.75	0.21
VB 0402	Brussels sprouts, raw	RAC	0.44	2.24	0.99	2.67	1.17	6.23	2.74	0.32	0.14	4.19	1.84	2.58	1.14
VB 0041	Cabbages, head, raw	RAC	0.22	8.97	1.97	27.12	5.97	1.44	0.32	24.96	5.49	4.55	1.00	11.23	2.47
VC 0046	Melons, except watermelon, raw (Cantaloupe)	RAC	0.013	9.20	0.12	11.95	0.16	14.63	0.19	8.99	0.12	7.86	0.10	2.46	0.03
VO 2045	Subgroup of tomatoes, raw (incl processed) (Lycopersicum spp. Only)	RAC	0.058	64.74	3.75	68.31	3.96	36.05	2.09	82.09	4.76	54.50	3.16	11.69	0.68
VL 0483	Lettuce, leaf, raw	RAC	1.43	14.50	20.74	11.76	16.82	13.14	18.79	19.50	27.89	4.81	6.88	2.23	3.19
VL 0502	Spinach, raw	RAC	0.22	2.20	0.48	1.76	0.39	13.38	2.94	2.94	0.65	5.53	1.22	0.02	0.00
VR 0577	Carrots, raw	RAC	0.02	26.26	0.53	27.13	0.54	10.07	0.20	16.49	0.33	44.69	0.89	8.75	0.18
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0.01	225.03	2.25	234.24	2.34	71.48	0.71	177.55	1.78	234.55	2.35	37.71	0.38
SO 2091	Subgroup of Sunflower seeds, raw (incl processed)	RAC	0	23.43	0.00	29.34	0.00	1.24	0.00	14.00	0.00	6.48	0.00	6.91	0.00
HS 0790	Pepper (black, white, pink, green)	RAC	0.455	0.31	0.14	0.41	0.19	0.12	0.05	0.34	0.15	0.70	0.32	0.89	0.40
Total intake (µg/person)=				50.7		46.3		30.1		50.0		33.7		12.6	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (µg/person)=				4800		4800		4400		4800		4800		4800	
%ADI=				1.1%		1.0%		0.7%		1.0%		0.7%		0.3%	
Rounded %ADI=				1%		1%		1%		1%		1%		0%	

Annex 3

METALAXYL (138)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.08 mg/kg bw						
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person								
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake	
FC 0004	Subgroup of Oranges, sweet, sour, raw	RAC	0.013	1.18	0.02	1.11	0.01	14.28	0.19	0.05	0.00	1.08	0.01	
JF 0004	Subgroup of Oranges, juice (single strength, incl. concentrated)	PP	0.02	0.08	0.00	0.26	0.01	12.61	0.25	0.14	0.00	0.33	0.01	
FP 0226	Apple, raw (incl juice, incl cider)	RAC	0	66.71	0.00	2.19	0.00	65.63	0.00	188.34	0.00	1.38	0.00	
FP 0230	Pear, raw	RAC	0	0.07	0.00	0.14	0.00	9.45	0.00	0.01	0.00	0.14	0.00	
FB 0269	Grapes, raw (incl must, incl dried, excl juice, excl wine)	RAC	0.182	0.16	0.03	0.92	0.17	19.73	3.59	0.02	0.00	0.40	0.07	
JF 0269	Grape juice (from wine grapes)	PP	0.073	0.01	0.00	0.01	0.00	0.41	0.03	0.01	0.00	NC	-	
-	Grape wine (incl vermouths) (from wine-grapes)	PP	0.138	0.31	0.04	0.23	0.03	60.43	8.34	0.52	0.07	31.91	4.40	
VA 2031	Subgroup of bulb onions	RAC	0.02	9.83	0.20	22.30	0.45	34.69	0.69	9.65	0.19	2.39	0.05	
VB 0042	Subgroup of Flowerhead Brassica, raw	RAC	0.275	0.02	0.01	0.02	0.01	4.86	1.34	0.01	0.00	NC	-	
VB 0402	Brussels sprouts, raw	RAC	0.44	0.88	0.39	0.69	0.30	2.89	1.27	0.01	0.00	NC	-	
VB 0041	Cabbages, head, raw	RAC	0.22	3.82	0.84	2.99	0.66	49.16	10.82	0.01	0.00	NC	-	
VC 0046	Melons, except watermelon, raw (Cantaloupe)	RAC	0.013	0.19	0.00	0.10	0.00	4.98	0.06	0.01	0.00	NC	-	
VO 2045	Subgroup of tomatoes, raw (incl processed) (Lycopersicum spp. Only)	RAC	0.058	15.50	0.90	5.78	0.34	71.52	4.15	2.00	0.12	12.50	0.73	
VL 0483	Lettuce, leaf, raw	RAC	1.43	0.29	0.41	0.03	0.04	6.71	9.60	0.01	0.01	NC	-	
VL 0502	Spinach, raw	RAC	0.22	0.17	0.04	0.01	0.00	0.81	0.18	0.01	0.00	NC	-	
VR 0577	Carrots, raw	RAC	0.02	2.07	0.04	3.00	0.06	25.29	0.51	0.05	0.00	NC	-	
VR 0589	Potato, raw (incl flour, incl frozen, incl starch, incl tapioca)	RAC	0.01	23.96	0.24	13.56	0.14	213.41	2.13	104.35	1.04	8.56	0.09	
SO 2091	Subgroup of Sunflower seeds, raw (incl processed)	RAC	0	0.99	0.00	0.22	0.00	32.01	0.00	12.12	0.00	0.48	0.00	
HS 0790	Pepper (black, white, pink, green)	RAC	0.455	0.05	0.02	1.12	0.51	0.24	0.11	0.14	0.06	0.18	0.08	
Total intake (µg/person)=					3.2		2.7		43.3		1.5		5.4	
Bodyweight per region (kg bw) =					60		60		60		60		60	
ADI (µg/person)=					4800		4800		4800		4800		4800	
%ADI=					0.1%		0.1%		0.9%		0.0%		0.1%	
Rounded %ADI=					0%		0%		1%		0%		0%	

Annex 3

METCONAZOLE (313)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.04 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day		Intake as µg/person/day									
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
FS 0013	Subgroup of Cherries, raw	RAC	0.07	0.92	0.06	9.15	0.64	0.01	0.00	0.61	0.04	0.06	0.00	6.64	0.46
FS 0014	Subgroup of Plums, raw	RAC	0.04	2.40	0.10	8.60	0.34	0.06	0.00	2.52	0.10	0.58	0.02	4.16	0.17
DF 0014	Plums, dried (prunes)	PP	0.092	0.09	0.01	0.06	0.01	0.01	0.00	0.18	0.02	0.04	0.00	0.06	0.01
FS 2001	Subgroup of peaches, raw (incl dried apricots)	RAC	0.045	8.01	0.36	5.87	0.26	0.18	0.01	8.19	0.37	1.64	0.07	22.46	1.01
FB 0020	Blueberries, raw	RAC	0.14	0.01	0.00	0.03	0.00	0.01	0.00	0.01	0.00	0.01	0.00	0.01	0.00
FI 0327	Banana, raw (incl plantains) (incl dried)	RAC	0.1	5.23	0.52	6.94	0.69	99.45	9.95	32.47	3.25	48.30	4.83	24.70	2.47
VA 0381	Garlic, raw	RAC	0.05	2.29	0.11	5.78	0.29	0.11	0.01	3.69	0.18	1.65	0.08	3.91	0.20
-	Onions, dry, raw	RAC	0.05	29.36	1.47	37.50	1.88	3.56	0.18	34.78	1.74	18.81	0.94	43.38	2.17
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0	0.68	0.00	NC	-	NC	-	0.39	0.00	0.22	0.00	0.49	0.00
VD 2065	Subgroup of dry beans, raw (incl processed)	RAC	0.04	78.20	3.13	60.68	2.43	35.89	1.44	80.34	3.21	75.90	3.04	87.62	3.50
VD 0541	Soya bean, dry, raw (incl flour, incl paste, incl curd, incl sauce, excl oil)	RAC	0.01	0.63	0.01	1.09	0.01	0.40	0.00	1.40	0.01	1.68	0.02	0.48	0.00
OR 0541	Soya oil, refined	PP	0.005	12.99	0.06	10.43	0.05	3.63	0.02	13.10	0.07	10.70	0.05	13.10	0.07
VD 2066	Subgroup of dry peas, raw	RAC	0.0425	9.09	0.39	3.35	0.14	1.06	0.05	9.48	0.40	15.11	0.64	10.58	0.45
VR 0596	Sugar beet, raw	RAC	0.02	NC	-	NC	-	NC	-	NC	-	0.01	0.00	NC	-
-	Sugar beet, sugar	PP	0.012	0.02	0.00	NC	-	0.01	0.00	0.09	0.00	0.07	0.00	12.63	0.15
VR 2071	Subgroup of tuberous and corm vegetables, raw (incl processed)	RAC	0	63.11	0.00	316.33	0.00	651.91	0.00	72.06	0.00	84.88	0.00	132.70	0.00
GC 0653	Triticale, raw (incl flour)	RAC	0.035	NC	-	NC	-	NC	-	0.01	0.00	0.39	0.01	NC	-
GC 0654	Wheat, raw (incl bulgur, incl fermented beverages, incl germ, incl wholemeal bread, excl white flour products, excl white bread)	RAC	0.035	0.01	0.00	1.13	0.04	0.01	0.00	0.05	0.00	0.74	0.03	0.01	0.00
CP 1211	Wheat, white bread	PP	0.021	0.25	0.01	0.63	0.01	0.12	0.00	0.43	0.01	1.39	0.03	0.22	0.00
CF 1211	Wheat, white flour (incl white flour products: starch, gluten, macaroni, pastry)	PP	0.008	301.24	2.41	268.64	2.15	30.21	0.24	222.51	1.78	134.73	1.08	343.12	2.74
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl flour, incl oil, incl beer, incl germ, incl starch)	RAC	0.01	29.81	0.30	44.77	0.45	108.95	1.09	52.37	0.52	60.28	0.60	75.69	0.76
GC 0447	Sweet corn on the cob, raw (incl frozen kernels, incl canned kernels)	RAC	0.01	0.14	0.00	0.94	0.01	5.70	0.06	2.61	0.03	1.94	0.02	0.22	0.00
GS 0659	Sugar cane, raw	RAC	0.0205	38.16	0.78	NC	-	12.58	0.26	0.34	0.01	17.79	0.36	42.78	0.88
-	Sugar cane, molasses	PP	0.027	NC	-	NC	-	NC	-	NC	-	0.01	0.00	NC	-
-	Sugar cane, sugar (incl non-centrifugal sugar, incl refined sugar and maltose)	PP	0.002	61.52	0.12	86.27	0.17	18.80	0.04	80.02	0.16	66.39	0.13	56.32	0.11
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0	4.06	0.00	3.27	0.00	7.01	0.00	13.93	0.00	14.01	0.00	9.36	0.00

Annex 3

METCONAZOLE (313)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.04 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day		Intake as µg/person/day									
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
SO 0495	Rape seed, raw	RAC	0.02	0.02	0.00	NC	-	NC	-	0.01	0.00	0.75	0.02	0.01	0.00
OR 0495	Rape seed oil, edible	PP	0.032	0.35	0.01	0.44	0.01	0.19	0.01	0.97	0.03	3.28	0.10	0.77	0.02
SO 2091	Subgroup of Sunflower seeds, raw (incl processed)	RAC	0.089	7.43	0.66	36.06	3.21	1.15	0.10	8.77	0.78	5.74	0.51	13.63	1.21
OR 0691	Cotton seed oil, edible	PP	0.004	3.22	0.01	1.54	0.01	1.01	0.00	0.74	0.00	1.12	0.00	2.93	0.01
SO 0697	Peanuts, nutmeat, raw (incl roasted, incl butter, excl oil)	RAC	0.04	0.46	0.02	1.21	0.05	6.64	0.27	2.71	0.11	1.26	0.05	1.84	0.07
OR 0697	Peanut oil, edible	PP	0.056	0.36	0.02	0.01	0.00	2.57	0.14	0.07	0.00	2.29	0.13	0.36	0.02
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat)	RAC	0	31.20	0.00	72.44	0.00	20.88	0.00	47.98	0.00	33.08	0.00	36.25	0.00
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0	3.29	0.00	6.14	0.00	0.82	0.00	1.57	0.00	2.23	0.00	1.07	0.00
MO 0105	Edible offal (mammalian), raw	RAC	0.037	4.79	0.18	9.68	0.36	2.97	0.11	5.49	0.20	3.84	0.14	5.03	0.19
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0	289.65	0.00	485.88	0.00	26.92	0.00	239.03	0.00	199.91	0.00	180.53	0.00
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	14.63	0.00	29.76	0.00	8.04	0.00	129.68	0.00	25.04	0.00	35.66	0.00
PF 0111	Poultry fat, raw (incl rendered)	RAC	0	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.10	0.00	0.10	0.00
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.019	0.12	0.00	0.12	0.00	0.11	0.00	5.37	0.10	0.24	0.00	0.10	0.00
PE 0112	Eggs, raw, (incl dried)	RAC	0	7.84	0.00	23.08	0.00	2.88	0.00	14.89	0.00	9.81	0.00	14.83	0.00
-	-	-		-	-	-	-	-	-	-	-	-	-	-	-

Total intake (µg/person)=

10.7

13.2

14.0

13.1

12.9

16.7

Bodyweight per region (kg bw) =

60

60

60

60

60

60

ADI (µg/person)=

2400

2400

2400

2400

2400

2400

%ADI=

0.4%

0.6%

0.6%

0.5%

0.5%

0.7%

Rounded %ADI=

0%

1%

1%

1%

1%

1%

METCONAZOLE (313)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.04 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
FS 0013	Subgroup of Cherries, raw	RAC	0.07	1.40	0.10	4.21	0.29	0.04	0.00	2.93	0.21	1.50	0.11	NC	-
FS 0014	Subgroup of Plums, raw	RAC	0.04	3.75	0.15	3.33	0.13	5.94	0.24	2.64	0.11	2.50	0.10	0.06	0.00
DF 0014	Plums, dried (prunes)	PP	0.092	0.61	0.06	0.35	0.03	0.05	0.00	0.35	0.03	0.49	0.05	0.13	0.01
FS 2001	Subgroup of peaches, raw (incl dried apricots)	RAC	0.045	13.03	0.59	16.29	0.73	8.29	0.37	12.95	0.58	5.35	0.24	0.04	0.00
FB 0020	Blueberries, raw	RAC	0.14	0.04	0.01	0.23	0.03	0.01	0.00	0.83	0.12	0.33	0.05	NC	-
FI 0327	Banana, raw (incl plantains) (incl dried)	RAC	0.1	25.76	2.58	23.65	2.37	23.83	2.38	24.37	2.44	19.43	1.94	101.55	10.16
VA 0381	Garlic, raw	RAC	0.05	0.98	0.05	1.49	0.07	12.88	0.64	3.74	0.19	2.05	0.10	1.14	0.06
-	Onions, dry, raw	RAC	0.05	19.69	0.98	29.83	1.49	24.64	1.23	31.35	1.57	9.72	0.49	12.59	0.63
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0	5.07	0.00	0.83	0.00	0.17	0.00	3.70	0.00	NC	-	NC	-
VD 2065	Subgroup of dry beans, raw (incl processed)	RAC	0.04	107.87	4.31	119.29	4.77	45.91	1.84	201.31	8.05	224.04	8.96	104.90	4.20
VD 0541	Soya bean, dry, raw (incl flour, incl paste, incl curd, incl sauce, excl oil)	RAC	0.01	0.47	0.00	0.77	0.01	9.12	0.09	8.05	0.08	0.04	0.00	6.06	0.06
OR 0541	Soya oil, refined	PP	0.005	19.06	0.10	21.06	0.11	5.94	0.03	33.78	0.17	40.05	0.20	13.39	0.07
VD 2066	Subgroup of dry peas, raw	RAC	0.0425	5.01	0.21	3.76	0.16	1.82	0.08	3.44	0.15	3.49	0.15	5.15	0.22
VR 0596	Sugar beet, raw	RAC	0.02	0.01	0.00	NC	-	0.01	0.00	0.01	0.00	NC	-	NC	-
-	Sugar beet, sugar	PP	0.012	0.01	0.00	NC	-	0.01	0.00	NC	-	NC	-	NC	-
VR 2071	Subgroup of tuberous and corm vegetables, raw (incl processed)	RAC	0	226.09	0.00	234.58	0.00	161.10	0.00	185.04	0.00	234.85	0.00	100.25	0.00
GC 0653	Triticale, raw (incl flour)	RAC	0.035	0.01	0.00	0.17	0.01	0.29	0.01	0.01	0.00	NC	-	NC	-
GC 0654	Wheat, raw (incl bulgur, incl fermented beverages, incl germ, incl wholemeal bread, excl white flour products, excl white bread)	RAC	0.035	1.00	0.04	0.11	0.00	0.05	0.00	0.84	0.03	0.06	0.00	0.06	0.00
CP 1211	Wheat, white bread	PP	0.021	1.30	0.03	0.46	0.01	0.06	0.00	0.22	0.00	2.44	0.05	0.77	0.02
CF 1211	Wheat, white flour (incl white flour products: starch, gluten, macaroni, pastry)	PP	0.008	198.08	1.58	193.03	1.54	106.24	0.85	185.09	1.48	168.67	1.35	131.59	1.05
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl flour, incl oil, incl beer, incl germ, incl starch)	RAC	0.01	18.51	0.19	26.18	0.26	26.04	0.26	39.99	0.40	7.36	0.07	64.58	0.65
GC 0447	Sweet corn on the cob, raw (incl frozen kernels, incl canned kernels)	RAC	0.01	11.43	0.11	3.71	0.04	0.74	0.01	13.63	0.14	3.07	0.03	1.50	0.02
GS 0659	Sugar cane, raw	RAC	0.0205	NC	-	NC	-	4.27	0.09	0.01	0.00	NC	-	3.24	0.07
-	Sugar cane, molasses	PP	0.027	NC	-	NC	-	0.08	0.00	NC	-	NC	-	NC	-
-	Sugar cane, sugar (incl non-centrifugal sugar, incl refined sugar and maltose)	PP	0.002	92.24	0.18	95.72	0.19	24.12	0.05	77.39	0.15	117.73	0.24	100.67	0.20
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0	8.52	0.00	8.94	0.00	15.09	0.00	9.60	0.00	14.57	0.00	26.26	0.00
SO 0495	Rape seed, raw	RAC	0.02	NC	-	NC	-	0.01	0.00	NC	-	NC	-	NC	-

Annex 3

METCONAZOLE (313)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.04 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
OR 0495	Rape seed oil, edible	PP	0.032	12.52	0.40	7.63	0.24	3.00	0.10	6.01	0.19	NC	-	NC	-
SO 2091	Subgroup of Sunflower seeds, raw (incl processed)	RAC	0.089	23.43	2.09	29.34	2.61	1.24	0.11	14.00	1.25	6.48	0.58	6.91	0.61
OR 0691	Cotton seed oil, edible	PP	0.004	1.68	0.01	0.66	0.00	1.13	0.00	1.18	0.00	0.89	0.00	0.37	0.00
SO 0697	Peanuts, nutmeat, raw (incl roasted, incl butter, excl oil)	RAC	0.04	3.26	0.13	2.22	0.09	5.38	0.22	4.85	0.19	1.54	0.06	1.82	0.07
OR 0697	Peanut oil, edible	PP	0.056	1.02	0.06	0.23	0.01	1.81	0.10	0.42	0.02	5.23	0.29	0.01	0.00
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat)	RAC	0	140.03	0.00	150.89	0.00	79.32	0.00	111.24	0.00	120.30	0.00	51.27	0.00
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0	6.44	0.00	15.51	0.00	3.79	0.00	8.29	0.00	18.44	0.00	8.00	0.00
MO 0105	Edible offal (mammalian), raw	RAC	0.037	15.17	0.56	5.19	0.19	6.30	0.23	6.78	0.25	3.32	0.12	3.17	0.12
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0	388.92	0.00	335.88	0.00	49.15	0.00	331.25	0.00	468.56	0.00	245.45	0.00
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	73.76	0.00	53.86	0.00	23.98	0.00	87.12	0.00	53.38	0.00	84.45	0.00
PF 0111	Poultry fat, raw (incl rendered)	RAC	0	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.71	0.00	NC	-
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.019	0.33	0.01	0.72	0.01	0.27	0.01	0.35	0.01	0.80	0.02	NC	-
PE 0112	Eggs, raw, (incl dried)	RAC	0	25.84	0.00	29.53	0.00	28.05	0.00	33.19	0.00	36.44	0.00	8.89	0.00
-	-	-		-	-	-	-	-	-	-	-	-	-	-	-
Total intake (µg/person)=				14.5		15.4		8.9		17.8		15.2		18.2	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (µg/person)=				2400		2400		2200		2400		2400		2400	
%ADI=				0.6%		0.6%		0.4%		0.7%		0.6%		0.8%	
Rounded %ADI=				1%		1%		0%		1%		1%		1%	

METCONAZOLE (313)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.04 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
FS 0013	Subgroup of Cherries, raw	RAC	0.07	0.01	0.00	0.01	0.00	5.96	0.42	0.01	0.00	NC	-
FS 0014	Subgroup of Plums, raw	RAC	0.04	0.07	0.00	0.01	0.00	15.56	0.62	0.01	0.00	NC	-
DF 0014	Plums, dried (prunes)	PP	0.092	0.01	0.00	0.01	0.00	0.37	0.03	0.01	0.00	NC	-
FS 2001	Subgroup of peaches, raw (incl dried apricots)	RAC	0.045	0.02	0.00	0.01	0.00	10.76	0.48	0.01	0.00	NC	-
FB 0020	Blueberries, raw	RAC	0.14	NC	-	NC	-	0.20	0.03	NC	-	NC	-
FI 0327	Banana, raw (incl plantains) (incl dried)	RAC	0.1	44.80	4.48	118.17	11.82	25.25	2.53	454.49	45.45	310.23	31.02
VA 0381	Garlic, raw	RAC	0.05	0.82	0.04	2.06	0.10	3.79	0.19	0.03	0.00	0.29	0.01
-	Onions, dry, raw	RAC	0.05	9.01	0.45	20.24	1.01	30.90	1.55	9.61	0.48	2.11	0.11
VP 2060	Subgroup of beans with pods (all commodities within this group)	RAC	0	NC	-	NC	-	NC	-	NC	-	NC	-
VD 2065	Subgroup of dry beans, raw (incl processed)	RAC	0.04	41.93	1.68	19.42	0.78	108.31	4.33	66.18	2.65	42.47	1.70
VD 0541	Soya bean, dry, raw (incl flour, incl paste, incl curd, incl sauce, excl oil)	RAC	0.01	2.89	0.03	0.21	0.00	0.48	0.00	3.16	0.03	0.26	0.00
OR 0541	Soya oil, refined	PP	0.005	2.32	0.01	2.54	0.01	18.70	0.09	2.51	0.01	6.29	0.03
VD 2066	Subgroup of dry peas, raw	RAC	0.0425	4.43	0.19	11.36	0.48	4.22	0.18	9.36	0.40	1.21	0.05
VR 0596	Sugar beet, raw	RAC	0.02	0.01	0.00	NC	-	NC	-	NC	-	NC	-
-	Sugar beet, sugar	PP	0.012	0.56	0.01	0.24	0.00	NC	-	NC	-	5.13	0.06
VR 2071	Subgroup of tuberous and corm vegetables, raw (incl processed)	RAC	0	250.41	0.00	208.74	0.00	213.64	0.00	602.70	0.00	388.95	0.00
GC 0653	Triticale, raw (incl flour)	RAC	0.035	0.01	0.00	NC	-	NC	-	NC	-	NC	-
GC 0654	Wheat, raw (incl bulgur, incl fermented beverages, incl germ, incl wholemeal bread, excl white flour products, excl white bread)	RAC	0.035	0.05	0.00	0.01	0.00	0.05	0.00	0.01	0.00	0.97	0.03
CP 1211	Wheat, white bread	PP	0.021	0.43	0.01	0.41	0.01	1.56	0.03	0.11	0.00	0.07	0.00
CF 1211	Wheat, white flour (incl white flour products: starch, gluten, macaroni, pastry)	PP	0.008	44.78	0.36	86.96	0.70	214.05	1.71	20.31	0.16	103.60	0.83
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl flour, incl oil, incl beer, incl germ, incl starch)	RAC	0.01	116.66	1.17	10.52	0.11	38.46	0.38	76.60	0.77	34.44	0.34
GC 0447	Sweet corn on the cob, raw (incl frozen kernels, incl canned kernels)	RAC	0.01	3.63	0.04	20.50	0.21	8.78	0.09	0.02	0.00	0.17	0.00
GS 0659	Sugar cane, raw	RAC	0.0205	5.62	0.12	50.91	1.04	NC	-	11.04	0.23	0.10	0.00
-	Sugar cane, molasses	PP	0.027	NC	-	NC	-	NC	-	NC	-	NC	-
-	Sugar cane, sugar (incl non-centrifugal sugar, incl refined sugar and maltose)	PP	0.002	28.13	0.06	55.38	0.11	78.09	0.16	18.04	0.04	45.60	0.09
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0	4.39	0.00	135.53	0.00	6.11	0.00	0.72	0.00	317.74	0.00
SO 0495	Rape seed, raw	RAC	0.02	NC	-	0.01	0.00	NC	-	NC	-	NC	-

Annex 3

METCONAZOLE (313)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.04 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
OR 0495	Rape seed oil, edible	PP	0.032	0.07	0.00	0.03	0.00	4.62	0.15	0.03	0.00	NC	-
SO 2091	Subgroup of Sunflower seeds, raw (incl processed)	RAC	0.089	0.99	0.09	0.22	0.02	32.01	2.85	12.12	1.08	0.48	0.04
OR 0691	Cotton seed oil, edible	PP	0.004	1.28	0.01	0.05	0.00	0.45	0.00	0.42	0.00	0.15	0.00
SO 0697	Peanuts, nutmeat, raw (incl roasted, incl butter, excl oil)	RAC	0.04	7.14	0.29	0.45	0.02	1.87	0.07	6.22	0.25	0.53	0.02
OR 0697	Peanut oil, edible	PP	0.056	5.02	0.28	0.05	0.00	0.17	0.01	0.29	0.02	NC	-
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat)	RAC	0	29.18	0.00	50.89	0.00	121.44	0.00	22.58	0.00	72.14	0.00
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0	1.05	0.00	1.14	0.00	18.69	0.00	0.94	0.00	3.12	0.00
MO 0105	Edible offal (mammalian), raw	RAC	0.037	4.64	0.17	1.97	0.07	10.01	0.37	3.27	0.12	3.98	0.15
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0	108.75	0.00	70.31	0.00	436.11	0.00	61.55	0.00	79.09	0.00
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	3.92	0.00	12.03	0.00	57.07	0.00	5.03	0.00	55.56	0.00
PF 0111	Poultry fat, raw (incl rendered)	RAC	0	NC	-	NC	-	0.32	0.00	NC	-	NC	-
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.019	0.10	0.00	0.70	0.01	0.97	0.02	0.10	0.00	NC	-
PE 0112	Eggs, raw, (incl dried)	RAC	0	3.84	0.00	4.41	0.00	27.25	0.00	1.13	0.00	7.39	0.00
-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total intake (µg/person)=				9.5		16.5		16.3		51.7		34.5	
Bodyweight per region (kg bw) =				60		60		60		60		60	
ADI (µg/person)=				2400		2400		2400		2400		2400	
%ADI=				0.4%		0.7%		0.7%		2.2%		1.4%	
Rounded %ADI=				0%		1%		1%		2%		1%	

Annex 3

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PENDIMETHALIN (292)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.1 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
FC 0001	Group of Citrus fruit, raw (incl citrus fruit juice, incl kumquat commodities)	RAC	0.005	34.91	0.17	16.51	0.08	17.23	0.09	104.48	0.52	35.57	0.18	98.49	0.49
FB 2005	Subgroup of Caneberries, raw	RAC	0.05	0.42	0.02	1.05	0.05	0.01	0.00	0.02	0.00	0.02	0.00	1.24	0.06
FB 2006	Subgroup of Bush berries, raw (including processed)	RAC	0.05	0.53	0.03	1.31	0.07	0.40	0.02	1.66	0.08	0.01	0.00	0.99	0.05
FB 0269	Grapes, raw (incl must, incl dried, incl juice, incl wine)	RAC	0.05	16.25	0.81	28.96	1.45	2.87	0.14	24.22	1.21	9.33	0.47	68.64	3.43
FB 0275	Strawberry, raw	RAC	0.05	0.70	0.04	2.01	0.10	0.04	0.00	1.36	0.07	0.37	0.02	2.53	0.13
VA 0381	Garlic, raw	RAC	0	2.29	0.00	5.78	0.00	0.11	0.00	3.69	0.00	1.65	0.00	3.91	0.00
-	Onions, dry, raw	RAC	0	29.36	0.00	37.50	0.00	3.56	0.00	34.78	0.00	18.81	0.00	43.38	0.00
VA 0384	Leek, raw	RAC	0.02	0.18	0.00	1.59	0.03	0.03	0.00	0.28	0.01	0.01	0.00	3.21	0.06
-	Onions, dry, raw	RAC	0.095	29.36	2.79	37.50	3.56	3.56	0.34	34.78	3.30	18.81	1.79	43.38	4.12
VB 0042	Subgroup of Flowerhead Brassica, raw	RAC	0	2.54	0.00	0.49	0.00	0.01	0.00	3.57	0.00	7.79	0.00	3.12	0.00
VL 0483	Lettuce, leaf, raw	RAC	0.062	0.53	0.03	0.36	0.02	0.16	0.01	6.21	0.39	1.90	0.12	6.05	0.38
VL 0401	Broccoli, Chinese, raw (i.e. kailan)	RAC	0.05	0.42	0.02	0.08	0.00	0.01	0.00	0.60	0.03	NC	-	0.52	0.03
VL 0472	Cress, garden, raw	RAC	0.05	0.06	0.00	0.10	0.01	0.06	0.00	0.15	0.01	NC	-	0.07	0.00
VL 0480	Kale (Borecole, Collards), raw	RAC	0.05	0.57	0.03	5.77	0.29	0.11	0.01	0.92	0.05	5.25	0.26	2.12	0.11
VL 0485	Mustard greens, raw (i.e. Indian mustard, Amsoi, mustard cabbage)	RAC	0.05	0.03	0.00	0.31	0.02	0.01	0.00	0.05	0.00	0.47	0.02	0.11	0.01
VL 0494	Radish leaves, raw	RAC	0.05	0.26	0.01	0.45	0.02	0.28	0.01	0.68	0.03	NC	-	0.33	0.02
VL 0495	Rape greens, raw	RAC	0.05	0.03	0.00	0.31	0.02	0.01	0.00	0.05	0.00	NC	-	0.11	0.01
VL 0496	Rucola, raw (i.e. Arrugula, Rocket salad, Roquette)	RAC	0.05	1.27	0.06	2.25	0.11	1.39	0.07	3.38	0.17	13.81	0.69	1.63	0.08
VP 0062	Beans without pods: (Phaseolus spp.) (succulent seeds), raw	RAC	0.05	1.56	0.08	0.60	0.03	0.49	0.02	1.18	0.06	0.90	0.05	7.79	0.39
VP 0064	Peas without pods (Pisum spp) (succulent seeds)	RAC	0.01	1.97	0.02	0.51	0.01	0.02	0.00	0.79	0.01	3.68	0.04	3.80	0.04
VD 2065	Subgroup of dry beans, raw (incl processed)	RAC	0.05	78.20	3.91	60.68	3.03	35.89	1.79	80.34	4.02	75.90	3.80	87.62	4.38
VD 0541	Soya bean, dry, raw (incl paste, incl curd, incl oil, incl sauce)	RAC	0.05	72.79	3.64	59.05	2.95	20.55	1.03	74.20	3.71	61.12	3.06	73.24	3.66
VD 0072	Peas (dry) (Pisum spp), raw	RAC	0.05	1.62	0.08	3.22	0.16	0.92	0.05	1.50	0.08	2.90	0.15	0.17	0.01
VR 0577	Carrots, raw	RAC	0.0625	9.51	0.59	30.78	1.92	0.37	0.02	8.75	0.55	2.80	0.18	6.10	0.38
VS 0624	Celery	RAC	0.02	2.14	0.04	3.79	0.08	2.35	0.05	5.69	0.11	0.02	0.00	2.75	0.06
VS 0621	Asparagus, raw	RAC	0.05	0.01	0.00	0.01	0.00	0.01	0.00	0.03	0.00	0.07	0.00	0.21	0.01
GC 0654	Wheat, raw (incl bulgur, incl fermented beverages, incl germ, incl wholemeal bread, incl white flour products, incl white bread)	RAC	0.01	381.15	3.81	341.55	3.42	38.35	0.38	281.89	2.82	172.83	1.73	434.07	4.34

Annex 3

PENDIMETHALIN (292)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.1 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
CM 0649 (GC 0649)	Rice, husked, dry (incl polished, incl flour, incl starch, incl oil, incl beverages)	REP	0	45.40	0.00	14.99	0.00	84.88	0.00	111.73	0.00	194.75	0.00	93.12	0.00
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl flour, incl oil, incl beer, incl germ, incl starch)	RAC	0.05	29.81	1.49	44.77	2.24	108.95	5.45	52.37	2.62	60.28	3.01	75.69	3.78
GS 0659	Sugar cane, raw (incl sugar, incl molasses)	RAC	0	99.68	0.00	86.27	0.00	31.38	0.00	80.36	0.00	84.18	0.00	99.10	0.00
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0.05	4.06	0.20	3.27	0.16	7.01	0.35	13.93	0.70	14.01	0.70	9.36	0.47
SO 0702	Sunflower seed, raw (incl oil)	RAC	0.05	7.40	0.37	35.86	1.79	1.15	0.06	8.76	0.44	5.45	0.27	13.62	0.68
HH 0738	Mint, raw	RAC	0.077	0.50	0.04	0.01	0.00	NC	-	NC	-	NC	-	NC	-
HH 0740	Parsley leaves, raw (incl dried)	RAC	0.305	0.60	0.18	1.07	0.33	0.66	0.20	1.60	0.49	NC	-	0.77	0.23
DH 1100	Hops, dry	RAC	0.05	0.01	0.00	0.04	0.00	0.01	0.00	0.01	0.00	NC	-	0.01	0.00
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.026	24.96	0.65	57.95	1.51	16.70	0.43	38.38	1.00	26.46	0.69	29.00	0.75
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.051	6.24	0.32	14.49	0.74	4.18	0.21	9.60	0.49	6.62	0.34	7.25	0.37
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.009	3.29	0.03	6.14	0.06	0.82	0.01	1.57	0.01	2.23	0.02	1.07	0.01
MO 0105	Edible offal (mammalian), raw	RAC	0.026	4.79	0.12	9.68	0.25	2.97	0.08	5.49	0.14	3.84	0.10	5.03	0.13
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.006	289.65	1.74	485.88	2.92	26.92	0.16	239.03	1.43	199.91	1.20	180.53	1.08
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	14.63	0.00	29.76	0.00	8.04	0.00	129.68	0.00	25.04	0.00	35.66	0.00
PF 0111	Poultry fat, raw (incl rendered)	RAC	0	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.10	0.00	0.10	0.00
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0	0.12	0.00	0.12	0.00	0.11	0.00	5.37	0.00	0.24	0.00	0.10	0.00
PE 0112	Eggs, raw, (incl dried)	RAC	0	7.84	0.00	23.08	0.00	2.88	0.00	14.89	0.00	9.81	0.00	14.83	0.00
Total intake (µg/person)=				21.6		27.7		11.1		24.8		18.9		30.2	
Bodyweight per region (kg bw) =				60		60		60		60		60		60	
ADI (µg/person)=				6000		6000		6000		6000		6000		6000	
%ADI=				0.4%		0.5%		0.2%		0.4%		0.3%		0.5%	
Rounded %ADI=				0%		0%		0%		0%		0%		1%	

Annex 3

PENDIMETHALIN (292)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.1 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
FC 0001	Group of Citrus fruit, raw (incl citrus fruit juice, incl kumquat commodities)	RAC	0.005	114.42	0.57	62.91	0.31	26.97	0.13	96.72	0.48	96.22	0.48	563.19	2.82
FB 2005	Subgroup of Caneberries, raw	RAC	0.05	0.56	0.03	1.43	0.07	0.14	0.01	1.23	0.06	1.14	0.06	0.01	0.00
FB 2006	Subgroup of Bush berries, raw (including processed)	RAC	0.05	1.31	0.07	5.50	0.28	0.01	0.00	2.57	0.13	0.82	0.04	2.15	0.11
FB 0269	Grapes, raw (incl must, incl dried, incl juice, incl wine)	RAC	0.05	142.23	7.11	105.77	5.29	7.87	0.39	52.44	2.62	109.22	5.46	10.96	0.55
FB 0275	Strawberry, raw	RAC	0.05	4.49	0.22	5.66	0.28	0.02	0.00	6.63	0.33	5.75	0.29	0.05	0.00
VA 0381	Garlic, raw	RAC	0	0.98	0.00	1.49	0.00	12.88	0.00	3.74	0.00	2.05	0.00	1.14	0.00
-	Onions, dry, raw	RAC	0	19.69	0.00	29.83	0.00	24.64	0.00	31.35	0.00	9.72	0.00	12.59	0.00
VA 0384	Leek, raw	RAC	0.02	4.01	0.08	4.41	0.09	0.72	0.01	0.54	0.01	16.41	0.33	0.03	0.00
-	Onions, dry, raw	RAC	0.095	19.69	1.87	29.83	2.83	24.64	2.34	31.35	2.98	9.72	0.92	12.59	1.20
VB 0042	Subgroup of Flowerhead Brassica, raw	RAC	0	9.50	0.00	6.77	0.00	NC	-	3.21	0.00	9.36	0.00	0.75	0.00
VL 0483	Lettuce, leaf, raw	RAC	0.062	14.50	0.90	11.76	0.73	13.14	0.81	19.50	1.21	4.81	0.30	2.23	0.14
VL 0401	Broccoli, Chinese, raw (i.e. kailan)	RAC	0.05	NC	-	NC	-	9.03	0.45	NC	-	NC	-	0.12	0.01
VL 0472	Cress, garden, raw	RAC	0.05	0.10	0.01	NC	-	1.27	0.06	0.13	0.01	0.21	0.01	0.10	0.01
VL 0480	Kale (Borecole, Collards), raw	RAC	0.05	NC	-	NC	-	14.54	0.73	NC	-	NC	-	2.32	0.12
VL 0485	Mustard greens, raw (i.e. Indian mustard, Amsoi, mustard cabbage)	RAC	0.05	NC	-	NC	-	NC	-	NC	-	NC	-	0.13	0.01
VL 0494	Radish leaves, raw	RAC	0.05	NC	-	NC	-	NC	-	3.78	0.19	NC	-	0.48	0.02
VL 0495	Rape greens, raw	RAC	0.05	NC	-	NC	-	1.93	0.10	NC	-	NC	-	0.12	0.01
VL 0496	Rucola, raw (i.e. Arrugula, Rocket salad, Roquette)	RAC	0.05	NC	-	NC	-	NC	-	1.09	0.05	0.38	0.02	2.40	0.12
VP 0062	Beans without pods: (Phaseolus spp.) (succulent seeds), raw	RAC	0.05	2.21	0.11	5.25	0.26	4.17	0.21	1.61	0.08	16.95	0.85	0.17	0.01
VP 0064	Peas without pods (Pisum spp) (succulent seeds)	RAC	0.01	10.72	0.11	1.99	0.02	2.72	0.03	4.26	0.04	4.23	0.04	NC	-
VD 2065	Subgroup of dry beans, raw (incl processed)	RAC	0.05	107.87	5.39	119.29	5.96	45.91	2.30	201.31	10.07	224.04	11.20	104.90	5.25
VD 0541	Soya bean, dry, raw (incl paste, incl curd, incl oil, incl sauce)	RAC	0.05	106.33	5.32	117.78	5.89	42.12	2.11	195.70	9.79	222.52	11.13	80.47	4.02
VD 0072	Peas (dry) (Pisum spp), raw	RAC	0.05	3.80	0.19	1.25	0.06	0.90	0.05	2.33	0.12	2.70	0.14	3.83	0.19
VR 0577	Carrots, raw	RAC	0.0625	26.26	1.64	27.13	1.70	10.07	0.63	16.49	1.03	44.69	2.79	8.75	0.55
VS 0624	Celery	RAC	0.02	7.68	0.15	2.85	0.06	NC	-	3.34	0.07	16.83	0.34	4.04	0.08
VS 0621	Asparagus, raw	RAC	0.05	0.84	0.04	2.08	0.10	7.11	0.36	1.01	0.05	1.69	0.08	0.04	0.00
GC 0654	Wheat, raw (incl bulgur, incl fermented beverages, incl germ, incl wholemeal bread, incl white flour products, incl white bread)	RAC	0.01	253.07	2.53	244.73	2.45	134.44	1.34	235.10	2.35	216.39	2.16	167.40	1.67

Annex 3

PENDIMETHALIN (292)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.1 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
CM 0649 (GC 0649)	Rice, husked, dry (incl polished, incl flour, incl starch, incl oil, incl beverages)	REP	0	20.96	0.00	16.04	0.00	339.67	0.00	75.51	0.00	16.86	0.00	86.13	0.00
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl flour, incl oil, incl beer, incl germ, incl starch)	RAC	0.05	18.51	0.93	26.18	1.31	26.04	1.30	39.99	2.00	7.36	0.37	64.58	3.23
GS 0659	Sugar cane, raw (incl sugar, incl molasses)	RAC	0	92.24	0.00	95.72	0.00	28.47	0.00	77.39	0.00	117.73	0.00	103.90	0.00
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0.05	8.52	0.43	8.94	0.45	15.09	0.75	9.60	0.48	14.57	0.73	26.26	1.31
SO 0702	Sunflower seed, raw (incl oil)	RAC	0.05	23.40	1.17	29.33	1.47	1.24	0.06	13.85	0.69	6.48	0.32	6.91	0.35
HH 0738	Mint, raw	RAC	0.077	NC	-	NC	-	NC	-	NC	-	NC	-	NC	-
HH 0740	Parsley leaves, raw (incl dried)	RAC	0.305	1.43	0.44	2.14	0.65	NC	-	2.54	0.77	0.78	0.24	1.14	0.35
DH 1100	Hops, dry	RAC	0.05	NC	-	NC	-	0.02	0.00	0.02	0.00	NC	-	NC	-
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.026	112.02	2.91	120.71	3.14	63.46	1.65	88.99	2.31	96.24	2.50	41.02	1.07
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.051	28.01	1.43	30.18	1.54	15.86	0.81	22.25	1.13	24.06	1.23	10.25	0.52
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.009	6.44	0.06	15.51	0.14	3.79	0.03	8.29	0.07	18.44	0.17	8.00	0.07
MO 0105	Edible offal (mammalian), raw	RAC	0.026	15.17	0.39	5.19	0.13	6.30	0.16	6.78	0.18	3.32	0.09	3.17	0.08
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.006	388.92	2.33	335.88	2.02	49.15	0.29	331.25	1.99	468.56	2.81	245.45	1.47
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	73.76	0.00	53.86	0.00	23.98	0.00	87.12	0.00	53.38	0.00	84.45	0.00
PF 0111	Poultry fat, raw (incl rendered)	RAC	0	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.71	0.00	NC	-
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0	0.33	0.00	0.72	0.00	0.27	0.00	0.35	0.00	0.80	0.00	NC	-
PE 0112	Eggs, raw, (incl dried)	RAC	0	25.84	0.00	29.53	0.00	28.05	0.00	33.19	0.00	36.44	0.00	8.89	0.00
Total intake (µg/person)=				37.0		37.7		17.3		41.7		46.7		25.9	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (µg/person)=				6000		6000		5500		6000		6000		6000	
%ADI=				0.6%		0.6%		0.3%		0.7%		0.8%		0.4%	
Rounded %ADI=				1%		1%		0%		1%		1%		0%	

PENDIMETHALIN (292)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.1 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day									
				Intake = daily intake: µg/person									
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
FC 0001	Group of Citrus fruit, raw (incl citrus fruit juice, incl kumquat commodities)	RAC	0.005	21.16	0.11	2.94	0.01	58.52	0.29	0.44	0.00	5.13	0.03
FB 2005	Subgroup of Caneberries, raw	RAC	0.05	0.01	0.00	7.30	0.37	2.29	0.11	0.01	0.00	NC	-
FB 2006	Subgroup of Bush berries, raw (including processed)	RAC	0.05	0.82	0.04	4.05	0.20	5.94	0.30	0.43	0.02	2.66	0.13
FB 0269	Grapes, raw (incl must, incl dried, incl juice, incl wine)	RAC	0.05	0.60	0.03	1.26	0.06	103.25	5.16	0.74	0.04	44.23	2.21
FB 0275	Strawberry, raw	RAC	0.05	0.01	0.00	0.01	0.00	3.35	0.17	0.01	0.00	0.01	0.00
VA 0381	Garlic, raw	RAC	0	0.82	0.00	2.06	0.00	3.79	0.00	0.03	0.00	0.29	0.00
-	Onions, dry, raw	RAC	0	9.01	0.00	20.24	0.00	30.90	0.00	9.61	0.00	2.11	0.00
VA 0384	Leek, raw	RAC	0.02	0.02	0.00	1.44	0.03	1.22	0.02	0.01	0.00	NC	-
-	Onions, dry, raw	RAC	0.095	9.01	0.86	20.24	1.92	30.90	2.94	9.61	0.91	2.11	0.20
VB 0042	Subgroup of Flowerhead Brassica, raw	RAC	0	0.02	0.00	0.02	0.00	4.86	0.00	0.01	0.00	NC	-
VL 0483	Lettuce, leaf, raw	RAC	0.062	0.29	0.02	0.03	0.00	6.71	0.42	0.01	0.00	NC	-
VL 0401	Broccoli, Chinese, raw (i.e. kailan)	RAC	0.05	0.01	0.00	0.01	0.00	NC	-	0.01	0.00	NC	-
VL 0472	Cress, garden, raw	RAC	0.05	0.09	0.00	0.07	0.00	NC	-	0.06	0.00	0.13	0.01
VL 0480	Kale (Borecole, Collards), raw	RAC	0.05	0.79	0.04	0.62	0.03	NC	-	0.01	0.00	NC	-
VL 0485	Mustard greens, raw (i.e. Indian mustard, Amsoi, mustard cabbage)	RAC	0.05	0.04	0.00	0.03	0.00	NC	-	0.01	0.00	NC	-
VL 0494	Radish leaves, raw	RAC	0.05	0.44	0.02	0.32	0.02	NC	-	0.30	0.02	0.59	0.03
VL 0495	Rape greens, raw	RAC	0.05	0.04	0.00	0.03	0.00	NC	-	0.01	0.00	NC	-
VL 0496	Rucola, raw (i.e. Arrugula, Rocket salad, Roquette)	RAC	0.05	2.17	0.11	1.57	0.08	NC	-	1.47	0.07	2.93	0.15
VP 0062	Beans without pods: (Phaseolus spp.) (succulent seeds), raw	RAC	0.05	0.30	0.02	3.13	0.16	4.11	0.21	0.01	0.00	NC	-
VP 0064	Peas without pods (Pisum spp) (succulent seeds)	RAC	0.01	0.21	0.00	0.02	0.00	5.51	0.06	0.02	0.00	NC	-
VD 2065	Subgroup of dry beans, raw (incl processed)	RAC	0.05	41.93	2.10	19.42	0.97	108.31	5.42	66.18	3.31	42.47	2.12
VD 0541	Soya bean, dry, raw (incl paste, incl curd, incl oil, incl sauce)	RAC	0.05	15.80	0.79	14.29	0.71	104.36	5.22	17.11	0.86	35.20	1.76
VD 0072	Peas (dry) (Pisum spp), raw	RAC	0.05	1.53	0.08	2.52	0.13	3.52	0.18	3.56	0.18	0.74	0.04
VR 0577	Carrots, raw	RAC	0.0625	2.07	0.13	3.00	0.19	25.29	1.58	0.05	0.00	NC	-
VS 0624	Celery	RAC	0.02	3.66	0.07	2.65	0.05	4.84	0.10	2.47	0.05	4.94	0.10
VS 0621	Asparagus, raw	RAC	0.05	0.01	0.00	0.01	0.00	0.17	0.01	0.01	0.00	NC	-
GC 0654	Wheat, raw (incl bulgur, incl fermented beverages, incl germ, incl wholemeal bread, incl white flour products, incl white bread)	RAC	0.01	57.20	0.57	110.47	1.10	272.62	2.73	25.82	0.26	132.04	1.32

Annex 3

PENDIMETHALIN (292)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.1 mg/kg bw						
Codex Code		Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
					G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
CM 0649 (GC 0649)	Rice, husked, dry (incl polished, incl flour, incl starch, incl oil, incl beverages)	REP	0	52.55	0.00	286.02	0.00	18.64	0.00	19.67	0.00	75.09	0.00	
GC 0645	Maize, raw (incl glucose & dextrose & isoglucose, incl flour, incl oil, incl beer, incl germ, incl starch)	RAC	0.05	116.66	5.83	10.52	0.53	38.46	1.92	76.60	3.83	34.44	1.72	
GS 0659	Sugar cane, raw (incl sugar, incl molasses)	RAC	0	33.75	0.00	106.29	0.00	78.09	0.00	29.09	0.00	45.70	0.00	
TN 0085	Group of Tree nuts, raw (incl processed)	RAC	0.05	4.39	0.22	135.53	6.78	6.11	0.31	0.72	0.04	317.74	15.89	
SO 0702	Sunflower seed, raw (incl oil)	RAC	0.05	0.94	0.05	0.22	0.01	32.01	1.60	12.12	0.61	0.48	0.02	
HH 0738	Mint, raw	RAC	0.077	NC	-	NC	-	NC	-	NC	-	NC	-	
HH 0740	Parsley leaves, raw (incl dried)	RAC	0.305	1.03	0.31	0.74	0.23	1.87	0.57	0.70	0.21	1.39	0.42	
DH 1100	Hops, dry	RAC	0.05	NC	-	NC	-	0.04	0.00	NC	-	NC	-	
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.026	23.34	0.61	40.71	1.06	97.15	2.53	18.06	0.47	57.71	1.50	
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.051	5.84	0.30	10.18	0.52	24.29	1.24	4.52	0.23	14.43	0.74	
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.009	1.05	0.01	1.14	0.01	18.69	0.17	0.94	0.01	3.12	0.03	
MO 0105	Edible offal (mammalian), raw	RAC	0.026	4.64	0.12	1.97	0.05	10.01	0.26	3.27	0.09	3.98	0.10	
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.006	108.75	0.65	70.31	0.42	436.11	2.62	61.55	0.37	79.09	0.47	
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	3.92	0.00	12.03	0.00	57.07	0.00	5.03	0.00	55.56	0.00	
PF 0111	Poultry fat, raw (incl rendered)	RAC	0	NC	-	NC	-	0.32	0.00	NC	-	NC	-	
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0	0.10	0.00	0.70	0.00	0.97	0.00	0.10	0.00	NC	-	
PE 0112	Eggs, raw, (incl dried)	RAC	0	3.84	0.00	4.41	0.00	27.25	0.00	1.13	0.00	7.39	0.00	
Total intake (µg/person)=					13.2		15.8		36.2		11.6		29.6	
Bodyweight per region (kg bw) =					60		60		60		60		60	
ADI (µg/person)=					6000		6000		6000		6000		6000	
%ADI=					0.2%		0.3%		0.6%		0.2%		0.5%	
Rounded %ADI=					0%		0%		1%		0%		0%	

PYRASULFOTOLE (321)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.01 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day		Intake as ug/person/day									
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
GC 0650	Rye, raw (incl flour)	RAC	0.02	0.13	0.00	19.38	0.39	0.10	0.00	0.12	0.00	0.03	0.00	2.15	0.04
GC 0653	Triticale, raw (incl flour)	RAC	0.02	NC	-	NC	-	NC	-	0.01	0.00	0.39	0.01	NC	-
GC 0654	Wheat, raw (incl meslin)	RAC	0.02	0.01	0.00	1.12	0.02	NC	-	0.01	0.00	0.56	0.01	NC	-
-	Wheat, bulgur	PP	0.02	NC	-	NC	-	NC	-	0.03	0.00	NC	-	NC	-
CF 1210	Wheat, germ	PP	0.016	NC	-	NC	-	0.01	0.00	0.01	0.00	0.14	0.00	0.01	0.00
CP 1212	Wheat, wholemeal bread	PP	0.02	0.01	0.00	0.01	0.00	0.01	0.00	0.01	0.00	0.03	0.00	0.01	0.00
CP 1211	Wheat, white bread	PP	0.02	0.25	0.01	0.63	0.01	0.12	0.00	0.43	0.01	1.39	0.03	0.22	0.00
-	Wheat, Fermented Beverages (Korean jakju and takju)	PP	0.02	NC	-	NC	-	NC	-	NC	-	NC	-	NC	-
CF 1211	Wheat, white flour (incl white flour products: starch, gluten, macaroni, pastry)	PP	0.02	301.24	6.02	268.64	5.37	30.21	0.60	222.51	4.45	134.73	2.69	343.12	6.86
GC 0640	Barley, raw (incl malt extract, incl pot&pearled, incl flour & grits, incl beer, incl malt)	RAC	0.02	19.91	0.40	31.16	0.62	5.04	0.10	3.10	0.06	9.77	0.20	4.31	0.09
GC 0647	Oats, raw (incl rolled)	RAC	0.02	0.05	0.00	7.05	0.14	0.10	0.00	1.71	0.03	0.96	0.02	0.04	0.00
GC 0651	Sorghum, raw (incl flour, incl beer) (i.e. Chicken corn, Dari seed, Durra, Feterita)	RAC	0.091	4.34	0.39	0.01	0.00	16.25	1.48	15.82	1.44	10.97	1.00	2.92	0.27
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat)	RAC	0.02	31.20	0.62	72.44	1.45	20.88	0.42	47.98	0.96	33.08	0.66	36.25	0.73
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.02	3.29	0.07	6.14	0.12	0.82	0.02	1.57	0.03	2.23	0.04	1.07	0.02
MO 0105	Edible offal (mammalian), raw	RAC	0.084	4.79	0.40	9.68	0.81	2.97	0.25	5.49	0.46	3.84	0.32	5.03	0.42
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.01	289.65	2.90	485.88	4.86	26.92	0.27	239.03	2.39	199.91	2.00	180.53	1.81
PM 0110	Poultry meat, raw (incl prepared)	RAC	0.02	14.63	0.29	29.76	0.60	8.04	0.16	129.68	2.59	25.04	0.50	35.66	0.71
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.02	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.10	0.00	0.10	0.00
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.02	0.12	0.00	0.12	0.00	0.11	0.00	5.37	0.11	0.24	0.00	0.10	0.00
PE 0112	Eggs, raw, (incl dried)	RAC	0	7.84	0.00	23.08	0.00	2.88	0.00	14.89	0.00	9.81	0.00	14.83	0.00
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Total intake (ug/person)=

11.1

14.4

3.3

12.5

7.5

11.0

Bodyweight per region (kg bw) =

60

60

60

60

60

60

ADI (ug/person)=

600

600

600

600

600

600

%ADI=

1.9%

2.4%

0.6%

2.1%

1.2%

1.8%

Rounded %ADI=

2%

2%

1%

2%

1%

2%

Annex 3

PYRASULFOTOLE (321)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.01 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as ug/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
GC 0650	Rye, raw (incl flour)	RAC	0.02	3.21	0.06	35.38	0.71	0.21	0.00	6.50	0.13	1.49	0.03	NC	-
GC 0653	Triticale, raw (incl flour)	RAC	0.02	0.01	0.00	0.17	0.00	0.29	0.01	0.01	0.00	NC	-	NC	-
GC 0654	Wheat, raw (incl meslin)	RAC	0.02	NC	-	NC	-	NC	-	0.01	0.00	NC	-	NC	-
-	Wheat, bulgur	PP	0.02	NC	-	NC	-	0.01	0.00	NC	-	NC	-	NC	-
CF 1210	Wheat, germ	PP	0.016	0.97	0.02	0.10	0.00	0.03	0.00	0.01	0.00	NC	-	0.04	0.00
CP 1212	Wheat, wholemeal bread	PP	0.02	0.03	0.00	0.01	0.00	0.01	0.00	0.01	0.00	0.05	0.00	0.02	0.00
CP 1211	Wheat, white bread	PP	0.02	1.30	0.03	0.46	0.01	0.06	0.00	0.22	0.00	2.44	0.05	0.77	0.02
-	Wheat, Fermented Beverages (Korean jakju and takju)	PP	0.02	NC	-	NC	-	NC	-	4.36	0.09	NC	-	NC	-
CF 1211	Wheat, white flour (incl white flour products: starch, gluten, macaroni, pastry)	PP	0.02	198.08	3.96	193.03	3.86	106.24	2.12	185.09	3.70	168.67	3.37	131.59	2.63
GC 0640	Barley, raw (incl malt extract, incl pot&pearled, incl flour & grits, incl beer, incl malt)	RAC	0.02	36.18	0.72	53.45	1.07	9.39	0.19	35.25	0.71	46.68	0.93	15.92	0.32
GC 0647	Oats, raw (incl rolled)	RAC	0.02	7.50	0.15	6.26	0.13	0.15	0.00	4.87	0.10	3.16	0.06	2.98	0.06
GC 0651	Sorghum, raw (incl flour, incl beer) (i.e. Chicken corn, Dari seed, Durra, Feterita)	RAC	0.091	NC	-	NC	-	1.44	0.13	1.15	0.10	NC	-	7.12	0.65
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat)	RAC	0.02	140.03	2.80	150.89	3.02	79.32	1.59	111.24	2.22	120.30	2.41	51.27	1.03
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.02	6.44	0.13	15.51	0.31	3.79	0.08	8.29	0.17	18.44	0.37	8.00	0.16
MO 0105	Edible offal (mammalian), raw	RAC	0.084	15.17	1.27	5.19	0.44	6.30	0.53	6.78	0.57	3.32	0.28	3.17	0.27
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.01	388.92	3.89	335.88	3.36	49.15	0.49	331.25	3.31	468.56	4.69	245.45	2.45
PM 0110	Poultry meat, raw (incl prepared)	RAC	0.02	73.76	1.48	53.86	1.08	23.98	0.48	87.12	1.74	53.38	1.07	84.45	1.69
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.02	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.71	0.01	NC	-
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.02	0.33	0.01	0.72	0.01	0.27	0.01	0.35	0.01	0.80	0.02	NC	-
PE 0112	Eggs, raw, (incl dried)	RAC	0	25.84	0.00	29.53	0.00	28.05	0.00	33.19	0.00	36.44	0.00	8.89	0.00
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total intake (ug/person)=				14.5		14.0		5.6		12.9		13.3		9.3	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (ug/person)=				600		600		550		600		600		600	
%ADI=				2.4%		2.3%		1.0%		2.1%		2.2%		1.5%	
Rounded %ADI=				2%		2%		1%		2%		2%		2%	

Annex 3

PYRASULFOTOLE (321)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.01 mg/kg bw						
Codex Code		Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: ug/person							
					G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
GC 0650	Rye, raw (incl flour)	RAC	0.02	0.03	0.00	0.01	0.00	13.95	0.28	0.01	0.00	0.88	0.02	
GC 0653	Triticale, raw (incl flour)	RAC	0.02	0.01	0.00	NC	-	NC	-	NC	-	NC	-	
GC 0654	Wheat, raw (incl meslin)	RAC	0.02	NC	-	NC	-	NC	-	NC	-	0.97	0.02	
-	Wheat, bulgur	PP	0.02	0.01	0.00	NC	-	NC	-	NC	-	NC	-	
CF 1210	Wheat, germ	PP	0.016	0.04	0.00	0.01	0.00	0.01	0.00	0.01	0.00	NC	-	
CP 1212	Wheat, wholemeal bread	PP	0.02	0.01	0.00	0.01	0.00	0.03	0.00	0.01	0.00	0.01	0.00	
CP 1211	Wheat, white bread	PP	0.02	0.43	0.01	0.41	0.01	1.56	0.03	0.11	0.00	0.07	0.00	
-	Wheat, Fermented Beverages (Korean jakju and takju)	PP	0.02	NC	-	NC	-	NC	-	NC	-	NC	-	
CF 1211	Wheat, white flour (incl white flour products: starch, gluten, macaroni, pastry)	PP	0.02	44.78	0.90	86.96	1.74	214.05	4.28	20.31	0.41	103.60	2.07	
GC 0640	Barley, raw (incl malt extract, incl pot&pearled, incl flour & grits, incl beer, incl malt)	RAC	0.02	11.58	0.23	2.33	0.05	46.71	0.93	3.72	0.07	16.26	0.33	
GC 0647	Oats, raw (incl rolled)	RAC	0.02	0.37	0.01	0.07	0.00	2.79	0.06	0.10	0.00	NC	-	
GC 0651	Sorghum, raw (incl flour, incl beer) (i.e. Chicken corn, Dari seed, Durra, Feterita)	RAC	0.091	89.16	8.11	2.02	0.18	NC	-	35.38	3.22	NC	-	
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat)	RAC	0.02	29.18	0.58	50.89	1.02	121.44	2.43	22.58	0.45	72.14	1.44	
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.02	1.05	0.02	1.14	0.02	18.69	0.37	0.94	0.02	3.12	0.06	
MO 0105	Edible offal (mammalian), raw	RAC	0.084	4.64	0.39	1.97	0.17	10.01	0.84	3.27	0.27	3.98	0.33	
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.01	108.75	1.09	70.31	0.70	436.11	4.36	61.55	0.62	79.09	0.79	
PM 0110	Poultry meat, raw (incl prepared)	RAC	0.02	3.92	0.08	12.03	0.24	57.07	1.14	5.03	0.10	55.56	1.11	
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.02	NC	-	NC	-	0.32	0.01	NC	-	NC	-	
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.02	0.10	0.00	0.70	0.01	0.97	0.02	0.10	0.00	NC	-	
PE 0112	Eggs, raw, (incl dried)	RAC	0	3.84	0.00	4.41	0.00	27.25	0.00	1.13	0.00	7.39	0.00	
-	-	-		-	-	-	-	-	-	-	-	-	-	
Total intake (ug/person)=					11.4		4.1		14.8		5.2		6.2	
Bodyweight per region (kg bw) =					60		60		60		60		60	
ADI (ug/person)=					600		600		600		600		600	
%ADI=					1.9%		0.7%		2.5%		0.9%		1.0%	
Rounded %ADI=					2%		1%		2%		1%		1%	

Annex 3

PYRAZIFLUMID (322)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.02 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.36	13.49	4.86	26.63	9.59	15.05	5.42	16.28	5.86	6.47	2.33	47.88	17.24
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.036	0.32	0.01	3.07	0.11	0.07	0.00	5.00	0.18	0.29	0.01	5.57	0.20
FP 0230	Pear, raw	RAC	0.36	2.16	0.78	6.24	2.25	0.05	0.02	4.07	1.47	1.16	0.42	5.34	1.92
FP 0307	Persimmon, Japanese, raw (i.e. Kaki fruit)	RAC	0.36	1.91	0.69	0.01	0.00	1.94	0.70	1.96	0.71	NC	-	0.25	0.09
FB 0269	Grapes, raw (incl must, incl wine, excl dried, excl juice)	RAC	0.57	13.94	7.95	26.46	15.08	2.79	1.59	18.58	10.59	8.54	4.87	59.95	34.17
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	1.14	0.51	0.58	0.51	0.58	0.01	0.01	1.27	1.45	0.12	0.14	2.07	2.36
JF 0269	Grape juice	PP	0.057	0.14	0.01	0.29	0.02	0.05	0.00	0.30	0.02	0.24	0.01	0.05	0.00
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0005	24.96	0.01	57.95	0.03	16.70	0.01	38.38	0.02	26.46	0.01	29.00	0.01
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.002	6.24	0.01	14.49	0.03	4.18	0.01	9.60	0.02	6.62	0.01	7.25	0.01
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.002	3.29	0.01	6.14	0.01	0.82	0.00	1.57	0.00	2.23	0.00	1.07	0.00
MO 0105	Edible offal (mammalian), raw	RAC	0.005	4.79	0.02	9.68	0.05	2.97	0.01	5.49	0.03	3.84	0.02	5.03	0.03
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.0001	289.65	0.03	485.88	0.05	26.92	0.00	239.03	0.02	199.91	0.02	180.53	0.02
Total intake (µg/person)=				15.0		27.8		7.8		20.4		7.8		56.1	
Bodyweight per region (kg bw) =				60		60		60		60		60		60	
ADI (µg/person)=				1200		1200		1200		1200		1200		1200	
%ADI=				1.2%		2.3%		0.6%		1.7%		0.7%		4.7%	
Rounded %ADI=				1%		2%		1%		2%		1%		5%	

Annex 3

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PYRAZIFLUMID (322)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.02 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as µg/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.36	41.14	14.81	56.49	20.34	26.64	9.59	31.58	11.37	51.94	18.70	3.05	1.10
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.036	14.88	0.54	11.98	0.43	0.15	0.01	9.98	0.36	30.32	1.09	3.47	0.12
FP 0230	Pear, raw	RAC	0.36	8.79	3.16	8.44	3.04	12.37	4.45	9.60	3.46	10.27	3.70	0.23	0.08
FP 0307	Persimmon, Japanese, raw (i.e. Kaki fruit)	RAC	0.36	0.01	0.00	0.30	0.11	3.59	1.29	0.15	0.05	0.02	0.01	NC	-
FB 0269	Grapes, raw (incl must, incl wine, excl dried, excl juice)	RAC	0.57	128.64	73.32	97.04	55.31	7.74	4.41	43.94	25.05	88.68	50.55	8.80	5.02
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	1.14	3.09	3.52	1.51	1.72	0.03	0.03	1.38	1.57	4.26	4.86	0.42	0.48
JF 0269	Grape juice	PP	0.057	0.56	0.03	1.96	0.11	0.02	0.00	2.24	0.13	2.27	0.13	0.34	0.02
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0005	112.02	0.06	120.71	0.06	63.46	0.03	88.99	0.04	96.24	0.05	41.02	0.02
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.002	28.01	0.06	30.18	0.06	15.86	0.03	22.25	0.04	24.06	0.05	10.25	0.02
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.002	6.44	0.01	15.51	0.03	3.79	0.01	8.29	0.02	18.44	0.04	8.00	0.02
MO 0105	Edible offal (mammalian), raw	RAC	0.005	15.17	0.08	5.19	0.03	6.30	0.03	6.78	0.03	3.32	0.02	3.17	0.02
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.0001	388.92	0.04	335.88	0.03	49.15	0.00	331.25	0.03	468.56	0.05	245.45	0.02
Total intake (µg/person)=				95.6		81.3		19.9		42.2		79.2		6.9	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (µg/person)=				1200		1200		1100		1200		1200		1200	
%ADI=				8.0%		6.8%		1.8%		3.5%		6.6%		0.6%	
Rounded %ADI=				8%		7%		2%		4%		7%		1%	

Annex 3

PYRAZIFLUMID (322)

International Estimated Daily Intake (IEDI)

ADI = 0 - 0.02 mg/kg bw

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: µg/person							
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake
FP 0226	Apple, raw (incl cider, excl juice)	RAC	0.36	66.67	24.00	2.06	0.74	55.83	20.10	188.29	67.78	1.38	0.50
JF 0226	Apple juice, single strength (incl. concentrated)	PP	0.036	0.03	0.00	0.10	0.00	7.19	0.26	0.03	0.00	NC	-
FP 0230	Pear, raw	RAC	0.36	0.07	0.03	0.14	0.05	9.45	3.40	0.01	0.00	0.14	0.05
FP 0307	Persimmon, Japanese, raw (i.e. Kaki fruit)	RAC	0.36	0.41	0.15	0.32	0.12	0.02	0.01	0.58	0.21	12.51	4.50
FB 0269	Grapes, raw (incl must, incl wine, excl dried, excl juice)	RAC	0.57	0.57	0.32	0.69	0.39	98.34	56.05	0.73	0.42	44.12	25.15
DF 0269	Grapes, dried (= currants, raisins and sultanas) (from table-grapes)	PP	1.14	0.01	0.01	0.13	0.15	1.06	1.21	0.01	0.01	0.03	0.03
JF 0269	Grape juice	PP	0.057	0.01	0.00	0.01	0.00	0.41	0.02	0.01	0.00	NC	-
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0005	23.34	0.01	40.71	0.02	97.15	0.05	18.06	0.01	57.71	0.03
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.002	5.84	0.01	10.18	0.02	24.29	0.05	4.52	0.01	14.43	0.03
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.002	1.05	0.00	1.14	0.00	18.69	0.04	0.94	0.00	3.12	0.01
MO 0105	Edible offal (mammalian), raw	RAC	0.005	4.64	0.02	1.97	0.01	10.01	0.05	3.27	0.02	3.98	0.02
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.0001	108.75	0.01	70.31	0.01	436.11	0.04	61.55	0.01	79.09	0.01
Total intake (µg/person)=				24.6		1.5		81.3		68.5		30.3	
Bodyweight per region (kg bw) =				60		60		60		60		60	
ADI (µg/person)=				1200		1200		1200		1200		1200	
%ADI=				2.0%		0.1%		6.8%		5.7%		2.5%	
Rounded %ADI=				2%		0%		7%		6%		3%	

SPIROPIDION (323)**International Estimated Daily Intake (IEDI)****ADI = 0 - 0.02 mg/kg bw**

Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day		Intake as ug/person/day									
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
VC 0424	Cucumber, raw	RAC	0.34	8.01	2.72	30.66	10.42	1.45	0.49	19.84	6.75	0.27	0.09	34.92	11.87
VC 2040	Subgroup of Melons, Pumpkins and Winter squashes	RAC	0.25	42.62	10.66	46.85	11.71	4.21	1.05	67.02	16.76	12.84	3.21	110.47	27.62
VO 0448	Tomato, raw	RAC	0.245	41.73	10.22	75.65	18.53	10.66	2.61	82.87	20.30	24.75	6.06	200.93	49.23
-	Tomato, canned (& peeled)	PP	0.15	0.20	0.03	0.31	0.05	0.02	0.00	1.11	0.17	0.11	0.02	1.50	0.23
-	Tomato, paste (i.e. concentrated tomato sauce/puree)	PP	0.46	2.34	1.08	1.33	0.61	1.57	0.72	4.24	1.95	0.34	0.16	2.83	1.30
JF 0448	Tomato, juice (single strength, incl concentrated)	PP	0.19	0.29	0.06	0.29	0.06	0.01	0.00	0.38	0.07	0.05	0.01	0.14	0.03
VO 0051	Subgroup of peppers, raw (incl dried sweet peppers, excl dried chili peppers), excl okra	RAC	0.49	8.48	4.16	13.74	6.73	10.13	4.96	11.29	5.53	9.52	4.66	26.36	12.92
VD 0541	Soya bean, dry, raw (Glycine soja)	RAC	0.49	0.58	0.28	0.05	0.02	0.37	0.18	0.03	0.01	1.65	0.81	0.30	0.15
-	Soya paste (i.e. miso)	PP	0.098	NC	-	NC	-	NC	-	NC	-	NC	-	NC	-
-	Soya curd (i.e. tofu)	PP	0.051	NC	-	NC	-	NC	-	NC	-	NC	-	NC	-
OR 0541	Soya oil, refined	PP	0.01	12.99	0.13	10.43	0.10	3.63	0.04	13.10	0.13	10.70	0.11	13.10	0.13
-	Soya sauce	PP	0.02	0.01	0.00	0.02	0.00	0.01	0.00	0.34	0.01	0.03	0.00	0.01	0.00
-	Soya flour	PP	0.79	0.05	0.04	0.86	0.68	0.02	0.02	1.02	0.81	0.01	0.01	0.15	0.12
VR 0589	Potato, raw (incl flour, incl frozen, incl tapioca, excl starch)	RAC	0.28	59.60	16.69	316.10	88.51	9.77	2.74	59.59	16.69	54.12	15.15	119.82	33.55
-	Potato, starch	PP	0.16	0.03	0.00	0.01	0.00	0.01	0.00	0.15	0.02	0.01	0.00	0.01	0.00
HS 0444	Peppers, chili, dried	PP	2.905	0.42	1.22	0.53	1.54	0.84	2.44	0.50	1.45	0.95	2.76	0.37	1.07
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0065	24.96	0.16	57.95	0.38	16.70	0.11	38.38	0.25	26.46	0.17	29.00	0.19
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.013	6.24	0.08	14.49	0.19	4.18	0.05	9.60	0.12	6.62	0.09	7.25	0.09
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.013	3.29	0.04	6.14	0.08	0.82	0.01	1.57	0.02	2.23	0.03	1.07	0.01
MO 0105	Edible offal (mammalian), raw	RAC	0.098	4.79	0.47	9.68	0.95	2.97	0.29	5.49	0.54	3.84	0.38	5.03	0.49
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.0057	289.65	1.65	485.88	2.77	26.92	0.15	239.03	1.36	199.91	1.14	180.53	1.03
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	14.63	0.00	29.76	0.00	8.04	0.00	129.68	0.00	25.04	0.00	35.66	0.00
PM 0110	Poultry meat, raw (incl prepared) - 90% as muscle	RAC	0.00041	13.17	0.01	26.78	0.01	7.24	0.00	116.71	0.05	22.54	0.01	32.09	0.01
PM 0110	Poultry meat, raw (incl prepared) - 10% as fat	RAC	0.00035	1.46	0.00	2.98	0.00	0.80	0.00	12.97	0.00	2.50	0.00	3.57	0.00
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.00035	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.10	0.00	0.10	0.00
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.0033	0.12	0.00	0.12	0.00	0.11	0.00	5.37	0.02	0.24	0.00	0.10	0.00

Annex 3

SPIROPIDION (323)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.02 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as ug/person/day							
				G01 diet	G01 intake	G02 diet	G02 intake	G03 diet	G03 intake	G04 diet	G04 intake	G05 diet	G05 intake	G06 diet	G06 intake
PE 0112	Eggs, raw, (incl dried)	RAC	0.00089	7.84	0.01	23.08	0.02	2.88	0.00	14.89	0.01	9.81	0.01	14.83	0.01
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total intake (ug/person)=					49.7		143.4		15.9		73.0		34.9		140.1
Bodyweight per region (kg bw) =					60		60		60		60		60		60
ADI (ug/person)=					1200		1200		1200		1200		1200		1200
%ADI=					4.1%		11.9%		1.3%		6.1%		2.9%		11.7%
Rounded %ADI=					4%		10%		1%		6%		3%		10%

SPIROPIDION (323)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.02 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as ug/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
VC 0424	Cucumber, raw	RAC	0.34	6.72	2.28	11.03	3.75	32.10	10.91	15.10	5.13	4.05	1.38	9.57	3.25
VC 2040	Subgroup of Melons, Pumpkins and Winter squashes	RAC	0.25	20.68	5.17	25.00	6.25	85.72	21.43	34.31	8.58	11.54	2.89	23.32	5.83
VO 0448	Tomato, raw	RAC	0.245	32.13	7.87	51.27	12.56	34.92	8.56	73.37	17.98	15.15	3.71	8.88	2.18
-	Tomato, canned (& peeled)	PP	0.15	7.57	1.14	2.66	0.40	0.30	0.05	0.97	0.15	7.31	1.10	0.41	0.06
-	Tomato, paste (i.e. concentrated tomato sauce/puree)	PP	0.46	4.96	2.28	3.20	1.47	0.15	0.07	1.61	0.74	6.88	3.16	0.52	0.24
JF 0448	Tomato, juice (single strength, incl concentrated)	PP	0.19	0.80	0.15	0.07	0.01	0.05	0.01	0.61	0.12	0.40	0.08	0.08	0.02
VO 0051	Subgroup of peppers, raw (incl dried sweet peppers, excl dried chili peppers), excl okra	RAC	0.49	6.39	3.13	15.53	7.61	19.09	9.35	10.36	5.08	8.29	4.06	4.53	2.22
VD 0541	Soya bean, dry, raw (Glycine soja)	RAC	0.49	0.02	0.01	0.33	0.16	6.64	3.25	3.94	1.93	NC	-	5.78	2.83
-	Soya paste (i.e. miso)	PP	0.098	NC	-	NC	-	NC	-	1.87	0.18	NC	-	NC	-
-	Soya curd (i.e. tofu)	PP	0.051	NC	-	NC	-	0.68	0.03	0.87	0.04	NC	-	NC	-
OR 0541	Soya oil, refined	PP	0.01	19.06	0.19	21.06	0.21	5.94	0.06	33.78	0.34	40.05	0.40	13.39	0.13
-	Soya sauce	PP	0.02	0.45	0.01	0.29	0.01	2.93	0.06	4.35	0.09	0.09	0.00	0.70	0.01
-	Soya flour	PP	0.79	0.22	0.17	0.27	0.21	0.29	0.23	0.17	0.13	NC	-	NC	-
VR 0589	Potato, raw (incl flour, incl frozen, incl tapioca, excl starch)	RAC	0.28	225.03	63.01	226.35	63.38	71.26	19.95	173.36	48.54	234.55	65.67	37.71	10.56
-	Potato, starch	PP	0.16	NC	-	1.74	0.28	0.05	0.01	0.92	0.15	NC	-	NC	-
HS 0444	Peppers, chili, dried	PP	2.905	0.11	0.32	0.21	0.61	0.36	1.05	0.21	0.61	0.25	0.73	0.15	0.44

Annex 3

SPIROPIDION (323)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.02 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets as g/person/day				Intake as ug/person/day							
				G07 diet	G07 intake	G08 diet	G08 intake	G09 diet	G09 intake	G10 diet	G10 intake	G11 diet	G11 intake	G12 diet	G12 intake
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0065	112.02	0.73	120.71	0.78	63.46	0.41	88.99	0.58	96.24	0.63	41.02	0.27
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.013	28.01	0.36	30.18	0.39	15.86	0.21	22.25	0.29	24.06	0.31	10.25	0.13
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.013	6.44	0.08	15.51	0.20	3.79	0.05	8.29	0.11	18.44	0.24	8.00	0.10
MO 0105	Edible offal (mammalian), raw	RAC	0.098	15.17	1.49	5.19	0.51	6.30	0.62	6.78	0.66	3.32	0.33	3.17	0.31
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.0057	388.92	2.22	335.88	1.91	49.15	0.28	331.25	1.89	468.56	2.67	245.45	1.40
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	73.76	0.00	53.86	0.00	23.98	0.00	87.12	0.00	53.38	0.00	84.45	0.00
PM 0110	Poultry meat, raw (incl prepared) - 90% as muscle	RAC	0.00041	66.38	0.03	48.47	0.02	21.58	0.01	78.41	0.03	48.04	0.02	76.01	0.03
PM 0110	Poultry meat, raw (incl prepared) - 10% as fat	RAC	0.00035	7.38	0.00	5.39	0.00	2.40	0.00	8.71	0.00	5.34	0.00	8.45	0.00
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.00035	0.10	0.00	0.10	0.00	NC	-	0.10	0.00	0.71	0.00	NC	-
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.0033	0.33	0.00	0.72	0.00	0.27	0.00	0.35	0.00	0.80	0.00	NC	-
PE 0112	Eggs, raw, (incl dried)	RAC	0.00089	25.84	0.02	29.53	0.03	28.05	0.02	33.19	0.03	36.44	0.03	8.89	0.01
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total intake (ug/person)=				90.7		100.8		76.6		93.4		87.4		30.0	
Bodyweight per region (kg bw) =				60		60		55		60		60		60	
ADI (ug/person)=				1200		1200		1100		1200		1200		1200	
%ADI=				7.6%		8.4%		7.0%		7.8%		7.3%		2.5%	
Rounded %ADI=				8%		8%		7%		8%		7%		3%	

SPIROPIDION (323)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.02 mg/kg bw							
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day				Intake = daily intake: ug/person							
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake		
VC 0424	Cucumber, raw	RAC	0.34	0.68	0.23	1.81	0.62	10.40	3.54	0.01	0.00	0.04	0.01		
VC 2040	Subgroup of Melons, Pumpkins and Winter squashes	RAC	0.25	5.04	1.26	6.54	1.64	38.26	9.57	11.70	2.93	NC	-		
VO 0448	Tomato, raw	RAC	0.245	12.99	3.18	4.79	1.17	58.40	14.31	0.92	0.23	0.09	0.02		
-	Tomato, canned (& peeled)	PP	0.15	0.07	0.01	0.08	0.01	2.42	0.36	0.07	0.01	NC	-		
-	Tomato, paste (i.e. concentrated tomato sauce/puree)	PP	0.46	0.58	0.27	0.22	0.10	2.21	1.02	0.24	0.11	3.10	1.43		
JF 0448	Tomato, juice (single strength, incl concentrated)	PP	0.19	0.05	0.01	0.01	0.00	0.42	0.08	0.01	0.00	0.02	0.00		
VO 0051	Subgroup of peppers, raw (incl dried sweet peppers, excl dried chili peppers), excl okra	RAC	0.49	8.97	4.40	14.13	6.92	25.14	12.32	0.91	0.45	NC	-		

Annex 3

SPIROPIDION (323)				International Estimated Daily Intake (IEDI)				ADI = 0 - 0.02 mg/kg bw						
Codex Code	Commodity description	Expr as	STMR mg/kg	Diets: g/person/day		Intake = daily intake: ug/person								
				G13 diet	G13 intake	G14 diet	G14 intake	G15 diet	G15 intake	G16 diet	G16 intake	G17 diet	G17 intake	
VD 0541	Soya bean, dry, raw (Glycine soja)	RAC	0.49	2.76	1.35	0.07	0.03	0.33	0.16	3.16	1.55	NC	-	
-	Soya paste (i.e. miso)	PP	0.098	NC	-	NC	-	NC	-	NC	-	NC	-	
-	Soya curd (i.e. tofu)	PP	0.051	NC	-	NC	-	NC	-	NC	-	NC	-	
OR 0541	Soya oil, refined	PP	0.01	2.32	0.02	2.54	0.03	18.70	0.19	2.51	0.03	6.29	0.06	
-	Soya sauce	PP	0.02	0.01	0.00	0.13	0.00	0.17	0.00	0.01	0.00	0.56	0.01	
-	Soya flour	PP	0.79	0.11	0.09	0.08	0.06	0.07	0.06	0.01	0.01	0.03	0.02	
VR 0589	Potato, raw (incl flour, incl frozen, incl tapioca, excl starch)	RAC	0.28	23.96	6.71	13.54	3.79	213.41	59.75	104.35	29.22	8.56	2.40	
-	Potato, starch	PP	0.16	0.01	0.00	0.01	0.00	NC	-	NC	-	NC	-	
HS 0444	Peppers, chili, dried	PP	2.905	0.58	1.68	1.27	3.69	1.21	3.52	0.12	0.35	NC	-	
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) -80% as muscle	RAC	0.0065	23.34	0.15	40.71	0.26	97.15	0.63	18.06	0.12	57.71	0.38	
MM 0095	MEAT FROM MAMMALS other than marine mammals, raw (incl prepared meat) - 20% as fat	RAC	0.013	5.84	0.08	10.18	0.13	24.29	0.32	4.52	0.06	14.43	0.19	
MF 0100	Mammalian fats, raw, excl milk fats (incl rendered fats)	RAC	0.013	1.05	0.01	1.14	0.01	18.69	0.24	0.94	0.01	3.12	0.04	
MO 0105	Edible offal (mammalian), raw	RAC	0.098	4.64	0.45	1.97	0.19	10.01	0.98	3.27	0.32	3.98	0.39	
ML 0106	Milks, raw or skimmed (incl dairy products)	RAC	0.0057	108.75	0.62	70.31	0.40	436.11	2.49	61.55	0.35	79.09	0.45	
PM 0110	Poultry meat, raw (incl prepared)	RAC	0	3.92	0.00	12.03	0.00	57.07	0.00	5.03	0.00	55.56	0.00	
PM 0110	Poultry meat, raw (incl prepared) - 90% as muscle	RAC	0.00041	3.53	0.00	10.83	0.00	51.36	0.02	4.53	0.00	50.00	0.02	
PM 0110	Poultry meat, raw (incl prepared) - 10% as fat	RAC	0.00035	0.39	0.00	1.20	0.00	5.71	0.00	0.50	0.00	5.56	0.00	
PF 0111	Poultry fat, raw (incl rendered)	RAC	0.00035	NC	-	NC	-	0.32	0.00	NC	-	NC	-	
PO 0111	Poultry edible offal, raw (incl prepared)	RAC	0.0033	0.10	0.00	0.70	0.00	0.97	0.00	0.10	0.00	NC	-	
PE 0112	Eggs, raw, (incl dried)	RAC	0.00089	3.84	0.00	4.41	0.00	27.25	0.02	1.13	0.00	7.39	0.01	
-	-	-		-	-	-	-	-	-	-	-	-	-	
Total intake (ug/person)=					20.5		19.1		109.6		35.7		5.4	
Bodyweight per region (kg bw) =					60		60		60		60		60	
ADI (ug/person)=					1200		1200		1200		1200		1200	
%ADI=					1.7%		1.6%		9.1%		3.0%		0.5%	
Rounded %ADI=					2%		2%		9%		3%		0%	

Annex 4

Annex 4: International Estimate of Short-Term Intakes (IESTIs) of pesticide residues

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ArfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FC 0303	Kumquats (all commodities)	highest utilisation: Total	0	0.59	1.000	JP	Gen pop, > 1 yrs	135	120.00	<25	NR	1	0.11 - 1.41	2% - 30%	2% - 30%	20% - 20%
FC 0204	Lemon (all commodities)	highest utilisation: Total	0.018 - 0.085	0.14	1.000	PRIMO-DE	child	P95	125.50	71	3	2a	0.05 - 2.32	1% - 50%	0% - 20%	1% - 50%
FC 0205	Lime (all commodities)	highest utilisation: Total	0.037 - 0.085	0.14	1.000	AU	Gen pop, > 2 yrs	579	259.21	49	3	2a	0.01 - 0.75	0% - 10%	0% - 10%	0% - 20%
FC 0003	Subgroup of Mandarins (incl mandarin-like hybrids)	Total		0.14	1.000	CA	Child, <6 yrs	84	316.63	124	3	2a	4.999	100%	40%	100%
FC 0003	Subgroup of Mandarins (incl mandarin-like hybrids)	raw, without peel		0.14	1.000	CN	Child, 1-6 yrs	151	586.75	124	3	2a	7.248	140%	70%	140%
FC 0003	Subgroup of Mandarins (incl mandarin-like hybrids) (all other commodities)	highest utilisation: Juice (pasteurised)	0.018 - 0.085		1.000	DE	Gen pop, 14-80 yrs	9	999.39	NR	NR	3	0.02 - 0.48	1 - 10%	0 - 10%	1 - 5%
FC 0004	Subgroup of Oranges, sweet, sour (incl orange-like hybrids)	Total		0.11	1.000	AU	Child, 2-6 yrs	1735	800.83	156	3	2a	6.440	130%	70%	130%
FC 0004	Subgroup of Oranges, sweet, sour (incl orange-like hybrids) (all other commodities)	highest utilisation: raw, without peel	0.011 - 0.052	0.11	1.000	DE	Child, 2-4 yrs	92	238.40	121	3	2a	0.03 - 3.28	1 - 70%	0 - 40%	1 - 70%
FC 0005	Subgroup of Pummelo and Grapefruits (incl Shaddock-like hybrids, among others Grapefruit) (all commodities)	highest utilisation: Total	0.019 - 0.044	0.074	1.000	PRIMO-DE	child	P90	253.56	270	3	2b	0.01 - 3.49	0% - 70%	0% - 40%	0% - 70%
FP 0226	Apple	Total		0.15	1.000	AU	Child, 2-6 yrs	1997	668.41	153	3	2a	7.688	150%	70%	150%

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FP 0226	Apple	raw with peel (incl consumption without peel)		0.15	1.000	CN	Child, 1-6 yrs	1314	403.39	255	3	2a	8.491	170%	70%	170%
FP 0226	Apple (all other commodities)	highest utilisation: dried	0.012 - 0.075	0.15	4.000	AU	Child, 2-6 yrs	154	26.07	6	NR	1	0.14 - 0.82	3 - 20%	2 - 10%	3 - 20%
FP 0230	Pear	Total		0.14	1.000	CA	Child, <6 yrs	175	498.28	255	3	2a	9.697	190%	50%	190%
FP 0230	Pear	raw with peel (incl consumption without peel)		0.14	1.000	CN	Child, 1-6 yrs	413	418.33	255	3	2a	8.055	160%	60%	160%
FP 0230	Pear (all other commodities)	highest utilisation: juice (pasteurised)	0.078	0.14	1.000	PRIMO-NL	child	P97.5	599.50	NR	NR	3	0.02 - 2.54	0 - 50%	0 - 20%	0 - 50%
FS 0013	Subgroup of Cherries	Total		0.99	1.000	PRIMO-DK	child	P97.5	269.00	<25	NR	1	12.105	240%	200%	240%
FS 0013	Subgroup of Cherries	raw		0.99	1.000	DE	Child, 2-4 yrs	24	187.50	<25	NR	1	11.494	230%	220%	230%
FS 0013	Subgroup of Cherries (all other commodities)	highest utilisation: canned/preserved	0.585	0.99	1.000	NL	Child, 2-6 yrs	E	58.90	5	NR	1	0.24 - 3.17	5 - 60%	5 - 30%	3 - 60%
FS 0014	Subgroup of Plums (all commodities)	highest utilisation: dried (prunes)	0.012 - 0.025	0.04 - 0.08	1.000	AU	Child, 2-6 yrs	13	447.59	10	NR	1	0.01 - 1.88	0% - 40%	0% - 10%	0% - 40%
FS 0240	Apricot	Total		0.25	1.000	PRIMO-DE	child	P95	264.86	50	3	2a	5.648	110%	30%	110%
FS 0240	Apricot (all other commodities)	highest utilisation: raw with peel (incl consumption without peel)	0.17	0.25	1.000	AU	Gen pop, > 2 yrs	77	1056.90	55	3	2a	0.07 - 4.35	1 - 90%	1 - 90%	1 - 90%
FS 2237	Japanese apricot (ume)	Total		0.25	1.000	JP	Child, 1-6 yrs	25	25.50	<25	NR	1	0.352	7%	3%	7%
FS 0245	Nectarine	Total		0.25	1.000	CA	Child, <6 yrs	37	353.22	128	3	2a	9.341	190%	60%	190%
FS 0245	Nectarine	raw with peel (incl consumption without peel)		0.25	1.000	NL	toddler, 8-20 m	6	183.60	131	3	2a	10.919	220%	60%	220%

Annex 4

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FS 0245	Nectarine (all other commodities)	highest utilisation: canned/preserved	0.17	0.25	1.000	NL	Child, 2-6 yrs	E	118.50	60	3	2a	0.07 - 3.24	1 - 60%	1 - 20%	1 - 60%
FS 0247	Peach	Total		0.25	1.000	PRIMO-NL	toddler	P100	183.60	131	3	2a	10.919	220%	100%	220%
FS 0247	Peach	raw with peel (incl consumption without peel)		0.25	1.000	JP	Child, 1-6 yrs	76	306.00	255	3	2a	13.161	260%	70%	260%
FS 0247	Peach (all other commodities)	highest utilisation: canned/preserved	0.17	0.25	1.000	PRIMO-NL	child	E	118.50	60	3	2a	0.07 - 3.24	1 - 60%	1 - 30%	1 - 60%
FB 0264	Blackberries	Total		1.4	1.000	PRIMO-UK	toddler	P97.5	155.40	<25	NR	1	15.004	300%	230%	300%
FB 0264	Blackberries	raw with skin		1.4	1.000	DE	Gen pop, 14-80 yrs	35	460.00	<25	NR	1	8.433	170%	170%	160%
FB 0264	Blackberries (all other commodities)	highest utilisation: canned/preserved	0.84	1.4	1.000	NL	Child, 2-6 yrs	E	61.70	<25	NR	1	0.15 - 4.69	3 - 90%	2 - 40%	3 - 90%
FB 0266	Dewberries (incl Boysenberry, Loganberry) (all commodities)	highest utilisation: Total	0	1.4	1.000	PRIMO-UK	toddler	P97.5	25.50	<25	NR	1	2.46 - 2.46	50% - 50%	40% - 40%	50% - 50%
FB 0272	Raspberries, red, black	Total		1.4	1.000	PRIMO-IE	child	P97.5	184.76	<25	NR	1	12.933	260%	150%	260%
FB 0272	Raspberries, red, black	raw with skin		1.4	1.000	NL	toddler, 8-20 m	E	59.40	4	NR	1	8.154	160%	100%	160%
FB 0272	Raspberries, red, black (all other commodities)	highest utilisation: jam	0.84	1.4	1.000	CA	Child, <6 yrs	443	78.10	NR	NR	3	0.46 - 4.27	9 - 90%	5 - 30%	9 - 90%
FB 0272	Raspberries, red, black	juice (pasteurised)	0.84		1.000	PRIMO-DE	child	P95	188.96	NR	NR	3	9.828	200%	-	200%
FB 0020	Blueberries	Total		1.2	1.000	CA	Child, <6 yrs	189	176.21	<25	NR	1	13.735	270%	220%	270%
FB 0020	Blueberries	raw with skin		1.2	1.000	DE	Gen pop, 14-80 yrs	70	388.00	<25	NR	1	6.097	120%	120%	100%

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FB 0020	Blueberries (all other commodities)	highest utilisation: canned/preserved	0.8	1.2	1.000	NL	Child, 2-6 yrs	E	61.70	<25	NR	1	0.12 - 4.06	2 - 80%	1 - 30%	2 - 80%
FB 0020	Blueberries	sauce/puree	0.8		1.000	NL	Child, 2-6 yrs	E	109.70	NR	NR	3	4.770	100%	40%	100%
FB 0021	Currants, black, red, white	Total		1.2	1.000	AU	Gen pop, > 2 yrs	322	797.60	<25	NR	1	14.285	290%	290%	190%
FB 0021	Currants, black, red, white	raw with skin		1.2	1.000	DE	Child, 2-4 yrs	7	150.00	<25	NR	1	11.146	220%	140%	220%
FB 0021	Currants, black, red, white (all other commodities)	highest utilisation: jam (incl jelly)	0.8	1.2	1.000	CA	Child, <6 yrs	443	78.10	NR	NR	3	0.52 - 4.06	10 - 80%	7 - 30%	10 - 80%
FB 0021	Currants, black, red, white	juice (pasteurised)	0.8		1.000	PRIMO-NL	child	E	525.80	NR	NR	3	22.861	460%	200%	460%
FB 0268	Gooseberry	Total		1.2	1.000	PRIMO-DE	child	P100	94.96	<25	NR	1	7.056	140%	110%	140%
FB 0268	Gooseberry	raw with skin		1.2	1.000	DE	Women, 14-50 yrs	10	338.10	<25	NR	1	6.013	120%	120%	20%
FB 0268	Gooseberry	composite foods; unspecified ind processed	0.8		1.000	NL	Child, 2-6 yrs	1	3.48	NR	NR	3	0.151	3%	1%	3%
FB 0273	Rose hips (all commodities)	highest utilisation: jam (incl jelly)	0.8	1.2	1.000	CA	Child, <6 yrs	443	78.10	NR	NR	3	0.51 - 4.06	10% - 80%	5% - 50%	10% - 80%
FB 0269	Grapes (all commodities)	highest utilisation: raw with skin	0.005 - 0.035	0.06 - 0.12	1.000	CN	Child, 1-6 yrs	232	366.72	637	3	2b	0.07 - 4.09	1% - 80%	1% - 40%	0% - 80%
FB 0275	Strawberry (all commodities)	highest utilisation: Raw with skin	0.06	0.2	1.000	NL	toddler, 8-20 m	52	166.73	18	NR	1	0.03 - 3.27	1% - 70%	0% - 40%	0% - 70%
FI 0326	Avocado (all commodities)	highest utilisation: Total	0	0.1	1.000	AU	Child, 2-6 yrs	182	229.90	171	3	2a	1.54 - 3.01	30% - 60%	30% - 30%	30% - 60%
VC 0421	Bitter melon (Balsam pear, Bitter cucumber, Bitter gourd)	Total		0.24	1.000	PRIMO-UK	toddler	P97.5	96.30	114	3	2b	4.782	100%	50%	100%
VC 0421	Bitter melon (Balsam pear, Bitter cucumber, Bitter gourd)	raw without peel		0.24	1.000	CN	Gen pop, > 1 yrs	1387	400.21	608	3	2b	5.414	110%	110%	70%

Annex 4

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VC 0421	Bitter melon (Balsam pear, Bitter cucumber, Bitter gourd)	cooked/boiled without peel		0.24	1.000	JP	Gen pop, > 1 yrs	833	147.90	130	3	2a	1.742	30%	30%	30%
VC 0422	Bottle gourd (Cucuzzi)	Total		0.24	1.000	PRIMO-UK	toddler	P97.5	96.30	114	3	2b	4.782	100%	50%	100%
VC 0422	Bottle gourd (Cucuzzi)	raw with skin		0.24	1.000	CN	Gen pop, > 1 yrs	519	453.00	325	3	2a	4.973	100%	100%	NC
VC 0423	Chayote (Christophine)	Total		0.24	1.000	US	Child, < 6 yrs	18	165.19	197	3	2b	8.203	160%	80%	160%
VC 0423	Chayote (Christophine)	raw with skin		0.24	1.000	CN	Child, 1-6 yrs	124	284.75	197	3	2a	10.108	200%	90%	200%
VC 0424	Cucumber	Total		0.24	1.000	PRIMO-CZ	child, 4-6 yrs	P97.5	280.55	391	3	2b	9.439	190%	80%	190%
VC 0424	Cucumber	raw with skin		0.24	1.000	CN	Child, 1-6 yrs	340	212.11	458	3	2b	9.465	190%	110%	190%
VC 0424	Cucumber (all other commodities)	highest utilisation: cooked/boiled (without skin)	0.12	0.24	1.000	NL	Gen pop, > 1 yrs	E	200.03	333	3	2b	0.05 - 2.19	1 - 40%	4 - 40%	1 - 40%
VC 0425	Gherkin (all commodities)	highest utilisation: raw with skin	0.12	0.24	1.000	JP	Child, 1-6 yrs	484	91.80	55	3	2a	0.09 - 2.87	2% - 60%	1% - 50%	2% - 60%
VC 0427	Loofah, Angled (Sinkwa, Sinkwa towel gourd)	Total		0.24	1.000	PRIMO-UK	toddler	P97.5	96.30	114	3	2b	4.782	100%	50%	100%
VC 0427	Loofah, Angled (Sinkwa, Sinkwa towel gourd)	raw without peel		0.24	1.000	TH	Child, 3-6 yrs	759	129.62	133	3	2b	5.458	110%	40%	110%
VC 0428	Loofah, Smooth	Total		0.24	1.000	-	-	-	-	-	-	-	-	-	-	-
VC 0428	Loofah, Smooth	raw without peel		0.24	1.000	CN	Child, 1-6 yrs	196	296.64	133	3	2a	8.369	170%	70%	170%
VC 0430	Snake gourd	Total		0.24	1.000	PRIMO-UK	toddler	P97.5	96.30	114	3	2b	4.782	100%	50%	100%
VC 0430	Snake gourd	raw without peel		0.24	1.000	TH	Child, 3-6 yrs	759	129.62	133	3	2b	5.458	110%	40%	110%
VC 0431	Squash, Summer (Courgette, Marrow, Zucchini, Zucchini)	Total		0.24	1.000	US	Child, < 6 yrs	252	149.52	186	3	2b	7.424	150%	80%	150%

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VC 0431	Squash, Summer (Courgette, Marrow, Zucchini, Zucchini) (all other commodities)	highest utilisation: raw with skin	0.12	0.24	1.000	NL	Gen pop, > 1 yrs	10	68.23	328	3	2b	0.08 - 0.75	2 - 10%	1 - 10%	1 - 10%
VC 0431	Squash, Summer (Courgette, Marrow, Zucchini, Zucchini)	cooked/boiled with skin (incl consumption without skin)		0.24	1.000	PRIMO-NL	toddler	P90	72.30	289	3	2b	5.104	100%	70%	100%
VC 0046	Melons, except watermelon (Cantaloupe)	Total		0.09	1.000	PRIMO-BE	toddler	P100	540.00	540	3	2b	8.191	160%	90%	160%
VC 0046	Melons, except watermelon (Cantaloupe)	raw without peel		0.09	1.000	CN	Child, 1-6 yrs	258	296.81	998	3	2b	4.967	100%	80%	100%
VC 0046	Melons, except watermelon (Cantaloupe) (all other commodities)	highest utilisation: juice (pasteurised)	0.05		1.000	BR	Gen pop, > 10 yrs	48	720.00	NR	NR	3	0 - 0.56	0 - 10%	0 - 10%	0 - 4%
VC 0432	Watermelon	Total		0.1	1.000	CA	Child, <6 yrs	171	953.64	4302	3	2b	18.543	370%	260%	370%
VC 0432	Watermelon	raw without peel		0.1	1.000	JP	Child, 1-6 yrs	56	448.80	3000	3	2b	8.631	170%	170%	170%
VC 0432	Watermelon	juice (pasteurised)	0.1		1.000	NL	Gen pop, > 1 yrs	0	NC	NR	NR	3	NC	NC	NC	NC
VO 2704	Goji berry (all commodities)	highest utilisation: Dried	0	0.17	3.000	AU	Child, 2-6 yrs	1	28.36	<25	NR	1	0.16 - 0.76	3% - 20%	0% - 0%	3% - 20%
VO 0448	Tomato	Total		0.17	1.000	CA	Child, <6 yrs	340	250.22	175	3	2a	6.227	120%	30%	120%
VO 0448	Tomato	raw with peel		0.17	1.000	CN	Child, 1-6 yrs	1117	263.76	175	3	2a	6.458	130%	50%	130%
VO 0448	Tomato (all other commodities)	highest utilisation: cooked/boiled (with peel)	0.064 - 0.1	- 0.17	1.000	NL	toddler, 8-20 m	31	81.77	86	3	2b	0.53 - 4.09	10 - 80%	5 - 40%	10 - 80%
VO 0444	Peppers, chili (all commodities)	highest utilisation: raw with skin	0.05	0.13	1.000	CN	Gen pop, > 1 yrs	1743	295.71	43	3	2a	0 - 0.93	0% - 20%	0% - 20%	0% - 3%

Annex 4

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all

850%
gen pop

1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VO 0445	Peppers, sweet (incl. pimiento) (Bell pepper, Paprika) (all commodities)	highest utilisation: raw with skin	0.05	0.13	1.000	CN	Child, 1-6 yrs	1002	169.85	170	3	2b	0.01 - 4.11	0% - 80%	0% - 30%	0% - 80%
VO 0440	Egg plant (Aubergine) (all other commodities)	highest utilisation: Total	0.1	0.17	1.000	AU	Child, 2-6 yrs	29	128.25	318	3	2b	0.11 - 3.44	2 - 70%	1 - 60%	2 - 70%
VO 0440	Egg plant (Aubergine)	raw with skin		0.17	1.000	CN	Child, 1-6 yrs	969	253.44	444	3	2b	8.010	160%	90%	160%
VO 0443	Pepino (Melon pear, Tree melon)	Total		0.17	1.000	PRIMO-NL	Gen pop	P95	424.02	340	3	2a	2.850	60%	60%	50%
VO 2713	Scarlet eggplant (gilo, Ethiopian eggplant) (all commodities)	highest utilisation: Total	0	0.17	1.000	PRIMO-NL	Gen pop	P95	424.02	28	3	2a	1.1 - 1.24	20% - 20%	20% - 20%	30% - 30%
VP 0061	Beans with pods (Phaseolus spp): (immature pods + succulent seeds)	Total		0.42	1.000	CA	Child, <6 yrs	261	203.31	<25	NR	1	5.670	110%	60%	110%
VP 0061	Beans with pods (Phaseolus spp): (immature pods + succulent seeds) (all other commodities)	highest utilisation: cooked/boiled	0.075	- 0.42	1.000	PRIMO-NL	toddler	P97.5	127.90	<25	NR	1	0.11 - 3.13	2 - 60%	1 - 40%	2 - 60%
VP 0061	Beans with pods (Phaseolus spp): (immature pods + succulent seeds)	frozen		0.42	1.000	NL	toddler, 8-20 m	E	116.60	1	NR	1	4.801	100%	60%	100%
VP 0522	Broad bean with pods (immature pods + succulent seeds) (Vicia spp)	Total		0.42	1.000	PRIMO-NL	toddler	E	116.60	<25	NR	1	4.801	100%	60%	100%
VP 0522	Broad bean with pods (immature pods + succulent seeds) (Vicia spp)	cooked/boiled		0.25	1.000	JP	Gen pop, > 1 yrs	216	153.00	<25	NR	1	0.689	10%	10%	2%

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VP 0542	Sword bean with pods (immature pods + succulent seeds) (Canavalia spp) (all commodities)	highest utilisation: cooked/boiled	0	0.25 - 0.42	1.000	CN	Gen pop, > 1 yrs	891	316.83	<25	NR	1	1.49 - 1.49	30% - 30%	30% - 30%	0% - 0%
VP 0062	Beans without pods: (Phaseolus spp.) (succulent seeds) (all commodities)	highest utilisation: Total	0.1	0.044 - 0.1	1.000	PRIMO-IE	child	P97.5	157.79	<25	NR	1	0 - 0.79	0% - 20%	3% - 8%	0% - 20%
VP 0523	Broad bean without pods (succulent seeds) (Vicia spp) (all commodities)	highest utilisation: Total	0.1	0.044 - 0.1	1.000	PRIMO-IE	child	P97.5	157.79	<25	NR	1	0 - 0.79	0% - 20%	3% - 8%	0% - 20%
VP 0541	Soya bean without pods (succulent seeds) (Glycine max) (all commodities)	highest utilisation: cooked/boiled	0	0.06 - 0.1	1.000	CN	Child, 1-6 yrs	195	260.25	<25	NR	1	0.79 - 0.97	20% - 20%	8% - 9%	20% - 20%
VR 0589	Potato (all commodities)	highest utilisation: Total	0	0	1.000	PRIMO-UK	infant	P97.5	191.10	216	3	2b	0 - 0	0% - 0%	0% - 0%	0% - 0%
VS 0623	Cardoon	Total		2.1	1.000	PRIMO-IT	adult	P97.5	98.60	100	3	2b	9.341	190%	190%	-
VS 0623	Cardoon	raw		2.1	1.000	NL	Gen pop, > 1 yrs	0	NC	732	3	2a	NC	NC	NC	NC
VS 0623	Cardoon	cooked/boiled		2.1	1.000	PRIMO-NL	Gen pop	E	200.00	100	3	2a	12.766	260%	260%	-
VS 0624	Celery	Total		2.1	1.000	PRIMO-BE	toddler	P100	133.20	462	3	2b	47.144	940%	400%	940%
VS 0624	Celery	raw		2.1	1.000	CN	Child, 1-6 yrs	454	180.29	861	3	2b	70.391	1410%	720%	1410%
VS 0624	Celery	cooked/boiled		2.1	1.000	PRIMO-NL	Gen pop	P97.5	444.30	607	3	2b	42.539	850%	850%	-
VS 0624	Celery	canned/preserved	0.845		1.000	NL	Child, 2-6 yrs	E	126.21	NR	NR	3	5.796	120%	70%	120%
VS 0624	Celery	juice (pasteurised)	0.845		1.000	AU	Gen pop, > 2 yrs	39	854.08	NR	NR	3	10.772	220%	220%	6%

Annex 4

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VS 0624	Celery	canned babyfood	0.845		1.000	NL	toddler, 8-20 m	20	16.01	NR	NR	3	1.327	30%	NC	30%
VS 0624	Celery	composite foods; unspecified ind processed	0.845		1.000	NL	Gen pop, > 1 yrs	2	3.95	NR	NR	3	0.051	1%	1%	NC
VS 0380	Fennel, bulb (Florence fennel)	Total		2.1	1.000	AU	Child, 2-16 yrs	2	160.41	328	3	2b	26.595	530%	340%	530%
VS 0380	Fennel, bulb (Florence fennel) (all other commodities)	highest utilisation: canned babyfood	0.845	2.1	1.000	NL	toddler, 8-20 m	4	54.32	NR	NR	3	0.05 - 4.5	1 - 90%	1 - 50%	90%
VS 0380	Fennel, bulb (Florence fennel)	cooked/boiled		2.1	1.000	PRIMO-NL	child	E	166.80	251	3	2b	57.111	1140%	490%	1140%
VS 0627	Rhubarb	Total		2.1	1.000	PRIMO-UK	toddler	P97.5	83.40	76	3	2a	34.092	680%	410%	680%
VS 0627	Rhubarb	raw		2.1	1.000	NL	Gen pop, > 1 yrs	0	NC	89	3	2a	NC	NC	NC	NC
VS 0627	Rhubarb	sauce/puree	0.845		1.000	PRIMO-NL	child	P100	149.67	NR	NR	3	6.873	140%	110%	140%
GC 0645	Maize (corn) (all commodities)	highest utilisation: flakes	0.00016 - 0.01	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0 - 0.34	0% - 7%	0% - 3%	0% - 7%
TN 0660	Almonds (all commodities)	highest utilisation: Total	0	0	1.000	CA	Child, <6 yrs	62	63.32	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0662	Brazil nut (all commodities)	highest utilisation: Total	0	0	1.000	PRIMO-UK	child, 4-6 yrs	P97.5	17.80	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0295	Cashew nut (all commodities)	highest utilisation: raw incl roasted	0	0	1.000	TH	child, 3-6 yrs	374	98.84	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0664	Chestnut (all commodities)	highest utilisation: Total	0	0	1.000	CN	Gen pop, > 1 yrs	807	475.25	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0665	Coconut (all commodities)	highest utilisation: raw (i.e. nutmeat)	0	0	1.000	TH	child, 3-6 yrs	826	423.40	383	3	2a	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0666	Hazelnut (all commodities)	highest utilisation: Total	0	0	1.000	PRIMO-IE	child	P97.5	65.42	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0669	Macadamia nut (all commodities)	highest utilisation: Total	0	0	1.000	PRIMO-DE	women, 14-50 yrs	P100	141.69	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%

FENPYROXIMATE (193)

Acute RfD= 0.005 mg/kg bw (5 µg/kg bw)

IESTI

Maximum %ARfD:

1410%
all850%
gen pop1410%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
TN 0672	Pecan (all commodities)	highest utilisation: Total	0	0	1.000	PRIMO-DE	child	P100	44.41	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0673	Pine nut (all commodities)	highest utilisation: Total	0	0	1.000	BR	Gen pop, > 10 yrs	47	200.00	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0675	Pistachio nut (all commodities)	highest utilisation: Total	0	0	1.000	PRIMO-IE	child	P97.5	115.86	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
TN 0678	Walnut (all commodities)	highest utilisation: Total	0	0	1.000	PRIMO-BE	toddler	P100	60.00	<25	NR	1	0 - 0	0% - 0%	0% - 0%	0% - 0%
SB 0716	Coffee bean (all commodities)	highest utilisation: extract (beverage)	0.025	0	0.180	CA	women, 15-49 yrs	2666	2088.65	NR	NR	3	0 - 0.14	0% - 3%	0% - 3%	0% - 1%
DH 1100	Hops, dry (all commodities)	highest utilisation: raw = dried	5.15	0	1.000	US	Gen pop, 0-85 yrs	3162	24.49	<25	NR	3	0.57 - 1.78	10% - 40%	10% - 40%	0% - 4%
DT 1114	Tea, green, black (black, fermented and dried) (all commodities)	highest utilisation: Total	0.015 - 1.4	0	1.000	PRIMO-IE	child	P97.5	30.60	<25	NR	3	0.61 - 2.14	10% - 40%	10% - 40%	10% - 40%
MM 0095	Meat from mammals other than marine mammals	Total	NA	NA	1.000	CN	Child, 1-6 yrs	4329	261.46	NR	NR	1	NA	20%	10%	20%
MM 0095	Meat from mammals other than marine mammals: 20% as fat	Total		0.13	1.000	CN	Child, 1-6 yrs	4329	52.29	NR	NR	1	0.421	8%	6%	9%
MM 0095	Meat from mammals other than marine mammals: 80% as muscle	Total		0.041	1.000	CN	Child, 1-6 yrs	4329	209.17	NR	NR	1	0.531	10%	8%	10%
MF 0100	Mammalian fats (except milk fats)	Total		0.13	1.000	PRIMO-UK	infant	P97.5	18.10	NR	NR	1	0.270	5%	5%	5%
MO 0105	Edible offal (mammalian)	Total		0.77	1.000	ZA	Gen pop, > 10 yrs	-	523.58	NR	NR	1	7.238	140%	140%	120%
ML 0106	Milks	Total	0.005		1.000	PRIMO-UK	infant	P97.5	1080.70	NR	NR	3	0.621	10%	5%	10%

Annex 4

FIPRONIL (202)

Acute RfD= 0.03 mg/kg bw (30 µg/kg bw)

IESTI

Maximum %ARfD:

20%
all10%
gen pop20%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FI 0327	Banana (incl Dwarf banana & Plantain) (all commodities)	highest utilisation: raw without peel	0	0	1.000	CN	Child, 1-6 yrs	286	455.81	767	3	2b	0 - 0	0% - 0%	0% - 0%	0% - 0%
VA 0385	Onion, bulb (all commodities)	highest utilisation: raw without skin	0.02	0.033	1.000	JP	Child, 1-6 yrs	748	102.00	244	3	2b	0.01 - 0.62	0% - 2%	0% - 1%	0% - 2%
VO 0448	Tomato (all commodities)	highest utilisation: raw with peel	0.008	0.008	1.000	CN	Child, 1-6 yrs	1117	263.76	175	3	2a	0.04 - 0.3	0% - 1%	0% - 0%	0% - 1%
VL 0460	Amaranth leaves (Bledo) (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P97.5	402.30	<25	NR	1	0.32 - 0.66	1% - 2%	1% - 1%	2% - 2%
VL 0464	Chard (Beet leaves, Silver beet) (all commodities)	highest utilisation: cooked/boiled	0	0.02919	1.000	PRIMO-NL	child	P100	81.80	105	3	2b	0.1 - 0.39	0% - 1%	0% - 1%	1% - 1%
VL 0465	Chervil (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P100	23.00	<25	NR	1	0.01 - 0.04	0% - 0%	0% - 0%	0% - 0%
VL 0469	Chicory leaves (green and red cultivars) (Sugar loaf) (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P100	143.00	440	3	2b	0.45 - 0.7	1% - 2%	0% - 1%	1% - 2%
VL 2752	Chrysanthemum, edible leaved (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P97.5	402.30	<25	NR	1	0.18 - 0.66	1% - 2%	0% - 1%	2% - 2%
VL 0470	Corn salad (Lambs lettuce) (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P100	50.00	<25	NR	1	0.02 - 0.08	0% - 0%	0% - 0%	0% - 0%
VL 0510	Cos lettuce (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-NL	child	P97.5	140.10	290	3	2b	0.27 - 0.67	1% - 2%	1% - 1%	2% - 2%
VL 0474	Dandelion (Laiteron, Pissenlit) (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P100	143.00	35	3	2a	0.05 - 0.35	0% - 1%	0% - 1%	1% - 1%
VL 0476	Endive (i.e. Scarole) (all commodities)	highest utilisation: cooked/boiled	0	0.02919	1.000	PRIMO-NL	toddler	P95	135.20	251	3	2b	0.63 - 1.16	2% - 4%	1% - 1%	2% - 4%
VL 0482	Lettuce, head (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-NL	child	P97.5	140.10	290	3	2b	0.29 - 0.67	1% - 2%	1% - 1%	2% - 2%
VL 0483	Lettuce, leaf (all commodities)	highest utilisation: Total	0	0.02919	1.000	CN	Child, 1-6 yrs	243	387.25	305	3	2a	0.17 - 1.81	1% - 6%	1% - 2%	2% - 6%

FIPRONIL (202)

Acute RfD= 0.03 mg/kg bw (30 µg/kg bw)

IESTI

Maximum %ARfD:

20%
all10%
gen pop20%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VL 0492	Purslane (all commodities)	highest utilisation: cooked/boiled	0	0.02919	1.000	PRIMO-NL	Gen pop	P100	271.20	<25	NR	1	0.03 - 0.12	0% - 0%	0% - 0%	0% - 0%
VL 0501	Sowthistle (all commodities)	highest utilisation: raw	0	0.02919	1.000	CN	Gen pop, > 1 yrs	1187	592.49	35	3	2a	0.36 - 0.36	1% - 1%	1% - 1%	0% - 0%
VL 0502	Spinach (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P97.5	402.30	<25	NR	1	0.13 - 0.66	0% - 2%	0% - 1%	0% - 2%
VL 0401	Broccoli, Chinese (i.e. kailan) (all commodities)	highest utilisation: raw	0	0.02919	1.000	CN	Child, 1-6 yrs	334	222.48	311	3	2b	0.73 - 1.21	2% - 4%	1% - 2%	2% - 4%
VL 0466	Chinese cabbage (type Pak-choi) (i.e. celery mustard) (all commodities)	highest utilisation: raw	0	0.02919	1.000	CN	Child, 1-6 yrs	1966	327.07	1548	3	2b	0.36 - 1.78	1% - 6%	1% - 3%	1% - 6%
VL 0472	Cress, Garden (all commodities)	highest utilisation: raw	0	0.02919	1.000	CN	Gen pop, > 1 yrs	1443	352.50	<25	NR	1	0.01 - 0.19	0% - 1%	0% - 1%	0% - 0%
VL 0468	Flowering white cabbage (Choisum) (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P97.5	148.10	186	3	2b	0.63 - 0.73	2% - 2%	1% - 2%	2% - 2%
VL 0480	Kale (Borecole, Collards) (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-DE	child	P100	142.12	672	3	2b	0.48 - 0.77	2% - 3%	1% - 2%	2% - 3%
VL 0481	Komatsuna	Total		0.02919	1.000	PRIMO-BE	toddler	P100	114.40	<25	NR	1	0.188	1%	0%	1%
VL 2781	Mizuna	Total		0.02919	1.000	PRIMO-BE	toddler	P100	114.40	<25	NR	1	0.188	1%	0%	1%
VL 0485	Mustard greens (Indian mustard, Amsoi, mustard cabbage, red mustards) (all commodities)	highest utilisation: raw	0	0.02919	1.000	CN	Child, 1-6 yrs	635	299.31	245	3	2a	0.35 - 1.43	1% - 5%	1% - 2%	2% - 5%
VL 0494	Radish leaves	Total		0.02919	1.000	PRIMO-DE	child	P100	142.12	<25	NR	1	0.257	1%	0%	1%
VL 0495	Rape greens (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-DE	child	P100	142.12	<25	NR	1	0.11 - 0.26	0% - 1%	0% - 0%	1% - 1%
VL 0496	Rucola (Arrugula, Rocket salad, Roquette, Roman	highest utilisation: Total	0	0.02919	1.000	PRIMO-DE	child	P100	43.44	<25	NR	1	0.03 - 0.08	0% - 0%	0% - 1%	0% - 0%

Annex 4

FIPRONIL (202)

Acute RfD= 0.03 mg/kg bw (30 µg/kg bw)

IESTI

Maximum %ARfD:

20%
all

10%
gen pop

20%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
	rocket) (all commodities)															
VL 0506	Turnip greens (Namenia, Tendergreen) (all commodities)	highest utilisation: Total	0	0.02919	1.000	PRIMO-BE	toddler	P100	114.40	35	3	2a	0.05 - 0.3	0% - 1%	0% - 1%	0% - 1%
VL 2832	Witloof chicory sprouts (Belgian endives) (all commodities)	highest utilisation: cooked/boiled	0	0.02919	1.000	PRIMO-NL	toddler	P97.5	160.65	124	3	2a	0.26 - 1.17	1% - 4%	1% - 1%	1% - 4%
VP 0061	Beans with pods (Phaseolus spp): (immature pods + succulent seeds) (all commodities)	highest utilisation: Total	0.008	0.0099	1.000	CA	Child, <6 yrs	261	203.31	<25	NR	1	0.01 - 0.13	0% - 0%	0% - 0%	0% - 0%
VD 0071	Beans (dry) (Phaseolus spp) (all commodities)	highest utilisation: Total	0.002	0	1.000	PRIMO-UK	infant	P97.5	159.00	<25	NR	3	0 - 0.04	0% - 0%	0% - 0%	0% - 0%
VD 0541	Soya bean (dry) (Glycine spp) (all commodities)	highest utilisation: Total	0.00411	0	1.000	CN	Child, 1-6 yrs	179	239.05	<25	NR	3	0 - 0.06	0% - 0%	0% - 0%	0% - 0%
VR 0574	Beetroot (all commodities)	highest utilisation: Total	0	0.00212	1.000	AU	Child, 2-6 yrs	53	314.08	135	3	2a	0 - 0.07	0% - 0%	0% - 0%	0% - 0%
VR 0575	Burdock, greater or edible	Total		0.00212	1.000	PRIMO-BE	toddler	P100	99.90	75	3	2a	0.030	0%	0%	0%
VR 0577	Carrot (all commodities)	highest utilisation: raw with skin	0	0.00212	1.000	CN	Child, 1-6 yrs	400	234.68	300	3	2b	0.02 - 0.09	0% - 0%	0% - 0%	0% - 0%
VR 0578	Celeriac (Turnip rooted celery) (all commodities)	highest utilisation: Total	0	0.00212	1.000	PRIMO-BE	toddler	P100	196.90	749	3	2b	0.01 - 0.07	0% - 0%	0% - 0%	0% - 0%
VR 0469	Chicory, roots (all commodities)	highest utilisation: Total	0	0.00212	1.000	AU	Gen pop, > 2 yrs	175	26.16	48	3	2b	0 - 0	0% - 0%	0% - 0%	0% - 0%
VR 0583	Horseradish (all commodities)	highest utilisation: Total	0	0.00212	1.000	PRIMO-DE	Gen pop	P97.5	79.50	220	3	2b	0 - 0.01	0% - 0%	0% - 0%	0% - 0%
VR 0587	Parsley, turnip-rooted (Hamburg roots)	Total		0.00212	1.000	PRIMO-NL	Gen pop	P97.5	96.60	140	3	2b	0.009	0%	0%	0%

FIPRONIL (202)

Acute RfD= 0.03 mg/kg bw (30 µg/kg bw)

IESTI

Maximum %ARfD:

20%
all10%
gen pop20%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VR 0588	Parsnip (all commodities)	highest utilisation: cooked/boiled (without skin)	0	0.00212	1.000	PRIMO-NL	child	E	133.30	227	3	2b	0 - 0.05	0% - 0%	0% - 0%	0% - 0%
VR 0494	Radish (all commodities)	highest utilisation: Total	0	0.00212	1.000	PRIMO-NL	child	E	64.40	172	3	2b	0 - 0.02	0% - 0%	0% - 0%	0% - 0%
VR 0590	Radish, black (all commodities)	highest utilisation: Total	0	0.00212	1.000	PRIMO-NL	child	E	64.40	172	3	2b	0 - 0.02	0% - 0%	0% - 0%	0% - 0%
VR 0591	Radish, Japanese (Chinese radish, Daikon) (all commodities)	highest utilisation: raw without skin	0	0.00212	1.000	CN	Child, 1-6 yrs	1187	293.37	1000	3	2b	0 - 0.12	0% - 0%	0% - 0%	0% - 0%
VR 0498	Salsify (Oyster plant) (all commodities)	highest utilisation: Total	0	0.00212	1.000	PRIMO-BE	toddler	P100	99.90	75	3	2a	0.03 - 0.03	0% - 0%	0% - 0%	0% - 0%
VR 0596	Sugar beet (all commodities)	highest utilisation: composite foods; unspecified ind processed	0.003	0	1.000	NL	Child, 2-6 yrs	2554	168.93	NR	NR	3	0.01 - 0.03	0% - 0%	0% - 0%	0% - 0%
VR 0497	Swede (Rutabaga) (all commodities)	highest utilisation: Total	0	0.00212	1.000	PRIMO-UK	infant	P97.5	90.00	500	3	2b	0 - 0.07	0% - 0%	0% - 0%	0% - 0%
VR 0506	Turnip, garden (all commodities)	highest utilisation: cooked/boiled (without peel)	0	0.00212	1.000	PRIMO-NL	child	E	133.31	176	3	2b	0.03 - 0.05	0% - 0%	0% - 0%	0% - 0%
VR 0573	Arrowroot (all commodities)	highest utilisation: Total	0	0.00212	1.000	AU	Child, 2-6 yrs	689	0.20	45	3	2b	0 - 0	0% - 0%	0% - 0%	0% - 0%
VR 0463	Cassava (Manioc) (all commodities)	highest utilisation: cooked/boiled (without peel)	0	0.00212	1.000	PRIMO-NL	Gen pop	E	250.00	356	3	2b	0.02 - 0.02	0% - 0%	0% - 0%	0% - 0%
VR 0585	Jerusalem artichoke (i.e. Topinambur) (all commodities)	highest utilisation: cooked/boiled (without peel)	0	0.00212	1.000	PRIMO-NL	child	E	133.30	56	3	2a	0.01 - 0.03	0% - 0%	0% - 0%	0% - 0%
VR 0589	Potato (all commodities)	highest utilisation: Total	0.00493	0.0296	1.000	PRIMO-UK	infant	P97.5	191.10	216	3	2b	0 - 1.95	0% - 7%	0% - 2%	0% - 7%
VR 0508	Sweet potato (all commodities)	highest utilisation: Total	0	0.00212	1.000	CA	Child, <6 yrs	91	358.61	546	3	2b	0.05 - 0.18	0% - 1%	0% - 0%	0% - 1%
VR 0504	Tannia (Tanier, Yautia) (all commodities)	highest utilisation: cooked/boiled (without peel)	0	0.00212	1.000	NL	Gen pop, > 1 yrs	E	249.97	170	3	2a	0.01 - 0.02	0% - 0%	0% - 0%	0% - 0%

Annex 4

FIPRONIL (202)

Acute RfD= 0.03 mg/kg bw (30 µg/kg bw)

IESTI

Maximum %ARfD:

20%
all10%
gen pop20%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VR 0505	Taro (Dasheen, Eddoe) (all commodities)	highest utilisation: cooked/boiled (without peel)	0	0.00212	1.000	CN	Child, 1-6 yrs	199	384.18	668	3	2b	0.02 - 0.15	0% - 1%	0% - 0%	0% - 1%
VR 0600	Yams (all commodities)	highest utilisation: Total	0	0.00212	1.000	PRIMO-UK	adult	P97.5	693.70	365	3	2a	0.02 - 0.04	0% - 0%	0% - 0%	0% - 0%
GC 0648	Quinoa	Total	0.008		1.000	PRIMO-FI	child 3 yrs	P97.5	75.69	<25	NR	3	0.040	0%	0%	0%
GC 0650	Rye (all commodities)	highest utilisation: flakes	0.008	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0.01 - 0.27	0% - 1%	0% - 0%	0% - 1%
GC 0653	Triticale	Total	0.008		1.000	PRIMO-UK	child, 4-6 yrs	P97.5	296.20	<25	NR	3	0.116	0%	0%	0%
GC 0654	Wheat (all commodities)	highest utilisation: flakes	0.008	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0.01 - 0.27	0% - 1%	0% - 0%	0% - 1%
GC 0640	Barley (all commodities)	highest utilisation: flakes	0.00536	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0 - 0.18	0% - 1%	0% - 0%	0% - 1%
GC 0641	Buckwheat (all commodities)	highest utilisation: flakes	0.00536	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0 - 0.18	0% - 1%	0% - 0%	0% - 1%
GC 0647	Oats (all commodities)	highest utilisation: flakes (rolled oats)	0.00536	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0 - 0.18	0% - 1%	0% - 0%	0% - 1%
GC 0649	Rice (all commodities)	highest utilisation: flour (cereals)	0.00012 - 0.0114	0	1.000	CN	Child, 1-6 yrs	416	268.93	NR	NR	3	0 - 0.01	0% - 0%	0% - 0%	0% - 0%
GC 0645	Maize (corn) (all commodities)	highest utilisation: flakes	0.004	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0 - 0.14	0% - 0%	0% - 0%	0% - 0%
GS 0659	Sugar cane (all commodities)	highest utilisation: thick juice	0.00304	0	1.000	CN	Gen pop, > 1 yrs	436	1817.52	NR	NR	3	0 - 0.1	0% - 0%	0% - 0%	0% - 0%
SO 0699	Safflower seed (all commodities)	highest utilisation: Total	0.008	0	1.000	PRIMO-DE	child	P95	49.74	<25	NR	3	0 - 0.02	0% - 0%	0% - 0%	0% - 0%
SO 0702	Sunflower seed (all commodities)	highest utilisation: Total	0.008	0	1.000	CA	women, 15-49 yrs	121	296.25	<25	NR	3	0 - 0.04	0% - 0%	0% - 0%	0% - 0%
MM 0095	Meat from mammals other than marine mammals	Total	NA	NA	1.000	CN	Child, 1-6 yrs	4329	261.46	NR	NR	1	NA	9%	7%	9%
MM 0095	Meat from mammals other than marine mammals: 20% as fat	Total		0.65651	1.000	CN	Child, 1-6 yrs	4329	52.29	NR	NR	1	2.128	7%	5%	7%

Annex 4

354

FIPRONIL (202)

Acute RfD= 0.03 mg/kg bw (30 µg/kg bw)

IESTI

Maximum %ARfD:

20%
all10%
gen pop20%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
MM 0095	Meat from mammals other than marine mammals: 80% as muscle	Total		0.04926	1.000	CN	Child, 1-6 yrs	4329	209.17	NR	NR	1	0.639	2%	2%	2%
MF 0100	Mammalian fats (except milk fats)	Total		0.65651	1.000	PRIMO-UK	infant	P97.5	18.10	NR	NR	1	1.366	5%	4%	5%
MO 0105	Edible offal (mammalian)	Total		0.32752	1.000	ZA	Gen pop, > 10 yrs	-	523.58	NR	NR	1	3.079	10%	10%	9%
ML 0106	Milks	Total	0.04321		1.000	PRIMO-UK	infant	P97.5	1080.70	NR	NR	3	5.367	20%	7%	20%
FM 0812	Cattle milk fat	Total	0.12		1.000	BR	Gen pop, > 10 yrs	441	150.00	NR	NR	3	0.279	1%	1%	1%
PM 0110	Poultry meat	Total	NA	NA	1.000	CN	Child, 1-6 yrs	175	347.00	NR	NR	1	NA	1%	1%	1%
PM 0110	Poultry meat: 10% as fat	Total		0.08343	1.000	CN	Child, 1-6 yrs	175	34.70	NR	NR	1	0.179	1%	0%	1%
PM 0110	Poultry meat: 90% as muscle	Total		0.01169	1.000	CN	Child, 1-6 yrs	175	312.30	NR	NR	1	0.226	1%	0%	1%
PF 0111	Poultry, fats	Total		0.08343	1.000	CA	Child, <6 yrs	66	49.38	NR	NR	1	0.242	1%	1%	1%
PO 0111	Poultry, edible offal (includes kidney and liver)	Total		0.04231	1.000	CN	Gen pop, > 1 yrs	421	345.63	NR	NR	1	0.275	1%	1%	1%
PE 0112	Eggs	Total		0.06141	1.000	PRIMO-UK	infant	P97.5	108.00	NR	NR	1	0.762	3%	1%	3%

Annex 4

METALAXYL (138)

Acute RfD= 0.5 mg/kg bw (500 µg/kg bw)

IESTI

Maximum %ARfD:

100%
all30%
gen pop100%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FC 0004	Subgroup of Oranges, sweet, sour (incl orange-like hybrids) (all commodities)	highest utilisation: Total	0.013 - 0.132	0.026	1.000	AU	Child, 2-6 yrs	1735	800.83	156	3	2a	0.01 - 1.52	0% - 0%	0% - 0%	0% - 0%
FP 0226	Apple	raw with peel (incl consumption without peel)	0	0	1.000	CN	Child, 1-6 yrs	1314	403.39	255	3	2a	0.000	0%	0%	0%
FP 0230	Pear	raw with peel (incl consumption without peel)	0	0	1.000	CN	Child, 1-6 yrs	413	418.33	255	3	2a	0.000	0%	0%	0%
FB 0269	Grapes (all commodities)	highest utilisation: raw with skin	0.073 - 0.182	0.884	1.000	CN	Child, 1-6 yrs	232	366.72	637	3	2b	1.16 - 60.27	0% - 10%	0% - 6%	0% - 10%
VA 0385	Onion, bulb (all commodities)	highest utilisation: raw without skin	0.02	0.02	1.000	JP	Child, 1-6 yrs	748	102.00	244	3	2b	0.01 - 0.37	0% - 0%	0% - 0%	0% - 0%
VB 0400	Broccoli (all commodities)	highest utilisation: cooked/boiled	0.275	1.21	1.000	PRIMO-NL	toddler	P97.5	160.70	286	3	2b	0.5 - 57.19	0% - 10%	0% - 4%	0% - 10%
VB 0404	Cauliflower (all commodities)	highest utilisation: cooked/boiled	0.275	1.21	1.000	PRIMO-NL	toddler	P97.5	142.00	749	3	2b	0.04 - 50.54	0% - 10%	0% - 6%	0% - 10%
VB 0402	Brussels sprouts (all commodities)	highest utilisation: cooked/boiled	0.44	0.77	1.000	PRIMO-NL	toddler	P90	103.80	<25	NR	1	0.02 - 7.84	0% - 2%	0% - 1%	0% - 2%
VB 0041	Cabbages, head (all commodities)	highest utilisation: raw	0.22	0.44	1.000	CN	Child, 1-6 yrs	287	255.54	1403	3	2b	0.16 - 20.9	0% - 4%	0% - 2%	0% - 4%
VC 0046	Melons, except watermelon (Cantaloupe) (all commodities)	highest utilisation: Total	0.013	0.026	1.000	PRIMO-BE	toddler	P100	540.00	540	3	2b	0 - 2.37	0% - 0%	0% - 0%	0% - 0%

METALAXYL (138)

Acute RfD= 0.5 mg/kg bw (500 µg/kg bw)

IESTI

Maximum %ARfD:

100%
all30%
gen pop100%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VO 2704	Goji berry (all commodities)	highest utilisation: Dried		0.234	3.000	AU	Child, 2-6 yrs	1	28.36	<25	NR	1	0.22 - 1.05	0% - 0%	0% - 0%	0% - 0%
VO 0448	Tomato (all commodities)	highest utilisation: raw with peel	0.058	0.234	1.000	CN	Child, 1-6 yrs	1117	263.76	175	3	2a	0.31 - 8.89	0% - 2%	0% - 1%	0% - 2%
VL 0483	Lettuce, leaf	Total		8.14	1.000	CN	Child, 1-6 yrs	243	387.25	305	3	2a	503.453	100%	30%	100%
VL 0483	Lettuce, leaf (all other commodities)	highest utilisation: raw	1.43	8.14	1.000	NL	Child, 2-6 yrs	91	140.10	118	3	2a	0.84 - 166.17	0 - 30%	0 - 10%	0 - 30%
VL 0502	Spinach (all commodities)	highest utilisation: Total	0.22	0.22	1.000	PRIMO-BE	toddler	P97.5	402.30	<25	NR	1	0.02 - 4.97	0% - 1%	0% - 1%	0% - 1%
VR 0577	Carrot (all commodities)	highest utilisation: raw with skin	0.02	0.02	1.000	CN	Child, 1-6 yrs	400	234.68	300	3	2b	0 - 0.87	0% - 0%	0% - 0%	0% - 0%
VR 0589	Potato (all commodities)	highest utilisation: Total	0.01	0.02	1.000	PRIMO-UK	infant	P97.5	191.10	216	3	2b	0.01 - 1.32	0% - 0%	0% - 0%	0% - 0%
SO 0699	Safflower seed (all commodities)	highest utilisation: Total	0		1.000	PRIMO-DE	child	P95	49.74	<25	NR	3	0 - 0	0% - 0%	0% - 0%	0% - 0%
HS 0790	Pepper, Black; White; Pink; Green	Total	0.455		1.000	AU	Gen pop, > 2 yrs	260	6.88	<25	NR	1	ND	-	-	-
DT 9999	Ginseng root, dried	Total		0.03	1.000	PRIMO-EFSA	adults	E	36.00	<25	NR	1	0.018	0%	0%	0%

Annex 4

METCONAZOLE (313)

Acute RfD= 0.04 mg/kg bw (40 µg/kg bw)

IESTI

Maximum %ARfD:

3%
all

1%
gen pop

3%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
GC 0653	Triticale	Total	0.035		1.000	PRIMO-UK	child, 4-6 yrs	P97.5	296.20	<25	NR	3	0.506	1%	1%	1%
GC 0654	Wheat (all commodities)	highest utilisation: flakes	0,008 - 0,067	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0,02 - 1,2	0% - 3%	0% - 1%	0% - 3%

PENDIMETHALIN (292)

Acute RfD= 1 mg/kg bw (1000 µg/kg bw)

IESTI

Maximum %ARfD:

1%
all0%
gen pop1%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FB 0269	Grapes (all commodities)	highest utilisation: raw with skin	0,05	0,05	1.000	CN	Child, 1-6 yrs	232	366.72	637	3	2b	0,16 - 3,41	0% - 0%	0% - 0%	0% - 0%
VA 0384	Leek (all commodities)	highest utilisation: raw	0,02	0,23	1.000	CN	Child, 1-6 yrs	401	149.40	176	3	2b	0 - 6,39	0% - 1%	0% - 0%	0% - 1%
VO 2704	Goji berry (all commodities)	highest utilisation: Dried	0	0,05	3.000	AU	Child, 2-6 yrs	1	28.36	<25	NR	1	0,05 - 0,22	0% - 0%	0% - 0%	0% - 0%
VO 0448	Tomato (all commodities)	highest utilisation: raw with peel	0	0,05	1.000	CN	Child, 1-6 yrs	1117	263.76	175	3	2a	0,53 - 1,9	0% - 0%	0% - 0%	0% - 0%
VO 0442	Okra (Lady's finger, Gombo) (all commodities)	highest utilisation: Total	0	0,05	1.000	US	Child, < 6 yrs	26	82.30	17	NR	1	0,27 - 0,28	0% - 0%	0% - 0%	0% - 0%
VO 0444	Peppers, chili (all commodities)	highest utilisation: raw with skin	0	0,05	1.000	CN	Gen pop, > 1 yrs	1743	295.71	43	3	2a	0,12 - 0,36	0% - 0%	0% - 0%	0% - 0%
VO 0445	Peppers, sweet (incl. pimiento) (Bell pepper, Paprika) (all commodities)	highest utilisation: raw with skin	0	0,05	1.000	CN	Child, 1-6 yrs	1002	169.85	170	3	2b	0,3 - 1,58	0% - 0%	0% - 0%	0% - 0%
VO 0440	Egg plant (Aubergine) (all commodities)	highest utilisation: raw with skin	0	0,05	1.000	CN	Child, 1-6 yrs	969	253.44	444	3	2b	0,71 - 2,36	0% - 0%	0% - 0%	0% - 0%
VO 0443	Pepino (Melon pear, Tree melon)	Total		0.05	1.000	PRIMO-NL	Gen pop	P95	424.02	340	3	2a	0.838	0%	0%	0%
VO 2713	Scarlet eggplant (gilo, Ethiopian eggplant) (all commodities)	highest utilisation: Total	0	0,05	1.000	PRIMO-NL	Gen pop	P95	424.02	28	3	2a	0,32 - 0,37	0% - 0%	0% - 0%	0% - 0%
VD 0541	Soya bean (dry) (Glycine spp) (all commodities)	highest utilisation: Total	0,05	0	1.000	CN	Child, 1-6 yrs	179	239.05	<25	NR	3	0 - 0,74	0% - 0%	0% - 0%	0% - 0%
GC 0654	Wheat (all commodities)	highest utilisation: flakes	0,01	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0,01 - 0,34	0% - 0%	0% - 0%	0% - 0%
GC 0649	Rice (all commodities)	highest utilisation: Total	0	0	1.000	CA	Child, <6 yrs	666	461.40	<25	NR	3	0 - 0	0% - 0%	0% - 0%	0% - 0%
GC 0645	Maize (corn) (all commodities)	highest utilisation: flakes	0,05	0	1.000	CA	Child, <6 yrs	1909	539.23	NR	NR	3	0,03 - 1,72	0% - 0%	0% - 0%	0% - 0%
GS 0659	Sugar cane (all commodities)	highest utilisation: thick juice	0	0	1.000	CN	Gen pop, > 1 yrs	436	1817.52	NR	NR	3	0 - 0	0% - 0%	0% - 0%	0% - 0%

Annex 4

PENDIMETHALIN (292)

Acute RfD= 1 mg/kg bw (1000 µg/kg bw)

IESTI

Maximum %ARfD:

1%
all

0%
gen pop

1%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
SO 0702	Sunflower seed (all commodities)	highest utilisation: Total	0,05	0	1.000	CA	women, 15-49 yrs	121	296.25	<25	NR	3	0,03 - 0,23	0% - 0%	0% - 0%	0% - 0%
HH 0740	Parsley, leaves (all commodities)	highest utilisation: Total	0,3	0,76	1.000	PRIMO-UK	vegetarian	P97.5	79.90	<25	NR	1	0,01 - 0,91	0% - 0%	0% - 0%	0% - 0%

PYRAZIFLUMID (322)

Acute RfD= 2 mg/kg bw (2000 µg/kg bw)

IESTI

Maximum %ARfD:

3%
all2%
gen pop3%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
FP 0226	Apple (all commodities)	highest utilisation: raw with peel (incl consumption without peel)	0.36	0.73	1.000	CN	Child, 1-6 yrs	1314	403.39	255	3	2a	1.83 - 41.32	0% - 2%	0% - 1%	0% - 2%
FP 0230	Pear (all commodities)	highest utilisation: Total	0.36	0.73	1.000	CA	Child, <6 yrs	175	498.28	255	3	2a	0.09 - 50.56	0% - 3%	0% - 1%	0% - 3%
FP 0307	Persimmon, Japanese (i.e. Kaki fruit) (all commodities)	highest utilisation: raw with peel (incl consumption without peel)	0	0.73	1.000	TH	Child, 3-6 yrs	20	264.88	228	3	2a	27.8 - 30.73	1% - 2%	1% - 1%	1% - 2%
FB 0269	Grapes (all commodities)	highest utilisation: raw with skin	0.57	0.98 - 1.96	1.000	CN	Child, 1-6 yrs	232	366.72	637	3	2b	3.1 - 66.82	0% - 3%	0% - 2%	0% - 3%
MM 0095	Meat from mammals other than marine mammals	Total	NA	NA	1.000	CN	Child, 1-6 yrs	4329	261.46	NR	NR	1	NA	0%	0%	0%
MM 0095	Meat from mammals other than marine mammals: 20% as fat	Total		0.002	1.000	CN	Child, 1-6 yrs	4329	52.29	NR	NR	1	0.006	0%	0%	0%
MM 0095	Meat from mammals other than marine mammals: 80% as muscle	Total		0.0005	1.000	CN	Child, 1-6 yrs	4329	209.17	NR	NR	1	0.006	0%	0%	0%
MF 0100	Mammalian fats (except milk fats)	Total		0.002	1.000	PRIMO-UK	infant	P97.5	18.10	NR	NR	1	0.004	0%	0%	0%
MO 0105	Edible offal (mammalian)	Total		0.005	1.000	ZA	Gen pop, > 10 yrs	-	523.58	NR	NR	1	0.047	0%	0%	0%
ML 0106	Milks	Total	0.0001		1.000	PRIMO-UK	infant	P97.5	1080.70	NR	NR	3	0.012	0%	0%	0%

Annex 4

SPIROPIDION (323)

Acute RfD= 0.3 mg/kg bw (300 µg/kg bw)

IESTI

Maximum %ARfD:

60%
all40%
gen pop60%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
VC 0424	Cucumber (all commodities)	highest utilisation: raw with skin	0.34	0.7	1.000	CN	Child, 1-6 yrs	340	212.11	458	3	2b	0.13 - 27.61	0% - 9%	0% - 6%	0% - 9%
VC 0046	Melons, except watermelon (Cantaloupe) (all commodities)	highest utilisation: Total	0.25	0.91	1.000	PRIMO-BE	toddler	P100	540.00	540	3	2b	0.01 - 82.82	0% - 30%	0% - 10%	0% - 30%
VC 0429	Pumpkins (all commodities)	highest utilisation: raw without peel	0.25	0.91	1.000	CN	Child, 1-6 yrs	561	322.71	1852	3	2b	0.41 - 54.6	0% - 20%	0% - 10%	0% - 20%
VC 0432	Watermelon (all commodities)	highest utilisation: Total	0.25	0.91	1.000	CA	Child, <6 yrs	171	953.64	4302	3	2b	78.54 - 168.74	30% - 60%	30% - 40%	30% - 60%
VO 0448	Tomato (all commodities)	highest utilisation: raw with peel	0.15 - 1.7	0.44 - 4.8	1.000	CN	Child, 1-6 yrs	1117	263.76	175	3	2a	1.29 - 26.59	0% - 9%	0% - 3%	0% - 9%
VO 0444	Peppers, chili (all commodities)	highest utilisation: raw with skin	0.49 - 4.15	1.2 - 7	1.000	CN	Gen pop, > 1 yrs	1743	295.71	43	3	2a	0.01 - 8.61	0% - 3%	0% - 3%	0% - 0%
VO 0445	Peppers, sweet (incl. pimiento) (Bell pepper, Paprika) (all commodities)	highest utilisation: raw with skin	0.49	1.2	1.000	CN	Child, 1-6 yrs	1002	169.85	170	3	2b	0.09 - 37.89	0% - 10%	0% - 5%	0% - 10%
VD 0541	Soya bean (dry) (Glycine spp) (all commodities)	highest utilisation: Total	0.01 - 0.79	0	1.000	CN	Child, 1-6 yrs	179	239.05	<25	NR	3	0.01 - 7.26	0% - 2%	0% - 2%	0% - 2%
VR 0589	Potato (all commodities)	highest utilisation: Total	0.16 - 0.67	0.7 - 2	1.000	PRIMO-UK	infant	P97.5	191.10	216	3	2b	0.11 - 64.58	0% - 20%	0% - 7%	0% - 20%
MM 0095	Meat from mammals other than marine mammals	Total	NA	NA	1.000	CN	Child, 1-6 yrs	4329	261.46	NR	NR	1	NA	0%	0%	0%
MM 0095	Meat from mammals other than marine mammals: 20% as fat	Total		0.021	1.000	CN	Child, 1-6 yrs	4329	52.29	NR	NR	1	0.068	0%	0%	0%
MM 0095	Meat from mammals other than marine mammals: 80% as muscle	Total		0.01	1.000	CN	Child, 1-6 yrs	4329	209.17	NR	NR	1	0.130	0%	0%	0%
MF 0100	Mammalian fats (except milk fats)	Total		0.021	1.000	PRIMO-UK	infant	P97.5	18.10	NR	NR	1	0.044	0%	0%	0%

SPIROPIDION (323)

Acute RfD= 0.3 mg/kg bw (300 µg/kg bw)

IESTI

Maximum %ArfD:

60%
all40%
gen pop60%
child

Codex Code	Commodity	Processing	STMR or STMR-P mg/kg	HR or HR-P mg/kg	DCF	Country	Population group	n	Large portion, g/person	Unit weight, edible portion, g	Variability factor	Case	IESTI µg/kg bw/day	% acute RfD rounded	% acute RfD rounded	% acute RfD rounded
MO 0105	Edible offal (mammalian)	Total		0.2	1.000	ZA	Gen pop, > 10 yrs	-	523.58	NR	NR	1	1.880	1%	1%	1%
ML 0106	Milks	Total	0.0055		1.000	PRIMO-UK	infant	P97.5	1080.70	NR	NR	3	0.683	0%	0%	0%
PM 0110	Poultry meat	Total	NA	NA	1.000	CN	Child, 1-6 yrs	175	347.00	NR	NR	1	NA	0%	0%	0%
PM 0110	Poultry meat: 10% as fat	Total		0.00035	1.000	CN	Child, 1-6 yrs	175	34.70	NR	NR	1	0.001	0%	0%	0%
PM 0110	Poultry meat: 90% as muscle	Total		0.00041	1.000	CN	Child, 1-6 yrs	175	312.30	NR	NR	1	0.008	0%	0%	0%
PF 0111	Poultry, fats	Total		0.00035	1.000	CA	Child, <6 yrs	66	49.38	NR	NR	1	0.001	0%	0%	0%
PO 0111	Poultry, edible offal (includes kidney and liver)	Total		0.00033	1.000	CN	Gen pop, > 1 yrs	421	345.63	NR	NR	1	0.002	0%	0%	0%
PE 0112	Eggs	Total		0.00089	1.000	PRIMO-UK	infant	P97.5	108.00	NR	NR	1	0.011	0%	0%	0%

Annex 5: Reports and other documents resulting from previous joint meetings of the FAO panel of experts on pesticide residues in food and the environment and the WHO core assessment group on pesticide residues.

1. Principles governing consumer safety in relation to pesticide residues. Report of a meeting of a WHO Expert Committee on Pesticide Residues held jointly with the FAO Panel of Experts on the Use of Pesticides in Agriculture. FAO Plant Production and Protection Division Report, No. PL/1961/11; WHO Technical Report Series, No. 240, 1962.
2. Evaluation of the toxicity of pesticide residues in food. Report of a Joint Meeting of the FAO Committee on Pesticides in Agriculture and the WHO Expert Committee on Pesticide Residues. FAO Meeting Report, No. PL/1963/13; WHO/Food Add./23, 1964.
3. Evaluation of the toxicity of pesticide residues in food. Report of the Second Joint Meeting of the FAO Committee on Pesticides in Agriculture and the WHO Expert Committee on Pesticide Residues. FAO Meeting Report, No. PL/1965/10; WHO/Food Add./26.65, 1965.
4. Evaluation of the toxicity of pesticide residues in food. FAO Meeting Report, No. PL/1965/10/1; WHO/Food Add./27.65, 1965.
5. Evaluation of the hazards to consumers resulting from the use of fumigants in the protection of food. FAO Meeting Report, No. PL/1965/10/2; WHO/Food Add./28.65, 1965.
6. Pesticide residues in food. Joint report of the FAO Working Party on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Agricultural Studies, No. 73; WHO Technical Report Series, No. 370, 1967.
7. Evaluation of some pesticide residues in food. FAO/PL:CP/15; WHO/Food Add./67.32, 1967.
8. Pesticide residues. Report of the 1967 Joint Meeting of the FAO Working Party and the WHO Expert Committee. FAO Meeting Report, No. PL:1967/M/11; WHO Technical Report Series, No. 391, 1968.
9. 1967 Evaluations of some pesticide residues in food. FAO/PL:1967/M/11/1; WHO/Food Add./68.30, 1968.
10. Pesticide residues in food. Report of the 1968 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Agricultural Studies, No. 78; WHO Technical Report Series, No. 417, 1968.
11. 1968 Evaluations of some pesticide residues in food. FAO/PL:1968/M/9/1; WHO/Food Add./69.35, 1969.
12. Pesticide residues in food. Report of the 1969 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Group on Pesticide Residues. FAO Agricultural Studies, No. 84; WHO Technical Report Series, No. 458, 1970.
13. 1969 Evaluations of some pesticide residues in food. FAO/PL:1969/M/17/1; WHO/Food Add./70.38, 1970.
14. Pesticide residues in food. Report of the 1970 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Agricultural Studies, No. 87; WHO Technical Report Series, No. 4574, 1971.
15. 1970 Evaluations of some pesticide residues in food. AGP:1970/M/12/1; WHO/Food Add./71.42, 1971.

16. Pesticide residues in food. Report of the 1971 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Agricultural Studies, No. 88; WHO Technical Report Series, No. 502, 1972.
17. 1971 Evaluations of some pesticide residues in food. AGP:1971/M/9/1; WHO Pesticide Residue Series, No. 1, 1972.
18. Pesticide residues in food. Report of the 1972 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Agricultural Studies, No. 90; WHO Technical Report Series, No. 525, 1973.
19. 1972 Evaluations of some pesticide residues in food. AGP:1972/M/9/1; WHO Pesticide Residue Series, No. 2, 1973.
20. Pesticide residues in food. Report of the 1973 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Agricultural Studies, No. 92; WHO Technical Report Series, No. 545, 1974.
21. 1973 Evaluations of some pesticide residues in food. FAO/AGP/1973/M/9/1; WHO Pesticide Residue Series, No. 3, 1974.
22. Pesticide residues in food. Report of the 1974 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Agricultural Studies, No. 97; WHO Technical Report Series, No. 574, 1975.
23. 1974 Evaluations of some pesticide residues in food. FAO/AGP/1974/M/11; WHO Pesticide Residue Series, No. 4, 1975.
24. Pesticide residues in food. Report of the 1975 Joint Meeting of the FAO Working Party of Experts on Pesticide Residues and the WHO Expert Committee on Pesticide Residues. FAO Plant Production and Protection Series, No. 1; WHO Technical Report Series, No. 592, 1976.
25. 1975 Evaluations of some pesticide residues in food. AGP:1975/M/13; WHO Pesticide Residue Series, No. 5, 1976.
26. Pesticide residues in food. Report of the 1976 Joint Meeting of the FAO Panel of Experts on Pesticide Residues and the Environment and the WHO Expert Group on Pesticide Residues. FAO Food and Nutrition Series, No. 9; FAO Plant Production and Protection Series, No. 8; WHO Technical Report Series, No. 612, 1977.
27. 1976 Evaluations of some pesticide residues in food. AGP:1976/M/14, 1977.
28. Pesticide residues in food – 1977. Report of the Joint Meeting of the FAO Panel of Experts on Pesticide Residues and Environment and the WHO Expert Group on Pesticide Residues. FAO Plant Production and Protection Paper 10 Rev, 1978.
29. Pesticide residues in food: 1977 evaluations. FAO Plant Production and Protection Paper 10 Suppl., 1978.
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Annex 6: Livestock dietary burden calculations

FENPYROXIMATE (193)

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Corn, field stover	AF/AS	4.1	HR	83	4.94	15	25	40		0.741	1.23	1.98	
Corn, field forage/silage	AF/AS	1.3	HR	40	3.25		75	60			2.44	1.95	
Citrus dried pulp	AB	1.8	STMR	91	1.98	10				0.198			
Corn, field asp gr fn	CM/CF	0.86	STMR	85	1.01	5				0.051			
Corn, field grain	GC	0.01	STMR	88	0.01	70			75	0.008			0.0085
Corn, field milled bypds	CM/CF	0.0037	STMR	85	0.00				5				0.0002
Corn gluten feed	CM/CF	0.0015	STMR	40	0.00				20				0.0008
Total						100	100	100	100	0.997	3.67	3.93	0.00949

DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Bean vines	AL	7.5	HR	35	21.43		20				4.286		
Corn, field stover	AF/AS	4.1	HR	83	4.94	15	20	40		0.741	0.988	1.98	
Corn, field forage/silage	AF/AS	1.3	HR	40	3.25	30	40	60	50	0.975	1.300	1.95	1.63
Citrus dried pulp	AB	1.8	STMR	91	1.98	10	20			0.198	0.396		
Corn, field grain	GC	0.01	STMR	88	0.01	45			50	0.005			0.01
Total						100	100	100	100	1.92	6.969	3.93	1.63

POULTRY BROILER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Corn, field grain	GC	0.01	STMR	88	0.01	75	70		70	0.00852	0.0080		0.00795
Corn, field milled bypds	CM/CF	0.0037	STMR	85	0.00	25	30			0.0011	0.0013		
Corn, field hominy meal	CM/CF	0.0015	STMR	88	0.00			20				0.000340909	
Total						100	100	20	70	0.00961	0.00926	0.000341	0.00795

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Corn, field grain	GC	0.01	STMR	88	0.01	75	70		80	0.00852	0.00795		0.0091
Corn, field milled bypds	CM/CF	0.0037	STMR	85	0.00	25	30			0.00109	0.0013		
Corn gluten feed	CM/CF	0.0015	STMR	40	0.00				10				0.0004
Corn, field hominy meal	CM/CF	0.0015	STMR	88	0.00			20				0.000340909	
Total						100	100	20	90	0.00961	0.00926	0.000341	0.00947

FENPYROXIMATE (193)

ESTIMATED MEAN DIETARY BURDEN

BEEF CATTLE														MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)				
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP	
Corn, field stover	AF/AS	2.05	STMR/STMR-P	83	2.47	15	25	40		0.3705	0.617	0.99		
Citrus dried pulp	AB	1.8	STMR/STMR-P	91	1.98	10	5	30		0.1978	0.099	0.59		
Corn, field asp gr fn	CM/CF	0.86	STMR/STMR-P	85	1.01	5				0.0506				
Corn, field forage/silage	AF/AS	0.38	STMR/STMR-P	40	0.95		70	30			0.665	0.29		
Corn, field grain	GC	0.01	STMR/STMR-P	88	0.01	70			75	0.0080			0.0085	
Corn, field milled bypds	CM/CF	0.0037	STMR/STMR-P	85	0.00				5				0.0002	
Corn gluten feed	CM/CF	0.0015	STMR/STMR-P	40	0.00				20				0.0008	
Total						100	100	100	100	0.627	1.38137098	1.8663584	0.00949037	

DAIRY CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Bean vines	AL	2.8	STMR/STMR-P	35	8.00		20	0			1.6	0.00	
Corn, field stover	AF/AS	2.05	STMR/STMR-P	83	2.47	15	20	40		0.370	0.4939759	0.99	
Citrus dried pulp	AB	1.8	STMR/STMR-P	91	1.98	10	20	30		0.198	0.3956044	0.59	

Annex 6

Corn, field forage/silage	AF/AS	0.38	STMR/STMR-P	40	0.95	30	40	30	50	0.285	0.38	0.29	0.475
Corn, field grain	GC	0.01	STMR/STMR-P	88	0.01	45			50	0.005			0.00568182
Total						100	100	100	100	0.858	2.8695803	1.87	0.481

POULTRY BROILER													
MEAN													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Corn, field grain	GC	0.01	STMR/STMR-P	88	0.01	75	70		70	0.01	0.0080		0.00795455
Corn, field milled bypdts	CM/CF	0.0037	STMR/STMR-P	85	0.00	25	30			0.0011	0.0013		
Corn, field hominy meal	CM/CF	0.0015	STMR/STMR-P	88	0.00			20				0.00034091	
Total						100	100	20	70	0.00961	0.00926	0.000341	0.00795

POULTRY LAYER													
MEAN													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Corn, field grain	GC	0.01	STMR/STMR-P	88	0.01	75	70		80	0.008523	0.00795455		0.00909091
Corn, field milled bypdts	CM/CF	0.0037	STMR/STMR-P	85	0.00	25	30			0.001088	0.0013		
Corn gluten feed	CM/CF	0.0015	STMR/STMR-P	40	0.00				10				0.000375
Corn, field hominy meal	CM/CF	0.0015	STMR/STMR-P	88	0.00			20				0.00034091	
Total						100	100	20	90	0.00961	0.00926	0.000341	0.00947

FIPRONIL (202)

Fipronil desulfinyl

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													
MAX													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP

Annex 6

Rice straw	AF/AS	0.31459	HR	90	0.35		10	60	55		0.035	0.21	0.192
Kale leaves	AM/AV	0.01784	HR	15	0.12		20				0.024		
Lespedeza forage	AL	0.0119	HR	22	0.05			20				0.011	
Wheat forage	AF/AS	0.01322	HR	25	0.05		10	20			0.005	0.011	
Pea vines	AL	0.0119	HR	25	0.05		20				0.01		
Potato culls	VR	0.0094	HR	20	0.05	30	30			0.014	0.014		
Barley forage	AF/AS	0.01322	HR	30	0.04		10				0.004		
Sorghum, grain forage	AF/AS	0.01322	HR	35	0.04	15				0.006			
Potato process waste	AB	0.0016	STMR	12	0.01	30				0.004			
Rice bran/pollard	CM/CF	0.00736	STMR	90	0.01	15			20	0.001			0.002
Soybean meal	SM	0.00612	STMR	92	0.01	5			25	0.000			0.002
Barley grain	GC	0.00134	STMR	88	0.00	5				0.000			
Total						100	100	100	100	0.0254	0.0921	0.2311	0.1955

DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.31459	HR	90	0.35		5	20	25		0.017	0.07	0.087
Kale leaves	AM/AV	0.01784	HR	15	0.12		20	40			0.024	0.048	
Lespedeza forage	AL	0.0119	HR	22	0.05	40		40		0.022		0.022	
Wheat forage	AF/AS	0.01322	HR	25	0.05	20	15			0.011	0.008		
Pea vines	AL	0.0119	HR	25	0.05		20				0.01		
Potato culls	VR	0.0094	HR	20	0.05	10	30			0.005	0.014		
Barley forage	AF/AS	0.01322	HR	30	0.04		10				0.004		
Oat forage	AF/AS	0.01322	HR	30	0.04	10				0.004			
Sorghum, grain forage	AF/AS	0.01322	HR	35	0.04	20			15	0.008			0.006
Corn, field forage/silage	AF/AS	0.005	HR	40	0.01				10				0.001
Rice bran/pollard	CM/CF	0.00736	STMR	90	0.01				10				8E-04
Soybean meal	SM	0.00612	STMR	92	0.01				40				0.003
Total						100	100	100	100	0.0489	0.0772	0.1391	0.0978

Annex 6

POULTRY BROILER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.0094	HR	20	0.05		10				0.005		
Rice bran/pollard	CM/CF	0.00736	STMR	90	0.01	10	10	20	5	0.001	8E-04	0.002	4E-04
Soybean meal	SM	0.00612	STMR	92	0.01	25	40	25	35	0.002	0.003	0.002	0.002
Barley grain	GC	0.00134	STMR	88	0.00	65	40	15	10	0.001	6E-04	2E-04	2E-04
Bean seed	VD	0.001275	STMR	88	0.00			40				6E-04	
Corn, field grain	GC	0.001	STMR	88	0.00				50				6E-04
Total						100	100	100	100	0.0035	0.0088	0.0041	0.0035

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat forage	AF/AS	0.01322	HR	25	0.05		10				0.005		
Pea vines	AL	0.0119	HR	25	0.05		10				0.005		
Potato culls	VR	0.0094	HR	20	0.05		10				0.005		
Beet, sugar tops	AM/AV	0.01	HR	23	0.04		5				0.002		
Rice bran/pollard	CM/CF	0.00736	STMR	90	0.01	10	5	20	20	0.001	4E-04	0.002	0.002
Soybean meal	SM	0.00612	STMR	92	0.01	25	25	25	30	0.002	0.002	0.002	0.002
Barley grain	GC	0.00134	STMR	88	0.00	65	35	15		0.001	5E-04	2E-04	
Bean seed	VD	0.001275	STMR	88	0.00			40				6E-04	
Corn, field grain	GC	0.001	STMR	88	0.00				50				0.00
Total						100	100	100	100	0.0035	0.0195	0.0041	0.0042

Fipronil desulfinyl

ESTIMATED MEAN DIETARY BURDEN													
BEEF CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.05843	STMR/STMR-P	90	0.06		10	60	55		0.006	0.039	0.036
Lespedeza forage	AL	0.003635	STMR/STMR-P	22	0.02			20				0.003	
Pea vines	AL	0.003635	STMR/STMR-P	25	0.01		20	20			0.003	0.003	
Potato process waste	AB	0.0016	STMR/STMR-P	12	0.01	30	40			0.004	0.005		
Cowpea forage	AL	0.003635	STMR/STMR-P	30	0.01		30				0.004		
Rice bran/pollard	CM/CF	0.00736	STMR/STMR-P	90	0.01	15			20	0.001226667			0.002
Soybean meal	SM	0.00612	STMR/STMR-P	92	0.01	5			25	0.000332609			0.002
Corn, field forage/silage	AF/AS	0.002	STMR/STMR-P	40	0.01	15				0.00075			
Potato culls	VR	0.0005	STMR/STMR-P	20	0.00	30				0.001			
Barley grain	GC	0.00134	STMR/STMR-P	88	0.00	5				0.000			
Total						100	100	100	100	0.0071	0.0184	0.0452	0.0390

Annex 6

DAIRY CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.05843	STMR/STMR-P	90	0.06		5	20	25		0.003	0.013	0.016
Lespedeza forage	AL	0.003635	STMR/STMR-P	22	0.02	40		60		0.006609091		0.01	
Pea vines	AL	0.003635	STMR/STMR-P	25	0.01	0	20			0	0.003		
Potato process waste	AB	0.0016	STMR/STMR-P	12	0.01	10	30			0.001333333	0.004		
Cowpea forage	AL	0.003635	STMR/STMR-P	30	0.01	0	15			0	0.002		
Crown vetch forage	AL	0.003635	STMR/STMR-P	30	0.01	0		20		0		0.002	
Rice bran/pollard	CM/CF	0.00736	STMR/STMR-P	90	0.01	15	20		10	0.001226667	0.002		8E-04
Soybean meal	SM	0.00612	STMR/STMR-P	92	0.01	10	10		60	0.000665217	7E-04		0.004
Corn, field forage/silage	AF/AS	0.002	STMR/STMR-P	40	0.01	25			5	0.00125			3E-04
Total						100	100	100	100	0.0111	0.0143	0.0253	0.0213

POULTRY BROILER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice bran/pollard	CM/CF	0.00736	STMR/STMR-P	90	0.01	10	10	20	5	0.001	8E-04	0.002	4E-04
Soybean meal	SM	0.00612	STMR/STMR-P	92	0.01	25	40	25	35	0.002	0.003	0.002	0.002
Potato culls	VR	0.0005	STMR/STMR-P	20	0.00		10				3E-04		
Barley grain	GC	0.00134	STMR/STMR-P	88	0.00	65	40	15	10	0.001	6E-04	2E-04	2E-04
Bean seed	VD	0.001275	STMR/STMR-P	88	0.00			40				6E-04	
Corn, field grain	GC	0.001	STMR/STMR-P	88	0.00				50				6E-04
Total						100	100	100	100	0.0035	0.0043	0.0041	0.0035

POULTRY LAYER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Pea vines	AL	0.003635	STMR/STMR-P	25	0.01		10				0.001		
Rice bran/pollard	CM/CF	0.00736	STMR/STMR-P	90	0.01	10	5	20	20	0.000817778	4E-04	0.002	0.002
Soybean meal	SM	0.00612	STMR/STMR-P	92	0.01	25	25	25	30	0.001663043	0.002	0.002	0.002
Beet, sugar tops	AM/AV	0.001	STMR/STMR-P	23	0.00		5				2E-04		
Potato culls	VR	0.0005	STMR/STMR-P	20	0.00		10				3E-04		
Barley grain	GC	0.00134	STMR/STMR-P	88	0.00	65	45	15		0.000989773	7E-04	2E-04	
Bean seed	VD	0.001275	STMR/STMR-P	88	0.00			40				6E-04	

Annex 6

POULTRY LAYER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Corn, field grain	GC	0.001	STMR/STMR-P	88	0.00				50				6E-04
Total						100	100	100	100	0.0035	0.0047	0.0041	0.0042

Fipronil MB46136

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.334	HR	90	0.37		10	60	55		0.037	0.223	0.204
Potato culls	VR	0.0268	HR	20	0.13	30	30	10		0.040	0.04	0.013	
Beet, sugar tops	AM/AV	0.02	HR	23	0.09		20				0.017		
Barley straw	AF/AS	0.0532	HR	89	0.06	10	20	30		0.006	0.012	0.018	
Potato process waste	AB	0.00505	STMR	12	0.04	30	20			0.013	0.008		
Corn, field forage/silage	AF/AS	0.01	HR	40	0.03	5				0.001			
Soybean meal	SM	0.0174	STMR	92	0.02	5			45	0.001			0.009
Rye grain	GC	0.004	STMR	88	0.00	20				0.001			
Total						100	100	100	100	0.0619	0.1151	0.2540	0.2126

DAIRY CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.334	HR	90	0.37		5	20	25		0.019	0.074	0.093
Potato culls	VR	0.0268	HR	20	0.13	10	30	10		0.013	0.04	0.013	
Beet, sugar tops	AM/AV	0.02	HR	23	0.09		30				0.026		
Kale leaves	AM/AV	0.01135	HR	15	0.08			40				0.03	
Barley straw	AF/AS	0.0532	HR	89	0.06	10	25			0.006	0.015		
Potato process waste	AB	0.00505	STMR	12	0.04	10	10			0.004	0.004		
Triticale straw	AF/AS	0.0241	HR	90	0.03			30				0.008	
Corn, field forage/silage	AF/AS	0.01	HR	40	0.03	35			25	0.009			0.006
Soybean meal	SM	0.0174	STMR	92	0.02	10			50	0.002			0.009

Annex 6

DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rye grain	GC	0.004	STMR	88	0.00	20				0.001			
Barley grain	GC	0.0027	STMR	88	0.00	5				0.000			
Total						100	100	100	100	0.0353	0.1040	0.1259	0.1085

POULTRY BROILER														MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)				
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP	
Potato culls	VR	0.0268	HR	20	0.13		10				0.013			
Soybean meal	SM	0.0174	STMR	92	0.02	25	40	25	35	0.005	0.008	0.005	0.007	
Cassava/tapioca roots	VR	0.00212	HR	37	0.01		10				6E-04			
Rye grain	GC	0.004	STMR	88	0.00	35	40	50		0.002	0.002	0.002		
Triticale grain	GC	0.004	STMR	89	0.00	40				0.002				
Wheat grain	GC	0.004	STMR	89	0.00			25	10			0.001	4E-04	
Corn, field grain	GC	0.002	STMR	88	0.00				55				0.001	
Total						100	100	100	100	0.0081	0.0234	0.0081	0.0083	

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.0268	HR	20	0.13		10				0.013		
Beet, sugar tops	AM/AV	0.02	HR	23	0.09		5				0.004		
Barley straw	AF/AS	0.0532	HR	89	0.06		5				0.003		
Soybean meal	SM	0.0174	STMR	92	0.02	25	25	25	30	0.005	0.005	0.005	0.006
Cassava/tapioca roots	VR	0.00212	HR	37	0.01		5				3E-04		
Rye grain	GC	0.004	STMR	88	0.00	35	35	35		0.002	0.002	0.002	
Triticale grain	GC	0.004	STMR	89	0.00	40				0.002			
Wheat grain	GC	0.004	STMR	89	0.00		15	40			7E-04	0.002	
Corn, field grain	GC	0.002	STMR	88	0.00				70				0.00
Total						100	100	100	100	0.0081	0.0280	0.0081	0.0073

Fipronil MB46136

ESTIMATED MEAN DIETARY BURDEN													
BEEF CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.089585	STMR/STMR-P	90	0.10		10	60	55		0.01	0.06	0.055
Potato process waste	AB	0.00505	STMR/STMR-P	12	0.04	30	40	5		0.012625	0.017	0.002	
Barley straw	AF/AS	0.02225	STMR/STMR-P	89	0.03	10	20	35		0.0025	0.005	0.009	
Soybean meal	SM	0.0174	STMR/STMR-P	92	0.02	5	20		45	0.000945652	0.004		0.009
Potato culls	VR	0.00361	STMR/STMR-P	20	0.02	30	10			0.005415	0.002		
Corn, field forage/silage	AF/AS	0.004	STMR/STMR-P	40	0.01	5				0.0005			
Rye grain	GC	0.004	STMR/STMR-P	88	0.00	20				0.000909091			
Total						100	100	100	100	0.0229	0.0374	0.0706	0.0633

Annex 6

DAIRY CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.089585	STMR/STMR-P	90	0.10		5	20	25		0.005	0.02	0.025
Potato process waste	AB	0.00505	STMR/STMR-P	12	0.04	10	30			0.004208333	0.013		
Barley straw	AF/AS	0.02225	STMR/STMR-P	89	0.03	10	25			0.0025	0.006		
Triticale straw	AF/AS	0.02	STMR/STMR-P	90	0.02	0		50		0		0.011	
Soybean meal	SM	0.0174	STMR/STMR-P	92	0.02	10	25	15	60	0.001891304	0.005	0.003	0.011
Potato culls	VR	0.00361	STMR/STMR-P	20	0.02	10	15	10		0.001805	0.003	0.002	
Corn, field forage/silage	AF/AS	0.004	STMR/STMR-P	40	0.01	35		5	15	0.0035		5E-04	0.002
Rye grain	GC	0.004	STMR/STMR-P	88	0.00	20				0.000909091			
Barley grain	GC	0.0027	STMR/STMR-P	88	0.00	5				0.000153409			
Total						100	100	100	100	0.0150	0.0313	0.0362	0.0377

POULTRY BROILER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Soybean meal	SM	0.0174	STMR/STMR-P	92	0.02	25	40	25	35	0.005	0.008	0.005	0.007
Potato culls	VR	0.00361	STMR/STMR-P	20	0.02		10				0.002		
Rye grain	GC	0.004	STMR/STMR-P	88	0.00	35	50	50		0.002	0.002	0.002	
Triticale grain	GC	0.004	STMR/STMR-P	89	0.00	40				0.002			
Wheat grain	GC	0.004	STMR/STMR-P	89	0.00			25	10			0.001	4E-04
Corn, field grain	GC	0.002	STMR/STMR-P	88	0.00				55				0.001
Total						100	100	100	100	0.0081	0.0116	0.0081	0.0083

POULTRY LAYER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Barley straw	AF/AS	0.02225	STMR/STMR-P	89	0.03		5				0.001		
Wheat straw	AF/AS	0.02	STMR/STMR-P	88	0.02		5				0.001		
Soybean meal	SM	0.0174	STMR/STMR-P	92	0.02	25	25	25	30	0.004728261	0.005	0.005	0.006
Potato culls	VR	0.00361	STMR/STMR-P	20	0.02		10				0.002		
Beet, sugar tops	AM/AV	0.00297	STMR/STMR-P	23	0.01		5				6E-04		
Rye grain	GC	0.004	STMR/STMR-P	88	0.00	35	35	35		0.001590909	0.002	0.002	
Triticale grain	GC	0.004	STMR/STMR-P	89	0.00	40				0.001797753			

Annex 6

POULTRY LAYER														MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)				
Wheat grain	GC	0.004	STMR/STMR-P	89	0.00		15	40			7E-04	0.002		
Corn, field grain	GC	0.002	STMR/STMR-P	88	0.00				70				0.00	
Total						100	100	100	100	0.0081	0.0118	0.0081	0.0073	

FIPRONIL (202) Total

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.34749	HR	90	0.39		10	60	55		0.039	0.232	0.212
Potato culls	VR	0.02816	HR	20	0.14	30	30	10		0.042	0.042	0.014	
Beet, sugar tops	AM/AV	0.03	HR	23	0.13		20				0.026		
Potato process waste	AB	0.006202	STMR	12	0.05	30	40	5		0.016	0.021	0.003	
Barley straw	AF/AS	0.0432	HR	89	0.05	10		25		0.005		0.012	
Soybean meal	SM	0.020898	STMR	92	0.02	5			45	0.001			0.01
Corn, field forage/silage	AF/AS	0.005	HR	40	0.01	5				0.001			
Rye grain	GC	0.006	STMR	88	0.01	20				0.001			
Total						100	100	100	100	0.0657	0.1276	0.2605	0.2226

DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.34749	HR	90	0.39		5	20	25		0.019	0.077	0.097
Potato culls	VR	0.02816	HR	20	0.14	10	30	10		0.014	0.042	0.014	
Beet, sugar tops	AM/AV	0.03	HR	23	0.13		30				0.039		
Kale leaves	AM/AV	0.01135	HR	15	0.08			40				0.03	
Potato process waste	AB	0.006202	STMR	12	0.05	10	30			0.005	0.016		
Barley straw	AF/AS	0.0432	HR	89	0.05	10	5			0.005	0.002		
Triticale straw	AF/AS	0.034076	HR	90	0.04			30				0.011	
Soybean meal	SM	0.020898	STMR	92	0.02	10			60	0.002			0.014
Corn, field forage/silage	AF/AS	0.005	HR	40	0.01	35			15	0.004			0.002

Annex 6

DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rye grain	GC	0.006	STMR	88	0.01	20				0.001			
Barley grain	GC	0.00402	STMR	88	0.00	5				0.000			
Total						100	100	100	100	0.0323	0.1186	0.1329	0.1120

POULTRY BROILER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.02816	HR	20	0.14		10				0.014		
Soybean meal	SM	0.020898	STMR	92	0.02	25	40	25	35	0.006	0.009	0.006	0.008
Rye grain	GC	0.006	STMR	88	0.01	35	50	50		0.002	0.003	0.003	
Triticale grain	GC	0.006	STMR	89	0.01	40				0.003			
Barley grain	GC	0.00402	STMR	88	0.00				10				5E-04
Bean seed	VD	0.00391	STMR	88	0.00			25				0.001	
Corn, field grain	GC	0.003	STMR	88	0.00				55				0.002
Total						100	100	100	100	0.0108	0.0266	0.0102	0.0103

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.02816	HR	20	0.14		10				0.014		
Beet, sugar tops	AM/AV	0.03	HR	23	0.13		5				0.007		
Barley straw	AF/AS	0.0432	HR	89	0.05		5				0.002		
Wheat straw	AF/AS	0.034076	HR	88	0.04		5				0.002		
Soybean meal	SM	0.020898	STMR	92	0.02	25	25	25	30	0.006	0.006	0.006	0.007
Rye grain	GC	0.006	STMR	88	0.01	35	35	35		0.002	0.002	0.002	
Triticale grain	GC	0.006	STMR	89	0.01	40				0.003			
Cassava/tapioca roots	VR	0.00212	HR	37	0.01		5				3E-04		
Barley grain	GC	0.00402	STMR	88	0.00		10				5E-04		
Bean seed	VD	0.00391	STMR	88	0.00			40				0.002	
Corn, field grain	GC	0.003	STMR	88	0.00				70				0.002
Total						100	100	100	100	0.0108	0.0338	0.0098	0.0092

FIPRONIL (202) Total

ESTIMATED MEAN DIETARY BURDEN													
BEEF CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.104425	STMR/STMR-P	90	0.12		10	60	55		0.012	0.07	0.064
Potato process waste	AB	0.006202	STMR/STMR-P	12	0.05	30	40	5		0.015505	0.021	0.003	
Barley straw	AF/AS	0.03225	STMR/STMR-P	89	0.04	10	20	35		0.003623596	0.007	0.013	
Soybean meal	SM	0.020898	STMR/STMR-P	92	0.02	5	20		45	0.001135761	0.005		0.01
Potato culls	VR	0.00443	STMR/STMR-P	20	0.02	30	10			0.006645	0.002		
Rye grain	GC	0.006	STMR/STMR-P	88	0.01	20				0.001363636			
Corn, field forage/silage	AF/AS	0.002	STMR/STMR-P	40	0.01	5				0.00025			
Total						100	100	100	100	0.0285	0.0463	0.0849	0.0740

DAIRY CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Rice straw	AF/AS	0.104425	STMR/STMR-P	90	0.12		5	20	25		0.006	0.023	0.029
Potato process waste	AB	0.006202	STMR/STMR-P	12	0.05	10	30			0.005168333	0.016		
Barley straw	AF/AS	0.03225	STMR/STMR-P	89	0.04	10	25			0.003623596	0.009		
Triticale straw	AF/AS	0.03	STMR/STMR-P	90	0.03	0		50		0		0.017	
Soybean meal	SM	0.020898	STMR/STMR-P	92	0.02	10	25	15	60	0.002271522	0.006	0.003	0.014
Potato culls	VR	0.00443	STMR/STMR-P	20	0.02	10	15	10		0.002215	0.003	0.002	
Rice hulls	CM/CF	0.00725	STMR/STMR-P	90	0.01	0		5		0		4E-04	
Rye grain	GC	0.006	STMR/STMR-P	88	0.01	20			15	0.001363636			0.001
Corn, field forage/silage	AF/AS	0.002	STMR/STMR-P	40	0.01	40				0.002			
Total						100	100	100	100	0.0166	0.0394	0.0459	0.0437

POULTRY BROILER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Soybean meal	SM	0.020898	STMR/STMR-P	92	0.02	25	40	25	35	0.006	0.009	0.006	0.008
Potato culls	VR	0.00443	STMR/STMR-P	20	0.02		10				0.002		
Rye grain	GC	0.006	STMR/STMR-P	88	0.01	35	50	50		0.002	0.003	0.003	

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Triticale grain	GC	0.006	STMR/STMR-P	89	0.01	40				0.003			
Barley grain	GC	0.00402	STMR/STMR-P	88	0.00				10				5E-04
Bean seed	VD	0.00391	STMR/STMR-P	88	0.00			25				0.001	
Corn, field grain	GC	0.003	STMR/STMR-P	88	0.00				55				0.002
Total						100	100	100	100	0.0108	0.0147	0.0102	0.0103

POULTRY LAYER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Barley straw	AF/AS	0.03225	STMR/STMR-P	89	0.04		5				0.002		
Wheat straw	AF/AS	0.03	STMR/STMR-P	88	0.03		5				0.002		
Soybean meal	SM	0.020898	STMR/STMR-P	92	0.02	25	25	25	30	0.005678804	0.006	0.006	0.007
Potato culls	VR	0.00443	STMR/STMR-P	20	0.02		10				0.002		
Beet, sugar tops	AM/AV	0.00396	STMR/STMR-P	23	0.02		5				9E-04		
Rye grain	GC	0.006	STMR/STMR-P	88	0.01	35	35	35		0.002386364	0.002	0.002	
Triticale grain	GC	0.006	STMR/STMR-P	89	0.01	40				0.002696629			
Barley grain	GC	0.00402	STMR/STMR-P	88	0.00		15				7E-04		
Bean seed	VD	0.00391	STMR/STMR-P	88	0.00			40				0.002	
Corn, field grain	GC	0.003	STMR/STMR-P	88	0.00				70				0.002
Total						100	100	100	100	0.0108	0.0153	0.0098	0.0092

METALAXYL (138)

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat forage	AF/AS	3.3	HR	25	13.20		20	100			2.64	13.2	
Barley forage	AF/AS	3.3	HR	30	11.00		10				1.1		
Trefoil forage	AL	3.3	HR	30	11.00		20				2.2		
Sorghum, grain forage	AF/AS	3.3	HR	35	9.43	15				1.414			
Corn, field forage/silage	AF/AS	3.3	HR	40	8.25		50				4.125		
Rice straw	AF/AS	4.65	HR	90	5.17				55				2.842
Citrus dried pulp	AB	1.34	STMR	91	1.47	10				0.147			
Potato culls	VR	0.02	HR	20	0.10	30				0.030			
Total						55	100	100	55	1.592	10.07	13.2	2.842

DAIRY CATTLE													
DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat forage	AF/AS	3.3	HR	25	13.20	20	20	60		2.640	2.64	7.92	
Barley forage	AF/AS	3.3	HR	30	11.00		10				1.1		
Oat forage	AF/AS	3.3	HR	30	11.00	10		40	5	1.100		4.4	0.55
Trefoil forage	AL	3.3	HR	30	11.00	40	40			4.400	4.4		
Sorghum, grain forage	AF/AS	3.3	HR	35	9.43	30			35	2.829			3.3
Corn, field forage/silage	AF/AS	3.3	HR	40	8.25		30		10		2.475		0.825
Total						100	100	100	50	10.969	10.62	12.32	4.675

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POULTRY BROILER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Carrot culls	VR	0.02	HR	12	0.17		10				0.017		
Total							10				0.017		

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat forage	AF/AS	3.3	HR	25	13.20		10				1.32		
Trefoil forage	AL	3.3	HR	30	11.00		10				1.1		
Cabbage heads, leaves	AM/AV	0.22	HR	15	1.47		5				0.073		
Carrot culls	VR	0.02	HR	12	0.17		10				0.017		
Total							35				2.51		

METALAXYL (138)

ESTIMATED MEAN DIETARY BURDEN													
BEEF CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Cabbage heads, leaves	AM/AV	0.44	STMR/STMR-P	15	2.93		20				0.587		
Corn, field stover	AF/AS	1.94	STMR/STMR-P	83	2.34	15	25	40		0.350602	0.584	0.935	
Sorghum, grain stover	AF/AS	1.94	STMR/STMR-P	88	2.20			60				1.323	
Barley straw	AF/AS	1.94	STMR/STMR-P	89	2.18		5				0.109		
Rice straw	AF/AS	1.94	STMR/STMR-P	90	2.16				55				1.186
Citrus dried pulp	AB	1.34	STMR/STMR-P	91	1.47	10	5			0.147253	0.074		
Trefoil forage	AL	0.192	STMR/STMR-P	30	0.64		20				0.128		
Corn, field forage/silage	AF/AS	0.192	STMR/STMR-P	40	0.48		25				0.12		
Potato culls	VR	0.01	STMR/STMR-P	20	0.05	30				0.015			
Total						55	100	100	55	0.513	1.602	2.258	1.186

DAIRY CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Cabbage heads, leaves	AM/AV	0.44	STMR/STMR-P	15	2.93		20	0			0.587	0	
Corn, field stover	AF/AS	1.94	STMR/STMR-P	83	2.34	15	20	40		0.350602	0.467	0.935	
Rye straw	AF/AS	1.94	STMR/STMR-P	88	2.20	0			5	0			0.11
Sorghum, grain stover	AF/AS	1.94	STMR/STMR-P	88	2.20	0		60		0		1.323	
Barley straw	AF/AS	1.94	STMR/STMR-P	89	2.18	0	10			0	0.218		
Rice straw	AF/AS	1.94	STMR/STMR-P	90	2.16	0			20	0			0.431
Citrus dried pulp	AB	1.34	STMR/STMR-P	91	1.47	10	20			0.147253	0.295		
Wheat forage	AF/AS	0.192	STMR/STMR-P	25	0.77	5				0.0384			
Oat forage	AF/AS	0.192	STMR/STMR-P	30	0.64	10				0.064			
Trefoil forage	AL	0.192	STMR/STMR-P	30	0.64	40	30			0.256	0.192		
Sorghum, grain forage	AF/AS	0.192	STMR/STMR-P	35	0.55	20			15	0.109714			0.082
Corn, field forage/silage	AF/AS	0.192	STMR/STMR-P	40	0.48	0			10	0			0.048
Total						100	100	100	50	0.965969	1.759	2.258	0.672

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POULTRY BROILER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Carrot culls	VR	0.02	STMR/STMR-P	12	0.17		10				0.017		
Total							10				0.017		

POULTRY LAYER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Cabbage heads, leaves	AM/AV	0.44	STMR/STMR-P	15	2.93		5				0.147		
Sorghum, grain stover	AF/AS	1.94	STMR/STMR-P	88	2.20		10				0.22		
Trefoil forage	AL	0.192	STMR/STMR-P	30	0.64		10				0.064		
Carrot culls	VR	0.02	STMR/STMR-P	12	0.17		10				0.017		
Total							35				0.448		

PENDIMETHALIN (292)

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Grass hay	AF/AS	1030	HR	88	1170.45	15	50		40	175.568	585.227273		468.2
Carrot culls	VR	0.38	HR	12	3.17		15	5			0.475	0.158	
Alfalfa hay	AL	2.1	HR	89	2.36	15			10	0.354			0.236
Kale leaves	AM/AV	0.25	HR	15	1.67		20				0.33333333		
Bean vines	AL	0.33	HR	35	0.94			60				0.6	
Almond hulls	AM/AV	0.42	STMR	90	0.47			10				0.047	
Grape pomace, wet	AB	0.05	STMR	15	0.33			20				0.067	
Wheat straw	AF/AS	0.17	HR	88	0.19			5				0.01	
Corn, field forage/silage	AF/AS	0.05	HR	40	0.13		15				0.01875		
Corn, field grain	GC	0.05	STMR	88	0.06	70			50	0.0			0.0
Total						100	100	100	100	176.0	586.1	0.85	468.4

DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Grass hay	AF/AS	1030	HR	88	1170.45	45	60		70	526.705	702.272727		819.3
Carrot culls	VR	0.38	HR	12	3.17	10	15	5		0.317	0.475	0.158	
Alfalfa hay	AL	2.1	HR	89	2.36	20	25		25	0.472	0.58988764		0.59
Kale leaves	AM/AV	0.25	HR	15	1.67			40				0.667	
Bean vines	AL	0.33	HR	35	0.94			55				0.519	
Almond hulls	AM/AV	0.42	STMR	90	0.47	10				0.047			
Corn, field grain	GC	0.05	STMR	88	0.06	15			5	0.009			0.003
Total						100	100	100	100	527.5	703.3	1.34	819.9

Annex 6

POULTRY BROILER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Carrot culls	VR	0.38	HR	12	3.17		10				0.31666667		
Bean seed	VD	0.05	STMR	88	0.06		20	70			0.01136364	0.04	
Corn, field grain	GC	0.05	STMR	88	0.06	75	70		70	0.043	0.03977273		0.04
Pea seed	VD	0.05	STMR	90	0.06	20				0.011			
Soybean seed	VD	0.05	STMR	89	0.06	5				0.003			
Wheat grain	GC	0.05	STMR	89	0.06			30				0.017	
Total						100	100	100	70	0.057	0.37	0.057	0.040

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Carrot culls	VR	0.38	HR	12	3.17		10				0.31666667		
Kale leaves	AM/AV	0.25	HR	15	1.67		5				0.08333333		
Wheat straw	AF/AS	0.17	HR	88	0.19		10				0.01931818		
Bean seed	VD	0.05	STMR	88	0.06		20	70			0.01136364	0.04	
Corn, field grain	GC	0.05	STMR	88	0.06	75	55		80	0.043	0.03125		0.045
Pea seed	VD	0.05	STMR	90	0.06	20				0.011			
Soybean seed	VD	0.05	STMR	89	0.06	5				0.003			
Wheat grain	GC	0.05	STMR	89	0.06			30				0.017	
Total						100	100	100	80	0.057	0.46	0.057	0.045

PENDIMETHALIN (292)

ESTIMATED MEAN DIETARY BURDEN													
BEEF CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Grass hay	AF/AS	462.5	STMR/STMR-P	88	525.57	15	50		40	78.83523	262.8		210.2
Alfalfa hay	AL	0.97	STMR/STMR-P	89	1.09	15			10	0.163483			0.109
Carrot culls	VR	0.0625	STMR/STMR-P	12	0.52		15	5			0.078	0.026	
Almond hulls	AM/AV	0.42	STMR/STMR-P	90	0.47			10				0.047	
Kale leaves	AM/AV	0.05	STMR/STMR-P	15	0.33		20				0.067		
Grape pomace, wet	AB	0.05	STMR/STMR-P	15	0.33			20				0.067	
Bean vines	AL	0.05	STMR/STMR-P	35	0.14			60				0.086	
Corn, field forage/silage	AF/AS	0.05	STMR/STMR-P	40	0.13		15	5			0.019	0.006	
Corn, field grain	GC	0.05	STMR/STMR-P	88	0.06	70			50	0.040			0.028
Total						100	100	100	100	79.0	262.9	0.23	210.4

DAIRY CATTLE													
DAIRY CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Grass hay	AF/AS	462.5	STMR/STMR-P	88	525.57	45	60	0	70	236.5057	315.3	0	367.9
Alfalfa hay	AL	0.97	STMR/STMR-P	89	1.09	20	40		25	0.217978	0.436		0.272
Carrot culls	VR	0.0625	STMR/STMR-P	12	0.52	10		5		0.052083		0.026	
Almond hulls	AM/AV	0.42	STMR/STMR-P	90	0.47	10		10		0.046667		0.047	
Kale leaves	AM/AV	0.05	STMR/STMR-P	15	0.33	0		30		0		0.1	
Grape pomace, wet	AB	0.05	STMR/STMR-P	15	0.33	0		20		0		0.067	
Bean vines	AL	0.05	STMR/STMR-P	35	0.14	0		35		0		0.05	
Corn, field grain	GC	0.05	STMR/STMR-P	88	0.06	15			5	0.008523			0.003
Total						100	100	100	100	236.8	315.8	0.29	368.2

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POULTRY BROILER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Carrot culls	VR	0.0625	STMR/STMR-P	12	0.52		10				0.052		
Bean seed	VD	0.05	STMR/STMR-P	88	0.06		20	70			0.011	0.04	
Corn, field grain	GC	0.05	STMR/STMR-P	88	0.06	75	70		70	0.04	0.04		0.04
Pea seed	VD	0.05	STMR/STMR-P	90	0.06	20				0.01			
Soybean seed	VD	0.05	STMR/STMR-P	89	0.06	5				0.00			
Wheat grain	GC	0.05	STMR/STMR-P	89	0.06			30				0.017	
Total						100	100	100	70	0.057	0.103	0.057	0.040

POULTRY LAYER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Carrot culls	VR	0.0625	STMR/STMR-P	12	0.52		10				0.052		
Kale leaves	AM/AV	0.05	STMR/STMR-P	15	0.33		5				0.017		
Wheat straw	AF/AS	0.05	STMR/STMR-P	88	0.06		10				0.006		
Bean seed	VD	0.05	STMR/STMR-P	88	0.06		20	70			0.011	0.04	
Corn, field grain	GC	0.05	STMR/STMR-P	88	0.06	75	55		80	0.042614	0.031		0.045
Pea seed	VD	0.05	STMR/STMR-P	90	0.06	20				0.011111			
Soybean seed	VD	0.05	STMR/STMR-P	89	0.06	5				0.003			
Wheat grain	GC	0.05	STMR/STMR-P	89	0.06			30				0.017	
Total						100	100	100	80	0.057	0.12	0.057	0.05

PYRASULFOTOLE (321)

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat forage	AF/AS	0.19	HR	25	0.76		20	100			0.152	0.76	
Barley forage	AF/AS	0.19	HR	30	0.63		10				0.063		
Wheat asp gr fn	CM/CF	0.44	STMR	85	0.52	5				0.026			
Barley hay	AF/AS	0.5	HR	100	0.50	15				0.075			
Barley grain	GC	0.02	STMR	88	0.02	50	70		70	0.011	0.016		0.016
Wheat milled bypds	CM/CF	0.014	STMR	88	0.02	30			30	0.005			0.005
Total						100	100	100	100	0.117	0.231	0.76	0.021
DAIRY CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat forage	AF/AS	0.19	HR	25	0.76	20	20	60		0.152	0.152	0.456	
Barley forage	AF/AS	0.19	HR	30	0.63		10				0.063		
Oat forage	AF/AS	0.19	HR	30	0.63	10		40	5	0.063		0.253	0.032
Barley grain	GC	0.02	STMR	88	0.02	45	40		40	0.010	0.009		0.009
Wheat milled bypds	CM/CF	0.014	STMR	88	0.02	25	30		45	0.004	0.005		0.007
Total						100	100	100	90	0.230	0.229	0.709	0.048
POULTRY BROILER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Barley grain	GC	0.02	STMR	88	0.02	75	70	15	10	0.017	0.016	0.003	0.002
Rye grain	GC	0.02	STMR	88	0.02			35				0.008	
Wheat grain	GC	0.02	STMR	89	0.02			50				0.011	
Wheat milled bypds	CM/CF	0.014	STMR	88	0.02	25	20		5	0.004	0.003		8E-04
Total						100	90	100	15	0.021	0.019	0.023	0.003

Annex 6

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat forage	AF/AS	0.19	HR	25	0.76		10				0.076		
Barley grain	GC	0.02	STMR	88	0.02	75	90	15		0.017	0.02	0.003	
Rye grain	GC	0.02	STMR	88	0.02			20				0.005	
Wheat grain	GC	0.02	STMR	89	0.02			20				0.004	
Wheat milled bypdts	CM/CF	0.014	STMR	88	0.02	25		20	30	0.004		0.003	0.005
Total						100	100	75	30	0.021	0.096	0.016	0.005

PYRASULFOTOLE (321)

ESTIMATED MEAN DIETARY BURDEN													
BEEF CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat asp gr fn	CM/CF	0.44	STMR/STMR-P	85	0.52	5				0.025882			
Barley hay	AF/AS	0.21	STMR/STMR-P	100	0.21	15		100		0.0315		0.21	
Oat hay	AF/AS	0.21	STMR/STMR-P	100	0.21		20				0.042		
Barley forage	AF/AS	0.051	STMR/STMR-P	30	0.17		10				0.017		
Barley grain	GC	0.02	STMR/STMR-P	88	0.02	50	70		70	0.011364	0.016		0.016
Wheat milled bypdts	CM/CF	0.014	STMR/STMR-P	88	0.02	30			30	0.004773			0.005
Total						100	100	100	100	0.073519	0.075	0.21	0.021

DAIRY CATTLE													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Barley hay	AF/AS	0.21	STMR/STMR-P	100	0.21	20	0	50		0.042	0	0.105	
Oat hay	AF/AS	0.21	STMR/STMR-P	100	0.21	10	20	50	5	0.021	0.042	0.105	0.011
Barley forage	AF/AS	0.051	STMR/STMR-P	30	0.17	0	10			0	0.017		
Barley grain	GC	0.02	STMR/STMR-P	88	0.02	45	40		40	0.010227	0.009		0.009
Wheat milled bypdts	CM/CF	0.014	STMR/STMR-P	88	0.02	25	30		45	0.003977	0.005		0.007
Total						100	100	100	90	0.077205	0.073	0.21	0.027

Annex 6

POULTRY BROILER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Barley grain	GC	0.02	STMR/STMR-P	88	0.02	75	70	15	10	0.02	0.016	0.003	0.002
Rye grain	GC	0.02	STMR/STMR-P	88	0.02			35				0.008	
Wheat grain	GC	0.02	STMR/STMR-P	89	0.02			50				0.011	
Wheat milled bypds	CM/CF	0.014	STMR/STMR-P	88	0.02	25	20		5	0.00	0.003		8E-04
Total						100	90	100	15	0.02	0.019	0.023	0.003

POULTRY LAYER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Wheat hay	AF/AS	0.21	STMR/STMR-P	100	0.21		10				0.021		
Barley grain	GC	0.02	STMR/STMR-P	88	0.02	75	90	15		0.017045	0.02	0.003	
Rye grain	GC	0.02	STMR/STMR-P	88	0.02			20				0.005	
Wheat grain	GC	0.02	STMR/STMR-P	89	0.02			20				0.004	
Wheat milled bypds	CM/CF	0.014	STMR/STMR-P	88	0.02	25		20	30	0.003977		0.003	0.005
Total						100	100	75	30	0.021023	0.041	0.016	0.005

Annex 6

PYRAZIFLUMID (322)

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Apple pomace, wet	AB	0.9	STMR	40	2.25		20	20			0.45	0.45	
Total							20	20			0.45	0.45	
DAIRY CATTLE													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Apple pomace, wet	AB	0.9	STMR	40	2.25	10	10	10		0.225	0.225	0.225	
Total						10	10	10		0.225	0.225	0.225	
POULTRY BROILER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
No feed items applicable!													
POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
No feed items applicable!													

ESTIMATED MEAN DIETARY BURDEN

POULTRY LAYER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
No feed items applicable!													

Annex 6

SPIROPIDION (323)

ESTIMATED MAXIMUM DIETARY BURDEN													
BEEF CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.9	HR	20	4.50	30	30	10		1.350	1.35	0.45	
Tomato pomace,wet	AB	4.1	STMR	95	4.32			10				0.432	
Potato process waste	AB	0.34	STMR	12	2.83	30	40			0.850	1.133		
Soybean asp gr fn	SM	0.83	STMR	85	0.98	5				0.049			
Soybean meal	SM	0.62	STMR	92	0.67		20	10	65		0.135	0.067	0.438
Soybean seed	VD	0.415	STMR	89	0.47	5	10	20	15	0.023	0.047	0.093	0.07
Soybean hulls	SM	0.28	STMR	90	0.31	10				0.031			
Total						80	100	50	80	2.303	2.665	1.042	0.508

DAIRY CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.9	HR	20	4.50	10	30	10		0.450	1.35	0.45	
Tomato pomace,wet	AB	4.1	STMR	95	4.32			10				0.432	
Potato process waste	AB	0.34	STMR	12	2.83	10	30			0.283	0.85		
Soybean meal	SM	0.62	STMR	92	0.67	10	25	15	60	0.067	0.168	0.101	0.404
Soybean seed	VD	0.415	STMR	89	0.47	10	10	20	10	0.047	0.047	0.093	0.047
Potato dried pulp	AB	0.36	STMR	88	0.41		5				0.02		
Total						40	100	55	70	0.847	2.436	1.076	0.451

Annex 6

POULTRY BROILER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.9	HR	20	4.50		10				0.45		
Soybean meal	SM	0.62	STMR	92	0.67	25	40	25	35	0.168	0.27	0.168	0.236
Soybean seed	VD	0.415	STMR	89	0.47	20	20	15		0.093	0.093	0.07	
Potato dried pulp	AB	0.36	STMR	88	0.41		20				0.082		
Total						45	90	40	35	0.262	0.895	0.238	0.236

POULTRY LAYER													MAX
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.9	HR	20	4.50		10				0.45		
Soybean meal	SM	0.62	STMR	92	0.67	25	25	25	30	0.168	0.168	0.168	0.202
Soybean seed	VD	0.415	STMR	89	0.47	20	15	15		0.093	0.07	0.07	
Potato dried pulp	AB	0.36	STMR	88	0.41		15				0.061		
Total						45	65	40	30	0.262	0.75	0.238	0.202

Annex 6

SPIROPIDION (323)

ESTIMATED MEAN DIETARY BURDEN													
BEEF CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Tomato pomace,wet	AB	4.1	STMR/STMR-P	95	4.32			10				0.432	
Potato process waste	AB	0.34	STMR/STMR-P	12	2.83	30	40			0.85	1.133		
Potato culls	VR	0.24	STMR/STMR-P	20	1.20	30	30	10		0.36	0.36	0.12	
Soybean asp gr fn	SM	0.83	STMR/STMR-P	85	0.98	5				0.048824			
Soybean meal	SM	0.62	STMR/STMR-P	92	0.67		20	10	65		0.135	0.067	0.438
Soybean seed	VD	0.415	STMR/STMR-P	89	0.47	5	10	20	15	0.023315	0.047	0.093	0.07
Soybean hulls	SM	0.28	STMR/STMR-P	90	0.31	10				0.031111			
Total						80	100	50	80	1.313249	1.675	0.712	0.508

DAIRY CATTLE													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Tomato pomace,wet	AB	4.1	STMR/STMR-P	95	4.32		0	10			0	0.432	
Potato process waste	AB	0.34	STMR/STMR-P	12	2.83	10	30			0.283333	0.85		
Potato culls	VR	0.24	STMR/STMR-P	20	1.20	10	30	10		0.12	0.36	0.12	
Soybean meal	SM	0.62	STMR/STMR-P	92	0.67	10	25	15	60	0.067391	0.168	0.101	0.404
Soybean seed	VD	0.415	STMR/STMR-P	89	0.47	10	10	20	10	0.046629	0.047	0.093	0.047
Potato dried pulp	AB	0.36	STMR/STMR-P	88	0.41	0	5			0	0.02		
Total						40	100	55	70	0.517354	1.446	0.746	0.451

POULTRY BROILER													
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.24	STMR/STMR-P	20	1.20		10				0.12		
Soybean meal	SM	0.62	STMR/STMR-P	92	0.67	25	40	25	35	0.17	0.27	0.168	0.236
Soybean seed	VD	0.415	STMR/STMR-P	89	0.47	20	20	15		0.09	0.093	0.07	
Potato dried pulp	AB	0.36	STMR/STMR-P	88	0.41		20				0.082		
Total						45	90	40	35	0.26	0.565	0.238	0.236

Annex 6

POULTRY LAYER													MEAN
Commodity	CC	Residue (mg/kg)	Basis	DM (%)	Residue dw (mg/kg)	Diet content (%)				Residue Contribution (ppm)			
						US- CAN	EU	AU	JP	US-CAN	EU	AU	JP
Potato culls	VR	0.24	STMR/STMR-P	20	1.20		10				0.12		
Soybean meal	SM	0.62	STMR/STMR-P	92	0.67	25	25	25	30	0.168478	0.168	0.168	0.202
Soybean seed	VD	0.415	STMR/STMR-P	89	0.47	20	15	15		0.093258	0.07	0.07	
Potato dried pulp	AB	0.36	STMR/STMR-P	88	0.41		15				0.061		
Total						45	65	40	30	0.261737	0.42	0.238	0.202



An Extra Joint Meeting of the Food and Agriculture Organization of the United Nations (FAO) Panel of Experts on Pesticide Residues in Food and the Environment and the World Health Organization (WHO) Core Assessment Group on Pesticide Residues (JMPR) was held virtually from 6 to 17 September and the 4th and 7th of October, 2021. The Meeting was held in pursuance of recommendations made by previous Meetings and accepted by the governing bodies of FAO and WHO that studies should be undertaken jointly by experts to evaluate possible hazards to humans arising from the occurrence of pesticide residues in foods. During the meeting the FAO Panel of Experts was responsible for reviewing pesticide use patterns (use of good agricultural practices), data on the chemistry and composition of the pesticides and methods of analysis for pesticide residues and for estimating the maximum residue levels that might occur as a result of the use of the pesticides according to good agricultural use practices. The WHO Core Assessment Group was responsible for reviewing toxicological and related data and for estimating, where possible and appropriate, acceptable daily intakes (ADIs) and acute reference doses (ARfDs) of the pesticides for humans. This report contains information on ADIs, ARfDs, maximum residue levels, and general principles for the evaluation of pesticides. The recommendations of the Joint Meeting, including further research and information, are proposed for use by Member governments of the respective agencies and other interested parties.

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